CAPITAL MARKETS AND FINANCIAL ADVISORY SERVICES (CMFAS) EXAMINATIONS

STUDY GUIDE

CM-SIP

CAPITAL MARKETS – SPECIFIED INVESTMENT PRODUCTS - DERIVATIVES AND COLLECTIVE INVESTMENT SCHEMES (SIP)



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The Institute of Banking and Finance

January 2026 (Version 1.1)

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Acknowledgements

IBF would like to express its gratitude to all members of the Capital Markets and Financial Advisory Services (CMFAS) Examinations Board and CMFAS Examinations Industry Panel - Derivatives for their contributions and support in the development of the CMFAS Study Guides and Examinations.

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Preface

Capital Markets – Specified Investment Products – Derivatives and Collective Investment Schemes (SIP)

The objective of the CMFAS Capital Markets – Specified Investment Products – Derivatives and Collective Investment Schemes (CM-SIP) Examination is to test candidates on their knowledge and understanding of the characteristics of Specified Investment Products (SIPs). Under the Securities and Futures Act 2001 (SFA), SIPs include exchange-traded and over-the-counter derivatives products such as futures, options, warrants, binary options, structured notes, and so on.

Candidates are required to pass the relevant modules of the CMFAS examination pertaining to the regulated activity that they intend to conduct. Once they have passed the relevant CMFAS examinations, candidates must ensure that their notification to act as appointed representatives are lodged with the Monetary Authority of Singapore (MAS) on the Public Register via the Representative Notification Framework (RNF) before they can commence any regulated activities. For details, please refer to the MAS Notice on Competency Requirements for Representatives of Holders of Capital Markets Services Licence and Exempt Financial Institutions under the Securities and Futures Act (SFA 04-N22).

Candidates who have passed the CMFAS Examinations are encouraged to continue their learning journey by attending IBF accredited programmes, attaining the required skills leading to IBF Certification. For more information, please visit www.ibf.org.sg.

Organisation of the Study Guide

The Study Guide consists of 12 Chapters, starting with an overview of the derivatives markets. Each Chapter begins with a list of learning objectives and ends with a chapter summary.

The emphasis in each Chapter is to ensure that candidates have a good understanding of simple to complex derivative products, as well as their applications. Examples and case studies are used where appropriate to enhance candidate's understanding of key learning points and application of the topics discussed.

A summary of each Chapter is provided below:

Chapter 1:	**Overview of Derivatives** An overview of the derivatives markets and common characteristics of derivatives and structured products.
Chapter 2-10:	**Futures, Forwards and Swaps** **Futures Strategies** **Options** **Structured Warrants and Daily Leveraged Certificates** **Barrier Options, Binary Options & Callable Contracts**

Structured Deposits and other Structured Products
Structured Notes
Structured Funds
Contracts for Differences (CFDs)

Provides an understanding of the characteristics, features, valuation concepts and techniques of futures, options, warrants, structured products, structured notes, structured funds and structured exchange traded funds**.**

Chapter 11: **Key Products and Investment Risks for Derivatives and Structured Products**

An appreciation of the risks of investing in different types of derivatives and structured products and related risk management methods.

Chapter 12: **Case Studies**

Provides case studies on the various securities and futures products to demonstrate the application of investment tools and analysis.

We have included a set of Review Questions with the answers highlighted in bold to support candidate's learning.

A list of essential readings is also provided in Appendix F. Candidates should ensure that they complete the essential readings before attempting the examination.

The Appendices are provided for candidates' reference and to enhance their understanding of the important concepts on SIPs.

Study Guide Updates

The Study Guide is updated at appropriate intervals to reflect changes and developments in the financial industry. Candidates should ensure that they have the latest version of the Study Guide before sitting for the examination. Please refer to Updates to the Study Guides at www.ibf.org.sg for the latest updates.

Important Notes about the Examination

The CM-SIP Examination is conducted at the IBF Assessment Centre.

The examination includes questions that test candidates' knowledge, understanding and application of SIPs.

The examination comprises 80 multiple-choice questions and multiple response questions, with a duration of 2 hours and 30 minutes. It also includes 6 case studies of 18 questions (6 case studies with 3 questions each). The passing mark is 70%.

Candidates should note the following:

(i) For calculation questions, candidates should assume that:

(a) 365-day count convention applies for Singapore Government Securities (e.g., Singapore government bonds, bills, etc.); and

(b) 360-day count convention applies for all other products (unless otherwise stated).

(ii) Candidates are advised to bring along a non-programmable financial calculator for use during the examination. A formulae sheet will be provided during the examination.

For more information on the examination rules, regulations, and other administrative procedures, please refer to the IBF website at www.ibf.org.sg.

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Chapter 1:
Overview of Derivatives

Learning Objectives
The candidate should be able to: ✓ Describe common underlying asset classes for derivatives and structured products. ✓ Describe the different types of derivatives and their uses. ✓ Describe the characteristics of exchange-traded and over-the-counter derivatives. ✓ Discuss the development of the derivatives market in Singapore and beyond.

1.1 Introduction

A derivative is a financial instrument whose value is linked to the value or volatility of an underlying reference or asset. Derivatives and structured products can be used to implement investment views and to design a broader set of investment opportunities. The underlying reference can be a particular security, an asset, an interest rate, a combination of securities, a basket, an index, or other derivatives. Derivatives can either be traded on centralized and regulated exchanges or between counterparties in the less regulated **over-the-counter (OTC)** markets.

Futures, **forwards**, **swaps**, and **options** are the basic types of derivatives available to investors. Besides standard derivatives, there are also a wide range of more complex or "exotic" derivatives structured by combining similar and different types of derivatives. By varying the features (e.g., strike price, tenure, currency, conditions for execution, etc.) and combining different derivatives, the issuer is able to create a wider range of potential payoff outcomes and risk profiles to attract different investors holding different views.

Derivatives are used for **hedging** (insuring against risk), **speculating** (taking a view on the future direction of the market), **arbitraging** (taking advantage of perceived price differentials between two or more markets or assets to make a profit), as well as in **repackaging** by combining instruments to meet specific investment needs, and to create desired synthetic positions.

Complex or "exotic" derivatives are created by combining different types of derivatives and adding features to a traditional derivatives product. **Barrier options, knock-in and knock-out products**, **swaptions, structured notes** and **structured funds** are examples of complex or exotic derivatives.

Investors also use derivatives to trade, and hedge based on volatility, or their expectations on the degree of price movement – whether prices will be range bound, have wild price swings, or move steadily in one direction for a period of time. With the increasing popularity on trading the volatility of indices, we are also seeing the creation of **leveraged exchange-traded notes (ETNs)** and **leveraged exchange-traded funds (ETFs)** allowing investors and traders to take advantage of an augmented impact of rising or falling percentage volatility. Many leveraged ETFs and ETNs actively use futures and options to replicate or track the direction of the underlying indices or the inverse direction of underlying indices, making such funds and notes effectively "derivative instruments".

1.2 Underlying Asset Classes for Derivatives and Their Uses

When reduced to their building blocks, most derivatives are referenced from or crafted out of five basic asset classes comprising foreign exchange, money market instruments, fixed income, equities and commodities. Indices, baskets, funds or portfolio mixes of these asset classes are extensions and representations of these building blocks. Table 1.2 below shows the underlying asset classes together with the common derivative instruments and structures associated with the assets.

Table 1.2 - Underlying Asset Classes and Related Securities and Derivatives

Underlying Assets	Commonly Traded Securities / Indices	Typical Derivatives / Structured Products
Currency	• The bulk of foreign exchange (FX) trading occurs mainly between major currencies (e.g., USD, EUR, JPY, GBP, CHF) in the spot and forward markets.	• *Currency swaps, options, futures and forwards of major currencies*, account for most of the global volume. • *Non-deliverable forwards (NDFs)* are used as proxies for currencies with strong capital controls. • *OTC currency options, structured deposits and structured notes.*
Money Market, Short Term Treasury Instruments & their respective Interest Rates	• Government T-Bills (1, 3, 6, 9, 12 months). • Short-term interbank instruments. • Commercial papers. • Negotiable certificate of deposits. • Repurchase agreements.	• *Treasury futures and options.* • *Interest rate swaps (e.g., LIBOR swaps).* • *OTC structured deposits and structured notes.*
Fixed Income (Debt)	• Sovereign and corporate debt instruments.	• *Bond swaps, bond forwards, bond futures and bond options.*

Underlying Assets	Commonly Traded Securities / Indices	Typical Derivatives / Structured Products
	• Bond indices.	• *Warrants[1].* • *Credit default swaps.* • *OTC structured notes.*
Equity	• Listed stocks. • Equity or stock indices.	• *Index futures, options, equity warrants.* • *Exchange-traded funds (ETFs) and notes (ETNs)[2].* • *OTC equity-linked notes and participation structures.*
Commodities	• Commodities. • Commodities-linked stocks. • Commodities indices.	• *Futures, forwards and options.* • *ETFs.* • *OTC commodity-linked structured products.*

Derivatives and structured products can allow investors to:

a. Hedge or insure against the risk of unfavourable price movements or potential falls in the value of underlying assets or portfolio of assets held (e.g., using futures, forwards, options);

b. Adopt a view on the future direction of the underlying asset without having to invest directly in it;

c. Augment potential profits beyond what can be gained purely from exposure to the underlying asset (e.g., options, warrants);

d. Change the payoffs profile (e.g., cashflow sequence, size, tenure, currency, risks), which would otherwise be unavailable with the underlying asset (e.g., swaps, futures, forwards, options);

e. Profit from the fall in the value of the underlying assets or indices by taking short positions (e.g., through options and futures);

f. Secure the right without the obligation to decide when to buy or sell the underlying asset using a small investment outlay (e.g., through long options and warrants). A short call/put would represent an obligation, but not the right should the buyer chose to exercise);

g. Arbitrage or take advantage of price differentials between two or more markets to make a profit.

h. Change the nature of an asset or liability to meet specific needs that cannot be met from the standardized financial instruments available in the markets; and

[1] A debt warrant is similar to an equity or company warrant, except that it is attached to a bond or other debt security, giving the holder the right to purchase more bonds or debt securities from the same issuer at a stated price. Equity or company warrants (not debt warrants) are covered in detail in Chapter 5.

[2] ETFs and ETNs are not derivatives but often actively use derivatives instruments in enhancing or hedging. For example, leveraged ETFs and volatility ETFs use futures and options to replicate or track indices.

i. Create synthetic positions without incurring the costs of buying or selling the underlying assets.

1.2.1 Money Market and Other Short Term Interest Rate Instruments

These are short-term debt instruments issued by governments, financial institutions and corporations that are highly liquid and less risky. They make up more than 95% of the trading of futures and options traded on the global exchanges and according to the Bank of International Settlements, they have consistently accounted for more than three-quarters of OTC derivatives trades between 2015 to 2019.

The minimum denominations of these securities are relatively large, and the size of the transactions can be substantial. The maturities of money market securities range from 1 day to 1 year (often less than 90 days). The bulk of the money market is made up of interbank lending and trading in securities which includes **Treasury bills, commercial papers, banker's acceptances, negotiable certificates of deposits** and **repurchase agreements**. Institutional investors are the dominant investors for these instruments.

Many of the money market instruments are benchmarked against interbank rates (e.g., SIBOR, LIBOR, HIBOR, TIBOR, EURIBOR) within the respective markets, with the most popular interbank rate being the LIBOR[3] (London Interbank Offer Rate). LIBOR is an average interbank lending rate calculated using the submission of actual borrowing rates contracted between banks and can be subjected to manipulation if submission banks engage in collusion. It is illegal to collude to manipulate interbank rates and indices in order to engineer and trigger outcomes affecting the derivatives market and some traders and dealers were prosecuted for such actions in the past.

LIBOR as a reference benchmark was phased out and ceased to be used in 2021[3]. It has been replaced by the Secured Overnight Finance Rate (SOFR) as a benchmark interest rate for dollar-denominated derivatives and loans.

1.2.2 Foreign Exchange

Cash investments in local or foreign currencies represent the lowest common denominator for all assets. Foreign exchange derivatives make up the second largest category of derivatives traded globally.

Currency forwards and futures are the most basic form of foreign exchange derivatives which generate different payoff outcomes for investors depending on expected currency movement over time. **Swaps** and **options** can be used to change the underlying currency or interest payoffs given different conditional triggers (exchange rate reaching a particular strike price). The trading of currency forwards, futures, swaps and options is largely dominated by the institutional investors.

In emerging markets where capital flows are tightly controlled and where the currency is not freely convertible and there is low liquidity, hedging against the local currency risk from the emerging market may be difficult. This is partly solved through the use of a **non-deliverable forwards (NDFs)** contract which is created as a proxy to the emerging market currency. NDFs are traded OTC in offshore financial centres and the contracts are settled in foreign currencies other than the NDF.

[3] Cases of misconduct by banks involving their LIBOR submissions have regulators globally signalling transition away from the LIBOR, which underpins most money market as well as derivative contracts. In July 2017, Andrew Bailey announced that the FCA would no longer seek to ask panel banks to submit quotes for LIBOR by the end of 2021, hence discontinuing LIBOR as a global benchmark rate.

1.2.3 Fixed Income Securities

These are borrowings by issuers, mainly made up of governments, government-linked entities or public institutions, financial institutions and large multinational corporations (e.g., **investment grade bonds**, **junk bonds**, etc.). A holder, or investor, of a debt security is essentially providing a loan of a certain amount called the principal to the issuer. In return, the holder receives periodic interest payments, and the principal is returned to him or her at maturity. Differences in size, frequency, and timing of the cash flow result in a return for the holder that is commensurate with the risk taken and opportunity costs incurred. Fixed income derivatives include futures, swaps, and options which provide investors with alternative cash flows, pay off profiles and modified risk exposures compared to the underlying fixed income instrument.

While the financial industry continuously innovates with new products to populate the investment frontier, investment banks in the 1980s were also engaged in the popularising creative repackaging of debt for resale. Banks purchased long-term fixed income securities with higher yield and used these as underlying assets against which they repackaged and issued smaller denominations of marketable fixed income securities. For instance, coupon payments of long-term **Treasury bonds** were stripped to form the underlying asset base for the issuance of **zero-coupon bonds**.

Similarly, portfolios of real estate loans were bought and used as the underlying asset base for issuing **mortgage-backed securities (MBS).** Consumer loans and student loans also became the underlying assets for **collateralised debt obligations (CDOs).** So even without derivatives, there was creative repackaging and issuance of fixed income assets which created new products with varying payoff profiles. Interpreted broadly, various CDOs and stripped zero-coupon bonds are “derived” from other bonds making them fixed-income derivatives.

By the late 1990s, we saw the active promotion of credit derivative instruments. Investment banks started actively repackaging and modifying the credit risk of fixed income securities with the help of credit insurance and **credit default swaps (CDS)** issued by insurance companies. Investment banks also bought select high-yield securities (junk bonds of low credit rating i.e., CCC rating and below), mixed in securities with better credit ratings, and added CDS and credit insurance to create a new cocktail of fixed income instruments for resale.

The credit quality of the underlying asset and the credit management process ultimately plays a critical role in determining the risk and value of the downstream repackaged product. This was dramatically illustrated in the 2008 Global Financial Crisis, when the default of real estate mortgage loans and CDOs, triggered the collapse of Fannie Mae and Freddie Mac in the US, the bailout and sale of several global financial institutions, forced capital restructuring for the global banking community, and the subsequent massive rise in regulatory control and cost.

1.2.4 Equities

Equities provide investors with a chance to participate in the profitability of companies. When earnings prospects and growth are positive, the share or stock’s price rises over time and when prospects are negative, the stock price declines.

Equities are considered more volatile than other traditional asset classes. The ability of equity derivatives to take short positions[4], or grant a right to sell, is particularly useful to hedge the downside volatility of stocks.

[4] While it is possible to take short positions in derivatives in general, shorting put options is not a hedge against downside risks.

Equity futures contracts allow investors to buy (long) or sell (short) equities or shares for future delivery at an agreed price which can be higher or lower than the current price. Equity (long) call options and warrants provide investors with the choice to participate in the upside of the underlying equity with a smaller outlay of funds thereby augmenting potential profits while limiting losses to the small outlay. Equity (long) put options allow investors to take short positions.

For equity-linked derivative structures, investors can try to enhance returns by selling put options to collect the premium from willing counter parties when the market is flat. This allows investors to secure much higher yields than normally afforded by typical deposit instruments in the market, but the higher yield comes with the accompanying risk of having to own underperforming equity securities should they fall to critical trigger price levels. We will review these instruments in Chapters 7 and 8.

1.2.5 Commodities

Commodities are the original asset class that spurred the creation of derivative products in the first place. Unlike financial derivatives, a significant portion of commodity derivatives trading comes from regular hedging needs and opportunistic demands from businesses in the agriculture, mining and manufacturing sectors – representing real underlying demand from businesses.

Demand and supply conditions of the underlying commodities determine the long run direction of each commodity but in the short term, trading can be influenced by other factors such as the movement of the USD. Supply shocks can also make commodities quite volatile, e.g., the Organisation of the Petroleum Exporting Countries (OPEC)' decisions on oil prices. Hence the ability to take short positions with derivatives is also very useful for a commodity investor.

By convention, most commodities futures are denominated in USD. When the USD appreciates against other major currencies, commodity prices tend to move in the opposite direction (and vice versa). In cases when there are extreme shocks to currencies or the equity market, safe-haven commodities like gold, silver and platinum will tend to rise. Futures and options for such safe-haven commodities have become increasingly popular as a means to hedge against or take advantage of the volatility.

1.2.6 Trading with Indices

Investing in index-linked products provides investors with an alternative way of gaining exposure to an asset class, market or particular security. There are futures, option contracts, ETFs and ETNs pegged against equity indices, commodity indices as well as popular currency pairs. Such indirect exposure will allow the investors to:

a. Adopt a cost-effective investment exposure (passively over the long term);

b. Hedge and protect profits or minimize losses on an existing outstanding position;

c. Take a speculative short-term exposure to take advantage of a short-term opportunity;

d. Augment the exposure through leverage (typically 2-3x a reference index) using **ETFs, ETNs, options, futures** and **warrants** on the index of particular underlying asset classes; and

e. Speculate on the potential volatility (or lack of volatility) of a security or the index of an asset class.

In the last few years, we have seen the increasing popularity of leveraged and volatility ETNs and ETFs offered on exchanges catering to traders and investors who are seeking potentially higher returns from the volatility and from leveraging. The most popular of these volatility indices is the Chicago Board of Exchange Volatility Index (CBOE VIX or VIX), which measures the market's perceived volatility of the S&P500 Index in either direction using the price movement of call and put options. VIX options and futures trades are carried out on the CBOE every day. When volatility rises, the VIX rises and when there is less market volatility, VIX falls in value. VIX options and futures enable investors to trade on the volatility regardless of the direction of the market.

To increase the attractiveness for traders, various leveraged ETNs and ETFs have been introduced on the VIX (e.g., ETNs such as the XIV, ZIV VIIX, VIIZ, TVIX, TVIZ, or ETFs such as SVXY, UVXY). These products allow investors to buy or short volatility, i.e., trade on the rise or fall in the percentage volatility of the S&P500.

It is important to highlight that such structures on volatility are extremely risky. For example, on 5 Feb 2018 when the VIX spiked up and more than doubled in a single day, the short volatility ETF issued by Pro-shares, SVXY and XIV issued by Credit Suisse plummeted by more than 80% in a single day, wiping out a few billion dollars overnight on just two instruments.

Trading on both counters were halted temporarily but at the open there was no recovery for traders holding the XIV ETN since a termination event (price level) had been triggered and the issuing bank closed out the synthetic product to protect itself from downside losses forcing traders caught in the wrong bet to realise losses. In the subsequent weeks, in response to pressure to address the highly risky leveraged position created by its product, Pro-Shares, a major issuer of ETFs, cut its leveraging of short and long volatility ETFs (SVXY and UVXY) by half. When the VIX rises by 1%, however, the 3x leveraged products on other types of ETFs remained in play for traders.

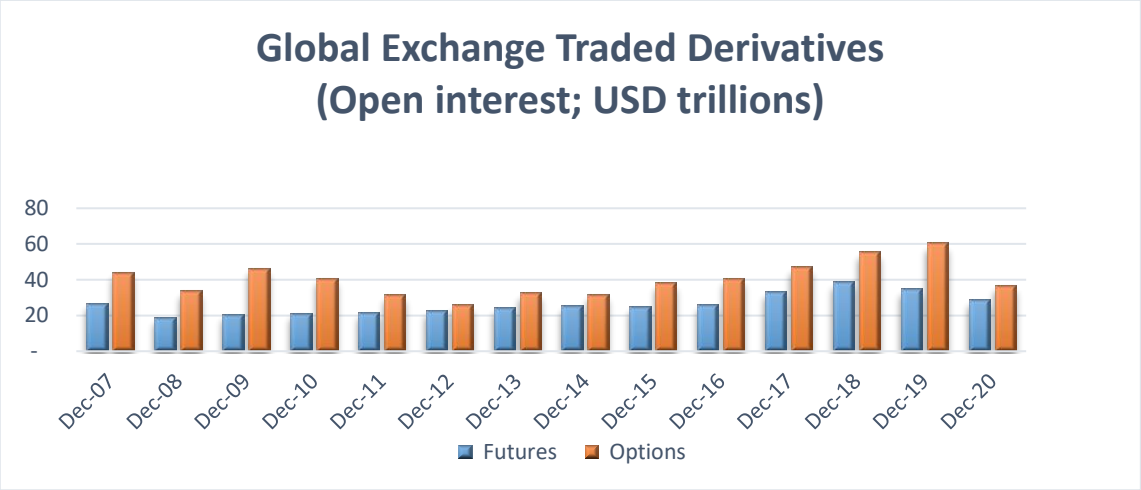
1.3 Exchange Traded Derivatives

Derivatives can either be traded on centralized and regulated exchanges such as the Singapore Exchange Derivatives Trading (SGX-DT), ICE Futures Singapore (IFSG) or between counterparties in the OTC markets. Exchange-traded derivatives are standardized products (mainly futures and options) which are traded on centralized, regulated exchanges (e.g., trading floors or computerized trading). The characteristics of each product are specified according to standard conventions, such as the contract size, underlying asset, delivery or expiration dates and delivery arrangements.

After expiry, each contract will be settled, either by physical delivery or by a cash settlement. Once a trade is executed between a buyer and seller on the exchange, it is recorded with the exchange's **clearing house**. The contracts are ultimately not between the original buyer and the original seller, but between the holders at expiry and the clearing house, who acts as a guarantor that the trade will be settled as originally intended, using pooled initial margins from both sides of the trade.

An example of the Open Interest of Global Exchange-Traded Derivatives is shown in Figure 1.3.

Figure 1.3 - Open Interest of Global Exchange-Traded Derivatives (2007- 2020)



Source: Bank of International Settlement (BIS), BIS Derivatives Statistics[5]

1.4 Over-the-Counter Traded Derivatives

OTC derivatives are traded directly between two parties, without going through an exchange. Transactions are seldom terminated or assigned (i.e., transferred to a third party) prior to maturity unless it is agreed by both counterparties. Banks, other large financial institutions, fund managers and corporations are the main participants on OTC transactions. Historically, once an OTC trade is done, the two parties settle the trade bilaterally. Prior to the credit crisis, which started in 2007, OTC derivatives markets were largely unregulated. In 2018, OTC derivatives trading was estimated to be more than five times the value of derivatives traded on global exchanges[6].

Currency and interest rate forwards and swaps make up the bulk of OTC derivatives. OTC currency forwards and swap contracts are largely driven by trades between international banks. Corporates who need to hedge the underlying currency positions of their cash flows, and some sophisticated investors, may also take part as counterparties but they form the minority.

Credit default swaps (CDS) are the next largest product group traded in the OTC markets. Indeed, up until 2007, CDS was growing in popularity with institutions using credit insurance to enhance the credit rating and pricing of bonds. This "selling off risk", or insurance, became a lucrative business. Insurance companies and financial institutions that sold credit insurance managed to generate good revenues as long as there was no major systemic downturn of the financial sector. However, when mortgage-backed securities started to collapse, CDS, while providing profits for those who bought the product, became a disaster for those who sold it due to the obligation to make payouts for having provided insurance against default.

The key differences between exchanges and OTC markets are shown in Table 1.4.

[5] BIS Derivatives Statistics (https://stats.bis.org).

[6] Source: Bank of International Settlements Statistics Warehouse (https://www.bis.org/statistics/derstats.htm).

Table 1.4 - Key differences between Exchanges and OTC Markets

	Exchanges	OTC Markets
Characteristics	• Standardised contracts • Clearing house acts as the central counterparty to: - match trades - eliminate credit risk • Margin required as collateral • Transparency • Regulated	• Non-standardised, customisable contracts • Bilateral agreements • Master agreement (ISDA) • Credit risk between transacting parties (ISDA CSA is sometimes used to mitigate the risk) • Largely unregulated (before 2008)

1.4.1 Regulation of OTC Derivatives Trading

Post-mortem reviews of the 2008 Global Financial Crisis attributed a good portion of the blame to the proliferation of products and a lack of monitoring and regulation of OTC financial derivatives. This was especially the case for credit derivatives, which were sold by investment banks to institutional clients and cascaded down to private and retail clients. Financial derivatives which originally began as tools for risk management, became instruments for speculative gains. Several famous investment experts had warned of the potential risk of derivatives to the financial system, the most famous quote being from Warren Buffet in 2002 who said, "In my view, derivatives are financial weapons of mass destruction, carrying dangers that, while now latent, are potentially lethal[7]."

In the aftermath of the international financial crisis, the leaders of the G20 countries, in September 2009, agreed to push for changes and reforms in the global financial system. This included having greater transparency in the trading of derivatives, which required getting more standardized OTC derivative contracts to be traded on exchanges or electronic trading platforms, clearing through central counterparties, reporting to trade repositories, and subjecting non-centrally cleared contracts to higher capital requirements.

1.4.1.1 Europe

In 2012, the European Market Infrastructure Regulation[8] (EMIR) was introduced to set the framework for regulating OTC derivatives in Europe. In line with G20 commitments, EMIR required European financial institutions and their counterparties to measure, report, monitor and clear obligations and manage operational risk, counterparty credit risk in relation to OTC derivative transactions not cleared through a clearing house.

1.4.1.2 United Kingdom

In 2013, the UK passed new legislation for the Financial Services (Banking Reform) Act[9], requiring major universal banks to separate and ring fence their retail and commercial banking activities from investment banking.

[7] Warren Buffet – Chairman's Letter to shareholders of Berkshire Hathaway Inc. 2002 Annual Report.
[8] "Questions and Answers on the proposal to amend the European Market Infrastructure Regulation (EMIR)"- European Commission 2012; https://ec.europa.eu/commission/presscorner/detail/en/MEMO_17_1583.
[9] The Financial Services (Banking Reform) Act (2013); http://www.legislation.gov.uk/ukpga/2013/33/pdfs/ukpga_20130033_en.pdf.

1.4.1.3 United

In the US, the Dodd-Frank Wall Street Reform and Consumer Protection Act[10] was introduced in 2010 where the role of Commodity Futures Trading Commission was enhanced to also cover the regulation of swap trades. Subsequently, in 2013, the Federal Reserve Board announced that banking entities, covered by Section 619 of the Dodd-Frank Wall Street Reform and Consumer Protection Act, would be required to fully conform their activities and investments to the statute and regulations by July 21, 2015. Section 619, commonly referred to as the "Volcker Rule," prohibits insured depository institutions and its affiliates from engaging in short-term proprietary trading of certain securities, derivatives, and other financial instruments for the firm's own account (subject to certain exemptions, including market making and risk-mitigating hedging). It also imposes limits on banking entities' investments in hedge funds and private equity[11].

1.4.1.4 Singapore

Singapore has also taken steps to align industry players with G20 recommendations. These include the increased standardization of derivative contracts, increased central clearing, more reporting to trade repositories and higher capital requirements for non-centrally cleared contracts to reflect their greater level of counterparty risk.

In adopting the appropriate clearing regime, the Monetary Authority of Singapore (MAS) takes into consideration the following factors:

a. Level of systemic risks posed by the product;

b. Depth and liquidity of the market for the product; and

c. Availability of OTC Central Counterparties (CCPs) regulated by MAS as approved clearing houses (ACHs) or recognised clearing houses (RCHs).

From 1 October 2018, the MAS required all SGD and USD fixed and floating rate swaps to be cleared through MAS-regulated CCPs in order to make derivatives trading safer and reduce counterparty risks. These are the more widely traded interest-rate derivatives in Singapore.

For a start, MAS focused on banks that have the largest share of OTC derivatives activities in Singapore that required clearing to address the largest counterparty credit risks in the Singapore OTC derivative markets. Banks whose gross notional outstanding OTC derivatives exceeding S$20 billion are required to clear their trades through MAS-regulated CCPs. These banks accounted for 90% of the OTC derivatives contracts in terms of outstanding notional amount in Singapore. Banks are exempted from the clearing obligations as long as they did not exceed a maximum threshold of S$20 billion gross notional outstanding derivatives contracts booked in Singapore for each of the last four calendar quarters[12].

In line with other jurisdictions such as the US, EU, and Japan, MAS exempted intra-group transactions from the scope of clearing obligations. Such transactions do not transfer risks in or out of a corporate or group. MAS has also exempted public bodies from clearing obligations, including all central banks and governments, as well as international multilateral organisations such as the Bank for International Settlements, the International Monetary Fund and the World Bank. The policy functions of public bodies may require them to

[10] Securities and Exchange Commission, Dodd Frank Act, Section 619.
[11] US Commodity Futures Trading Commission 2010; https://www.cftc.gov/LawRegulation/DoddFrankAct/index.htm.
[12] MAS Response to Feedback Received on Draft Regulations for Mandatory Clearing of Derivatives Contract, May 2018; https://www.mas.gov.sg/publications/consultations/2015/consultation-paper-on-draft-regulations-for-mandatory-clearing-of-derivatives-contracts.

maintain flexibility to respond to a variety of circumstances, which may be constrained if they are subject to clearing obligations.

When implemented, the clearing of interest rate swaps (IRS), which constitutes more than 90% of interest rate derivative contracts booked in Singapore (by gross notional amount outstanding), would significantly reduce systemic risks in the Singapore financial system.

Clearing obligations are more enforceable for IRS trades when both counterparties book and clear their transactions in Singapore. Clearing obligations become more difficult to enforce for some cross-border transactions when the foreign regulators have yet to (or do not) require counterparties to be obliged to seek central clearance. The interaction of jurisdictions' rules on cross-border transactions remains an ongoing challenge. The US and Japan have commenced mandatory clearing; the EU has put in place its mandatory clearing regime which has yet to commence.

In relation to transactions which were traded in Singapore but booked elsewhere, such as foreign subsidiaries or local branches of foreign banks, MAS maintains oversight of such operations through its consolidated supervision of local banking groups. MAS engages banks regularly on their policies regarding booking of OTC derivative contracts and appropriate risk controls as part of banking supervision. The MAS will also assess the risk based on trade data reported to our licensed trade repositories, which includes OTC derivative contracts booked or traded in Singapore.

With the global step up of regulatory requirements and costs, banks have become noticeably more selective on the types of derivatives business. Many banks systematically scaled down their investment banking activities including proprietary trading, capital-intensive activities and higher-risk exotic and credit derivatives businesses from 2008-2016. In 2019, OTC derivatives trading recovered to a certain extent, growing more rapidly than trading in exchange-traded derivatives[13].

1.4.2 Markets in Financial Instruments Directive II

Markets in Financial Instruments Directive II (MiFID II) is a new set of foundational regulations and directives used for the governing of capital markets in the European Union (EU). Banks that are EU entities (domiciled in EU) and their branches are subject to MiFID II. Banks that deal with EU investors (individual or corporate) are also subject to MiFID II. In Singapore, there are many branch operations of EU financial institutions who are required to meet the guidelines for MiFID II.

Although Singapore's local brokerage firms and local banks are not bounded by such EU directives, they are likely to be called upon to support the compliance processes of their European counterparties who have to report on a wide range of financial transactional data so to help them fulfil reporting requirements. As good corporate citizens in a global financial centre, we can expect financial institutions and exchanges in Singapore to assist EU banks in meeting their requirements.

The bulk of the review requirements under MiFID II falls on the shoulders of the senior managers of financial institutions and compliance teams. Much of the additional workflow and documentation requirements should be automated in systems but this will require both operation and sales staff to ensure that the right verification procedures and data are all conducted and recorded in alignment with the requirements under MiFID II. As an investment professional, you should be aware of your own organisation's obligations or decisions on how they wish to align with MiFID II regulations and your role and responsibilities relating any new rules.

[13] Bank of International Settlements Quarterly Review, September 2019.

Some examples of additional support which will be required of financial institutions under MiFID II are discussed below:

Role or Activity	Requirement
You execute a trade for a local client which involves the purchase of EU securities (e.g., BMW, LVMH) from an EU-domiciled counterparty.	The EU counterparty will require supporting transaction data to complete their MiFID II reporting requirements. They will request for some transactional information records, including details about the Singapore-based client and the financial institution. Transactions and post-trade reporting requirements under MiFID II may need to be reported T+1. Post-trade systems will need to reflect the within 15 minutes from trade execution and certain pre-and post-trade information transparency requirements that need to be adhered to under the EU guidelines. *Recommended Action: The local financial institution should review the operational impact for executing and reporting of trades to see if it meets the MiFID II reporting requirements.*
You have EU clients. **For example, you are a local fund manager working for a private bank and you have some EU-based UHNW individuals and institutional clients**.	APAC-based financial institutions (especially private banks and fund managers) are required under MiFID II to protect the client by classifying the client properly (e.g., as a retail client, professional client, or elective professional client, etc.) based on his knowledge, experience, and skill. The logic follows that the professional investor will be more suited to make certain types of investments that a retail investor would not. An individual who does not meet the criteria as a professional investor can choose to be an elective professional investor if he meets the specified quantitative and qualitative requirements set. *Recommended Action: In Singapore, MAS guidelines regarding client classification and client and product suitability are in place. It would make sense to see how much of the information and requirements are similar and if the reporting processes can be consolidated. If it is required under MiFID II obligations to report, the additional information will be added to the systems and processes for the various institutions.*
You work for a financial institution that is a market-maker for several listed securities in the EU and you have a trading desk in Singapore to carry out such activities for APAC clients.	There are extensive regulatory requirements under MiFID II relating to market-making activity which require the market-maker to provide price quotes at specific times and to follow specific control requirements. When consolidated, the OTC trades of the Singapore branch or subsidiary must not trigger the specified limits of the European financial institution.

Role or Activity	Requirement
	Recommended Action: Market-makers for EU securities serving APAC clients should work with their head offices in Europe to ensure that they are able to meet this requirement.
You are a distributor of EU funds in Singapore to APAC clients.	Under the MiFID II requirements, as a fund distributor for EU funds, among other reporting requirements, you have to: a. Ensure that distribution is only to the target market; b. Provide sales information to the EU-issuer and assist in meeting their after-sales responsibilities; and c. Provide a statement on suitability to the EU fund investor on how the investment advice meets their preferences, risk-return objectives and other aspects of their client investment profile. You have to assist the EU issuer in satisfying the reporting and due diligence requirements regarding: a. Reporting on your status and role (e.g., non-independent i.e.; restricted advisor or independent) as distributor; b. Providing evidence that you have carried out KYC and KYP (know-your-client and product) requirements under MiFID II; and c. Disclose of inducements to clients and have demonstrated how these inducements have benefited the client. *Recommended Action: As above.*
You work for a subsidiary of an EU financial institution that conducts both underwriting and secondary placement activities for financial instruments.	Under MiFID II, you are required to identify, manage and disclose conflicts of interest for corporate finance desks before accepting a mandate. This includes disclosures to clients of specific conflicts of interest regarding fees, underwriting and placing operations, targeted investors, and clients and individuals involved. Firms must evaluate the contractual clauses for corporate finance services against MiFID II guidelines. For example: a. When financial institutions have a contract with clients which preclude business with other clients, or ties a client into restrictive clauses, such as accepting ancillary products or services; or b. Financial institutions must manage conflicts arising from serving multiple clients, where your behaviour in serving one client could adversely affect the outcomes and the way you serve other clients.

Role or Activity	Requirement
	Recommended Action: You should review your practices for the investment banking business and identify how you have documented your processes for engagement with clients on contractual activities and how well the management team and other relevant stakeholders (especially legal and compliance) have set in place processes that will help your financial institution meet the MiFID II guidelines.

1.5 Structured Products

Structured products are typically made up of a combination of one or more derivatives and an underlying investment. The underlying can be single securities, a group (or basket) of securities, or indices. These products are popularly sold as structured deposits, structured notes or structured funds offering investors opportunities to optimize the yield or to participate in movement of the underlying.

Structured products are often OTC products and are structured to create greater yield for existing securities held by the client or may give the client the potential participation in the movement of an underlying security, index or basket of securities. There are different combinations and outcomes that can be designed to attract investors with the accompanying risks.

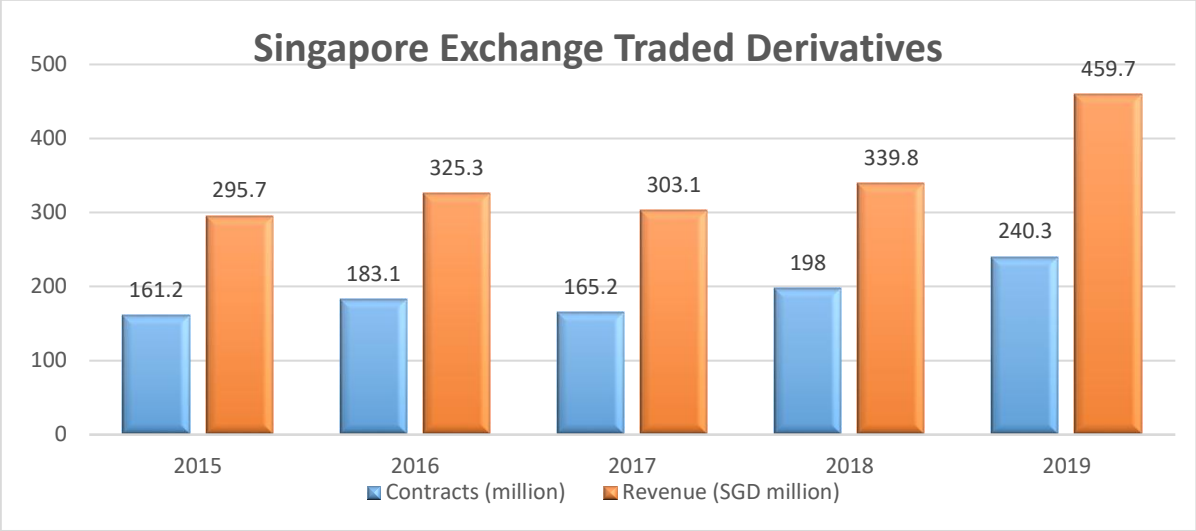
In exchange for returns that are better than the market, investors will bear the risk of having to convert to or own securities if the underlying reaches (or as long as the price does not reach) a particular trigger price level. The structure may often require investors selling put options to the counterparty, which may result in the accumulation of more of the underlying securities as it moves lower - so there are inherent risks that investors must be comfortable bearing when investing in particular types of structured products. They can also involve leveraging that would augment returns or losses.

1.6 The Singapore Derivatives Market

Singapore has a vibrant and growing derivatives market and is a major hub for derivatives trading in Asia. SGX-DT, the derivatives arm of SGX, is a Pan-Asian derivatives exchange and has become the world's most liquid offshore market for the benchmark equity indices of China, India, Japan and Taiwan. It is also the leading commodities exchange for contracts on iron ore and rubber and provides Eurodollar, Euroyen, and Japanese government bond and stock index futures trading services for investors in Asia. It also offers interest rate futures contracts trading and hedging services, including hedging loans and deposits, and interest rate swaps denominated in SGD, USD, and JPY; and intra-commodity, inter-commodity, and inter market trading and arbitrage services.

Figure 1.6(a) provides an overview of the Singapore Exchange Traded Derivatives volume and revenue.

Figure 1.6(a) - Singapore Exchange Traded Derivatives



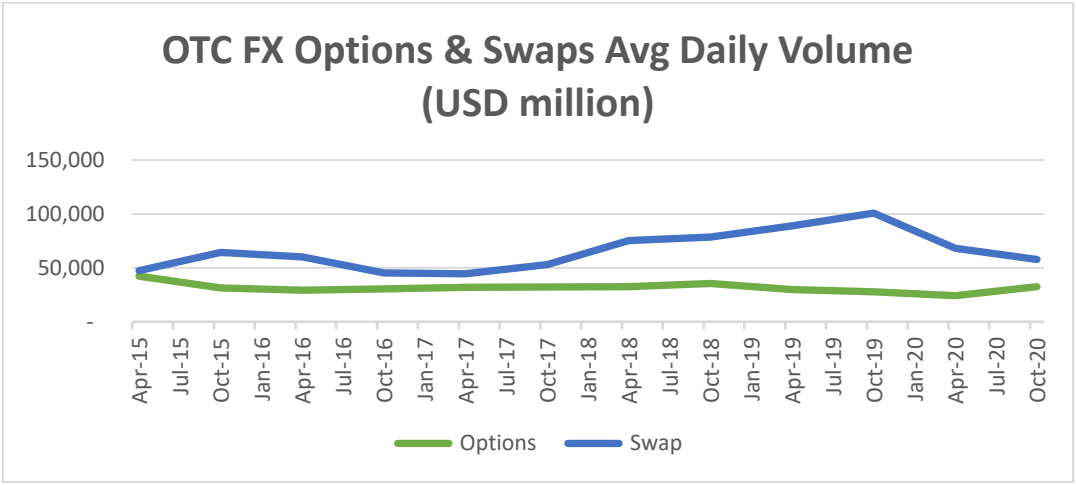
Source: SGX Annual Reports 2015-2019

The volume of exchange-traded derivatives in Singapore has bounded back in the last two years, after falling in 2017.

Singapore, being the leading foreign exchange hub for Asia, and third largest in the world (after New York and London), has benefited from the synergistic relationships from having grown its foreign exchange derivatives business. Singapore, however, lags behind in the other asset classes, when compared to the major exchanges in New York and London.

The commencement of two new derivatives exchange - ICE Futures Singapore and Asia Pacific Exchange - have also added breadth to the derivatives markets in Singapore. Figure 1.6(a) provides an overview of the OTC FX Options and Swaps Volume in Singapore for period 2015 to 2020.

Figure 1.6(b) - OTC FX Options & Swaps Volume in Singapore (USD million)



Source: Singapore Foreign Exchange Market Committee

Based on the October 2020 semi-annual survey conducted by the Singapore Foreign Exchange Market Committee (SFEMC)[14], the average daily reported turnover in OTC foreign exchange derivatives was US$91 billion (forex swaps and options) and this reflected a decrease compared to the previous period. The decline in trading volumes coincided with the global economic downturn brought about by the COVID 19 pandemic.

1.7 Summary

1. A **derivative** is a financial instrument whose value is linked to the value or volatility of an underlying reference or asset. Derivatives can either be traded on centralized and regulated exchanges or between counterparties in the less regulated **over-the-counter (OTC)** markets.
2. Derivatives and **structured products** can allow investors to:

    i. **Hedge** or insure against the risk of unfavourable price movements – potential fall in the value of underlying assets or portfolio of assets held;

    ii. Adopt a view on the future direction of the underlying and take position without having to invest directly in it;

    iii. Augment potential profits generated otherwise gained purely from exposure to the underlying asset;

    iv. Change the payoffs profile (e.g., cashflow sequence, size, tenure, currency, risks), otherwise unavailable in the underlying asset;

    v. Profit from the fall in the value of the underlying assets or indices by taking short position);

    vi. Secure the right without the obligation to decide when to purchase or sell the underlying asset using a small investment outlay;

    vii. **Arbitrage** or take advantage of price differentials between two or more markets to make a profit;

    viii. Change the nature of an asset or liability to meet specific needs that cannot be met from the standardized financial instruments available in the markets; and

    ix. Create **synthetic positions** without incurring the costs of buying or selling the underlying assets.

3. **Futures, forwards, swaps**, **options and warrants** are considered the most basic or "plain vanilla" derivatives. Complex or "exotic" derivatives are created by combining different types of derivatives or by adding features to a traditional derivatives product. The underlying reference for derivatives can be a particular security, a combination of securities, an asset, an interest rate, other derivatives, or an index. At the most basic level, the asset building block for such derivatives include foreign exchange, money market interest rate instruments, fixed income, equities, commodities, or a mix or funds of theses asset classes or indices.

4. Participants offering derivatives and structured products have varying motives. They use derivatives to hedge and manage risk of existing investment positions, take speculative positions, or capitalise on perceived arbitrage opportunities between assets and markets.

[14] SFEMC Semi-Annual Surveys April 2015 – Oct 2020 (https://www.sfemc.org/statistics.html).

5. Money market and short-term interest rates instruments make up the bulk of derivatives trading worldwide. These tend to be low-risk, highly liquid, short-term debt instruments (e.g., **Treasury bills, commercial papers, banker's acceptances, negotiable certificates of deposits** and **repurchase agreements**) issued by governments and financial institutions. FX trading activities form the second largest category of derivatives traded globally. Similar to the underlying, trading in currency swaps, options, futures and forwards are mostly carried out between the major global currencies (USD, EUR, JPY, GBP, AUD).

6. Growth in trading of **non-deliverable forwards** used as proxies for many emerging market currencies has been significant largely because of the strong capital controls set in place for such currencies. Fixed income derivatives include futures, swaps, and options are mainly yield-oriented instruments which provide investors with alternatives cash flows pay off and return profiles to meet their varied objectives.

7. **Credit default swaps (CDS)** are high risk derivative instruments used to take on the risk of loan default.

8. The ability of equity and commodity derivatives to take short positions is particularly useful to hedge the downside volatility of stocks and commodities.

9. A significant portion of commodity derivatives trading comes from regular hedging needs and opportunistic demands from businesses in the agriculture, mining, and manufacturing sectors.

10. By convention, most commodities futures are denominated in USD. When US interest rates rise and the USD appreciate against other major currencies, commodity futures prices tend to move in the opposite direction.

11. When there are extreme shocks in the market, the futures, and options of safe-haven commodities (precious metals) can see price increases due to higher demand for hedging against currency and equity market downside volatility.

12. **Exchange-traded derivatives** are standardized products, traded on exchanges (trading floors or computerized trading) and recorded with the exchange's **clearing house**. After expiry, each contract will be settled, either by physical delivery (typically for commodity underlying assets) or by a cash settlement (typically for financial underlying assets).

13. OTC derivatives are instruments traded directly between two parties, without going through an exchange. Such OTC derivatives transactions are estimated to be eight times larger than the value traded on the global exchanges.

14. In the aftermath of the global financial crisis, the leaders of the G20 countries agreed to push for changes and reforms to the global financial system. This included having greater transparency in the trading of derivatives, which required getting more standardized OTC derivative contracts to be traded on exchanges or electronic trading platforms, clearing through central counterparties, reporting to trade repositories, and having non-centrally cleared contracts subject to higher capital requirements.

15. Singapore being the leading foreign exchange hub for Asia and third largest in the world, after London and New York, has benefited from the synergistic relationships with the growth of its foreign exchange derivatives business. Being one of Asia leading financial centres, the growth in banking in Singapore has also made it one of the largest OTC derivatives hubs in the world.

Chapter 2:
Futures, Forwards & Swaps

Learning Objectives
The candidate should be able to: ✓ Discuss the origins and evolution of futures and futures exchanges. ✓ Explain the economic importance of futures markets. ✓ Know the difference between futures and other assets and financial instruments. ✓ Describe the general features in futures contracts. ✓ Explain the differences between futures and forward contracts. ✓ Explain specific items within the contract specification for a futures contract. ✓ Explain and carry out basic calculation on notional value and margin requirements. ✓ Describe the different types of financial and commodity futures. ✓ Explain the role of futures exchanges and how they operate. ✓ Describe the settlement process of futures exchanges, including the Singapore Exchange Derivatives Trading Ltd (SGX-DT). ✓ Show how leverage can impact the performance returns of futures. ✓ Explain futures pricing models and how arbitrage opportunities arise. ✓ Explain and apply the concept of basis in futures contracts. ✓ Explain how to use interest rate swaps, foreign exchange swaps, currency swaps, equity swaps and credit default swaps.

2.1 Introduction - Futures and Forwards

Futures are standardized contracts for the purchase and sale of financial instruments or physical commodities on regulated futures exchanges with price determined for future delivery. Originally these were just private

non-homogeneous contracts made directly between buyers and sellers. Transactions involving such future deliveries over-the-counter were known as forward contracts. With greater usage of contracts of particular specification of quality, size, and time of delivery, standardization of the more popular contract terms soon took place. In addition, with the desire for a credit-worthy market-maker or counterparty, margin requirements were set in, and the futures exchange was born.

A futures exchange is a marketplace for standardized financial and commodity derivatives where the buying and selling of underlying products are contracted for future delivery. The world's largest and most diverse derivatives exchange is the Chicago Mercantile Exchange (CME) which handles 3 billion contracts worth approximately USD $1 quadrillion annually. CME is the result of the merger of the CME with the Chicago Board of Trade (CBOT).

The second largest futures exchange is the Intercontinental Exchange (ICE) which was created out of a period of mergers and acquisitions among exchanges and clearing houses between the US and Europe with the intent to create more competitive entities and that can provide the reach across global markets. ICE is also well supported by large Wall Street investment banks who needed ICE to be the clearing house for credit default swaps. Key to the success of ICE is their ability to leverage on the advances in technology which allowed for the integration of platforms and processes across exchanges and clearing houses in various markets in different locations. With the support of the investment banks, ICE is able to raise capital for acquisition of major entities like the NYSE and to spin-off Euronext in an IPO.

ICE also acquired the Singapore Mercantile Exchange (SMX) to create ICE Futures Singapore (IFSG). A third derivatives exchange, Asia Pacific Exchange Pte Ltd (APEX), a Chinese-investor backed exchange, was granted approval by the MAS in 2018 to offer commodity and financial derivatives. Table 2.1 provides a list of the major global futures exchanges.

Table 2.1 – Major Global Futures Exchanges

<table>
<tr>
<td>CME Group
• Chicago Board of Trade (CBOT)
• Chicago Mercantile Exch (CME)
• NYMEX Holdings, Inc.
• COMEX
• Kansas City Board of Trade</td>
<td rowspan="3">ICE – Intercontinental Exchange
• ICE Futures US
• ICE Futures, Europe (Int’l Petroleum Exchange, LIFFE)
• ICE Endex (APX Endex)
• ICE Futures, Canada (Winnipeg Commodities Exchange)
• ICE Futures Singapore
• ICE Clear Europe (London)
• ICE Clear Netherlands
• ICE Clear US (NYBOT)
• NYSE (ETF, Equities, Bond Options)</td>
<td>SGX - Singapore Exchange Derivatives Trading</td>
</tr>
<tr>
<td>BM&F Bovespa SA
(Commodity futures, options, funds, etc.)</td>
<td>EUREX - Deutsche Borse</td>
</tr>
<tr>
<td>Hong Kong Exchange and Clearing Limited
• Stock Exchange of HK Ltd
• Hong Kong Futures Exchange
• London Metal Exchange</td>
<td>National Stock Exchange of India</td>
</tr>
<tr>
<td rowspan="3"></td>
<td rowspan="3">Euronext
(Belgium, France, Netherland, Portugal, UK)</td>
<td>Moscow Exchange</td>
</tr>
<tr>
<td>China
• China Financial Futures Exchange
• Shanghai Futures Exchange
• Dalian Commodities Exchange
• Zhengzhou Commodities Exchange
• Guangzhou Futures Exchange</td>
</tr>
<tr>
<td>Japan Exchange Group</td>
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<td></td>
<td></td>
<td>Korea Exchange</td>
</tr>
</table>

The biggest futures exchange in Europe is Eurex, owned by Deutsche Borse. Euronext is the second largest futures market in Europe offering investors access to the futures markets in Belgium, France, the Netherland, Portugal and the UK. Most of the other large futures exchanges have grown out as extensions of their equity markets. In South America, B3 S.A. (formerly BM&F Bovespa) of Sao Paolo, Brazil is the largest futures market serving the region.

In Asia, SGX in Singapore operates the largest derivatives market with its futures, swaps and options benefitting from its position as the largest foreign exchange centre in Asia. The SGX has elevated its position in Asia by offering investors futures contracts on all the major equities indices for all the major markets in Asia. In HK, in addition to its equity market and futures exchange, Hong Kong Exchange and Clearing Limited also owns the London Metal Exchange.

In China, the Shanghai Futures Exchange, Dalian Commodities Exchange, and Zhengzhou Commodities Exchange began as non-profit pure commodities futures exchanges. The Chinese commodities exchanges serve the agriculture and manufacturing sectors in China. Together, as a joint venture, they started the China Financial Futures Exchange. China is still developing its financial markets and sorting out the challenges in its banking sector. The Chinese Yuan was only approved by the IMF as world reserve currency in 2015 and the government is taking its time to gradually allow new financial products and services to be introduced. However, given the size of the economy, the potential growth for the financial futures market is enormous.

Given the interconnectivity of today's futures markets, investors can trade in futures through multiple platforms and service providers such as banks, broking firms and internet-based brokerage services.

2.1.1 Futures versus Forwards

Forward contracts were the predecessor to **Futures** contracts. The differences, advantages and disadvantages of futures and forward contracts are listed in Table 2.1.1.

Table 2.1.1 – Differences between Futures contracts and Forward contracts

FUTURES	FORWARDS
• Futures contract**s** are standardized according to quality, quantity, delivery time and place. • As the futures exchange is the counterparty for the futures contract, there is no counterparty risks between the actual buyer and seller of futures contracts (i.e., counterparty risks will only be realised in the event the exchange itself fails). Therefore, futures contracts have lower counterparty risks compared to forward contracts. • Futures contracts are mark-to-market at the end of each trading day and participants have to cover margin shortfalls according to the rules of the futures exchange. • Futures contracts are usually offset (or 'closed out') on the exchange before maturity. Open or	• Forward contracts are usually bilaterally negotiated between the buyer and seller, and the agreed terms of the forward contract are not standardized. • Forward contracts are typically not subject to mark-to-market adjustments and have no interim partial settlement requirements. • Forward contracts can be settled by physical delivery or cash settlement based on the difference between the contracted price and market price at maturity. • Forward contracts are traded OTC, with no 'guarantee' of an active secondary market. Settlement terms may vary from contract to contract.

FUTURES	FORWARDS
'live' contracts are settled based on the settlement price (and procedures) at contract maturity. • Globally, most futures trading now occur through electronic trading platforms linked to the exchanges with pit trading activity being phased out.	• Investors with forward contracts are exposed to counterparty risks and would be exposed to losses if any of the direct parties they deal with default on the contracts. However bilateral counterparty risks can be eliminated for forwards which are cleared by a central clearing counterparty.

Forward contracts may be more useful for market players who require customized business and investment solutions. For example, this could involve taking a position or hedging in an exotic currency or a delivery period which is not readily available in the futures markets. Specific needs can be fulfilled through a customized contract with a financial institution or another commercial counterparty.

Understanding the features of futures and forwards are important, so that investors know how to use them appropriately to meet investment objectives under different market conditions. For example, the directional and leverage effects of futures can help express the investor's degree of bearishness or bullishness about the market.

2.1.2 Economic Importance and Impact of the Futures Market

The futures market is both highly active and central to the global financial marketplace. It is an important source of vital market information and often serves as an indication of market sentiment.

2.1.2.1 Price Discovery

The futures market reflects the current price of a contract for a product to be delivered in the future. The futures market therefore provides the platform for ongoing price discovery and price adjustment where buyers and sellers contract and influence prices at the current point in time for future delivery given the available information and predictions about the future. How market participants absorb and react to the information, constantly changes the prices of commodities and assets – a process known as **price discovery.**

Such information and predictions can include future economic performance, potential political developments, and forecast on environmental and climatic changes which may or may not actually come about.

2.1.2.2 Risk Management

The futures market is also a place where investors manage risk. Risks are reduced because market prices are transparent and readily available, therefore investors know how much they need to buy or sell. For consumers and buyers of commodities, hedging with futures contracts helps to reduce the final cost to consumers by lowering the risk of manufacturers raising the prices of key supplies due to the volatility in the spot market. Business managers and financial investors also use suitable futures contracts as part of their planning to reach their investment objectives.

2.1.2.3 Augmenting Volatility

Participants can use futures contracts to leverage on long or short positions. In times of major financial distress in the equities, bond or commodities markets, the leveraging impact from the futures and options markets

can significantly augment volatility. For example, the downward pricing pressure from shorting futures usually adds downward selling pressure on the spot market and underlying assets. Hedge funds and traders can also take advantage of such situations to make profits from short selling. The value of underlying assets tends to fall faster than normal thereby destroying market value over a very short time frame.

With the advent of computer-driven trading activity and electronic platforms, the rapid pace of the impact of wealth destruction during major shocks can have an extended rippling impact on other asset classes and the economy. Hence, governments have had to intervene in both the futures and spot markets in extreme situations to contain or dampen the rippling impact on the rest of the market.

2.2 General Features of Futures Contracts

The following are the general characteristics of trading and investing in futures:

a. Futures are **standardized contracts** for the purchase and sale of financial instruments or physical commodities, on **a regulated futures exchange**, for future delivery on the **delivery date**.

b. The **price movement** of futures contracts is directly influenced by the price movement and value of the underlying asset. The price of the futures contract is determined by the current demand and supply forces which in turn are influenced by the composite information and market expectations about the future.

c. A futures contract has a **finite lifespan** which ends on the expiration date. Hence, the market direction and time horizon are important considerations for the futures investor.

d. Futures contracts carry a **notional value** (monetary value upon conversion). The notional value of the futures contract depends on the contract size or multiple and the market price of the futures contract at any point in time. At expiration, parties have to fulfil their obligation – one to deliver and the other to receive delivery. The futures contract is converted to its notional value at the point in time. Unlike out-of-the money options or warrants at expiry which go to zero value, there is a notional value for futures contract at the point of expiration.

e. Futures contracts have **lower counterparty risks** compared with many other instruments because their counterparty is the futures exchange backed by margin requirements.

f. Capitalizing on the **ability to leverage**, investors only need to place a fraction of the value of the futures contract as an **initial margin** with the futures exchange to gain exposure to total futures contract position. Hence, they have the potential to augment gains and losses. The use of leverage is also an important aspect of the futures investor's investment strategy and trading decisions (although various other forms of financing are available for investing in other financial instruments and assets).

g. The value of futures contracts is **mark-to-market** at the end of the trading session each day and any margin shortfall required by the futures exchange must be covered by the trader or investor.

h. In most trade executions, buyers and sellers do not deliver or receive the underlying but often **square off (close out) their positions** with the exchange to realize gains or losses.

i. The specific characteristics involving the typical contract size and units will depend on the type of futures contract invested. These can be found in greater detail in the **contract specifications** or **term sheet** of the particular futures contract. Traders utilize the characteristics of futures contract and the underlying asset, together with other derivatives to craft and deploy sophisticated strategies to generate different payoff

and risk profiles given the expected outlook. Futures can be used to protect underlying investment positions as well to make outright bets on the direction of the market, e.g., by taking a short position on the market, sector or currency.

2.2.1 Calculating Notional Value for Different Types of Futures Contracts

Futures contracts carry **notional value** (i.e., the financial value when the contracts are converted to cash at any one time of day). The notional value of a futures contract depends on the futures contract price and the multiplier for a particular contract.

Calculating Notional Value

1. **Suppose the S&P 500 Mini Index stands at 2,358.50 on 31st March 2020. The multiplier for E-mini S&P 500 futures contracts is fixed at 50x.**

   In April, an investor is looking at the contract for the June 2020 E-mini S&P 500 futures contract that has a bid / ask price of 2,361.00 / 2,361.25.

   The Notional Value of an equities futures contract = Futures contract price x Multiplier (or Contract Size). If a purchase is made at the price of 2,361.25, then the notional value would be USD 50 x 2,361.25 = USD 118,062.50.

   The multiples for the futures contracts can be different for each type of futures contract. For the E-mini NASDAQ 100 Jun 2020 futures contract, the multiple is 20x. Suppose this contract on 4th April was done at a price of 5,429.25. The actual notional value would then be USD 20 x 5,429.25 = USD 108,585.

2. **In Singapore, assume that the SGX FTSE China A50 Sep 2020 contract is trading at 10,300 on 4 April.**

   The multiple is 1x (i.e.; contract size is USD 1), so the notional value of the FTSE A50 contracts is exactly the same as the contracted price of USD 10,300.

3. **Suppose the EUR/USD futures quote for June 2020 was last traded on the CME at 1.07045 in April.**

   The contract unit size is EUR 125,000. The notional value would simply be EUR 125,000 or USD 133,806.25.

2.2.2 Leverage

Leverage is an important feature of futures, and the key concepts and definitions relating to futures leverage or margining are explained below:

a. **Initial Margin** - When buying or selling futures contracts, an investor does not need to pay for the entire contract at the time the trade is initiated. Instead, the investor makes a small upfront payment called the initial margin, to initiate a position.

b. **Maintenance Margin** – This is the minimum amount that must be maintained on deposit by the customer with the broker at all times. If the investors' net equity is above this minimum amount, there will be no

request for additional funds even though there has been a negative market movement against the futures position.

c. **Additional Margin** - Based on mark-to-market procedures of an exchange, each trading account is credited or debited at the end of each trading day and checked to ensure that the trading account maintains the appropriate margin for all open positions. If the balance in the margin account falls below the maintenance level, an additional margin will be required. The account must be restored to the initial margin level immediately or within a stipulated time.

In Singapore, SGX provides participants with maintenance margin and initial margin information on a daily basis through a posted margin schedule on all futures contracts traded. The margins will vary if market conditions change significantly (e.g., if markets become extremely volatile).

How Margins are Used in Futures Trading

Consider a trade in the SGX Nikkei 225 Index futures contract traded on SGX[15]. The value of this contract is JPY500 times the level of the Nikkei 225 index. So, when the level of the index is 22,500, the value of the futures contract is JPY 11,250,000 (JPY 500 X 22,500).

To initiate a trade, an investor only needs to post an **initial margin** of JPY 731,250 (based on an initial margin of 6.5%, the amount required is 6.5% x JPY 11,250,000 = JPY 731,250).

What happens if the level of the Nikkei 225 changes?

If the Nikkei rallies to 24,075 (a 7% increase), the contract is worth JPY 12,037,500 (JPY 500 X 24,075). The investor only posted an initial margin of JPY 731,250, but now has a JPY 787,500 profit (JPY 12,037,500 less JPY 11,250,000 or more than 100% gain with respect to the initial margin of JPY 731,250). This ability to achieve a large profit with a relatively modest market move in the underlying index is due to leverage and is one of the reasons why people like to trade futures.

What happens if the Nikkei 225 falls in value?

If the Nikkei falls 160 points to 22,340, the contract is now worth JPY 11,170,000, and our investor has a loss of JPY 80,000 (JPY 11,170,000 less JPY 11,250,000). Each day, the exchange will compare the value of the futures contract to the investor's account, and either add profits or subtract losses to their margin balance. The exchange requires this balance to stay above certain minimum levels (**maintenance margin** or 5%), which in the case of the Nikkei, is JPY 562,500. Here, the investor has a reduction in margin balance of JPY 80,000 (current margin is JPY 731,250 less JPY 80,000 = JPY 651,250. This is above the maintenance margin of JPY 562,000.)

2.2.3 Contract Specifications and Term Sheets

A contract specification is a non-binding agreement that sets the basic terms and conditions under which an investment will be made. The key information required in the contract is indicated below:

- Description and standard size or denomination of the underlying asset;
- Contract size and value;

[15] Refer to SGX Website (https://www2.sgx.com/) for details of the SGX Nikkei 225 Index Futures.

- Tick size;
- Trading hours;
- Price limits;
- Last trading day;
- Settlement basis;
- Final settlement price;
- Mark-to-market procedures;
- Margin call procedures.

A list of the main futures contracts' specifications traded on SGX-DT is provided in Appendix C of this Study Guide.

Table 2.2.3(a) sets out the key terms used in futures contract specifications.

Table 2.2.3(a) – Understanding the Terms Used in Futures Contract Specifications

Contract Standardization	• Each futures contract has a standardized size that does not change in terms of the currency and contract size as a function of the underlying asset or index. • Example: The E-mini S&P 500 equity index futures contract is denominated in USD and the size is always USD50 times the price of S&P 500 index.
Contract Value	• The contract value is calculated by multiplying the size of the contract by the current price. • Example: The E-mini S&P 500 contract is USD 50 times the price of the index. If the index is trading at 1,850, the value of 1 E-mini contract would be USD 92,500.
Tick Size	• The minimum price change in a futures contract is measured in ticks, i.e.; a tick is the smallest amount that the price of a particular contract can move. Tick sizes can vary from contract to contract. • Example: A tick in the E-mini S&P 500 futures contract is equal to ¼ of an index point. Since an index point is USD50, 1 tick is USD12.50.
Price Limits	• A price limit is the maximum amount the contract price can move in 1 day based on the previous day's settlement price. • Some futures markets impose limits on daily price fluctuations. These limits are set by the exchange and help to regulate dramatic price swings. • When a futures contract settles at its limit bid or offer, the limit may be widened to facilitate transactions for the next trading session. This may help futures prices return to a level that is reflective of the current market environment.
Mark-to-Market	• Futures contracts are mark-to-market daily as an important safety measure. At the end of each trading day, the exchange sets a settlement price based on the day's closing price range for each contract.

	• Each trading account is credited/debited based on the day's profits/losses and checked to ensure that it has the appropriate margin for all open positions.
Margin Call	• If an investor adds to his position or sustains a loss, and his account no longer meets the performance requirements, he will receive a margin or a performance bond call from his Licensed Representative. • The margin call will require the investor to either add money to his account or reduce his positions until the minimum margin or performance bond requirements are satisfied. • Trading firms may suspend trading privileges or close accounts that are unable to meet their minimum margin or performance bond requirements.

As an example, the specifications of an equity index futures contract, the MSCI Singapore Free Index (SiMSCI) futures contract, are shown in Table 2.2.3(b):

Table 2.2.3(b) – Contract Specifications of the SiMSCI Futures Contract

Contract Size	SGD200 x Singapore MSCI (SiMSCI) futures price.
Contract Months	2 nearest serial months and 4 quarterly contract months.
Trading Hours	• 8.30 am – 5.15 pm (T session). • 6.15 pm – 10.55 pm (T + 1 session).
Minimum Price Fluctuation (per tick)	0.1 index point (SGD20).
Daily Price Limits	• When the price moves by 15% in either direction from the previous day's settlement price, trading at or within the price limit of +/- 15% is allowed for the next 10 minutes. After this cooling-off period, there shall be no price limits for the rest of the trading day. • There shall be no price limits on the last trading day of the expiring contract.
Last Trading Day	The 2nd last business day of the contract month. Note: A business day is defined as a day on which the Singapore equity market is open for trading.
Settlement Basis	Cash settlement.
Final Settlement Price (FSP)	The final settlement price shall be the value of SiMSCI computed based on the Special Quotation methodology applied on each component stock of SiMSCI on the day following the last trading day (FSP Day).

2.3 Types of Futures

The rapid growth in various types of futures instruments originally resulted from farmers, miners, manufacturers and businesses having the need to better manage risks and uncertainty. With futures contracts, they could fix prices for a period of time which made it easier to plan and make decisions. Futures contracts can be broadly classified into the following broad categories:

a. **Commodity futures** – agriculture, energy & metals.

b. **Financial futures** – equity, interest rates & foreign exchange.

c. **Others** - innovation has led to the creation of futures contracts in new areas (e.g., weather).

Table 2.3 lists the different types of futures contracts on the different exchanges.

Table 2.3 - Futures Contracts on Different Exchanges

<table>
<tr><th colspan="3">COMMODITIES</th><th colspan="3">FINANCIAL</th><th>OTHERS</th></tr>
<tr><th>Agriculture</th><th>Energy</th><th>Metals</th><th>Equity</th><th>Forex</th><th>Interest Rates</th><th>Other/Exotics</th></tr>
<tr><td rowspan="2">Soft
Sugar
Cotton
Cocoa
Coffee
Palm Oil
Rubber
Potatoes
Orange Juice</td><td>Crude Oil
Brent, WTI
Light Sweet Crude</td><td>Precious
Gold
Silver
Platinum
Palladium</td><td rowspan="3">Market
S&P 500
Nasdaq 100
DJIA
Nikkei 225
FTSE 100
China 50
DAX, CAC
Nifty 50
Ibovesta
MSCI Indices</td><td>Major Currencies
(To USD)</td><td>US Treasuries
2, 3, 5, 10 year Notes, Bonds</td><td>Freight Rates
Wet/Dry Freight
Dry Bulk Freight (by carrier size)</td></tr>
<tr><td rowspan="2">Petrochemicals
Butane, Ethane Propane, Benzene,
Paraxylene
Polypropylene
Polyethylene</td><td>Base Metals
Copper
Steel
Aluminium
Lead
Zinc
Tin
Nickel</td><td>Cross Rates
(Non-USD)</td><td>MAC Swaps
2, 5, 7, 10, 20, 30 years</td><td>Weather
Heating and Cooling Degree Days (HDD/CDD)</td></tr>
<tr><td rowspan="2">Grains
Corn/Maize
Soya Beans, Sunflower
Wheat, Oats,
Canola, Barley,
Sorghum, Rice</td><td>Emerging Market
Currencies to USD</td><td>STIRS EUR
1 month / 30 days, 2, 3, 5, 30 year Eurobonds</td><td>Real Estate
By Sector and Location</td></tr>
<tr><td>Refined Oil
Gasoline
Heating Oil
Fuel Oil
Naphtha</td><td>Ferrous Metals
Hot-roll Coils
Iron Ore 58%</td><td>Sector
Utilities
Consumer
Technology
Financials</td><td>Crypto-Currencies
Bitcoin, Ethereum</td><td>LIBOR, EURIBOR
Interest Rate Futures</td><td></td></tr>
</table>

COMMODITIES			FINANCIAL			OTHERS
Agriculture	**Energy**	**Metals**	**Equity**	**Forex**	**Interest Rates**	**Other/Exotics**
Livestock Cattle Hogs Piglets	**Coal** Coking Coal Thermal Coal, By source	62%, Rebars, Billets, Scrap	Healthcare Industrial Real Estate, (REIT) Energy			
Dairy Milk, Cheese Butter	**Electricity** By Source & Location	**Minor Metals** Cobalt Molybdenum				
Biofuel Ethanol Biodiesel	**Wind Power** (EEX)		**Volatility** VIX VSTOXX VHSI			
Forestry Timber Pulp	**Natural Gas** By source					

Today, there is a wide range of futures contracts traded on futures exchanges in the world. Some of the exchanges will also carry specific commodities futures contract particularly relevant to their markets. Singapore has no large agriculture or mining commodities base. SGX has compensated for this by distinguishing itself over the years, as an exchange that carries futures contracts that cover selected equity market indices relevant to the Asian region – such as the MSCI Indonesia, Malaysia, Thailand, Philippines, FTSE China 50, MSCI Taiwan and the SGX India Futures, as well as Asian currency futures and commodities futures that are relevant to Asian businesses. Singapore offers derivatives contracts in iron ore, hot-roll steel coils, rubber, petrochemicals, refined oil products, natural gas, and freight.

2.3.1 Interest Rates Futures

When banks lend or borrow money over a period, they are exposed to the risk of interest rate movements. If they wish to hedge their position, they can partially or fully hedge their position through interest rate futures contracts or options on futures contracts. More importantly, interest rate futures and options on such futures, allow investors and traders to take positions based on how they believe short-term or long-term interest rates might move across the yield curve given government fiscal and monetary policies and their view of specific economic performance statistics. Interest rate futures contracts are available in most futures exchanges especially in the US, UK, and Singapore.

2.3.1.1 Short-Term Interest Rates Futures

Some of the more popular short-term futures contracts in this category are US Treasury bills futures, Eurodollar futures and Euroyen futures.

The underlying asset of such interest rate futures contracts would typically be:

a. A time deposit in a Eurocurrency (a currency that is lent or borrowed outside its country of origin), where the interest rate of such a deposit or fund offered for a specified period to a first-class bank is used to determine the price of a futures contract for the same period; or

b. A government treasury bill, where the interest rate at which an investor is willing to invest in the Treasury bill for the period forms the basis of the futures price of Treasury bill futures.

The price of short-term interest rate futures is determined by the following formula:

$$P = 100 \times (1 - R)$$

where R = annualized interest rate for that period.

Price of A Short-Term Interest Rate Futures Contract

Suppose the applicable interest rate (R) is 3.50% per annum. The price (P) of the futures contract will be:
P = 100 x (1 - 0.035) = 96.50

2.3.1.2 Long-Term Interest Rates Futures

These futures are used by market players who want to manage their interest rate exposures or to take a position based on their view of the market. For example, a borrower with outstanding debt may use interest rate contracts to protect himself from a rise in interest rates. Or a fixed income asset manager may sell interest rate futures to protect the value of his bond portfolio. If interest rates rise, the value of the futures will fall (as interest rates and bond prices move in opposite directions). The profit realized from closing out of the futures position can offset the decline in value of the bond portfolio.

The pricing of interest rate futures contracts is determined by the following formula:

$$\left\{1 + \frac{d_1 r_1}{360}\right\} \times \left\{1 + \frac{d_2 r_2}{360}\right\} = \left\{1 + \frac{DR}{360}\right\}$$

$$r_2 = \frac{360}{d_2} \times \left\{\frac{(360 + DR)}{(360 + d_1 r_1)} - 1\right\}$$

where:
d_1 = number of days in the 1st period
d_2 = number of days in the 2nd period
D = number of days for the entire period
r_1 = interest rate for the 1st period
r_2 = interest rate for the 2nd period
R = interest rate for the entire period

Pricing of Interest Rate Futures Contracts

Calculate the implied forward rate for the 2nd period of the futures contract.

Spot date to 15 Sep = 45 days
Spot date to 15 Dec = 135 days
45-day interest rate = 1.25%
135-day interest rate = 1.75%

Compute the implied forward rate for the period from 15 September to 15 December.

$$\left(1+\frac{45 \times 0.0125}{360}\right) \times \left(1+\frac{90 \times r_2}{360}\right) = \left(1+\frac{135 \times 0.0175}{360}\right) = 1.006563$$

Solve for r_2 (or the implied forward rate): $r_2 = 2.00\%$.

One of the more popular interest rate futures on the SGX-DT is the 10-year Mini Japanese Government Bond futures contracts, which is discussed in the example below.

Calculating Profits & Losses – 10-Year Mini JGB

Each contract for a 10-Year 6% Coupon Japanese Government Bond has a face value of JPY 10 million. Each contract is valued at JPY 100,000 multiplied by the contract price.

Assume that on 7h April, the quote for 10-Year 6% coupon Mini-JGB Futures is shown below:

Contract	Last	Bid	Ask	Open	High	Low	Total Volume	Open Interest	Daily Settlement Price
Jun-17	150.57	150.5	150.57	150.45	150.63	150.42	1,348	16,116	150.57

In April, a bond fund manager has a portfolio which includes some Japanese Government Bonds. Japanese bond yields have rebounded from -0.29% to 0.06% currently. He expects Japanese bond yields to start rising with US government bonds yields over the next 3 months ahead of a second- rate hike which will probably be in June or July.

The fund manager decides to sell 100 June contracts of 10-Yr 6% Coupon Mini-JGB Futures at 150.57. His predictions were correct, and the US yields led the overall lifting of bond yield, but the price decline was modest. He was able to buy back and close out his futures position at 150.15. How much profit did the fund manager make?

His profit would be calculated as 100 x JPY 100,000 x (150.57 – 150.15) = JPY 4,200,000.

2.3.1.3 US Treasury Futures

US Treasuries are amongst the most liquid financial products in the world. US Treasury futures and options lend themselves to a wide variety of risk management and trading applications, including hedging, income enhancement, duration adjustments, interest rate speculation and spread trades. US Treasury futures and options contracts mirror the, 2, 5, 10 and 30-year US Treasury bond tenures and are the more popular of the long-term interest rate futures contracts on the CME. On the SGX, the mini-Japan Government Bond Futures are one of the more popular futures on long term government bonds.

Each US Treasury futures contract has a face value at maturity of $100,000 with the exceptions of 2-year and 3-year US Treasury futures contracts which have face value at maturity of $200,000. Prices are quoted in points per $2,000 for the 2-year and 3-year contract and points per $1,000 for all other US Treasury futures. More information on US Treasury Futures can be found on the website of the CME Group.

Calculating Profit and Loss – US Treasury Note Futures

Profit Example

A fund manager believes that the US economy is likely to benefit from the planned infrastructure spending and the Federal Reserve will raise interest rates at least one more time by June or September. He believes that Treasury yields for 5 years and 10 years will move up ahead of future hikes.

The fund manager sells 10 contracts of Jun 2017 5-year Treasury Note Futures at 117.245. He was correct and the economy continues to improve with rising GDP growth from infrastructure spending and job creation in the US. The 5-year Treasury rises, and the June 2017 5-year T-Note futures price declines. The fund manager buys back the contract at 116.000.

What is the profit from this example?

Calculation = 10 x (117.2450 - 116.00) x 1,000 =$ 12,450 profit.

How many ticks does this gain represent?

Recalculating using ticks: The tick size for 5-year contract is 1/4 of 1/32nd of 1 point.

The dollar value for minimum tick of the 5-year contract is $7.8125 [derived from: 0.25 x 0.03125 x 0.01 x face value of $100,000].

Number of ticks for the profit on this trade = [(117.245 – 116.00) * $1,000] / 7.8125 = 159.36 ticks.

Profit and loss calculation using ticks = 10 x 159.36 ticks x 7.8125 = $12,450.

Loss Example

It is Thursday, 7April 2017. There is a news report about a surprise US missile strike on Syrian air bases, which further pushed down yields and caused bond prices to rise. This morning, the 10-year yield fell to 2.31%, the lowest level since the U.S. Federal Reserve March rate hike when the 10-year yield was at 2.6%.

A trader notices that the June 2017 10-year T-Note futures have responded to the report by posting only a "modest rise" in his opinion, from 125 8/32 to 125 12/32. He believes that more buyers will be coming in through over the next few hours.

The trader buys 10 contracts of June 2017 10-year Treasury Note Futures at 125 12/32 in early trading on Friday. Unfortunately, the trader was wrong, and prices reversed later in the trading day. The June 2017 10-year T-Note futures fell to 124 8/32 at the end of the day.

What is the mark-to-market gain at the end of the day?
The loss = 10 x (124 8/32 – 125 12/32) x $1,000 = - $11,250 loss.
Recalculating using ticks: The tick size for 10-year contract is 1/2 of 1/32nd of 1 point.
The $ value for minimum tick of the 10-year contract is $15.625 [derived from: 0.5 x 0.03125 x 0.01 x face value of $100,000].

How many ticks does this loss represent?

Number of ticks for the loss on the trade = [(124 8/32 – 125 12/32) x $1,000] / 15.625 = -72 ticks.

Loss on this trade = 10 Contracts x -72 Ticks x $15.625 = - $11,250.

2.3.2 Currency Futures

Currency futures are contracts to buy or sell a specified amount of a particular currency in relation to another currency at a set price and at a future date. Financial institutions, investment managers, corporations and private investors use currency futures to manage the risks or take advantage of profit opportunities arising from currency rate fluctuations. Currency futures were first introduced at the Chicago Mercantile Exchange (now the CME Group) in 1972 after the fixed exchange rate system and gold standard were discarded.

Naturally, the volume of currency futures and other currency derivatives activity would be affected by the volume of underlying activity in the spot market. In 2016, the Bank of International Settlements (BIS) Triennial Survey reported that global foreign exchange turnover was USD 5.1 billon per day in 2016, a fall of 5.5% from USD 5.4 billion in 2013. Part of the decline in global FX activity was attributed to the lower growth in global trade and capital flows which remained below the pre-Global Financial Crisis (GFC levels).

More significant was the structural change in trading patterns favouring the greater use of FX for hedging purposes rather than risk-taking purposes, a consequence of the reduction in proprietary trading and step-up in risk controls for FX derivatives. Banks have also continued to consolidate and cut costs over the last decade. According to the same BIS survey, the number of dealer banks willing to warehouse risk declined, and non-bank market-makers were gaining a stronger foothold as providers of liquidity, sometimes bypassing traditional bank intermediaries. These shifts all contributed to the dip in overall FX trading activity. Despite the overall decline, the trading in most FX derivatives, particularly FX swaps, continued to grow. The survey also revealed that a number of **emerging market economy (EME)** currencies, especially the Renminbi had gained market share. Singapore continued to take the top spot for FX trading activity amidst the general decline which bodes well for the growth of FX-derivatives business here.

2.3.2.1 Types of Foreign Exchange Futures Contracts Offered

Currency futures contracts mirror demand and activity in the underlying currency market. Hence you can expect the futures activity will generally be higher for the major currency pairs.

On the CME, the currency futures have been grouped according to currency majors against the USD, EME currencies against the USD, cross-rates not involving the USD, and the E-Micro Forex Futures contracts that trade at 1/10th the size of regular currency futures contracts. On Euronext, trading focuses on the more active currencies within the majors (EUR/USD, GBP/USD, and GBP/EUR).

On SGX-DT, Asian currencies trade against the USD, amongst themselves, as well as against the non-deliverable forward for the Renminbi /Chinese Yuan (CNY), Indonesian Rupiah (IDR), Taiwan Dollar (TWD), Indian Rupee (INR), Korean Won (KRW), Malaysian Ringgit (MYR), and the Philippine Peso (PHP). SGX remains one of the few places where you can gain exposure for hedging, trading and investing in such Asian currencies. Singapore's foreign currency derivatives market has benefited from the growing demand for the currencies of Asia's emerging market economies. China, Indonesia and India contributed significantly to the increase in global trade and consumption. As these huge economies expand over the next few years, the SGX foreign currency derivatives market stands to benefit from the rising demand for exposure and risk management activities for these currencies.

2.3.2.2 Interest Rate Parity Theory

The theory states that the forward (futures) premium or discount between two currencies should be equal to the difference in the domestic interest rates for securities of the same maturity (except for the effects of small transaction costs). If the interest rate parity relationship is violated, arbitrageurs can make risk-free profits in foreign exchange markets and their actions will force futures and spot exchange rates back into alignment.

The relationship between the spot rate and forward (futures) rate is as follows:

$$F = S \times \frac{1 + R_c\left(\frac{n}{360}\right)}{1 + R_b\left(\frac{n}{360}\right)}$$

where:
F = forward (futures) rate
S = spot rate
R_b = annualized interest rate of base currency
R_c = annualized interest rate of counter-currency
N = number of days

Interest Rate Parity

Base currency: British Pound (GBP)
Counter currency: US Dollar (USD)

Spot price:	GBP/USD = USD 1.2422	
$R_b = R_{UK}$:	0.70444%	(e.g., 1-year GBP LIBOR)
$R_c = R_{US}$:	1.79900%	(e.g., 1-year USD LIBOR)

What is the theoretical price for a 1-year forward contract?

$$F = S \times \frac{1 + R_C(n/360)}{1 + R_b\left(\frac{n}{360}\right)}$$

$$= 1.2422 \times \frac{(1 + 0.01799 \times (360/360))}{(1 + 0.00704 \times (360/360))}$$

Hence, GBP 1 = USD 1.2557
Forward price: GBP/USD 1.2557

2.3.3 Equity Index Futures

Equities index futures seek to replicate the future performance of an underlying equity market index and offer investment opportunities for investors who wish to capitalize on broad market movements in the regional and international share markets. They serve as hedging and trading instruments for investors to protect against, or profit from, price fluctuations without having to change their actual dollar investment in the stocks.

An index is simply a representative "basket" or portfolio of stocks or commodities. An equity index is a performance benchmark of an equity market and measures the performance of a selected basket of shares which represent the performance of the entire market. Various global equity market indices serve as benchmarks for the performance of their respective national and regional equity markets. These indices are normally a mathematical composite that reflect the general performance of the overall market it represents. They can be weighted by price (as in the DJIA) or by market capitalization as in the (MSCI indices).

With the diversification and reduction of concentration risk being common practice in many investment portfolios, the purchase of futures indices is another way of raising or lowering exposure to specific markets or sectors. Buying the index is also considered a cost-effective means of getting exposure to the equities market.

In general, the various stock exchanges carry futures contracts related to their own market indices. However, there are some exchanges that have recognized the globalization and the internationalization of markets, and these exchanges have positioned themselves as regional or global futures exchanges by offering futures contracts on other markets. Likewise, in Singapore, SGX has sought to consolidate its position as a regional platform for trading futures, options and structured warrants for Asian markets.

SGX also tries to improve its trading activity in equities futures by offering equity index futures that are particularly relevant for fund portfolio managers whose portfolio performances are benchmarked against certain major indices, such as the MSCI. The creation of various futures contracts on key country-specific indices is useful for fund managers who need to rapidly respond to changes and rebalance their portfolio, hedge positions and take advantage of short-term country-specific opportunities that emerge in Asia markets.

Over the years, there have been successful launches of popular futures contracts – the most popular has been the SGX FTSE China A50 Futures contracts, a key SGX contract which continues to enjoy high trading volume every day. Table 2.3.3 shows the SGX Future Contracts on Equity Indices.

Table 2.3.3 – SGX Future Contracts on Equity Indices

Table 2.3.3 - SGX Futures Contracts on Equity Indices		
SGX MSCI Singapore SGX Straits Times		
Some futures on Asian Equity Indices listed on SGX		
SGX FTSE China A50 SGX FTSE China H50 SGX FTSE Malaysia	SGX FTSE Philippines SGX FTSE Taiwan SGX FTSE Thailand	SGX Nikkei 225 SGX USD Nikkei 225 SGX Mini Nikkei 225 SGX Nifty 50

2.3.3.1 Differences between Equity Index Futures and Stocks

Differences between equity index futures and a portfolio of stocks are shown in Table 2.3.3.1.

Table 2.3.3.1 – Differences between Equity Index Futures and a Portfolio of Stocks

Features	Equity Index Futures	Portfolio of Stocks
Underlying asset	Cash index	Ownership of shares in a company
Settlement	Daily mark-to-market	Fixed settlement based on transacted price
Risk	Leverage can magnify gains as well as losses by several fold	Market volatility of index
Short Selling	• No uptick rule • No borrowing of shares	Short seller needs to borrow shares and must pay dividends to share owner (long)
Dividends	No dividends on futures	Dividends to owner of shares (long)

2.3.3.2 Pricing Equity Index Futures

The fair value price of an equity index is represented by the following formula:

Futures price = Spot (cash) price + Financing cost - Income from stock

Or **Futures price = Spot price + Interest - Dividend**

$$F = S \times \left[1 + r\frac{t}{360}\right] - D$$

where:
F = fair value of futures
S = equity index or spot price
r = annualized financing rate (money market yield)
t = time to expiry
D = value of dividends paid before expiry

Equity index futures do not trade at precisely the fair value level aligned with the spot or cash value of the associated equity index all the time. This difference between the futures and spot values is known as the **basis**. The fair value of an equity index futures contract is normally expected to be positive such that futures prices are more than the spot prices (F > S). This is attributed to the fact that under normal market conditions, financing costs, as reflected in short-term interest rates such as the London Interbank Offered Rate (LIBOR) or Singapore Interbank Offered Rate (SIBOR), normally exceed dividend yields. Due to a loose monetary policy by central banks in recent years after the 2007-2008 financial crisis, the reverse has been true.

If the equity futures prices are to rally much above their fair market value, an astute arbitrageur may act to buy the equity portfolio and sell equity index futures in an attempt to capitalize on that mispricing. These arbitrageurs may attempt to trade in a basket or subset of the stocks included in an equity index or use electronic trading systems to trade in all or virtually all of the constituents of a particular equity index as part of the arbitrage transaction. In the process of buying stocks and selling futures, the arbitrageur may bid up the stocks or push futures prices down to re-establish equilibrium.

2.3.4 Single Stock Futures

A Single Stock Futures (SSF) contract is expected to track the price movement on the underlying stock. On the SGX, a buyer agrees to pay a specified price at a predetermined date, while a seller is obligated to deliver the value of the underlying stock at the same predetermined date. On SGX, SSFs are cash settled and there is no physical delivery of the underlying stock.

SSFs can be used by portfolio managers to protect against adverse price movements of the underlying stock and to diversify risks using margin financing, and there are no shareholder related privileges benefits linked to SSF contracts. The leveraging involved with SSF contracts usually amplifies gains and losses, making such products riskier than the underlying asset.

Some of the advantages and disadvantages associated with SSFs are discussed below.

Advantages:

a. SSF contracts are leveraged instruments with margin and are usually based on several thousand shares, the leverage effect amplifies potential gains and losses;

b. Transactions costs are lower than buying the underlying stocks since the amount of outlay is lower. with margins placed in the SSF contracts;

c. For the same reason, offshore investors have less currency exposure since they are only investing a smaller portion in the futures margin position instead of the total underlying value;

d. The ease of short selling futures allows investors to take advantage of market downturns and participate with gains on shorting particular stocks; and

e. Once an exchange agrees to include a particular SSF contract, it will have to ensure that there is liquidity, transparency, fairness, and timeliness in the market-making process with reasonable bid-offer spreads. Investors should also be able to close out their positions, typically on computer-based electronic platforms. The exchange guarantees the clearing house support from designated market-makers.

One of the main disadvantages of SSF contracts is that they can invite potential short selling of stocks, which can reduce underlying asset prices and cause downward spirals during periods of uncertainty and market shocks. The market's ability to accept increased downward volatility is an important consideration before new futures contracts are introduced.

SSF contracts are currently available on the overseas exchanges such as HKSE, ICE and EUREX. While equity index futures are becoming more popular than trading in SSFs these days, the trading of futures on the more popular equity counters on an exchange is still actively carried out.

2.3.5 Commodity Futures

Investors and businesses that have to manage risk or want or take advantage of perceived opportunities in certain commodities can do so using commodity futures. Contract specifications for commodity futures tend to be more unique because of their contract size and their conventional measures. Trading of commodity futures has gone well beyond the traditional purpose of hedging for price changes at future delivery for an underlying commodity. Today we see a great deal more strategic and speculative investments using commodity futures and attempts made by major hedge funds to move commodity prices and indices for the sake of gains on ETFs, leveraged ETFs or other futures and options positions.

Investors seeking pure exposure to commodities will recognize that commodity futures and options on commodity futures represent one of the purest forms of exposure to the underlying commodity next to buying the underlying physical commodity itself. Taking delivery of the underlying physical commodity will lead to problems with storage, security, and spoilage. Buying indirect exposure by investing in the stock of agriculture or mining companies mean investing in the management and profitability of these companies – representing a different form of exposure. Hence, investors and traders are left to look to the futures market as the means of gaining direct pure exposure to specific commodities.

In Singapore, SGX has launched several types of commodity futures contracts that are particularly relevant to agriculture, oil refining, and mining sectors in the region. The aim is to continue its strategy of being the hub of commodity futures trading activity for Asia.

2.3.5.1 Gold and Other Precious Metal Futures

Gold and other precious metals futures are one of the most popular commodity futures contracts being traded. Precious metal futures (especially gold futures) are often used as safe-haven investments during times of financial or political uncertainty in the world. Silver and platinum futures will typically move in tandem with gold but liquidity for both silver and platinum is lower.

Prices of gold and other precious metal futures rise during periods of higher market volatility. For example, pre-Global Financial Crisis in 2007, gold traded between USD650-700/ounce. With extreme volatility during the Global Financial Crisis of 2008-2012 and uncertainty for the USD and major asset classes, gold futures prices trebled in value to USD 1,800/ounce before receding in the subsequent years when volatility tempered.

When there is a fear of a fall in value of key global currencies, investors holding such currencies will switch to other safer currencies for the short period or to gold until the uncertainty disappears. For example, during Brexit, the dramatic fall in GBP was accompanied by the interest in gold futures (see Table 2.3.5.1).

Table 2.3.5.1 – Gold as a Safe Haven Currency

Comparing Fall in GBP/USD against Rise in Gold Futures	Before Announcement (23rd June 2017)	After Announcement (24th June 2017)	Two Week After (7th July 2017)
GBP/USD	1.4877	1.3679 (-8.05%)	1.2908 (- 13.24%)
Gold Futures USD/Troy ounce	1263.10	1322.40 (+4.69%)	1362.10 (+7.84%)

When the potential military conflicts and wars involving major global powers, there is a tendency for flight to safety. For example, in early April 2017 in the Middle East, there was an escalation of tension between the US and Syria-Russia on the usage of banned-chemical weapons and the US missile response. At the same time US-North Korean nuclear-testing tensions also escalated. During this period, gold futures prices moved up 4% to USD 1,290.70/ounce just before Easter. When tensions eased, the price of gold retraced and weakened.

Calculating Notional Profit for Gold Futures

On 10 April 2017, there were US military tensions with Syria, North Korea, and Russia. Gold for June delivery gained 0.39% on escalating concerns. At the start of European trading, gold futures prices for Jun 2017 traded on the ICE was at USD 1,259.10/Troy ounce.

A gold trader believed that tensions in the Middle East between the US-Syria and Russia were about to escalate and bought 100 contracts of Jun 2017 gold futures. Each contract unit = 100 Troy ounce. Gold rose USD 1,288.35/Troy ounce before Good Friday on 13 April.

How much was his notional profit gained at the end of the day?

Total Notional Profit: 100 x (USD1,288.50 – USD1,259.10) x 100 = USD 294,000.

Conversely when interest rates rise, the prices of gold and other precious metals come under pressure because these commodities do not provide interest yield or dividends. When risk is low, investors are more willing to be exposed to other risky assets, hence safe haven assets will fall. The other major development affecting gold futures is the growing interaction with gold-related ETFs. According to the World Gold Council, the largest growth in demand for gold in 2016 (+660%) came from ETFs. There will be growing interaction between the futures of precious metals and their ETFs.

Besides the influence from the financial markets, gold enjoys underlying demand from global jewellery industry. China and India are the largest demand sources for gold for jewellery-making, and more than half of the trading in physical gold is estimated to come from Asia (also mainly China and India). When the demand for the import of precious metals is interrupted or blocked there will also be underlying downward pressure on the commodity.

Gold futures on the CME and ICE are quoted in 100 troy ounces. In Singapore, the SGX Gold contract is dormant and not traded.

2.3.5.2 Base Metal and Ferrous Metal Futures

The demand and prices of base metals and ferrous metals depend on economic growth and the underlying demand for such metal futures comes from major traders, miners, manufacturers who desire actual physical delivery of such metals.

When global economies are expanding, or if there is strong infrastructure build up, demand for base metals firm up. Hence, base and ferrous metals in general rose significantly during the period of major infrastructure development and expansion in China especially from 2002-2012. The China slowdown from 2012-2016 amidst overcapacity led to a sharp fall in ferrous and other base metals. The expected increase in demand for steel and base metals as a result of proposed fiscal expenditure on US infrastructure in 2017-2018 led to an upward drift in ferrous metal and base metal futures prices.

The pricing of steel across markets is not uniform and there are trade disputes between the US and China with regards to trade barriers and anti-dumping allegations. These can disrupt the price discovery process in the futures market for such products. For steel producers in Asia (e.g., India, China, Australia), having a transparent and dependable steel futures market is important. The US President has announced steel and aluminium tariffs to protect local US manufacturers as these two metals have caused disparity in the pricing of steel and aluminium compared to the same products outside the US. Naturally, the new tariffs will widen the difference in pricing across exchanges in the US compared to other markets outside the US. There are different types and grades of steel varying in form and composition that will appeal to different players. For example, steel from China and other base metal prices quoted in the SGX are usually quite different from that of the steel price quotes on the CME which feature steel from US companies.

Iron ore and steel are key commodity futures contracts for the SGX in Singapore. The London Metal Exchange is the largest global metal exchange and continues to be a main exchange for iron ore and steel futures. We have also seen the rise of steel futures contracts traded on the Chinese exchanges as well, hence the competition is very keen. Singapore is positioned as a key clearing centre for iron ore contracts that are traded elsewhere (especially China).

Several types of iron ore and steel contracts traded on the SGX are listed below:

a. SGX TSI Iron Ore CFR China (62% FE Fines) Index
b. SGX HRC CFR ASEAN Steel
c. SGX Platts Iron Ore CFR China (Lump Premium) Index
d. SGX MB Iron Ore CFR China (58% FE Fines) Index
e. SGX TSI Iron Ore CFR China (58% FE Fines) Index

2.3.5.3 Crude Oil Futures

Crude oil futures remain a focal point for many investors because of the dramatic price volatility of crude oil over the last 10 years and continued volatility. Trading to a high of USD155/barrel just before the Global Financial Crisis, it dramatically plunged to USD48/barrel in 2009, rebounded and hovered above USD100/barrel for most of 2010-2013 before sliding to new lows of USD29.60 in 2015 amidst structural changes in the industry and competition from fracking. Futures contracts may have helped crude oil producers lock in higher prices on some occasions, but for extended price declines, there is little protection for the producers. To stabilize crude oil prices, there was an agreement amongst OPEC member countries to limit the supply which eventually helped to keep crude oil range between USD55-65/barrel for the later part of 2018 ahead of the listing of Saudi Aramco. Crude oil futures remain one of the most actively traded instruments.

2.3.5.4 Energy – Natural Gas, Other Refined Products and Coal Futures

The volatility of various energy futures is relatively well correlated. Like crude oil, the other traditional energy futures products (e.g., LNG, jet fuel, gas oil, kerosene, coal) have been relatively volatile over the last decade. For some downstream refined products, there may be a slight lag effect before the full impact of price changes moves through the inventory of downstream refined products. Coal, though somewhat unhealthy for the environment, remains a viable substitute in many developing economies.

Singapore is a major petrochemical refining base and the SGX also provides futures contracts for diverse petrochemical and refined products produced by regional players, as well as Indonesian coal futures.

2.3.5.5 Energy - Electricity Futures

Energy futures have been available in the US for many years. With multiple sources of production (LNG, coal, hydroelectricity, etc.) in different regions of the US, and multiple users (towns, cities, industries, etc.), the need to optimize non-peak hours and fluctuating demand have resulted in the development of the electricity futures.

On the SGX, there is a monthly and a quarterly electricity futures contract, but electricity futures is still relatively new and caters to specific needs of industrial users in Singapore. The local electricity market is thermal in nature with more than 90% dependent on natural gas. The electricity generating companies offer every half-hour to sell the electricity into the spot market, matching the prevailing electricity demand at that period but prices may fluctuate during periods of tight supply due to plant outages or network constraints. Electricity consumers who purchase electricity directly from the spot market are therefore exposed to these price fluctuations. Electricity futures contracts can be used to offset expensive electricity for consumers or to secure higher proceeds to offset cheaper electricity during off-peak periods.

2.3.5.6 Agriculture Futures

Finally, there is a wide range of agriculture futures contracts (grain, soft commodity, livestock, dairy products, and forest products) that have existed for several decades driven mainly by the underlying supply and business needs of farmers, producers, wholesale consumers, as well as investors and speculators.

2.4 Contract Settlement and Clearing Process and Orders on the SGX

2.4.1 Modes of Settlement

A buyer or seller of futures contracts can settle in the following ways:

2.4.1.1 Delivery or Hold to Expiry

All futures contracts have an expiry date. At that date, the buyer will accept delivery of the underlying asset and the seller liquidates his position by delivering the asset at the agreed contract price.

Commodity contracts will state the physical delivery to an approved warehouse of the underlying commodity. Currencies and some financial instruments such as the S&P 500 index, simply call for cash settlement. Every futures contract specifies the last day of trading before the expiry date.

Investors must know the expiry dates because as the date approaches, liquidity slowly decreases as traders begin to roll their positions to the next available contract month.

2.4.1.2 Offset Position

This is the simplest and usually the most common option for financial futures. If the investor enters the market by buying two S&P 500 futures contracts, he could offset his position by selling two contracts. If he enters the market by selling two contracts, he will offset the sale by purchasing two contracts. To limit the risk of holding a position overnight, many traders exit all positions and are flat (no position) at the end of every trading day.

2.4.1.3 Roll Position

Some longer-term traders do not want to give up their market exposure when the current contract expires, so they transfer or roll the position to the new contract month.

For example, an investor who is long with S&P 500 December contracts may want to retain his position beyond December. As the expiry date approaches, he could simultaneously sell the December contract and buy the next March contract. By offsetting his position in the December contract, he takes an equivalent long position in the March contract, effectively rolling his long position from the December contract to the March contract. Note that for all outstanding positions held at the close of a trading day, the exchanges will carry out a mark-to-market calculation to establish any shortfall that must be covered if margin requirements are triggered.

2.4.2 Minimum Price Movement for Contracts in SGX-DT – Understanding Tick Sizes

A tick is defined as:

a. The minimum price movement of a trading instrument;

b. The minimum increment in which prices can change; and

c. The worth of each price movement in terms of dollars.

Price movements of different trading instruments vary. The tick sizes for various derivatives traded on SGX can be found on the SGX website.

2.4.3 Listing of Derivatives Contracts on SGX-DT

Derivatives are traded on SGX[16] under the following categories:

a. **Foreign exchange** – e.g., USD/SGD FX Futures & SGX AUD/USD FX Futures.

b. **Equity Indices** – e.g., MSCI Singapore (SiMSCI) Futures & Nikkei 225 Index Futures.

c. **Interest Rates** – e.g., Eurodollar Futures and Euroyen LIBOR Futures.

d. **Commodities** – e.g., PLATTS Singapore Fuel Oil 380cst Index Futures Contract & SICOM TSR20 Rubber Futures.

e. **Dividend Indices** – e.g., Nikkei Stock Average Dividend Point Index Futures.

A list of the main futures contracts specifications traded on SGX-DT is provided in Appendix C.

2.4.4 Derivatives Clearing Process on SGX-DT

Clearing is the process where positions carried on Clearing Members' books are margined and marked to the latest market prices. At the end of each day, all outstanding positions are marked to their respective daily closing prices.

The Singapore Exchange Derivatives Clearing Limited (SGX-DC), a wholly owned SGX subsidiary, provides clearing for:

a. Products listed on SGX-DT;

[16] More information derivatives traded on SGX-DT is available on the SGX website.

b. OTC commodity trades; and

c. OTC financial derivatives trades registered via industry-used trade registration system.

Clearing Members are required to report their open positions for each contract at the end of the day. To further mitigate SGX-DC's exposure from significant overnight and intra-day price changes, SGX-DC performs intra-day mark-to-market daily.

SGX performs three intra-day margin cycles each day, once in the late morning, once in the afternoon for current day trades and positions; and once immediately after the end-of-day settlement cycle that includes trades for next day clearing. Participants buying futures can refer to the margin schedule for the week. Margin schedule requirements are updated regularly or as required during period of high volatility or market shocks. The SGX makes the decision and can increase requirements during such periods of higher risk (so it is not a strict case of applying a simple formula). At each intra-day cycle, trades and positions are mark-to-market and margin requirement re-calculated and the margin requirements and shortfalls for outstanding positions will be made known to relevant parties.

The following are settled on trades executed for current day clearing and positions that are brought forward from previous day:

a. Settlement variations for futures and OTC swaps;
b. Premiums for options;
c. Coupons for OTC financial instruments;
d. Price alignment interest for OTC financial instruments; and
e. Clearing fees and other trade related fees.

Similarly, ICE Futures Singapore through its subsidiary, ICE Clear, provides clearing for the financial, currency and commodity contracts that are traded by its members.

2.4.5 Mutual Offset System (MOS)

SGX-DC and the Chicago Mercantile Exchange (CME) have a long-standing mutual offset arrangement since 1984 that allows market participants to initiate positions in one exchange for allocation to the other exchange on a real-time basis.

Currently the following SGX-DT products are eligible for mutual offset with CME:

a. Nikkei 225 Index Futures (NK);
b. USD Nikkei 225 Index Futures (NU);
c. SGX FTSE Emerging Market Index Futures (FEM); and
d. SGX FTSE China H50 Index Futures (FCH).

2.4.6 Potential Role of Singapore as a Global Clearing Centre

SGX-DC is registered as a designated clearing organization for the US Commodity Futures Trading Commission (CFTC). This allows SGX-DC to provide clearing for US persons with swaps. Singapore's Central Counterparty (CCP) process was assessed to be equivalent by the European Commission which paved the way for Singapore to provide clearing services to European participants. SGX-DC and the Central Depository (Pte) Ltd (CDP) have been recognized by the European Securities and Markets Authority which allows the two Singapore entities to offer services in the EU.

The hope is for Singapore to be positioned as a high-quality global clearing centre for futures and derivatives and there is a potential to develop it as a global clearing centre. As the internet and disruptive technology change the way we perform transactions, the clearing process will also need to continue to adapt to changes to come.

2.5 Futures Pricing Model

2.5.1 Spot Price

The spot price is the current market price for the underlying asset or commodity.

2.5.2 Cost of Carry Model

The relationship between futures prices and spot prices can be summarized in terms of **cost of carry**. Cost of carry includes financial costs (e.g., interest costs on bonds, interest expenses on margin accounts and interest on loans used to purchase a security), and economic costs (e.g., opportunity costs from taking the initial position).

The net cost of carry involves all costs that an investor may incur to hold a similar position in the cash market, less the returns that he would have received from this position. Costs typically include financing charges at the prevailing rate of interest because he may have borrowed to finance a similar position in the cash market, and if not, he may have lost interest on the capital invested to keep his position open.

In contrast, for the futures market, the investor only has to deposit a fraction of the value of his position in the form of a margin. The returns that he receives could be dividends or bonuses from shares in the cash market. In the case of an index future, his returns may be gauged by the average return that an index delivers.

The cost of carry model assumes that arbitrage between the cash market and the futures market eliminates all imperfections in pricing. For simplicity, it assumes that:

a. The contract is held till maturity, so that a fair price can be arrived at;

b. There are no transaction costs or margin requirements;

c. There are no restrictions on short selling; and

d. Investors can borrow and lend at the same interest rate.

Once all distortions in the futures price have been eliminated by arbitrage trading, the fair futures price will be:

Futures price = Spot price + Net cost of carry of the asset (from today to the date on which the contract expires)

The cost of carry model is defined as:

$$F_{0,t} = S_o (1 + C_{0,t})$$

where:

$F_{0,t}$	=	Futures price
S_o	=	Spot price
$C_{0,t}$	=	Percentage cost required to store (or carry) to time t

The cost of carrying or storing includes:

i Storage costs;
ii Insurance costs;
iii Transportation costs;
iv Financing costs.

When this theory is modified to reflect real world conditions by discarding the assumptions or including other variables, it becomes more complex. However, the main point is that there are costs and benefits involved in keeping an open position in a cash market, and the futures prices change accordingly.

2.5.3 Cash-and-Carry Arbitrage

Table 2.5.3 shows how the cost of carry model can execute two arbitrage strategies (i.e., cash-and-carry and reverse cash-and-carry) in the debt, physical and futures markets.

Table 2.5.3 – Executing Cash-and-Carry and Reverse Cash-and-Carry Arbitrage Strategies

Market	Arbitrage Strategies	
	Cash & Carry Arbitrage **Futures price > theoretical fair value**	**Reverse Cash & Carry Arbitrage** **Futures price < theoretical fair value**
Debt	◦ Borrow funds ◦ Payback funds with interest	◦ Lend short sale proceeds to earn interest ◦ Payoff any borrowing cost for asset short sold
Physical	◦ Buy underlying asset & store ◦ Deliver against futures	◦ Borrow underlying asset and short sell asset ◦ Return underlying assets borrowed
Futures	◦ Sell futures	◦ Buy futures and accept delivery of futures

Evaluating the Opportunity for Arbitrage

An investor is evaluating a potential arbitrage opportunity in gold.

The market information is shown below:

- One contract = 100 oz.
- The spot price for gold is USD 1,275/oz.
- The 1-year futures price for gold is USD 1,305/oz.
- The loan interest rate for 1 year is 2.25%.

Is there any arbitrage opportunity?

Firstly, determine the fair value of the forward/futures price.
The theoretical fair value of the 1 year forward/futures price is given by:

$F_{0,t} = S_o (1 + C_{0,t}) = 1,275 (1 + 0.0225) = 1,303.69$

The futures price of USD 1,305 is higher than the theoretical fair value of USD 1,303.69. The arbitrage action calls for the sale of the futures and the simultaneous purchase of the spot gold.

The investor borrowed amount = 1 x USD 1,275 x 100 = USD 127,500 to purchase 100oz of gold at spot price of USD1275/oz. The investor sells 100oz (1 contract) gold futures and receives = 100 x USD 1,305 = USD 130,500. He then pays back the amount borrowed USD130,368.75, which is the initial amount borrowed (USD 127,500) and interest cost (0.0225 x USD 127,500 = USD 2,868.75).

The arbitrage profit = USD 130,500.00 – USD130,368.75 = $131.25 (a negligible 0.1% arbitrage opportunity).

How do we improve the opportunity for Cash-and-Carry arbitrage profit?

The 1-year futures price for gold has to be higher and/or spot price has to be lower. The borrowing interest cost also has to be significantly lower.

Reverse Cash-and-Carry Arbitrage

In this scenario, spot gold is at USD 1,285/oz.

Is there any arbitrage opportunity?

Firstly, determine the fair value of the forward/futures price.

The theoretical fair value of the 1 year forward/futures price is given by:

$F_{0,t} = S_o (1 + C_{0,t}) = 1,285 (1 + 0.0225) = 1,313.91$

The futures price of USD 1,305 is lower than the theoretical fair value of USD1,313.91. The arbitrage action calls for the purchase of the futures and the simultaneous short sale of the spot gold.

To facilitate the short sale of spot gold, the investor borrows 100oz of gold at 1.00% (borrowing cost) for a year. He then sells 100z of gold at spot price and receives 100 x USD 1,285 = USD 128,500 and uses the borrowed gold for spot delivery.

He lends proceeds from the sale at 2.25% and receives USD 2,891.25 interest in 1 year. In 1 year, the total amount accumulated = USD 128,500 + USD 2,891.25 = USD 131,391.25. The investor also buys the 100 oz of gold 1-year futures at USD 1,305/oz = USD 130,500.00.

In 1 year, the investor receives 100 oz of gold to cover the gold borrowed = USD 130,500. Proceeds surplus = USD 131,391.25 – USD 130,500.00 = USD 891.25. He pays 100 oz of gold and cost on borrowed gold of USD 128,500 x 0.01 = -USD 1,285.00.

Once the cost of borrowing gold is taken into consideration, the overall 'arbitrage loss' = USD 891.25 – USD 1,285 = -USD 393.75 (or -0.306% loss). Hence, there was no arbitrage opportunity.

To improve opportunity for Reverse Cash-and-Carry arbitrage profit, the following has to change:

- Cost of borrowing gold has to be much lower or free while lending rate for proceeds has to be higher; or
- Futures price for gold needs to be lower and/or spot gold has to be higher.

Arbitrage opportunities rarely exist or last for more than milliseconds. The advent of computer-based trading has driven efficiency beyond human levels and any differential that would have allowed for any small arbitrage disappears.

2.5.4 Expectancy Model of Futures Pricing

This model implies futures prices are expected spot prices of an asset in the future. If there are more traders who expect the futures price of an asset to rise in the future than those who expect it to fall, the current futures price of that asset will be positive. This theory suggests that it is not the relationship between the cash market price and the futures price that is relevant, but the relationship between the expected spot price on the date of expiry of the contract and the futures price that matters. Both the cost-of-carry and the expectations models explain some sort of the movement in futures prices.

2.6 Basis in Futures Contracts

The difference between the futures price and the spot price is called the basis. When markets are in contango (typically during stable or bullish conditions), the basis is usually positive, meaning the futures price is higher than the spot price due to carrying costs such as financing and storage.

Conversely, when markets move into backwardation (often during bearish or stressed conditions), the basis can become negative, meaning the futures price is lower than the spot price.

Since a futures contract is settled at the cash market price on its expiry date, the futures price and the spot price will converge as expiration approaches.

2.6.1 Factors Affecting Basis

The size and duration of basis in the market of a futures contract is influenced by:

a. Cost / return of carry;

b. Time to maturity;

c. Yield curve changes;

d. Relative liquidity of cash & futures markets;

e. Market rates versus administered rates;

f. Expectations of market participants;

g. Differences in coupons between fixed income instruments.

If the asset to be hedged and the underlying asset of the futures contract are exactly the same, the basis will be zero at expiry of the futures contract. Before expiry, the basis may be positive or negative. In practice however, the futures contract and its underlying asset may not exactly match the asset being hedged.

Cost / return of carry. The cost/return of carry depends on the difference between cost of funds and the yield on the underlying assets. The greater the net cost/return of carry, the greater the basis.

Time to maturity. The greater the mismatch between the maturity dates of the cash and futures contracts, the greater the basis. But as the expiry date draws near, the net financing cost declines and forces the basis toward zero. The narrowing of the basis is called **convergence**. The futures and spot prices converge until the two become the same at the expiry date and hence, the basis approaches zero.

Yield curve changes. Changes in the shape of the yield curve also affect the basis. Depending on whether the basis was positive (i.e., futures price is lower than cash price in a **backwardation** relationship) or negative (i.e.; futures price is higher than cash price in a **contango** relationship), a steepening in the yield curve will cause the basis to widen in the former and to narrow in the latter.

Relative liquidity of cash and futures markets. Sometimes position adjustments are reflected first through the futures market, while the cash market remains unchanged, thus affecting the basis. As we have mentioned that where the futures market leads the cash market for some commodities, for example, US Treasury bonds, any news that affects the commodities will be first felt in the futures market. The basis will be affected temporarily because of the lag in the cash market.

Market rates versus administered rates. Market rates are more volatile than administered rates. Depending on the futures contracts used, the basis between the underlying instrument and the futures contracts will be affected.

Expectations of market participants. Whenever market sentiment swings to one extreme or another, the basis between cash and futures will widen. Sometimes the basis may even go beyond the implied value.

Differences in coupons between fixed income instruments. Changes in market rates affect the prices of two fixed income instruments bearing different coupons differently. Consequently, the basis will move in different manner.

2.6.2 Understanding Basis, Contango and Backwardation in Futures Trading

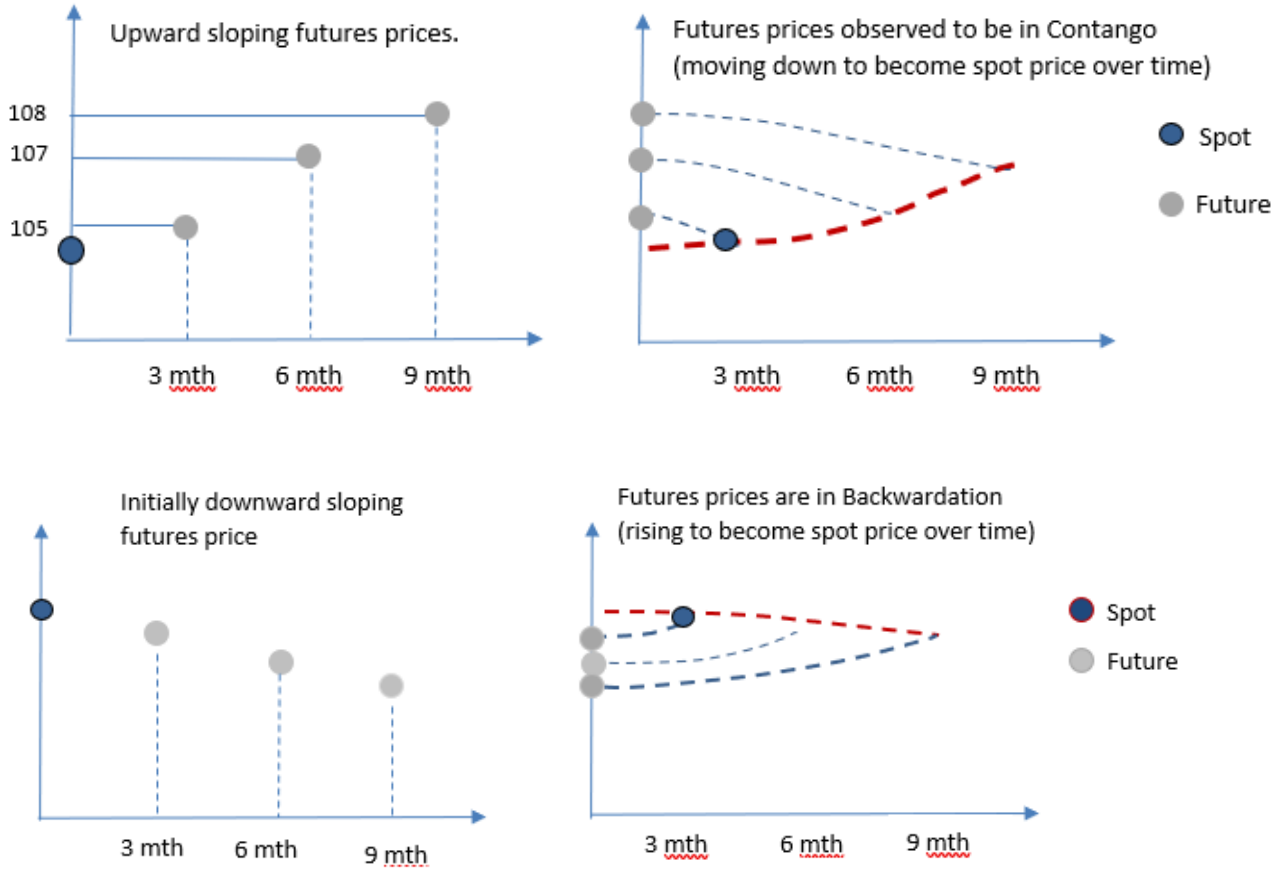
Basis in a hedging situation can be defined as follows:

Basis = Spot price of asset to be hedged - Futures price of contract used

If the spot value of asset to be hedged and the value of the underlying asset in the futures contract are exactly the same, the basis would be zero upon expiration of the futures contract.

Contango theory suggests that traders in the market are willing to pay more than the theoretically expected price for that commodity in the future. At expiry, the spot price = expected futures price. When the futures prices for a commodity are in contango, we can observe over time that the futures delivery price is falling to converge toward the spot price at expiry of the contract.

Figure 2.6.2(a) – Backwardation and Contango of Futures Prices

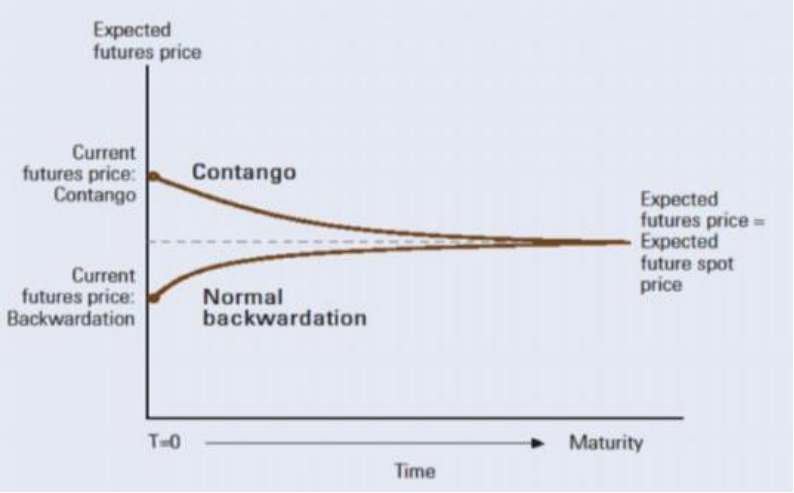


Backwardation is the term used when we see the futures delivery price rise over time to meet the spot price. Initially, the spot price is higher than the futures price because there could be a supply shortage in the near term before being normalized over time with more supply in the longer term. The shortage in the short term causes an inverted price schedule.

In colloquial terms, when traders talk about futures and say that futures prices are in **contango**, they are referring to futures prices to be upward sloping. This implies that there are more investors going long and prices are expected to increase. When they say that futures prices are in backwardation, they are saying that futures prices are inverted against spot prices. Some traders expecting lower prices over time will try to hedge their position by selling forward. When investors are rolling over futures positions to take longer term positions when futures are in backwardation, they create positive rolling yield since the futures prices are rising. Figure 2.6.2(a) shows both the backwardation and contango of future prices.

If the asset to be hedged and the asset underlying the futures contract are exactly the same, the basis would be zero at the expiration of the futures contract. Prior to the expiration, the basis may be positive (backwardation) or negative (contango). Figure 2.6.2(b) below shows how as the futures contract moves to the expiration date, the futures prices will converge towards the spot price.

Figure 2.6.2(b) – Futures Price Convergence



In practice, the futures contract and its underlying asset may not exactly match the asset being hedged because:

- The underlying asset in the futures contract is not completely identical;
- The hedge may be uncertain about the exact date the asset will be bought or sold; and
- The hedge may require the futures contract to be sold before the delivery month.

These imperfections give rise to **basis risk**.

2.7 Swaps

Swaps are customized OTC agreements between two counterparties, usually financial institutions. There are swap agreements sold on exchanges, but these tend to be more standardized agreements between the exchange and the investor. Most of swap transactions occur in the OTC markets.

In Singapore, forex forwards and forex swaps accounted for 70-75% of foreign exchange during 2016-2017, while the currency interest rate swaps account for about 60-66% of all foreign currency derivatives, thus they are a major part of the activity here. Equity and bond swaps are less actively traded.

The swap market is one of the largest and most liquid markets in the world, and its participants are mainly financial institutions who use swaps to hedge or take outstanding net directional positions. Swaps allow investors and traders to switch between floating and fixed interest rates, make currency interest payments, or manage currency flows. The most actively traded swaps include:

a. "Plain vanilla swaps" or interest rate swaps;
b. Foreign exchange swaps;
c. Currency swap;
d. Equity swaps;
e. Credit default swaps.

2.7.1 Interest Rate Swaps

Interest rate swaps (IRS) are used by investors to hedge, manage and speculate on the movement of interest rates. Initially, the demand came from corporates, but today, the major players are financial institutions which use it as part of their overall tool to manage and take advantage of interest rate movements.

An IRS is an agreement based on a notional principal amount, the term, payment frequency, where one party agrees to receive fixed interest rate payment in exchange for making floating interest rate payments to a counterparty based on the same notional amount. At maturity, the parties will settle the differences in amount depending on whether the interest rate widens or narrows. The party with the larger payment simply pays the other party the difference.

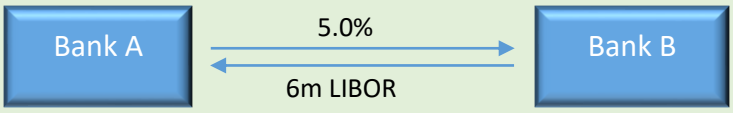
When there are no additional variations to the fixed-vs-floating rate agreement, we call it a "plain-vanilla interest rate swap" or a "vanilla swap" for short. Vanilla swaps make up the majority of interest rate swap contracts in most market.

Investors that actively use interest rate swaps are typically corporates, and financial institution (e.g., bond traders, portfolio managers, hedge funds, etc.). In some markets, due to counterparty risk, only banks with better credit strength would be the main market-makers (main counterparty) for the swap market. If a bank receives and executes a very large swap transaction, it can offset the swap through other offsetting swap trades with other banks to shed their risk. Likewise, if a swap agreement turns unfavourable and is expected to worsen, the position can be offset by swap arrangement with another counterparty. While the parameters would have changed for the new swap agreement, the offset may only be partial.

At the time of a swap transaction, the fixed rate offered will be equivalent to expected floating rate value as implied by the forward LIBOR curve. The fixed rate required is the compensation paid to the counterparty for the uncertainty of having to pay the short-term LIBOR rate over the term of the agreement. This is the **swap rate**. As the expectations for LIBOR changes, the swap rate will also change. The **LIBOR swap rate curve is created by plotting the rates** at future time periods to convert fixed rates to floating rates and floating rates to fixed rates. The swap rate is also sometimes quoted as a spread over the equivalent (similar term) local government yield. This is the **swap spread**.

Plain Vanilla Swap

Bank A enters into a plain vanilla swap agreement with Bank B with the following details:



Bank A pays Bank B a fixed rate of 5% per annum on a principal of $100 million.

In return, Bank B pays Bank A a floating rate of 6-month LIBOR for the same principal of $100 million.

The tenor of the swap is 3 years, with payment exchanges every 6 months.

Swaps start date is March 5, 2014.

The first exchange of payments would take place on 5 September 2014, 6 months after the initiation of the swap agreement. Bank A would pay Bank B $2.5 million. This is the interest on the $100 million for 6 months at 5%. Bank B would pay Bank A interest on the $100 million principal at the 6-month LIBOR rate prevailing 6 months prior to 5 September 2014 – that is on 5 March 2014. Suppose that the 6-month LIBOR rate on 5 March 2014, is 4.2%. Bank B pays Bank A 0.5 x 0.042 x $100 million = $2.1 million. Note that there is no

uncertainty about the first exchange of payments because it is determined by the LIBOR rate at the time the swap begins.

The second exchange of payments would take place on 5 March 2015, a year after the initiation of the swap agreement. Bank A would pay $2.5 million to Bank B. Bank B would pay interest on the $100 million principal to Bank A at the 6-month LIBOR rate prevailing 6 months prior to 5 March 2015 – that is, on 5 September 2014. Suppose that the 6-month LIBOR rate on 5 September 2014, is 4.8%. Bank B pays 0.5 x 0.048 x $100 million = $2.4 million to Bank A.

In total there are six exchanges of payment on the swap. The fixed payments are always $2.5 million. The floating rate payments on a payment date are calculated using the 6-month LIBOR rate prevailing 6 months before the payment date.

The following table provides a complete example of the payments made under the swap for the various 6-month LIBOR rates. The table shows the swap cashflows from the perspective of Bank A. Note that the $100 million principal is used only for the calculation of interest payments. The principal itself is not exchanged.

Essentially, Bank A is synthetically 'borrowing' at a fixed rate of 5% for 3 years and 'lending' at a 6-month floating rate based on LIBOR.

Cashflows to Bank A ($ million)

Date	LIBOR	Floating Cash Flow	Fixed Cash Flow	Net Cash Flow
Mar 5, 2014	4.20%			
Sep 5, 2014	4.80%	+2.10	−2.50	−0.40
Mar 5, 2015	5.30%	+2.40	−2.50	−0.10
Sep 5, 2015	5.50%	+2.65	−2.50	+ 0.15
Mar 5, 2016	5.60%	+2.75	−2.50	+0.25
Sep 5, 2016	5.90%	+2.80	−2.50	+0.30
Mar 5, 2017		+2.95	−2.50	+0.45

2.7.2 Foreign Exchange Swaps

Due to terminology, it is easy to get confused between a foreign exchange swap and a currency swap but there are differences. In a foreign exchange swap, two parties agree to buy and sell a currency pair for a specific amount at a specific agreed **spot** rate today and to **reverse the trade** by selling and buying back the same amount of currency at a later date at an agreed **forward** rate. So, there are **two legs** in a foreign exchange swap.

Foreign Exchange Swap to Lock in Forward Rate

A trader has EUR 5 million and requires USD in 6 months and wants to lock in the forward rate. The trader will sell EUR 5 million at the current EUR/USD spot rate and agree to buy back the EUR 5 million in 6 months at the EUR/USD forward rate.

2.7.3 Currency Swaps

In a currency swap, two parties agree to exchange a specific series of payments in one currency for a series of payments in another currency. The payments that are exchanged are essentially the interest and principal payments of a loan denominated in one currency for a loan of an equal amount of another currency.

Currency Swap

A Singapore bank borrows USD, and the US bank borrows SGD:

- The Singapore bank will pay for the US bank's debt in Singapore dollars i.e., principal and interest payments made in SGD
- The US bank will pay for the Singapore bank's USD debt i.e., the principal and interest payments in USD.

For this to work, both banks have agreed on a fixed or floating rate, the amount of borrowings and the term/maturity date.

2.7.4 Commodity Swaps

A commodity swap operates quite similar to a vanilla interest rate swap where the "floating" market price of the underlying commodity is traded for a fixed price in a specified period. SGX offers standardized swap agreements for various commodities alongside futures contracts.

2.7.5 Equity Swaps

For equity swaps, the two counterparties will agree to exchange a series of future cash flows periodically over time with at least one leg of the agreement involving equity returns flows or equity index returns (the latter being the more popular form of this swap). A couple of examples are cited below:

- An equity index return leg vs one leg pegged to (LIBOR + x%) over a period of 6, 9 or 12 months; or
- An equity index return leg vs a fixed rate of return over 12 months.

Equity Swap

It is 31 July 2017. A fund manager has enjoyed a strong performance for her passively managed fund which tracks the S&P 500 index, which is already up 10.2% year-to-date. She now expects the S&P 500 to track sideways for the rest of the year and that there will be a modest correction in the next two years. The fund manager is thinking of different ways to hedge her risks. She enters a swap contract with an investment bank as the counterparty to create a "partial" hedge.

Passive Fund Details

Portfolio size	= $400 million
Notional principal of swap	= $200 million
S&P 500 index value as of 31st July 2017	= 2,470.20

Investment bank agrees to pay 4% return fixed.
Fund manager agrees to pay percentage change in S&P500 + 2% return.

In the event of a negative return, the investment bank agrees to pay the fixed rate plus any loss on the index return for the 6-month period. Payment is made every 6 months for 4 periods (Jan 2018, Jul 2018, Jan 2019, Jul 2019) maturing in July 2019. The payouts would be calculated as follows:

Scenarios for S&P500	Fund Manager Pays	Investment Bank Pays	Net Payment & Fund Manager's Exposure
Jan 2018 S&P 500 rises +4% to 2,569.00	S&P500 change in % + 2% = 6% (i.e.; 4% +2% = 6%) $200m x 180/360 x 6% = -($6m)	Fixed Payment of 4%: $200m x 180/360 x 4% = +$4m	Portfolio of $400m rises by 4% = $416m Subtract net payment of: -$6m + $4m = -$2m Total value of fund = $416m - $2m = $414m Positive Return reported = + 3.50%
From 2,569.00 on Jan 2018 the S&P500 falls -8% to 2,363.48 on Jul 2018.	S&P500 change in % + 2% = -6% (i.e.; -8% + 2% = -6%) No payment since return is negative. (Instead, the fund manager now receives an additional 6%)	Fixed payment: $200m x 180/360 x 4% = +$4m Floating payment: = $200m x 180/360 x 6% = +$6m Total payment = $10m (It is as if the fund manager had sold off its position and received a fixed rate)	Portfolio falls -8% from $414m Jan 2018 to $380.88m on Jul 2018 = $380.88m Add net payment of: $4m + $6m = $10.00m Total value of fund = $380.88m + $10m = $390.88m Return reported over previous period = (390.88/414.0 -1) x 100 = -5.58% (negative)
From 2,363.48 in Jul 2018, the S&P500 rebounds +11% to 2,623.47 on Jan 2019	S&P500 change in % + 2% = +13% (i.e.; 11% + 2% = 13%) $200m x 180/360 x 13%= -($13m)	Fixed Payment: $200m x 180/360 x 4% = +$4m	Portfolio rises 11% from $390.88m Jul 2018 to $433.88m on Jan 2019 = $433.88m Subtract net payment of: -$13m + S4m = -($9m) Total value of fund = $433.88m -$9m =$424.88m Return reported over the previous period = 424.88/390.88 -1 = +8.70% (positive)

From 2,623.47 S&P500 rises +5% to 2,754.64 by Jul 2019	S&P500 change % + 2% = 7% (i.e.; 5% + 2% = 7%) $200m x 180/360 x 7% = -($7m)	Fixed payment: = $200m x 180/360 x 4% = +$4m	Portfolio rises by 5% from: $424.88m Jan 2018 to $446.12m on Jul 2019 = $446.12m Subtract net payment of: -$7m + $4m = - $3m Total value of fund = $443.12m Return reported over the previous period: 443.12/424.88 - 1 = + 4.28% (positive)

From a cash flow perspective:

'000m	Jul 2017		Jan 2018		Jul 2018		Jan 2019		Jul 2019	
Portfolio Value	$400.00	+4%	$416.00	- 8%	$380.88	+11%	$433.88	+5%	$446.12	
Swap Cash Flows			-2.0		+10.00		- 9.00		- 3.00	
Portfolio End Value		**+3.5%**	**$414.00**	**-5.58%**	**$390.88**	**+8.70%**	**$424.88**	**+4.28%**	**$443.12**	

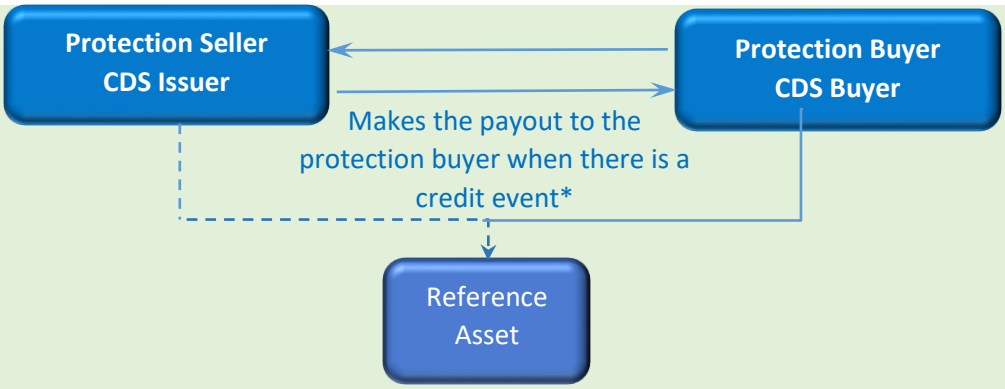
2.7.6 Credit Default Swaps

Credit default swaps (CDS) provide insurance against credit risk of a particular entity/obligor. The entity/obligor is known as the reference entity. A CDS acts like an insurance agreement where the seller of the swap agrees to compensate the buyer in the event, that the reference entity's respective bond or loan, also known as reference asset, defaults. The buyer of the insurance obtains the right to sell bonds issued by the reference entity for their face value when a credit event happens, and the seller of the insurance agrees to buy the bonds for their face value when a credit event occurs. The total face value of the bonds that can be sold is known as the credit default swap's notional principal. In exchange, the seller of the insurance receives a fee or a regular spread. CDS are bought over-the-counter or through CDS auctions and can be traded. There are no limits to the size of CDS contract, but the notional value is typically USD 10-20 million.

Credit Default Swap

Suppose an investor wants to buy a 3-year protection for a USD 20 million bond exposure to an oil-related logistics company at 160bps. The quarterly payment would be a charge of 160/4 = 40 basis points which would be (40/10,000 x USD 20 million) = USD 80,000 per quarter and annual cost is USD 320,000.

Pays a spread "x" basis points per annum



*In the case where physical settlement is specified: when a credit event happens, the protection seller will buy the reference entity's defaulted bonds at face value from the protection buyer.

CDS becomes a more speculative product when others who do not own the loan are also allowed to bet on a possible default – these are called "naked CDS". Issuing CDS to earn premiums for the purpose of insurance can be dangerous when done excessively. For a period before the Global Financial Crisis, the insurance company AIG sold huge amounts of CDS insuring mortgaged-backed securities that had triple-A rating. It was doing well and reaping good income from the issuance of CDS (mostly naked CDS) against the same pool reference assets. The overall potential collateral payback in the worst-case scenario was larger than the total sum of the reference assets. However, since these were triple-A rated mortgage-backed securities, the insurer thought it to be relatively safe.

A CDS can also be considered like a put option written on a bond. If the bond value falls, the buyer will see an increase in the value of the put as compensation. When mortgage-backed securities began to go into distress, their asset values dropped, and credit-rating agencies downgraded both the mortgaged-backed securities and AIG's own debt. Since AIG's credit rating was lowered, it triggered the requirement in the provisions for AIG to payout against all the CDS that it had issued. The CDS required payouts against insufficient AIG assets in the near term and it had insufficient offsetting trades against these swaps, so AIG collapsed.

2.7.6.1 CDS Spreads

The premium paid by the buyer for the protection to the issuer is called the CDS spread. It is usually quoted in basis points per annum of the contract's value (notional or otherwise) and is usually paid quarterly. Note that the term "spread" in the context of CDS is different from many other instruments – there is no reference benchmark against which this spread is measured, nor is there any risk-free rate we are measuring against. This is a simple premium charged directly against the value of the referenced asset.

CDS spreads are a measure of risks. When the risk increases the CDS spread will widen. Often the speed and degree to which credit spreads widen provides an indication of the risks associated with a particular event. For example, when there were concerns about the sovereign debt of Portugal, Ireland, Greece, and Spain, CDS spread increased significantly and when the risks subsided, so did the spread for the CDS.

2.7.6.2 Credit Event

A credit event would typically be (amongst others):

a. Bankruptcy;
b. Failure to Make Payment;
c. Debt Restructure (adverse change);

d. Repudiation;
e. Moratorium;
f. Obligation Acceleration; or
g. Obligation Default.

The bankruptcy of the underlying company issuing the bond and its failure to make payments, are considered as credit events.

Debt Restructuring refers to a change in the terms of the debt, which causes the debt to be less favorable to debt holders. Common examples of debt restructuring include a decrease in the principal amount to be paid, a decline in the coupon rate, a postponement of payment obligations, a longer maturity time, or a change in the priority ranking of payment.

When a credit event occurs, a notice will be issued by the buyer or seller to specify the protection amount that will be paid by seller either through physical settlement or cash settlement. For physical settlement, the issuer of the CDS (the protection seller) will buy the distressed loans or bond from the protection buyer. This is the obligation of the CDS issuer (protection seller).

2.7.6.3 Cash Settlement

For cash settlement, the counterparties simply determine the difference between the notional value of the CDS (i.e., the face value of the reference asset) at the initiation of the agreement against the final value (i.e., the value of the defaulted reference asset) after the credit event occurs. Payment is made by the protection seller to the protection buyer based on the difference.

2.7.6.4 Physical settlement

For physical settlement, the protection buyer has the right to sell bonds issued by the reference entity at face value to the protection seller. The protection seller is obligated to buy the bonds of the defaulted reference entity at face value.

2.8 Summary

1. Futures are standardized contracts for the purchase and sale of financial instruments or physical commodities, on regulated futures exchanges, and for future delivery.
2. Futures contracts are among the oldest exchanged traded derivatives instruments. They developed as a natural extension from forward contracts.
3. Futures markets serve as a platform for ongoing **price discovery** and price adjustment which buyers and sellers are willing to agree at the current point in time for future delivery given the available information and predictions about the future.
4. The futures market also provides **risk management** – risk is reduced because participants can lock in prices for future settlement.
5. The futures market can **augment the downward volatility and pressure** on underlying assets because traders and speculators can actively short the market during major market corrections using futures.
6. Futures contracts have the following features:
    - are **standardized contracts** for the purchase and sale of financial instruments or physical commodities, on a **regulated futures exchange**, for **future delivery**.

- are directly influenced by the **price movements** and value of their underlying assets.
- have **finite lifespans** ending on their expiration dates.
- carry a **notional value (monetary value upon conversion)**.
- have **low counterparty risk**. Their counterparty is the Futures Exchange.
- have the **ability to leverage** using an **initial margin.**
- **mark-to-market** at the end of the trading session.
- usually **square their positions** with the exchange to realize gains or losses without having to take delivery.

7. When using margin financing, investors have to comply with **mark-to-market** procedures, which involves putting up an **initial margin** (a small up-front payment to initiate a position), monitoring the amount of the **maintenance margin** (the minimum amount to be maintained on deposit with the broker at all times), and if the balance falls below this minimum amount, putting up an **additional margin** (to return the account to the initial margin level immediately or by a stipulated time when the broker makes a margin call). If the balance in the margin account falls below the maintenance level, an additional margin call is issued to the investor and the account must be returned to the initial margin level immediately or by a stipulated time.
8. Forward contracts are non-standardized contracts negotiated directly between a buyer and a seller on mutually agreed terms, are traded over-the-counter (OTC), and settlement terms may vary from contract to contract. Forward contracts are generally non-transferrable, and investors are exposed to counterparty risk.
9. Every futures contract contains exact **contract specifications** with regards to the underlying asset. Some basic information that the investor must know for each contract include: the description and standard size or denomination of the underlying asset, contract value, tick size, price limits, mark-to-market procedures and margin call procedures.
10. A buyer/seller of futures contracts can settle in the following ways:
    - **Delivery or Hold to Expiry** - At the expiration date, the buyer will accept delivery of the underlying asset (and the seller liquidates his position by delivering the asset) or make a cash settlement as per the contract.
    - **Offset Position** - The investor, who is long, can offset his position by selling contracts of the same type and quantities (and vice versa for an investor with a short position).
    - **Roll Position** - For longer-term traders, they can maintain their market exposure when the current contract expires by transferring or rolling their position to the new contract month.
11. A **term sheet** is a non-binding agreement covering the more important aspects of a deal, without going into every minor detail. It lays the groundwork by ensuring the parties involved are in agreement on most major aspects of the deal, giving them the opportunity to make clarifications. This reduces the chances of any misunderstanding as well as saves time and legal costs before a formal contract is drawn up.
12. Futures can be broadly classified into **commodity futures** (agriculture, energy & metals) and **financial futures** (equity, interest rates & foreign exchange). Innovation by market players has led to the creation of futures contracts in new areas such as electricity, real estate and weather.

13. **Interest rate futures** allow investors and traders to take positions based on how they believe the short-term rates or long-term interest rates might move across the yield curve given government fiscal and monetary policy actions and their view of specific economic performance statistics.

14. **Currency futures** are contracts to buy or sell a specified amount of a particular currency in relation to another currency at a set price and at a future date. Financial institutions, investment managers, corporations and private investors use currency futures to manage the risks or take advantage of profit opportunities arising from currency rate fluctuations.

15. **Interest rate parity** theory states that the forward (futures) premium or discount between two currencies should be equal to the difference in the domestic interest rates for securities of the same maturity. If this relationship is violated, arbitrageurs will be able to make risk-free profits in foreign exchange markets and their actions will force futures and spot exchange rates back into alignment.

16. **Equities index futures** seek to replicate the future performance of an underlying equity market index and as hedging and trading instruments for equity investors and especially fund managers who benchmark their portfolio performances against an equities index. Investing and trading in such index futures can allow managers to create a similar intended outcome without having to directly adjust components within their portfolio.

17. Equity index futures do not trade at precisely the fair value level aligned with the spot or cash value of the associated equity index all the time and this difference is the basis. The fair value of an equity index futures contract is normally expected to be positive such that futures prices is more than the spot price. This is attributable to the fact that under normal market conditions, financing costs normally exceed dividend yields.

18. A **Single Stock Futures** contract is expected to track the price movement on the underlying stock, its used by portfolio managers to protect against adverse price movements and diversify risks using margin financing.

19. **Commodity futures** provide one of the purest forms of exposure to the underlying commodity next to buying the physical commodity itself.

20. The derivatives traded on SGX include the following categories: foreign exchange, equity Indices, interest rates, commodities, and dividend indices. The price movements of the different derivative instruments vary. A tick is the minimum price movement of a trading instrument and the tick value is each price movement in terms of dollars.

21. **Clearing** is the process in which positions carried on Clearing Members' books are margined and marked to the latest market prices. At the end of each day, all outstanding positions are marked to their respective daily closing prices. A mutual offset system allows market participants to initiate positions in one exchange for allocation to the other on a real-time basis. SGX-DC and the Chicago Mercantile Exchange (CME) currently have mutual offset arrangement for some products.

22. The spot market price is the price at which cash transactions for the physical or actual commodity occur. The relationship between futures prices and spot prices can be summarized in terms of cost of carry. Cost of carry includes financial costs and economic costs. The cost of carry model assumes that arbitrage between the cash market and the futures market eliminates all imperfections in pricing. Two arbitrage strategies that are associated with the model are the **Cost-and-Carry Arbitrage** and **Reverse Cost-and-Carry Arbitrage.**

23. If distortions in the futures price are eliminated by arbitrage trading, the fair futures price will be the sum of the spot price and the net cost of carry of the asset.

24. The **expectancy model** of futures pricing states that the futures price is the expected spot price of an asset in the future. The theory suggests that it is not the relationship between the cash market price

and the futures price that is relevant, but the relationship between the expected spot price on the date of expiry of the contract and the futures price.

25. The difference between the actual futures price and the spot price is called the **basis**. When the markets are not bullish, not volatile or are in an upward secular run, the basis is usually positive but at times when the markets turn bearish, the basis can turn negative. If the asset to be hedged and the underlying asset of the futures contract are exactly the same, the basis will be zero at expiry of the futures contract. Before expiry, the basis may be positive or negative. In practice however, the futures contract and its underlying asset may not exactly match the asset being hedged.

26. When the futures prices for a commodity are in **contango**, traders in the market are willing to pay more than the theoretically expected price for that commodity in the future. We can observe over time the futures delivery price is falling to converge toward the spot price at expiry of the contract. **Backwardation** is the term used when we see the futures delivery price rise over time to meet the spot price.

27. An **interest rate swap (IRS)** is an agreement based on a notional principal amount, the term, payment frequency, where one party agrees to receive fixed interest rate payment in exchange for making floating interest rate payment to a counterparty based on the same notional amount. At each payment date, the parties will settle the difference between the fixed and floating interest rates.

28. When we refer to a **foreign exchange swap,** the two parties agree to buy and sell a currency pair for a specific amount at a specific agreed spot rate today and to reverse the trade by selling and buying back the same amount of currency at a later date at an agreed forward rate.

29. When we refer to a **currency swap,** we are talking about two parties agreeing to exchange a specific series of payments in one currency for a series of payments in another currency. The payments that are exchanged are essentially the interest and principal payments of a loan denominated in one currency for a loan of an equal amount of another currency.

30. For **equity swaps**, the two counterparties will agree to exchange a series of future cash flows periodically over time with at least one leg of the agreement involving equity returns or dividend flows, or an equity index return, the latter being the popular form of this swap.

31. **Credit default swaps (CDS)** provide insurance against credit risk. A CDS acts like an insurance agreement where the seller of the swap agrees to compensate the buyer in the event that the reference entity's respective bond or loan defaults. In exchange, the issuer of the CDS (protection seller) receives a fee or a regular spread.

Chapter 3:
Futures Strategies

Learning Objectives
The candidate should be able to: ✓ Understand and explain the participants and their roles in the futures market. ✓ Explain the various types of futures orders, futures trades and trading strategies. ✓ Describe and explain the different types of spread trades. ✓ Explain how to identify and manage financial risks using futures. ✓ Describe and demonstrate how to develop and manage hedging programmes for various cash positions. ✓ Describe and demonstrate how to develop and manage hedging for short-term and long-term interest rate risks. ✓ Explain how to manage a hedge programme for equity risk exposures. ✓ Describe and demonstrate arbitrage strategies using short-termand long-term interest rate futures.

3.1 Market Participants

The main participants in the futures markets are:

a. Hedgers;
b. Speculators;
c. Market Makers;
d. Proprietary Trading Firms;
e. Arbitrageurs;
f. Portfolio Managers;
g. Hedge Funds; and
h. Central Banks, Sovereign Funds and Regulators.

As volatilities in financial markets increase, investors and market participants who are driven by different motivations look to derivatives for hedging or maximizing returns. It is important to understand their motivations, and that participants can take on different perspectives of a speculator, hedger and arbitrageur

at different times. Knowing which group of participants is driving the market can help in deciding whether to go with or against the market direction.

For traders and portfolio managers, a risk-increasing or risk-reducing trade will affect their choice of instrument and strategy. The increase in hedge funds and greater allocations to alternative assets, has contributed to the rise of innovative trading strategies that use derivatives. Growth in wealth management has also resulted in financial institutions offering more structured products to affluent clients, and most of these products are constructed using futures and other derivatives.

3.1.1 Hedgers

Hedgers have a position in the underlying asset or commodity. They use futures to reduce or limit the risk associated with an adverse price change. Producers, such as farmers, often sell futures on the crops they raise, to hedge against a drop in commodity prices. This makes it easier for producers to do long-term planning.

Similarly, industrial users such as food processing plants buy futures to secure their input costs. This allows them to base their business planning on a fixed cost for core supplies (e.g., corn or wheat), manage price risk and stabilize the cost passed on to customers. Other examples include airlines hedging fuel costs or jewellery manufacturers hedging the cost of gold and silver.

3.1.2 Speculators (Individual Traders)

Many speculators are individuals trading with their own funds. Traditionally, individual traders have been characterized as individuals wishing to express their opinion about, or gain financial advantage from, the price direction of a particular market. Electronic trading has levelled the playing field for the individual trader by improving access to price and trade information. The speed and ease of trade execution, combined with the application of modern risk management techniques, gives individual traders access to markets and strategies that were once reserved for institutions.

3.1.3 Market Makers

Market makers are trading firms that have contractually agreed to provide liquidity to the markets, continually providing both bids (an expression to buy) and offers (an expression to sell), usually in exchange for a reduction in trading fees. Increasingly, the electronic market makers as a group provide much of the market liquidity that allows large transactions to take place without effecting a substantial change in price. Market makers often profit from capturing the spread, the small difference between the bid and offer prices over a large number of transactions, or by trading related futures markets that they view to have price or profit opportunities.

3.1.4 Proprietary Trading Firms

Proprietary trading firms, also known as prop shops, profit as a direct result of their traders' activity in the marketplace. These firms supply their traders with the education and capital required to execute a large number of trades per day. By using the capital resources of the prop shop, traders gain access to more leverage than they would if they were trading on their own account. They also gain access to research and strategies developed by larger institutions.

Some large banks still have their own proprietary trading units for trading derivatives and their underlying assets. One of the most famous cases of bank losses from proprietary trading was the fall of Barings in 1995 due to a single proprietary trader's excessive risk taking on the Nikkei 225 Futures, which accumulated losses

that were twice the bank's capital. There have been a number of major cases of misconduct involving proprietary trading which surfaced during the Global Financial Crisis and thereafter. Due to poor controls and supervision for proprietary traders, some banks incurred huge losses trading derivatives and decided to close off their proprietary trading units.

Financial institutions actively hedge their sale of various derivatives instruments to different groups of investors (institutional and private clients) through futures and options. Those issuing exchange traded funds (ETFs) and exchange traded notes (ETNs) on stock exchanges also manage daily offsetting futures and options positions against their long and short ETF and ETN positions through their proprietary trading desk. Judging by the proprietary trading losses (e.g., from distressed debts, collateralised debt obligations (CDOs), credit default swaps (CDS), Delta One[17] instruments) that are still reported by several major international banks well after the Global Financial Crisis, the appetite for high risk-rewards will not go away. We should expect proprietary trading to remain a part of the banking landscape in future, even if such activities have pared down for a few seasons amidst global regulatory scrutiny.

3.1.5 Arbitrageurs

Arbitrageurs seek to make riskless profits from the disequilibrium between the cash and futures markets. They do not take directional bets on the market, but instead take advantage of price differences between related markets or have a fixed relationship, i.e., exploit the difference between the implied value and the market value. These players rely on computer systems and models to identify and capitalize on arbitrage opportunities. When there is disequilibrium due to mismatched demand and supply in one market, windows of opportunities usually close up very fast as computers identify such opportunities immediately and any disequilibrium is arbitraged away.

Arbitrageurs also help to improve liquidity in the futures market by helping to ensure that the market is efficient and the pricing relationship between the futures contract and its underlying instrument is consistent. Arbitrageurs normally work for institutions such as market makers and proprietary trading firms.

3.1.6 Portfolio Managers

A portfolio or investment manager is responsible for investing or hedging the assets of a mutual fund, ETF or closed-end fund. The portfolio manager implements the fund's investment strategy and manages the day-to-day trading. Futures are often used to increase or decrease the overall market exposure of a portfolio without disrupting the delicate balance of investments that may have taken a significant effort to build.

3.1.7 Hedge Funds

The diverse and highly liquid futures market offers hedge funds the ability to execute large transactions and adjust the market exposure of their portfolios without having to directly change the security holdings in their portfolio. A hedge fund is a managed investment portfolio that uses advanced investment strategies to maximize returns, either in an absolute sense or relative to a specified market benchmark. The name hedge fund is mostly historical, early hedge funds predominantly sought to offset the risk of a bear market by shorting the market. Today, hedge funds use different strategies to maximize returns. On occasion, hedge funds have

[17] Delta One trading desks provide over-the-counter (OTC) derivatives structures that are pegged one-to-one typically against the underlying, usually an index. The bank will trade on the underlying and its related derivatives (e.g. it can involve swaps with other banks) and seek to make arbitrage profits or profit from the performance of their holdings moving against the direction of the derivative structure sold. The trades are usually significantly large because margins are extremely fine. Since most of these trades are carried out OTC, they are not easy to monitor or regulate.

exploited the weakness of the market by shorting the futures market, collectively adding sharp downward pressure on both the futures and spot markets.

3.1.8 Sovereign Funds, Central Banks and Regulators

Due to the collective selling pressure from speculators and hedge funds, governments have sometimes found it necessary to offset the negative volatility by intervening in both the spot and futures markets to counterbalance speculative forces to stabilize the market. In severe situations, the tightening of liquidity, capital flows and trading activity have also been used to break the trading momentum in both the futures and spot markets. For example, during the Asian Currency Crisis in 1997, several Asian currencies and equity markets were driven downwards under persistent heavy selling pressure, but the speculation was broken when some Asian governments intervened in the spot and futures markets and introduced capital and currency exchange controls, suddenly locking the speculators into their short currency positions that they could not unwind for a period while incurring interest cost.

On a day-to-day basis, many governments also actively manage their currency within a trading band and actively participate in the currency spot and futures markets to influence the direction of their currency movement.

3.2 Types of Orders on the SGX

Most of the types of orders are common across different futures and stock exchanges (see Table 3.2).

Table 3.2 – Types of Orders

Order Type	Description
Limit Order	• Order with a specified price and quantity. • A limit order has a price (premium) on the order, showing how much the trader is willing to pay or wants to receive for 1 contract.
Market to Limit (MTL) Order	• MTL orders have a validity time like other orders and can thus be entered as Fill-and-Kill (FAK)/Fill-or-Kill (FOK) or be stored in the book as a Day order or Good-Till-Cancelled (GTC) order. • If the order is a store order, any remaining quantity in the order will be stored in the book at that price. • If there is no price at the opposite side of the order book, the order is stored as a limit order at 1 price tick better than the best price on the same side of the book. If no such price exists, the order will be rejected.
Market Order	• If it is entered during a state where orders are continuously matched, then the market order will be matched at the best possible price. • If it is entered during a state where orders are not continuously matched, then the market order takes the Equilibrium Price (EP) as its price. Accordingly, the order price will change if the EP changes.

Order Type	Description
Stop (Limit, MTL and Market) Order	• Once a certain Stop Order trigger condition is fulfilled, the Stop Order is converted to either Limit, Market or MTL order. • A Stop Order is not shown to the market before it is converted to the specific order. The Stop Order trigger condition can be defined by the following parameters: ◦ **Stop Series**: The Stop Series defines the instrument the Stop Price shall be compared to. ◦ **Stop Price**: The Stop Price defines what price level the price condition must meet to convert the order to a limit order. ◦ **Stop Price Reference Type**: The Stop Price Reference Type can be Bid Price, Ask Price or Last Price. This shows if the Stop Price will be compared to the Bid, Ask or Last Price. ◦ **Stop Price Condition**: The Stop Price Condition can be bigger than or equal to (>=) or less than or equal to (<=). This will indicate if the Stop Price must be bigger than or equal to (>=) or less than or equal to (<=) the Bid, Ask or Last Price to convert the order to a limit order. • When the Stop Order trigger condition is fulfilled, the Stop Order is converted to the specified order type as defined in the order itself.
Market-if-Touched (MIT) Order	• A MIT is an order to buy (or sell) a contract below (or above) the market. • This order is held in the system until the trigger price is touched and is then submitted as a market order. • A MIT order is similar to a Stop order, except that an MIT sell order is placed above the current market price, and a stop sell order is placed below. • MIT orders are not shown to the market before they are converted to the specific order.
Session State Order (Limit, MTL, Market, Stop and MIT)	• Session state orders (SSOs) trigger the entry of an order into the order book. SSOs will trigger only when the market transitions into a new session state. Market participants can query on the list of valid session states and select the appropriate session state. • SSOs and their triggered orders are not visible to other market participants until triggering occurs. The SSOs will be deleted at the end of the trading day if the market does not transition into the session state during the day (e.g., due to trading halts, invalid session state etc.). SSOs may not be GTC.

Source: SGX

3.2.1 Other Types of Orders - Packs & Bundles

Packs and bundles were introduced by CME Group for Eurodollar contracts. NYSE Euronext LIFFE's packs and bundles are available for Short-Term Interest Rate (STIR) Euribor and Sterling futures contracts[18].

A futures pack or bundle involves the purchase or sale of a series of futures representing a segment along the yield curve. It can be used to create or liquidate positions along the yield curve and allows users to gain

[18] London International Financial Futures and Options Exchange (LIFFE) is a futures exchange based in London.

exposure to longer term interest rates, without the risk of not being able to complete a particular leg of a spread or strategy (i.e., legging risk). The advantage is that transactions can be carried out at a single price or value, eliminating the need to enter multiple orders in each contract, reducing the overall cost of trading, and limiting the possibility that some orders may go unfilled.

A **futures pack** is a type of futures order enabling purchase of a predefined number of futures contracts in 4 consecutive delivery months. For example, Eurodollar packs are the simultaneous purchase of an equally weighted, consecutive series of 4 Eurodollar futures. All 4 contract months in the strip are executed in a single transaction, eliminating the inconvenience of partial fills.

A **futures bundle** is a type of futures order that allows investors to buy a predefined number of futures contracts in each consecutive quarterly delivery month for 2 or more years. An order can contain all the quarterly futures contracts within the standard 2, 3 or 5-year bundle periods (there are no 1-year bundles because these are effectively packs). For example, a Eurodollar bundle consists of the simultaneous sale or purchase of one each of a series of consecutive Eurodollar futures contracts. The first contract in any bundle is typically the first quarterly contract in the Eurodollar strip, but bundles can be constructed starting with any quarterly contract.

The price of a pack or bundle is quoted based on referring to the average net change in the value of all Eurodollar futures contracts included in the pack or bundle since the prior day's settlement price. They are quoted in increments of ¼ of 1 basis point (0.01%). The price quotation reflects the average net change based on the net price of each of the spread's constituent contracts.

3.3 Types of Trades & Strategies

The main investment and trading strategies used in the futures markets are:

a. Outright trades;
b. Hedging;
c. Basis trades; and
d. Spread trades.

a. **Outright trades** - An investor can simply decide to buy or sell a futures contract, thereby wagering that the price of the underlying asset will rise or fall. Outright trades are familiar to most equity investors and are easy to understand.

b. **Hedging** - The trader sells a futures contract to offset positions he holds in the cash market. For instance, if you have a large portfolio of U.S. stocks that you do not want to sell for tax reasons but fear a sharp market decline, you could sell S&P 500 index futures as a hedge against a market decline.

c. **Basis trades** - This is an arbitrage trading strategy where the trader takes opposing long and short positions in the two securities to profit from the convergence of their values. The strategy is called basis trading because it aims to profit off very small basis point changes in value between two securities.

   For example, a trader could be long (short) futures and short (long) the spot market in Treasuries. He could buy 10-year U.S. Treasury bond futures and short physical 10-year U.S. Treasury bonds, expecting the undervalued futures to appreciate relative to the overpriced bond, thus netting him a profit from his positions. For the trader to make a worthwhile profit, he would have to undertake a large amount of leverage to increase the size of his positions.

d. **Spread trades** - Spread trades are discussed in detail in Section 3.4.

3.4 Spread Trades

A spread combines both a long and a short position, executed at the same time in related futures contracts. The purpose of the strategy is to mitigate the risks of holding only a long or a short position.

Margin requirements tend to be lower due to the more risk adverse nature of spread trades. However, when spread trades are created with different commodities or with different maturity dates, they are imperfect hedges and the spreads may become vulnerable to both legs moving in a less inversely correlated fashion than expected, therefore resulting in potential losses. Table 3.4 sets out the various types of spread trades.

Table 3.4 - Types of Spread Trades

Spread Trade	Description
Intra-commodity	Spread is on the same commodity
Inter-commodity	Spread is on different commodities
Intra-market	The positions are traded on the same exchange
Inter-market	The legs of the spread trade on different exchanges
Intra-delivery	The contracts mature in the same delivery month
Inter-delivery	The contracts mature in different delivery months

Example of a Spread Trade

A customer is long March gold and short June silver. They both trade on the Chicago Board of Trade (CBOT). What type of spread trade is this?

This spread trade can be described as:

- **Inter-commodity** – gold and silver are both metal commodities, but they are not identical.
- **Intra-market** - both commodities trade on the CBOT; or
- **Inter-delivery** - one commodity is delivered in March and the other in June.

3.4.1 Calendar Spread

A **calendar spread** (also known as a **horizontal or time spread**) is a position established with futures contracts by simultaneously entering a long and short position on the same underlying asset but with different delivery months. The strategy can be executed in options or futures.

The strategy seeks to profit from the difference in the change in the near-term futures contract (1st leg) compared to the change in the distant futures contract (2nd leg). In a successful execution of a calendar spread, a trader makes more profit in one leg to offset losses in the other leg. So, it is the extent of the movement in each leg of the futures contract that will typically determine profit or loss.

Buying a calendar spread involves the following actions:

Type of Calendar Spread	Near Term Futures Contract (Leg 1)	Distant Futures Contract (Leg 2)
Bullish Calendar Spread	Long / Buy	Short / Sell

Type of Calendar Spread	Near Term Futures Contract (Leg 1)	Distant Futures Contract (Leg 2)
Bearish Calendar Spread	Short / Sell	Long / Buy

In currency pairs for example, a bullish USD means we are bearish on EUR and vice versa. Hence, whether we call it a **bullish or bearish calendar spread,** it depends on which currency pair you are considering. Therefore, with currency pairs, it is better to be clear and state which currency you are bullish about when establishing strategies.

More importantly, with regards to spread strategies, it is important to have a view as to which leg needs to strengthen or weaken in order to make a profit.

Bullish Calendar Spread and Futures Contracts Rise

Suppose it is 19 April 2020 and the S&P 500 index sits at 2,344.00. On the Chicago Mercantile Exchange (CME), the Jun 2020 E-mini futures contract is asking for 2,335.00 and its Dec 2020 E-mini futures contract is asking for 2,330.00. A trader believes that the 1Q 2020 earnings reporting season and infrastructure growth story will lead to a higher S&P 500 index level. However, he also wants to partially hedge on the risk of being wrong and the S&P 500 declining.

Suppose the initial margin requirement is $6,000 per nominal contract value of $100,000 (i.e., 6%) and the maintenance margin requirement is $5,000 per nominal value of the contract of $100,000 (i.e., 5%). Contract unit size = 50 x S&P 500 E-mini futures contract price.

The trader can buy 10 contracts of the Jun 2020 S&P 500 E-mini futures and sell 10 contracts of the Dec 2020 S&P 500 E-mini futures at the prices below. What is the initial margin of the strategy?

At 6% initial margin, the set-up of the strategy is $139,950 The calculation is shown below:

April 2020 (initial position)	No. of Contracts	S&P 500 E-mini Contract Value	Contract Value as at April 2020		Initial Margin 6%
Jun 2020 (1st Leg)	Buy 10	2,335	50 x 10 x 2,335	= $1,167,500	$70,050
Dec 2020 (2nd Leg)	Sell 10	2,330	50 x 10 x 2,330	= $1,165,000	$69,900
Total outlay in terms of the initial margin for the strategy					**$139,950**

Note: Assume worst case scenario, initial margin is not net off.

Scenario 1 – S&P 500 Index Rises by 4%

Let us suppose that the trader is correct. 1Q 2020 earnings season results surpassed expectations and in response, the S&P 500 index rises by about 4% to 2,438 in May 2020. The S&P 500 E-mini futures contract for June rises by 6% to 2,475, and December contract rises by 5% to 2,447.

What is the net profit position from the strategy?

May 2020	Close Out Trade	S&P 500 E-mini Contract Value	Contract Value as at May 2020		Return on Outlay
Jun 2020 (1st Leg)	Sell 10	2,475 ↑ +6%	50 x 10 x 2,475	= $1,237,500	
Dec 2020 (2nd Leg)	Buy 10	2,447 ↑ +5%	50 x 10 x 2,447	= $1,223,500	
Profit from closing long position on 1st leg = + $1,237,500 – $1,167,500				$70,000	
Loss from closing short position on 2nd leg = - $1,223,500 + $1,165,000				-$58,500	
Net profit from the strategy (excluding fees)				**Profit $11,500**	**+ 8.2%**

Scenario 2 – SP& 500 Index Rises by 2.5%

Suppose the same strategy is deployed, but this time 1Q 2020 results were mixed. The outlook for the next year is still very positive with expectations that the US government's policy of raising infrastructure spending will materialise in 6-9 months' time. The S&P 500 index rises by about only 2.5% to 2,402.60 in May 2020. CME's S&P 500 E-mini futures contract for June 2020 rises only by 3% to 2,405, and December 2020 contract rises by 3.5% to 2,412.

What is the net profit position from the strategy?

May 2020	Close Out Trade	S&P 500 E-mini Contract Value	Contract Value as at May 2020		Return on Outlay
Jun 2020 (1st Leg)	Sell 10	2,405 ↑ +3.0%	50 x 10 x 2,405	= $1,202,500	
Dec 2020 (2nd Leg)	Buy 10	2,412 ↑ +3.5%	50 x 10 x 2,412	= $1,206,000	
Profit from closing long position on 1st leg = + $1,202,500 – $1,167,500				$35,000	
Loss from closing short position on 2nd leg = - $1,206,000 + $1,165,000				-$41,000	
Net profit from the strategy (excluding fees)				**Loss - $6,000**	**- 4.3%**

Note that in this Bullish Calendar Spread strategy, we had both contracts going up. However, in one scenario, the June 2020 contract moved up more than the increase in the December 2020 contract, so profits were made. When the opposite happened, we had losses. In a correction, you can still make profits if the June 2020 futures contract is closed off with lesser losses than the gains from covering the short position on the December 2020 futures contracts.

Scenario 3 – S&P 500 Index Falls

Suppose the same strategy is deployed by a trader who was bullish about the market. However, he was wrong, and the S&P 500 index opened 4% lower. It continued its downtrend at mid-morning to 7% when the circuit breaker[19] kicked in.

Upon resumption, the trader managed to close off his long Jun 2020 E-mini futures contract early when it was 7.5% down from his purchase price. Volatility continued falling more throughout the day and the trader managed to close off his Dec 2020 E-mini futures short position when it was down 9%. At the close of the trading day, the S&P 500 recovered slightly by being down 8% after being down 10% at one point.

May 2020	Cost Price	Close Out Trade	S&P 500 E-mini Contract Value	Contract Value as at May 2020		Return on Outlay
Jun 2020 (1st Leg)	2,335	Sell 10	2,160 ↓ - 7.5%	50 x 10 x 2,160	= $1,080,000	
Dec 2020(2nd Leg)	2,330	Buy 10	2,120 - 9.0%	50 x 10 x 2,120	= $1,060,000	
Loss from closing long position on 1st leg = + $1,080,000 - $1,167,500					- $87,500	
Profit from closing short position on 2nd leg = - $1,060,000 + $1,165,000					+ 105,000	
Net profit from the strategy (excluding fees)					**Profit $17,500**	**+12.5%**

[19] "Circuit Breaker" in relation to a security or futures contract refers to such maximum allowable price advance or decline from a reference price for the security or futures contract during the trading phase.

Bearish Calendar Spread for E-micro EUR/USD Futures on CME		
Contract unit		12,500 EUR
Minimum price fluctuation (tick)		0.0001 USD/EUR
18 Dec and 17 Mar Futures contracts (as at 17 Nov)		EUR/USD 1.1810, 18 Dec 2017 EUR/USD 1.1866, 17 Mar 2018
Market View @ Nov 2017	:	The President of the European Central Bank is dovish and interest rates will not rise as early as in the US where the Feds are bent on a rate hike in December. The USD is expected to strengthen further with a potential second-rate hike in March next year. EUR/USD is expected to weaken over the next 3 months.
Strategy	:	Sell 100 EUR/USD contract on 18 Dec 2017 Buy 100 EUR/USD contracts on 17 Mar 2018
18 Dec and 17 Mar Futures contracts (as at 7 Dec)	:	EUR/USD 1.1750, 18 Dec 2017 EUR/USD 1.1828, 17 Mar 2018
Profit calculation as at 7 Dec if we close out the position	:	100 x Dec contract 12,500 x (Sold 1.1810 - Buy back 1.1750) = +7,500 100 x Mar contract 12,500 x (Sold 1.1828 - Bought at 1.1866) = -4,750 **Profit from the spread for 100 Dec and Mar contracts** +2,750
Gains from shorting December contract and buying back generated $7,500 while March contract bought at 1.1866 was sold at EUR/USD 1.1828, incurring a loss amount of - $4,750. The profit gained was $2,750.		

To profit from a bullish calendar spread, the following conditions must hold:

a. The near-term futures contract bought must rise more than the distant contract sold; or

b. The near-term futures contract bought falls less than the distant contract sold.

Likewise, a bearish calendar spread is a strategy where the near-term futures contract is shorted, and more distant futures contract is bought. The following conditions must hold for a bearish calendar spread to yield profits:

a. The near-term futures contract sold needs to fall more than the distant futures contract bought; or

b. The near-term futures contract sold needs to rise less than the distant futures contract bought.

Table 3.4.1 illustrates the profit profiles for a bullish and bearish calendar spread.

Table 3.4.1 - Profit Profiles for Bullish and Bearish Calendar Spreads

Profit for Bullish Calendar Spread	
Near Term Contract Bought (1st Leg)	**Distant Contract Sold (2nd Leg)**
Rises More	(1) Rises less (2) Does not rise (3) Falls
(1) Falls less (2) Does not fall	Falls by more

Profit for Bearish Calendar Spread	
Near Term Contract Sold (1st Leg)	**Distant Contract Bought (2nd Leg)**
(1) Rises less (2) Does not rise (3) Falls	Rises more
Falls by more	(1) Falls by less (2) Does not fall

3.4.2 Butterfly Spread

A butterfly spread is a neutral trading strategy that contains 4 legs consisting of 3 different contracts combining bull and bear spreads. Investors or traders can buy or sell the butterfly spread depending on how they expect the different futures contracts to move.

In a **bull butterfly spread**, the investor seeks to profit from a rise in the underlying asset's price while in a **bear butterfly spread**, a profit is made from a fall in the price. A near term bull spread and a longer-term bear spread may be combined, or vice versa. The trader will buy in 2 different months, and then sell twice in 1 month. The month that is sold twice is expected to underperform the other 2 months that are bought individually.

In executing such a strategy, the trader believes that the mid-term futures contract prices will drop more while near term futures contract and distant futures prices will not fall as much. To generate profits from setting up a butterfly spread, the near-term contract is expected to strengthen (become more positive or less negative) relative to the distant spread (or wing).

Overall profits are generated when the spread between the near term and mid-term contract is higher than the loss sustained from the negative spread between the mid-term and distant contract. Alternatively, there is also the possibility of both spreads being positive for the investor, but this happens less frequently.

3.4.2.1 Buying a Butterfly Spread

Buying a butterfly spread means:

a. Buying 1 of the nearest delivery month (Leg 1);

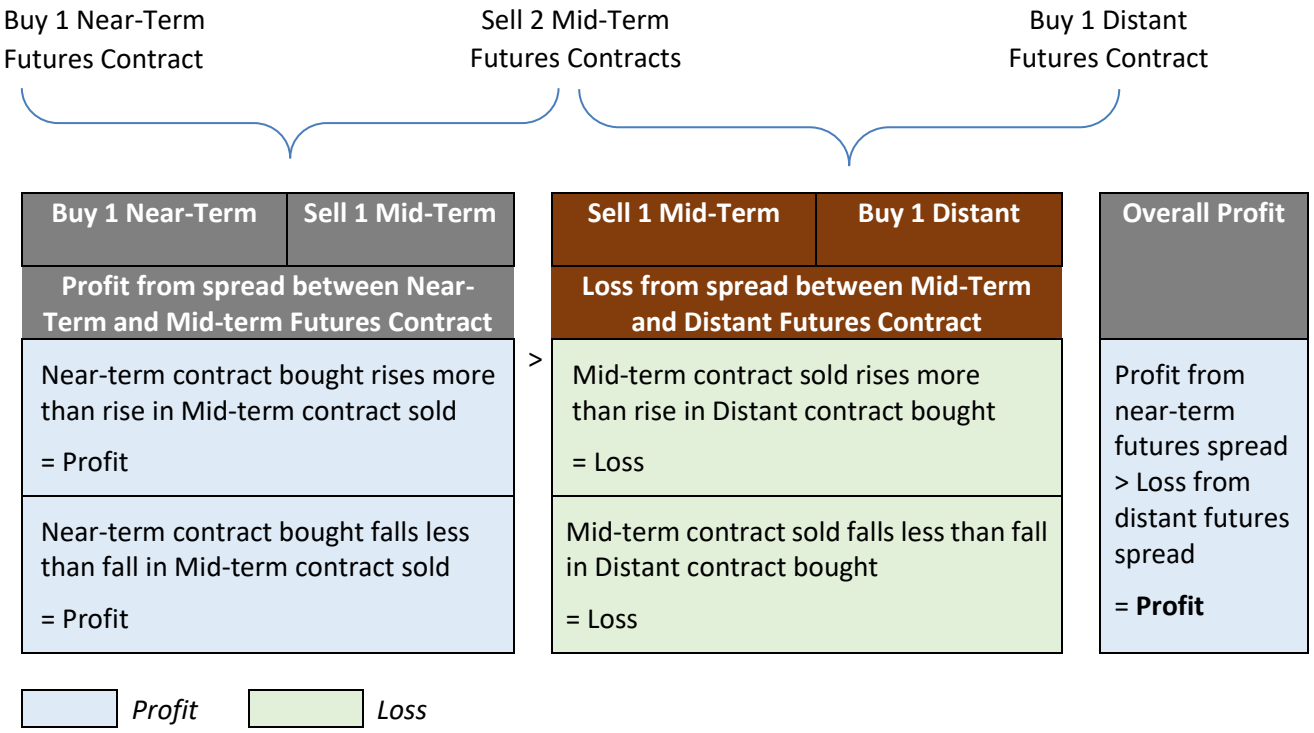
b. Selling 2 of the second nearest delivery month (Leg 2); and

c. Buying 1 of the furthest delivery month (Leg 3).

The ratio for purchase of a butterfly spread is +1: -2: +1 .

Figure 3.4.2.1 shows the profit from buying a butterfly spread.

Figure 3.4.2.1 - Profit from Buying a Butterfly Spread



This strategy is executed for commodities, where seasonal changes cause demand and supply of futures contracts. Near-term futures contracts may rise more significantly on strong demand compared to mid- term contracts and distant contracts. However, the spread is wider for the first pair of contracts compared to the second pair of contracts thereby generating profits.

Some traders of financial futures may also execute the strategy to take advantage of policy changes by the US Federal Reserve, such as interest rate hikes, or announcements of fiscal policies, bringing increased demand and price movement for financial futures in the near-term contracts compared to the mid-term contracts with minimal impact on the distant contracts.

Some commodities like precious metals will also react to expectations of how the USD and interest rates will change in the coming months. Traders in gold futures may prefer to take spreads between the differences in movement of the various futures contracts rather than take outright directional exposure.

Buying a Butterfly Spread

After the US Federal Reserve raised interest rates in December 2019, a trader decided to set up a butterfly spread because he expected short-term political uncertainty to drive up gold futures prices. However, with a potential of a second interest rate hike in June or July 2020, gold futures prices in June and July may somewhat be dampened making things more unpredictable.

Also, since gold can be quite volatile, he decided to take a spread position rather than a directional market exposure. He has the potential to close off his position just before the start of the 1st French election on 21 April 2020 when gold prices have risen amidst uncertain election results.

Assuming that the trader is roughly correct, this is how buying a butterfly spread works.

	Set up prices on 6 Jan 2020 Per Troy oz	Close position on 21 Apr 2020 Per Troy oz	Change Profit / (Loss) on contract position per Troy oz	Close position on 24 Apr 2020 Per Troy oz	Change Profit or (Loss) on contract position per Troy oz
Buy 1x Apr Gold Futures Contract	1,176.3	1,287.4	+111.1	1,275.8	+99.5
Sell 2x Jun Gold Futures Contract	1,179.0	1,289.1	- 110.1	1,277.5	-98.5
Buy 1x Aug Gold Futures Contract	1,181.7	1,292.4	+110.7	1,280.8	+99.1
1 contract = 100 Troy oz.					
Assume Butterfly Spread was closed out on 21 Apr 2020 before French Election Profits (Loss) from Near-Term Spread = (+111.1 - 110.1) Profit (Loss) from Distant Spread = (-110.1 + 110.7)					 +1.00 +0.60
Overall Spread					+1.60
Profit upon closing out on 21 Apr 2020 = 100 x 1.60 x 100 (i.e.; 100 Troy oz x 1.60 x 100 contracts)					$16,000
Assume Butterfly Spread is closed out on 24 Apr 2020 after French Election Profits (Loss) from Near-Term Spread = (+99.5 - 98.5) Profit (Loss) from Distant Spread = (-98.5 + 99.1)					 +1.00 + 0.60
Overall Spread					+1.60
Profit upon closing out on 24 Apr 2020 = 100 x 1.60 x 100 (i.e.; 100 Troy oz x 1.60 x 100 contracts)					$16,000
NOTE: Although gold prices came off after the French elections, **the spread was actually maintained on 24 April after the elections**. Such strategies simply look at the differences between the spreads and are less affected by the directional price movement of the underlying asset. So, whether gold futures prices go up or down, the strategy still allows the trader to earn a profit from the spread. In this case, profit was made on both legs of the butterfly spread on both dates although this does not always happen.					

3.4.2.2 Selling a Butterfly Spread

The reverse can also happen. Traders can also expect the mid-term futures contracts to rise more dramatically than the near-term futures contract and for the distant futures contracts to rise by less. In that case, selling the butterfly spread would make better sense.

Selling a butterfly spread means:

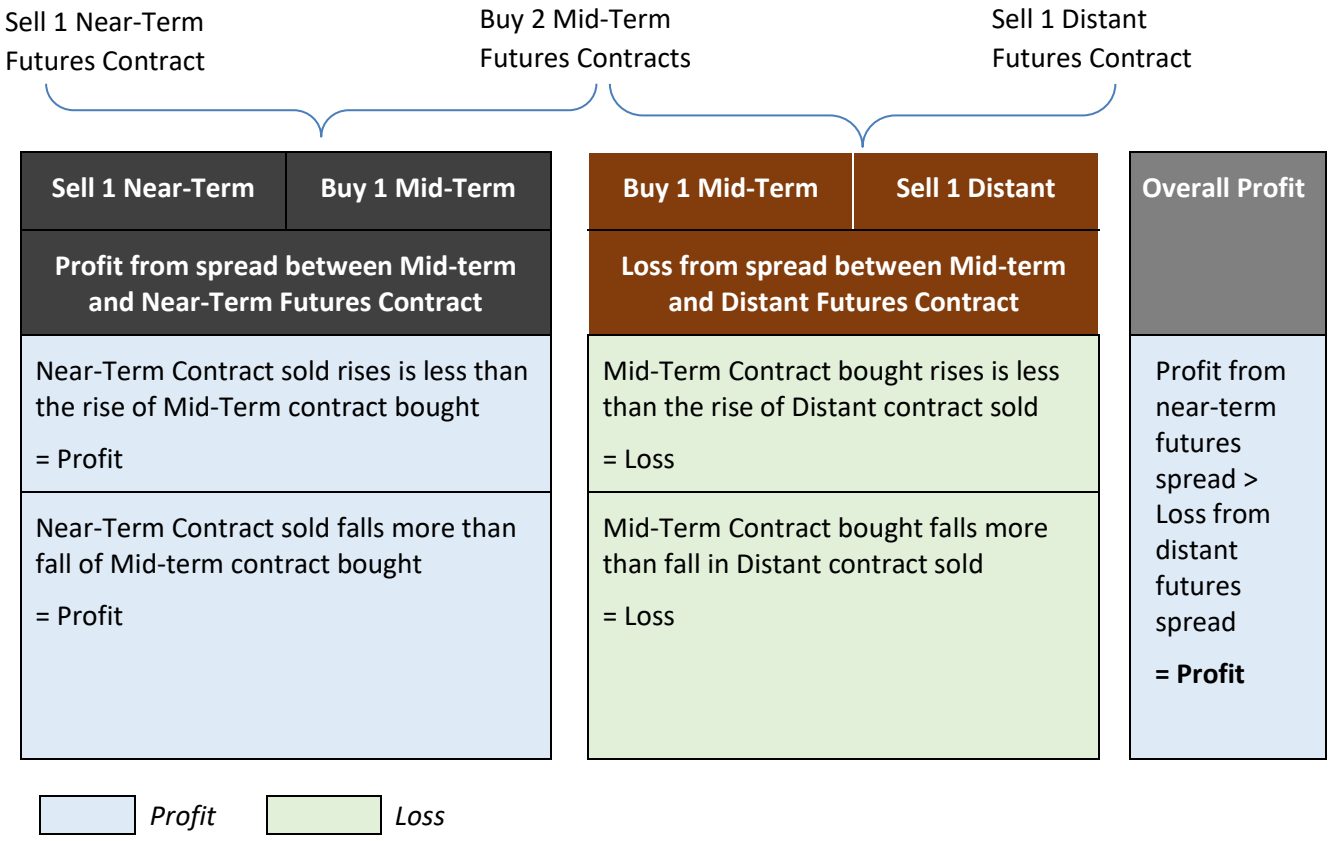
a. Selling 1 futures contract with near term delivery;

b. Buying 2 similar futures contract of mid-term delivery; and

c. Selling 1 futures contract with distant delivery.

Figure 3.4.2.2 shows the profit from selling a butterfly spread.

Figure 3.4.2.2 - Profit from Selling a Butterfly Spread



Selling A Butterfly Spread

It is 8 February 2020. An investor believes the yields will rise sharply in the near term, causing Eurodollar futures contracts to fall in the near term but he expects yields will dip back down because the economic data will disappoint and there will be new demand for longer maturities especially within the next 6-9 months.

Set up Sell Butterfly Spread in Feb 2020:		**8 Feb 2020**	**21 Apr 2020**
Short 1 x Eurodollar Jun 2020	=	98.8000	98.7350 ↓
Long 2 x Eurodollar Dec 2020	=	98.5325	98.5550 ↑
Short 1 x Eurodollar Jun 2021	=	98.3250	98.3800 ↑

Close Sell Butterfly Spread:

Close out short position 1 x Eurodollar Jun 2020	=	+ 0.0650
Close out long position 1 x Eurodollar Dec 2020	=	+ 0.0225
		+ 0.0875
Close out long position 1 x Eurodollar Dec 2020	=	+ 0.0225
Close out short position 1 x Eurodollar Jun 2021	=	- 0.0550
		- 0.0325

The overall profit is: 0.0875 – 0.0325 = +0.055 or 5.5 ticks.

Profit & Loss Calculation:

One interest rate basis point = 0.01 price points or 1 tick = USD 25 per contract.

Profit = Net gain = 5.5 ticks x USD 25 per tick = USD 137.50 for the basic structure.

3.4.2.3 The Double Butterfly

A basic extension of the butterfly strategy is the **double butterfly strategy** where one butterfly is bought, and another butterfly is sold in staggered fashion. In the double butterfly strategy, the contract used for the second and third legs of the first butterfly are identical to the first and second legs of the second butterfly. See Table 3.4.2.3.

Table 3.4.2.3 - Creating the Double Butterfly Strategy with Futures Contracts

Strategy Execution				
Buy First Butterfly Buy 1 Butterfly Futures Contract	Buy 1 near-term contract	Sell 2 mid-term contracts	Buy 1 distant contract	
Sell Second Butterfly • Sell a Butterfly Futures Contract matching the two mid-term sell futures contract date of the initial Buy a butterfly strategy. • Purchase 2 mid-term **futures** contracts at the same date as the last buy futures contract of the initial butterfly.		Sell 1 near-term contact	Buy 2 mid-term contracts	Sell 1 distant contract
Overall structure of the Double Butterfly	**Buy 1 near-term contract**	**Sell 3 mid-term contracts**	**Buy 3 mid-term contract**	**Sell 1 distant contract**

3.4.3 Condor Spread

A **condor spread** is similar to a butterfly spread because it also combines a bear and a bull spread. The difference is that there is no common middle month, as the expiration dates of the futures contracts are all different. A condor spread has a combination of 4 contracts with equally distributed delivery months.

The spread ratio is always +1 : -1 : -1 : +1. Hence, **buying a condor spread** means:

a. Buying 1 of the nearest delivery month (Leg 1);

b. Selling 1 of the second nearest delivery month (Leg 2);
c. Selling 1 of the third nearest delivery month (Leg 3); and
d. Buying 1 of the furthest delivery month (Leg 4).

Buying a Condor Spread

Current Market		**8 Feb 2017**
Eurodollar May 2020	=	98.8450
Eurodollar Jun 2020	=	98.8000
Eurodollar Sep 2020	=	98.6675
Eurodollar Mar 2021	=	98.4350

Strategy: 1 Bull Spread and 1 Bear Spread

Near - Mid Term		8th Feb	20th April	Nett	
Long May 2020	+1	98.8450	98.7875	-0.0575 loss	-5.75 Ticks
Short Jun 2020	-1	98.8000	98.7300	+0.0700 gain	+7.00 Ticks
				0.0125 gain	**+1.25 Ticks**

Mid Term - Distant		8th Feb	20th April	Nett	
Short Sep 2020	-1	98.6675	98.6250	+0.0425 gain	+4.25 Ticks
Long Mar 2021	+1	98.4350	98.4650	+0.0300 gain	+3.00 Ticks
				0.0725 gain	**+7.25 Ticks**

Net Gain:	All positions =	0.0850 gain	+8.50 Ticks

One interest rate basis point = 0.01 price point = USD 25 per contract.
0.0850 price points = 8.5 x 25 = **USD 212.50**

Value of one tick for Eurodollar transactions:
= 1/100 of 1% change in USD 1 million notional 90-day deposit
= 0.01 x 0.01 x $1 million x 90/360 = $25

There is a gain of 8.5 ticks = 8.5 x 25 = **USD 212.50**

3.4.4 Other Futures Spread Strategies

There are other types of futures spread strategies with specific focus on certain futures products.

3.4.4.1 TED Spread

A TED spread is the difference between the price of the 3-month US Treasuries futures contracts and 3-month Eurodollar futures contracts which have the same expiration month. Since the 3-month futures contracts for US Treasuries are no longer traded on the CME, the TED spread is now commonly known as the difference between the 3-month US Treasury bill rate and the 3-month USD LIBOR.

The TED spread is used as an indicator of credit risk because US T-bills are considered risk free, while the rate associated with the Eurodollar futures reflects the credit ratings of corporate borrowers. When the TED spread

increases, default risk is seen to be increasing and investors will prefer safer investments. As the spread shrinks, the default risk is seen to be decreasing.

3.4.4.2 Inter-Product Spread or Inter-Commodity Spread

Inter-commodity spreads strategy aims to take advantage of the value differential between two or more related commodities. The commodities would typically have some relationship and are influenced by similar factors (e.g., interest rates). So, for example, a trader could be looking for the spread difference between two financial products (e.g., US Treasury bills and the Eurodollar) when interest rates move.

Inter-product spreads can also involve the spread between two related commodities (e.g., gasoline and heating oil) when crude oil prices move. Instead of taking exposure in one product, the trader seeks to take advantage of the greater sensitivity and reaction of one product compared to another product.

3.5 Hedging and Portfolio Management

The original objective in the creation of forwards and futures was for hedging against the future delivery price of an underlying commodity. The same principle can be applied in hedging or partially hedging against the fall in value of investment portfolios.

3.5.1 Identifying and Measuring Risk

The objective of hedging is to reduce risk. If an adverse price movement occurs, losses will be contained or profits will be protected. However, if the price moves favourably, profits will be capped. Hedging does not eliminate price risks but converts price risks to basis risks between the underlying instrument and the hedging instrument. It confines the final price to a determinable range.

In portfolio management, hedging is used to:

a. Smoothen cash flows;
b. Simplify financial planning;
c. Reduce working capital requirements;
d. Allow for more efficient product pricing; and
e. Manage inventory more efficiently.

Before actual hedging is done, the investor should evaluate various factors to identify and measure the risks to be hedged. If the likelihood of the risk is high and the potential impact of the risk exposure is very significant relative to the cost of establishing the hedge, and the hedge is dependable, then it may make sense getting the hedge in place. However, there will be some subjectivity in the deciding if the need and benefits justify the costs of hedging.

The following questions set out in Table 3.5.1 may be helpful when considering hedging decisions:

Table 3.5.1 - Factors and Questions to Consider when Hedging

Factors and Questions	Considering the Need for Hedging
1) Probability of Negative Change i. What is the probability of the changes in prices/rates happening?	If the events causing the downside volatility risk are highly likely or if the probability of the volatility based on track record is high, then the need to hedge is

Factors and Questions	Considering the Need for Hedging
ii. What is the likelihood of events leading to greater volatility will occur?	higher.
2) <u>Probable Size of Negative Change</u> i. What is the likely size of the change? ii. What was the past record?	If the size of the change is small, a small partial hedge may be preferred, or it may not be needed at all. However, if the probable size of change on average is quite large in the past, perhaps a hedge is needed.
3) <u>Overall Impact of Negative Change</u> i. What is the risk value associated with not hedging - small or large?	Suppose the likelihood is small but if it happens that the impact is catastrophic, then it may be a risk worth hedging against. A "black swan[20]" type of risk does not often happen, but they cause mayhem when they do occur.
4) <u>Transaction Costs</u> i. What are the transaction costs associated with hedging (e.g., brokerage, financing, initial & variation margin)? ii. What is the cost of hedging compared with the risk value?	If the cost of setting up the hedge is too high given the probability of the event or size of the potential loss, then it may not make sense setting up the hedge for the portfolio. For example, small diversified global investment portfolios are exposed to currency risk. For portfolio that are too small, it may not make sense to fully hedge currency risk.
5) <u>Percentage of Value to be Hedged</u> i. What percentage of the principal value of the asset (or liability) to be hedged does this transaction cost constitute?	Given the probability, it may make sense to hedge only a percentage of the portfolio which is more cost efficient.
6) <u>Assumed Basis Risk</u> i. What is the basis risk for the hedge?	Basis risk is the likelihood that a hedging does not fully work. Basis risk occurs when the underlying asset and the hedging instrument do not properly offset each other. In setting up the hedge, it is important to review how well the hedge works and how much basis risk you are comfortable with.

Although the overall risk assessment is subjective and is not fully quantifiable, some traders and investors have tried to come up with a risk value and a hedging cost. If the risk value is higher than the hedging cost, the recommendation would be to perform the hedging activity. Here is a rough illustration of how this review might come about.

Hedging: Identifying & Measuring Risk

1. Probability of Negative Change in Prices/Rates?
- What is the probability of this negative change?

[20] A black swan is a type of event that occurs beyond the boundaries of normally expected situations and are almost impossible to predict. The term was first used by Nassim Nicholas Taleb, a former Wall Street trader, writer and finance academic.

= 45%

2. Probable Size of Negative Change in Prices/Rates?
- What is the probable size of this change?
= 10% to 11.5%

3. Risk Value
- What is the risk value associated with NOT hedging?
= 5.18% (i.e., 45% x 11.5%)

4. Transaction Costs
- What are the transaction costs associated with the hedge?
  - Brokerage = $15.00
  - Financing: Initial margin
  (assume initial level = 1,500 @ 12% for 3 months) = $45.00 (i.e., 1,500 x 12% x 3/12)
  - Financing: Additional margin
  (assume maximum adverse move = 45 ticks) = $33.75
  - Total Transaction Margin = $93.75

5. Percentage of Value to be Hedged
-What percentage of the principal value of the asset (or liability) to be hedged does this transaction cost constitute? Assume the principal is $1 million.
= 0.009375% (i.e., 93.75 / 1 million)

6. Assumed Basis Risk
- What is the basis risk for the hedge?
= 3.0%

7. Cost of Hedging versus Risk Value
  - Risk Value = 5.18%
  - Hedging Costs = 3% + 0.009375% = 3.009375%

Risk Value > Hedging Cost
→ Decision is to **HEDGE**

3.5.2 Developing an Effective Hedge Program

Once a decision to hedge is taken, the investor can develop the hedge program by considering the following factors:

a. **Risk exposure** - Evaluate the risk exposure in assets and liabilities (by group, by maturity or profit centre).

b. **Objective -** Set hedge objective in terms of the acceptable risk level or tolerance[21].

c. **Hedgeability** - Determine hedgeability using price correlation analysis (and cross hedges[22], if necessary).

[21] Risk as measured by beta, volatility, value at risk or other units of risk measure. Risk is covered in greater detail in Chapter 11.
[22] A cross hedge refers to hedging a security with another instrument where the two are positively correlated and have similar price movements.

d. **Hedge vehicle** - Determine the hedge instrument or vehicle, which should be an instrument with a high degree of correlation with the target security.

e. **Target** - Determine the target rate you want to lock in for the hedge (i.e., the rate or the price).

3.5.3 Determining the Hedge Instrument

When seeking an appropriate hedge instrument, it is essential to find one that is highly correlated with the target security. If the target security is different from the futures contract, the yield relationship and the correlation between the two securities must be carefully evaluated. The correlation between the two securities can be measured using regression analysis.

Some examples:

a. A long-term government bond is better hedged with the 30-year T-bond futures than with the 3-month Eurodollar futures because long-term government bond rates are more highly correlated with 30-year T-bond rates than Eurodollar rates.

b. T-bills are better hedged with matching T-bills futures (if available) than Eurodollar futures. In times of financial distress, T-bills futures prices will rise faster than Eurodollar futures, and the Eurodollar futures hedge becomes ineffective.

c. US T-bond futures are used to hedge the interest rate risk of US corporate bond portfolios. Credit quality and sector risks are not hedged; and

d. Economic factors have different impacts on different sectors of the economy and during periods of financial distress, investors will seek out safer investments. The switching of funds from, for example corporate bonds to Treasury bonds, will result in the widening of credit spreads on corporate bonds and their prices to fall relative to that of the government bonds. In this instance, the basis risks in selling Treasury bond futures to hedge corporate bond exposures will be greater as the basis between the two widens, thus rendering the hedge less effective.

Futures and options are frequently used as hedging instruments. However, there are some limitations when using futures and exchange-traded options for hedging:

i. Round number of contracts;
ii. Margin maintenance;
iii. Time horizon and expiry dates; and
iv. No exact hedge instruments.

3.5.4 Determining the Target Rate for the Hedge

An investor might wish to lock in a target rate or value for the hedge, such as the currency rate, interest rate or underlying asset price.

A minimum variance hedge is one which is held to the futures delivery date and is an example of a hedge that locks in the futures interest rate. On the other hand, if the hedge is to be lifted before delivery, the hedger cannot be assured of locking in a spot or future rate. This is due to the changes in basis.

The target rate for a hedge is calculated as follows:

Target Rate for Hedge = Futures Rate + Target Rate Basis

Or

(i) V_t **= Ending security price + Futures gain =** $S_t + (F - F_t)$

(ii) $V_t = F + (S_t - F_t)$ **= Initial futures price + Ending basis**

Where:

V_t = Value of the hedged position at time t
F = Initial futures price
S_t = Security or spot price at time t
F_t = Futures price at time t

The target basis concept explains why a hedge that is held till expiry locks in the futures rate. A hedge that is lifted before expiry just substitutes price risk for basis risk. Alternatively, investors can define the target rate and work backwards to find the hedge ratio that gives the desired target. However, they will have more risk than if they had chosen a hedge ratio that equates the target price to the futures price.

3.5.5 Hedges for Currently Held or Anticipated Positions

A hedge can be constructed for currently held positions or anticipated ones. It can also be used for a known or unknown period of time. Given all the possibilities, it helps to categorize the types of hedges.

3.5.5.1 For a currently held cash position

(a) **Weak form cash hedge (inventory hedge)**

This hedge aims to minimize the price variance of an existing asset portfolio that is to be held for an indefinite period of time. An example is the hedge placed on a bond dealer's inventory. A short position in a futures contract is used to offset any price movement in the cash position. The use of nearby futures contracts on securities similar to those in the inventory is the most effective as it maximizes the price covariance.

(b) **Strong form cash hedge (immunization)**

Financial institutions and fund managers use immunization to protect their portfolios. Immunization is essentially a strategy that matches the durations of assets and liabilities. To manage exposure to interest rate fluctuations, institutions will immunize the impact of interest rates on the portfolio value by calibrating the interest sensitivity of the portfolio.

In the strong form cash hedge, the portfolio's time horizon is finite or known. The hedging goal is to protect the portfolio by minimizing the variance in the expected total return on the portfolio for a given investment period.

To immunize portfolio returns, a cash and futures portfolio is created and maintained so that it has the same interest rate sensitivity as a zero-coupon (pure discount) bond with an initial maturity equal to the investment period. The zero-coupon bond's initial value and its interest rate sensitivity are tracked at each point in time. Whenever the interest rate sensitivity of the cash portfolio is less than that of the zero-coupon bond, futures must be purchased to augment the price sensitivity of the cash portfolio. Conversely, whenever the cash portfolio is more interest rate sensitive than the zero-coupon bond, futures must be sold to reduce this level of sensitivity.

3.5.5.2 For an anticipated cash position

(a) **Weak form anticipated hedge**

The goal here is to minimize the variance in the rate of return on an asset for a specified holding period, based on cashflows to be received at an unknown date. In this instance, the futures contracts purchased with the same interest rate sensitivity as the asset to be acquired. The uncertain timing of the cash receipt reduces the hedge's effectiveness. Nevertheless, futures can be used to narrow the range of possible outcomes.

(b) **Strong form anticipated hedge**

Strong form anticipated hedges apply whenever a known amount of cash will be received at a certain date. The aim is to minimize the variance of the acquisition price for the cash securities or one that most closely realises market-forecasted price. This hedge requires acquiring future delivery to the anticipated dollars of bonds as the anticipated cash inflow. Since margin calls on the long futures position must be financed, the appropriate hedge ratio is not an exact dollar match.

3.5.6 Hedge Ratio

The hedge ratio defines the expected movement in value of the cash instrument to be hedged, given a particular movement in the value of the futures contract, which is to serve as the hedge instrument. In technical terms, the hedge ratio is the ratio of futures to a spot position (or vice versa) that minimizes or eliminates risk. Equal dollar exposure in the futures contracts may not create the optimum or delta-neutral hedge[23]. The hedge ratio is the actual weighting of the hedge vehicle to the target security. A proper determination of the hedge ratio is essential to replicate the price movement of the target security.

The hedge ratio is calculated with the following equation:

$$\textbf{Hedge Ratio } (h) = \frac{\textbf{Change in Security Price}}{\textbf{Change in Futures Price}}$$

$$h = \frac{\Delta V - \Delta S}{\Delta F}$$

Where:

V = S + h x F
V = Value of hedged position
S = Security price
F = Futures price

For a complete or delta-neutral hedge (ΔV = 0), the hedge ratio will be:

$$h = -\frac{\Delta S}{\Delta F}$$

[23] A delta-neutral portfolio will consist of positions in capital markets products which have offsetting positive and negative deltas so that when market movements occur, the net changes or overall delta is zero.

$$\textbf{Number of contracts} = \textbf{- Hedge ratio x } \frac{\textbf{Loan Value}}{\textbf{Contract Size}}$$

Calculating the Hedge Ratio

Calculate the hedge ratio and the number of 90-day Eurodollar contracts required to hedge the interest rate on a 180-day loan for USD 100 million. Assume that the interest rate on the loan is correlated one-to-one with the 90-day Eurodollar rate of the nearest futures contract.

Bank loan amount = USD 100 million
Bank loan due = 180 days from today

Hedge: Delta neutral hedge (assume interest rate on loan 1:1 correlation with Eurodollar rate), where change in value of hedged position would be zero.

$$\text{Hedge Ratio (h)} = \frac{\text{Change in Value of Hedged Position - Change in Security Price}}{\text{Change in Futures Price}}$$

$$= \frac{0 - \text{Change in Security Price}}{\text{Change in Futures Price}}$$

$$h = -\frac{\Delta S}{\Delta F}$$

Consider the change in the security price (180-day USD loan) for a 1 percent change in USD interest rates for a notional value of USD 1 million. A rough approximation would be:
(0.01 x 180/360) x 1 million = USD 5,000

The 90-day Eurodollar futures has a notional value of USD 1 million. Each tick represents one basis point (0.0001) and has a value of USD 25. For 1 percent change (i.e.; 100 basis points) in USD interest rates, the 90- day Eurodollar futures value changes by 25 x 100 = USD 2,500.

i.e.,

h = - (5,000/2,500) = -2

$$\text{Number of contracts} = \text{- Hedge ratio x } \frac{\text{Loan Value}}{\text{Contract Size}}$$

$$= -2 \text{ x } \frac{\text{USD 100 million}}{\text{USD 1 million}}$$

= sell 200 contracts

Note that this stack hedge of selling a single 90-day Eurodollar futures contract to offset the impact of interest rate rises on the 180-day loan carries considerable basis risks. Even though it is assumed that there is perfect correlation between the loan rate and the 90-day Eurodollar rate, there is considerable timing mismatch: the interest income from the loan is receivable at the end of the loan period whereas any change in the value of the Eurodollar futures contract is realised immediately via the margin account with the futures broker.

Moreover, as the futures contracts have a maturity of 90 days (versus the loan of 180 days), the stack hedge has to be rolled over to the next futures delivery month when the current one matures. A new hedge ratio has to be recalculated.

(See following section).

3.5.7 Types of Hedges

3.5.7.1 Strips and Stacks

The futures buyer must also consider and decide whether to use strips or stacks:

a. **Strips** use successive futures contract months to match delivery dates on the futures contracts with the rollover dates or tenor on a cash loan.

b. **Stacks** use the deferred contract months to match the rollover dates or tenor on a cash loan.

3.5.7.2 Time Horizon

Under normal market conditions, the nearby contract is preferred due to its liquidity and statistically high correlation to the movement of the underlying instrument. When there is a mismatch of the time horizon and expiry date, basis risk occurs. If the hedging horizon extends beyond the expiry of the nearby contract, the hedger has to decide whether to use the nearby contract and rollover into a deferred one or use a deferred contract from the beginning.

Both choices will have basis risk when the hedge is terminated. When there is no exact futures contract for hedging an exposure, the correlation between the hedge vehicle and the instrument to be hedged must be properly studied to achieve the desired results. When the maturity date of the underlying instrument does not coincide with the maturity date of the hedge instrument, two types of hedges can be considered:

a. **Interpolative hedge** - used when time horizon straddles 2 expiry dates; and

b. **Extrapolative hedge** – used when time horizon stretches beyond the tenor of the last traded contract.

3.5.8 Hedging Long-Term Interest Rate Risk

Unlike the use of interest rate futures for hedging, which is straightforward, using bond futures to hedge a bond portfolio is more complicated because:

a. The conversion factor for deliverable bonds may require the underlying security to be hedged by a number of futures contracts that is not a round number. The extra contracts purchased also must be closed out before the delivery date; and

b. Convergence is less perfect. For example, in the case of US T-bonds futures, there are certain options embedded for the sellers and these include delivery timing and choice of deliverable bonds. As a result, the futures pricing is normally lower than that implied by the cash and carry arbitrage.

To hedge long-term interest rate risk, the following formula is used:

$$\textbf{Hedge ratio} = \frac{\textbf{PVBP of hedge security}}{\textbf{PVBP of most deliverable bond}} \times \textbf{Conversion factor for most deliverable bond}$$

where PVBP = Change in price for a single point change in yield

Hedging Long-Term Interest Rate Risk

A fixed income fund manager, Alice Tan wants to hedge a USD 20 million bond position with Treasury futures.

Suppose that:
- The conversion factor for the cheapest-to-deliver issue is 0.91.
- The price value of a basis point of the cheapest-to-deliver issue (at the settlement date) is 0.6895.
- The price value of a basis point of the bond to be hedged is 0.5954.

Alice needs to determine:
1. What is the hedge ratio?
2. How many Treasury bond futures contracts should be sold to hedge the bond?

Question 1 - What is the hedge ratio?

We assume a fixed yield spread between the bond to be hedged and the cheapest-to-deliver bond.

The hedge ratio is:

$$\text{Hedge ratio} = \frac{\text{PVBP of hedge security}}{\text{PVBP of most deliverable bond}} \times \text{Conversion factor of most deliverable bond}$$

$$= \frac{0.5954}{0.6895} \times 0.91$$

$$= 0.7858071 \approx \mathbf{0.79.}$$

Question 2 - How many Treasury bond futures contracts should be sold to hedge the bond?

$$\text{Number of contracts} = -\text{Hedge ratio} \times \frac{\text{Total value to be hedged}}{\text{Par value of futures contract}}$$

$$= -0.7858071 \times \frac{\text{USD } 20{,}000{,}000}{\text{USD } 100{,}000} = -157.1614213 \text{ contracts}$$

→ Alice should **SELL 157 contracts.**

3.5.9 Hedging Equity Risks

There are two types of risks in any equity portfolio:

a. Systemic / market risks; and

b. Specific / non-market risks.

Equity index futures can only hedge against market risks. The fund manager can offset the portfolio's market exposure by selling futures contracts when long in the equity market (short hedge) and buying futures

contracts when short in the equity market (long hedge). When deciding the number of contracts to use, two factors should be considered:

i. **Beta** - Beta is used to measure the portfolio risk or volatility in comparison to the index. To establish a suitable hedge, the fund manager must decide whether the portfolio is to be more or less volatile than the index. This is usually done through regression analysis to determine the historical betas of the respective stocks in the portfolio.

ii. **Modified Portfolio Value (MPV) -** The modified portfolio value is the value of the actual portfolio multiplied by the weighted beta of the stock portfolio. The weighted beta is the sum of the product of each stock's beta and percentage share of the stock within the portfolio. This forms the base hedge.

To hedge equity risk, the following formula is used:

$$\mathbf{N = \frac{V_P}{F \times T} \times \beta}$$

Where:
N = Number of contracts
V_P = Current value of portfolio
F = Current futures quote
T = Value per tick
β = Beta of portfolio
(Formula assumes that the maturity of the futures contract is close to the maturity of the hedge.)

Hedging Equity Risk (1)

John Lim, a fund manager, wants to hedge his equity portfolio over the next 3 months. He intends to use 3-month futures contracts to do the hedge.

Here is the market & portfolio information:

- Current value of S&P Index as at Sept = 1,000
- December S&P 500 futures price as at Sept = 1,010
- Value of stock portfolio = USD 5,050,000
- Beta of stock portfolio = 1.50
- Value of each S&P 500 futures contract = USD250

How many futures contracts does John Lim need to short (sell) to hedge his portfolio?

$$N = \frac{V_P}{F \times T} \times \beta$$
$$= \frac{\text{USD } 5{,}050{,}000}{\text{USD } 250 \times 1{,}010} \times 1.50$$

= 20.0 x 1.50
→ **SELL = 30.0 contracts.**
Hence, John sold 30 Dec S&P futures contracts priced at 1,010 with each contract valued at USD 250.

Hedging Equity Risk (2)

It is now December. The financial markets have seen a correction in recent months. The equity markets have fallen, and the current market information is as follows:

- Current value of S&P Index (December) = 900
- Dec S&P 500 futures price (December) = 902 (Note: John sold Dec S&P futures in Sept at 1,010)

Assuming there is no dividend payout on the S&P 500 index, what is the overall value of John Lim's fund in December (considering his equity portfolio and futures contract hedging position)?

Value of Stock Portfolio S&P 500	- In September = 1,000	
	- In December = 900	
	Change = (100)	(10% decline)
Stock Portfolio Beta	= 1.50	
Market decline	= -10.0%	
Expected decline for equity portfolio	= -15.0%	
Stock Portfolio – September	= USD 5,050,000	
Expected 3-month decline in value	= (USD 757,500)	(15% decline)
Expected value – December	= USD 4,292,500	

In December, calculate the value of hedge with futures contracts:

S&P 500 Dec futures price	= 1,010	
S&P 500 Dec futures price (December)	= 902	
Difference	= 108	
Change in S&P 500 index value	= 108	
Value per S&P 500 contract	= USD 250	
No. of hedging contracts	= 30	
Short futures position GAIN	= USD 810,000	(i.e., (1,010 - 902) x 30 x 250)

Overall Fund Value (assume no transaction costs)

Value of equity portfolio	= USD 4,292,500
Gain from hedging contracts	= USD 810,000
	= USD 5,102,500

3.5.10 Structuring the Hedge

The hedge's objective is to achieve equivalent dollar movement in cash and futures. After deciding to execute a hedging strategy and considering the five factors (risk exposure, objective, hedgeability, hedge vehicle and target), the next step is to choose the timing and contract delivery months. Then maturity and sensitivity adjustments should be made to ensure that the hedge fits the risk of the underlying exposure.

3.5.11 Managing the Hedge

The effectiveness of the hedge refers to the risk of a hedged position relative to an unhedged position. If the hedged position is 90% effective, over the long run the hedged position will have only 10% of the risk (i.e.; standard deviation) of an unhedged position. The absolute risk of the hedge is expressed as a standard deviation.

For example, it might be determined that the hedged position has a standard deviation of 10 basis points. Assuming a normal distribution of hedging errors, the user will then obtain the target rate plus or minus 10 basis points approximately 67% of the time (i.e., one standard deviation away).

Most hedges require very little active monitoring during their life span. However, changes in volatilities and yield spread relationships, for instance, may necessitate changing the hedge ratio. After a hedge is lifted, it is evaluated to determine the sources of error, hopefully to gain insights for subsequent hedges. Normally the main sources of error are due to the projected value of the basis at the lift date and the parameters estimated for cross hedges.

3.5.12 Further Examples of Hedging

Hedging Currency Exposure

Yankee Trading Co has exported goods to Canada and expects to receive USD 1 million 3-months from now.

On 1 January, the market rates are:
- CAD/USD spot = 0.9125
- April currency futures = 0. 9235

The finance manager of Yankee hedges his position on 1 January. On 1 April, the market rate is spot = 0.9485. Yankee receives the USD 1 million and the manager closes out his hedge.

What is the net result?

As the manager has locked in the CAD proceeds at the futures rate of 0.9235, he converts it to CAD 1,082,837 (USD 1 million x 1/0.9235). If he had not hedged, the appreciation of the Canadian currency would have reduced the sum in USD terms. He would have obtained only CAD 1,054,296 (USD 1 million x 1/0.9485), a difference of CAD 28,541 (before transaction costs).

Fixed Deposit Hedge

On 5 January, a corporate money manager expects to receive USD 25 million on 5 March. He plans to invest those funds in a 90-day time deposit with a bank. He expects interest rates to decline.

What should he do? What is the effective yield he achieved?

Current date: 5 January
Amount: USD 25 million (funds expected on 5th March)

Market information:

Date	Cash Market	Futures Market		ED Mar*
5 Jan	-	97.75	=	2.25%
5 Mar	3-month deposit = 1.80%	98.16	=	1.84%
	Differential =	0.41		0.41%

* Refers to the Eurodollar March futures contrac

Strategy:
5 Jan Buy 25 ED contracts @ 97.75.
5 Mar Sell 25 ED contracts @ 98.16 and deposit the funds for 3 months at 1.80% p.a.

Effective Yield (for 3 months):

Deposit funds for 3-months @ 1.80% p.a.	1.80%
Buy (@97.75) and Sell (@98.16) 25 ED contracts	0.41%
	2.21%

→ With this hedging strategy, yield of 2.21% can be locked in.

Locking in Deposit Yield

The treasurer of LMN Corporation is notified that USD 1 million will be available for investment in 3-month Eurodollar deposit on 10 September. His forecast is that interest rates will decline over the next few months. What should he do to lock in the current yield?

Current date: 27 July
Amount: USD 1 million (funds expected on 10 September)

Market information

Date	Cash Market	Futures Market		ED Sep
27 July	-	98.62	=	+1.38%
10 September	3-month deposit = 0.5%	99.40	=	-0.60%
				0.78%

If funds are NOT HEDGED:
Interest earned (from 10 September):
USD 1 million X 0.5% X 90/360 = USD 1,250.

If funds are HEDGED:
Profit earned from futures:
USD 1 million X 0.78% X 90/360 = USD 1,950.

Interest earned (from 10th Sep):
USD 1,001,950 X 0.5% X 90/360 = USD 1,252.44.

Annualised Yield for 3 months:

$$\left[\frac{\text{USD 1,950 + USD 1,252.44}}{\text{USD 1,000,000}}\right] \times \frac{360}{90} = 1.28\%$$

→ With the hedge, yield of 1.28% can be locked in (close to the current level of 1.38%).

Floating Rate Borrowing Cost

Assume you have issued a USD 10 million 3-year floating rate note pegged at 1% above 3-month LIBOR. The next interest rate fixing is 19 March. The 3-month LIBOR today is 1.85%, but you expect interest rates to move up in the meantime.

How do you protect against higher interest rates fixing?

Funding	:	Issue 3-year floating rate debt
Amount	:	USD 10 million
Cost	:	3-month LIBOR + 1%
Current rate	:	LIBOR = 1.85%
Next date	:	19 March (for fixing interest rate)

Market information

Date	Cash Market	Futures Market		Rate Equivalent
30 January	3-month floating rate = 1.850%	98.15	=	1.85%
	On 19th March, the 3month LIBOR is fixed at 2.25%. Hence, the 3-month floating rate of the debt is LIBOR + 1% = 2.25%			
19 March	+ 1% = 3.25%.	97.75	=	2.25%
		-0.40		-0.40%

Strategy:

30 January	-	Sell 10 ED March contracts @ 98.15.
19 March	-	Buy back 10 ED March contracts @ 97.75.

Effective Borrowing Rate:

- Rate on floating rate debt	3.25%
- Gain on futures	-0.40%
	2.85%

→ With this hedging strategy, the effective borrowing rate of 2.85% can be locked in. Without the hedge, the 3-month borrowing rate would be 3.25%.

Hedge Stock Portfolio (Equity Risk)

An investor has a Japanese equities portfolio worth JPY 530 million (about USD 5 million) as of 16 March 2018. The investor believes that in the next few months, if there was a correction, the Nikkei 225 index could retrace by 7-8% to just below the 20,000 level. However, he wants to hold on to his Japanese stocks for the rest of the year. He was advised to hedge his exposure using Nikkei 225 futures for the next 3 months due to the expected higher volatility in the global market. The beta of the portfolio is 1. The initial margin based on the SGX margin schedule on 16 March is JPY 495,000 per contract.

How many contracts will he need to set up a full hedge for his existing portfolio if he uses June 18 Nikkei Futures priced currently at 21,455?

Suppose he is correct, and the Nikkei falls back to 8%. What would be the profit or loss, if any?

What is the initial margin amount required to set up the hedge?

On 16 March 2018

Nikkei 225 = 21,676.51
June Nikkei Futures = JPY 21,455 (current price)
Current Portfolio = JPY 530 million
Value per contract of the Nikkei futures = JPY 500



1. Calculating the Number of Contracts.

$$N = \frac{V_P}{F \times T}$$

$$= \frac{530{,}000{,}000}{21{,}455 \times 500}$$

= 49.40 or approximately 50 contracts for a full hedge.

Assume Nikkei falls to 19,942.39 down by 8%.

Given the portfolio beta of 1, the portfolio should fall by the same percentage.

Nikkei 225 @	16 March 2018	Current Portfolio =	JPY 530,000,000	
Nikkei 225 @	June 2018		JPY 487,600,000	Assume an 8% fall over
		Fall in value =	JPY 42,400,000	the period.

Value of Hedge – Futures Contracts

When the Nikkei 225 index falls by 8%, the June 2018 Nikkei 225 futures contract is at 19,735.00. The hedge is executed as follows:

SELL:50 Jun 2018 Nikkei 225 futures contracts on 16 March		JPY 536,375,000	(i.e., 500 x 21,455.00 x 50)
BUY:50 Jun 2018 Nikkei 225 futures contract in June		- JPY 493,375,000	(i.e., 500 x 19,735.00 x 50)
		JPY 43,000,000	
June Nikkei 225 futures contract sold in March (less Nikkei 225 futures contract bought in June)		1,720	(i.e., 21,455 – 19,735.00)
Value per contract (JPY)	X	JPY 500	
No. of hedging contracts	X	50.0	contracts
Short futures position GAIN	=	JPY 43,000,000	

Initial Margin and Transaction Costs

The initial outlay and the cost of set up is low. The initial margin based on the SGX margin schedule on 16 March is JPY 495,000 per contract. For 50 contracts, the initial margin would be JPY 24,750,000. (USD 233,466.78).

Assuming a 0.25% execution fee, the transaction cost is about USD 583.67, plus a clearing fee of USD 0.90 per contract costing $45 for 50 contracts will mean incurring total transaction costs of USD 628.67 or JPY 66,646.

**Assume that USD 1 = JPY106.0108*

Implementing Hedge Fund Strategies

Hedge fund managers often employ a combination of long and short positions to generate superior risk-adjusted returns. One common strategy is the **"130/30" long-short strategy** where the fund manager identifies stocks which are superior or potential outperformers, and includes them in the portfolio, up to 130% of the fund's total assets under management (AUM). The additional positions are funded by shorting (up to 30%) stocks that have been identified as potential underperformers.

To the extent that the fund's objective is to outperform the equity index (e.g., the S&P500), the manager can use futures as part of the core fund holdings to capture the returns of the S&P500 using index futures. Using equity index futures can also give the manager the flexibility to manage and deploy the cash inflows and withdrawals quickly and efficiently.

3.6 Portfolio Management Techniques

Portfolio management is about setting investment policy, matching investments to meet financial objectives, asset allocation decisions and balancing risk against desired performance. It is also about assessing the strengths, weaknesses, opportunities and possible risks when making decisions about portfolio exposures to different asset classes (e.g., traditional assets vs alternative assets), different markets (e.g., domestic versus international markets) and different strategies (e.g., growth/aggressive versus safe/conservative strategies).

Constantly rebalancing the mix of assets requires the ability to perform this adjustment efficiently and economically. Using futures to allocate assets can be more effective and less costly because:

- Brokerage costs for futures transactions are cheaper;
- With margining, the cash outlay is smaller, and the transactions can be tailored to the user's cash flow needs;
- The transaction time is shorter as buyers and sellers are trading against the exchange as a counterparty;
- There is less impact on the market since the futures market is more liquid; and
- It is less disruptive to the whole portfolio management process.

Futures are also useful in controlling losses. For example, unwinding a bad position in the cash market may aggravate the losses due to illiquidity or other factors. Instead, the user can limit losses and hedge the risks with futures. When prices improve, he can unload the cash and futures position at the same time.

In certain countries, where the accounting system permits, futures are also used to delay loss realisation. For example, during financial reporting periods, securities that are carried at cost can be hedged against further losses using futures. After the reporting is completed, the losses are realised, and the futures contracts are unwound.

Synthetic strategies can also be employed. Depending on the availability of the instruments and their derivatives as well as transactions costs, synthetic instruments can be created using financial engineering, which could achieve even better results.

In addition to futures, options can also be used for hedging and portfolio management. When hedging equity risk exposure, investors should distinguish between market (systematic) risk and non-market (specific) risk. Market risk can be hedged with equity index options, and non-market risk with individual stock options. Options offer investors advantages such as greater flexibility, better timing and return profiles. Options will be discussed in detail in Chapter 4.

3.6.1 Examples of Portfolio Management Applications

Example – Asset Allocation of Bonds & Stocks

A fund manager is currently holding a portfolio of USD 50 million, with 50% equities / 50% in bonds. He wishes to change the composition to 40% equities / 60% bonds. Some of the stocks he is holding are illiquid and he needs time to dispose them. He also feels that the current bond yields are very attractive and wishes to lock them in.

He could solve the problem by buying bonds futures now, since futures only require margin payments. Subsequently, when the stocks are liquidated and the cash is received, he can then sell off the futures and buy the securities at the same time.

Example – Fixed Income Portfolio

A fund manager is holding some corporate bonds and is concerned that interest rates may rise soon. He wishes to shorten the maturities of his holdings where the volatility is smaller, and the impact of rising rates could be avoided. However, there are no suitable short-term securities available in the market.

He sells T-bond futures, which have a high correlation with his corporate bonds, and he would be effectively converting his bond holdings into a synthetic money market instrument.

Example – Cash Management & Interest Rate Expectations

Jessie Goh, a fund manager, expects interest rates to rise in the coming months. She intends to convert 10% of her USD 50 million semi-annual coupon bond holdings into short-term securities. She will sell bond futures and convert the bonds into synthetic money market instruments.

Here is the market & portfolio information:

Coupon rate	=	Annualised 8.0%	Last semi-annual coupon paid 3 months ago
Bond purchase (July)	=	101	
Bond futures price (September	=	108 13/32	
Conversion factor	=	1.0	
Initial investment:			
Bond purchase price	=	USD 101.00	
Accrued interest	=	USD 2.00	8%/2 x 100 par value x 3/6 months
		USD 103.00	

Sales proceeds:		
- Futures price	=	108.406
- Conversion factor	=	1.00
- Accrued interest (8%/2 x 100 par value x 5/6 months)	=	USD 3.33
		USD 111.736
Gain = Sales Proceeds – Initial Investment	=	USD 8.736
Investment Returns:		
- Gains	=	USD 8.736
- Initial investment	=	USD 103.00
- Return	=	8.48%

How many futures contracts does Jessie Goh need to short (sell) to hedge her portfolio?

$$N = \frac{V_P}{F \times T} = \frac{\text{USD } 5{,}000{,}000}{108.40625 \times \text{USD } 1{,}000}$$

$$= \frac{\text{USD } 5{,}000{,}000}{108{,}406.25}$$

= 46.12

She needs to sell **46 contracts.**

3.7 Arbitrage

Markets are not always efficient, giving rise to arbitrage opportunities every now and then. An arbitrageur is a market participant who profits from price differences that arise when the same, or extremely similar, security, currency, or commodity, is traded on two or more markets. Arbitraging involves the simultaneous buying in one market and selling in the other market whenever prices are out of line.

Arbitrageurs can make risk-free profits by exploiting the disequilibrium between the futures and cash markets. To benefit from these opportunities, the use of powerful computers and sophisticated trading programmes are essential in the fast-paced global market environment.

An example of an arbitrage opportunity is the simultaneous buying (selling) of equity index futures (i.e., S&P 500) while selling (buying) the underlying stocks of the index, capturing profit as the temporarily inflated basis between these two baskets. The point where profitability exists is usually given by the number of points the futures must be over or under the underlying basket for an arbitrage opportunity to exist.

Another example is in the pricing of bond strips. The arbitraging process begins with the calculation of strips, followed by a comparison of market prices of the underlying securities or related instruments. Whenever there is a discrepancy between the strips and the market prices, there is a possibility for arbitrage. Further evaluation of the arbitrage transaction costs (brokerage, financing, initial and additional margin) is necessary to ascertain if the strategy will yield returns that are justifiable, and if the risks (like basis risks), can be effectively managed.

3.7.1 Arbitrage Using Interest Rate Futures

3.7.1.1 Arbitrage between Cash and Futures

As futures are derivatives of the underlying cash markets, the arbitrage process will ensure the prices in these two markets are in line.

Eurodollar Arbitrage Opportunity (Discrepancy Between the Implied and Market Price)

Calculate the implied forward rate for the 2nd period of the futures contract.

Spot date	:	27 Jul
Spot date to 15 September	:	50 days
15 September to 15 December	:	91 days
50 days interest rate	:	1.00% bid / **1.0625%** ask
141 days interest rate	:	**1.25%** bid / 1.3125% ask

Step 1 – Compute the theoretical futures price

Based on the pricing of futures contracts, compute the implied forward rate for the period from 27 July to 15 September.

$$\left[1+\frac{50 \times 0.010625}{360}\right] \times \left[1+\frac{91 \times r_2}{360}\right] = \left[1+\frac{141 \times 0.0125}{360}\right] = 1.004896$$

Solve for r_2 (or the implied forward rate) and theoretical futures price:

r_2 = 1.351%.

Theoretical Sep futures price = 98.65 (i.e., 100 x (1 – 0.0135))

Currently the September Eurodollar futures price is 98.72. Since there is a difference between the market price and the theoretical price, there is an arbitrage opportunity.

Step 2 – Execute the arbitrage operation

- Lend 141 days @ 1.25%
- Borrow 50 days @ 1.0625%
- Sell September Eurodollar futures @ 98.72
- Arbitrage profit = (9,872 – 9,865) x USD 25 per contract = USD 175

3.7.1.2 Arbitrage between Forward Rate Agreements and Futures

A forward rate agreement (FRA) is the over-the-counter (OTC) equivalent of futures. Since they are quoted OTC, FRAs can be tailored to the exact needs of the users in terms of the maturity date and contract amount.

Arbitrage between futures and FRAs is risk-free only when the value dates for the two correspond. Even then it must be remembered that using futures requires more careful liquidity management due to margin calls. In the interbank market, a direct quotation will have a spread of 1-3 basis points while the futures normally has a spread of 1 basis point. This gives the price-maker an opportunity to arbitrage between futures and FRAs.

When the arbitrage is between futures and FRAs with different value dates, there will always be residual basis risks or fixing risks. To minimize these risks, care should be taken to ensure that no calendar events (e.g., the

financial year-end, share settlement date, etc.) or no release of important economic data (e.g., unemployment figures, consumer price index, etc.) will take place between the 2 dates.

Sometimes, disequilibrium in these markets may be due to institutional constraints such as limits to deal in only one market and internal restrictions on hedging between these instruments. Therefore, whenever there are certain buying or selling activities in one market driving the prices in one direction, parties who are nimble and unhampered by restrictions can capitalize on arbitrage opportunities more easily.

3.7.1.3 Arbitrage between Futures and Interest Rate Swaps

An Arbitrage between Futures and Interest Rate Swaps (IRS) is a transaction in which two parties agree to make periodic payments to each other, calculated on the basis of specified interest rates and a hypothetical (or notional) principal amount. Typically, the payment made by one party is calculated using a floating rate of interest (such as LIBOR), while the payment made by the other party is determined based on a fixed interest rate or a different floating rate.

Nowadays arbitrage opportunities in the Eurodollar market are rare and the margin is slim. The most popular IRS is traded over the International Monetary Market (IMM) dates with quarterly fixing of the floating rate. Arbitraging is possible whenever the market price differs from the strips. When this happens, the arbitrageur can buy or sell the IRS, and sell or buy the futures contracts that are used to calculate the strips.

Alternatively, arbitraging can also incorporate some speculative element. For example, in a 1-year IRS with quarterly payments, a floating rate payer feels that interest rates will only go up in the second half of the IRS period. He can sell the latter 2 futures contracts instead of selling all 4 successive contracts.

3.7.2 Arbitrage using Options

Besides using futures, market players can also take outright positions, hedge and execute arbitrage strategies using options. Managers of large and complex portfolios, which hold investments covering several asset classes, can use futures and options concurrently to achieve their investment objectives.

Option arbitrageurs essentially take advantage of temporary discrepancies in the option pricing structure whereby option prices trade out-of-line with the price of the underlying instrument or with the price of other options. The details of options and option strategies will be covered in Chapter 4.

3.8 Summary

1. The main players in the futures markets are:

    - Hedgers;
    - Speculators;
    - Market Makers;
    - Proprietary Trading Firms;
    - Arbitrageurs;
    - Portfolio Managers;
    - Hedge Funds; and
    - Central Banks, Sovereign Funds and Regulators.

2. With greater market volatility, participants are driven by different motivations and would look to derivatives to execute strategies for hedging or maximizing returns. Hedge funds and alternative asset managers have contributed to more innovative trading strategies, many of which involve the use of derivatives. Financial institutions create and sell a wide range of structured products, most of which are constructed using futures and other derivative products.

3. The types of orders available on SGX are:

    - Day Order.
    - Good-Till-Cancelled (GTC) Order.
    - Limit Order.
    - Fill-or-Kill (FOK) Order.
    - Fill-and-Kill (FAK) Order.
    - Market to Limit (MTL) Order.
    - Market Order.
    - Stop (Limit, MTL and Market) Order.
    - Market-if-Touched (MIT) Order.
    - Session State Order (Limit, MTL, Market, Stop and MIT).

4. The main investment and trading strategies deployed in the futures markets are outright trades, hedging, basis trades and spread trades.

5. **Basis trading** is an arbitrage trading strategy in which the trader takes opposing long and short positions in the two securities, so as to profit from the convergence of their values and make a profit. The strategy is called basis trading because it aims to profit off very small basis point changes in value between the two securities.

6. A spread combines both a long and a short position put on at the same time in related futures contracts. The intention of the strategy is to mitigate the risks arising from holding only a long or a short position. There are various types of spread positions: intra-commodity, **inter-commodity, intra-market, inter-market, inter-delivery** and **intra-delivery**.

7. A **calendar spread** position is established with futures contracts by simultaneously entering a long and short position on the same underlying asset but with different delivery months. It is also referred to as a horizontal or time spread.

8. A **butterfly spread** is a neutral trading strategy which combines bull and bear spreads. In a bull spread, the investor seeks to profit from a rise in the price of the underlying while and in a bear strategy, a profit is made from a price fall. A butterfly spread may be in the form of a near term bull spread and a longer-term bear spread combination, or vice versa if bearish. The butterfly spread contains 4 legs, in which the trader will buy 2 different months, and then sell the middle month twice.

9. A **condor spread** is similar to a butterfly spread as this strategy also combines a bear and a bull spread. The difference is that there is no common middle month, as the expiration dates of the futures contracts are all different. A condor has a combination of 4 contracts with equally distributed delivery months.

10. The **TED spread** is the difference between the price of the 3-month futures contracts for U.S. Treasuries and 3-month contracts for Eurodollars, both with the same expiration month. Since the 3-month futures contracts for U.S, Treasuries are no longer traded on the CME, the TED spread is now commonly known as the difference between 3-month US Treasury bill rate and the 3-month USD LIBOR. The TED spread is

used as an indicator of credit risk because U.S. T-bills are considered risk-free, while the rate associated with the Eurodollar futures reflects the credit ratings of corporate borrowers. When the TED spread increases, default risk is seen to be rising.

11. The objective of hedging is to reduce risk. If adverse price movements occur, losses can be contained, or profits can be protected. However, if the price moves favourably, the profits will be capped. Hedging does not eliminate price risks but converts price risk to basis risk between the underlying instrument and the hedging instrument, by confining the final price to a determinable range.

12. In **portfolio management**, hedging can be used to smoothen cash flow, simplify financial planning, reduce working capital requirements, allow for more efficient product pricing and manage inventory more efficiently.

13. Before actual hedging is done, the investor will consider and evaluate various factors to identify and measure the risks to be hedged. The assessment will require answers to the following questions:
    - What is the probability of this change?
    - What is the probable size of this change?
    - What is the risk value associated with NOT hedging?
    - What is the transaction cost associated with the hedge?
    - What percentage of the principal value of the asset (or liability) to be hedged does this transaction cost constitute?
    - What is the basis risk for the hedge?
    - What is the cost of hedging compared with the risk value?

14. When a decision to hedge is taken, the investor can develop the hedge program by considering the following:
    - Evaluate the risk exposure in assets and liabilities (by group, by maturity or profit centre).
    - Set hedge objective in terms of the acceptable risk level or tolerance.
    - Determine hedgeability using price correlation analysis (and cross hedges, if necessary).
    - Determine the hedge instrument or vehicle, which should be an instrument with a high degree of correlation with the target security.
    - Determine the target rate to be locked in for the hedge (i.e., the rate or the price).

15. While futures and options are frequently used as hedging instruments, there are some limitations such as the round number of contracts, margin maintenance, time horizon and expiry dates and no exact hedge instrument.

16. When seeking an appropriate hedge instrument, it is essential to find one that is highly correlated with the target security. A cross-hedge can be used in which a security is hedged with another instrument which is different, but they are positively correlated and have similar price movements. The relationship and the correlation between the two securities must be carefully evaluated and can be measured using regression analysis.

17. An investor might wish to lock in a target rate or value for the hedge, such as the currency rate, interest rate or underlying asset price. The target basis concept explains why a hedge that is held till expiry locks in the futures rate. A hedge that is lifted before expiry merely substitutes price risk for basis risk. The target

rate for a hedge is calculated by adding the futures rate and the target rate basis.

18. **Immunization** is a hedging strategy that matches the durations of assets and liabilities. To manage exposure to interest rate fluctuations, institutions will immunize the impact of interest rates on the portfolio value by calibrating the interest sensitivity of the portfolio. A hedge can be constructed for currently held positions or anticipated ones, or for a known or unknown time period.

19. The **hedge ratio** defines the expected movement in value of the cash instrument to be hedged, given a particular movement in the value of the futures contract, which is to serve as the hedge instrument. The hedge ratio is the actual weighting of the hedge vehicle to the target security. It is the ratio of futures to a spot position (or vice versa) that achieves an objective, such as minimizing or eliminating risk. A delta-neutral hedge is one where the overall delta of the portfolio is zero.

20. The futures buyer must decide whether to use **strips or stack**. Strips use successive futures contract months to match delivery dates on the futures contracts with the rollover dates or tenor on a cash loan. Stacks use the deferred contract months to match the rollover dates or tenor on a cash loan. Both choices will have basis risk at the time of termination of the hedge.

21. When the maturity date of the underlying instrument does not coincide with the expiry dates of the hedge instrument, two types of hedges can be considered: interpolative hedge (used when time horizon straddles two expiry dates), and extrapolative hedge (used when time horizon stretches beyond the tenor of the last traded contract).

22. There are two types of risks in any equity portfolio: **systemic (market) risks** and **specific (non-market) risks**. Equity index futures can only hedge against market risks, by selling futures contracts when long in the equity market (short hedge) and buying futures contracts when short in the equity market (long hedge). In deciding the number of contracts to use, there are two factors to consider: the beta and the modified portfolio value. The investor then decides the contract delivery months and monitors the necessary maturity and sensitivity adjustments of the hedging positions.

23. The absolute risk of the hedge is expressed as a standard deviation. The effectiveness of the hedge refers to the risk of a hedged position relative to an unhedged position. Most hedges require very little active monitoring during their life span. However, changes in volatilities and yield spread relationships may necessitate changing the hedge ratio.

24. Portfolio management is about setting investment policy, matching investments to meet financial objectives, asset allocation decisions and balancing risk against desired performance. It is also about assessing the strengths, weaknesses, opportunities and possible risks when making decisions about portfolio exposures to different asset classes, different market and different strategies. The need to constantly rebalance the mix of assets efficiently and economically can be achieved using futures.

25. Markets are not always efficient, and they give rise to arbitrage opportunities every now and then. Arbitrageurs profit from price differences that arise when a security, commodity or asset, is traded on two or more markets. Arbitrage strategies often involve the simultaneous buying in one market and selling in the other market whenever prices between the derivatives (such as futures or options) and the underlying asset (such as stock or bonds) are out of line.

Chapter 4: Options

Learning Objectives
The candidate should be able to: ✓ Understand the basic concepts of options and how they are traded. ✓ Explain option features of call and put options. ✓ Describe the major factors affecting the price sensitivity of options. ✓ Explain the rights, obligations, benefits and risks of using options. ✓ Calculate the breakeven, profit and loss from an option contract. ✓ Explain how traders use options strategies. ✓ Explain how to use neutral, bullish and bearish trading strategies based on the investor's outlook. ✓ Explain how to use options to hedge an investor's underlying position. ✓ Describe options on basic underlying assets and indices (e.g., equities, bonds, interest rates, currencies, futures and swaps).

4.1 What is an Option?

An option is a contract between two parties – the option writer (seller/issuer) and the option holder (buyer) – that gives the holder of the option the right to buy (or to sell) the underlying from (to) the option writer. For our learning purposes, we will initially apply the concept with equities as the underlying asset before we discuss options on other underlying asset classes.

There are two types of options – calls and puts.

Call Options

a. **Buyer** – The call option buyer has the right but not obligation to buy the underlying asset at a contracted strike price within a specific period of time. If the option holder decides to exercise the call option, he will have to pay the seller the contracted price (strike price) and take possession of the underlying asset.

b. **Seller** – The call option seller is obliged to deliver the underlying asset at the strike price if the buyer exercises the option, i.e., the call option seller is forced to take a short position if the option is exercised.

Put Options

a. **Buyer** – The put option buyer has the right but not obligation, to sell the underlying asset at a contracted strike price within a specific period. When a put option is exercised, the option holder will deliver the underlying asset and receive payment at the contracted price (strike price).

b. **Seller** – The put option seller is obligated to take delivery of the underlying asset at the strike price if the buyer chooses to exercise the option.

Options originated with equity options. In 1973, the Chicago Board Option Exchange (CBOE) began trading listed call options on 16 underlying stocks. Prior to exchange-traded options, options were only transacted over-the counter (OTC), mainly among financial institutions. Today, there are options on a wide range of underlying assets including equities, bonds, interest rates, currencies and various financial indices. There are also options on futures contracts and options on options (known as **compound options**) and with product innovation, there are newer categories of underlying assets including real estate, weather, market volatility and exchange traded funds. Other types include options with unusual underlying instruments, variations in strike price calculations, diverse payoff mechanisms, or contingent expiry conditions.

In Singapore, the options market has grown and developed more actively in the OTC market through structured products offered by numerous financial institutions to other financial institutions and to accredited investors. Hence, investors will require greater scrutiny of counterparty risk from time to time.

4.2 Option Features and Definitions

4.2.1 Style of Option

Options are either European-style or American-style options. An option contract has a finite life and will lapse at the expiration date. **European options** can only be exercised on the expiration date, while holders of **American options** can exercise their right at any time before expiration[24].

4.2.2 Strike Price

The strike price is the fixed or contracted price at which the call option buyer has the right to purchase the underlying asset, or the put option buyer has the right to sell the underlying asset. The **strike price** is also referred to as the **exercise price**, which is a key variable to determine the option's intrinsic value. The closer the strike price is to the underlying asset value, the more valuable the option because the greater the odds that the option will expire with a positive intrinsic value (in-the-money).

4.2.3 Expiration Date

All options will have a maturity date or expiration date. The longer the time to maturity, the higher the time value. Exchange-traded options in the US typically expire in the 1st month of each quarter and would typically belong to any of the following expiration cycles in Table 4.2.3:

[24] For all equity options listed in the U.S., the expiration date falls on the 3rd Friday of the expiration month (except when that Friday is also a holiday, in which case it will be brought forward by 1 day to Thursday).

Table 4.2.3 - Option Cycles for Options

Cycle Number	Cycle Name	Expiration Months
1st month of every quarter	JAJO	January, April, July, and October
2nd month of every quarter	FMAN	February, May, August and November
3rd month of every quarter	MJSD	March, June, September and December

4.2.4 Class and Series

Options written on the same asset (e.g., a particular stock, index, futures) and issuer are said to be from the same **class**. Options of the same class, which have the same exercise price and expiration month, are in the same **series** (e.g., all ABC January 50 calls).

4.3 Option Price

The **option price** that the option buyer pays to the seller is also commonly called the **option premium**. Since options provide the buyer the right, and not the obligation, to buy or sell the underlying asset, an option's price and volatility is influenced by the price movement of the underlying asset or index. The value of this right rises or falls with the value of the underlying instrument.

An option's price consists of two parts:

Option Price = Intrinsic Value + Time Value

4.3.1 Intrinsic Value

Intrinsic value relates to the "**moneyness**" i.e., the potential profit or loss from the exercise of an option at a point in time. Intrinsic value is the value of the option when an option buyer exercises it immediately.

Theoretically, intrinsic value is the difference between the current market price and the strike price. This difference is the amount by which the option is in-the-money. As the option holder has the right but not the obligation to buy or sell the underlying asset, the intrinsic value of an option cannot be negative.

By definition, an option has intrinsic value when there are profits– when it is **in-the-money (ITM).** When an option is just **at-the-money (ATM)** or **out-of-the-money (OTM)**, the intrinsic value is zero.

a. Intrinsic Value of a Call Option = **Max (0, Current Market Price of Underlying – Option Strike Price)**

b. Intrinsic Value of a Put Option = **Max (0, Option Strike Price – Current Market Price of Underlying)**

Please refer to Table 4.3.1 on the "moneyness" of call and put options.

Table 4.3.1 – "Moneyness" of Call and Put Options

Call Options	Put Options
$C = S_T - X$ if $S_T > X$ In-the-money $C = 0$ if $S_T = X$ At–the-money $C = 0$ if $S_T < X$ Out-of-the-money Where: C = Value of call option S_T = Value of the underlying asset X = Exercise price of the asset	$P = X - S_T$ if $X > S_T$ In-the-money $P = 0$ if $X = S_T$ At–the-money $P = 0$ if $X < S_T$ Out-of-the-money Where: P = Value of put option S_T = Value of the underlying asset X = Exercise price of the asset

Calculating the Intrinsic Value of a Call Option

The exercise price of a call option is $50. The price of the call is $6, and it expires in 3 months. The current share price is $54.

Hence:
- Intrinsic value = $4 ($S_T - X$ = $54 - $50 = $4)
- Time value = $2 (Option price – Intrinsic value = $6 - $4 = $2)

At expiration, if the share price declines to $48:
- Intrinsic value of the call option = $0 ($S_T < X$ so it is out-of-the-money).
- Time value is always zero at expiration.

Calculating the Intrinsic Value of a Put Option

The exercise price of the put option is $50. The price of the put option is $9 and expires in 1 month. The current share price is $44.

Hence:
- Intrinsic value is $6
- ($X - S_T$ = $50 - $44 = $6)
- Time value = $3 (Option price – Intrinsic value = $9 - $6 = $3)

At expiration, if the share price rises to $53:
- Intrinsic value of put option = $0 ($X < S_T$ so it is out-of-the-money).
- Time value is always zero at expiration.

4.3.2 Time Value of Options

The option price is generally higher than its intrinsic value before expiry. This difference between the actual option price and its intrinsic value is the **time value**, which is calculated with the following equation:

Time Value of Option = Option Price – Intrinsic Value

The option buyer hopes that there will be an opportunity to make a profit before it expires. This can occur when the option is in-the-money (i.e., when the intrinsic value is positive).

a. For the call option buyer, it is when the share price rises above the strike price:

Intrinsic Value = Current Market Price – Option Strike Price > 0

b. For the put option buyer, it is when the share price falls below the strike price:

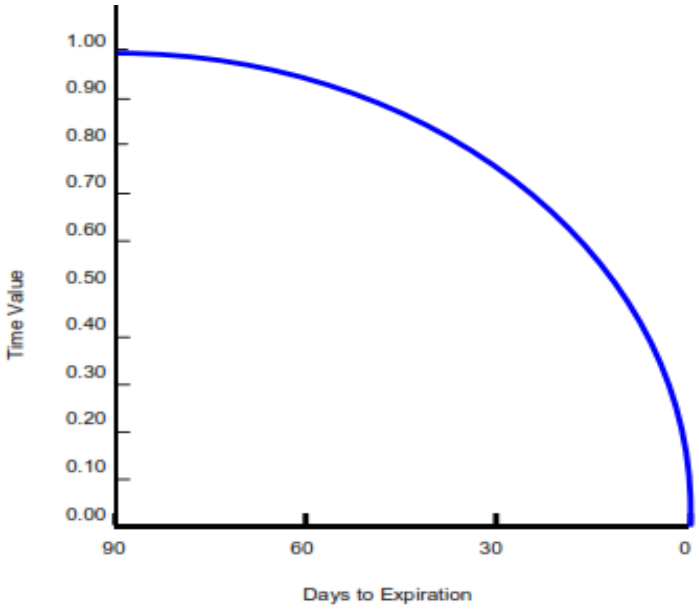
Intrinsic Value = Option Strike Price – Current Market Price > 0

The time value of an option is generally positive. At expiry, the time value will be zero. The value of the option at expiration is exactly its intrinsic value:

Option Price = Intrinsic Value (as Time Value → 0)

The longer the time to an option's expiration, the greater is its time value. Even an option which seems relatively worthless today (ATM or OTM = zero intrinsic value) could be in-the-money by the time the expiration date comes around (see Figure 4.3.2).

Figure 4.3.2 – Time Value of Options



4.3.3 Summary Table of Comparison of Call Option vs Put Options

Table 4.3.3 summarises the major differences between call options and put options:

Table 4.3.3: Comparison of Call Options vs Put Options

Criteria	Call Option		Put Option	
	Buyer	Seller	Buyer	Seller
Obligations	Pays the option premium.	Delivers the underlying instrument	Pays the option premium.	Takes delivery of the underlying asset at the strike price.

Criteria	Call Option		Put Option	
	Buyer	Seller	Buyer	Seller
	Pays for underlying asset at strike price if call is exercised.	at the agreed strike price.	Delivers the underlying asset if put is exercised.	
Benefits	Potential capital gain from the rise of the underlying asset less call premium paid. Leverage from the small outlay for potential capital gains. Rise in the option value if it is exchange traded, when the price of the underlying asset rises.	Additional yield/gains from call premium.	Potential capital gain from the fall of the underlying asset less put premium paid. Leverage from the small outlay for potential capital gains. Rise in the option value if it is exchange traded, when the price of the underlying asset falls.	Additional yield/gains from put premium.
Maximum Gain	Capital gain from the underlying asset less option price. $U_{Mkt} - U_{Strike}$ – Call	Call option premium received.	Capital gain from the underlying asset less option price. $U_{Strike} - U_{Mkt}$ – Put	Put option premium received.
Risk & Maximum Loss	Loss from the value of call option price if unexercised. Counterparty risk.	For uncovered call, the need to buy the underlying asset at the higher market price than the strike price. For covered calls, the loss of the opportunity of potential capital gains (since underlying asset has to be delivered. Counterparty risk.	Loss from the value of put option price if unexercised. Counterparty risk.	Capital loss from taking delivery of the underlying asset at the strike price while it is falling below the strike price. Counterparty risk.

4.4 Factors Influencing Equity Option Prices (Premiums)

There are six major factors which affect the price of equity options:

a. Underlying share price;
b. Strike price / exercise price;
c. Time to expiration;
d. Volatility of underlying share;
e. Current risk-free interest rates; and
f. Expected Dividends.

Table 4.4 summarises the relationship of these factors with the equity option price:

Table 4.4: Factors Influencing Equity Options Prices (Premiums)

Option Price Sensitivity	Call Option	Put Option
Price of underlying share ↑	↑	↓
Exercise price ↑	↓	↑
Time to expiration ↑	↑	↑
Volatility of underlying share ↑	↑	↑
Risk-free interest rate ↑	↑	↓
Expected dividends ↑	↓	↑

4.4.1 Changes in the Underlying Share Price

The option price will change when the value of the underlying share moves. The change in share price will have opposite effects on calls and puts. For calls, when the value of the underlying share rises, the call value will increase while the put price will fall. A decrease in the underlying share's value will have the opposite effect.

4.4.2 Strike Price (or Exercise Price)

This impacts the intrinsic value of the option. The lower the exercise price, the higher the price of a call option, all other factors being equal. For a put option, the higher the exercise price, the higher the put option value. An option's premium (intrinsic value plus time value) generally increases as the option becomes deeper in-the-money and decreases as the option becomes more deeply out-of-the-money.

4.4.3 Time to Expiration

This affects the time value component of an option's premium. The longer the time to expiration, the greater the price of the option. As expiration approaches, the option's time value erodes for both puts and calls, as less time remains for the underlying share price to rise or fall. As the time to expiration decreases, the time value approaches zero and the option's value approaches its intrinsic value.

4.4.4 Price Volatility of the Underlying Share

Volatility is a measure of risk or the variability of price of the underlying asset or security. Higher volatility indicates greater expected fluctuations and can significantly affect the time value portion of an option's premium. Higher volatility also suggests higher option prices because there is a higher chance for the underlying share price to move in favour of the option buyer and be in-the-money (for both puts and calls).

4.4.5 Current Risk-Free Interest Rate

Interest rates affect the cost to carry (shares of) an underlying security. Cost of carry is the opportunity cost, which here is the interest foregone from an alternative investment (e.g., risk-free Treasury bills). All other factors constant, the higher the interest rate, the greater the opportunity cost of directly buying the underlying share. There is a positive relationship between interest rates and the call option value.

4.4.6 Expected Dividends of Underlying Share during the Option's Life

For equity options, dividends will result in a reduction of the price of the underlying share when it goes ex-dividend. A drop in the share price will cause the call option value to decline. For put options, the reverse is true.

4.5 Payoff Profiles

Depending on the price paid for the option or option strategy used, the payoff profile for options will reflect the maximum gains and losses and breakeven points if the price of the underlying asset reaches the option strike price.

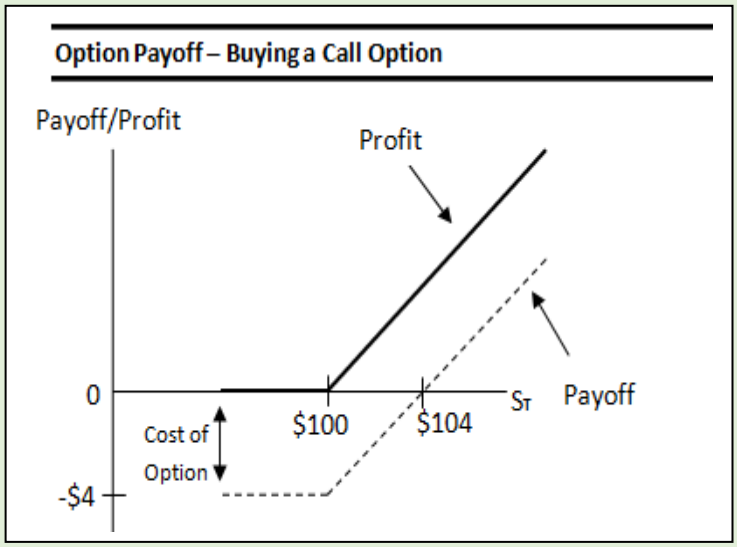
Exploration Payoff Profile for Buying an Equity Call Option

Suppose an investor bought a call option on a share a month ago. The current market data is as follows:

- Price of the call option (P) bought 1 month ago = $4
- The option's exercise price (X) is $100
- The underlying share price (S) is $110

Draw the payoff diagram and explain the following:

i. Payoff and profit
ii. Breakeven point
iii. Maximum gains
iv. Maximum loss
v. Obligations of the call buyer



Payoff to Call Buyer:

$C = S_T - X$ if $S_T > X$ In-the-money
$C = 0$ if $S_T = X$ At–the-money
$C = 0$ if $S_T < X$ Out-of-the-money

i. **Payoff and profit:** Payoff to option buyer is $10 (i.e., $110 -$100), and the profit after deducting the option price is $6 as long as the underlying stock price stays at the current level.

ii. **Breakeven:** The call option holder only breaks even if the price of the underlying share rises above the exercise price by an amount equal to the price of the option. As the price of the call option is $4 and

Restricted

its exercise price is $100, the breakeven point for the call option buyer is when the price of the underlying share is exactly $104.

iii. **Maximum gain:** To a call option holder, the maximum gain is the payoff of the option less the price paid for it. This can, theoretically, be unlimited.

iv. **Maximum loss:** As the option buyer would have already paid for the option, the maximum loss is the price paid for the option. The payoff cannot be negative as the option is exercised only if $S_T > X$. If $S_T < X$, exercise does not occur, and the option expires with zero value.

v. **Obligations:** Except for paying the option price, the buyer has no other obligation unless he exercises the option, and thereafter to make the appropriate payment based on the strike price.

Expiration Payoff Profile for Selling an Equity Call Option

Suppose an investor sells a call option. The current market data is as follows:

- Price of the call option (P) is $4
- The option's exercise price (X) is $100
- Current underlying share price (S) is $90

Show the option payoff diagram and explain the following:

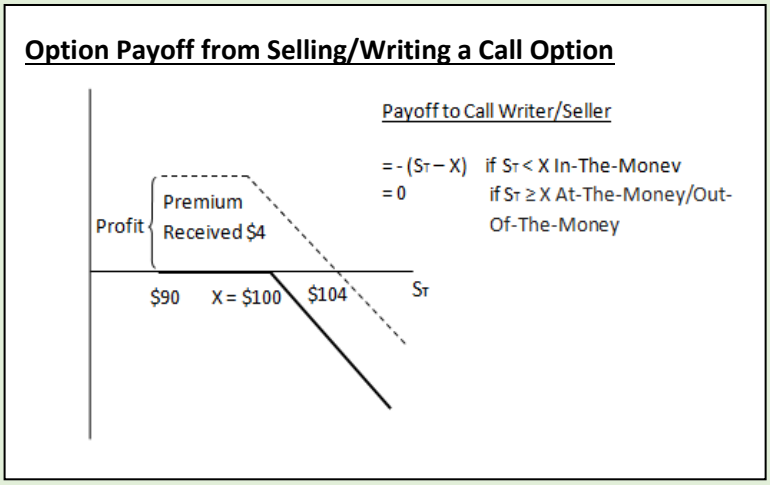
i. Payoff and profit
ii. Breakeven point
iii. Maximum gain
iv. Maximum loss
v. Obligations of the call seller / writer

Payoff to Call Writer/Seller:

$C = -(S_T - X)$ if $S_T < X$ In-the-money

$C = 0$ if $S_T \geq X$ At–the-money/ Out-of-the-money

Option Payoff from Selling/Writing a Call Option



i. Payoff and profit: As long as the current share price ($90) stay below $100 which is the exercise price of the option, there is no reason for the buyer of the option to exercise, hence the payoff is zero. Since the call writer receives the call premium of $4 (as long as the option is out-of-the-money), the profit is $4. If the share price rises and goes beyond the strike price, the buyer will exercise the call option. From an initial $4 profit, the profit will become a loss once the price rises beyond $104.

ii. Breakeven: The call option writer breaks even (net profit = 0) if the underlying share price rises above the exercise price by an amount equal to the option price. The breakeven point is where the negative payoff ($100 - $104) is exactly equal to the call option premium received ($4). Note that the breakeven point for both the call option buyer and writer is exactly the same, i.e., $104.

iii. Maximum gain: The maximum gain for a call writer is equal to the price of the option sold, i.e.; the $4 call premium. This only happens when the underlying share price is equal to or less than the option's exercise price.

iv. **Maximum loss:** The maximum loss for a call option writer is, theoretically, unlimited. This happens when the underlying share price rises above the exercise price plus the option price; the higher the underlying share price, the greater the loss to the option writer.

v. **Obligations**: The writer of the call has the obligation to deliver the shares at the exercise price X = $100, when the option is exercised.

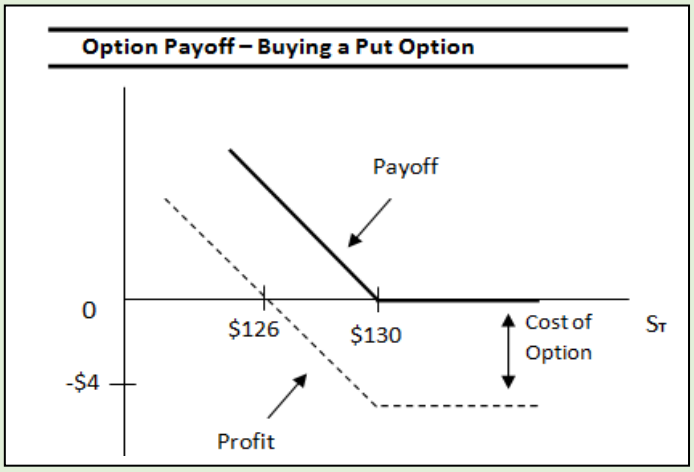
Expiration Payoff Profile for Buying an Equity Put Option

An investor buys a put option on a share a month ago. The market data is as follows:

- Price of the put option (P) purchased 1 month ago is $4
- The option's exercise price (X) is $130
- The underlying share price (S) at expiry is $120

Show the option payoff diagram and explain the following:

i. Breakeven point
ii. Payoff and profit
iii. Maximum gain
iv. Maximum loss
v. Obligations of the put buyer



Payoff to Put Buyer:

$P = X - S_T$ if $X > S_T$ In-the-money

$P = 0$ if $S_T = X$ At–the-money

$P = 0$ if $X < S_T$ Out-of-the-money

A put option gives the holder the right to sell an asset at the exercise price. Here, the holder will not exercise the option unless the underlying asset price is less than the exercise price. The solid line illustrates the payoff at maturity to the put option holder. If the share price at expiry is above the exercise price, the put option has no value. If the share price falls below the exercise price, the put value at expiry increases. The broken line represents the put option holder's profit at expiry, net of the initial cost of the put.

i. **Payoff and profit:** The payoff to the put option buyer is $X - S_T$ = $130 - $120 = $10 but the net profit after deducting the $4 option price = $10 - $4 = $6.

ii. **Breakeven:** The put option holder only breaks even (net profit = 0) if the price of the underlying share falls below the exercise price by an amount equal to the price of the option. In this example, the price of the put option is -$4 and the option's exercise price is $130, so the breakeven point for the put option buyer is when the price of the underlying is $126.

iii. **Maximum Gain:** Similar to buying call options, the maximum gain to a put option holder is the option payoff less the option price. The further the underlying asset price falls below the exercise price, the

greater the put option's payoff. So theoretically, the underlying asset value could fall to zero, and the maximum gain in this case on exercising the put would be \$130 – 0 - \$4 = \$126.

iv. **Maximum Loss:** The option buyer would have already paid for the option, which is \$4 in this example. Therefore, the maximum loss is the option price for an option that is not executed.

v. **Obligation:** There is no obligation of the put buyer except when he exercises the put.

Upon execution of the put, the put seller forces the buyer to take delivery of underlying at the strike price. (The put seller must be able to deliver underlying by going into the market to secure it at a cheaper level).

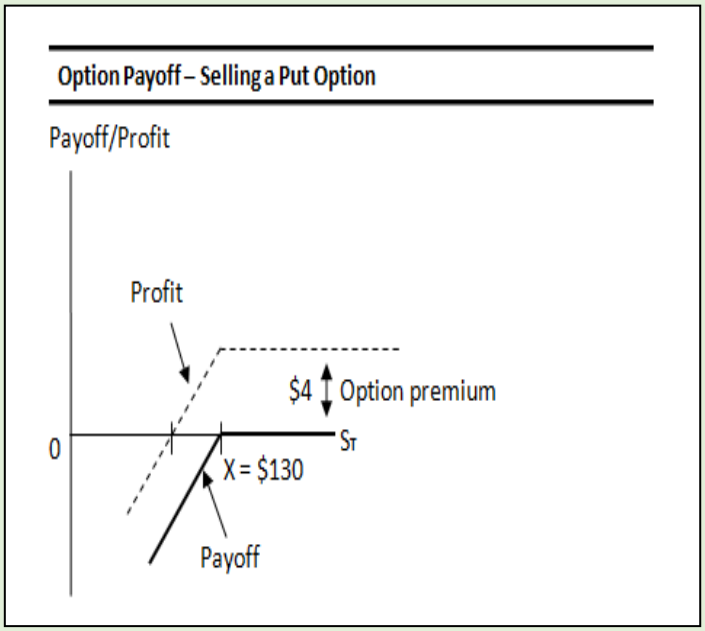
Payoff Profile for Selling an Equity Put Option at Expiration

An investor sells a put option on a share that expires in 1 month. The market data is as follows:

- Price of the put option sold 1 month ago = \$4
- The option's exercise price X is \$130
- Current value of the asset is S_{T1} = \$140
- Value of underlying share at expiry is S_{T2} = \$120

Show the option payoff diagram and explain the following:

i. Payoff amount and profit
ii. Breakeven point
iii. Maximum gain
iv. Maximum loss
v. Obligations of the put seller / writer



Payoff to Put Seller / Writer:
$P = -(X - S_T)$ if $S_T < X$ (In-the-money)
$P = 0$ if $S_T \geq X$ (At–the-money/Out-of-the-money)

The payoff to the put writer is exactly the mirror image of the put holder.

- **Payoff and Profit:** The payoff is zero when the underlying asset price S_T (which in this example is \$140) is above the strike price \$130. The writer receives the put premium of \$4 which is also the maximum profit. When the S_T falls below the strike price X, the buyer of the put can exercise the option. The profit will be reduced and turned into losses for the put writer as the underlying asset keeps falling in value.
- **Breakeven:** The put option writer breaks even (net profit = 0) if the price of the underlying falls below the exercise price by an amount that is equal to the price of the option. So, when the underlying asset

price falls to $126, the payoff = $126 - $130 = -$4 which offsets the put option price. Note that the breakeven point for both the put option buyer and put option writer is exactly the same.

- **Maximum Gain:** The maximum gain for a put option writer is equal to the option premium i.e., the price of the option sold = $4 in this example. This only happens in the situation when the price of the underlying asset is equal to or greater than the option's exercise price. Similar to the call writer, the put writer can offset his position by buying an identical contract in the market.
- **Maximum Loss:** The maximum loss for a put option writer can be significant. Theoretically, the maximum loss is when the underlying asset price falls to zero. $X - S_T - P = \$130 - \$0 - \$4 = \$126$. The buyer of the put would exercise the put option to sell the asset to the writer of the put at $130 but this would be offset against the option premium.
- **Obligation:** The option writer has the obligation to accept the underlying asset at the strike price, when exercised by the buyer of the put option.

4.6 Put-Call Parity Theory

4.6.1 Definition

Put-call parity is the relationship between European call and put prices for the same underlying share, strike price and expiration month. The put-call parity formula is:

$$\mathbf{C + PV\,(X) = P + S}$$

Where:

C = Current price or market value of the European call
X = Option strike price
PV(X) = Present value of the strike price of European call discounted from the expiration date at the risk- free rate
P = Current price or market value of the European put
S = Current market value of the underlying share

The formula states that a portfolio, comprising of a call option and cash equal to the present value of the option's strike price, has the same expiration value as a portfolio comprising of the corresponding put option and the underlying share. For European options, when the expiration values of the 2 portfolios are the same, their present values must also be the same. This equivalence is put-call parity. Put-call parity does not apply to American options because they can be exercised before expiry, and the present values of the 2 portfolios may differ.

This principle can be applied to create synthetic securities. Suppose that with put-call parity: C + PV(X) = P + S. Therefore, a synthetic bond or share can be created:

a. A synthetic bond: PV(X) = S + P – C
b. A synthetic share: S = PV(X) + C – P

In creating synthetic securities, it is assumed that:

i. Dividends are zero;

ii. Strike prices are the same for calls and puts;

iii. Expiration dates are the same for calls and puts; and

iv. The number of shares of stock must equal the number of shares represented by the options.

Put-call parity demonstrates two very important concepts. Firstly, it is always possible to replicate one of the investments with the other three. Secondly, it shows that options can be priced from a relative standpoint.

4.6.2 Arbitrage

Put-call parity shows that using combinations of put and call options can create positions which are equivalent to holding the shares itself. If market conditions create a divergence in the value, an arbitrage opportunity would exist. In an arbitrage situation, arbitrageurs will jump in and execute profitable, risk-free trades resulting from price anomalies in the financial markets, until any departure from put-call parity is eliminated.

Through the put-call parity principle, one can understand how financial institutions are able to value options, create synthetic options on the same underlying security, and identify the drivers of options and derivatives trading in the financial markets.

4.6.3 Synthetic Positions

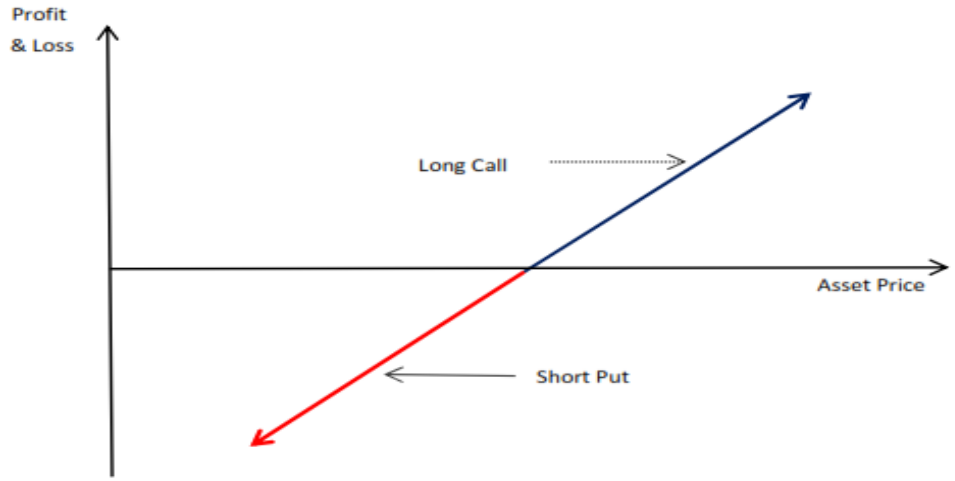
It is possible to construct a position in the underlying share with synthetic positions. Synthetic shares positions replicate the risk and payoff of either being long or short in the underlying asset. Synthetic options imitate the risk-reward profile of "real" options by using a combination of calls, puts and the underlying share or security. Option-arbitrage strategies involve synthetic positions, where the risk-return profile of any position can be exactly replicated by using combinations of calls, puts and the underlying share. Here are some examples of synthetic structures for stocks/shares and equity options.

4.6.3.1 Synthetic Long Stock

A synthetic long stock using options simulates the payoff of a long stock position. Construction of the synthetic position requires a long at-the-money call and a short at-the-money put on the underlying share, with the same expiration date. It is a low-cost alternative to purchasing the share outright, giving the investor the same unlimited profit and downside risk profile.

Figure 4.6.3.1 shows the synthetic long stock position.

Figure 4.6.3.1 – Synthetic Long Stock



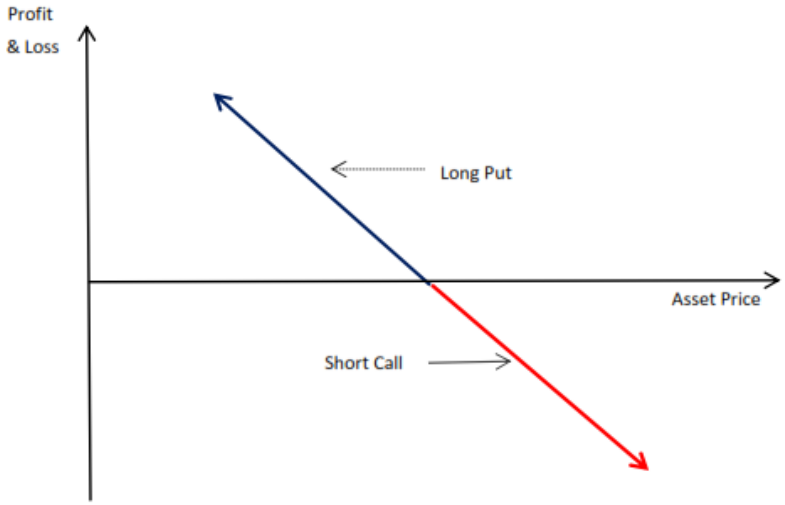
Long Call + Short Put = Synthetic Long Stock

4.6.3.2 Synthetic Short Stock

A synthetic short stock position using options also simulates the payoff of a short stock position. The position can be constructed by shorting at-the-money calls and buying an equal number of at-the-money puts on the underlying share with the same expiration date. This strategy gives the investor unlimited downside risk, with the upside being the full extent of the decline in the underlying share price.

Figure 4.6.3.2 shows the synthetic short shock position.

Figure 4.6.3.2 – Synthetic Short Stock

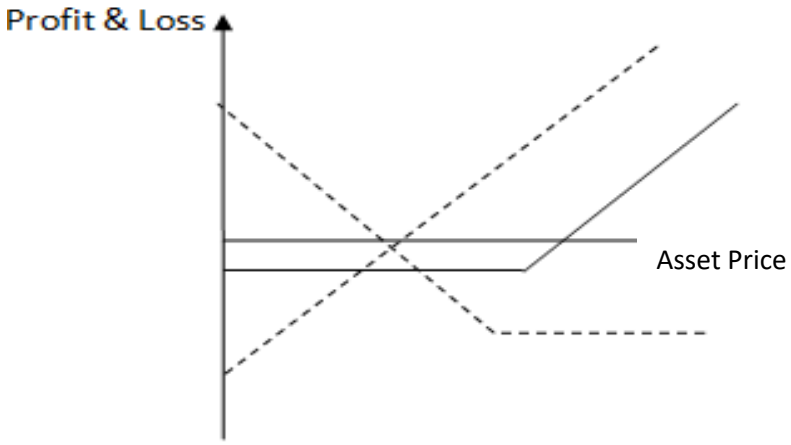


Long Put + Short Call = Synthetic Short Stock

4.6.3.3 Synthetic Long Call

Figure 4.6.3.3 shows how a synthetic long call can be constructed with a long position in the underlying asset and a long put:

Figure 4.6.3.3 – Synthetic Long Call

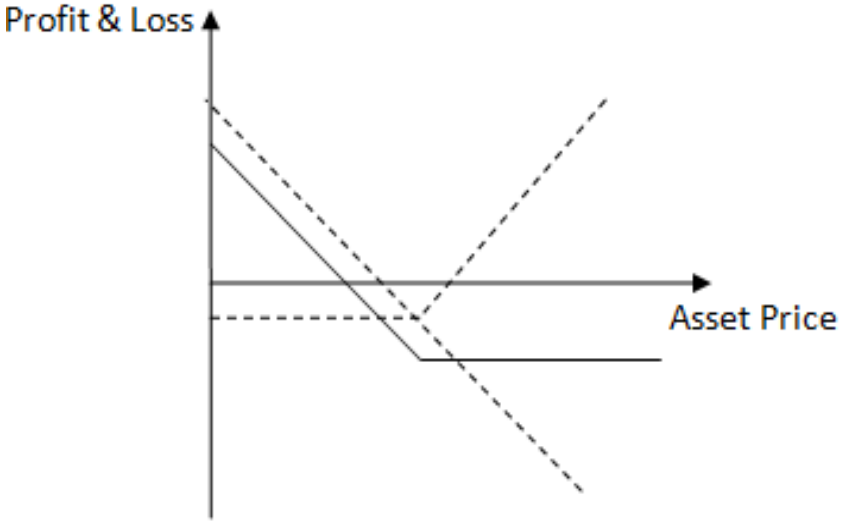


Long Underlying + Long Put = Synthetic Long Call

4.6.3.4 Synthetic Long Put

Figure 4.6.3.4 shows how a synthetic long put can be constructed with a short position in the underlying asset and a long call:

Figure 4.6.3.4– Synthetic Long Put

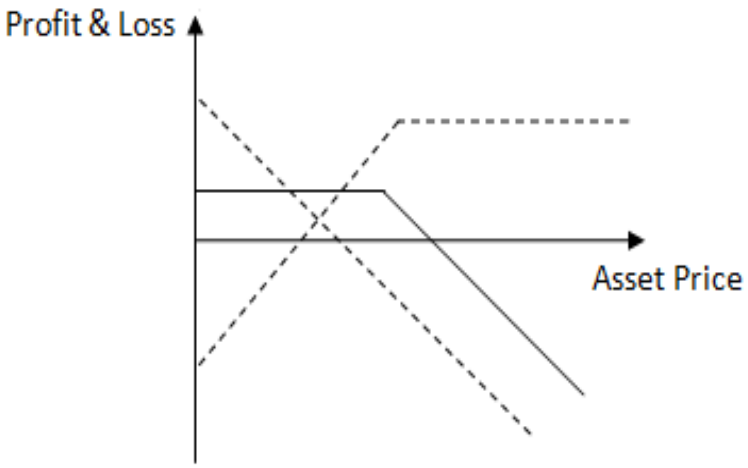


Short Underlying + Long Call = Synthetic Long Put

4.6.3.5 Synthetic Short Call

Figure 4.6.3.5 shows how a synthetic short call can be constructed with a short position in the underlying asset and a short put:

Figure 4.6.3.5 – Synthetic Short Call

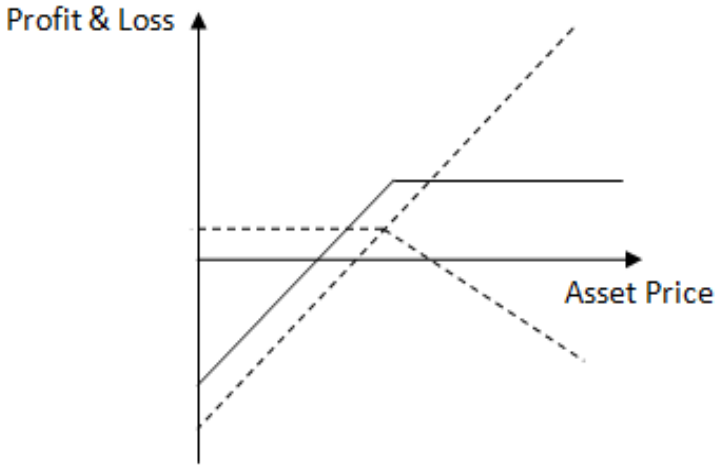


Short Underlying + Short Put = Synthetic Short Call

4.6.3.6 Synthetic Short Put

Figure 4.6.3.6 shows how a synthetic short put can be constructed with a long position in the underlying asset and a short call:

Figure 4.6.3.6 – Synthetic Short Put



Long Underlying + Short Call = Synthetic Short Put

4.7 Option Valuation

4.7.1 Black-Scholes Formula[25]

In 1973, Professors Fischer Black, Myron Scholes and Robert Merton came up with a formula for calculating the value of options. Known as the **Black-Scholes formula**, it states that the value of an option is determined by:

a. Underlying share price;
b. Options strike price;
c. Time until expiration;
d. Implied volatility;
e. Dividend status; and
f. Interest rates.

The Black-Scholes formula was the first widely used model for option pricing. It can be used by anyone who invests and trades in options to calculate an option's theoretical value, by plugging in the values of the six variables. While the Black-Scholes model does not perfectly describe real-world options trading, it is still extensively used.

Another common method is the binomial pricing model, especially for pricing American-style options due to the early exercise feature. By using pricing models, investors can compute and anticipate an option's premium under various scenarios. This helps them to better predict the outcome of their trading strategies for changes in interest rates, dividend distributions, volatility, or other factors that influence the option's actual value.

Note: For the purposes of the CM-SIP or CM-CMP Examinations, candidates will **not** be required to calculate option prices using the Black-Scholes Formula. However, candidates will be required to understand and explain how the key variables in the formula are used by market-makers and product specialists to provide estimated price quotes. Candidates should also be aware of the practical challenges and limitations of hedging. Hence, they should read Section 4.7 Option Valuation with a focus on explaining usage and application.

4.7.2 Volatility

Volatility is an important factor in deciding what kind of options to buy or sell. The official mathematical value of share volatility is denoted as "σ" (the annualized standard deviation of a share's daily price changes). There are two types of volatility:

a. **Statistical (Historical) Volatility** – measures actual asset price changes over a specific period; and
b. **Implied Volatility** – measures how much the market expects asset prices to move for an option price (the volatility that is implied by the market).

4.7.3 The Option Greeks

Understanding volatility and managing risk positions requires monitoring the various option "Greeks". The Greeks are a collection of statistical values that give investors an overall view of what drives option premiums and the changes in pricing model inputs. Statistical data on past performance can be used as inputs to extrapolate future share price movements. These values can help decide what options strategies to use.

[25] https://www.investopedia.com/terms/b/blackscholes.asp

Each type of volatility risk is represented by a Greek letter (with the exception of Vega). The sensitivities or relationships are expressed as a change in a parameter as a function of a change in a 2nd parameter:

a. **Delta** – Change in option price vs change in the price of the underlying asset.

b. **Gamma** – Change in delta vs change in the price of the underlying asset.

c. **Theta** – Change in option price vs change in time to maturity.

d. **Vega** – Change in option price vs change in volatility.

e. **Rho** – Change in option price vs change in interest rates.

4.7.4 Delta

Delta is the main sensitivity among the 5 Greeks. It is the 1st order relationship between the option price and the underlying price, and is denoted as follows:

Delta = Change in option price/Change in underlying price

$$\Delta = \partial V/\partial S$$

where V is the option price and S is the underlying price.

For a call option, a delta of 0.50 means a ½ dollar rise in the option premium for every dollar increase in the share price. For a put option contract, the premium rises as share prices fall. As options near expiration, in-the-money options approach a delta of 1.00. Delta is not a fixed percentage. Changes in the share price and time to expiration affect the delta value.

Using Delta to Calculate Changes in Option Prices

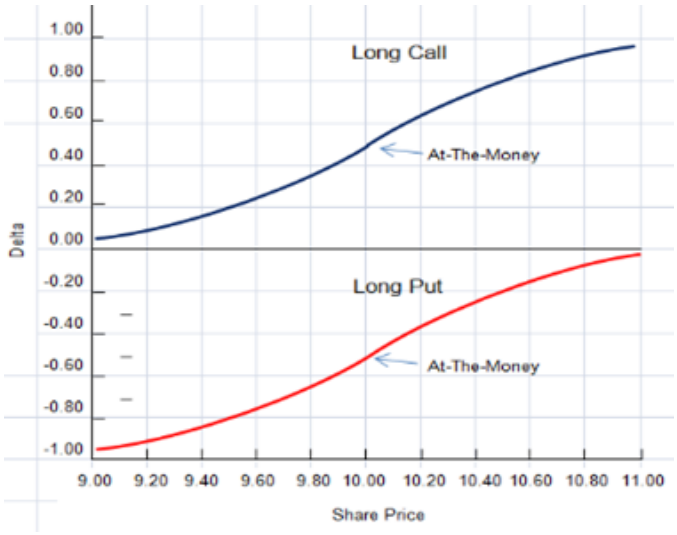
The delta for at-the-money call option on Share XYZ is 0.50. It is expected that the option price changes by $0.50 for every $1.00. Therefore, for a S$2.00 increase in share price, the option price changes by (0.50 x 2) = $1.00. The call option increases by $1.00 and the put option decreases by $1.00.

Figure 4.7.4 shows the delta for a long call and a long put position:

a. The delta of call option for shares can range from 0 to 1. Every increase in the underlying asset price results in a rise in the value of the call option while a decrease in the underlying price would result in a decrease in the call option value. The delta of the put options is negative and has a value between -1 to 0. An increase in the underlying asset price results in a decrease in the value of the puts.

b. In-the-money options will move more than out-of-the-money options. Towards expiration, the delta for in-the-money call options will tend to 1 (in-the-money put options will tend to -1) while the delta of out-of-the-money options will tend to zero.

Figure 4.7.4 – Option Deltas



Traders use delta to construct hedges to offset the risk assumed by buying and selling options.

Delta Hedging

If a trader is short 1,000 call options, he must buy a certain number of the underlying asset to hedge against an increase in the underlying asset's price, and the amount to buy is largely determined by delta.

If the option delta = 0.642, and the underlying share price increases by $1, the dealer will lose $642 on the short position on options (-1,000 x 64.2 cents). To offset this loss, he will have to take a long position of 642 units of the underlying asset to gain $642 (642 x $1).

This process is known as **delta hedging**. As the underlying asset price changes, delta changes. As time to expiry declines, delta changes as well.

Since delta constantly changes, delta hedging is a dynamic process and hence is commonly referred to as dynamic hedging.

4.7.5 Gamma

Gamma indicates an absolute change in delta. Gamma is the sensitivity of delta to changes in the underlying asset price. Therefore, it is the 2nd order relationship between the option price and its underlying asset price:

$$\textbf{Gamma} = \partial^2 V / \partial S$$

All Gamma values are positive because the values change in the same direction as delta (i.e., a higher Gamma means a higher change in delta and vice versa). The signs change with positions or trading strategies because the underlying risk will vary for buyers and sellers of options. All net buying (long) strategies will have positive Gamma and net selling (short) strategies will have negative Gamma. For example, a short call seller has a negative Gamma position, with downside exposure. The call buyer is positive Gamma, with potential for upside gains.

Gamma tells us how fast delta changes when the underlying moves, which can vary for an option series across various strike prices and different expiration months. Gamma is highest near the option's strike price and

decreases as the option goes deeper into or out-of-the money. Options that are very deeply into or out-of-the-money have Gamma values close to 0. The profile of Gamma for both long call and long put positions are similar, as depicted in Figure 4.7.5.

Figure 4.7.5 – Option Gamma



Gamma has important implications for risk management in options. By incorporating Gamma into the risk analysis, the investor knows that only considering the delta positions is insufficient. If there are two different options with the same delta value, the investor cannot be certain that their outcomes would be similar if there is a market shift. The option with the higher Gamma will have higher risk to the option seller, because if prices move unfavourably, the option with the higher Gamma will experience greater losses.

Gamma is highest when the option is at-the-money and close to expiry. Higher Gammas mean greater potential gain for option buyers. When Gamma is high, delta changes rapidly and does not provide a good estimate of sensitivity to the underlying asset price changes. If the market moves unfavourably, investors with a net short option position can suffer significant losses.

4.7.6 Theta

Theta is the sensitivity of an option's premium to change in time. The loss of value of an option over time is referred to as time decay. The quantitative measure of the rate at which the time value of an option is eroded is known as Theta. Theta is negative for both call and put options (assuming no short positions) and is usually displayed as a 1-day or 7-day measure. For example, a 1-day Theta of -0.0250 indicates the option's theoretical value changes by -0.0250 or minus 2.5 cents a day. Results may not be exact due to rounding.

At-the-money options experience more significant dollar losses over time than in-the-money or out-of-the-money options, with the same underlying share and expiration date. This is because at-the-money options have the most time value built into the premium and hence, have the most value to lose as the expiration date gets closer.

4.7.7 Vega

Vega is the sensitivity of option value to changes in implied volatility. Vega indicates an absolute change in option value for a 1% change in volatility. For example, a Vega of 0.090 indicates that the option's theoretical value increases by 0.090 if the implied volatility increases by 1.0%. Alternately, the option's theoretical value decreases by 0.090 if the implied volatility decreases by 1.0%.

Vega is larger as the option is closer to being at-the-money. Hence, the option price becomes most sensitive to volatility when the option is at-the-money. Vega for both call and put options are positive – increased volatility leads to higher option prices. Volatility cannot be easily obtained because it is volatility over the entire life of the option, and not past or current volatility. Nonetheless, a good estimate for (future) volatility can be obtained from its past volatility. The current estimate of volatility, implied volatility, is the market's consensus estimate of the underlying's rate of return volatility. This is obtained by "working backwards" and replacing the Black-Scholes price with the option's market price in the Black-Scholes equation to derive the volatility. This assumes the market value is the true reflection of the option's value.

4.7.8 Rho

Rho is the sensitivity of option value to changes in risk-free interest rates. This is the continuously compounding rate of return on the risk-free security whose maturity corresponds to the option's life. Rho indicates the absolute change in option value for a 1% change in the interest rate. For example, a Rho of 0.060 indicates that the option's theoretical value will increase by 0.060 if the interest rate decreases by 1%.

The call option price has a positive correlation with the risk-free rate, whereas the put option price has a negative correlation. Both types of options are not very sensitive to this variable, especially European-style options.

4.7.9 Practical Problems with Hedging using Delta and Gamma for Options and Warrant

Delta is at best an estimate of changes in the price of the option (or warrant) given the change of the underlying asset price because actual price changes will differ with the changing market forces of demand and supply. Gamma, being the second derivative, which depends on delta estimates, will be prone to even more inaccuracies.

In less efficient markets or markets which depend on a few market-makers to provide "fair" price quotes, the same market-makers may be tempted to manipulate delta and Gamma estimates to create price quotes which favour themselves. Unfortunately, the potential for conflicts of interests exist in such markets and other traders and investors would be placed at an unfair disadvantage, unless there is more competition, transparency, and comparison of quotes.

In practice, most traders conduct delta-hedging on daily basis and hedge less using Gamma and Vega because it is difficult and expensive to find suitable alternatives or similar options to create the Gamma and Vega hedging. Over time most options will move out-of-the-money or in-the-money, so Gamma and Vega will have very little influence. Furthermore, as adjustments are made based on Delta, Gamma and Vega will also change.

Volatilities and trading volumes will also spike up for the option, warrant, or futures contract at critical points when the price of the underlying asset moves rapidly, and therefore it will become very difficult to estimate hedging for Delta, Gamma and Vega.

4.8 Differences between Exchange-Traded and OTC-Traded Options

Options can be traded both on an exchange and OTC basis. Both trading structures have their pros and cons, which essentially involve a trade-off between a greater guarantee of contract performance and greater customization of the option contract. The differences between exchange-traded and OTC-traded options are listed in Table 4.8.

Table 4.8 – Differences between Exchange-Traded and OTC-traded Options

Exchange Traded Options	OTC Traded Options
• Originated and traded on an organised and regulated exchange. Overseen by an extensive government and industry regulation. • Standardized terms (e.g., type, notional size, expiration date) in the form of a public transaction. • Settled through a clearing house associated with the exchange. This guarantees performance of the option contract. • Daily mark-to-market prices available because contracts are standardized.	• Does not trade on an exchange or through a formalised trading system. Less regulation from government and industry bodies. • Contracts are not standardized. Option terms are tailor-made and can be fully customised to suit the needs of parties involves. Transactions are private. • No clearing house, leading to weaker performance guarantees. Counterparty risk needs to be managed and the selection of a reputable or financially sound counterparty or broker becomes important. • Daily mark-to-market procedures may not be readily available.

4.9 Understanding Option Contract Specifications

Table 4.9 sets out an example of the contract specification for a typical option contract. Although the key features of all option contracts are generally the same, specific features may differ depending on the agreement between the counterparties, or whether the contract is OTC or exchange traded. For example, OTC options may be customised with specific strike prices and contract months and there may not be any price or position limits.

Table 4.9 – Example of an Option Contract Specification and the Terms Explained

Option Term Sheet		Explanation
Ticker Symbol	CS – Calls	This is the contract's ticker symbol when it's traded on the exchange.
Contract Size	One MSCI Singapore index futures contract	This states the underlying asset. The investor then knows that this is an option on the Singapore equity index futures.
Contract Months	2 nearest serial months and June quarterly cycle (Mar, Jun, Sep, Dec)	This option has 6 expiration dates – Mar, Jun, Sep, Dec plus the 2 nearest serial months. For example, in June 2018, an option buyer will see options listed for Jun, Jul, Aug, Sep, Dec 2018 and Mar 2019. Jul & Aug 2018 represent the nearest 2 serial months outside the quarter.
Trading Hours	T Session 8.15am-8.30am (Opening) 8.30am-5.15pm	T Session is the equivalent of the regular trading session while T+1 is equivalent to extended trading session. This denotes the market trading hours for this particular contract.

Option Term Sheet		Explanation
	T+1 Session 6.00pm-6.15pm (Opening) 6.15pm-2.00am	Some exchanges may have a cancellation or matching period which in this case is 8.15am-8.30am and 6.00pm-6.15pm.
Trading Hours on Last Trading Day	8.30am-5.15pm	Trading hours for expiring contracts may be set slightly differently from non-expiring contracts.
Last Trading Day	2nd last business day of the expiring contract month	The last trading day for expiring contracts is usually set differently from contracts that are not yet expiring. In this example, the last trading day is the 2nd last business day rather than the usual last business day for non-expiring contracts.
Tick Size	0.1 index point = SGD 20	This shows the monetary value associated with each tick movement.
Strike Price Intervals	5 index points intervals	Exchange-traded options have pre-defined strike prices. OTC options will have strike prices as agreed by both counterparties.
Exercise Procedure	European style or American style	European options can only be exercised on the expiration date. American options can be exercised any time from issuance until expiration.
Settlement at Expiration	Cash settled	This states whether the contract is cash-settled or delivered. (In-the-money options will be automatically exercised at expiration).
Daily Price Limits	5% up or down	Exchange rules may halt or suspend trading when price limits are breached, e.g., in extremely volatile markets. These "circuit breakers" are meant to slow down price movements from cascading into a sharp market plunge or increase.
Position Limits	10,000 contracts net on same side of the market, and in all contract months combined	Exchange rules may specify the position limit that each investor can take. This limits the risk that the exchange must bear for each participating investor.

4.10 Options on Other Underlying Assets

4.10.1 Equity Index Options

Equity index options were first traded on the CBOE in 1983. These options allow investors to trade on general equity market (index) movements, just like individual equity options. An investor who is bullish on a market can buy a call on the equity index, whereas an investor who is bearish on the same market could either write the call on the equity index or purchase a put on the index.

Unlike equity options, which require physical delivery of the underlying stock upon exercise, the exercise of index options is settled in cash. The cash settlement is equal to the difference between the closing price of the index (on settlement date) and the strike price of the option multiplied by a predetermined multiplier. As an example, information on the S&P 500 index options is shown in Table 4.10.1.

Table 4.10.1 - S&P 500 Index Options

Product Name	Symbol	Underlying Value	Contract Multiplier	Exercise Style
S&P 500® Options	SPX	Full Value of S&P 500	$100	European
Mini-SPX Options	XSP	1/10th of S&P 500	$100	European

4.10.2 Bond Options

A bond option gives the option buyer the right but not the obligation to buy or sell a bond at a pre-agreed price on or before the option's expiration date. Unlike equity options, bond options are typically OTC-traded. Therefore, bond options prices are not readily available like on a public exchange and the pricing and valuation of bond options is more complex than that of equity options.

As the price of a bond approaches its par value towards maturity, the uncertainty of the bond's price disappears. Therefore, the expiration of a bond option is usually set significantly before the bond's maturity date. Holding long bond option positions allows the investor to participate in the bond market at his/her discretion instead of being compelled to hold long bond positions. Investors with large bond portfolios can use bond options to hedge against future interest rate movements. An investor typically buys a bond call option if he expects interest rates to fall, thereby causing the bond's price to increase. Conversely, an investor buys a bond put option if he expects interest rates to increase and bond prices to drop.

Example – Bond Options

Bank A buys a call option on US Treasury bonds from Bank B. The bond option's strike price is $102 while bond's current spot price is $101. Bank A pays an option premium to Bank B. At the option's maturity, Bank A either exercises the option if the spot price of the bond is above the option's strike price or lets the option expire.

4.10.3 Interest Rate Options

Market participants can hedge or take advantage of the expected direction of interest rate movements by buying options on interest rate futures. Unlike futures which represent an immediate contractual obligation, the option holder can decide if he wants to exercise the option. When exercised, the option holder takes on a long or short position on the futures contract – making it an obligation.

Such options allow borrowers who are worried about rising interest rates, to fix interest rates at a particular call strike price to avoid paying higher interest in the future. It also allows lenders or investors to fix interest rates at a particular put strike level, to protect earnings at a particular level when interest rates are falling.

Interest rate options are offered OTC as well as traded on futures and option exchanges. Interest rate options provide liquidity and flexibility to manage risk efficiently across the entire yield curve, which is particularly useful with expectations of interest rate movements.

Globally, one of the largest volumes of interest rate option trading occurs for US Treasuries and Eurodollar on the CME. Euribor futures and options are traded actively on ICE. In Singapore, options on the futures for 10-year Mini Japanese Government Bond and Euroyen (TIBOR) are traded on the SGX.

As with other types of options, prices of interest rate options are affected by factors like time to expiry, volatility, strike price relative the current price of the underlying asset (e.g., the particular treasury interest rate, or particular interbank offer rate such as USD LIBOR).

4.10.3.1 Options on Eurodollar Futures

One of the most actively traded exchange-listed interest rate option contracts in the world is the option on Eurodollar futures. The liquidity of Eurodollar options offers traders and hedgers an opportunity to take advantage of their views on the direction of LIBOR rates.

The price of the Eurodollar futures contract is an index value constructed as 100 minus the yield expressed as a percent. The reason for this arrangement is that a long position (a purchased contract) gains as the index rises, implying that the add-on yield (LIBOR) falls. The long position gains as the underlying index rises as LIBOR falls. Conversely, a short position gains as the index falls and LIBOR rises.

The settlement for Eurodollar futures options follows the cash settlement procedure adopted for the underlying Eurodollar futures contracts.

For the futures contracts, each tick or price movement = USD 1 million x 1/10,000 x 90/360 = 25 basis points[26].

The options do not have to be issued with the same expiry as the futures contracts. For example, the following interest rate option is a Mid-Curve Eurodollar Option which allows the trading of an option expiring earlier (in this case it is 6 months ahead of its futures contract).

Example – Mid Curve Eurodollar Option	
Product	Eurodollar 1 Year Mid-Curve Option, traded on CME

[26] For the nearest expiring or "spot" month in Eurodollar futures and options (serial or quarterly) traded on the CME, the minimum price fluctuation is 1/4 of a basis point or a "1/4 tick," which is USD 6.25. For all other CME Eurodollar contracts, the minimum price fluctuation is 1/2 of one basis point, or a "1/2 tick," which is USD 12.50.
Source: https://www.infinityfutures.com/documents/cmeeurooptbrochure.pdf

Underlying Asset	December 2019, Quarterly Eurodollar Futures
USD 6-month LIBOR Rate	2.8635%
Contract Size and Face Value	USD 1 million
Options style	American Style (can be exercised anytime)
Value Per Basis Point	USD 1 million x 1/10,000 x 90/360 = 25 basis points
Option Expiration	June 2019

As at 16 November 2018, the Futures Price for Eurodollars maturing in December 2019 is 96.93. Strike prices are listed in intervals of 25 basis points (0.25).

Call and Put Prices as at 16 November 2018

Call Price	Strike Price	Put Price
51.50	9,637.50	4.00
42.00	9,650.00	6.75
33.25	9,662.50	10.50
26.25	9,675.00	15.75
20.25	9,687.50	22.50
15.50	9,700.00	29.75
12.00	9,712.50	38.75
9.25	9,725.00	48.25
7.25	9,737.50	58.75
5.75	9,750.00	69.75

It is Oct 2018. The U.S. Federal Reserve funds rate is 2.25% currently and 1-month LIBOR is currently trading at 2.30088%.

Source: CME quote at close of 18 October 2018.

Suppose you own a floating rate investment pegged to 6-month-LIBOR quoted at 2.8635%. You are worried that the U.S. Federal Reserve may delay the rate hike which may cause the floating rate to fall back, affecting LIBOR.

Assume you want to guarantee a return of 2.75%, a 50-basis point rise. The strike price for which you want to protect this investment = 1.00 – 0.0275 = 97.250. You would then have to pay a cost of 9.25 basis points to set up this hedge. So, the realised minimum return would be 2.7500 – 0.0925 = 2.6575%.

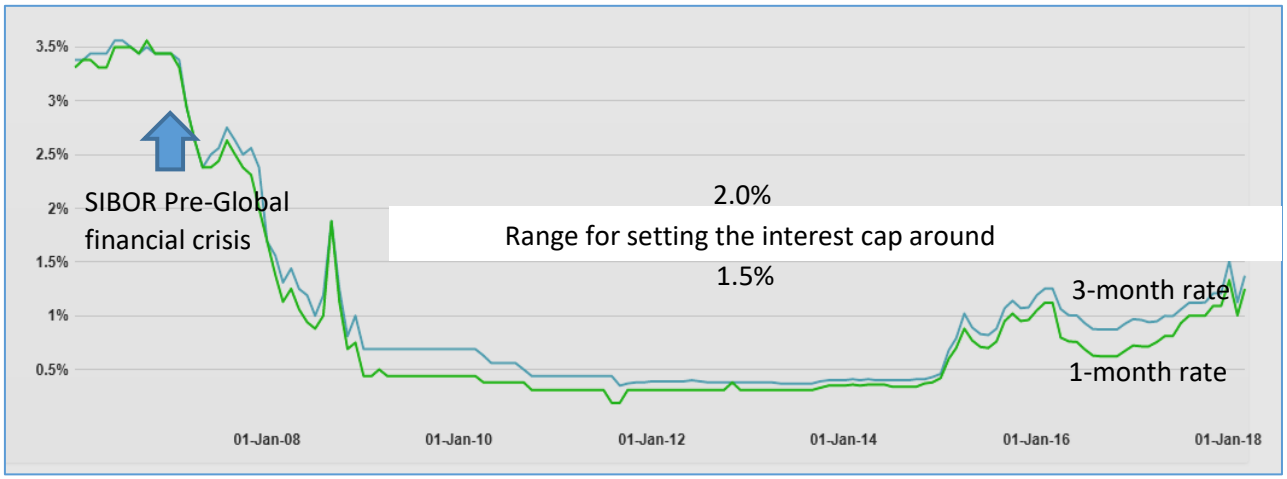
Likewise, if you are a borrower and you are worried that the interest rates will increase beyond the expected 50 basis point hike, then you would be buying a put option on the futures but note that the cost is higher at 48.25 basis points. The cost would be 2.7500 + 0.4825 = 3.2325%, the higher cost reflecting the greater probability of the interest rates rising.

4.10.3.2 Interest Rate Caps

Borrowers who want to set limits on variable interest rates to pay for their borrowings will often use interest rate caps. An interest rate cap is an OTC derivative that protects the holder from the underlying interest rate (e.g., LIBOR, SIBOR) rising above the specified cap rate. Payments may be made whenever the underlying interest exceeds the cap rate. The maturity of the interest rate cap is set up to match the underlying exposure to the interest rate increase.

Payment to the holder of the interest rate cap may be made on a monthly, quarterly, semi-annual or annual basis. Payment in each period will depend on the difference between the interest rate during the period and the cap rate. Interest rate caps involve buying call options. Please refer to Figure 4.10.3.2 for the historical 1-month and 3-month SIBOR rates for period of 2008 to 2018.

Figure 4.10.3.2 – Historical 1-Month and 3-Month SIBOR Rates (2008-2018)



Example – Interest Rate Cap

It is the end of March 2018 and let us assume that the SIBOR rate is at 1.375%. Further interest rate hikes by the U.S. Federal Reserve are expected to translate to higher interest rates and financing costs across different markets. In Singapore, SIBOR rates are expected to rise as well.

To fund a land purchase, a local property developer recently took on a 3-year SGD 200 million loan pegged to the 3-month SIBOR. Though he is already well-buffered for the further increases in the SIBOR rate in 2018, further rate hikes in the US in 2019 may move SIBOR well above 2.00%, so the property developer decided to buy a 3-year SGD 200 million notional interest rate cap linked to the 3-month SIBOR with a strike at 2.00% to account for more rate hikes in 2019 and 2020.

PAYMENT FROM INTEREST RATE CAP AT 2%

Quarter	$SIBOR_{3mth}$	STRIKE	spread	Interest
31-Mar-18	1.3750%	2.00%	0	-
30-Jun-18	1.3750%	2.00%	0	-
30-Sep-18	1.6250%	2.00%	0	-
31-Dec-18	1.8750%	2.00%	0	-
31-Mar-19	1.8750%	2.00%	0	-
30-Jun-19	2.1250%	2.00%	0.00125	62,500
30-Sep-19	2.3750%	2.00%	0.00375	187,500
31-Dec-19	2.3750%	2.00%	0.00375	187,500
31-Mar-20	2.1250%	2.00%	0.00125	62,500
30-Jun-20	2.1250%	2.00%	0.00125	62,500
30-Sep-20	2.1250%	2.00%	0.00125	62,500
31-Dec-20	2.3750%	2.00%	0.00375	187,500
31-Mar-21	2.3750%	2.00%	0.00375	187,500

Note: Assumed rate movement of 3-month SIBOR (not actual).

Once the interest rate cap is set up, if SIBOR exceeds 2.00%, there will be a payment from the interest rate cap in each quarter that will help to compensate for his outstanding loans interest payments.

If the 3-month SIBOR rises in the subsequent months in the manner reflected in the table (on the right), then the first payment from the interest rate cap will come on June 2019 when the 3-month SIBOR is at

2.125%, generating a payment of 90/360 x (2.125%-2.00%) x $200 million for the actual 90 days when SIBOR is above the cap rate. The calculation works out to be $62,500 received to cover the incremental interest above 2% for the quarter.

4.10.3.3 Interest Rate Floors

Interest rate floors provide protection in the opposite direction to interest rate caps, where investors with floating rate investments may desire protection from the reference floating rate (e.g., LIBOR) falling below a minimum desire rate of return (e.g., if LIBOR falls below the floor rate, a payment will be triggered). Such interest rate floors are used mainly by investors to protect their rate of return. Interest rate floors involve buying put options.

4.10.3.4 Interest Rate Collars

A borrower may be willing to sell an interest rate floor to collect the premium to fund the purchase of an interest rate cap. The resultant structure is called an interest rate collar. This limits the range of possible cash flows. The borrower using collar is buying a call option at one strike price partly funded by a put option at a lower strike price.

Likewise, a fund manager may be willing to sell a cap to give up some upside yield to fund the structure. The fund manager sells a call option to fund the purchase of put option. The interest rate floor protects the downside risk of the underlying interest rate falling below their desired floor rate. The setting up of collars will lower the costs of setting up the protection but it requires giving up potential gains.

4.10.4 Currency Options

Currency options are traded in all major derivatives exchanges in the world. Currency options with the highest volume and liquidity are for the developed market currencies such as the US dollar, European Euro, British pound, Swiss franc, Canadian dollar, Japanese yen, Australian dollar, and New Zealand dollar.

A currency option is a contract which grants the option buyer the right, but not the obligation to buy or sell a predetermined amount of foreign currency at a fixed exchange rate on or before a specified date. Currency options are structurally similar to currency futures contracts. A currency call option is similar to a long position in currency futures, whereas a currency put is similar to a short position in currency futures. Unlike other asset classes, the key difference between currency options and other types of options is the two-sided exposure that an investor takes on when they use a currency call or put option. Buying a USD/JPY call option is effectively buying a JPY/USD put option.

Currency options can be used for speculative purposes or to hedge an existing currency exposure. For example, corporations with foreign subsidiaries are likely to face currency exposure when repatriating profits back to their home country and currency options can be used to hedge against such risks. The easiest way to hedge an underlying currency position using options is to buy either a call to protect a short position or a put to protect a long position.

4.10.5 Options on Futures

Options can be priced off futures as well as cash instruments. Options on cash instruments are exercised to obtain a position in a cash asset, whereas options on futures are exercised to obtain a futures position.

Options on futures tend to enjoy greater liquidity than the options that require actual delivery of a cash instrument. For institutions undertaking cross-hedging, this is important. For example, if an institution needs to cross-hedge a non-deliverable bond by selling calls on T-bond futures, it may have to swap out of the original bond and into a deliverable bond to satisfy an exercise.

All option contracts on SGX are futures options and are American-style and can be exercised any time prior to expiry. If an option is in-the-money at expiry, it is SGX's policy to exercise the option unless there are specific instructions to the contrary.

The SGX Nifty Index futures and options are derivatives on the Indian equity market. The Nifty Index is a well-diversified market capitalisation weighted index comprising 50 large and highly liquid securities, representing about 70% of the free float market capitalization, on the National Stock Exchange (NSE) of India. The contracts are cash settled, with the final settlement price based on the on the average weighted prices of the individual component stocks of the index. The futures can also be settled by mutual offset with CME.

4.10.6 Swaptions

A swaption is an option on a swap. It gives the buyer the right to enter into a (receive or pay) swap agreement with a counterparty at a specified swap rate (otherwise known as the strike price). Majority of the swaption contracts traded are related to interest-rate swaps.

a. A buyer of a **receiver swaption** is buying an option to enter into an interest rate swap which allows the buyer to receive a pre-determined fixed rate and pay a floating rate over a specified period of time.

b. A buyer of a **payer swaption** is buying an option to enter into an interest rate swap which allows the buyer to pay a pre-determined fixed rate and receive a floating rate over a specified period of time.

For swaption contracts, the counterparties of such OTC transaction agree on:

i. The premium of the swaption contract to be paid by the buyer;

ii. The expiry of the swaption (typically a few days before the underlying swap contract starts).

iii. Details of the underlying swap such as:

- The notional amount applied (which should be similar to the underlying swap).
- The fixed rate (or strike price) and floating rate index for each leg in the swap; and
- The coupon payment frequency.
- The day count convention on each leg of the underlying swap.

4.11 Basic Option Strategies

Calls, puts and the underlying shares can be combined to create various trading and investment strategies.

4.11.1 Call Buying Strategies

Buying a call option is one of the simplest and most popular strategies used by option investors. A call option provides potential exposure to the underlying asset. Hence, investors buying call options are typically bullish

on a share or stock, to take advantage of leverage or to hedge against the possible increase in the underlying asset. The specific motivations vary for each investor, such as:

a. **Capital investment** – An uncovered long position in call options is an alternative to buying the underlying share. The price of the option is a fraction of the purchase price of the underlying share.

b. **Risk exposure** – The position has limited downside risk (which is the call option price if it is never executed or resold) while enjoying upside returns in tandem with the underlying share price.

c. **Cash extraction** – An option can also be used as part of a cash extraction strategy when the investor, who has the underlying asset and needs cash, sells it and buys call options to maintain upside exposure to the underlying asset; and

d. **Hedging** – An option can be used to hedge an outstanding short position in the underlying asset. For example, if an investor who has a long EUR/USD position, he buys Euros and sells US Dollars. If he is worried about a sudden dollar appreciation, he can buy a put on EUR/USD or a call on USD/EUR, which are effectively the same.

Hedging a Short on the Underlying Asset

Assume the following scenario:

- An investor has short sold the underlying stock at $12.00.
- At the same time, he bought a call with a strike price of $12.40.
- The premium paid for the call option was $0.80.

To hedge short on the underlying asset:

Sell stock	=	S_0	=	$12.00
Buy call (strike)	=	X	=	$12.40
Pay call premium	=	C_0	=	$0.80
Share price @ time (T)	=	S_T		(T = expiration date)

Profit = Max $\{0, S_T - X\} - S_T + S_0 - C_0$

Scenarios:

- Breakeven Point

$S_T = S_0 - c_0$

S_0	=	$12.00
C_0	=	$0.80
$S_T = S_0 - C_0$	=	$11.20

- Maximum Loss

If $S_T > X$, then loss = $S_0 - X - C_0$

Considering of the original short underlying the respective cash flows should be:-

1. (So – S_T)
2. –c, to pay for the call premium
3. Max (S_T – X, 0) if exercise

Max loss is, So – S_T – c + S_T – X, hence, 12 – 0.8 – 12.4 = (1.20)

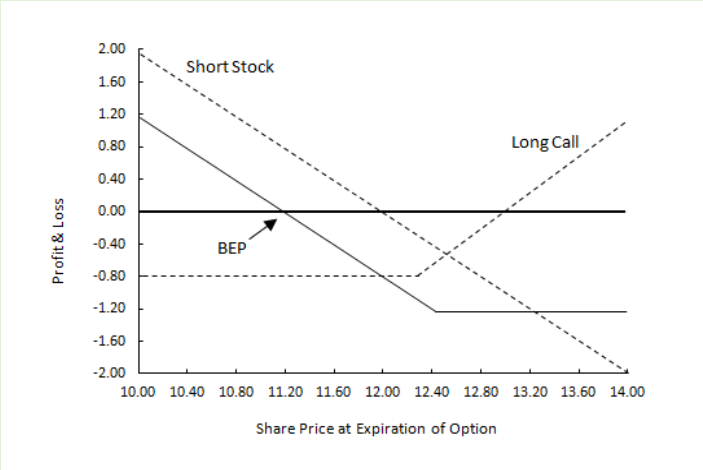
S_0	=	$12.00
X	=	$12.40
c_0	=	$ 0.80
$S_0 - X - c_0$	=	$(1.20)

- Maximum Gain **If S_T = 0, maximum gain would be $S_0 - S_T - C_0$ = \$11.20**

S_0	=	\$12.00
S_T	=	\$0.00
c_0	=	\$0.80
$S_0 - S_T - c_0$	=	\$11.20

Assume now S_T = \$8: $S_0 - S_T - c_0$ = \$3.20
Assume now S_T = \$0: $S_0 - S_T - c_0$ = \$11.20
Hence, Maximum Gain = Breakeven Point – S_T = \$11.20 when S_T = \$0

The graph below illustrates how hedging works:



Equity calls can also be combined with non-equity underlying securities. For example, in a convertible bond, a long call option is embedded with a fixed income instrument. The investor has the downside protection of the bond coupon and potential upside participation of the call option when the bond is converted to equity.

4.11.2 Call Writing Strategies

An investor who writes or sells a call is obliged to deliver the underlying stock to the call buyer at the strike price, when the option is exercised. The writer of the call option immediately receives the premium which is the price of the call as a cash inflow. The investor could be writing an uncovered or a covered call.

1. **Uncovered** – By writing uncovered (naked) calls, the investor would merely be extracting the option premium. However, he would risk facing unlimited losses if the underlying share's price moves up sharply during the life of the option. In an uncovered call, if the underlying asset price moves up, he would have to purchase the underlying asset from the market at a higher price in order to fulfil the delivery obligations.

2. **Covered** – A covered call position involves the simultaneous purchase of the underlying share with the sale of a call option. The call writer has covered or hedged his position by buying the underlying asset. For investors holding the underlying asset, writing a covered call is a yield enhancement strategy.

Example of a Covered Call

The investor bought the underlying stock at $12.00. At the same time, he sold a call with a strike price of $12.40. The premium received for the call option was $0.80.

To execute a covered call strategy:

- Buy stock = S_0 = $12.00
- Write call (strike) = X = $12.40
- Call price or premium = C_0 = $0.80

Share price @ time (T) = S_T *(T = expiration date)*

Profit = S_T – Max {0, S_T- X} -S_0 + C_0

Scenarios:

Breakeven Point: $S_T = S_0 – C_0$ = $12.00 - $0.80 = $11.20

Maximum Gain: If $S_T > X$, then Profit = $X – S_0 + C_0$ = $12.40 - $12.00 + $0.80 = $1.20

Maximum Loss: If $S_T \leq X$, then Profit = $S_T – S_0 + C_0$

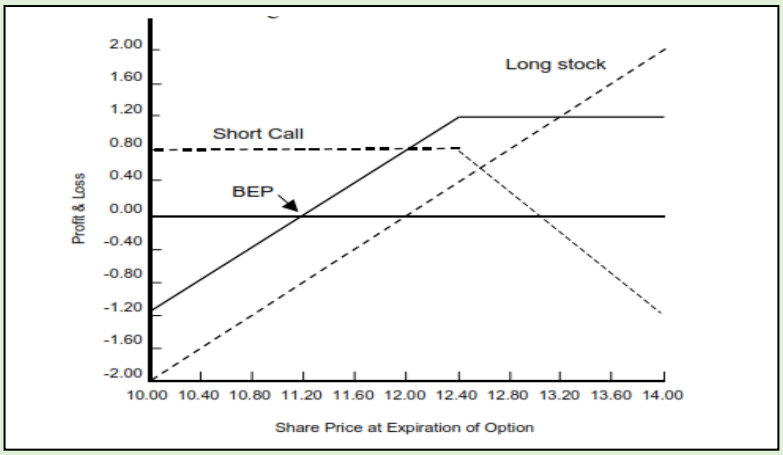
The maximum loss occurs when the price of the underlying asset owned falls to zero, i.e.; $S_T = 0$.
The call option sold is not exercised because the underlying has fallen in value.

Original value of the stock S_0 = $12.00 but final value S_T = 0

Seller of call option received C_0 = $0.80

Maximum Loss = $S_T – S_0 + C_0$ = 0 - $12.00 + $0.80 = - $11.20

Example of a Covered Call



4.11.3 Put Buying Strategies

Buying a put is a popular strategy used by investors who are bearish on the underlying share as it gives them the right, but not the obligation, to sell the underlying share at the strike price. An investor could buy a put to assume an uncovered position or for hedging.

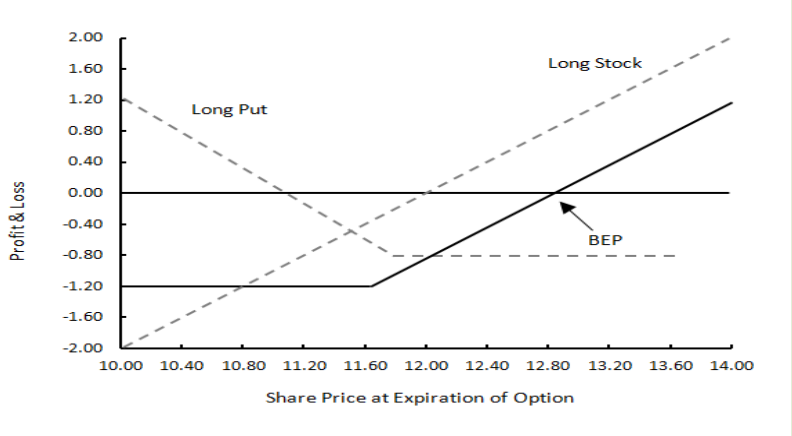
a. **Uncovered** – Buying an uncovered put is an alternative to shorting the underlying share. The holder will profit from a decline in the price of the underlying asset.

b. **Hedging** – Combining a long put with a long position in the underlying share is known as a **protective put** strategy or **portfolio insurance.** It limits the downside of the investor's portfolio, while still being able to enjoy the upside afforded by an increase in the underlying asset, by paying a fixed premium for the put.

Example of a Protective Put

The investor has bought the underlying share at $12.00 and bought a put with a strike price of $11.60. The premium paid for the put option was $0.80.

Example of a Protective Put



To execute a protective put strategy:

- Buy stock	=	S_0	=	$12.00
- Buy put (strike)	=	X	=	$11.60
- Pay premium	=	p_0	=	$0.80
Share price @ time (T)	=	S_T		*(T = expiration date)*

Profit = S_T + Max {0, X – S_T} -S_0- p_0

Scenarios:

Breakeven Point: **S_T = S_0 + P_0** = $12.00 + $0.80 = $12.80

Maximum Loss: If S_T = X, then **Profit = X – S_0 – P_0** = $11.60 - $12.00 - $ 0.80 = -$(1.20)

Maximum Gain: If S_T > X, then **Profit = S_T – S_0 – P_0 =** S_T - $12.00 - $0.80 (theoretically unlimited less $12.80)

4.11.4 Put Writing Strategies

By selling or writing a put, the seller gives the buyer the right (but not the obligation) to sell the underlying asset at the strike price, to the seller. The put writer receives a premium for writing the put, and earning a return is often the main motivation for using this strategy. The put writer may be in an uncovered position or have a "short" put embedded in a structured product.

a. **Uncovered** – Uncovered put writing increases income by the premium collected. Some investors may write puts at a strike price equal to their target buying price for gaining entry into the stock. For example, some financial institutions will offer their private clients the opportunity to be the seller of such put options to the bank as their counterparty. Clients usually earn a yield significantly better than what is available in the market for the risk they are taking. If the stocks drop to a certain level, they have to pick up the stock. Hence, such clients must be fully comfortable owning the underlying stock if the strike price is reached. In some cases, the price risk can be considerable as the investor is exposed to the downside volatility of the underlying stock. There are several variations to such options.

b. **Yield enhancement** – When combined with the purchase of a fixed rate note, put writing enhances the yield of the note. Structuring a fixed income instrument with an embedded short put would result in a **bull equity-linked note**.

The example below shows how put options are used in a structured product, with the prospect of offering the investor an enhanced return. This outcome will be realised if market conditions perform according to the investor's expectations, but he should also be aware and prepared if the market scenario turns out differently.

Bull Equity-Linked Note (ELN) (with embedded put option)

- Tenor: 1-month note
- Issue price: $9,900 (a 1% discount on a $10,000 denominated note)
- An equivalent note (without embedded short put) would yield 5% per annum
- Embedded put option:
  - Underlying stock = ABC Ltd
  - Current market share price = $10
  - Put strike price = $9 (90% of current price of $10)

Scenarios at Note Maturity:

Scenario 1: When $S_T \geq \$9.00$

The noteholder receives the full-face value of the instrument, $10,000. His return would be $10,000/$9,900 = 1.01% for 1 month. When quoted on an annualised basis this 1.01% would be equivalent to $(1.0101)^{12}-1$ = 0.128165 or 12.82% p.a.

ELN Specifications

Face value	=	$10,000
Issued at discount	=	$ (100)
Issue price	=	$9,900
Tenor	=	1 month
Interest rate (for 1 month)	=	1.01% (Note issued at 1% discount)
Rate of return (annualised)	=	12.82%

Scenario 2: When S_T < $9.00

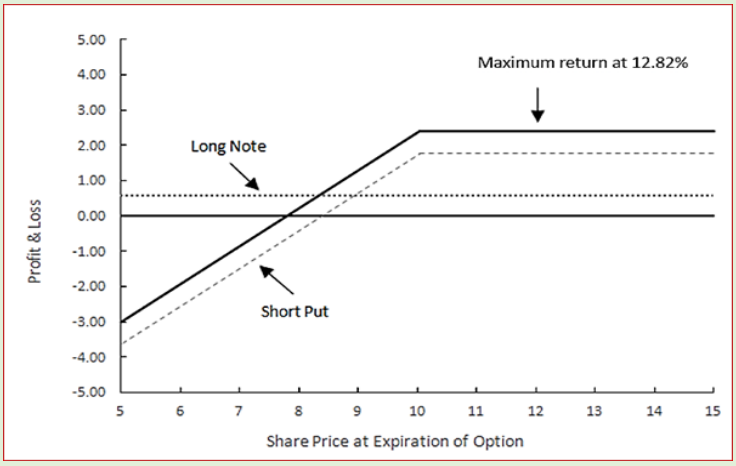
ABC's share price is less than the $9 strike price. The noteholder receives 1,111 shares, which is equivalent to the face value of note divided by the exercise price (i.e., $10,000/$9) within the settlement period for the underlying stock. Depending on the price of the underlying asset price in the future, the investor may suffer a loss.

Suppose ABC Ltd's share price falls to $8.80/share. What is the unrealised return of the ELN purchase?

Value of the shares multiplied by total number of shares = ($8.80 x 1,111) = $9,776.80
The return would be ($8.80 x 1,111)/9,900 -1 = -1.24% or a loss of 1.24%

However, the investor in buying this ELN, should be willing to hold to ABC shares purchased at the $9 price level and to hold it over time until the desired target price and return is reached.

Profit and Loss Payoff of Bull Equity-Linked Note



Comparing the ELN to Other Instruments:

In fixed income market, for a similar plain vanilla interest rate note, yield	=	5.00% p.a. (with no embedded put option)
Interest earned for this ELN	=	1.01%
Expected return for this ELN	=	12.82% p.a.
ELN higher returns from embedded put	=	7.82% p.a. (enhanced payoff from writing put)

4.12 Market Outlook Driven Option Strategies

4.12.1 Neutral Strategies

Neutral options trading strategies are used when the options trader does not have a view (or is uncertain) on whether the underlying share price will rise or fall from its current price levels. They are also called **non-directional strategies** as the potential to profit does not depend on whether the underlying share price will go up or down. The correct neutral strategy to use depends on the expected volatility of the underlying share price:

a. **Bullish on volatility** – The trader is uncertain about the direction but by adopting these neutral trading strategies he can profit when the underlying share price experiences big moves upwards or downwards.

b. **Bearish on volatility** – The trader does not expect any major movement. By adopting these neutral trading strategies, he can profit when the underlying share price moves in a very tight narrow range or does not move at all.

4.12.2 Bullish Strategies

Bullish options trading strategies are used when the trader expects the underlying share price to move up. To select the optimum trading strategy, it is necessary to assess how high the share price can go and the timeframe in which the rally will occur.

4.12.3 Bearish Strategies

Bearish options trading strategies are used when the trader expects the underlying share price to move down. To select the optimum trading strategy, it is necessary to assess how low the share price can go and the timeframe in which the fall will happen.

4.12.4 Creating Option Spreads

Another approach is to create a call spread or a put spread. An option spread is created by the simultaneous purchase and sale of options of the same class on the same underlying security, but with different strike prices and/or expiration dates. A spread can be constructed using:

a. Call options – referred to as **call spreads.**

b. Put options – referred to as **put spreads.**

If an option spread is designed to profit from a rise in the price of the underlying security, it is called a **bull spread.** Conversely, a **bear spread** is a spread where a favourable outcome is attained when the price of the underlying security goes down. Option buyers consider using spreads to reduce the net cost of entering a trade. A naked option seller can use spreads instead to lower margin requirements. This can also increase his buying power while simultaneously putting a cap on the maximum loss potential.

4.12.5 Straddle

Also known as a “long straddle” or “buy straddle”, this strategy involves simultaneously buying a put and a call of the same underlying stock, strike price and expiration date.

To construct a long straddle:

a. Buy 1 ATM Call; and

b. Buy 1 ATM Put.

Other features of straddles include:

i. **Upside → Unlimited profit potential.** By having long positions in both call and put options, straddles can achieve large profits regardless of the underlying share price direction, if the move is strong enough.

ii. **Downside → Limited risk.** The maximum loss for long straddles occurs when the share price on expiration date is trading at the strike price of the options bought. At this price, both options expire worthless, and the options trader loses the entire initial debit taken to enter the trade.

iii. **Market View → Neutral strategy.** Used when the options trader thinks that the underlying securities will experience significant volatility in the near term.

A Straddle for ABC Corporation

Current share price at T_0: S_o = $20 (September)

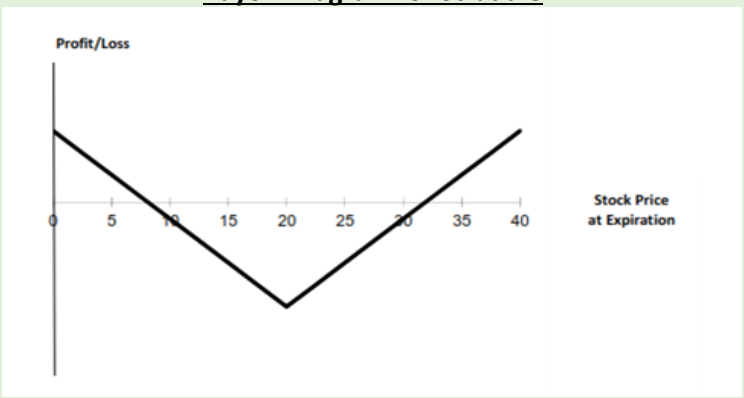
Strategy Execution:

- Long 1 ATM Call @ Strike = $20 (OCT 20)
- Long 1 ATM Put @ Strike = $20 (OCT 20)
- Cash outlay = Call premium + Put premium

Payoff at Expiration (at T_1)

If S_1 = $20	Both call and put options expire worthless Investment loss = Cash outlay
If S_1 > $20	Call option is in-the-money and put option expires worthless. Profit = S_1 – $20 – Cash outlay (Potentially unlimited upside)
If S_1 < $20	Put option is in-the-money and call option expires worthless. Profit = $20 – S_1 – Cash outlay (Maximum payout = $20 - $0)

Payoff Diagram for Straddle



4.12.6 Strangle

A strangle is also known as the "long strangle". It involves simultaneously buying a slightly out-of-the-money put and a slightly out-of-the-money call of the same underlying stock and expiration date.

To construct a long strangle:

a. Buy 1 OTM Call; and

b. Buy 1 OTM Put.

Other features of strangles include:

i. **Upside → Unlimited profit potential.** Large gains for the long strangle option strategy are attainable when the underlying share price makes a very strong move, either upwards or downwards.

ii. **Downside → Limited risk.** Maximum loss for the long strangle strategy is reached when the underlying stock price on expiration date is trading between the strike prices of the options bought. At this price, both options expire worthless, and the options trader loses the entire initial debit taken to enter the trade.

iii. **Market View → Neutral strategy.** Taken when the trader thinks that the underlying stock will experience significant volatility in the near term.

A Strangle for ABC Corporation

Current share price at T_0: S_o = \$20 (September)

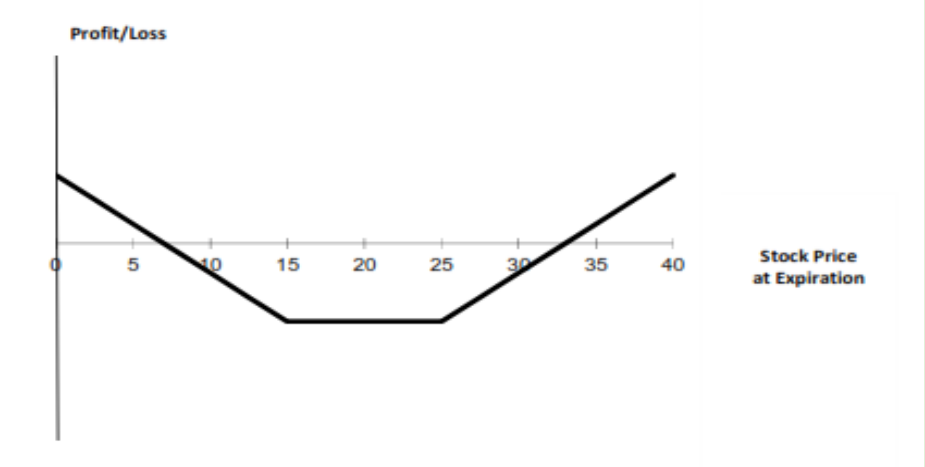
Strategy Execution:

- Long 1 OTM Call @ Strike = \$25 (OCT 25)
- Long 1 OTM Put @ Strike = \$15 (OCT 15)
- Cash outlay = Call premium + Put premium

Payoff at Expiration (at T_1)

If \$15 < S_1 < \$25	Both call and put options expire worthless Investment loss = Cash outlay
If S_1 > \$25	Call option is in-the-money and put option expires worthless Profit = S_1 – \$25 – Cash outlay (Potentially unlimited upside)
If S_1 < \$15	Put option is in-the-money and call option expires worthless Profit = \$15 – S_1 – Cash outlay (Maximum payoff = \$15 - \$0)

Payoff Diagram for a Strangle



A Short Strangle for ABC Corporation

Let us look at the reverse situation – a short strangle. The short strangle is an attractive strategy for investors who expect volatility in the market to decrease and for the underlying asset price to stay relatively range bound with moderate fluctuations within the band.

Suppose the investor / option writer sells the option on 1 Sep.

Strategy Execution:

- Short 1 OTM Call @ Strike = $25 (OCT 25)
- Short 1 OTM Put @ Strike = $15 (OCT 15)
- Profit Received = Call premium + Put premium

As long as the underlying asset price stays within the two strike prices (i.e., below $25 and above $15), the option writer gets to keep the full value of the premium. When the volatility is low, such a strategy would be attractive.

Let us suppose that the Call price is $2 and the Put price = $1. The initial combined premium value = $3.

If S_1 = \$14	• Put option is in-the-money and call option expires worthless. • Put holder has an incentive to exercise the put option with strike price \$15. • The put writer is obligated to accept the stock at \$15 and sell the stock in the market at \$14, realising a loss of \$1. • The call writer still has a profit of \$2 from the strategy. Actually, he could have entered the market once the price reached \$15 to buy a put to close his position so as to protect his position.
If S_1 = \$26	• Call option is in-the-money and put option expires worthless. • The call option holder exercises the call option. • For the call writer, there would be another \$1 lost since he has to buy the underlying asset in the market at \$26 to deliver the stock at \$25 to the call option holder. He would have entered the market to buy the stock once it crosses \$15 to close his position.

Refer to Chapter 4: Appendix (Non-Examinable Content) A4.7 Protecting and Re-Centering Options Spread Position Over Time to see how a trader can manage his/her position on a strangle.

4.12.7 Bull Call Option Spreads

A bull call spread is a vertical spread created by the simultaneous purchase and sale of calls on the same underlying security, but with different strike prices. It is constructed by buying an in-the-money call option while simultaneously writing a higher striking out-of-the-money call option of the same underlying security and the same expiration month.

To construct a bull call spread:

a. Buy 1 ITM Call (with a lower strike price); and

b. Sell 1 OTM Call (with a higher strike price).

Other features of bull call spreads include:

i. **Upside → Limited upside profits.** Maximum gain is reached for the bull call spread strategy when the share price moves above the higher strike price of the two calls. It is equal to the difference between the strike prices of the two call options minus the initial debit taken to enter the position.

ii. **Downside → Limited downside risk**. The bull call spread strategy will result in a loss if the share price declines at expiration. Maximum loss cannot be more than the initial debit taken to enter the spread position.

iii. **Market View → Moderately bullish market view**. The strategy is employed when the options trader thinks that the price of the underlying asset will go up moderately in the near term. By shorting the out-of-the-money call, the trader reduces the cost of establishing the bullish position but forgoes the chance of making a large profit in the event that the underlying asset price increases dramatically.

A Bull Call Spread for ABC Corporation

Current share price at T_0: S_o = \$21 (September)

Strategy Execution

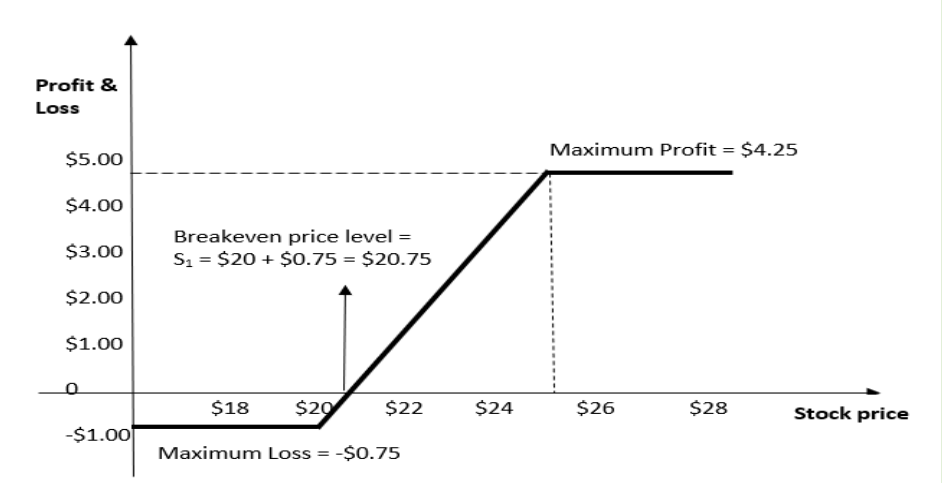
- Long 1 ITM OCT Call @ Strike = \$20, Call price C_1= \$1.50.
- Short 1 OTM OCT Call @ Strike = \$25, Call price C_2 = \$0.75.
- Cash outlay to set up the bull call spread = $C_2 – C_1$ = + \$0.75 - \$1.50 = -\$0.75

The Breakeven Price Level is when the Net Profit – Cash Outlay for set up = 0, i.e., (S_{BEP} - \$20) – \$0.75 = 0 and solving for the Breakeven Price S_{BEP} = \$20.75.

Payoff at Expiration (at T1)

If S_1 < \$20	Both call options expire worthless Investment loss = Cash outlay which is \$0.75 - \$1.50 = Loss of \$0.75
If \$20 < S_1 < \$25	The OCT \$20 call option is in-the-money while the OCT \$25 call option expires worthless Profit = S_1 – \$20 – Cash outlay Maximum profit = \$25 – \$20 – Cash outlay = \$5 – 0.75 = \$4.25
If S_1 > \$25	Both the call options expire in-the-money Investor gains on the long OCT \$20 call option but only up to \$25 since the call sold at strike of \$25 will be exercised. Profit = \$25 – \$20 – Initial outlay This is also the maximum profit possible = \$5 – 0.75 = \$4.25

Profit/Loss Diagram for a Bull Call Spread



4.12.8 Bull Put Option Spreads

The bull put spread is a vertical spread and is also known as the bull put credit spread (as a credit is received upon entering the trade). Bull put spreads can be implemented by selling a higher striking in-the-money put

option and buying a lower striking out-of-the-money put option on the same underlying stock with the same expiration date.

To construct a bull put spread:

a. Buy 1 OTM Put (with a lower strike price)

b. Sell 1 ITM Put (with a higher strike price)

i. **Upside → Limited upside profit.** If the share price closes above the higher strike price on expiration date, both options expire worthless. The strategy earns the maximum profit which is equal to the credit taken in when entering the position.

ii. **Downside → Limited downside risk.** If the share price drops below the lower strike price on expiration date, then the strategy incurs a maximum loss equal to the difference between the strike prices of the two puts minus the net credit received when putting on the trade.

iii. **Market View →** Strategy is employed when the options trader thinks that the price of the underlying asset will only go up moderately in the near term.

A Bull Put Spread for ABC Corporation

Current share price at T_0: S_o = \$24.00 (September)

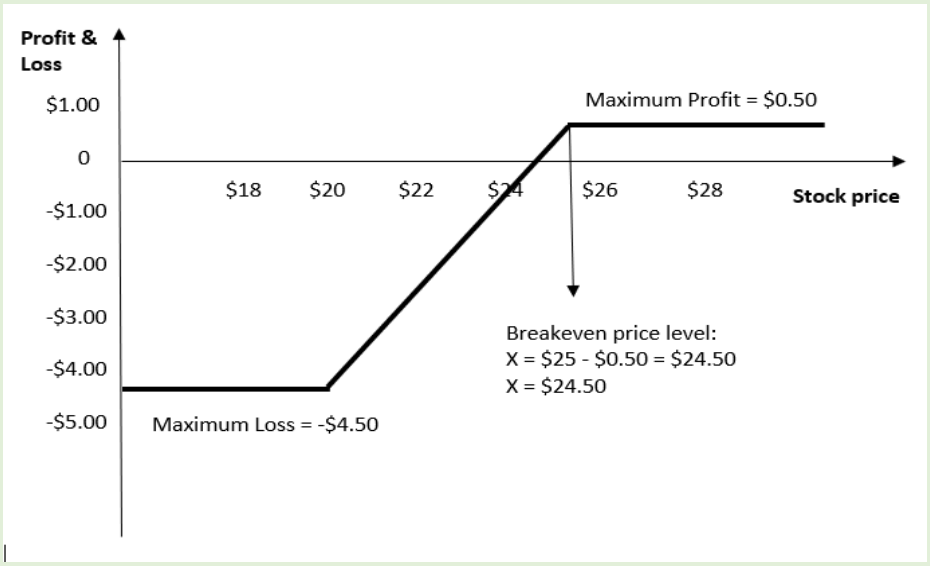
Strategy Execution

- Long 1 OCT OTM Put @ Strike X_1= \$20, Put price P_1 = \$0.75
- Short 1 OCT ITM Put @ Strike X_2 = \$25, Put price P_2 = \$1.25.
- Cash received = ITM Put price – OTM Put price i.e.; $P_2 – P_1$ = + \$1.25- \$0.75 = 0.50.

Payoff at Expiration (at T_1)

If S_1 > \$25	• Both put options are worthless. • Investment gain = Proceeds from bull put spread ($P_2 – P_1$) = \$0.50.
If \$20 < S_1 < \$25	• The OCT \$25 put option sold is in-the-money ($S_1 < X_2$ but remains above X_1) but the OCT \$20 put option expires worthless. • Loss = - (X_2 + S) + Proceeds of initial put spread. For example, if S_1 falls to \$23, then the payoff would be - \$2 + \$0.50 = - \$1.50. • Maximum loss = - (\$25 - \$20) + \$0.50 = - \$5 + \$0.50 = - \$4.50.
If S_1 < \$20	• Both put options expire in-the-money. • Maximum loss = - (\$25 - \$20) + \$0.50 = - \$5 + 0.50 = - \$4.50 (since you will also be able to exercise your put option with strike price at \$20).

Profit and Loss Diagram for a Bull Put Spread



Both the bull call and bull put spreads are **vertical spreads** as they combine long and short options with the same expiration dates and can be executed when investors expect a moderately bullish market. The difference in the two strategies is in the net cash position at the time they are executed. The bull call spread results in a net debit position, requiring a cash outlay, while a bull put spread yields a net credit position, giving the investor a positive cash flow when initiating the trade.

4.12.9 Bear Call Option Spreads

The bear call spread option strategy is also known as the bear call credit spread, as a credit is received upon entering the trade. Bear call spreads can be implemented by buying call options of a certain strike price and selling the same number of call options of lower strike price on the same underlying security expiring in the same month.

To construct a bear call spread:

a. Buy 1 OTM Call (with a higher strike price)

b. Sell 1 ITM Call (with a lower strike price)

Other features of bull call spread include:

i. **Upside → Limited downside profit.** The maximum gain attainable using the bear call spread strategy is the credit received upon entering the trade. To reach the maximum profit, the share price needs to close below the strike price of the lower striking call sold at expiration date where both options would expire worthless.

ii. **Downside → Limited upside risk.** If the share price rises above the strike price of the higher strike call at the expiration date, then the bear call spread strategy suffers a maximum loss equal to the difference in strike prices between the two options minus the original credit taken in when entering the position.

iii. **Market View →**The bear call spread trading strategy is employed when the options trader thinks that the price of the underlying asset will go down moderately in the near term.

A Bear Call Spread for ABC Corporation

Current share price at T_0: S_o = \$18 (September)

Strategy Execution

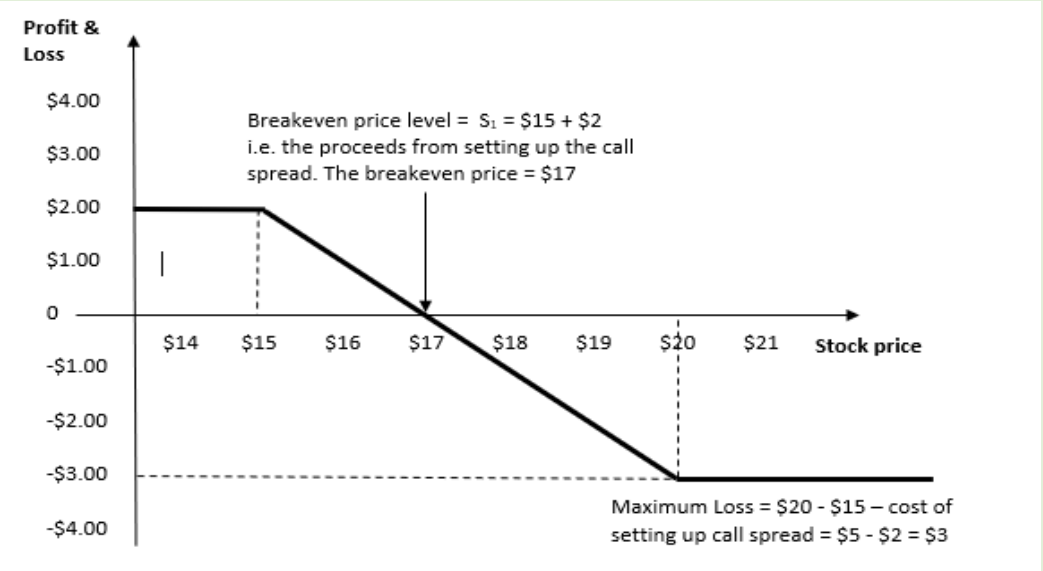
- Long 1 OTM Oct Call @ Strike = \$20, Call price C_1 = \$1.50
- Short 1 ITM Oct Call @ Strike = \$15, Call price C_2 = \$3.50

Cash received = ITM Call price – OTM Call price = $C_2 - C_1$ = \$3.50 - \$1.50 = \$2

Payoff at Expiration (at T_1)

If S_1 < \$15	• Both call options expire worthless. • Payoff = Cash received (based on our call premium assumptions, this amount would be \$2.00).
If \$15 < S_1 < \$20	• The OCT 15 call option is ITM, but the OCT \$20 call option expires worthless • Payoff/Loss = (S_1 – \$15) – Cash received. • Maximum gain/loss = (\$20 – \$15) – Cash received = \$5 - \$2 = \$3.
If S_1 > \$20	• Investor gains on the long OCT 20 call option but loses on the short OCT 15 call option. Maximum net loss is - \$3. • Loss = \$20 – \$15 – Cash received = \$5 – \$2 = \$3.

Payoff Diagram for Bear Call Spread



4.12.10 Bear Put Option Spreads

A bear put spread is a vertical spread which is also known as the bear put debit spread (as a debit is taken upon entering the trade). It is implemented by buying a higher striking in-the-money put option and selling a lower striking out-of-the-money put option of the same underlying security with the same expiration date.

To construct a bear put spread:

a. Buy 1 ITM Put (with a higher strike price)

b. Sell 1 OTM Put (with a lower strike price)

Other features of bear put spreads include:

i. **Upside → Limited downside profit.** To reach maximum profit, the share price needs to close below the strike price of the out-of-the-money put on the expiration date. Both options expire in-the-money but the higher strike put that was purchased will have higher intrinsic value than the lower strike put that was sold. Thus, maximum profit for the bear put spread strategy is equal to the difference in strike prices minus the debit taken when the position was entered.

ii. **Downside → Limited upside risk.** If the share price rises above the in-the-money put option strike price at the expiration date, then the bear put spread strategy suffers a maximum loss equal to the debit taken when putting on the trade.

iii. **Market View →** By shorting the out-of-the-money put, the options trader reduces the cost of establishing the bearish position but forgoes the chance of making a large profit in the event that the underlying asset price plummets.

A Bear Put Spread for ABC Corporation

Bear Put Spread
Current share price at T_0: S_o = \$18 (September)

Strategy Execution

- Long 1 Oct ITM Put, Strike @ \$20, Put price P_1 = \$3
- Short 1 Oct OTM Put, Strike @ \$15, Put price P_2 = \$1
- Cash outlay = ITM Put Bought – OTM Put Sold = - \$3 + \$1 = - \$2

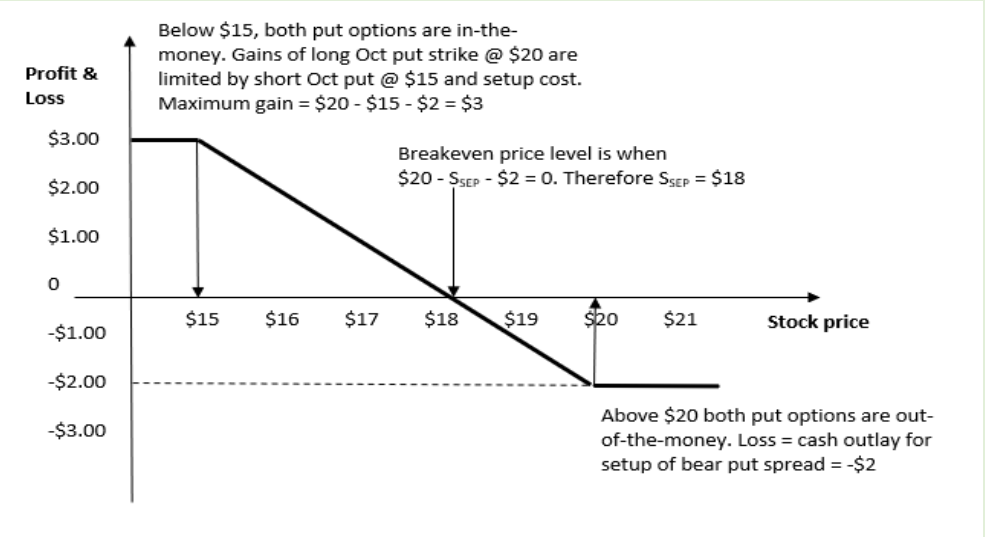
Payoff at Expiration (at T_1)

If S_1 > \$20	• Both put options expire worthless. • The loss incurred would be the cash outlay of setting up the bear put spread. • The loss would be the cash outlay = -\$3 + \$1 = -\$2 loss.
If \$15 < S_1 < \$20	• The Oct put option with strike @ \$20 is in-the-money, but the Oct put option with strike @\$15 expires worthless. • Payoff = \$20 – S_1 – Cash outlay. • Maximum gain = \$20 – \$15 – \$2 = \$3.

If S1 < $15

- Both the OCT put strike @$15 and the OCT put strike @ $20 are in-the-money.
- The gain from long Oct Put @$20 will be limited by the short Oct Put @$15 and set up cost.
- Maximum gain = $20 - $15 - $2 = $3.

Payoff Diagram for Bear Put Spread



Both the bear call and bear put spreads are vertical spreads which require the purchase and sale of the same type of option (either call/call or put/put) with the same expiration dates. These strategies can be used when investors expect the market to be mildly bearish. The difference in the two strategies is that at the time of executing the trade, the bear call spread places the investor in a net credit position while the bear put spread results in a net debit position, requiring a cash outlay.

4.12.11 Other Market Driven Strategies

We could continue to add various combinations of calls and put options to create different payoff outcomes given expected price trends. The previous sections described the various methods that can be used to capitalize on anticipated changes in the volatility of the underlying asset price. Short options are vehicles to profit from stable prices, time decay and falling volatility. Long options will bring profits when prices trend heavily over a short time period and volatility rises.

The butterfly and condor strategies are essential neutral strategies while the seagull (see Chapter 4: Appendix (Non-Examinable Content) A4.6 Seagull Strategy) is a mildly bearish strategy involving four or more call and put options providing additional payoff outcomes. The butterfly spread is a combination of a bull spread and a bear spread. It is a limited profit, limited risk options strategy.

Butterfly spreads are constructed using 4 options:

a. Buy 1 ITM option;

b. Sell 2 ATM options;

c. Buy 1 OTM option.

The table below summarizes the various strategies that can be adopted, depending on the direction of the market and the direction of the underlying asset's volatility:

	Bullish	Bearish	Undecided / Neutral
Volatility Up	Long Call	Long Put	Long Straddle or Strangle
Volatility Down	Short Put	Short Call	Short Straddle or Strangle
Volatility Neutral	Bull Spread	Bear Spread	Butterfly or Condor

4.13 Hedging Strategies with Options

Hedging enables market participants to transfer part, or all of the risk associated with holding a position in the underlying instrument from one party to another. Options can be used to act like a traditional insurance policy and were originally intended to function as such. Today, the market has evolved such that other traders – arbitrageurs, speculators and spreaders – outnumber the hedgers. Many hedgers come to the marketplace with either a natural or a chosen long or short position, and now wish to lay off part or all of the risk. As with insurance, there is a cost associated to hedging. The cost may be directly apparent (e.g., involves an initial cash outlay), or indirect (e.g., opportunity costs of potential profits in alternative situations or additional risk exposure) in some circumstances.

4.13.1 Using Options to Protect an Equity Portfolio

Buying protective puts (portfolio insurance) is the most obvious way to hedge a long portfolio. Once the investor selects the option (for a specific expiration date and strike price), the level of risk assumed, and the cost incurred (premium paid) is known. Writing covered calls is another simple and commonly used strategy by investors, by selling options against their long position. However, this strategy does not offer limited downside risk compared to buying protective puts. Income generated from selling the call options can mitigate a decline in the portfolio value, but the premiums received may not give much of a buffer when there is a sharp fall in the market.

Example – Covered Call

The stock of ABC Corporation is trading at $20, and the fund manager sells a call with a $30 strike for $1.50 premium. The fund will forego participation in the upside movement of the stock above $30. As long as the stock trades below $30, the fund collects a premium of $1.50 as the option would not be exercised.

If the manager wants to hold certain stocks in the portfolio, he can effectively lower the cost of these stocks by selling calls and collecting the option premiums. The main risk exposure is that if the market falls and he want to liquidate, the lower market price may not be adequately covered by the option premiums received.

Conversely if the share price trades higher above $30, it will be "called away" as the calls will be exercised. The fund will not enjoy the full upside of the price appreciation, or it will have to buy at a much higher price to add the stock back into its core holdings. Hence, this is a strategy that investors can use if their investment outlook is for a stable or range bound market.

4.13.2 Using Options to Manage Currency Exposure

The easiest way to hedge an underlying currency position using options is to buy either a call to protect a short position or a put to protect a long position.

Hedging Currency Risk for an Importer

A Japanese trading company has to pay USD 2 million for its imports in 3 months' time. Currently, the USD/JPY rate is at 100.00. The finance manager thinks that the USD/JPY rate will decrease within the next few months, to below 95.00. For protection in case her view is wrong, she will buy a USD call JPY put with strike at 105.00 and expiry in 3 months' time.

Hedging Currency Risk for an Exporter

A Japanese exporter has US dollar receipts, which he can hedge by buying a USD put / JPY call[27]. In the event the USD weakens against the JPY, he can exercise the USD put / JPY call option.

Current USD / JPY rate = 100.00 Premium for a 3-month USD put JPY call with strike at USD 1 = Yen 95.00.			
In 3 months' time:	**Scenario 1**	**Scenario 2**	**Scenario 3**
USD / JPY	100.00	110.00	90.00
Option Premium	JPY 1	JPY 1	JPY 1
Effective Exchange Rate	99.00	109.00	94.00

In scenario 1, he would sell USD / JPY at 100.00 minus JPY 1 for the option premium. It is similar for scenario 2. For scenario 3 where the USD / JPY is at 90.00, he will exercise his USD put option at 95.00 strike and minus the option premium of JPY 1, giving an effective rate of 94.00.

4.13.3 Zero-Cost Options

If a hedger wishes to assume a very limited risk exposure, he can do so by simply purchasing an option. If he also wants to avoid putting up his cash associated with such a strategy, it is possible for him to be cash neutral by simultaneously combining the purchase of the option with the sale of another option. A **collar** is established by buying a protective put while writing an out-of-the-money covered call. The call option is sold with a strike price where the premium received is equal to the premium paid for the put purchased, and this is called a **zero-cost collar** (costless collar).

Depending on the volatility of the underlying asset, the call strike can range from 30% to 70% out-of-money, enabling the writer of the call to still enjoy a limited profit if the share price increases.

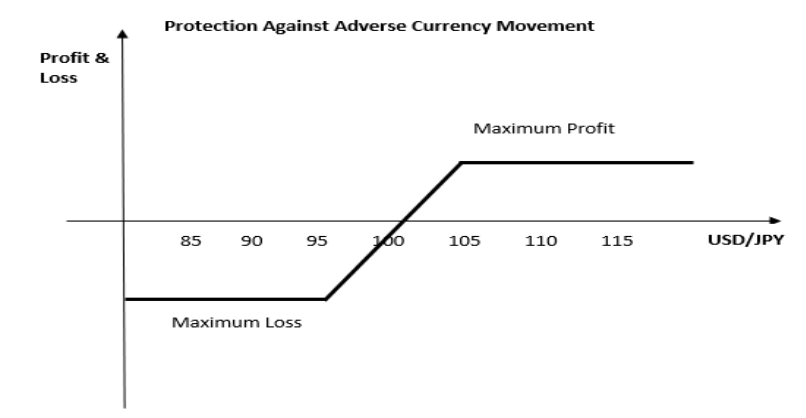
[27] Buying an option contract differs from engaging in a futures or forward contract. The exporter will engage in a futures / forward contract to sell USD against the JPY if his view is that the USD will drop against the JPY and book the exchange rate today. However, if he thinks that the USD will trade higher against the JPY, he will not do anything but wait for the USD/JPY rate to reach a higher level before selling.

Zero-Cost Collar Currency Hedge for an Exporter

Let's take the example of the same exporter earlier with USD/JPY at 100.00. If his view is that USD/JPY rate will go lower, he can buy a USD put / JPY call at 95.00 and simultaneously sell a USD call / JPY put at 105.00. His premium for buying a USD put at 95.00 can be reduced by the sale of the USD call at 105.00.

Alternatively, he can adjust the strike prices such that the premiums paid equals to the premiums received, thereby enjoying a zero-cost option strategy. If the market trades to below 95.00, he can exercise his USD put and be protected at 95.00. If the market trades above 105.00, he would have sold off his USD / JPY at 105.00 with the USD call option being exercised.

Zero-Cost Collar Currency Hedge



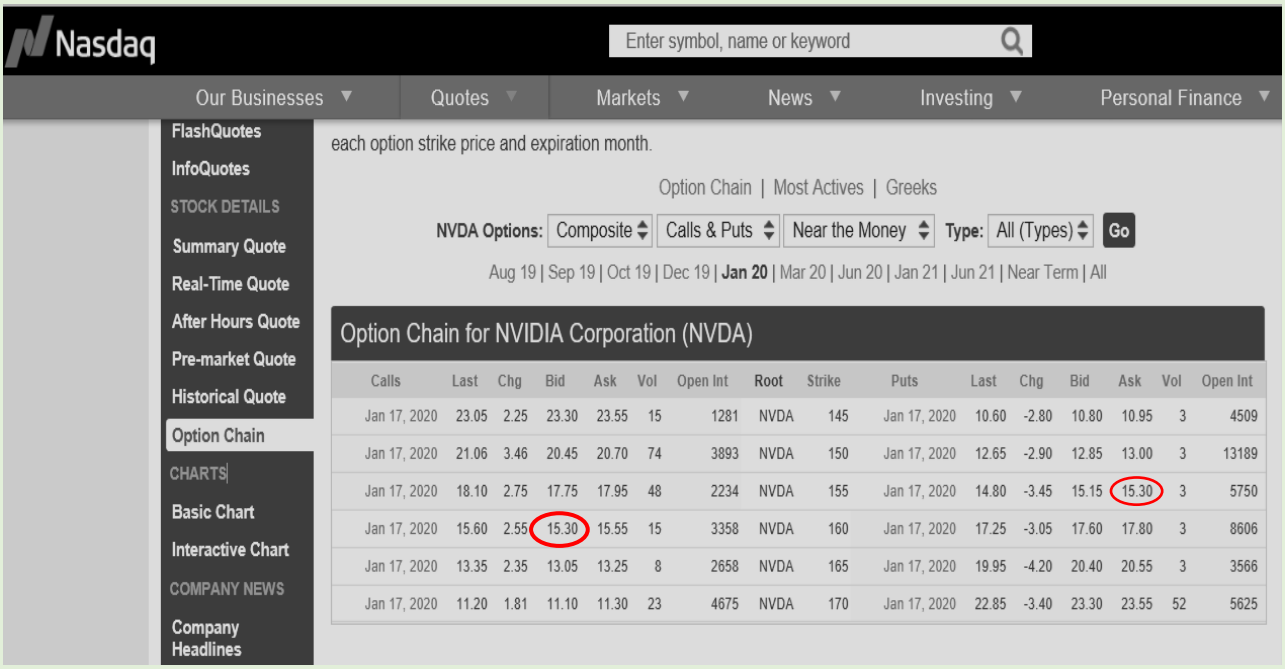
Zero cost or costless collars can also be established to fully protect existing long stock positions with little or no cost, since the premium paid for the protective puts is offset by the premiums received for writing the covered calls.

Costless Collar to Protect Profits from Underlying Shares

Assume that an investor bought shares of NVIDIA Corporation (NVDA) at the start of the year which have since risen by 18%, and he wanted to hold onto his gains. He decided to set up a costless collar to protect his profits. While he believed in the potential of NVDA stock for the longer term, he was worried that the US-China trade tensions may send the stock downwards. There could also be some performance challenges for the year.

As the investor was looking at the opening of the trading day, there was a new announcement that the US trade tariff for select electronic and high-tech equipment and parts will be postponed till December. This caused the prices of certain stocks to surge, including NDA shares, which immediately jumped by 3% to $156.00 (assume that prices are quoted in US dollars).

The existing quotes on the option chain on Nasdaq for NVDA are shown below:



Source: NASDAQ quotes as of 13 August 2019

Strategy Execution:

- Sell 1,000 OTM Call @ Strike = $160.00 (17 Jan 2020), C_1 = $15.30
- Buy 1,000 near ATM Put @ Strike = $155.00 (17 Jan 2020), P_1 = $15.30[28]

This option strategy:

- Has an outstanding long position 1,000 shares of NVDA = $156,000.
- Is essentially a combination of a protective put and a covered call strategy.
- Earns premiums for the covered call while the protective put provides downside price protection.

For a zero-cost collar (or costless collar), premium received for writing the covered calls offsets the premium paid for the protective puts. The cash outlay is zero.

The cash outlay to pay for the 1,000 put options bought ($15,300) is offset by $15,300 proceeds from the sale of the call options.

Payoff at Expiration (17 Jan 2020)

The proceeds of the call option strike at $160 (USD15.30 (bid price)) can be used to pay for the put option strike at $155 price of $15.30 (ask price).

[28] In this example, we assume that there are no dividends or interest. Therefore, the price of the OTM Call ≈ price of near ATM put.

If S_1 < \$155 (say it falls to \$125)

- Put option is in-the-money and the call option expires worthless.
- Payoff (from put option) = \$155– S_1 = \$155 –\$125 = \$30.
- Gains from the puts will offset most of the losses from the long stock position.
  - Loss of stock position = \$156,000 – \$125,000 = \$31,000.
  - Exercising put option = \$155,000 – \$125,000 = \$30,000.
  - The put was not exactly at-the-money so there is a loss of \$1,000. (If the put strike was at-the-money for the price paid, there would be no losses).

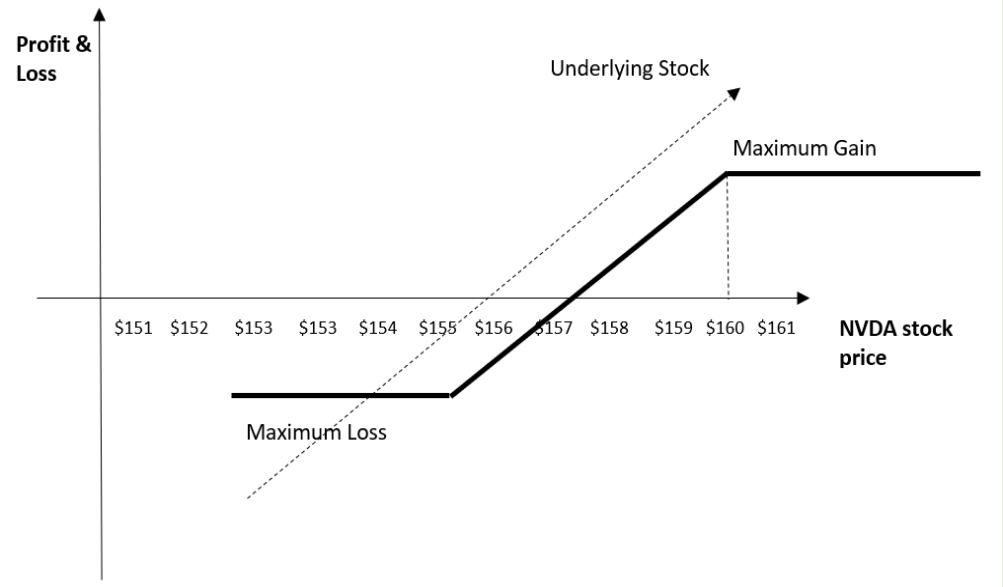
If \$155 < S_T < \$160

- Both call and put options expire worthless.
- S_T > put strike at \$155 so the investor would not exercise the put.
- S_T < call strike at \$160 so the investor would not exercise the call.
- Payoff from options = \$0.
- If the underlying share price rises from the current price of \$156 to just below the call strike at \$159.95, there are gains for the underlying shares.

If S_1 > \$160

- Call option sold is in-the-money and the put option expires worthless.
- The underlying stock will be delivered to the call buyer.

Payoff Diagram at Expiration



4.13.4 Summary of Options Strategies

Table 4.13.4 summarises how the basic options strategies are applied. Other strategies can be found in Chapter 4: Appendix (Non-Examinable Content) Appendix A4.

Table 4.13.4 - Summary of Basic Options Strategies

	YIELD ENHANCEMENT	BULLISH STRATEGIES	MARKET NEUTRAL	PROTECTIVE HEDGING	BEARISH STRATEGIES
Buying a call		✓			
Selling a call	✓		✓		✓
Buying a put					✓
Selling a put	✓	✓	✓		
Buying A Protective Put *(buy a put to cover bullish underlying long position)*		✓		✓	
Buying A Protective Call *(buy a call to cover short position on underlying)*				✓	✓
Writing A Covered Call *(sell a call and hold underlyir to deliver if executed)*	✓		✓		
Writing A Collar *(sell a call to fund put purchase while holding the underlying)*	✓		✓	✓	
Buying an ELN	✓	✓			
Bull-Call Spread *(moderately bullish)*		✓		✓	
Bull Put Spread *(moderately bullish)*		✓		✓	
Bear-Call Spread *(moderately bearish)*				✓	✓
Bear-Put Spread *(moderately bearish)*				✓	✓

	YIELD ENHANCEMENT	BULLISH STRATEGIES	MARKET NEUTRAL	PROTECTIVE HEDGING	BEARISH STRATEGIES
Long Straddle *(high volatility strategy)*			✓		
Long Strangle *(high volatility strategy)*			✓		
Short Strangle *(low volatility strategy)*	✓		✓		
Butterfly Spread *(mildly negative strategy)*			✓		
Condor Spread *(mildly negative strategy)*			✓		

4.14 Summary

1. An **option** is a contract between two parties, an **option holder** and an **option writer**. In the world of finance, options are derivative financial instruments on an underlying asset or another financial security.

2. Options are traded on a wide range of assets and securities, including equities, bonds, interest rates, currencies, commodities, futures and various financial indices. With product innovation and the globalisation of markets, there are now new categories of options, which include real estate, weather, market volatility and exchange traded funds.

3. Standard options are referred to as **plain vanilla options**, and the non-standard variety are called **exotic or specialty options.**

4. There are two types of options – **calls** and **puts**. An investor can be a buyer or seller of both types of options.

5. Call option buyers have the right but not obligation to buy the underlying at a contracted strike price, within a specific period of time. Conversely, the call option seller is obligated to deliver the underlying asset at the strike price if the buyer chooses to exercise the option.

6. Put option buyers have the right but not obligation to sell the underlying asset at the strike price. The put option seller is obligated to take delivery of the underlying asset at the strike price if the buyer chooses to exercise the option.

7. The key features of options are:

    i. **Strike price (or exercise price)** – The contracted price at which the buyer of a call option has the right to purchase the underlying asset, and the holder of a put option has the right to sell the underlying asset;

ii. **Expiration date** – Options will lapse as they have a finite life; and

iii. **Style** – European-style options can only be exercised on the expiration date while American-style options allow holders to exercise their right at any time before expiration.

8. As a derivative product, the value and trading dynamics of an option are dependent on the price of the underlying asset. An option's price consists of two parts: intrinsic value and time value.

9. **Option Price = Intrinsic Value + Time Value.** Intrinsic value is the difference between the market price and the strike price. This difference is the amount by which the option is in-the-money.
   - For a call: Intrinsic Value = Current Market Price – Option Strike Price.
   - For a put: Intrinsic Value = Option Strike Price – Current Market Price.

10. **Moneyness** refers to the potential profit or loss from the immediate exercise of an option. It describes the relationship between the strike price of an option and the current trading price of its underlying security. Terms such as *in-the-money (ITM), out-of-the-money (OTM)* and *at-the-money (ATM)*, describe the moneyness of options.

11. The time value of an option is generally positive. At expiration, the time value will be zero. The value of the option at expiration is exactly its intrinsic value. **Option Price = Intrinsic Value** (as Time Value → 0)

12. There are six major factors which affect the price of equity options:
    - Underlying share price;
    - Strike price (Exercise price);
    - Time to expiration;
    - Volatility of underlying stock;
    - Current risk-free interest rates; and
    - Expected dividends.

13. The breakeven, maximum gain and maximum loss positions of trading options and executing various strategies can be understood by looking at the four basic option positions:
    - Buying a call option (long a call option).
    - Selling a call option (short a call option).
    - Buying a put option (long a put option).
    - Selling a put option (short a put option).

14. **Put-call parity** is the relationship that exists between European call and put prices for the same underlying stock, strike price and expiration month. The put-call parity formula is : **C + PV (X) = P + S.**

    The put-call parity demonstrates two very important concepts. Firstly, it is always possible to replicate one of the investments with the other three. Secondly, it shows that options can be priced from a relative standpoint. This principle can be applied to create synthetic securities.

15. The Black-Scholes formula was the first widely used model for option pricing. The formula states that the value of an option is determined by six factors:
    - Underlying share price;

- Options strike price;
- Time until expiration;
- Implied volatility;
- Dividend status; and
- Interest rates.

16. By using pricing models, investors can compute and anticipate an option's premium under various scenarios. This helps them to better predict the outcome of their trading strategies for changes in interest rates, dividend distributions, volatility changes, or other factors that influence the option's actual value.

17. Understanding volatility and managing risk positions requires monitoring the various option "Greeks". Each type of volatility risk is represented by a Greek letter and the sensitivities are expressed as a change in a parameter as a function of a change in a second parameter:
    - **Delta** – Change in option price vs change in the price of the underlying asset.
    - **Gamma** – Change in delta vs change in the price of the underlying asset.
    - **Theta** – Change in option price vs change in time to maturity.
    - **Vega** – Change in option price vs change in volatility.
    - **Rho** – Change in option price vs change in interest rates.

18. Investors can execute the option trading strategies based on the market outlook:
    - **Neutral** strategies (also called non-directional strategies) – employed when investor does not have a view on whether the underlying share price will rise or fall from its current price levels. The potential to profit does not depend on whether the underlying share price will go upwards or downwards.
    - **Bullish** strategies – employed when investor expects the underlying share price to move upwards.
    - **Bearish** strategies – employed when the options trader expects the underlying share price to move downwards.

19. The **straddle**, also known as a "long straddle" or "buy straddle", involves simultaneously buying of a put and a call of the same underlying stock, strike price and expiration date. To construct a long straddle:
    - Buy 1 ATM Call;
    - Buy 1 ATM Put.

20. A **strangle**, also known as the "long strangle", involves simultaneously buying of a slightly out-of-the-money put and a slightly out-of-the-money call of the same underlying stock and expiration date. To construct a long strangle:
    - Buy 1 OTM Call;
    - Buy 1 OTM Put.

21. An option **spread** is created by the simultaneous purchase and sale of options of the same class on the same underlying security, but with different strike prices and/or expiration dates. A spread can be constructed using call options (referred to as call spreads) or put options (referred to as put spreads).

22. The three basic classes of spreads are:

- **Vertical spreads** – constructed using options of the same class, same underlying security, same expiration month, but at different strike prices.
- **Horizontal (or calendar spreads)** – constructed using options of the same underlying security, same strike prices but with different expiration dates.
- **Diagonal spreads** – created using options of the same underlying security but different strike prices and expiration dates (a diagonal spread is a combination of vertical and calendar spreads).

23. The **butterfly spread,** and the **condor spread** are neutral strategies while the **seagull strategy** is used when the outlook is mildly negative for the underlying. The butterfly spread is a combination of a bull spread and a bear spread. It is a limited profit, limited risk options strategy. Butterfly spreads are constructed using 4 options:
    - Buy 1 ITM option;
    - Sell 2 ATM options; and
    - Buy 1 OTM option.

24. Equity index options allow investors to trade on general equity market (index) movements, such as the S&P 500 index. Unlike individual equity options, which require physical delivery of the underlying stock upon exercise, the exercise of index options is settled in cash. The amount of cash settlement is equal to the difference between the closing price of the index (on settlement date) and the strike price of the option multiplied by a predetermined multiplier.

25. A bond option buyer has the right but not the obligation to buy or sell a bond at a pre-set price on or before the option's expiration date. Long bond options positions allow the investor to participate in the bond market at his discretion instead of being forced to hold long bond positions. Investors with large bond portfolios can use bond options to hedge against interest rate movements.

26. Interest rates options are related to bond options. However, these types of options focus on interest rates alone. An interest rate option gives the option buyer the right but not the obligation to make (call) or receive (put) a known interest rate payment.

27. Options on futures can be exercised to obtain a futures position. Options on futures tend to enjoy greater liquidity than those that require actual delivery of a cash instrument. For institutions undertaking cross-hedging, this is important.

28. A swaption is an option on a swap. It gives the right of the buyer to enter into a (buy or sell) swap agreement with a counterparty at a specified rate, given the strike price (usually set to zero) at a future date. Majority of the swaptions market is related to interest-rate swaps.

29. A currency or foreign exchange option is a contract which grants the option buyer the right, but not the obligation to buy or sell a predetermined amount of foreign currency at a fixed exchange rate on or before a specified date. Currency options are structurally similar to currency futures contracts. Currency options can be used for speculative purposes or to hedge an existing currency exposure.

30. Investors can use equity options as a hedging tool to protect their equity portfolio. Buying protective puts (portfolio insurance) is the most obvious way in which the investor selects the option (for a specific expiration date and strike price), the level of risk assumed, and the cost incurred (premium paid) is known. Writing covered calls is another strategy by investors, by selling options against their long position. However, this strategy does not offer limited downside risk compared to buying protective puts.

31. A collar is constructed by buying a protective put while writing an OTM covered call. If the call option is sold with a strike price where the premium received is equal to the premium paid for the put purchased, it is called a zero-cost collar (costless collar).

Chapter 4: Appendix (Non-Examinable Content)

A4.1 Butterfly Spread

The butterfly spread is a neutral strategy that is a combination of a bull spread and a bear spread. It is a limited profit, limited risk options strategy. Butterfly spreads are constructed using 4 options:

- Buy 1 ITM option;
- Sell 2 ATM options; and
- Buy 1 OTM option.

There are 3 strike prices involved in a butterfly spread and it can be constructed using calls or puts.

A4.2 Long Call Butterfly Spread

This spread is a neutral market strategy using call options. Using calls with 3 different strike prices, the long butterfly can be constructed by:

- Buying 1 lower striking ITM call;
- Writing 2 ATM call; and
- Buying another higher striking OTM call.

The main features of a long call butterfly spread include:

1. **Upside → Limited profit.** Maximum profit for the long butterfly spread is attained when the underlying share price remains unchanged at expiration. At this price, only the lower striking call expires in-the-money.

2. **Downside → Limited risk.** Maximum loss for the long butterfly spread is limited to the initial debit taken to enter the trade plus commissions.

3. **Market View →** Long butterfly spreads are entered when the investor thinks that the underlying stock will not rise or fall much by expiration.

Example A4.1: A Long Call Butterfly Spread for ABC Corporation

Current share price (at T_0): S0 = $20 (September)

Strategy Execution

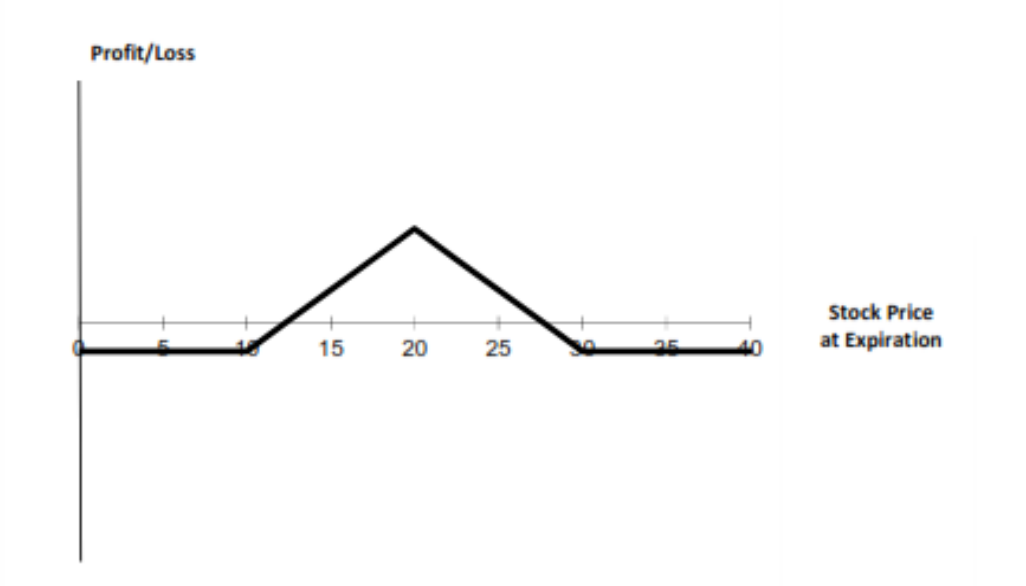
- Long 1 ITM Call @ Strike = $10 (OCT 10)
- Short 2 ATM Call @ Strike = $20 (OCT 20)
- Long 1 OTM Call @ Strike = $30 (OCT 30)

Cash outlay = ITM Call premium + OTM Call premium – ATM Call premiums

Payoff at Expiration (at T_1)

If S_1 < $10	All the call options expire worthless Investment loss = Cash outlay
If $10 < S1 < $20	The long OCT 10 call option expires in-the-money The short OC T20 call options expire worthless The long OCT 30 call option expires worthless Payoff = S_1- $10 – Cash outlay Maximum possible gain = $20 - $10 – Cash outlay
If $20 < S_1 < $30	The long OCT 10 call option expires in-the-money The short OCT 20 call options expire in-the-money The long OCT 30 call option expires worthless The gain from the long OCT 10 call is reduced by the loss from the short OCT 20 calls Payoff = [(S_1– $10) – 2 x ($S_1$– $20) – Cash outlay] Maximum possible loss = Cash outlay, which occurs when S_1 = $30
If S_1 > $30	All the call options expire in-the-money The positive payoffs from the 2 long calls are fully cancelled out by the negative losses of the 2 short calls Investment loss = Cash outlay

Payoff Diagram for a Long Call Butterfly Spread



A4.3 Long Put Butterfly Spread

This spread is a neutral market strategy using call and put options. Using puts with 3 different strike prices, the long butterfly can be constructed by:

- Buying 1 lower OTM put;
- Selling 2 ATM puts; and
- Buying 1 higher ITM put.

Other features of bull put butterfly spread include:

1. **Upside → Limited profit.** Maximum gain for the long-put butterfly is attained when the underlying share price remains unchanged at expiration. At this price, only the highest striking put expires in-the-money.

2. **Downside → Limited risk.** Maximum loss for the long-put butterfly is limited to the initial debit taken to enter the trade plus commissions.

3. **Market View** → Long butterfly spreads are entered when the investor thinks that the underlying stock will not rise or fall much by expiration.

Example A4.2: A Long Put Butterfly Spread for ABC Corporation

Current share price at T_0: S_0 = $25 (September)

Strategy Execution

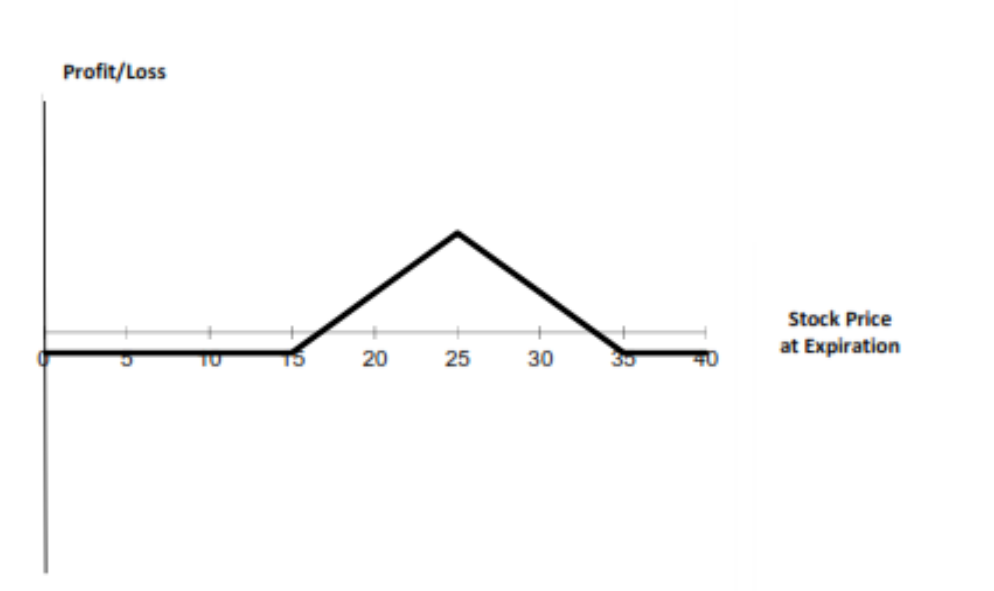
- Long 1 OTM Put @ Strike = $15 (OCT 15)
- Short 2 ATM Puts @ Strike = $25 (OCT 25)
- Long 1 ITM Put @ Strike = $35 (OCT 35)

Cash outlay = OTM Put premium + ITM Put premium – ATM Put premiums

Payoff at Expiration (at T_1)

If $S_1 < \$15$	• All the put options expire in-the-money • The positive payoffs from the 2 long puts are fully cancelled out by the negative losses of the 2 short puts • Investment loss = Cash outlay
If $\$15 < S_1 < \$25$	• The long OCT 15 put option expires worthless • The short OCT 25 put options expire in-the-money • The long OCT 35 put option expires in-the-money • Payoff = $[(\$35 – S_1) – 2 \times (\$25 – S_1) – \text{Cash outlay}]$ • Maximum possible gain = \$10 – Cash outlay, when share price = \$25
If $\$25 < S_1 < \$35$	• The long OCT 15 put option expires worthless • The short OCT 25 put options expire worthless • The long OCT 35 put option expires in-the-money • Payoff = $(\$35 – S_1) – \text{Cash outlay}$ • Maximum possible loss = Cash outlay, which occurs when $S_1 = \$35$
If $S_1 > \$35$	• All the put options expire worthless • Investment loss = Cash outlay

Payoff Diagram for Long Put Butterfly Spread



Restricted

Investors can take long butterfly call or put spread positions if they think the underlying stock will not rise or fall much by expiration. They are market neutral strategies, with a limited profit payoff based on taking limited risk. Both the long call and long put butterfly spreads will result in a debit position at the time of executing the trade, which is also the maximum amount of possible loss for the investor. Both the call and the put butterfly spreads involve 4 options, with 3 strike prices.

A4.3 Vertical, Horizontal & Diagonal Spreads

The three basic classes of spreads are the vertical spread, the horizontal spread and the diagonal spread. They are categorized by the relationships between the strike prices and expiration dates of the options involved.

i. **Vertical spreads** are constructed using options of the same class, same underlying security, same expiration month, but at different strike prices.

ii. **Horizontal (or calendar spreads)** are constructed using options of the same underlying security, same strike prices but with different expiration dates. It can be created by selling short-dated options and buying a long-dated one expressing a view that the volatility may increase, or vice-versa.

iii. **Diagonal spreads** are created using options of the same underlying security but different strike prices and expiration dates. A diagonal spread is a combination of vertical and calendar spreads.

A4.4 Condor Spread

This is a variation of the butterfly spread. It is similar in that it has 4 options, but all 4 have different strike prices (instead of only 3 different strike prices in a butterfly spread). The Condor Spread can be set up using either calls or puts (see Table A4.4).

Table A4.4 - Setting up Long or Short Condor Spreads

Long Condor Using Calls	Short Condor Using Calls
Buy a call with lowest strike price	Sell a call with lowest strike price
Sell a call with second lowest price	Buy a call with second lowest strike price
Sell a call with second highest price	Buy a call with second highest price
Buy a call with higher strike price	Sell a call with highest strike price

Long Condor Using Puts	Short Condor Using Puts
Buy a put with lowest strike price	Sell a put with lowest strike price
Sell a put with second lowest strike price	Buy a put with second lowest strike price
Sell a put with second highest strike price	But a put with second highest strike price
Buy a put with highest strike price	Sell a put with highest strike price

A4.5 Ratio Hedge

This is another variation of the above spreads. For a ratio hedge or ratio spread, the investor buys/sells option contracts in specified ratios. For example, in a 2:1 ratio spread, the investor will sell 2 options for every 1 long option (calls in a call ratio spread and puts in a put ratio spread).

Ratio hedges are a market neutral strategy where the investor expects little movement in the underlying share price in the near term. However, it is important to be aware of the risk-return profile of this strategy – the upside is limited but there is potentially unlimited risk on the downside, due to the net short position when the ratio position is established. There can be many different combinations of ratios, with different payoff profiles.

A4.6 Seagull Strategy

In using this strategy, the investor will have a moderately bearish view on the underlying asset and expects the volatility in trading to decrease. When taking such a position, the investor is expecting:

- Limited upside;
- Unlimited downside; and
- Zero or very low outlay required to establish the put option.

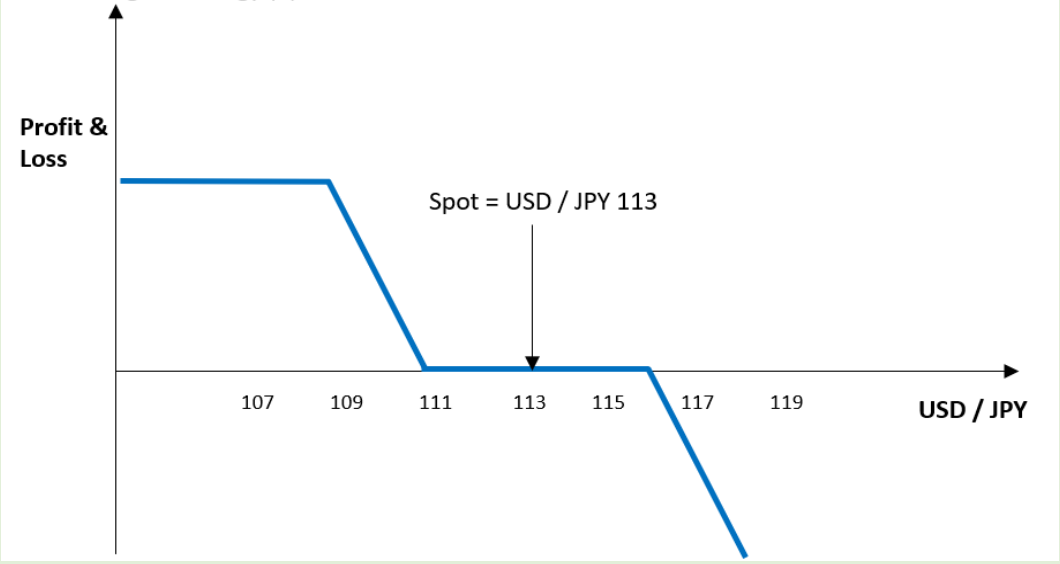
Example A4.6: Buying A Bearish Seagull

Current spot price for USD/JPY = 113

You have a moderately bearish view on the USD relative to JPY and have decided to establish a 3-month bearish seagull trade.

Strategy Execution

- Sell Strangle
  - Sell USD/JPY put option with strike (put strike) @ 109
  - Sell USD/JPY call with call strike @ 115
- Buy USD/JPY put strike @ 111
- Cash outlay = 0, ITM put option supported by OTM put and call strangle



Assume we are right, and the spot price falls to USD/JPY 110 after 1 month:

Option 1 – No more downside risk

(So, we can close out the position and take profit)

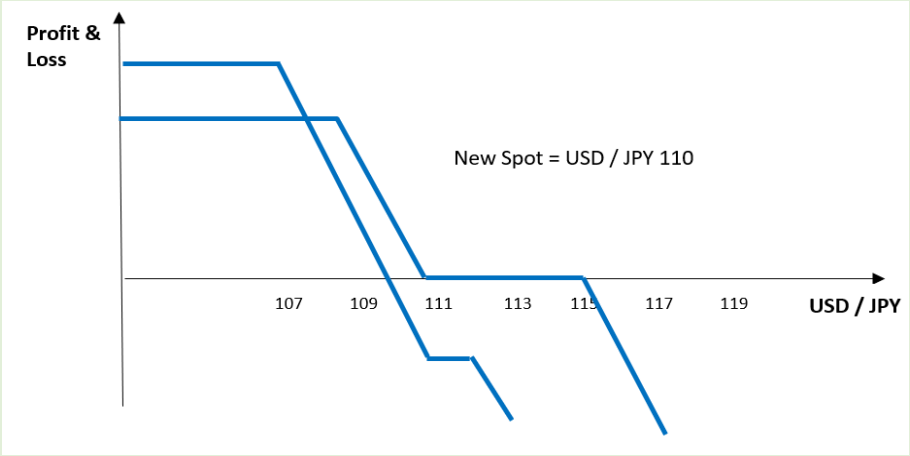
- Buy back Strangle:
  - USD/JPY call strike @ 115 is even cheaper since the USD/JPY call is more OTM.
  - USD/JPY put strike @ 109 is still OTM.
- Sell back USD/JPY put strike @ 111 which is well ITM.
- Closing out the positions should result in profits.

Option 2 – Further downside is expected so you re-establish the Seagull Strategy

- Keep the put option with strike @ 111 that was bought earlier.
- Close out and re-centre/re-establish Strangle position for 2 months:

  Close Strangle USD/JPY Call@115 and Put@109:

  - Buy back call @ strike 115
  - Buy back put @ strike 109
  - Cost: Pay 1.50% (based on the assumption of a 0.75% for buying the call and 0.75% for buying the put option)
- Re-establish Strangle: Sell OTM Call @112 and Sell OTM Put@ 107:
  - Sell call @ strike @ 112 (2 months)
  - Sell put @ strike @ 107 (2 months)
  - Proceeds: 1.35% (based on assumption of selling the call at 0.70% and 0.65% for selling the put option).
- Closing out the Strangle is likely to be costlier than proceeds received from new strangle but the net position will be small.
- Net Position = - 0.15% loss, but you are still holding the put strike @111 which is in-the-money to take advantage of further declines.



A4.7 Protecting and Re-Centering Options Spread Position Over Time

In actual practice when an options position is set up, a trader or investor will need to monitor the existing positions against the underlying asset and track the options available to protect profits or mitigate losses. The following is an example of how a short strangle position is being managed over a 3-month period and the price of the underlying asset moves.

Short Strangle*:* The short strangle is an attractive strategy for an investor who expects volatility to decrease and for the underlying asset price to stay relatively range bound with moderate fluctuations within the band. If the investor is managing the short strangle position over 3 months and is correct with regards to falling volatility, then he or she should expect the price (premiums) to decline, and they can protect their position.

A4.8 Summary of Options Strategies Covered in this Appendix

	YIELD ENHANCEMENT	BULLISH STRATEGIES	MARKET NEUTRAL	PROTECTIVE HEDGING	BEARISH STRATEGIES
Seagull Spread (moderately bearish)				✓	✓
Long Butterfly (low volatility strategy)			✓		
Short Butterfly (high volatility strategy)			✓		
Long Condor (low volatility strategy)			✓		
Short Condor (high volatility strategy)			✓		
Ratio Spread			✓		

Chapter 5: Structured Warrants and Daily Leveraged Certificates

Learning Objectives
The candidate should be able to: ✓ Understand the differences between company issued warrants and structured warrants. ✓ Explain the features, benefits and risks associated with structured warrants. ✓ Identify and compute the various elements in the valuation of structured warrants. ✓ Understand how adjustments for corporate actions are made for structured warrants. ✓ Identify various types of exotic warrants – index, currency translated, basket and yield-enhanced securities. ✓ Understand the uses, benefits and risks related to Daily Leveraged Certificates (DLCs) and responsibility of investment representatives towards clients with regards to trading this product.

5.1 Introduction

Besides futures and options, Specified Investment Products (SIPs) are products that have structures, features, and risks that may be more complex and include derivatives such as structured warrants, daily leveraged certificates (DLCs) and synthetic exchange traded funds (ETFs).

Under the MAS guidelines to enhance safeguards to retail investors, investment representatives must assess whether their clients or investors have the relevant knowledge and experience before they can invest in SIPs. Investors who wish to trade SIPs need to complete a Customer Account Review (CAR) with their respective broker or representative. The criteria used to qualify the investor in the customer account review consist of educational qualifications, work experience and investment experience. Investors will need to satisfy one of these three criteria to qualify in order to trade in SIPs.

5.2 Introduction to Structured Warrants

Structured warrants are financial instruments that are created by third parties, typically financial institutions, allowing investors to gain exposure to the performance of the underlying asset for a fraction of its price. Similar to options, structured warrants offer investors the potential to make a profit from bull market (call warrants) or bear market (put warrants) situations.

Like many other derivatives, the value of a warrant is influenced by the performance of the underlying assets or instruments it is linked to, which may be:

a. Individual stocks,
b. Equity indices,
c. Investment funds, or
d. Commodities.

Structured warrants have a short finite lifespan, usually less than a year (mostly 3-6 months). Like an option, the "structured" call warrant holder has the right to buy the underlying asset at a predetermined price, on or before the expiry date, depending on the exercise style of the warrant[29]. Some structured warrants are listed and traded on the SGX-ST trading platform[30]. Different types of structured warrants are designed for different investors who will have varying objectives and risk profiles. For example, a zero-strike warrant (essentially a synthetic stock) has an exercise price of zero, which means that it is always in-the-money (ITM). The cash settlement and warrant price are equal to the underlying security's closing price.

5.2.1 European vs American Style Call and Put Warrants

There are two main types of warrants – **European-style warrants** which can only be exercised at expiry, and **American-style warrants** that can be exercised throughout the life of the warrant. Warrants are, less popular in the US SGX-listed structured warrants follow the European-style warrants. For the purposes of the CM-SIP Examination on structured warrants, assume that you are always dealing with European-style warrants.

A European-style call warrant gives you the right to purchase the underlying security at a fixed strike price at the expiry date of the warrant. Hence if the price of the call warrant rises above its strike price at expiry, you earn the difference between the market price and the strike price. Investors that buy a call warrant would have a bullish view about the underlying asset. For the European-style call warrant, if the price of the underlying security rises towards the exercise price during the life of the call warrant, the value of the warrant will also rise, however there is no execution before the expiry. If the price is volatile and falls back below the call warrant exercise price, the value of the call warrant will fall, but there is no realization of the loss.

Conversely, a European style put warrant gives you the right to sell the underlying security at a fixed strike price at the expiry date of the warrant. If the put warrant's strike price is greater than the market price at expiry, you will make a profit from the difference. Buyers of put warrants will have a negative view about the underlying asset. At expiry, the settlement is made in cash and there is typically no delivery of the underlying asset. The settlement of structured warrants takes place according to the calculations below:

- **Profit For Call Warrants:** (Settlement Price of Underlying at Expiry – Exercise Price of Warrant) ÷ Conversion Ratio of Warrant.

[29] Warrants are usually call options. There are put warrants which give the holder the right to sell the underlying security at a predetermined price to the issuer.
[30] Structured warrants traded on SGX-ST can be found on https://www.sgx.com/securities/structured-warrant-screener.

- **Profit For Put Warrants:** (Exercise Price of Warrant – Settlement Price of Underlying at Expiry) ÷ Conversion Ratio of Warrant.

5.2.2 Structured Warrants vs Standard Company Warrants

In this Chapter, we will mainly be covering structured warrants. Standard company warrants, often issued as an incentive (sweetener) for bond issues are covered in Chapter 10 in the CM-EIP Examination. Structured warrants are different from company warrants in the following ways:

a. Structured warrants can be either "call" or "put" warrants, whereas company warrants are only call warrants. Hence structured warrants can be used in bull and bear market conditions.

b. Structured warrants have shorter durations, usually not more than 6 months.

c. Unlike company warrants, the exercise of structured warrants will not result in dilution of earnings in the company as no new shares will be issued. Structured warrants are cash-settled on maturity i.e., without the delivery of the underlying shares.

d. Structured warrants are issued by third parties and do not provide any new capital to the underlying companies; and

e. Designated market makers act as counterparties for structured warrants, hence investors in structured warrants are exposed to counterparty risks relating to the issuer and market-makers.

5.2.3 Benefits of Structured Warrants

While structured warrants are created by financial institutions with a profit motive in mind, they are created based on the market demand for alternative means of gaining exposure to the movement of asset prices. Below are some of the benefits of investing in structured warrants:

a. Call warrants provide leverage exposure to the upward movement of the underlying asset at a fraction of cost of the underlying asset itself.

b. Put warrants provide leverage exposure to downward movement of the underlying asset, which can offer portfolio managers some protection against downward movement of a portfolio.

c. Losses are limited to the value of the warrants purchased.

d. Structured warrants provide alternative means of gaining access and exposure to diverse markets, equities, baskets of equities, and select market indices without having to make huge investment commitments.

e. Since structured warrants are purchased with payment made upfront, there is no need for margin-calls situations (unlike options and **contracts for differences** (CFDs)).

Structured warrants are popular and actively traded in the stock exchanges of Singapore, Hong Kong, London and Mumbai. They are less popular in the US, where American options are the more popular instrument.

5.3 Key Features and Terms of a Warrant Issue

5.3.1 Conversion Ratio

Conversion ratio is the number of warrants needed to be exercised to buy or sell one unit of the underlying security. For example, if the conversion ratio to buy the shares of ABC is 5:1, the warrant holder needs 5 warrants to purchase 1 share.

5.3.2 Exercise or Strike Price

The conversion may also be expressed in terms of an exercise price (in addition/or instead of a ratio). The exercise price is the price at which the holder can buy or sell the underlying asset.

5.3.3 Expiry Date

The expiration date of a structured warrant is the last day on or before which the warrant has to be exercised.

5.3.4 Issue Price

The warrant issue price is the price at which the structured warrant is initially sold/issued by the issuer.

5.3.5 Physical Delivery or Cash Settlement

Warrants may be settled by physical delivery of shares or by cash. When warrants are exercised, call (put) warrant holders can expect to receive (or are expected to deliver) the underlying share (or cash) to issuer. The settlement style is made known when the warrant is first issued. Structured warrants listed on the Singapore Exchange Securities Trading Ltd (SGX-ST) are settled in cash.

5.3.6 Gearing

Structured warrants are usually priced at a fraction of the share price, allowing investors to trade more warrants than the underlying asset for the same investment outlay. Warrants therefore, offer benefits of gearing. A small gain in the underlying share price may lead to a larger gain in the call warrant price. Conversely, a fall in the underlying share price may lead to a larger loss in the warrant price.

Gearing indicates how many more warrants can be bought for a certain amount of capital, compared to buying the underlying share. For example, if the gearing of the warrant is 8x, the investor can have 8 times more underlying exposure than if they purchased the underlying with the same amount of capital.

5.4 Factors Affecting the Price of a Structured Warrant

The price of a structured warrant is the speculative value which investors are willing to pay for the warrant. It depends on a number of factors as discussed below.

5.4.1 Life of the Warrant

With all other things being equal, the longer the time before the warrants expire, the greater the warrant price. With more time, out-of-the-money (OTM) warrants may become attractive when the underlying asset prices moves over time. As the life of the warrant gets shorter, it will lose its speculative value. On the day before the warrants expire, they are only worth their intrinsic value. On the day after the warrants expire, they are worthless.

5.4.2 Performance of the Underlying Asset

Warrants are affected by the price of the underlying asset. Call warrants are worth more if the price of the underlying shares rises, and conversely, put warrants are worth more if the price of the underlying shares falls.

5.4.3 Exercise Price Level

The higher the exercise price for a call warrant, the less likely it will be exercised. Therefore, the call warrant will be worth less. The higher the exercise price for a put warrant the more likely it will be exercised, therefore the put warrant is more valuable with a higher exercise price.

5.4.4 Price Volatility of the Underlying Asset

The more volatile the price of the underlying asset (e.g., stock or index), the more likely the warrant will appreciate during a given time period. Therefore, investors are willing to pay larger premium for such a warrant.

The market will price the warrant based on the volatility of the underlying share. The term "implied volatility" refers to the market quotation for pricing the warrant. Buyers of warrants will be "long" volatility, and conversely sellers or writers of warrants are "short" volatility. Holding all other variables constant, if an investor purchases a warrant at an implied volatility of 7% and sells the warrant at an implied volatility of 8%, he will make a profit because the higher volatility will result in a higher warrant price. Conversely, a seller at 7% volatility will have a gain on his warrant position if volatility falls below 5%, all other things being equal.

5.4.5 Dividends

Although warrant holders are not entitled to dividends, the declaration of dividends will have a mild effect, with call warrants "losing out" (i.e., the call warrant's price becomes slightly lower) and put warrants "benefiting" (i.e., the put warrant's price becomes slightly higher).

5.4.6 Interest Rate

The higher the interest rate, the greater the hedging cost for call warrants written by the issuer, who has to delta hedge by buying back the shares, resulting in a higher price for call warrants. Conversely, hedging put warrants will be cheaper because of higher interest rates, resulting in lower put warrant prices.

5.4.7 Leverage

Warrants with greater leverage opportunities (conversion ratio) than others are able to command higher prices. Table 5.4.7(a) summarises the factors that influence structured warrants.

Table 5.4.7(a) – Summary Table of Factors Influencing Structured Warrants

Factors	Call	Put
Life of the warrant ↑	↑	↑
Exercise price ↑	↓	↑
Underlying asset price ↑	↑	↓
Volatility ↑	↑	↑
Dividends expectations (mild effect) ↑	↓	↑
Leverage (conversion ratio) ↑	↑	↑
Interest rate ↑	↑	↓

Table 5.4.7(b) summarises the potential realised gain for an ITM structured warrant for an investor. An investor can choose to either exercise the ITM warrants or sell them earlier in the market before maturity. Note that the realised gain from exercising structured warrant for an investor will include transaction costs.

Table 5.4.7(b) – Potential Gains from In-The-Money Warrants

	Call Warrants	Put Warrants
Scenario	Underlying share price rises above the warrant exercise price	Underlying share price falls below the warrant exercise price
Realised Gain from selling the warrant	• Prevailing price call warrant is sold less: - Sales transaction cost - Purchase price for the call warrant - Transaction cost of purchase	• Prevailing price put warrant is sold less: - Sales transaction cost - Purchase price for the put warrant - Transaction cost of purchase
Realised Gain from exercising the warrant	• (Current share price sold – Call warrant exercise price) x Conversion ratio less: - Transaction price for purchase of call - Transaction price for sale of shares	• (Put warrant exercise price – Current share price bought) x Conversion ratio less: - Transaction price for purchase of put - Transaction price for purchase of shares

5.4.8 Naming Convention on the SGX for Structured Warrants

Table 5.4.8 lists examples of actual structured warrants listed on SGX.

Table 5.4.8 – Examples of Structured Warrants Listed on SGX (as at April 2022)

										Market Information				Warrant Information				
SGX Stock Co	Security Name	Underlying	Type	Issuer	CCY	Exercise Pric	Underlyin	WPS	Expiry Date	Bid Vol ('0	Bid Price (	Ask Price (	Ask Vol ('0	Traded Vol	Delta (%	I.V. (%)	Premiur	E.G. (x)
9BXW	ALIBABA MB ECW220705	ALIBABA GROUP HOL	C	MB	HKD	125.00	96.55	45	07/5/2022	0	0	0	0	200.6	0	7.5312	29.467	0
HIMW	ALIBABA MB ECW221004	ALIBABA GROUP HOL	C	MB	HKD	100.00	96.55	25	10/4/2022	0	0	0	0	1201	0	0.6381	3.5733	0
YQOW	ALIBABA MB EPW220705	ALIBABA GROUP HOL	P	MB	HKD	105.00	96.55	45	07/5/2022	0	0	0	0	1700	-100	0	8.7519	0
HHKW	ASCENDASREIT MBECW22	ASCENDAS REIT	C	MB	SGD	3.50	2.82	3.5	11/1/2022	0	0.013	0.016	0	0	16.556	28.481	25.727	10.261
XJ6W	BYD MB ECW221004	BYD COMPANY LIMIT	C	MB	HKD	240.00	236.60	45	10/4/2022	0	0	0	0	400	0	0.2501	1.437	0
ZJVW	BYD MB EPW221004	BYD COMPANY LIMIT	P	MB	HKD	150.00	236.60	45	10/4/2022	0	0	0	0	200	-0	8.9129	-36.602	0
IUEW	CAPITALANDINVMBECW22	CAPITALANDINVEST	C	MB	SGD	4.20	4.12	5	10/28/2022	0	0.083	0.086	0	0	52.784	39.248	12.015	5.2402
MCAW	CITYDEV MB ECW220701	CITYDEV	C	MB	SGD	8.00	8.03	7.5	07/1/2022	0	0.084	0.088	0	300	54.666	48.221	7.472	6.9678
9A3W	COMFORTDEL MBECW221	COMFORTDELGRO	C	MB	SGD	1.88	1.50	2.5	10/14/2022	0	0.021	0.023	0	0	24.59	40.848	28.833	7.0257
9ZVW	DBS MB ECW220627	DBS	C	MB	SGD	35.00	34.05	15	06/27/2022	0.2	0.056	0.088	100	481.3	40.005	23.688	5.257	16.216
UKDW	DBS MB ECW220701	DBS	C	MB	SGD	32.00	34.05	15	07/1/2022	0	0.169	0.176	0	0	76.065	23.68	1.4244	10.217
ZOBW	DBS MB ECW221028	DBS	C	MB	SGD	33.50	34.05	15	10/28/2022	0	0.21	0.225	0	40	57.432	30.438	7.6358	6.2081
Y2BW	DBS MB EPW220627	DBS	P	MB	SGD	28.80	34.05	15	06/27/2022	0	0.008	0.012	0	0	-6.6249	30.154	-15.066	18.798
9SNW	DBS MB EPW220701	DBS	P	MB	SGD	31.00	34.05	15	07/1/2022	0	0.035	0.037	0	50	-20.666	31.255	-7.4156	13.403
MNTW	DBS MB EPW221215	DBS	P	MB	SGD	33.00	34.05	15	12/15/2022	0	0.205	0.3	1	140	-39.915	34.354	5.9471	4.4198
MWSW	GALAXY MB ECW220705	GALAXY ENTERTAINM	C	MB	HKD	64.58	45.80	9.93541	07/5/2022	0	0	0	0	0	0	10.018	41.004	0
8Q1W	GALAXY MB EPW220705	GALAXY ENTERTAINM	P	MB	HKD	39.74	45.80	19.87082	07/5/2022	0	0	0	0	200	-0	4.4581	-13.227	0
PZGW	GANFENGLIT MB ECW220	GANFENG LITHIUM CO	C	MB	HKD	138.00	89.80	65	07/5/2022	0	0	0	0	0	0	12.521	53.675	0
HVQW	GEELY MB ECW221004	GEELY AUTOMOBILE H	C	MB	HKD	18.00	12.48	6	10/4/2022	70	0.015	0	0	300	21.913	58.513	48.237	5.4695
Y9EW	GEELY MB EPW221004	GEELY AUTOMOBILE H	P	MB	HKD	10.00	12.48	6	10/4/2022	0	0	0	0	0	-0	4.2013	-19.872	0
OQTW	GENTINGSMBECW221028	GENTING SING	C	MB	SGD	1.05	0.81	3.5	10/28/2022	0	0.01	0.012	0	120	25.963	46.117	33.951	6.0085
VRTW	HSI 18200MBEPW220629	HANG SENG INDEX	P	MB	HKD	18,200	20,748	1200	06/29/2022	0	0.068	0.074	0	1200	-20.259	45.058	-10.096	9.2723
A89W	HSI 19000MBEPW220629	HANG SENG INDEX	P	MB	HKD	19,000	20,748	1200	06/29/2022	0	0.092	0.1	0	0	-26.856	42.179	-5.4688	9.0849
NPHW	HSI 19400MBEPW220530	HANG SENG INDEX	P	MB	HKD	19,400	20,748	1200	05/30/2022	13	0.045	0.05	120	40891.7	-23.488	37.488	-5.0511	16.244
DIYW	HSI 20200MBEPW220530	HANG SENG INDEX	P	MB	HKD	20,200	20,748	1200	05/30/2022	1	0.091	0.099	10	3168.6	-37.846	39.197	0.2828	12.943
W6EW	HSI 20400MBECW220629	HANG SENG INDEX	C	MB	HKD	20,400	20,748	1200	06/29/2022	0	0.17	0.199	0	22	58.2	29.436	3.7851	10.655
QQIW	HSI 21000MBEPW220530	HANG SENG INDEX	P	MB	HKD	21,000	20,748	1200	05/30/2022	0	0.141	0.145	0	0	-52.952	36.435	5.7451	11.688
WLTW	HSI 21200MBECW220530	HANG SENG INDEX	C	MB	HKD	21,200	20,748	1200	05/30/2022	43	0.085	0.091	30	4937.2	42.609	34.515	4.9097	15.601
NBQW	HSI 21400MBECW220629	HANG SENG INDEX	C	MB	HKD	21,400	20,748	1200	06/29/2022	10	0.106	0.12	0	50265	42.113	30.279	6.5484	12.365
PVDW	HSI 22000MBECW220530	HANG SENG INDEX	C	MB	HKD	22,000	20,748	1200	05/30/2022	30	0.051	0.057	15	14577	28.72	35.929	7.673	17.526
PBBW	HSI 22600MBEPW220530	HANG SENG INDEX	P	MB	HKD	22,600	20,7	1200	05/30/2022	0	0.3	0.33	0	0	-83.238	31.79	18.566	8.6351
Y4CW	HSI 22800MBECW220530	HANG SENG INDEX	C	MB	HKD	22,800	20,748	1200	05/30/2022	5	0.028	0.033	15.4	5162.7	17.872	36.417	10.79	19.864
OQNW	HSI 23000MBEPW220629	HANG SENG INDEX	P	MB	HKD	23,000	20,748	1200	06/29/2022	0	0.395	0.41	0	0	-75.324	34.835	23.546	5.9348

- 100% +

Source: SGX

5.5 Trading of Structured Warrants

5.5.1 Interpreting the Trading Name of Structured Warrants on SGX

On the SGX, to make trading easier, the names of structured warrants have been coded. Let's take two examples from Table 5.4.8, i.e., DBS MB ECW220627 and HSI 21000MBEPW220530.

a. **Underlying Instrument, DBS and HSI** – Refers to the underlying assets. If the underlying asset is an index, as in the case of the Hang Seng Index (HSI), the warrant's exercise price (21,000) is indicated immediately after the underlying asset's name.

b. **Issuer (MB)** – The next two letters refer to the financial institution, e.g., Macquarie Bank.

c. **Exercise Style (e)** – European Style. If it is an American-style warrant, there is no prefix to the Type of Warrant.

d. **Type of Warrant (CW)** – Call Warrant. The notation for a put warrant will be "PW" and for a discount certificate it will be "DC".

e. **Expiration Dates** – For example "27 June 2022" 220627 & "30 May 2022" 220530 are written in "YYMMDD" convention.

The name coding on other exchanges will differ from those on the SGX.

5.5.2 Investing in Call and Put Warrants

Let us use an example of a structured call warrant (code: DBS MB ECW171002).

Investing in Call Warrants of DBS GP HLDGS (underlying code: D05)

An investor bought longer-dated structured call warrants in December 2016, expecting that the rise in interest rates will prove beneficial to financial stocks. He bought a call warrant issued by Macquarie Bank listed on the SGX.

The call warrant details are shown below:

DBS Grp Holdings	Share price	= $18.16 (@ 6 December 2016)
Exercise price	X	= $16.50
Initial warrant price	WP	= $0.325 (@ 6 December 2016)
Conversion ratio	N	= 8:1
Last Trade Date (Expiry Date)		= 25 September 2017 (2 October 2017 about 9 months)
Issuer		: Macquarie Bank Ltd

Initial Gearing Ratio	=	Share price / (Warrant price x N)
	=	$18.16 / ($0.325 x 8) = 6.98x

Price and Profit Details	3 Jan 2017 ↓ in Share price	18 Jan 2017 ↑ in Share price	2 Aug 2017 ↑ in Share price
Share price of DBS	$17.32	$18.20	$22.19
% change in DBS price	-4.63%	+0.22%	+22.19%
Warrant price (W_p)	$0.245	$0.305	$0.665
% change in warrant price	-24.62%	-6.15%	104.62%
Dollar gain (loss) for stock	-($0.84)	+ $0.04	+$4.03
Gain (loss) for each warrant given initial price $0.325	-$(0.08)	-$(0.02)	$0.34
Dollar gain (loss) for 8 warrants	-($0.64)	-($0.16)	+$2.72
Value of 1,000 DBS shares	$17,320	$18,200	$22,190
Value of 8,000 warrants (8000 x W_p)	$1,960	$2,440	$5,320

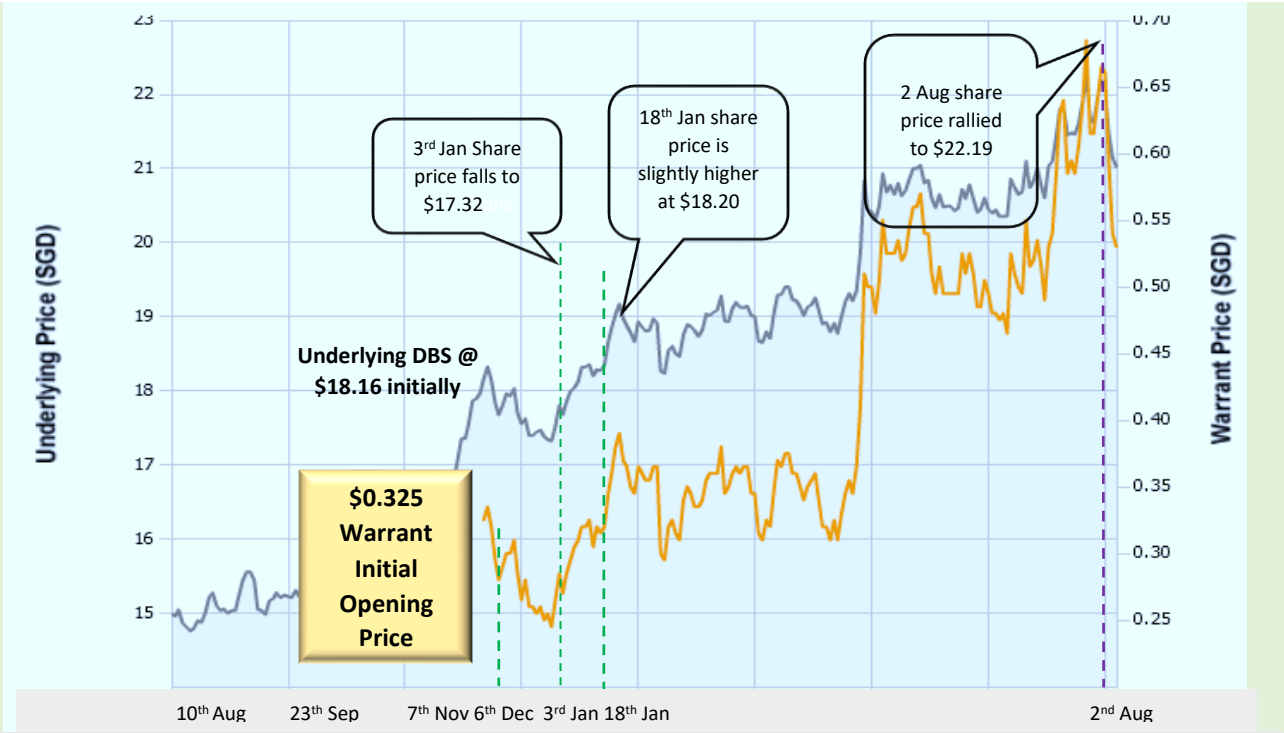


Chart Source: Macquarie Bank Ltd (Singapore Warrants)

A certain percentage change in the price of the underlying asset results in a larger percentage change in the warrant. When DBS' share price fell by 4.6%, the warrant price actually fell by 24.6%. The warrants investor could decide to realize the loss by selling the warrant on the market at $0.245 per warrant.

Note that although DBS' share price appreciated slightly to $18.20 on 18 August, the warrant price was still lower, partly due to the time value decay of the warrant and partly due to demand and supply for the day.

By 2 August, DBS' share price rose to $22.19 and the value of the warrant more than doubled to $0.665. Now the investor had two choices:

- He could realize his gains right now by selling his warrants on the market and secure a 105% gain on his structured warrant investment. So, if he bought 8,000 warrants for potential conversion into 1,000 DBS shares, he would obtain a gain of ($0.665 x 8,000) – ($0.325 x 8,000) = $5,320 - $2,600 = $2,720.
- He could also wait for another month or so until 25 September, expecting the DBS share price to stay at around the current level. Upon maturity, the cash settlement for the 8,000 warrants would be ($22.19 - $16.50) x 1,000 - $2,600 = $3,090. The investor could possibly realize a potentially higher return of 118.8%.

There is also the risk that the DBS share price corrects downwards towards maturity, and there is also the possibility of DBS shares reaching a higher price level by 25 September. If the share price falls below $21.82 by 25 September, the warrants investor would have a lower profit than if he had just sold the warrants outright on 2 August. If you look at the chart, the share price actually fell to around $21 by 8 August.

5.5.3 Investing in Put Warrants

An example of a put warrant is discussed below.

Investing in Put Warrants

Let's take another example, this time with the HSI. Suppose it is 9 August 2017 and the HSI stands at 27,854.9 points; an appreciation of 25.8% over 22,150.40 points at the start of the year. You believe that a significant correction is overdue, because of US-North Korea Nuclear tensions or further US interest rate hikes.

Yesterday you sold half of your Hong Kong equity portfolio which tracks the HSI, to lock in some gains for the year. For the other half of your portfolio, you have decided to buy some put warrants along with other derivatives to defend your Hong Kong stock portfolio returns.

You have other requirements for the put warrant, including:

- You would like the put strike price to be close to 27,000 and you will not buy anything below 26,000 since you think that the worst case would be a 7-10% decline from the current 27,854 level.
- You are looking for a 4-6-month HSI put warrant that is maturing some time in February 2018; and
- You do not mind if the warrant is listed on SGX or HKSE.

There are several banks offering put options and you have a choice of put warrants on the HSI traded on the SGX or Hong Kong Stock Exchange (HKSE). After screening the criteria, you have shortlisted the following put warrants from SGX and HKSE:

SGX Listed Put Warrants	Issue Date	Maturing	Strike Price (HKD)	Original Price	8 Aug Price	Issuer/Warrant Agent
HSI 26200 MB EPW170928 (CJUW)	20 Jul 2017	28 Sep 2017	26,200	0.133	0.049	Macquarie Capital Securities
HSI 26388 VT EPW170928 (CKDW)	04 Aug 2017	28 Sep 2017	26,388	0.076	0.062	Bank Vontobel AG/ The Central Depository Pte Ltd
HSI 27000 MB EPW170928 (CKIW)	07 Aug 2017	28 Sep 2017	27,000	0.122	0.108	Macquarie Capital Securities

HKSE Listed Put Warrants	Issued	Maturing	Strike Price (HKD)	Original Price	9 Aug Price	Warrant Agent
16444 UB-HSI @EP1712D	14 Jul 2017	28 Dec 2017	26,000	0.05	0.05	UBS Securities
28400 UB-HSI @EP1801C	08 Aug 2017	30 Jan 2018	26,400	0.124	0.122	UBS Securities
28404 UB-HSI @EP1802D	08 Aug 2017	27 Feb 2018	27,000	0.133	0.127	UBS Securities

HKSE Listed Put Warrants	Issued	Maturing	Strike Price (HKD)	Original Price	9 Aug Price	Warrant Agent
16621 SG-HSI @EP1712G	14 Jun 2017	28 Dec 2017	26,000	0.148	0.067	Societe Generale
27704 SG-HSI @EP1801E	03 Aug 2017	30 Jan 2018	26,400	0.120	0.097	Societe Generale
25339 SG-HSI @EP1802F	26 Jul 2017	27 Feb 2018	26,000	0.128	0.094	Societe Generale
27725 SG-HSI @EP1802H	03 Aug 2017	27 Feb 2018	27,000	0.129	0.108	Societe Generale
23689 JP-HSI @EP1712F	20 Jul 2017	28 Dec 2017	26,300	0.119	0.077	JP Morgan Chase
25685 JP-HSI @EP1802D	26 Jul 2017	27 Feb 2018	27,000	0.156	0.108	JP Morgan Chase
27646 HS-HSI @EP1802C	03 Aug 2017	27 Feb 2018	26,868	0.117	0.106	Hang Seng
27642 HS-HSI @EP1801D	03 Aug 2017	30 Jan 2018	26.368	0.120	0.100	Hang Seng
16649 GS-HSI @EP1712E	14 Jun 2017	12 Dec 2018	26,200	0.248	0.071	Goldman Sachs
24932 GS-HSI @EP1801D	25 Jul 2017	30 Jan 2018	26,500	0.115	0.097	Goldman Sachs
28585 GS-HSI @EP1802B	08 Aug 2017	27 Feb 2018	27,200	0.205	0.205	Goldman Sachs

Source: SGX, Macquarie Capital Securities, HKSE, Societe Generale

Out of the 4 put warrants at or above the strike level of 27,000 available as at 8-9 August (highlighted above), you find that the Macquarie warrant is too short dated for your purpose as it matures in September 2017. The rest of the warrants on the SGX on the date do not offer this strike level.

On the HKSE, both UBS and JP Morgan offer put warrants at the strike level of 27,000 maturing in February 2018, but JP Morgan's warrant seems to be less expensive, and you will have to look at the volatility and delta to see if there is any major difference. However, if the strike level is most important for you, you may decide to go with Goldman's put warrant which has the highest strike price at the 27,200.

Put warrants and the underlying asset price move in the opposite directions. Due to the large gearing effect, there is a larger percentage increase in the warrant price when the underlying asset price falls (and a similar percentage change for a price move in the other direction). When the asset price (e.g., share price or index price) remains flat, the warrant price declines because of time decay of the warrant.

Investing in Goldman Sachs Put Warrants of HSI (28585 GS-HSI @EP1802B)

Continuing from the example above, suppose that you've decided to invest in the Goldman Sachs put warrant (28585 GS-HSI @EP1802B) on this occasion. The details are given below:

Current HSI Level	S	=	27,757.09 (9 August)
Exercise price	X	=	27,200
Warrant price	WP	=	HKD 0.205
Conversion ratio	N	=	8,000 warrants per share
Last Trade Date (Expiry Date)		=	21 February 2018 (27 February 2018 - about 6 months)

Gearing ratio = Index price / (Warrant price x N)
= 27,757.09 / (HKD 0.205 x 8,000) = 16.93

Assumed Prices after 50 days	Scenario 1: ↓ in Share price	Scenario 2: Reaching breakeven	Scenario 3: ↑ in Share price
HSI level	25,814.09	24,981.38	29,422.51
Assumed % change in price	-7%	- 10%	+6%
28585 GS-HSI Put Warrant Price	HKD 0.31	HKD 0.36	HKD 0.11
Assume % change in Warrant Price	51.2%	+75.6%	-46.3%
Absolute fall of HSI from 27,757.09	1,943.0	2,775.71	- 1,665.43
Difference from strike price 27,200	-1,385.91	-2,218.62	2,222.51
HKD Gain on 8,000 warrants	+ HKD 840	+ HKD 1,240	HKD 760

Scenario 1

Assume a 7% fall in the Hang Seng Index from 27,757.09 to 25,814.09, the put warrant has appreciated by 51.2%, and there is a gain of HKD 840 (i.e., (HKD 0.31 – HKD 0.205) x 8,000 warrants). You now have a choice to realize the profit immediately or to hold.

If the index stays roughly at this same level until maturity, i.e., 7% down from the time you bought the put warrants, you can get a cash settlement of 27,200 - 25,814.09 = HKD 1,385.91. Given your initial outlay of HKD 1,640 (i.e., 8,000 x HKD 0.205), you will still have an overall loss of HKD 254.09.

Could you sell the warrant to some other buyer in the last few days of trading closer to maturity? Yes, but the warrant price will be falling over time for the same index level due to time decay as we move towards warrant expiry. Even if the index level is at 25,814.09 close to maturity, you will probably still make losses even if you sell the warrants because they would have fallen below cost closer to HK 0.12 – HKD 0.13 toward maturity due to time decay.

It is interesting to note that in this scenario, you could sell the put warrant on the exchange for HKD 0.31 to another investor instead of waiting for maturity and settlement. For that new investor to break even, the HSI would have to fall to 27,200 - (HKD 0.31 x 8,000) = 24,720 points, for him to breakeven at settlement.

Scenario 2

Since the put warrants cost HKD 0.205 each, for you to breakeven the index must fall below the strike price warrant value = 27,200 - (8,000 x HKD 0.205) = 25,560 (a 7.9% correction from the current HSI level) before you can make a profit from conversion.

Suppose after 50 days, the HSI corrects 10% to 24,981.38 and the put warrant value has risen to HKD 0.36. Assuming that prices stay at these levels until maturity, you stand to gain (HKD 2,218.62 - HKD 1,640) = HKD 578.62 and you will also have to hope that the HSI does not recover above the 25,560 breakeven point at maturity. It would make better sense to sell the put warrants in the market, for the profit of (HKD 0.36 - HKD 0.205) x 8,000 = HKD 1,240.

Scenario 3

Now suppose you are wrong, and the HSI is riding high on renewed belief in the Chinese economy and the HSI climbs a further 6% to 29,422.51 at the 50-day mark. Should you sell now or wait?

If you sell now, you are realizing a loss on the warrant of (-HKD 0.205 + HKD 0.11) x 8,000 = - HKD760 but at least you recoup HKD 0.11 per warrant. If you wait till maturity and the value continues to stay within this range, the warrant value will fall dramatically towards zero at expiry.

5.5.4 Delta

The delta of a warrant is the rate at which its price changes with respect to changes in the underlying asset's price. It is the price sensitivity of the warrant give the price movement of the underlying.

$$\textbf{Delta} = n \times \frac{\partial WP}{\partial S}$$

where:
Delta = Rate at which warrant price changes with change in share price
∂WP = Change in warrant price
∂S = Change in share price
n = Conversion ratio

Call deltas are always positive and put deltas are always negative. For example, at-the-money (ATM) warrants have a delta of around 0.5. This means that a 1 cent change in the underlying asset's price asset will result in a 0.5 cent change in the warrant's price. Warrants which are deeply ITM have a delta of close to 1, i.e., a 1 cent change in the underlying asset's value will result in a 1 cent change in the warrant value.

Delta is important when evaluating a warrant's strike price. Investors expecting a small move in the underlying asset price should buy warrants which are ATM or slightly ITM, while investors expecting a large move can buy OTM warrants. For example, when the release of a company's financial results is imminent, some investors hedging their long position on the underlying may decide to buy some cheaper OTM put options on the particular stock to offset potential negative reactions if the company results release fall below market expectations.

As useful as delta may be, it is only an estimate at a point in time in the life of a warrant. The value of delta will change given changing market conditions, so at best it is an estimate on the likely price movement of the warrant given the change in the underlying stock. Delta will also change with time decay.

Investing in Goldman Sachs Put Warrants of HSI *(28585 GS-HSI @EP1802B)*

In the case of our earlier example, it was found the delta for DBS Structured Call Warrants issued by Macquarie Capital (DBS MB eCW1710020) had a delta greater than 90% (or > 0.90). When there is speculative demand, there can be greater change in the warrant price than the change in the underlying stock price for very brief periods before rational thinking brings the sensitivity back below 1.

For the Goldman Sachs Put Warrants on the HSI (28585 GS-HSI @EP1802B) as at 9 August 2017, the delta was -45%. Most brokerage firms warrant agents and issuers will provide such information on their website. Some key statistics are shown below:

DBS MB eCW1710020 on 8 August	
Effective Gearing	6.65 x

28585 GS-HSI Put Warrants on 9 August	
Effective Gearing	7.58 x

Gearing	7x		**Gearing**	16.73x
Delta	>90 % (95%)		**Delta**	-45.33%
Moneyness	29% ITM		**Moneyness**	0.871% OTM

5.5.5 Effective Gearing

Effective gearing of a warrant is calculated by:

Effective gearing of a warrant = Delta x Gearing

Using the data of the 2 companies from the above example, the effective gearing of the call and put warrants are calculated:

Calculation of Effective Gearing

For DBS MB eCW1710020

Assuming that Delta	=	0.95 (or 95%)
Gearing ratio	=	7x
Effective Gearing	=	Delta x Gearing
	=	0.95 x 7 = 6.65 times

For 28585 GS-HSI @EP1802B

Assuming that Delta	=	-0.4533 (or 45.33%)
Gearing ratio	=	16.92x
Effective Gearing	=	Delta x Gearing
	=	-0.4533 x 16.92 = -7.67 times

5.5.6 Implied Volatility

Implied volatility is the market's consensus estimate or view regarding the warrant's underlying price volatility for the duration of the life of the warrant. Besides considering theoretical price factors, it also captures the prevailing sentiment and technical factors, such as the market demand and supply conditions, for the warrant.

Implied Volatility = *f* (share price, warrant price, strike price, maturity, interest rate and dividend yield)

Implied volatility can be used to compare the warrant value. In general, for similar warrants, the higher the implied volatility the more expensive the warrant.

5.6 Warrant Valuation

5.6.1 Intrinsic Value (IV)

This is the difference between the exercise price of a warrant and the market price of the underlying asset. Only in-the-money warrants have intrinsic value.

Call Warrant: IV = MAX {0, (S - X)/n}

Put Warrant: IV = MAX {0, (X - S)/n}

Where n is the conversion ratio.

5.6.2 Conversion Price (CP)

This is the total price which the investor pays by converting the warrants into the underlying security and is also known as the **breakeven price**.

Call Warrant: CP = X + nWP

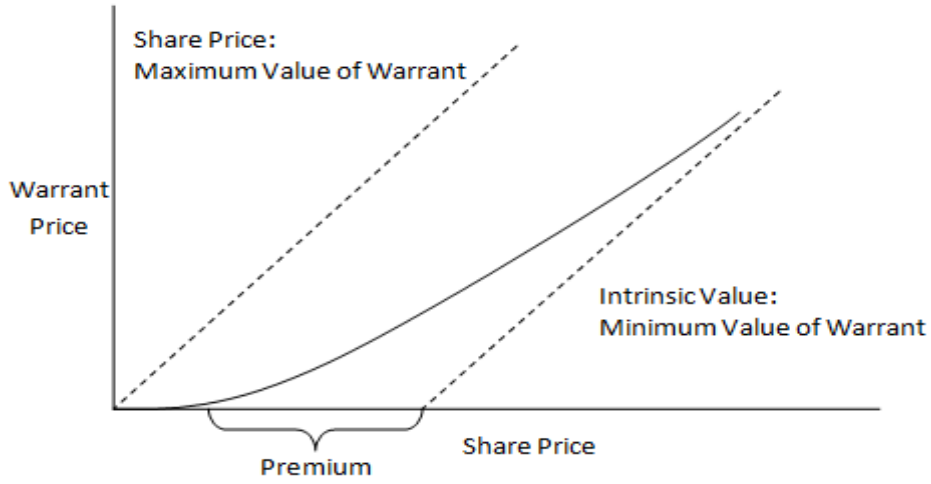
Put Warrant: CP = X – nWP

Where **n is the conversion ratio.**

5.6.3 Premium

Premium is the difference between the warrant price and intrinsic value and is different from option premium or option price. Warrant premium is mainly the time value of the warrant. (See Chapter 4 for the factors affecting time value of options). Figure 5.6.3 shows the profile of a call warrant's price, its minimum/intrinsic value, and its maximum/underlying value.

Figure 5.6.3 – Call Warrant Premium



Call and put warrant premiums are calculated using the following formula:

Call Warrant Premium ($) = nWP + X – S

Put Warrant Premium ($) = nWP - X + S

Premium is usually expressed as a percentage of the underlying share price. By definition, premium is the percentage by which the underlying share price needs to have moved at maturity for the investor to break even.

Hence, the premiums are calculated as follows:

Call Premium (%) = [(nWP + X - S)/ S] x 100

Put Premium (%) = [(nWP - X + S)/ S] x 100

Using the data from the 2 earlier examples, the intrinsic values, conversion prices and premiums of the call and put warrants are computed as follows:

Intrinsic Values, Conversion Prices and Premiums of Call and Put Warrants

For DBS MB eCW1710020 Call Warrants

DBS Group Holdings	Share price	=	$18.16 (on 8th Aug 2017)
Exercise price	X	=	$16.50
Initial warrant price	WP	=	$0.325 (initially)
Conversion ratio	N	=	8:1

Intrinsic Value (IV)
IV = MAX {0, (S-X)/n}
IV = MAX {0, ($18.16 - $16.50)/8} = $0.2075

Conversion Price (CP)
CP = X + nWP (assuming that the warrant price will fall from initial price closer to maturity)
CP = $16.50 + (8 x $0.325) = $19.10(Note that at actual maturity conversion CP = $16.50 since WP will be 0)

Premium (%)
Premium (%) = [(nWP + X – S) / S] x 100 = [(8 x $0.325) + $16.50 – $18.16] / $18.16] x 100 = 5.18%

For 28585 GS-HSI @EP1802B HSI Put Warrants

Assume HSI Level	S	=	24,981.38 (assuming Scenario 2 from previous example)
Exercise price	X	=	27,200
Warrant price	WP	=	$0.205
Conversion ratio	N	=	8,000 warrants per share

Intrinsic Value (IV)
IV = MAX {0, (X-S)/n}
IV = MAX {0, (27,200 - 24,981.38)/8,000} = $0.2773 (Note that if we had assumed scenario 3, this value = 0)

Conversion Price (CP)
CP = X – nWP
CP = 27,200 - (8,000 x $0.205) = $25,560

Premium (%)
Premium (%) = [(nWP - X + S) / S] x 100
Premium (%) = [(8,000 x $0.205) – 27,200 + 24,981.38]/24,981.38) x 100 = -2.32%

5.7 Roles of the Warrant Holder & Issuer

The listing document contains the terms and conditions of the structured warrant, which spells out the contractual rights and obligations of both the warrant issuer and warrant holder.

1. **Right and Obligation of Warrant Holder –** The warrant holder has the right but not the obligation to buy or sell a stated number of shares of an underlying security at a specified price (exercise or strike price) within a specified time period.

2. **Right and Obligation of Warrant Issuer –** Generally, the issuer will have the obligation to deliver the underlying instrument and receive the exercise price if the holder exercises a call or receive the underlying instrument and pay the exercise price if the holder exercises a put. Most, if not all, structured warrants on SGX-ST settle in cash instead of delivery of the underlying.

5.8 Role of Designated Market Makers in Structured Warrants Trading

Under the listing requirements, warrant issuers who commit to make a market for the structured warrants they issued do not need to comply with the minimum placement and holder size requirement for listed equity securities, and the minimum issue size requirement is SGD 2 million (previously SGD 5 million).

The listing document will state whether the warrant issuer has committed to make a market in the structured warrants. By committing to make a market, the designated market-maker (DMM) appointed by the warrants issuer has to provide competitive bid and offer prices for the structured warrants (during trading hours) as prescribed by the SGX to create adequate liquidity. The maximum spread (the difference between a DMM's bid and offer quotes) and minimum lot size that DMMs have to provide are set out by the structured warrants issuers in the listing documents of the respective structured warrants. DMMs provide liquidity by posting bid/offer quotes for the structured warrants for which they make a market. Appointed market-makers can include large international banks and brokers who can react to the changing market conditions to provide liquidity and generate demand for the structured warrants.

5.9 Settlement of Warrants

Structured warrants trade and settle like ordinary shares and company warrants listed on the SGX-ST.

5.9.1 Physical Settlement

To exercise a call warrant, the holder will need to submit an exercise notice with the cash payment (based on the exercise price) to the warrant agent. The warrant agent will then inform the Central Depository Pte Ltd (CDP) and the issuer, and CDP will credit the account of the holder with the underlying securities and debit the issuer's account with the same.

For a put warrant, CDP will debit the account of the holder for underlying securities and credit the issuer's account, and the warrant agent will make a cash payment on behalf of the issuer to the holder. Regardless of the type of warrant, exercise expenses are payable by the holder to the warrant agent.

5.9.2 Cash Settlement

Most structured warrants are automatically exercised and settled in cash on the expiration date. The proceeds are calculated based on the difference between the market price of the underlying and the exercise price.

Call Warrant: Cash settlement per warrant = (S – X)/n

Put Warrant: Cash settlement per warrant = (X – S)/n

In the call warrant **DBS MB ECW1710020** example if the price is settled at $18.16. The cash settlement is:

Call Warrant Cash Settlement for DBS MB ECW1710020

DBS Group Holdings	Share Price	=	$18.16
Exercise price	X	=	$16.50
Conversion ratio	N	=	8:1
Cash settlement per warrant		=	(S – X) / n
		=	($18.16 - $16.50) / 8 = $0.2075

In the put warrant example, if the settlement occurs when the HSI is at 24,981.38, the cash settlement is:

Put Warrant Cash Settlement for 28585 GS-HSI @EP1802B HSI Put Warrants

Share price	S	=	24,981.38 (assume Scenario 2 from previous example)
Exercise price	X	=	27,200
Conversion ratio	N	=	8,000
Cash settlement per warrant		=	(X – S) / n
		=	(27,200 - 24,981.38) / 8,000 = $0.2773

5.9.3 Settlement Procedures

All structured warrants listed on SGX-ST have the Asian style of expiry settlement, where the last trading day of a structured warrant is different from its expiry date[31]. For a structured warrant with an automatic exercise feature, its last trading day in the "Ready" and "Unit-Share" markets is at least 3 business days before its expiry date (such that the settlement or due date is at least a day before the expiry date).

Investors can only trade the structured warrant on or before the last trading day. The settlement price is based on the arithmetic average of the official closing price of the underlying for 5 market days prior to expiration date.

For example, for DBS MB ECW171002:

[31] The Asian option/warrant is a path-dependent contract where the payoff is calculated from the mean of the values at certain dates. A contract is called path-dependent when the payoff does not only depend on the terminal price of the underlying asset at maturity, but also on the path, or the values it has taken on to reach its final value.

a. Last trading day for the structured warrant in the "Ready" and "Unit-Share" markets is 25 September 2017, which is 5 trading days before expiry.

	Sun	Mo	Tu	We	Th	Fr	Sa
Sep	24	**25**	26	27	28	29	30
Oct	1	**2**	3	4	5	6	7

b. Expiry date or maturity date of the structured warrant with automatic exercise feature – 2 October 2017.

5.10 Corporate Actions and Adjustments for Structured Warrants

Structured warrants on equities are subject to adjustments arising from corporate actions of the underlying security. Corporate actions such as rights issues, bonus issues, stock splits, special dividends or consolidation may lead to a diluting or concentrative effect on the theoretical value of the underlying stock or security. This may result in adjustments to the conversion entitlement, the exercise price or other variables of the structured warrants effective on the ex-date of the corporate action.

Adjustments for some of the more common corporate actions are:

a. Dividends;
b. Share Buy Backs/Cancellations; or
c. Share Splits.

5.10.1 Adjustment for Dividends

Corporate adjustments are usually done for special dividends and not normal dividends.

5.10.1.1 Exercise Price

The exercise price and adjustment factor are calculated using the formula below:

New Exercise Price = Old Exercise Price x Adjustment Factor

Adjustment Factor = (P – SD – ND) / (P – ND)

where:
P = Last cum-date closing price of the underlying
SD = Special dividend per share on ex-date
ND = Normal dividend per share on ex-date

5.10.1.2 Conversion ratio

Conversion ratio is calculated using the formula below:

New Conversion Ratio = Old Conversion Ratio x Adjustment Factor

5.10.2 Adjustment for Share Buy Backs / Cancellations

5.10.2.1 Exercise Price

The new exercise price after adjusting for share buy backs or cancellations is given by the following formulae:

New Exercise Price = Old Exercise Price x Adjustment Factor 1 x Adjustment Factor 2

Adjustment Factor 1 = A / B

Adjustment Factor 2 = (P – CD) / P

where:
A = Existing shares
B = Number of shares on an ex-basis
P = Last cum-date closing price of underlying
CD = Cash distribution per share held immediately prior to the capital reduction / buyback

5.10.2.2 Conversion Ratio

Conversion ratio is calculated using the formula below:

New Conversion Ratio = Old Conversion Ratio x Adjustment Factor 1 x Adjustment Factor 2

The conversion ratio is the number of warrants needed to be exercised to buy or sell one unit of the underlying security.

5.10.3 Adjustment for Share Splits

5.10.3.1 Exercise Price

The new exercise price after adjusting for share splits is given by:

New Exercise Price = Old Exercise Price x Adjustment Factor

where:
Adjustment Factor = A / B
A = Existing shares
B = Number of shares on an ex-basis

5.10.3.2 Conversion Ratio

Conversion ratio is calculated using the formula below:

New Conversion Ratio = Old Conversion Ratio x Adjustment Factor

The conversion ratio is the number of warrants needed to be exercised to buy or sell one unit of the underlying security.

5.11 Exotic Warrant Structures and Features

Some of the exotic warrant structures include:

a. Index Warrants;
b. Currency Translated Warrants;
c. Basket Warrants;
d. Yield Enhanced Securities;

e. Commodity Warrants; and
f. Foreign Exchange Warrants.

5.11.1 Index Warrants

These warrants are settled on exercise via a cash payment. The amount of cash received is calculated by multiplying the difference between the exercise level (strike price) and the current index level or settlement index level by a multiplier to convert the index points into cash.

For example, for call warrants on MSCI-Singapore Free Index Futures:

Cash Settlement = Max {0, SGD1.00 x [(Settlement Reference Level - Exercise Level)/n]}

where the settlement reference level is based on the underlying index futures contract's final settlement price as determined by the derivatives exchange.

As the underlying index will have already been adjusted for corporate actions, adjustments are normally not required for index warrants.

5.11.2 Currency Translated Warrants

The settlement and trading currency (home currency of the investor) is different from the underlying asset's currency. For a structured warrant on the HSI traded on SGX-ST, the underlying currency (which is also the reference currency) is HKD. The settlement currency is SGD.

For each warrant:

Settlement amount = MAX {0, HKD1 x [(Reference Level – Exercise Level)/n]

x Exchange Rate (SGD / HKD)}

5.11.3 Warrants with Yield Enhancement

This type of warrant is designed to give the investor an attractive yield based on the current share price (or spot price). For marketing purposes, **capped warrants** are often grouped with other yield enhancement securities under the broader category called **discount certificates**.

In a capped warrant, the cap strike or exercise price represents the maximum price achievable with a discount certificate. Potential upside exposure is sacrificed in return for the ability to purchase the underlying asset at a discount to the market price. At maturity, if the underlying asset's closing price on expiration or valuation date(s) is at or above the exercise price, the holder will receive a cash settlement equal to the exercise price (i.e., the cap strike price). If the closing price is below the exercise price, then the holder will receive a cash settlement equal to the value of the underlying on the expiration or valuation date(s).

Warrant with Yield Enhancement	
Issue price of warrant	= $5.25 (89.0% of the reference spot)
Reference price of underlying (spot price)	= $5.90
Underlying security	= ABC Ltd
Conversion ratio (n)	= 1

Exercise Price/Cap Level	= $5.45 (92.4% of the reference spot)
Exercise Style	= European
Settlement	= Cash

On expiration:

If the closing price of the underlying asset is at or above the exercise price, the investor will receive $5.45.

If the closing price is below $5.45, the investor will receive the value of the underlying asset.

5.11.4 Basket Warrants

The underlying asset is a basket of (pre-defined) shares, which is likely to be based on a particular industry or sector (e.g., technology, biotech, banks / financial). A basket warrant may be settled with a physical delivery of the basket components or via cash.

5.11.5 Commodity Warrants

A commodity warrant is a warrant that has a commodity as its underlying asset. Oil and precious metals are popular underlying commodities for such warrants. Based on their market view, investors can trade warrants on the underlying commodities.

Commodity warrants structured and marketed by banks are generally cash settled. These warrants usually have a much longer expiry date than commodities futures and options. As commodities are traded globally and are linked to different benchmarks, it is important to verify the correct benchmark or index to be used at settlement. Commodity warrants can also be issued by resource-based companies or commodity firms. Depending on the warrant's structure, settlement can be by cash or physical delivery. If the warrant is settled by physical delivery, specific terms and conditions must be confirmed (e.g., delivery location, additional storage, transport costs).

5.11.6 Foreign Exchange Warrants

The price of a Foreign Exchange (FX) warrant is driven by the changes in the exchange rate between the two underlying currencies. Similar to currency options, FX warrant holders have the right to exchange an amount of one currency into another currency at a specified exchange rate before or on a specified date.

FX warrants are suited to investors who have a particular view on FX rates, either to hedge their currency exposure or speculate on currency movements. Before trading FX warrants, the investor has to identify which currency pair he wants to trade, of which one currency is considered bullish and the other currency bearish, relative to one another. For example, for the USD/JPY pair, if the investor is bullish on USD and bearish on Yen, he will buy USD/JPY call warrants. Another investor who believes that the Japanese Yen will appreciate against USD will buy USD/JPY put warrants.

FX warrants are cash settled and have features similar to other structured warrants. FX warrants are leveraged instruments where returns can be magnified when the FX market moves favourably. However, losses can be compounded if exchange rates move in the opposite direction from what the investor had anticipated. Investors should monitor FX movements closely and be aware of the macroeconomic factors which influence global and local economic activity, political developments, etc. Currencies of some countries which are

endowed with abundant natural resources can be very responsive to moves in commodity prices, such as Australia and Canada.

5.12 Daily Leveraged Certificates

Daily Leveraged Certificates (DLCs) are leveraged short-term exchange traded derivatives that were designed mainly for intra-day trading, offering traders and investors the opportunity to take amplified long or short positions on an underlying asset (typically an index). DLCs seek to achieve very high returns over a very short-time frame by magnifying the daily performance of the underlying benchmarks.

As DLCs can be highly leveraged, they are considered high-risk products and it is important that clients and investors are informed about the risks of these products before investing in DLCs. First introduced in Europe in 2012 where they are known as constant leverage products or factor certificates, DLCs made their first appearance in Asia on the SGX in 2017, where they allow traders and investors the opportunity to gain leveraged exposure to developed Asian market indices.

5.12.1 Features and Characteristics of Daily Leveraged Certificates

The features and characteristics of DLCs are summarized in Table 5.12.1.

Table 5.1.2.1 – Features and Characteristics of Daily Leveraged Certificates

Trading Board lot size	100 lots	**Currency Traded**	SGD (regardless of index)
Direction of Trade	Long or short	**Returns**	High daily leveraged returns. If held over a period, the compounded returns can be massive, but the losses can also wipe investors' capital out completely.
Leveraging	Up to 7x		
Maximum Tenure	3 years	**Risks**	(1) Potential for **full capital loss** (2) Issuer / counterparty credit risks (3) Liquidity – Issuer market-making inhibitions in extreme volatile periods
Traded on /Cleared by	SGX / SGX CDP		
Terms of Transacting	T+2 (similar to stocks on SGX)		
Fees charged	(1) **Brokerage commission:** 0.1 bps i.e., 0.001% of traded value (2) **Clearing fee:** 0.4 bps i.e., 0.004% of traded value (3) **Management fee:** Charged to product price if held overnight	**Airbag Feature**	Slows the rate of loss if the underlying asset for 3x (5x) leveraged product falls by 20% (10%) during the day.

	(may even include cost of hedging risk).		

5.12.1.1 Naming Convention for DLCs on SGX

Let's review this product using a real DLC example from the SGX with the product name "DLC SG 5x Long HSI 200714" with a security code "CIUW". The name for this instrument provides some helpful information – this is a DLC (Daily Leveraged Certificate) issued by SG (Societe Generale), which gives 5 times leverage and longs on the HSI. The expiry date is 14 July 2020.

5.12.1.2 Leveraging and Volatility

DLCs are bought mainly for the benefits of leveraging which has the potential for significantly amplifying gains. For example, a bullish investor can select a 3x Long DLC, which will rise in value by 3% for each 1% rise in the underlying index (before cost & fees). A bearish investor who expects the underlying index to fall can instead select the opposite - the 3x Short DLC, which will rise in value by 3% for every 1% fall in the underlying index (before cost & fees). If the market moves against an investor, the DLC will amplify losses in the same way. The investor's entire invested capital is at risk, but the maximum total loss will not be more than the total invested capital.

Depending on the desired risk and volatility of the speculator or trader, those seeking volatility may prefer a 5x leverage product over a low volatility product for a particular period and product. Those investing in a less volatile index like the Straits Times Index may also prefer 5x leverage but reduce it to 3x leverage when they are dealing with a more volatile index like the HSI.

5.12.1.3 Intraday-Trading or More

DLCs are meant to be traded intra-day. Speculators who want to take positions beyond one day will be exposed to the risk of compounded losses (or gains). DLCs are not suitable as a long-term investment.

Example of DLC (SG 5x Long HSI)

Here is an illustration from the reaction of the Hang Seng Index (HSI) to North Korean geopolitical worries on 10 and 11 August. You can see that in 2 days the DLC lost close to 17% of its value.

Date	HSI Level	HSI +/- %	Price of DLC SG 5x Long HSI	DLC Price Change %
4 Aug	27,576.59		3.06	
7 Aug	27,699.35	+0.45%	3.11	+1.63%
8 Aug	27,854.91	+0.56%	3.22	+3.54%
9 Aug	27,755.93	-0.36%	National Holiday	
10 Aug	27,462.73	-1.06%	3.00	- 6.83%
11 Aug	26,890.96	-2.08%	2.70	-10.00%

More than 17% lost in just 2 days from the correction of the HSI.

Compounded Gains (or Losses)		**-2.49%**		**-11.66%**

Over a short week of gains and losses, a net -2.49% correction on the HSI led to a compounded decline of 11.66% on the investment. A major correction of 8-10% of the HSI would lead to a 47-61% decline in this 5x DLC investment. For some of the more volatile indices like the HSI, a 2-3% movement either way in a week happens often enough and 8-10% movements in a month also happen occasionally.

Depending on the prevailing bullishness and bearishness of the market, one way of trading is to allow for compounded gains to continue for slightly longer periods with defined target returns to take profit and cut compounded losses immediately once losses cross any limits during the day. Momentum investors will follow the trend as their friend in this manner.

Traders and investors must be disciplined to cut their losses and not try to "average down" and hang on to the product over time, or they will risk incurring greater losses. When faced with book losses, some traders have bought the DLC holding the opposite direction to stem further losses, and they may seek to trade out of the position over time and volatility, but usually a better approach is to be watchful, disciplined and cut losses once limits have been breached. With DLCs, it is necessary to set stop loss limits.

DLCs are not a good investment product for adopting a buy-hold strategy, even if the investor takes long and short positions over time, because of the risk of full capital losses.

Example of the Compounding Effect of DLCs

Holders of DLCs over a 5-day period facing a directional move of 2% in the underlying asset will see the following compounded impact if they follow a buy-hold strategy:

Days Compounded	% Change in Underlying Index (Assume rises 2% each day for 5 days)	Percentage Price Change for DLCs			
		Long 3x DLC	Short 3x DLC	Long 5x DLC	Short 5x DLC
1 day	2.00%	6.00%	-6.00%	10.00%	-10.00%
2 days	4.04%	12.36%	-12.36%	21.00%	-21.00%
3 days	6.12%	19.10%	-19.10%	33.10%	-33.10%
4 days	8.24%	26.25%	-26.25%	46.41%	-46.41%
5 days	10.41%	33.82%	-33.82%	61.05%	-61.05%

Note that the compounding effect on the underlying index and the DLCs may not correspond as closely to what is mathematically expected as reflected in the table above because:

- The quoted price of the DLC is dependent on the market-maker, and the demand and supply conditions may slightly differ from the underlying index, however efficient markets might be.
- There is also the management cost charged to the DLC price over time; and
- The performances of the underlying index and the DLCs are reset at the end of each trading day.

5.12.1.4 Management Fee and Pricing Trends

Annual management fees are charged directly to the value of the certificate over time. The bank issuing the DLC will usually have some hedging cost to apply against the DLC exposure (perhaps with futures and options on the underlying exposure as counterparties and market makers holding a portfolio). The specialists who manage such DLC exposures, amongst other exposures on the bank's book, will risk manage their position and the benefit to them is that they can charge this cost to the DLC asset value.

Hence, even if the underlying asset stays flat over a period, the DLC certificate will see a small decline from the management cost. For the same underlying price level reached over the life of the DLC, we should see lower prices and investors should not expect the DLC to reach the same price level in parity with the underlying asset over time.

5.12.1.5 Liquidity and Market Makers

A designated market maker (i.e., one which is registered with SGX-ST) is required to provide competitive bid and offer quotations on a continuous basis, within the maximum spread, and for not less than the minimum quantity set out below:

a. **Maximum bid and offer spread** – 10 ticks or $0.20, whichever is greater; and

b. **Minimum quantity** subject to bid and offer spread – 10,000 DLCs.

The bid-offer price will depend on the quote provided by the market maker which is influenced by the demand and supply for the product at any one particular time during the normal trading hours. If there are no bids on screen at any time, traders and investors can call the issuer/designated market-maker for a quote. However, there are situations when the designated market-maker may choose not to/or may not be able to provide a quote. For example, the following instances have been quoted as possible reasons:

i. During the pre-market opening and 5 minutes following the opening of SGX-ST on any trading day;

ii. When trading in the underlying asset is suspended or limited in a material way for any reason;

iii. When trading in the DLC is suspended or limited in a material way for any reason;

iv. When the equity market experiences exceptional price movements and volatility;

v. When the DLC issuer and/or DMM faces technical problems affecting the ability of the DLC issuer and/or DMM to provide bid and offer prices; and

vi. When the DLC issuer is unable to hedge its exposure or unwind an existing hedge.

5.12.1.6 Airbag Feature

The airbag feature for DLC is a safeguard that could apply in a major market shock or crisis on a particular day, e.g., similar to the Global Financial Crisis or Asian Financial Crisis. It calls for a slowdown in the trading of a negatively affected DLC, if the underlying index has fallen by:

a. 20% or more in a day for a 3x leveraged DLC making losses; or
b. 10% or more in a day for a 5x leveraged DLC making losses.

If the airbag feature is triggered, an intra-day reset takes place over 30 minutes when trading in the DLC is suspended.

Upon reset, the performance of the DLC will depend on the lowest value of the underlying index for the Long DLC, or the highest value of the underlying index for the Short DLC, during the 15-minute period after the airbag is triggered.

Example – DLC Airbag Feature

Assume an investor bought a 5x Long DLC on Tuesday and the DLC had previously closed on Monday at a price of S$2.00 and the underlying index closed at 1,000 points. By Tuesday 11.00 am, the underlying index had dropped 10% (from 1,000 to 900 points), which translated into a 50% loss on the DLC, leaving it at a price of S$1.00.

At this point the airbag mechanism is triggered and the DLC issuer will ask SGX to suspend trading in the relevant DLC for 30 minutes from 11.00am to 11.30am.

The performance of the DLC after the suspension will be based on the Minimum Observed Level. Assuming the Minimum Observed Level equals the airbag trigger point (900 points), if the underlying index were to fall another 5% after trading resumes at 11.30am, the loss to the DLC would be 25% (i.e., 5 X 5%) of S$1.00 (instead of the previous close of S$2.00), which will leave the DLC at a value of S$0.75.

Without the airbag mechanism, an additional 5% fall in the underlying index would have left the DLC at a lower value of S$0.50 (total of 15% fall in underlying index multiplied by 5 times leverage would equate to a 75% loss on the DLC based on the previous close of S$2.00). In a more extreme case where the underlying index falls by an additional 10%, the DLC would be worthless. This is because the underlying index fall of 20% would equate to a 100% loss on the 5x Long DLC.

Source: SGX

Naturally, the airbag feature will only be triggered upon movements of the underlying index that results in losses (not gains). For certain DLCs, the airbag mechanism may not be available or triggered in certain circumstances. The SGX has highlighted to investors that they should read all listing documents provided by the issuer for the specific airbag mechanism for each product.

5.12.2 Benefits of DLCs at a Glance Compared to Other Structures

The following have been cited as key benefits of DLCs compared to other similar structures:

a. Augments potential returns and capitalizes on volatility with leverage of up to 5 times;

b. Allows for both long and short exposure, and allows for daily hedging;

c. Low capital outlay and loss limited to invested amount;

d. No margins required;

e. No implied volatility impact unlike options;

f. No time decay impact unlike options;

g. Transparent pricing and structure (except for exact management fee charged to assets); and

h. Does not require the tedious set up and maintenance of a fund structure like leveraged ETFs.

5.12.3 DLCs at Expiry

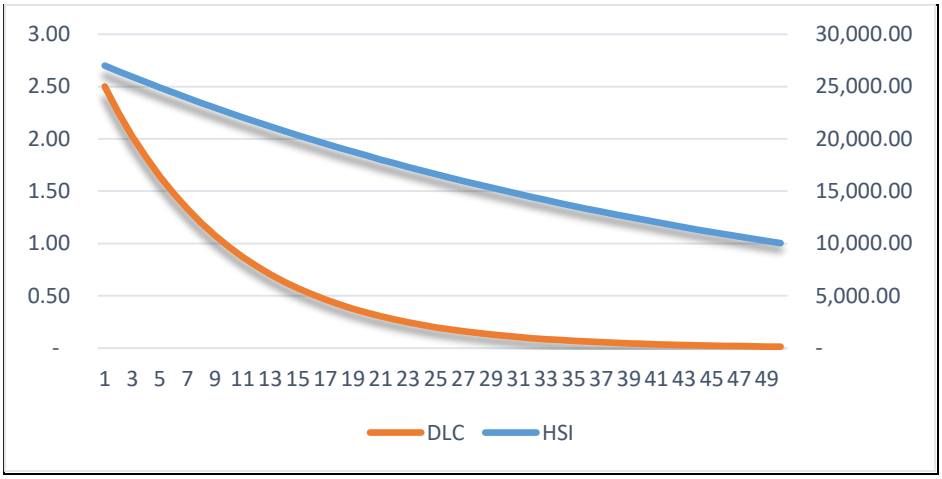
DLCs have a limited life with a maximum tenure of 3 years. At expiry, the final exercised value of the DLC is calculated and automatically paid to investors.

What happens if the DLC falls to zero before expiry? In the case of leveraged ETFs, the funds can reset the price through reverse splits of the fund unit when the fund price is too low. This will not be carried out in a DLC.

For example, if an index declines 2% per day for 50 days, a 5x DLC priced originally at $2.50 will be worth $0.01 and it may be difficult to trade at that level since management cost could wipe out the remaining value. So technically, it is conceivable that a DLC will shrink to zero before the end of its tenure but unlikely since markets will rise and fall and a 50-60% fall in an index does not happen that often (e.g., the DJIA fell more than 50% from 16,326 points in Oct 2007 to 8,150 in Feb 2009 during the Global Financial Crisis). It is possible and likely that some DLCs might fall to less than half their value before expiry and the remainder value will be automatically distributed proportionately to holders at that point.

Figure 5.1.2.3 shows an example of a DLC falling with the HIS.

Figure 5.12.3 – Example of a Long 5x DLC Value Falling with the HSI



5.12.4 Comparisons with Other Leveraged Products

DLCs and structured warrants are traded on the exchanges like the SGX-ST while CFDs are OTC products.

The leverage for CFDs can be much higher (up to 20 times) and they are margin-based products, whereas DLCs are not margin-based and are usually leveraged between to 3x-7x as of 2022. The leverage level for structured warrants can vary from 5 to 50 times depending on the underlying value and other optionality features.

Investors can potentially lose more than their invested capital when trading CFDs, whereas losses for DLCs and structured warrants are capped at the total invested capital.

For leveraged ETFs, there is a requirement of a fund structure which can involve investments in equities, futures, options and other instruments. The leveraging for an ETF does not usually rise to 5x but usually goes up to 3x on most exchanges.

5.13 Summary

1. A **warrant** is a derivative that gives the investor an option to buy or sell a stated number of the underlying at a strike price, within a specified time period. As warrants are already fully paid upfront, they are not subject to margin calls.

2. **Company warrants** are generally issued by listed companies to existing shareholders as a "sweetener" attached to an offering of bonds or rights issues for shares. They are essentially long-dated "American-style" call options, exercisable at any time during the life of the warrants. Warrants are detachable from the host instrument and may be listed and traded separately.

3. **Structured warrants** are issued by third party financial institutions based on various underlying instruments such as individual stocks, equity indices, investment funds and commodities. Structured warrants in Singapore are "European-style" options and can only be exercised on the expiry date.

4. A **zero-strike warrant** is a special type of warrant which is essentially a synthetic stock. A zero-strike warrant has an exercise price of zero, which means that it is always in-the-money. The cash settlement and price of this type of warrant is equal to the closing price of the underlying stock.

5. **Gearing** indicates how many more warrants can be bought with a certain amount of capital compared to buying the underlying share. Gearing enables investors to trade more warrants than the underlying for the same investment outlay.

6. The **delta** of a warrant is the rate at which its price changes with respect to changes in the price of the underlying asset. Call deltas are always positive and put deltas are always negative. At-the-money warrants typically have a delta of around 0.5 while those which are deep in-the-money have a delta of close to 1.

7. **Effective gearing** of a warrant is calculated simply by multiplying the delta of a warrant by its gearing**:** Effective gearing = Delta x Gearing.

8. Warrants may be settled by physical delivery of shares or by cash. The settlement style is made known when the warrant is issued. Structured warrants listed on the Singapore Exchange are settled in cash.

9. **Implied volatility** is the market's consensus estimate regarding the warrant's underlying price volatility for the duration of the life of the warrant. In addition to the theoretical price factors, it also captures the prevailing market sentiment. Implied volatility can be used to compare the warrant value, and for warrants with similar terms, the higher the implied volatility, the more expensive the warrant is.

10. **Intrinsic value** is the difference between the exercise price of a warrant and the market price of the underlying asset. Only in-the-money warrants have intrinsic value.

11. **Conversion ratio** is the number of warrants needed to be exercised to buy or sell one unit of the underlying security. Conversion price is the total price which the investor pays by converting the warrants into the underlying security, and it is also known as the breakeven price.

12. **Warrant premium** is the difference between the warrant price and intrinsic value (the term is different from the one used in options, where the premium is actually the price of the option). The warrant's premium is largely the time value of the warrant and is usually expressed as a percentage of the underlying share price.

13. The warrant holder has the right but not the obligation to buy or sell a stated number of shares of an underlying security at a specified price (exercise or strike price) within a specified time period. The warrant issuer has the obligation to deliver the underlying instrument and receive the exercise price if the holder exercises a call or receive the underlying instrument and pay the exercise price if the holder exercises a put. Structured warrants listed on SGX-ST are cash settled.

14. Structured warrant issuers appoint **designated market-makers (DMMs)** to make a market and provide liquidity for the structured warrants that have been issued. Structured warrants trade and settle like ordinary shares and company warrants are listed on SGX-ST. All structured warrants listed on SGX-ST have the "Asian style" of expiry settlement, where the last trading day of a structured warrant is different from its expiry date.

15. Corporate actions such as rights issue, bonus issue, stock splits, special dividends or consolidation may lead to a diluting or concentrative effect on the theoretical value of the underlying stock. This may result in adjustments to the conversion entitlement, the exercise price, or other variables of the structured warrants effective on the ex-date of the corporate action.

16. Exotic warrant structures include **index warrants, currency translated warrants, basket warrants, yield enhanced securities, commodity warrant*s*** and **foreign exchange warrants**. An example of a yield enhanced security which has a warrant is a **discount certificate**. The exercise price represents the maximum price achievable with a discount certificate. Potential upside exposure is sacrificed in return for the ability to purchase the underlying asset at a discount to the market price.

17. Based on their individual market views, investors can take positions using various available warrants and warrant-related securities to achieve their investment objectives. This would include:

    - Neutral view – Yield enhanced security;
    - Bullish view – Call warrant purchase, convertible bond, index linked note; or
    - Bearish view – Put warrant purchase.

18. **Daily Leveraged Certificates (DLCs)** are leveraged short-term exchange traded derivatives designed mainly for intra-day trading, offering traders and investors the opportunity to take amplified long or short positions on an underlying asset (typically an index). DLCs seek to achieve very high returns over a very short-time frame by magnifying the daily performance of the underlying benchmarks.

19. DLCs are high-risk products due to the leveraging and those dealing with clients trading in such instruments must highlight the risks associated with the product.

Chapter 6:
Barrier Options, Binary Options and Callable Contracts

Learning Objectives
The candidate should be able to: ✓ Describe and explain the different types of popular non-standard (exotic/complex) options and related structures transacted over-the-counter (OTC): • Barrier options; • Binary options; and ✓ • Basket options. ✓ Discuss the product features, investment views and payoff scenarios of some common type of barrier and binary option structures. ✓ Explain the features, benefits, investment returns and risk profiles of callable bull/bear contracts (CBBCs). Discuss the similarities and differences between knock-out products and structured warrants.

6.1 Introduction

There are a variety of exotic or complex options with flexible and innovative features which are used in structured products. The features can be varied to create strategies and solutions to meet the different demands of clients and to take advantage of opportunities available in the market. Such features include:

a. **Timing of option exercise** – Other than the final expiry date (European) or the ability to exercise it at any time before expiration (American), it can be at specific intervals before the expiration date (Bermudan) or triggered by the occurrence of an event.

b. **Uncertainty as to time of expiration** – This can be other than the final expiry date which is stipulated at the time of issuance, or it can be triggered by the occurrence of an event.

c. **Payoff computation** – The price level used to calculate the payoff can be a number other than the actual spot price.

d. **Conditional events or trigger points** – If the price level of the underlying asset touches or crosses a trigger point, it will affect the option's validity, and have an impact on the final payoff.

e. **Underlying assets** – The variety can extend beyond traditional asset classes and market indices to include alternative investments, such as real estate, and market indicators like volatility.

f. **Currency features** – Such features can be used to create payoff in a different currency, e.g., Quanto.

6.2 Barrier Options

Barrier options are one of the most popular exotic option structures used in structured products sold by banks. They are easy for clients to understand because of their simple "game-like" concepts and outcomes, which follow an "if then or else" approach.

For barrier options, the payoff to the investor depends not only on whether and how much the option is in-the-money at exercise, but also on whether the underlying asset has done one of the following:

a. Touching the barrier anytime during the life of the option,

b. Being above the barrier or below the barrier at expiry (or throughout the life of the option); or

c. Staying within two barriers or outside two barriers at expiry (or throughout the life of the structure).

For example, when a barrier is touched in a "knock-out" or "kick-out" option, the features and rights of the option are deactivated. In a "kick-in" or "knock-in" option, the features and rights of the option are activated if the barrier is touched.

Investment professionals should carefully review the terms offered by structured products to determine how the exact payoff will be calculated and the amount payable once the barrier event occurs, and/or what happens to the payoff if it doesn't occur, as there are different payoffs depending on the price movement of the underlying asset at option expiry or during the life of the option. Frequently, the features and payoffs of the options underlying structured products can be negotiated and customized.

The **barrier** is a price trigger level which, when reached, will result in an event (e.g., a payment of interest, a higher interest, a conversion of currency, selling or buying a stock, etc.). When the barrier / trigger price level is reached, a positive or negative event can occur. Depending on the underlying asset, the trigger level can reference a stock price, an index level, a currency exchange rate, a money market or a bond interest rate, a weighted average for a basket of stocks, and so on.

Barrier options are path dependent because they are either activated or deactivated / cancelled when a set trigger price level (the barrier) of the underlying asset is reached. Intuitively, barrier options should be cheaper than their vanilla counterparts because investors risk either being knocked out or not being knocked in.

6.3 "Knock-Out" Barrier Options

Knock-out barrier options are investment structures which terminate when the price of the underlying asset (e.g., equity, interest rates, indices, commodities, currencies, etc.) hits certain predetermined levels. Knock-out products can be constructed with one or more knock-out options. Investors can buy knock-out products to take an investment position with respect to both rising and falling prices. Products which benefit from increasing prices are described as call, long or bullish while those which gain from declining prices are often referred to as put, short or bearish.

A double knock-out option is cheaper than a single knock-out option because the double knock-out has two barrier prices, either of which could knock the option out of existence.

6.3.1 Single Barrier "Knock-Out" Option

In the simplest form of barrier options, we have the single barrier knock-out option where the option gets "knocked out" (or otherwise termed "kicked-out") once the underlying price reaches the barrier level. If the price of the underlying has to fall below the barrier level before getting knocked out and deactivated, it is called a down-and-out barrier. If it has to rise to the barrier before getting knocked out, it is called an up-and-out barrier option. In both cases, the key is that the option is deactivated when the barrier price level is reached or breached.

Some knock-out barrier options may give the investor a fixed rebate in the event the barrier is breached i.e., the underlying option is extinguished and instead the investor receives a fixed amount of money. See example in Table 6.3.1.

Table 6.3.1 – Example of Knock-Out Options

Example of Knock-Out Options

Investor A: Call Option	
Strike Price:	$8.00
Knock-out Price:	$9.00
Current Spot Price:	$7.00

Payoff Profile
Max = { $8.99 - $8.00 - Price $_{\text{Call Barrier}}$}
Min = { 0 - Price $_{\text{Call Barrier}}$}

Breakeven Point:
($8.99 -$8.00) - Price $_{\text{Call Barrier}}$ = 0
When the price of the call is $0.99.

During life of option if Spot Price touches $9.00:
- Call option is WORTHLESS.
- **Knocked out** once the barrier of $9.00 is reached.

Investor B: Put Option	
Strike Price:	$13.50
Knock-out Price:	$12.00
Current Spot Price:	$15.00

Payoff Profile
Max = { $13.50 - $12.01 - Price $_{\text{Put Barrier}}$}
Min = { 0 - Price $_{\text{Put Barrier}}$}

Breakeven Point
($13.50 -$12.01) - Price $_{\text{Put Barrier}}$ = 0
When the price of the put is $1.49.

During life of option if Spot Price touches $12.00:
- Put option is WORTHLESS.
- **Knocked out** once the barrier of $12.00 is reached.

For Investor A Who Buys the Knock-Out Call Option:

Investor A participates in the upside of the underlying asset through a call at the strike price of $8.00. Once the price of the underlying asset crosses $8.00, the investor has gains over the initial spot level of $7.00. However, if the underlying asset price rises to $9.00, the call option is cancelled, and the investor no longer participates in the upside gain as he was knocked out. Therefore, the call option with a barrier is cheaper than a normal call option because it can be knocked out at a certain price level.

Another way to look at the knock-out option is as a visible call option that Investor A bought with a strike price at $8, with an embedded call option sold by Investor A to the counterparty (usually the bank which offered this product) with the strike at $9. The premium received from the sale of the call was used to reduce the overall cost of the structure and making it cheaper than a standard call option. At $8, the investor participates in the upside of the underlying asset but at $9 and above, the counterparty bank exercises its call, and the investor will deliver the underlying asset, so he does not participate further in the upside beyond $9. His maximum gain is limited to $8.99-8.00 – call price.

Note, however, that unlike a bull spread described above, (bought a call option with a strike at $8 and sold a call option with a strike at $9), the long call option with the lower strike of $8 is immediately extinguished anytime the underlying asset trades at $9 or higher. Hence, at the point when the barrier (of $9) is breached, the investor loses the maximum intrinsic value of $0.99 (i.e., $8.99 - $8.00).

The payoff profile of the knock-out barrier option looks like this:



- which is different from the payoff profile of a bull spread using call options (refer to Section 4.12.7).

For Investor B Who Buys the Knock-Out Put Option:

As the buyer of the barrier put option, Investor B bought the right to sell the underlying asset at the strike of $13.50 and would report gains from buying back at the lower price. However, this right to participate in the shorting of the underlying asset is taken away from $12.00, the knock-out barrier level. The embedded put is unseen, and the investor has effectively sold within the barrier option which has a strike at $12.00. The premium received from the put makes the put barrier option cheaper than the normal put option. The investor does not enjoy the gains below $12.00, and his maximum gain is $13.50 - $12.01 – put price.

Note, however, that unlike a bear spread described above, (bought a put option with a strike at $13.50 and sold a put option with a strike at $12), the long put option with the higher strike of $13.50 is immediately extinguished anytime the underlying asset trades at $12 or lower. Hence, at the point when the barrier (of $12) is breached, the investor loses the maximum intrinsic value of $1.49 (i.e., $13.50 - $12.01).

The payoff profile of the knock-out barrier option looks like this:



- which is different from the payoff profile of a bear spread using put options (refer to Section 4.12.10).

6.3.2 Double Barrier Option

The double barrier option has two barrier levels (knock-out levels) instead of one. Double barrier options are also known as "double knock-out" options. Effectively, the investor is buying:

a. One barrier which is **HIGHER** than the strike price; and
b. Another barrier which is **LOWER** than the strike price.

Investors will get to participate in the movement of the underlying as the long as the price of the underlying stays within the two barriers.

Double Knock-Out Barrier Call Option

Strike Price (X)	:	$8.00
Spot Price, P_{spot}	:	$7.80
Knock-out Price 1	:	$7.00
Knock-out Price 2	:	$9.00

This double barrier call option strike price can be priced just above the current spot price of the underlying asset, e.g., at $8.00 while the spot price, P_{spot}, is at $7.80. It could have also been launched in-the-money, with the current spot price of the underlying trading at the strike price of $8.00, activating the visible call option immediately (the latter in-the-money is the more popular option).

Let us assume that the product is launched with $P_{spot} < X$:

i. If P_{spot} stays below X throughout then there is no reason to exercise, since a cheaper option is to get the underlying asset directly from the spot market below $8.00. The loss is the cost of barrier option that is not exercised at expiry.

ii. If at expiry, $8.00 < P_{spot} < $9.00, the option can be exercised i.e.; you can get up to a maximum gain of about $8.99 - $8.00 = $0.99.

iii. If at any point in the life of the barrier option, the underlying asset touches or passes the lower barrier of $7.00 or touches or passes the high barrier of $9.00, the knock-out will occur and the right to participate in the underlying is terminated. This double-barrier option becomes worthless.

Since it can be knocked out at both ends, the barrier option will be cheaper than a straight call option and put option combined with the same specifications.

6.4 Knock-in (or Kick-in) Barrier Options

For knock-in barrier options, the reverse happens i.e., the option becomes activated if the price of the underlying asset touches the knock-in barrier level. The terms or rules of the option become active from the time that the barrier is touched till the expiry, even if the price falls back. A rebate is usually paid to the buyer of the option at expiry if the knock-in barrier is never reached.

6.5 Summary and Application of Barrier Options

Unlike regular European call and put options, barrier options are path dependent options.

Knock-in barrier options only come into existence when certain conditions are met; typically, when the barrier is touched anytime during the life of the structure. Once this barrier is breached, the underlying call or put option becomes 'live'. Upon becoming 'live', the underlying call or put option carries the usual characteristics of a European call or put option.

In the case of knock-out barrier options, the underlying European call or put option is extinguished when the barrier is touched anytime during the life of the structure. Once the barrier is breached, the entire structure expires worthless. Table 6.5 provides an example on the behaviour of barrier options when barrier is exceeded.

Table 6.5 – Behaviour of Barrier Options When Barrier is Exceeded

Type of Barrier Option	When Barrier is Exceeded
Knock-Out	Deactivates option / the underlying option is extinguished
Knock-In	Activates option / the underlying option becomes 'live'

6.5.1 Advantages of Barrier Options

The advantages of a long position in barrier options are listed below:

a. Barrier options are cheaper and require a lower premium compared to standard options. The possibility exists that a knock-out barrier option will terminate before expiration (or a knock-in barrier option will never become active during its lifetime). Double knock-in option has higher knock-in probability than single knock-in option;

b. Higher returns on capital investment for the option buyer if barrier conditions are satisfied (i.e., there is no knock-out event or if the knock-in event occurs) as the investment sum is lower due to cheaper cost;

c. Greater selection of option products available for investors with diverse market views and with the benefit of leverage on capital invested; and

d. Knock-in barrier options are considered ideal for speculating large market moves, whereas knock-out barrier options are ideal for small moves in a sideways market.

6.5.2 Disadvantages of Barrier Options

The disadvantages of holding a long position in barrier options include:

a. High risk of loss of premium if barrier conditions are breached (i.e., knock-out event occurs or if there is no knock-in event); and

b. They are OTC products. The investor pays upfront for the option premium. They do not trade on an exchange and the investor is exposed to counterparty risk on the issuer, which arises when dealing in OTC products.

We will be covering knock-in and knock-out barriers again when we review various types of structured products in Chapter 7. For now, it suffices to say that the simplicity of the concept has helped, as clients can easily understand the payoff implication and possible outcomes. The knock-in and knock-out options are particularly actively used by banks in foreign currency derivative structures. Table 6.5.2 illustrates a simple example of a typical knock-out barrier option used in structured products sold.

Table 6.5.2 – Example of Structured Product with Embedded Knock-Out Option

Structured Product with Embedded Knock-Out Option

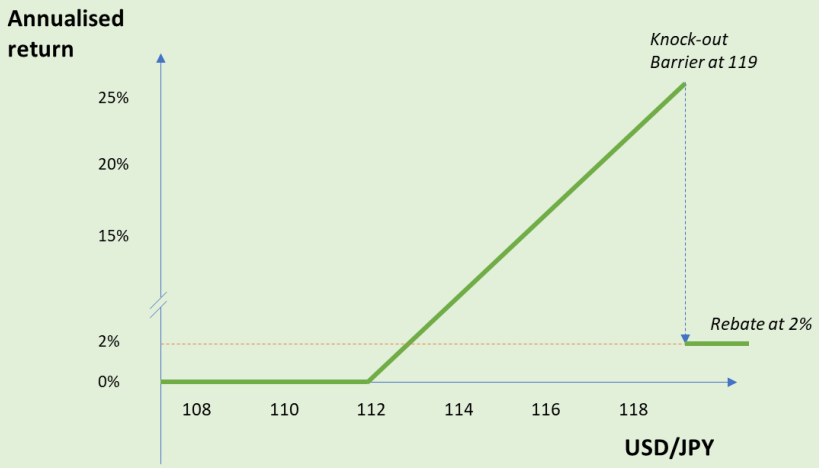
Your client is mildly bullish about the USD/JPY. She believes that the USD will strengthen slightly against JPY.

Current Spot	:	112.00
Strike Price	:	112.00
Knock-out Barrier	:	119.00
Minimum Yield	:	0%
Knock-out Rebate	:	2%
Money Market Rate	:	1.5%
Maturity/Expiry	:	3 months
Participation Rate	:	100%

The minimum size required for this trade is USD 500,000.

The client has money market deposits, and the interest is invested in a knock-out barrier option, with a rebate of 2% to make it more attractive on the downside for clients.

The option contracted is active at inception and the client buys the option structure. As long as the underlying spot trades below the knock-out barrier level of 119, the structure allows the client to participate in the movement of the currency from USD/JPY 112-118.99.



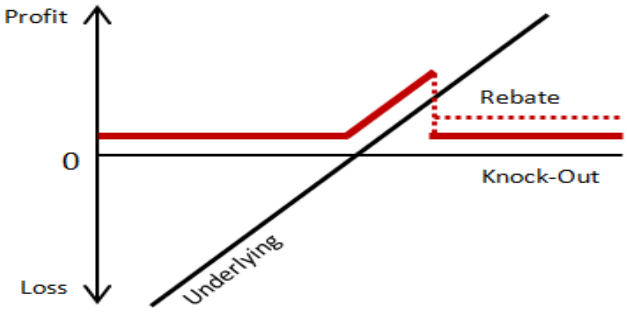
Once the underlying currency pair USD/JPY touches the barrier at 119, the right to participate in this upward movement of USD/JPY is terminated (deactivated/extinguished) and a fixed minimum annualized yield of 2% (rebate) is paid back together with the principal. This is an example of a structured product with an embedded up-and-out USD call /JPY put option.

6.6 Application of Barrier Options in Structured Products

6.6.1 Barrier Capital Preservation Certificates

This product contains an up-and-out barrier knock-out call. Investors refer to this product as "shark's fin" because of the shape of its payoff diagram. The investor receives a capped return at maturity if the knock-out event (the price hitting the barrier level) does not occur. If it does, the investor will receive the agreed capital amount. Figure 6.6.1 shows the payoff diagram for barrier capital preservation certificate.

Figure 6.6.1 - Payoff Diagram for Barrier Capital Preservation Certificate



Investment View

a. Rising underlying asset price.

b. Sharply falling underlying asset price possible.

c. Underlying asset price does not touch the barrier during its lifetime.

Characteristics

a. Minimal redemption at expiry equivalent to the capital preservation.

b. Level of capital preservation is defined as a percentage of the nominal (e.g., 100%). Capital preservation refers to the nominal only at maturity, and not to the purchase price.

c. Value of the product may fall below its capital preservation during the lifetime.

d. Participation in a positive performance of the underlying until knock-out.

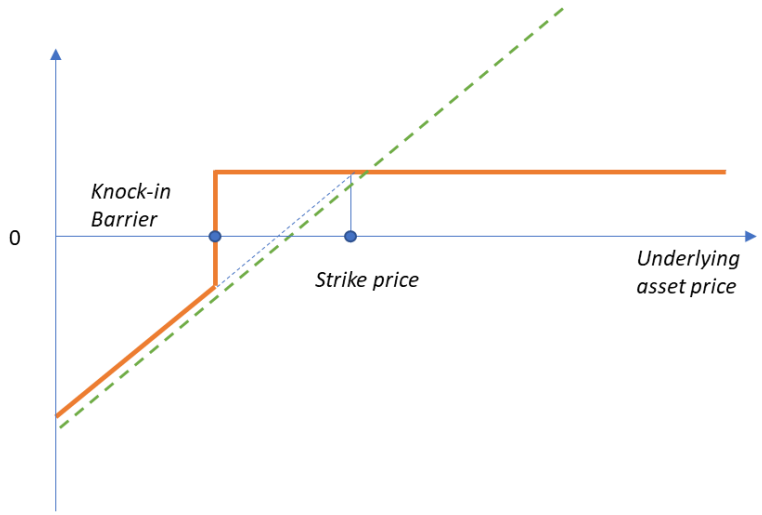
e. Possible payment of a rebate following a knock-out.

f. Capped upside.

6.6.2 Barrier Reverse Convertibles

The barrier reverse convertible is used to enhance the yield for investors when the underlying is expected to be flat or to rise or fall slightly. It offers conditional capital preservation through a kick-in barrier level that is set at specific percentage below the share price during issue of the instrument. The structure is also commonly called a reverse convertible note, comprising typically of a note, money-market instrument or a bond, and a put option sold by the investor to the issuer. Figure 6.6.2 shows the payoff diagram of a barrier reverse convertible.

Figure 6.6.2 - Payoff Diagram of Barrier Reverse Convertible



Investment View

a. Neutral view on direction of underlying asset.

b. Expect flat to very slight movement up or down with falling volatility.

c. Underlying asset price will not breach barrier during product lifetime.

d. Appeals to investors who are looking for income or enhanced yield.

Characteristics

i. By selling the knock-in put option, the investor is taking downside risk (i.e., if the underlying asset price is below the strike at expiry, and if the kick-in level was breached, the investors will lose capital). The principal is at risk if the barrier is breached / touched.

ii. The coupon is always paid regardless of market direction of the underlying asset price. Higher coupons will be required if there is more volatility in the underlying asset price.

iii. The maximum yield is earned as long as the price closes above the strike level or never touches the barrier kick-in level during the life of the investment. When this happens, the investor will be repaid 100% of his invested amount.

iv. However, if the barrier kick-in level is touched any time before maturity, the instrument is converted into a standard reverse convertible. Regardless of whether the kick-in level is touched or otherwise, the coupon is always paid out.

v. The risk to the investor is if the underlying asset falls enough in value to reach the kick-in level such that there is no longer capital preservation.

vi. Payout at maturity (only if barrier has been breached) =

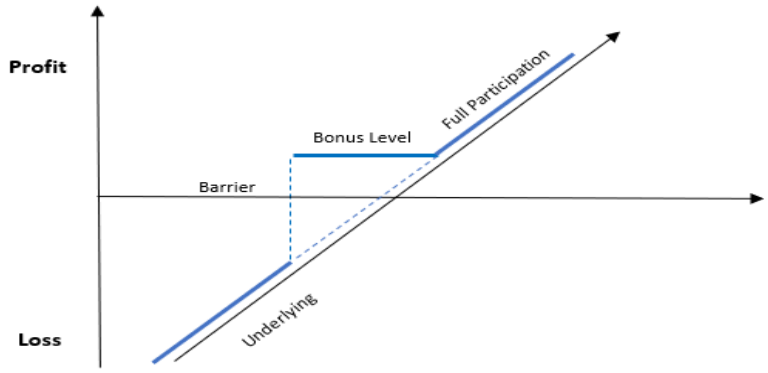
$$= \text{Min}\left\{100\%, 100\% - \frac{1}{\text{Strike Price}} \times \text{Max}\left[0, \text{Strike Price} - \left(\frac{\text{Underlying spot at maturity}}{\text{Underlying spot at inception}}\right)\right]\right\}$$

vii. There exists a popular knock-in variation where the investor is short a down-and-in barrier knock-in put option on the underlying asset with a strike at-the-money (i.e., the strike price is equal to the spot price).

6.6.3 Bonus Certificates

Investors in bonus certificates are looking for participation in the upside of the underlying with some protection on the downside. The expectation is for the underlying asset price to rise. The strike price or bonus level is typically set above the initial price of the underlying asset at the issue of the certificate, while the barrier knock-out level is set below the initial price. Figure 6.6.3 shows the payoff diagram for a bonus certificate.

Figure 6.6.3 - Payoff Diagram of Bonus Certificate



The barrier level is the price which the underlying must not touch or drop down during the lifetime of the certificate. If the underlying trades below the barrier level, the bonus mechanism and protection against loss is forfeited.

The bonus level refers to the redemption amount paid as long as the barrier is never touched during the lifetime of the certificate. If the underlying trades between the barrier level and the bonus level, the investor will receive the bonus paid on top of the principal amount invested. If the underlying closes above the bonus level, the investor will enjoy full participation the underlying upside.

Investment View

a. Investors are neutral or generally bullish that the underlying asset ~~that~~ will not trade below the barrier.

Characteristics

i. Offers exposure similar to direct investment in underlying, with benefit of a minimum return (the "Bonus") if underlying performance is flat or slightly negative. Bonus is available so long as the underlying asset is above the barrier.

ii. If the underlying asset is below the barrier, investors have 1:1 exposure to the downside.

iii. Barrier is usually observed continuously throughout the term of the product.

iv. Investors will forego the dividend payments of the underlying asset (which is partly used by the issuer to cover the cost of purchasing the put option from the investor).

v. If there are caps introduced to limit the upside, the bonus certificate will be priced cheaper of bonus minimum yield may be increased.

Risks

- The investor may suffer a loss equivalent to the negative performance of the underlying.
- The investor surrenders dividends in favour of the strategy.
- Credit risk of the issuer.

For the structurer/issuer, there are four variables he can play with to make it more attractive for different investors:

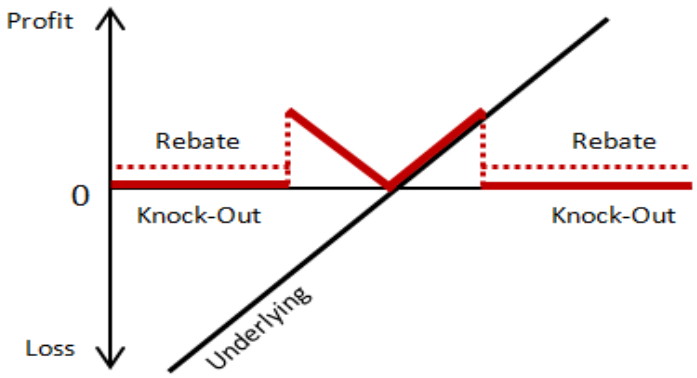
- The level of participation and capping upside;
- The strike price level or bonus level;
- The knock-out barrier level; and
- The maturity of the instrument.

There are trade-offs that the issuer and the investors have to make. Lowering the level of participation in the future upside and capping yields will allow for higher bonus levels, or further knock-out barrier levels. Lowering the bonus level and setting it closer to the initial price may make possible to bring the knock-out barrier closer or shortening the maturity. Likewise, lengthening the maturity will raise the risk for investors, therefore a higher bonus level may be allowable.

6.6.4 Barrier Capital Preservation Certificates (Straddle)

This product contains two knock-out options - barrier call and a barrier put. The investor will receive a return at maturity, which is capped, if no knock-out event occurs. There are 2 possible knock-out events, one for each of the options, and if it occurs, the investor receives the agreed capital amount. Figure 6.6.4 shows the payoff diagram of a barrier capital preservation certificate (straddle).

Figure 6.6.4 - Payoff Diagram of Barrier Capital Preservation Certificate (Straddle)



Investment View

a. No firm view on direction of underlying asset.

b. No large price swings (underlying asset price remains within a range).

c. Expect realized volatility to be higher than current implied volatility (i.e., strategy is cheap relative to likely payout).

Characteristics

i. Minimal redemption at expiry equivalent to the degree of capital preservation.

ii. Level of capital preservation is defined as a percentage of the nominal (e.g., 100%). Capital preservation refers to the nominal only, and not to the purchase price.

iii. Value of the product may fall below the nominal rate of capital preservation during its lifetime.

iv. The product contains an upper and lower knock-out barrier.

v. Participation in the rising or falling of the underlying asset until a barrier is breached.

vi. Possible payment of a rebate following a knock-out.

vii. Limited profit potential.

6.6.5 Positioning of the Barriers

The positioning of the barriers can make a big difference in the price of barrier options. In setting the terms of the options, including the price level for barrier, the clients' demands, and needs are always taken into consideration (otherwise there may be no interest generated for the products). The bank's exposure and profit potential as the counterparty are also important considerations.

There are some standard trade-offs in creating the option structures. Ceteris paribus, the closer the barrier is to the current price, the more likely the barrier will be reached, making the knock-in option more expensive. By the same token, ceteris paribus, the closer a barrier is to the current price, the cheaper the knock-out option will be.

For example, consider an up-and-in barrier call option. If investors have strong expectations of significant upward volatility in the price movement, they would be more willing to set the barrier to the farther end of the potential price movement to get a cheaper price for the option. If investors have less confidence with

regards to the upward movement of the underlying asset, the barrier should be set closer to the current price level of the underlying asset. Hence, clients can negotiate with bankers to set a preferred barrier but of course there will be a different price to every option.

6.7 Binary Options

Binary options (also called **digital options** or **one-touch options**) have two outcomes - paying out a specific yield or amount if the underlying asset reaches the pre-specified level, but zero if it does not. The one-touch option pays out as long as the price level is touched regardless of where the price level is at expiry.

The European variation, sometimes called the **all-or-nothing option** (or "cash-or-nothing" option), requires the underlying asset price to stay above the pre-specified price at expiry (for call options) or to be below the pre-specified price level at expiry (for put options) for payouts to be made. If they do not satisfy the criteria, nothing is paid.

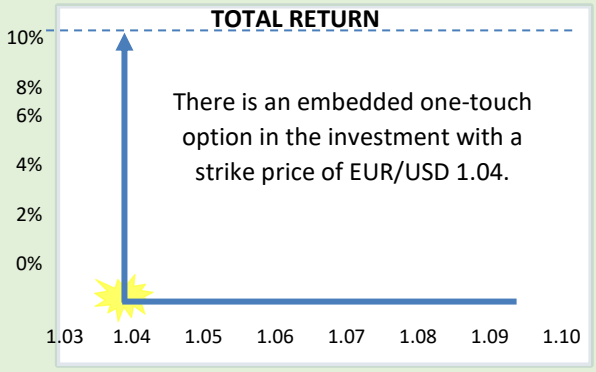
One-Touch Option

Your client holds a bearish view about EUR/USD (i.e., bearish about the EUR and bullish about USD). Due to upcoming elections in various parts of Europe, your client believes that he can profit from a temporary fall in EUR. As his banker, you can suggest the following solution to capitalize on this opportunity based on his view the bank takes the interest earned from his money market deposit to invest in a one-touch option for the possible sharp weakening of the EUR/USD.

If the EUR/USD really reaches the strike price of 1.04, the maximum yield of 10% will be paid out at maturity. If the EUR/USD does not touch the strike level, the minimum interest 1% will be paid. The principal will be repaid in full regardless of the outcome. Hence although the money market rate is 2.00%, the client is willing to sacrifice 1% for the possibility of a 10% return.

Current spot price : 1.08
One Touch Strike : 1.04
Maturity/Expiry : 3 months
Maximum Yield : 10.00%
Minimum Yield : 1.00%
Money Market Rate : 2.00%
Minimum amount to be invested: $100,000

The client is sacrificing about 1% to buy the one-touch option, while the bank believes there is a low probability that the strike level will be reached.



Another popular variation is a combination known as a **barrier binary option** where a higher fixed payment interest is provided when the underlying asset price touches a level, a lower fixed payment or interest paid if that does not happen. This variation can be a knock-in triggering the higher payment when the price rises to reach a predetermined level, or it may be a knock-out version which pays a lower amount once the underlying falls and touched a barrier. Adding another barrier will make the structure **a double barrier binary option (double no touch)** which requires the price level to stay within the barrier for a higher payment and to receive zero (or a lower payment) if either barrier is touched.

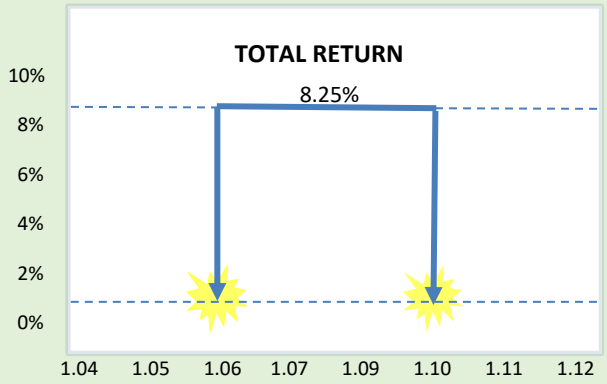
Another common structured product offered to clients is the **range currency deposit.** This product is typically offered by banks to clients who have excess deposits. As long as the currency pair stays within a certain range, clients will receive a better yield than otherwise available from the money market. If the currency breaches the range on either end, clients will get a lower yield.

Range Currency Deposit (Double Barrier Binary Option) (Double No-Touch)

The research team from your bank believes that EUR/USD will stay relatively range bound between EUR/USD 1.06 to 1.10 for the next 3-6 months.

Current spot price	: 1.09
Upper Barrier	: 1.10
Lower Barrier	: 1.06
Maturity/Expiry	: 3 months
Maximum Yield	: 8.25%
Minimum Yield	: 2.00%
Money Market Rate	: 1.50%

The minimum amount set as a requirement by the bank for this trade is USD 100,000.



Your client invests in this range currency deposit and earns an 8.25% annualized return as long the underlying EUR/USD stays within 1.06-1.10. If the EUR/USD touches or moves beyond 1.06 or 1.10, the returns will fall to 2.00%. So, there is a binary outcome here, either +8.25% or 2.00%.

In all the above variations, there are only two payment outcomes, i.e., the outcome is always binary. Some clients do not want to risk having a sharply lowered deposit return or in some cases, zero returns once the barriers are touched. In this case, what happens for 88 days out of 90 days when the EUR/USD stays within the 1.06-1.10 barrier, but for 1 day one of the barriers is touched? The answer is painful for the client, as he gets only 2% since it is a binary outcome.

Some clients prefer an accrual approach where the interest is accrued for days within the barriers earning the higher rate with the lower rate applied when the underlying asset falls outside the barrier. A range accrual option is more expensive than a double no-touch. Hence, in practice either the yield will drop, or the barrier (range) will narrow. In this example, let us suppose that for 60/90 days, EUR/USD falls within the barrier, and for 30/90 days, EUR/USD falls outside the barrier. The total return would be [(60/90 x 8.25%) + (30/90 x 2.00%)] = 5.50% + 0.67% = 6.17%.

6.8 Basket Options

Basket options are options on customized sets of underlying assets, typically a few indices or selected stocks. The pricing model for such options is similar to equity index options except there are fewer underlying assets. The correlation of the assets in the basket is calculated to gauge the volatility of the price of the basket of assets. The volatility of the price of the basket of assets will be lower than the weighted average of the different

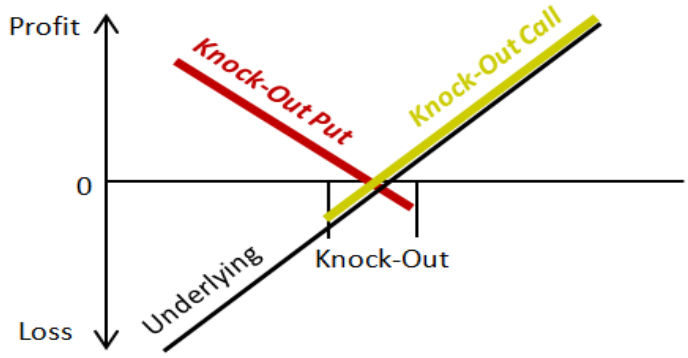
price volatility of the assets in the basket, so the basket option will be cheaper than the cost of individual options on each asset in the basket.

For clients who want to buy call options on a number of stocks, provided the call option is based on the performance on a basket of these selected stocks, it would be a cheaper alternative. However, there is correlation risk to the volatility of the price of the basket of assets and when markets turn sharply negative, correlation can increase significantly. A basket option can also be applied in the case of portfolio insurance where a put is purchased on a portfolio or basket of stocks held. This would be cheaper than buying individual put options.

6.9 Knock-Out Warrants

Knock-out warrants are similar to regular warrants except that they contain a knock-out or barrier feature. If the underlying asset touches the barrier, the warrant terminates. There are call warrants and put warrants for investors to trade a rising or falling market. In some markets, a knock-out warrant is called a callable bull/bear contract. Figure 6.9 shows the payoff diagram of a knock-out warrant.

Figure 6.9 - Payoff Diagram of Knock-Out Warrant



Investment View

a. Knock-Out (Call): Rising underlying asset value.

b. Knock-Out (Put): Falling underlying asset value.

Characteristics

i. A small investment that generates a leveraged performance relative to the underlying.

ii. Increased risk of total loss (limited to initial investment).

iii. Immediately expires if the barrier is breached during the product's life.

iv. Suitable for short-term speculation or hedging.

v. Flattening skew – sensitivity to volatility depends on the knock-out barrier and where the underlying is trading relative to the barrier.

vi. Continuous monitoring is required.

6.10 Callable Bull/Bear Contracts

A structured product that has gained popularity among retail investors in Asia is the **Callable Bull/Bear Contract (CBBC).** Based on the market outlook, investors can purchase either a:

a. **Bull contract:** Investor takes a bullish (positive) position on the underlying asset.

b. **Bear contract:** Investor takes a bearish (negative) position on the underlying asset.

Barrier options are embedded in CBBC investment products.

CBBCs are traded on the cash market of the exchange with settlement procedures similar to that of equities (T+2 days). They are issued by a third party, usually an investment bank, and are independent of the underlying asset and the exchange on which it is listed.

A CBBC tracks the performance of an underlying asset (assets) which could include:

i. Domestic single stocks;
ii. Domestic equity indices;
iii. Overseas single stocks;
iv. Overseas equity indices;
v. Currencies;
vi. Commodities; or
vii. Any other asset baskets as allowed by the stock exchange and deemed suitable by the issuer.

6.10.1 Price Movement of a CBBC (in relation to the underlying asset)

The pricing mechanism of a CBBC is simple and based on the underlying asset. There is greater transparency for CBBCs as compared to other structured products.

Price changes of a CBBC tend to closely follow the prices changes of the underlying asset as the delta of the CBBC is close to 1 ($\Delta \cong 1$). Hence, if the value of the underlying asset increases, the Bull CBBC (with conversion ratio of 1:1) will increase in value by approximately the same amount. For a Bear CBBC (with conversion ratio of 1:1), its value will decrease by approximately the same amount as the underlying asset. The conversion ratio is the number of underlying assets that each unit of CBBC is obligated to buy or sell. For example, a conversion ratio of 10:1 means that 10 units of a CBBC control 1 unit of the underlying asset.

6.10.2 Lifespan and Mandatory Call Event

CBBCs have a fixed lifespan (determined by the issuer) but they contain a mandatory call feature. This means that CBBCs may be called immediately by the issuer under certain conditions. The call will occur if the price of the underlying asset reaches a specified price level (the “call price”) at any time before expiry. Such a trigger event is referred to as a “Mandatory Call Event” (MCE). When this knock-out event occurs, the CBBC will expire early, and its trading terminates immediately.

An MCE is triggered before the expiry date if:

a. For a Bear Contract, the Spot Price touches or exceeds the Call Price.
b. For a Bull Contact, the Spot Price touches or falls below the Call Price.

6.10.3 Gearing Effect

CBBCs are leveraged products. Investors are only required to commit a cash sum which is a small proportion of the full value of the underlying asset. Hence, return on capital will be magnified compared to a direct investment in the underlying asset. Similarly, the percentage loss suffered by the investor will also be magnified in the event of adverse price movements in the underlying asset.

The effect of price movement on the CBBC investment can be calculated by finding the Effective Gearing. As a CBBC is a derivative-based product with delta close to 1 ($\Delta \cong 1$), the value of its gearing ratio is therefore equivalent to effective gearing. Hence, as a leveraged product, there is greater risk involved when investing in CBBCs. Before making an investment decision, the investor must be aware of the product's risk profile and assess his own investment objectives and risk appetite.

The gearing ratio of a CBBC investment is calculated as follows:

$$\textbf{Gearing Ratio} = \frac{\textbf{Underlying Asset Price}}{\textbf{Price of CBBC x Conversion Ratio}}$$

Effective gearing helps to quicken the calculation of the price movement of the CBBC in response to the price movement of the underlying asset. For instance, a CBBC with an effective gearing of 10 times means that for each 1% change in the price of the underlying asset, the theoretical price of the CBBC may increase or decrease by 10%.

Generally, the closer the strike price is to the spot price of a CBBC's underlying asset, the higher will be the gearing ratio of the CBBC. This means that the CBBC will experience greater price fluctuations when the asset's spot price is close to the strike price. Such a CBBC has a higher risk of being called.

6.10.4 Hedging Instrument (for Bear Contracts)

CBBCs can be used to hedge market risk. An investor holding assets with a view that they will grow in value in the long term might wish to protect his investment from short-term volatility. For instance, an investor has an equity portfolio that mirrors the STI. To hedge his position, he can buy CBBC bear contracts on the STI. If the Singapore equity market declines, a fall in his equity portfolio value will be offset by a gain in his CBBC position.

6.10.5 Categories of CBBC

There are two categories of CBBCs - N-CBBC and R-CBBC.

6.10.5.1 Category N-CBBC (N = No residual value)

For this CBBC, the call price is equal to its strike price. The CBBC holder will not receive any cash payment once the underlying asset price reaches or exceeds the call price (i.e., when MCE occurs and the CBBC is called).

6.10.5.2 Category R-CBBC (R = Residual value)

For this CBBC, the call price is different from the strike price. The CBBC holder may receive a small amount of cash payment (Residual Value) when the CBBC is called.

The characteristics, similarities and differences of the various types of CBBCs are summarised in Table 6.10.5.

Table 6.10.5 - Characteristics of Callable Bull/Bear Contracts (CBBCs)

Category	BULL CONTRACT		BEAR CONTRACT	
	R-Category Bull Contract	N-Category Bull Contract	R-Category Bear Contract	N-Category Bear Contract
View	Bullish	Bullish	Bearish	Bearish
Call Price & Strike Price	Call Price > Strike Price	Call Price = Strike Price	Call Price < Strike Price	Call Price = Strike Price
Mandatory Call Event Occurs (knock-out)	When Asset Price FALLS & Hits Call Price	When Asset Price FALLS & Hits Call Price	When Asset Price RISES & Hits Call Price	When Asset Price RISES & Hits Call Price

6.10.6 Factors Affecting CBBC Pricing & Valuation

The price of a CBBC is affected by several variables which can be broadly classified into theoretical and market factors.

6.10.6.1 Theoretical Factors

As a derivative product, the price of a CBBC is directly linked to the underlying asset. Financial costs are also involved in structuring the product. The theoretical factors determine the intrinsic value of the CBBC (the simple definition of intrinsic value is the difference between the underlying asset price and the strike price). The main variables under theoretical factors are the price movement of underlying asset, strike price and financial cost.

a. **Price Movement of Underlying Asset -** The major component of a CBBC is its intrinsic value. Hence, the movement in price of the underlying asset has a major impact on the CBBC price.

b. **Strike Price -** Assuming that the price of the underlying asset, maturity date and CBBC conversion ratio are unchanged; the higher the strike price, the lower will be the intrinsic value of a Bull CBBC contract (conversely, the higher the strike price, the higher will be the intrinsic value of a Bear CBBC contract).

c. **Financial Cost -** The issuer incurs funding and other costs in structuring the CBBC. It includes the issuer's cost of borrowing, adjustments for dividends (assuming the underlying asset is equity-related) and the issuer's profit margin.

The financial cost will be higher the longer the maturity of the CBBC and it declines over time as the CBBC moves closer to expiration date. It is included in the price of the CBBC at the time of issuance and details will be found in the Listing Document provided by the issuer.

In summary, the intrinsic value of a CBBC is the difference between the underlying asset price and the strike price (adjusted for the conversion ratio) after taking into account the financial cost. The greater the difference, the higher is the intrinsic value.

For Bull and Bear CBBCs, the intrinsic value is computed as follows:

CBBC Theoretical Price of:

$$\textbf{Bull Contract} = \frac{\textbf{(Underlying Asset Price-Strike Price) + Financial Cost}}{\textbf{Conversion Ratio}}$$

$$\textbf{Bear Contract} = \frac{\textbf{(Strike Price-Underlying Asset Price) + Financial Cost}}{\textbf{Conversion Ratio}}$$

6.10.6.2 Market Factors

Market and trading conditions influence the price of a CBBC. Market variables include the liquidity of the underlying asset, CBBC demand and supply and issuer factors.

i. **Liquidity of Underlying Asset -** An important characteristic of CBBCs is that delta is close to 1 ($\Delta \cong 1$; to be more precise, for Bull contracts $\cong +1$ and Bear contracts $\cong -1$). Issuers usually hedge their position against the CBBC that has been issued to investors. If the market for the underlying asset has poor liquidity, the issuers' hedging cost will rise and this in turn affects the price and liquidity of the CBBC.

ii. **Demand and Supply Factors -** Demand and supply factors for the specific CBBC on a day-to-day basis may impact the CBBC price.

iii. **Issuer Factors -** The issuer of the CBBC, usually an investment bank, makes a market and provides liquidity to the CBBC investors. In a stable market, the bid-ask spread is tight and transaction volumes are able to meet investor demand. If there is any market disruption, or if an issuer-specific development occurs that impacts the issuer's operations, it can adversely affect the trading of the CBBC.

6.10.7 Determination of Residual Value if Mandatory Call Event Occurs

For Category N-CBBCs, there is no residual value if a call event occurs.

For Category R-CBBCs, the residual value is determined as follows:

$$\textbf{Bull Contract} = \frac{\textbf{Settlement Price-Strike Price}}{\textbf{Conversion Ratio}}$$

$$\textbf{Bear Contract} = \frac{\textbf{Strike Price-Settlement Price}}{\textbf{Conversion Ratio}}$$

Where MCE Settlement Price: -

Bull contract: Not lower than the minimum trading price of underlying asset between the period of MCE up to the next trading session.

Bear contract: Not lower than the maximum trading price of underlying asset between the periods from the MCE up to the next trading session.

6.10.8 Valuation at Maturity (Expiry)

If a CBBC is not called before expiry, investors can hold it until maturity. Settlement at maturity is calculated with the following formulae:

Valuation at Maturity (Expiry)

$$\textbf{Bull Contract} = \frac{\textbf{Settlement Price-Strike Price}}{\textbf{Conversion Ratio}}$$

$$\textbf{Bear Contract} = \frac{\textbf{Strike Price-Settlement Price}}{\textbf{Conversion Ratio}}$$

Where MCE Settlement Price = Closing price of underlying asset on settlement day.

6.10.9 Capital Adjustments

If the CBBC is based on an underlying share (or an equity-related asset class), an adjustment is needed if there is a corporate action such as bonus issues, rights issues, share splits and reverse share splits. No action is needed for regular share dividends as it would have been considered by the issuer as part of the funding cost. Bonus shares, special and extraordinary dividends will require adjustments.

6.10.10 Other Issues Related to Trading on the Singapore Exchange

For definition of terms and details of specific dates (such as last trading day, expiry day, settlement day), the investor must be familiar with the rules of the exchange and the specific terms of the CBBC, which can be found in the Listing Document provided by the issuer.

6.10.11 Callable Bull/Bear Contracts (CBBCs) Examples & Illustrations

There are 2 categories of CBCCs:

a. N-CBBC; and
b. R-CBBC.

Several profit/loss calculations are shown in the examples below.

Category R – Bull Contracts

Company: ABC Ltd		
Strike Price:		$10.00
Call Price:		$11.00
Conversion Ratio		10 : 1
Financial Cost (annual rate)		10%
Time To Maturity (months)		12
Initial Spot Price of ABC:	=	$12.00
Price of Contract	=	[($12.00 - $10.00) + ($10.00 x 10.0% x 1)] ÷ 10
	=	$0.30

Scenario 1: Contract Sold Before Maturity		
Strike Price:		$10.00
Call Price:		$11.00
Conversion Ratio:		10: 1

Financial Cost (annual rate):		10%
Time To Maturity (months):		6 months
Current Spot Price of ABC:		$15.00
Price of Bull Contract:	=	[($15.00 - $10.00) + ($10.00 x 10% x 6/12)] ÷ 10
	=	$0.55
Rise in ABC Share price ($):	=	$15.00 - $12.00 = $3.00
Rise in ABC Share price %	=	[($15.00 - $12.00) ÷ ($12.00)] x 100%
	=	25.00%
Profit on Each Bull Contract ($):	=	$0.55 - $0.30
	=	$0.25
Profit on Each Bull Contract (%):	=	[($0.55 - $0.30) ÷ $0.30] x 100%
	=	83.33%

Scenario 2: Mandatory Call Event Triggered Before Maturity		
Strike Price:		$10.00
Call Price:		$11.00
Conversion Ratio:		10: 1
Financial Cost (annual rate):		10%
Time To Maturity (months):		6
Current Spot Price of ABC:		$10.50
		Price has fallen and closed below the Call Price of $11. The settlement price is $10.50.
Residual Value:		When a Mandatory Call Event occurs, the payoff for a Bull contract is the Residual Value:
	=	($10.50 - $10.00) ÷ 10
	=	$0.05
Fall in ABC Share price ($):	=	$12.00 - $10.50 = $1.50
Fall in ABC Share price %	=	[($12.00 - $10.50) ÷ ($12)] x 100%
	=	12.50%
Loss on Each Bull Contract ($):	=	$0.30 - $0.05
	=	$0.25
Loss on Each Bull Contract (%):	=	[($0.55 - $0.30) ÷ $0.30] x 100%
	=	83.33%

Scenario 3: Contract Held to Maturity		
Strike Price:		$10.00
Call Price:		$11.00
Conversion Ratio:		10 : 1
Financial Cost (annual rate):		10%
Time To Maturity (months):		0
Current Spot Price of ABC:		$15.00 => The settlement price is $15.00
Maturity Value of Bull Contract:	=	($15.00 - $10.00) ÷ 10
	=	$0.50
Rise in ABC Share Price ($):	=	$15.00 - $12.00 = $3.00

Rise in ABC Share Price (%):	=	[($15 - $12) ÷ ($12)] x 100%
	=	25.00%
Profit on Each Bull Contract ($):	=	$0.50 - $0.30
	=	$0.20
Profit on Each Bull Contract (%):	=	[$0.50 - $0.30] ÷ $0.30] x 100%
	=	66.67%

Category–R - Bear Contracts

Company: ABC Ltd		
Strike Price:		$11.00
Call Price:		$10.00
Conversion Ratio		1 : 1
Financial Cost (annual rate)		10%
Time To Maturity (months)		12
Initial Spot Price of ABC:	=	$9.00
Price of Contract	=	[($11.00 - $9.00) + ($11 x 10.0% x 1)] ÷ 1
	=	$3.10

Scenario 1: Contract Sold Before Maturity		
Strike Price:		$11.00
Call Price:		$10.00
Conversion Ratio:		1 : 1
Financial Cost (annual rate):		10%
Time To Maturity (months):		6
Spot Price of ABC Share:	=	$7.50
Price of Bear Contract:	=	[($11.00 - $7.50) + ($11.00 x 10% x 6/12)] ÷ 1
	=	$4.05
Fall in ABC Share Price ($):	=	$9.00 - $7.50
	=	$1.50
Fall in ABC Share Price (%):	=	[($9.00 - $7.50) ÷ ($9.00)] x 100%
	=	16.67%
Profit on Each Bear Contract ($):	=	$4.05 - $3.10
	=	$0.95
Profit on Each Bear Contract (%):	=	[($4.05 - $3.10) ÷ $3.10] x 100%
	=	30.65%

Scenario 2: Mandatory Call Event Triggered Before Maturity		
Strike Price:		$11.00
Call Price:		$10.00
Conversion Ratio:		1 : 1
Financial Cost (annual rate):		10%
Time To Maturity (months):		6
Spot Price of ABC Share:	=	$10.50
		Price has crossed and closed above the Call Price of $10.00. The settlement price is $10.50.
Residual Value:		When a Mandatory Call Event occurs, the payoff for a Bear contract is the Residual Value:
	=	($11.00 - $10.50) ÷ 1
	=	$0.50
Rise in ABC Share Price ($):	=	$10.50 - $9.00
	=	$1.50
Rise in ABC Share Price (%):	=	[($10.50 - $9.00) ÷ ($9.00)] x 100%
	=	16.67%
Loss on Each Bear Contract ($):	=	$3.10 - $0.50
	=	$2.60
Loss on Each Bear Contract (%):	=	[$3.10 - $0.50] ÷ $3.10] x 100%
	=	83.87%

Scenario 3: Contract Held to Maturity		
Strike Price:		$11.00
Call Price:		$10.00
Conversion Ratio:		1 : 1
Financial Cost (annual rate):		10%
Time To Maturity (months):		0
Current Spot Price of ABC:		$6.50 => The settlement price is $6.50
Maturity Value of Bear Contract:	=	($11.00 - $6.50) ÷ 1
	=	$4.50
Fall in ABC Share Price ($):	=	$9.00 - $6.50
	=	$2.50
Fall in ABC Share Price (%):	=	[($9.00 - $6.50) ÷ ($9.00)] x 100%
	=	27.78%
Profit on Each Bear Contract ($):	=	$4.50 - $3.10
	=	$1.40
Profit on Each Bear Contract (%):	=	[$4.50 - $3.10] ÷ $3.10] x 100%
	=	45.16%

6.11 Risks of CBBCs

1. **Potential Loss of Capital -** Buying CBBC securities requires an upfront cash investment. The investor may take a bearish or bullish view, but the market may not move according to expectations. The investor's capital is at risk, and he must be aware that he can potentially lose the full amount of the invested capital.

2. **Mandatory Call -** CBBCs may not be suitable for all investors and the investor must consider his investment objectives and risk appetite. If the CBBC is called when the Call Price is breached, the payoff for the N-category investor is zero and the Residual Value for the R-category investor may be small. Once the CBBC is called, it is irrevocable. The investor does not get to profit from or recover any losses if the price of the underlying asset bounces back.

3. **Gearing Effect –** As CBBCs are leveraged products, the investor is exposed to greater downside risk if the market conditions are unfavourable. Negative returns are magnified when compared to a direct investment in the underlying asset.

4. **Limited Life -** A CBBC typically has a limited lifespan (determined by issuer). The price fluctuates during the life of the instrument and may become worthless upon expiry or when a MCE occurs.

5. **Price Movement - Trading Close to Call Price -** While the CBBC closely follows the movement of the underlying asset price, under certain market conditions where there is a demand-supply imbalance or turbulence in the wider market, the delta being approximately equal to 1 ($\Delta \cong 1$) may not always hold. Furthermore, when the underlying asset is trading very near the call price, the price of the CBBC can be more volatile with wider bid-ask spreads and uncertain market liquidity.

6. **Price Movement – Volatility -** In volatile markets where the price of the underlying asset is fluctuating greatly, there is a greater likelihood that the CBBC will be knocked out.

7. **Liquidity and Market Disruption -** If there is market disruption where there is no trading taking place and price is unavailable on the underlying asset, the market for the CBBC will also be affected. If the underlying asset is a company's stock and the share has been suspended or is faced with any other action, it will have an adverse impact on the trading of the CBBC.

8. **Financial Cost -** This is included in the price of the CBBC at the time of issuance. If the issuer's cost of borrowing increases in the interbank market, this cost component goes up and raises the price of the CBBC. The amount of the financial cost is charged upfront and includes the entire period up to the expiry date. In the event of early termination, the CBBC investor will lose the full cost of funding even though the CBBC was held for a shorter duration.

9. **Counterparty Risk -** CBBCs are issued by third party investment banks. In the event the issuer defaults or encounters any difficulties, the investor has to face the risk of any exposure to the issuer.

6.12 Similarities and Differences between Knock-Out Products and Structured Warrants (or Other Products)

	CBBCs (Knock-Out Products)	Warrants
Underlying asset	Equities, market indices, commodities, currencies, others	Equities, market indices, commodities, currencies, others
Trading method	Traded on a stock exchange	Traded on a stock exchange
Bullish view	Bull contract	Call warrant
Bearish view	Bear contract	Put warrant
Strike price (Exercise price)	Yes	Yes
Call price	Yes	No
Mandatory call mechanism	Yes - Early termination when Bull contract: underlying assets $\leqslant$ call price Bear contract: underlying assets $\geqslant$ call price	No early termination
Implied volatility	No - insignificant to pricing	Yes – affects pricing
Maturity	Yes	Yes
Holding cost	Financial costs will be deducted daily	Time decay - time value decreases gradually as product approaches maturity
Margin requirement	No	No
Duration	Determined by the issuer	Determined by the issuer
Maximum loss	Limited to investment amount	Limited to investment amount

6.13 Summary

1. The different features in options structures can be varied to create strategies and solutions to meet the different demands of clients and to take advantage of opportunities available in the market. These will include:
   i. The **difference in the timing** for options to be exercised;
   ii. **Uncertainty about the time to expiration** when the options are exercised;

iii. **Payoff computation** – The price level used to calculate the payoff;

iv. **Conditional events or trigger points** – If the underlying price level touches or crosses a trigger point;

v. **Underlying assets** – The variety can extend beyond traditional asset classes; and

vi. **Currency features** – can be used to create payoff in a different currency.

2. For barrier options, the payoff to the investor will depend not only whether and how much the option is in-the-money at exercise, but also whether the underlying has done one of the following:

   i. Touching the barrier anytime during the life of the option,

   ii. Being above the barrier or below the barrier at expiry (or throughout the life of the option); or

   iii. Staying within two barriers or outside two barriers throughout the life of the structure.

3. **Barrier options are path dependent** because they are either activated or cancelled when a set trigger price level (the barrier) of the underlying asset is reached.

4. In the simplest form of barrier options, we have the **single barrier knock-out option** where the option gets "knocked out" (or otherwise termed kicked-out) once the underlying price reaches the barrier level. A **double barrier option** has two barrier levels (knock-out levels), with a barrier on either side of the strike price. For **knock-in barrier options** the reverse happens, i.e., the option becomes activated if the price of the underlying asset touches the knock-in barrier level.

5. Barrier options are cheaper and require a lower investment amount as compared to standard options, leading to higher potential returns on capital investment. With barrier options, there is a greater selection of options and structured products available for investors with diverse market views. A disadvantage of barrier options is the high risk of loss of investment capital if barrier conditions are breached when a knock-out event occurs. They are over-the-counter (OTC) financial products, exposing investors to issuer counterparty risk and there is a lack of liquidity.

6. **Barrier Capital Preservation Certificate** is called a **"shark's fin"** certificate because of the shape of its payoff diagram. The investor receives a capped return at maturity if the knock-out event (the price hitting the barrier level) does not occur. If it does, the investor will receive the agreed capital amount.

7. A **Barrier Reverse Convertible** is used to enhance the yield for investors when the underlying is expected to be flat or to rise or to fall slightly. It offers conditional capital preservation through a kick-in barrier level that is set at specific percentage below the share price during issue of the instrument.

8. Investors in **Bonus Certificates** are looking for participation in the upside of the underlying with some protection on the downside. The expectation is for the underlying to rise. The strike price or bonus level is typically set above the initial price of the underlying at the issue of the certificate, while the barrier knock-out level is set below the initial price.

9. In the **barrier capital preservation certificate,** there are two knock-out options - barrier call and a barrier put. The investor will receive a return at maturity, which is capped, if no knock-out event occurs. There are 2 possible knock-out events, one for each of the options, and if it occurs, the investor receives the agreed capital amount.

10. **Binary options** (also called **digital options** or **one-touch options**) have two outcomes - paying out a specific yield or amount if the underlying asset reaches the pre-specified level, but zero if it does not. The one-touch option pays out as long as the price level is touched regardless of where the price level is at expiry.

11. **Basket options** are options on customized sets of underlying assets, typically a few indices or selected stocks. The pricing model for such options is similar to equity index options except there are fewer underlying assets.

12. **Knock-out warrants** are similar to regular warrants except that they contain a knock-out or barrier feature. If the underlying asset touches the barrier, the warrant terminates. There are call warrants and put warrants for investors to trade a rising or falling market. In some markets, a knock-out warrant is called a callable bull/bear contract. The **callable bull/bear contract (CBBC)**is essentially a knock-out warrant. Based on the market outlook, investors can purchase a "bull" contract or a "bear" contract. Barrier options are embedded in CBBC investment products.

13. The pricing mechanism of a CBBC is simple and based on the underlying asset. Price changes of a CBBC closely follow the prices changes of the underlying asset as the delta of the CBBC is close to 1. There is greater transparency for CBBCs as compared to other structured products.

14. CBBCs have a fixed lifespan but they contain a mandatory call feature. The call will occur if the price of the underlying asset reaches a specified price level at any time before expiry. Such a trigger event is called a "Mandatory Call Event" and the CBBC will expire early, and its trading terminates immediately.

15. CBBCs are leveraged products and the return on capital can be magnified compared to a direct investment in the underlying asset. Similarly, the percentage loss suffered by the investor will also be magnified in the event of adverse price movements in the underlying assets. Hence, there is greater risk involved when investing in CBBCs.

16. CBBCs can be used by investors to hedge market risk. An investor may hold assets with a long-term view that they will grow in value but wants to protect his investment from market volatility in the near term.

17. There are two categories of CBBCs: N-CBBC (N = No residual value), where the holder will not receive any cash payment once MCE occurs and the CBBC is called, and R-CBBC (R = Residual value), where the holder may receive a small residual amount of cash payment when the CBBC is called.

18. The price of a CBBC is affected by several theoretical factors which determine the intrinsic value of the CBBC (intrinsic value is the difference between the underlying asset price and the strike price). The main variables under theoretical factors are the price movement of underlying asset, strike price and financial cost.

19. Market and trading conditions can also influence the price of a CBBC. Market factors include the liquidity of the underlying asset, CBBC demand and supply, and issuer factors.

20. If the CBBC is based on an underlying stock (or an equity-related asset class), an adjustment is made in the event of corporate actions such as bonus issues, rights issues, share splits and reverse share splits. No action is needed for regular share dividends as it would have been considered by the issuer as part of funding costs.

21. There are several risks to consider when investing in CBBCs: potential loss of capital, mandatory call, gearing effect, limited life, price movement close to call price, price volatility, liquidity and market disruption, financial cost, and counterparty risk.

Chapter 6: Appendix (Non-Examinable Content)

A6.1 Average Price (Rate) Options

For this type of option, the payoff depends on the average price of the underlying asset rather than a single observed data at expiry or when exercised. The average price can be calculated for the entire life of the option or for a specific period during the life of the option.

For the average price option, the average price for the underlying asset replaces the spot price. The payoff equation for such options are as follows:

Payoff for Call $_{(avg)}$ = Max {S_{avg} - X, 0}

Payoff for Put $_{(avg)}$ = Max {X - S_{avg}, 0};

Where: Call $_{(avg)}$ = the average rate or price of the call

and Put $_{(avg)}$ = the average rate or price of the put

and $S_{(avg)}$ = the average value of the underlying asset for the pre-specified period.

The benefit of average price options is that the cost of the option is usually less than an otherwise identical vanilla option. It is cheaper because the volatility of a moving average of price trend is typically less than the volatility of the individual observations of prices of the same price trend. The longer the period used for averaging, the greater the difference between the average price option and the vanilla option. Usually, the continuous sampling on a daily, monthly or quarterly is used and the arithmetic mean is calculated to determine the average price or rate (as in average rate currency options).

This average price option can be useful if there are a series of cash flow receipts or payments to be made and there is a need to offset the impact over the period with an average price rather than a tail end price at expiry. In such situations, using the tail end price at expiry may work to the investor's disadvantage.

Besides being cheaper than comparable vanilla options, when the average price options are used in issuing structured products, another benefit is that the product's payoff is less affected by unexpected spikes or plunges in the price of the underlying asset at the tail end near expiry. If a structured product includes an option where the pay-off depends upon the average of the asset price of the structure's life, the effect of any sudden, short term, change in the asset value near expiry will only affect the pay-off to the extent that it influences the average calculation. Hence, many banks use average pricing in their structuring of options.

Chapter 7: Structured Deposits and Other Structured Products

Learning Objectives
The candidate should be able to: ✓ Explain what structured products are and the reasons for their growth as investment vehicles. ✓ Explain the key components and features of structured deposits. ✓ Identify the main wrappers and types of structured products – deposits, notes, funds and investment-linked insurance policies. ✓ Understand the main strategies used in structured deposits. ✓ Discuss the risk-return trade-off for specific structured deposits. ✓ Explain the different client needs in using structured products. ✓ List the similarities and differences between different types of structured product categories.

7.1 What are Structured Products?

A structured product is a financial product that is designed to provide investors with more tailored investment alternatives linked to a specific risk /reward profile or hedging strategy options that addresses a view about the direction or volatility of an underlying asset or index. It is usually made up of a combination of two or more financial instruments, with one or more being a derivative structure created by an issuer (usually a financial institution).

The bulk of structured products are traded over-the-counter (OTC), with banks being counterparties to various types of individual and institutional investors. Over the years, some tailored structured products have become so popular and standardized that they have been featured on some exchanges.

There are a wide range of structured products but for the purposes of the CMFAS CM-SIP Examination, we will focus on the more common structures offered by financial institutions to clients.

Structured products can be applied to different underlying assets which can be an interest rate, a bond, an equity, an index, an exchange rate, or a commodity and can be wrapped with zero coupons into structured deposits, notes, or funds. We will feature those more commonly seen and traded.

Structured products are considered complex investments. Their risks and payoff structures may be less easily understood and are usually offered to the more experienced investors. In the past, losses from OTC derivatives and structured products have been the cause of lawsuits brought by retail and high net worth individuals against banks. In several cases, clients claimed that they did not understand, were not informed, or were even misinformed, about the risks associated with products. Hence, proper verification of clients' risk profiles and investment understanding, supported by documentation and acknowledgements from clients, has become standard practice amongst banks selling OTC structured products.

From the perspective of financial institutions, the expansion of derivatives over the last 30 years helped offset the narrowing of trading commissions for traditional asset classes offered by financial institutions. OTC structured products helped financial institutions increase the volume of transactional activity and generated higher returns than what would otherwise have been achieved from trading in traditional investments.

7.1.1 Client Suitability[32] for Structured Products

When helping clients invest in structured products, investment representatives should consider the following:

a. **Risk appetite** – This refers to an investor's ability and willingness to take risk.

b. **Investment experience and knowledge** - Investment representatives should check that investors have the relevant experience and knowledge to understand the structured products' features and risks before helping them to invest in any structured products. The investor must understand how the product works and review the possible outcomes under different market scenarios.

c. **Liquidity needs –** Investment representatives should review the liquidity needs of the clients and must highlight the consequences of early redemption when some products are not held till maturity.

d. **Investment horizon –** The value of certain structured products will only be fully realized if they are held to maturity. Some structured products have a longer time horizon than that which investors may be comfortable with. On the other hand, other derivatives structures may have a profile that display values decline over time due to time value decay and may actually become worthless at or near their expiry dates.

e. **Product risk –** Such risk relates to the product's features which can dramatically augment its payoff profile. Even though some structured products offer some capital preservation, there are other structured products where there is the possibility that an investor can sustain huge losses or build cumulative losses.

f. **Restrictions –** Structured products are classified as complex investments because they often contain embedded derivatives which need to be well understood by those who market these products, as well as the investors buying these products. Only investment representatives who are notified under the MAS Representative Notification Framework (RNF) are allowed to provide advice to investors who wish to trade in structured products, otherwise investors are required to have the relevant experience and knowledge to understand the features and risks of the structured products.

g. **Product disclosure –** Financial institutions and their representatives are expected to provide full disclosure of the potential risks to their clients. It is also important for the investors to understand how the returns

[32] On 7 October 2023, MAS, in conjunction with the Association of Banks in Singapore (ABS), Association of Financial Advisers (Singapore) (AFAS) and Life Insurance Association (LIA), launched a Basic Financial Planning Guide to help Singaporeans take steps to enhance their financial well-being. The Guide outlines a few rules of thumb, for individuals to start taking proactive steps to address their savings, insurance, and investment needs (Go.gov.sg).

are generated, by understanding how the underlying instruments perform under different market conditions.

The focus on promotion of structured products to lift transactional income had resulted in several cases of mis-selling and mismatches on client-product suitability for some risky structured products (e.g., accumulators, collateral debt obligations (CDOs)) in Singapore and Hong Kong.

Due to the incremental staff infrastructure and increased cost for documentation, compliance, regulatory reporting, risk monitoring, potential legal costs and bad publicity when things go awry, most financial institutions have cut back on promoting some of the more complex or riskier OTC structured products in recent years. Banks have also refocused on generating more recurring management fees compared to transactional income which can be more cyclical in nature and attract more regulatory and administrative costs.

7.1.2 Risks in Structured Products

There are different levels of risks associated with different structured products. Investors should always weigh their desire for additional returns against the potential and probable downside risk exposures for any structured investments.

At the low-risk end of the spectrum, some products offer no risk of capital loss (e.g., some structured deposits such as range accrual deposits), where only the opportunity cost of earning better yield is forgone in the worst-case scenario.

Going up the spectrum with the potential for higher returns, structured products (e.g., equity linked notes (ELNs), dual currency investments (DCIs)) will no longer offer any capital guarantee. Counterparty banks offering better yields will only do so if the probability of being knocked out is very high, and/or the risk of loss from having to take delivery of the underlying asset or converting to a weaker currency is also very high. There is no free lunch. Finally, risks are augmented when there are derivative instruments that have the potential of amplifying gains and losses by accumulating more of the underlying securities in response to larger price movements (e.g., accumulators).

In summary, structured products face various types of risks (note that the list below is not exhaustive):

a. **Market risk** – This refers to risk where the value of the underlying security falls in value. For the investor, this could mean losing the potential yield on an investment, sustaining book losses from having to acquire the underlying asset at prices above the current price or realizing losses by paying out the equivalent cash differential between the underlying asset price and the conversion price.

b. **Liquidity risk -** There is limited liquidity for structured deposits, as investors can only deal with the bank holding the structured deposits. There may be transaction costs or unwinding costs if the deposit is withdrawn early and the investor may not receive 100% of the invested money back.

c. **Reinvestment risk -** If a structured deposit is called or redeemed before maturity, the investor is exposed to reinvestment risk and may have to reinvest the funds at less attractive interest rates.

d. **Counterparty credit risk –** Investors receive the principal back if the structured deposit is held to maturity, provided the deposit-taking bank does not default. Unlike normal fixed deposits, structured deposits are not covered under the Deposit Insurance Scheme.

e. **Currency exposure** - As with all other financial instruments, the currencies that the structured products are denominated in would influence the overall return in relation to a client's reference currency. For foreign exchange (FX)-related structured products like DCIs, investors should note that there are embedded short put option on the alternative currency.

f. **Leverage effect** – The size of gains and losses can be augmented. For example, some structured products have "step-up" features which cause higher amounts of the underlying asset to be accumulated when there are large percentage falls in the underlying asset price.

7.1.3 Trade-off between Features in Structured Products

Structured products can be customized to some degree to fit the profile and needs of a client. Hence, the client will have to consider how much risk they are willing to take on in order to obtain a higher yield or return. Table 7.1.3 sets out the risks and return trade-offs that investors need to take into consideration when investing in structured products.

Table 7.1.3 – Risk and Return Trade-Off Considerations for Investors

Structured Product Features	Return / Yield	Risk and Return Trade-off Considerations
Investor sells a call to the issuer	Increase	The closer the call strike price to current price, the higher the likelihood for the issuer to be able to exercise the call. Therefore, there is a higher yield offered.
Investor sells a put to the issuer	Increase	The closer the put strike price to current price, the higher the likelihood for the issuer to be able to exercise the put. Hence, the higher the yield offered.
Investor buys a call from the issuer	Decrease	The greater the time to expiry of a call option, the more compensation the call buyer must pay the seller. This reduces the minimum yield of the structured product.
Investor buys a put from the issuer	Decrease	The greater the time to expiry of a put option, the more compensation the put buyer must pay the seller. This reduces the minimum yield of the structured product.
Time / maturity	Increase	In general, the longer the time to maturity, the greater the opportunity for the current price to reach the strike price. This increases the value of the structure.
Leverage effect	Increase	The greater the leverage effect, the greater the risks, the greater the yield/returns.

7.1.4 Participation Rate

The participation rate is another term often featured in certain structured products. Participation rate refers to the level of exposure to the movement of the underlying asset or index. It is not always the case that the investor gets a 100% participation rate on the underlying asset, for example:

a. If the discount sum, i.e., present value of the structure – face value of the structure (typically a zero-coupon bond), is less than the cost of the call option, the product issuer will buy fewer option contracts and the participation rate will be less than 100%; or

b. If the discount sum allows the issuer to purchase more of the call option contracts, the participation rate can exceed 100%.

Figure 7.1.4 illustrates an example of the participation rate of an ELN with 2 possible zero-coupon bonds.

Figure 7.1.4 – Participation Rate of an Equity-linked Structured Note with 2 Possible Zero-Coupon Bonds

Different Parameters for an Equity-Linked Structured Note with 2 Possible Zero-Coupon Bonds

S&P 500 Call Premium	=	$24.00	
Zero-Coupon Bond		**Bond A**	**Bond B**
Face Value	=	$100.00	$100.00
Discount Rate	=	5.00%	7.00%
Year to Maturity	=	5	5
Present Value	=	$78.35	$71.30
Discount Sum	=	$21.65	$28.70
Potential Participation		**Bond A**	**Bond B**
Discount Sum	=	$21.65	$28.70
Call Premium	=	$24.00	$24.00
Participation Rate	=	90.21%	119.58%

Calculating the Participation Rate:

Discounted Sum/Call Premium for Bond A:
21.65/24 = 0.9021 = 90.21%

Discounted Sum/Call Premium for Bond B:
28.70/24 = 1.1958 = 119.58%

7.1.5 Purpose and Uses of Over-the-Counter Structured Products

For investors, OTC structured products provide the following potential benefits:

a. **Customization:** Structured products can be customized to meet needs that cannot be fulfilled by standard traditional investments. Clients can negotiate with the counterparty bank to make adjustments to certain criteria and trade-offs so that the product becomes more attractive for them. A structured product can also be used to increase the efficiency of a portfolio by catering to diverse risk-return profiles, if there are a wide variety of financial instruments in the portfolio.

b. For example, an investor may wish to participate in the potential market upside and at the same time, he wants some principal preservation. Buying securities directly from the market will not satisfy the investor's needs, as the principal could be lost if securities prices decline. With a structured product that has underlying instruments of a bond and a call option on the securities, the investor can have his capital preserved and at the same time, benefits from the returns if securities prices advance.

c. **Optimizing Yield and Returns**: During periods of low interest rates, investors seek to improve the yield on their assets. Nothing is for free - so with the promise of higher returns, there are risks and trade-offs. For

some structured deposits, the risks may involve lower or zero yield. For other structured products with an even higher return potential, the risk could involve the client effectively selling a put option to the bank counterparty (e.g., which could mean agreeing to take delivery of securities at prices higher than market value which would mean capital losses). Hence, clients must decide if the risks are acceptable, given the probability and potential of higher returns.

d. **Capital Preservation**: In times of extreme market volatility, some investors look for structured products which are capital guaranteed. However, it must be highlighted that in many of such structured products, capital preservation is only achieved if the investments are held till maturity (i.e., early redemptions can result in losses). Finally, while it is true that structured products seek to preserve the underlying capital, there are no guarantees as there is always the counterparty risk which can result in a loss of capital (e.g., the Lehman crisis in 2008).

e. **Performance:** Some structured products are created to allow clients the opportunity to participate in the performance of certain underlying asset classes, index, or a unique customized basket of securities that they would otherwise have greater difficulty accessing in a cost-effective manner. When performance is the main focus, the question that clients should ask is whether direct exposure to the underlying asset is the better choice.

f. **Leverage Effect:** By incorporating options, clients gain leveraged exposure that can potentially augment gains by using a smaller investment outlay. However, in some structured products, the leverage effect can work against the clients when the market moves against the underlying position, resulting in the clients accumulating more of the underlying asset than desired and suffer greater losses (e.g., accumulators).

g. **Hedging:** Clients and investment managers can manage the risk exposure on existing investment positions by buying structured products that provide the right to purchase offsetting investment positions.

h. **An Alternative Rebalancing Activity**: Structured products can be used as a timely alternative action to directly purchase or sell an underlying asset as part of the asset allocation process. Taking a potential long or short position with a derivative can be timelier and more easily executed to help hedge or manage risk of existing investment positions. It can also be carried out without having to actually liquidate or rebalance any part of an existing investment portfolio, or to take advantage of opportunities and market trends.

7.1.6 Types of Structured Products

Structured products are available to investors in a variety of wrappers. Common wrappers include deposits, notes and bonds, funds, warrants, investment-linked policies (ILPs), exchange traded funds (ETFs) and exchange-traded notes (ETNs). Variations of wrappers also enable financial intermediaries to focus on their clients' needs in various market segments.

There are different ways of classifying and describing structured products. Structured products are known by different names or wrappers in different banks for their own marketing purposes, and such products can be structured for more than one purpose to satisfy the needs of different clients.

Table 7.1.6 lists some of the more popular structured products available to clients (although it is not an exhaustive list). In the next few chapters, we will look at the underlying asset classes based on a typical client's needs and by asset classes based on this table.

Table 7.1.6 – Popular Types of Structured Products

Asset Class / Purpose	Capital Preservation (Partial / Full)	Optimizing Yield/Returns	Capital Gain (Participation/Performance)
Foreign Currency Deposits and Precious Metals	• Currency Deposit with Call or Put Options: ○ With or without knock-out ○ With no-touch (lock-out) ○ Set up call or put spread • Deposit with Knock-Out Rebate Option • One-Touch Options • Step-up Digital Options • Range Currency Deposits: ○ Double-no touch ○ Digital spread	• Dual Currency Investments (DCIs): ○ Standard DCIs ○ Knock-out DCIs ○ Knock-in DCIs ○ Double No-Touch • Triple Currency Investment	• Performance Linked Basket/Notes: ○ Without loss limits ○ With loss limits • Accumulators • Decumulators
Equity	• Structured Deposit with Call or Put Options: ○ With knock-out and knock-out rebate ○ Worst-of call or best-of put ○ Digital options ○ Set up call or put spread	• Equity Linked Notes (ELNs): ○ Bull ELNs ○ Callable ELNs ○ EQ fixed coupon note (FCN) • Accumulators • Decumulators	• Performance Linked Basket/Notes: ○ Without loss limits ○ With loss limits • Fixed Coupon Note (FCN) • Twin Win Certificate
Fixed Income	Callable Range Accrual Note (CDRAN)[33]	• Callable Step-up Notes (CSUN) • CDRAN, Variable Maturity Range	

[33] CDRAN, VMRAN and CSUN are covered in Chapter 8 whilst CPPI and DPPI are covered in Chapter 9.

Asset Class / Purpose	Capital Preservation (Partial / Full)	Optimizing Yield/Returns	Capital Gain (Participation/Performance)
		• Accrual Note (VMRAN)	
Funds, ETFs and Others	Constant Proportion Portfolio Insurance (CPPI), Dynamic Proportion Portfolio Insurance (DPPI) [31]		• Performance Linked Basket/Notes: ○ Without loss limits ○ With loss limits

One point to note about actual investment outcomes is that the risk accepted by an investor may result in a totally different outcome from the original purpose for investing. For example, an investor may be looking for a better yield, but the risk may not be related to getting a lower yield - instead it may involve taking delivery of an underlying asset at an agreed price lower than the current market, thereby sustaining book losses against the capital invested. Therefore, being overly influenced by tempting returns may lead to the opposite consequence when investors ignore the risks, which may be a capital loss and participation in more downside exposure.

The risk-return trade-offs in structured products can be quite different from other asset classes. Under the wrong conditions and when poorly structured, the probability and size of losses may not justify the gains or the better yields. A competent investment representative will be aware when a product's pricing mechanism and structure works excessively to the disadvantage of clients and must be able to provide the appropriate advice and warning.

7.2 Structured Deposits with Capital Preservation

7.2.1 Structured Deposits

Structured deposits are a debt obligation of the bank. However, unlike structured notes, structured deposits are a type of deposit and are not debentures (i.e., unsecured debt). A structured deposit is essentially a combination of a deposit and an underlying financial instrument, where the return is linked to the performance of the underlying financial instrument. In Singapore, the distinguishing feature of structured deposits is that the principal sum, with or without interest, has to be repaid in full at maturity. This may not be so in other jurisdictions. For structured notes which are not principal guaranteed, investors may potentially lose part of or the whole principal sum.

Structured deposits can only be issued by banks and must meet the definition of "deposit" as defined under the Banking Act 1970. However, they are not covered by the Deposit Insurance Scheme under the Deposit Insurance Act 2005[34]. If an investor makes an early withdrawal of the deposit, part of the return and/or principal may be lost. The amount payable depends on the market value of the underlying financial instrument that the structured deposit is linked to, at the time of the early withdrawal. All investors need to observe the

[34] The Deposit Insurance Scheme was reconstituted under the Deposit Insurance and Policy Owners' Protection Schemes Act 2011, which came into operation on 31 December 2021.

requirements under the Financial Advisers Act (FAA) Guidelines on Structured Deposits[35], except for the following groups of investors:

a. Accredited investors, and

b. Institutional investors.

From a risk perspective, structured deposits are generally less risky than direct investments in underlying assets because the bank has to repay the full principal at maturity (or when redeemed before maturity applicable to structured deposits with early redemption features). Structured deposits are considered riskier than normal deposits because their returns depend on the performance of other instruments. In some scenarios, investors may only get back the principal with no returns. Table 7.2.1 lists out the features of structured deposits with nominal capital preservation.

Table 7.2.1 - Features of Structured Deposits with Nominal Capital Preservation (or Partial Capital Preservation Depending on Yield)

Structured Deposits with Nominal Capital Preservation	Features
Type of clients	Risk-averse, low risk tolerance.
Purpose	Seeking higher yield above deposits with some capital preservation.
Common underlying assets	Foreign currency investment (e.g., USD/JPY, EUR/USD).
Benefits / Gains	Slightly higher yield.
Risks and losses	• No capital loss if held till maturity. • Loss of interest (zero or lower interest rate on deposit may occur). • Counterparty risks. • Liquidity risk (investor is reliant on the OTC counterparty to liquidate).
Obligations	To provide specified amount of deposit for the structure.
Types of Structures	• Deposits with standard call or put, options. • Deposits with standard option with knock-out features. • Structured deposit with call (or put) spreads. • Range accrual deposits.

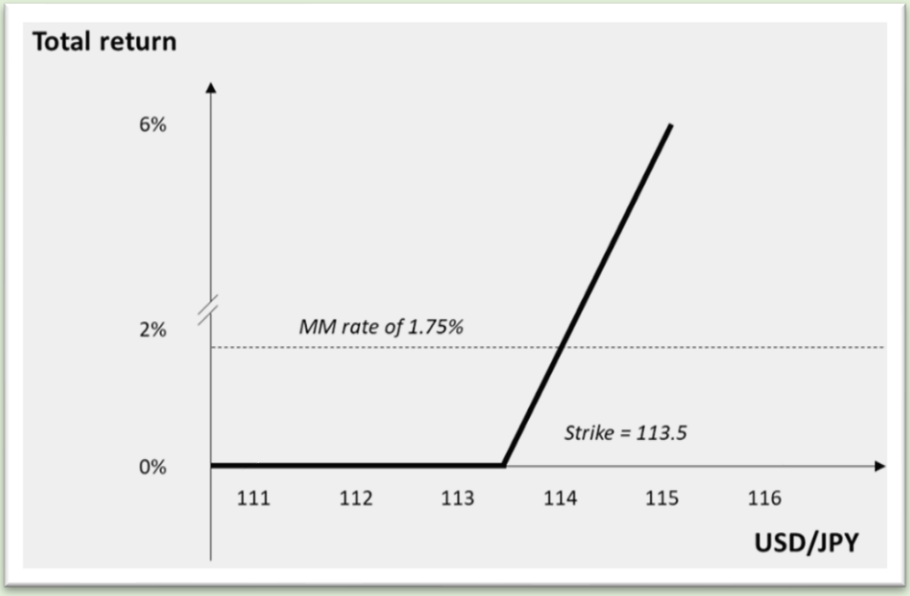
[35] Guidelines on Structured Deposits (FAA-G09) issued by MAS.

7.2.2 Deposit with Standard Call and Put Option

If a client holds a large amount of money market deposits in USD, for example, and is earning relatively low deposit rates, he can consider taking a view on the strength of the USD against an alternate currency (say the JPY) to potentially earn a higher rate of return from exchange gains.

Deposit with a long call option

FX view	:	Bullish USD/JPY
Current USD/JPY spot	:	112.43
Maturity / expiry	:	3 months
Strike price (USD call/JPY put)	:	113.50
Minimum yield	:	0.00%
Maximum yield	:	Unlimited
Equivalent money market rate per annum (3 months)	:	1.75%
Minimum amount to be invested	:	USD 250,000



Rather than placing their funds in a 3-month money market deposit which yields 1.75% per annum, the client can choose to place it in a structured deposit with upside participation in the performance of USD/JPY currency pair.

- The client places the USD funds with the counterparty bank for 3 months.
- Instead of paying the deposit interest to the client, the bank uses the interest to buy a 3-month USD call / JPY put option with a strike of 113.50.
- Note that the client has forgone the deposit interest in return for potential gains in the USD/JPY rate.
- At expiry (end of 3 months), if USD/JPY trades below 113.50 i.e., the option expires out-of-money, the client receives only the USD principal back with no interest.

- At expiry (end of 3 months), if USD/JPY trades above 113.50, the option is in-the-money, and the client receives the gain (prevailing market price of USD/JPY less the strike) in USD plus principal.
- As an example, if USD/JPY trades at 115.00 at expiry, for a deposit amount of USD250,000, the gain is JPY375,000 (i.e., [115.00 – 113.50] x 250,000). The gain of JPY375,000 in USD is 3,260.87 (i.e., 375,000/115). This is equivalent to a return of 1.3% (3,260.87/250,000) over a 3-month period or 5.2% (1.3% x 4) per annum (on a simple interest calculation).

7.2.3 Structured Deposit with Standard Option with a Knock-Out Feature

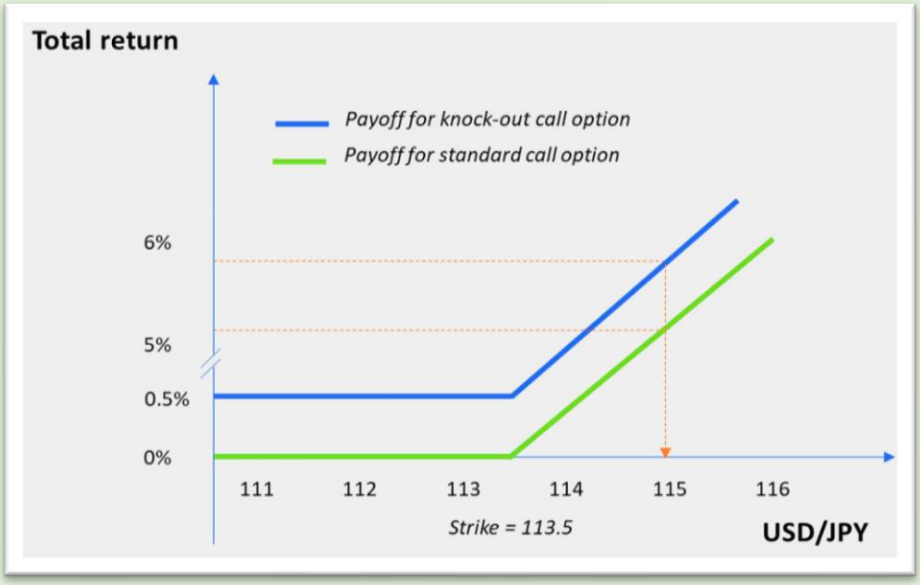
In the previous instrument in Section 7.2.2, we looked at where the currency pair USD/JPY was at expiry (at the end of 3 months) to determine return. There was less concern with the price movement and volatility of the USD/JPY during the life of the instrument before expiry.

Suppose the bank now introduces a new feature where the participation in the upside potential can be knocked out, if at any time during the 3 months the USD/JPY touches 109.50. This feature will affect payoff, volatility during the life of the structure, and risks.

As long as USD/JPY does not touch 109.50 any time during the life of the structure, everything else stays the same. If USD/JPY touches 109.50, the knock-out occurs immediately and the client only receives his principal. Since this works to the client's disadvantage, there must be a compensation for this additional risk and the bank must compensate the client with a higher possibility for potential yield or provide a higher minimum yield instead of zero. This is reflected in the example below.

Deposit with a Knock-Out Call Option

Forex view	:	Bullish USD/JPY
USD/JPY spot	:	112.43
Maturity / expiry	:	3 months
Strike price (USD call/JPY put)	:	113.50
Knock-out barrier	:	109.50
Minimum yield p.a. offered	:	0.50%
Maximum yield	:	Unlimited
Equivalent money market rate p.a. (3 months)	:	1.75%



In this example, the knock-out call option costs less than a standard call option with similar features (strike, tenor). This is because the option buyer is faced with the possibility that the option may be extinguished (knocked-out) before expiry.

With the lower price, the bank need not use the full amount of the interest forgone by the client to purchase the knock-out call option. Hence, part of the interest goes back to the client as the minimum yield of the structured deposit. Note that the client has forgone part of the deposit interest in return for potential gains in the USD/JPY rate.

Suppose the option was not knocked out (i.e., USD/JPY never traded at 109.50 or lower in the course of the 3 months:

- At expiry (end of 3 months), if USD/JPY trades below 113.50 i.e., the option expires out-of-money, the client receives the USD principal back and the minimum interest of 0.5% per annum.
- At expiry (end of 3 months), if USD/JPY trades above 113.50, the option is in-the-money, and the client receives the gain (prevailing market price of USD/JPY less the strike) in USD and 0.5% per annum plus principal.

However, if at any time within the 3-month period, USD/JPY reaches the 109.50 (knock-out level), the client will only get the minimum 0.5% return plus principal. For the client, this strategy is useful when the positive bullish trend prevails, and the downward volatility is low. The knock-out level is set quite far away from the initial spot price to avoid being easily knocked out.

Another variation of the knock-out feature is not to trigger any zero or a lower fixed return once the spot price reaches the knock-out level, but to have the entire pay-off profile to step down by a percentage (e.g., 2%), so that that there is still the potential for participation in the upside if the exchange rate rises significantly in the client's favour at maturity.

7.2.4 Structured Deposit with Bull Call (or Put) Spread

Spread strategies are useful when we believe that there will be only a small positive movement in a currency pair. For calls (puts), small upward (downward) movements are expected. This is similar to the bull-call spread discussed in Chapter 4. In this simple structure, we sell a call (put) with a higher (lower) strike price and buy a call (put) with a lower (higher) strike. Since this is a spread, the upside will be capped.

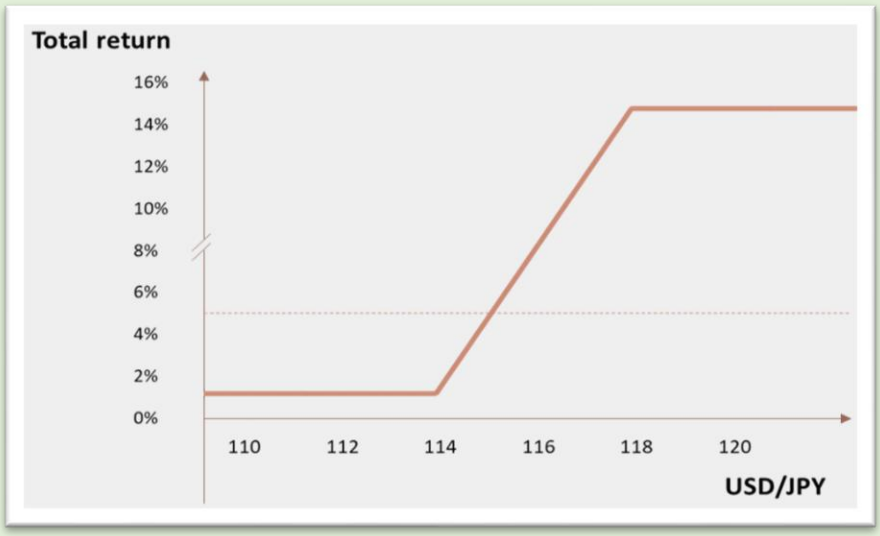
Deposit with a Standard Call Option with a Bull Call (or Put) Spread

A client believes that USD/JPY is likely to strengthen by a small percentage and she would like to participate in the upward move, but she also desires some capital preservation in case she is wrong. She has USD 250,000 to invest and would like to enhance her returns. She can invest in a structured deposit will a bull-call spread.

Forex view : Slightly bullish USD/JPY
USD/JPY spot : 112.00
Maturity / expiry : 3 months
Spread range (USD/JPY) : 114.00 - 118.00
Minimum yield offered : 1.00%
Maximum yield offered : 14.50%
Money market rate : 1.75%
Participation rate : 100%

This structure effectively involves:

- The client selling a call at USD/JPY 118.00; and
- The client buying a call at USD/JPY 114.00.



Restricted

At expiry:

- If USD/JPY > 118.00, the client's call with strike at 118.00 sold to the bank is exercised, so the client gives up any potential upside above 118.00. The client receives a maximum return of 14.50% from the bank counterparty.
- If 114.0 < USD/JPY < 118.00, profits will range from 1.00% - 14.50%, rising with the increase in the spot USD/JPY.
- If USD/JPY < 114.00, the client will still get the minimum of 1.00%.

Essentially, part of the deposit interest is used by the bank to fund the bull call spread, hence the client receives a minimum interest of 1% instead of the money market interest rate of 1.75%. In this case, we may infer that the cost of the bull call spread charged by the bank is 0.75% on an annualised basis.

Note that if USD/JPY is between 114 and 118, the bull call spread of the underlying currency pair ranges roughly from 0% to 14% annualised (or 0% to 3.5% over 3 months). Taking in the minimum return of 1%, the payoff for the client should theoretically range from 1% to 15% annualised. In this case, however, the payoff for the client ranges from 1% to 14.5% as indicated on the terms listed earlier. The difference represents the cost of the setting up the options-based strategy from the market and spread earned by the counterparty bank. Additionally, if the bank managed to purchase the bull call spread (from the market) at less than 0.75%, the difference is another spread earned by the bank.

Clients can negotiate with the bank for better trade-offs. The range of higher enhanced yields offered by the bank will depend on how close the strike price is to the initial spot price, the difference between the two strike prices, the volatility of the spot price, and market conditions.

If the client believes that the underlying currency pair is on a firm trend (e.g., USD/JPY is going up) and does not need the downside protection, then it makes more sense to take a direct stake in the underlying currency pair. However, if the intent is to protect the downside risk, then using the bull-call spread is a possible way of gaining exposure.

The benefit to the investor is the ability to secure a potential return that is higher than money market rates over 3 months while providing a minimum yield and preservation of nominal capital. So, the investor is paying for preservation, i.e., no losses. Also note that in this instrument, there are no knock-outs so the structure will not be affected even if there is high volatility.

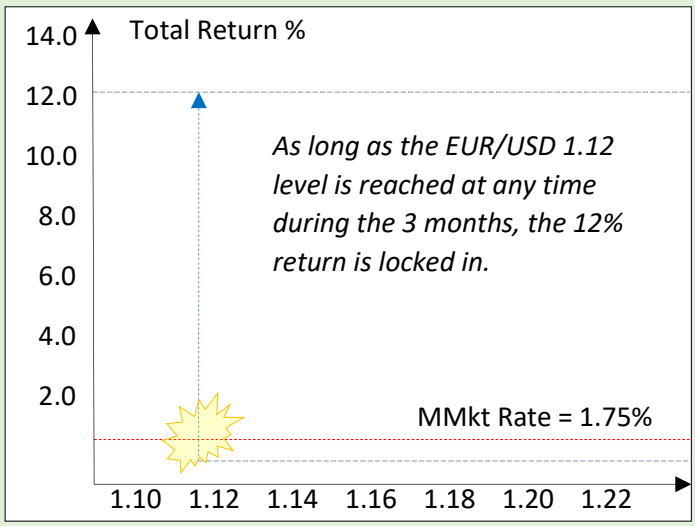
7.2.5 One-Touch Lock-in Option

Investors of this product are usually anticipating an event or situation to trigger a temporary spike or fall. If the strike price of the option is touched or crossed before expiry at any time, the maximum yield is paid at maturity. Otherwise, a minimum interest amount will be paid. The principal will be repaid in full regardless of the outcome.

One-Touch Lock-in Option

Suppose your client is very bearish about the EUR/USD (i.e., positive about USD but negative on EUR) due to an upcoming event in the next 3 months. Some counterparty banks believe that such a 5% fall from EUR/USD 1.18 to 1.12 within the 3-month period is unlikely.

Current spot price	:	1.18
One-touch strike price	:	1.12
Maturity / expiry	:	3 months
Maximum yield offered	:	12.00%
Minimum yield offered	:	0.75%
Money market (MMkt)rate	:	1.75%
Minimum amount to be invested:	:	$200,000



The client is sacrificing 1% to buy the one-touch option, which will trigger a 12% return if EUR/USD spot reaches the strike price of 1.12. As long as the bank has the position or is hedged, they can provide the 12% return as the counterparty. The notional capital is guaranteed.

7.2.6 Range Currency Deposit (Double Barrier Binary Option)

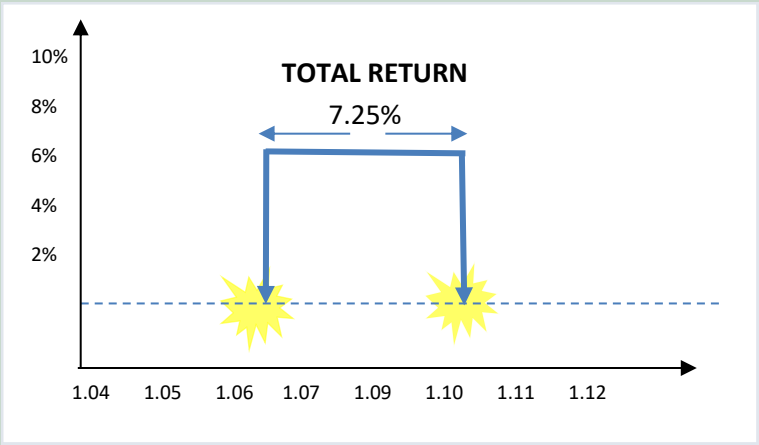
For a **range currency deposit** product, the investor typically believes that the currency pair will stay within a range. As long as the currency pair stays within a certain range, the investor will receive a better yield than otherwise available from the money market. If the currency breaches the range on either end, the investor will get a lower yield.

Range Currency Deposit (Double Barrier Binary Option) (Double No-Touch)

Your client believes that EUR/USD will stay relatively range bound between EUR/USD 1.06 - 1.10 for the next 3-6 months. The client can invest in a range currency deposit and earn 7.25% annualized return as long the underlying EUR/USD stays within 1.06 - 1.10.

Current spot price	: 1.09
Upper barrier	: 1.10
Lower barrier	: 1.06
Maturity / expiry	: 3 months
Maximum yield offered	: 7.25%

Minimum yield offered : 2.00%
Money market rate : 1.50%



If the EUR/USD touches 1.06 or 1.10, the return will fall to 2.00%. So, there is a binary outcome here – either 7.25% or 2.00%.

Clients investing in a range currency deposit may seek to negotiate a higher minimum return or a higher potential maximum return. As an investment representative of your financial institution, you can try to negotiate for higher returns by reducing the option price, and to do this you can ask the client to agree to:

- Narrow the range (however this means that the barriers are more likely to be touched);
- Skew the range to one side with the spot nearer to one barrier; and/or
- Extend the maturity.

There is also more room to adjust returns upwards if the price volatility of the underlying asset increases.

In the above variations, there are only two payment outcomes – hence the outcome is always binary. Some clients do not like the odds of having a sharply lower deposit return (or in some cases zero returns) once the barriers are touched. With consideration to the case above, what happens when 88 out of 90 days the EUR/USD stays within the 1.06 - 1.10 barriers, but for the remaining 2 days, the barrier is touched? The client will receive only a 2% return because of the terms of the investment. Hence, some prefer to opt for the range accrual currency deposit instead (see Section 7.2.7).

7.2.7 Range Accrual Currency Deposits

In a **daily range accrual currency deposit**, the maximum yield will apply as long as the spot trades within the band. Once the spot hits either barrier, the minimum yield will apply. Hence for days where the underlying asset stays within the range, it earns a higher rate but on days when it is outside the range, the lower yield will apply.

In the earlier example above, EUR/USD falls within the barrier for the first 88 days out of 90 days. The total return would be [(88/90 x 7.25%) + (2/90 x 2.00%)] = 7.09% + 0.04% = 7.13%.

Another variation would be to have different accrual periods (e.g., 13-weekly or 3-monthly observations) to see if the spot falls within the range. For the weekly accrual periods that fall within the range, the maximum

return is applied and for the weeks outside the barrier, the minimum return is applied. The overall return will depend on the average weighted weekly return given weeks outside the barrier and weeks within the barrier.

7.3 Structured Deposits Providing Yield Enhancement

Yield enhancement or return optimisation products are another category of structured products offered by many banks over-the counter to clients. Usually, clients holding existing foreign currency deposits, equities or fixed income that are parked in accounts and portfolios within the bank, often seek to enhance the return or yield on their investments. For many structured products offered, the potential for additional yield or capital gain will come with additional risks and no provision of capital preservation. The risks in such instruments often arise from having to write put or call options to the counterparty in the OTC transaction (refer to Table 7.3)

Table 7.3 - Features of Structured Deposits that Provide Yield Enhancement or Optimisation (Usually with No Capital Preservation or with Knock-Out Feature for Partial Downside Preservation)

Structured Deposits Providing Yield Enhancement or Optimisation	Features
Type of clients	Medium to high-risk appetite.
Purpose	Enhancing and optimizing yield above current deposit returns.
Common underlying asset	Foreign currency investment (e.g., USD/JPY, EUR/USD).
Benefits and gains	• Potentially significantly higher return on deposits. • Acquisition of alternate currency at lower exchange rate if converted. • Investors should be comfortable holding alternate currency and can plan to do so at a targeted lower exchange rate which is built into the strike level.
Risk and losses	• Loss from acquiring deposits at a higher exchange rate than market. • Loss of interest (zero or lower interest rate). • Counterparty risks. • Liquidity risk since the investor is reliant on the OTC counterparty.
Obligation	To provide specified amount of deposit for conversion if executed.
Types of structures	Various DCIs and triple currency investments.

7.3.1 Standard Dual Currency Investments

The **DCI,** also known as in the past as a **dual currency deposit (DCD),** is probably the most popular structured product in the local market for the last 20-25 years. The product consists of:

a. A short position in a call on the original reference currency (or a short position in a put on another currency); and

b. A long position in a money market deposit (or bond) denominated in the reference currency.

The option premium received enables the DCI to offer a higher interest rate than what is offered in-the-money market.

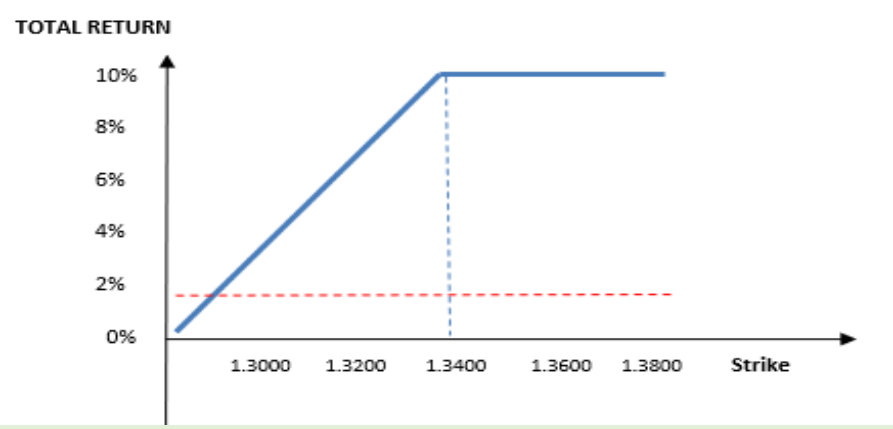
a. If the exchange rate strengthens or stay relatively flat and does not cross the strike price level at expiry, the principal plus interest is paid back in the **original reference currency** (or investment currency) of the client.

b. However, if the exchange rate weakens and has crossed the strike price at expiry, the principal plus interest is paid back to the client in the **alternate** currency. Hence there is a conversion to the weaker currency. The risk to the client is that he is forced to convert to the alternate currency and the currency's strength falls significantly.

A Standard Dual Currency Investment

Your client holds USD as his reference investment currency. He believes GBP/USD exchange rate will be flat or the USD will strengthen against the GBP in the next few months (e.g., due to interest rate hikes and domestic fiscal spending). He has decided to invest in a standard GBP/USD DCI.

GBP/USD Spot : 1.37
Strike Price : 1.34
Return offered by the bank : 10.00% p.a. (compared to the money market rate of 1.75%)



On expiry:

- If GBP/USD > 1.3400, the client will receive his initial investment plus interest of 10.00% in USD.
- If GBP/USD < 1.3400, the client will receive his initial investment plus interest converted at the strike level of 1.3400 in GBP and he may incur a forex loss.

It is important for bankers and investment representatives to understand how prices and payoffs are calculated, and that their clients are also clearly briefed. The product teams doing the calculations may know the size of the profit margins and fair rate of return, but teams who are selling or marketing the products may be oblivious to what the bank is actually making from the product sold. In the interest of transparency, some banks have automated spreadsheets that are able to generate daily pricing and returns, build in profits, and strike price levels for different structured products.

7.3.2 Knock-Out Dual Currency Investments (Disappearing DCI)

A **knock-out DCI** is simply a standard DCI with a knock-out barrier level set out-of-the money. Once the spot price touches the knock-out price level, the embedded option in the DCI will deactivate or disappear, hence the name "disappearing DCI." The investor will receive the principal and interest in the original invested reference currency regardless of where the spot level goes to eventually at expiry. Since this knock-out feature is beneficial to the client, all things being equal save the knock-out feature, the yield obtained is typically lower than the standard DCI. The knock-out DCI becomes interesting when the expected movement of the underlying currency pair is small, or range bound. Also, knock-out DCIs work better when there is a general expectation of a slightly softer base currency or investor reference currency typically USD.

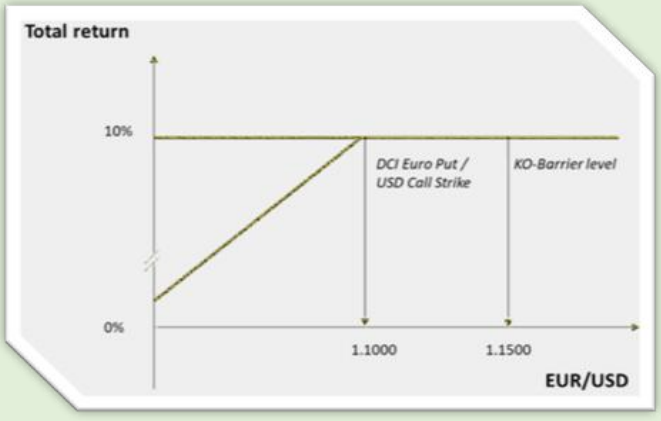
In the example below, the preference is for the EUR/USD to strengthen i.e., softer USD but not to the extent that it reaches the knock-out barrier.

A Knock-Out DCI (Disappearing DCI)

Client's view: EUR/USD downside limited. EUR to strengthen slightly.

Initial investment	: USD Deposit
EUR/USD strike	: 1.1000
Knock-out barrier level	: 1.1500
Current spot price	: 1.1200
Maturity / expiry	: 3 months
Knock-out DCI yield	: 8.50%
Standard DCI yield	: 9.00%
Money market rate	: 1.75%

USD deposit will be converted to EUR deposit at a rate of 1.1000 if EUR/USD is at or below 1.1000 at expiry.



Restricted

If the EUR/USD trades anytime at or above 1.1500 before the 3-month expiry, the investor will not have to face any currency conversion; he stays in his original reference currency (USD).

If we break down the components of the Disappearing DCI, it effectively comprises selling a call option on the base currency (USD) (or conversely speaking; selling a put option on EUR) with a knock-out barrier level (1.1500) set in the opposite direction to the strike price (1.100).

- If EUR/USD trades at or above 1.1500 (knock-out barrier) any time before the 3-month expiry, no currency conversion will take place. The investor's deposit remains in USD.
- If EUR/USD never touches the knock-out barrier level, then the DCI remains active. The final outcome will depend on whether EUR/USD stays above the 1.10 strike level at expiry.
- If EUR/USD closes at expiry below 1.10, then USD deposit will be converted to EUR.
- If EUR/USD strengthens significantly and closes above 1.10, deposit will remain in USD.

The closer the knock-out barrier level to the spot, the likelihood of being knocked out is higher, so the yield will be lower. The further the barrier level from the spot, the less likely for knock-out to occur and a higher yield will be offered. When there is increased volatility, the knock-out DCI is less sensitive than the standard DCI because the factors favouring knock-out will be beneficial against conversion.

7.3.3 Knock-In Dual Currency Investments (Appearing DCI)

A **knock-in DCI or appearing DCI** (also sometimes known as "kick-in DCI" depending on the financial institution or issuer) combines a money market deposit with a knock-in barrier level set in-the-money. The product mechanism is as follows:

a. If the spot rate does not breach this knock-in barrier, then the DCI is not activated, and the investor gets back his principal and interest in the original reference currency invested.

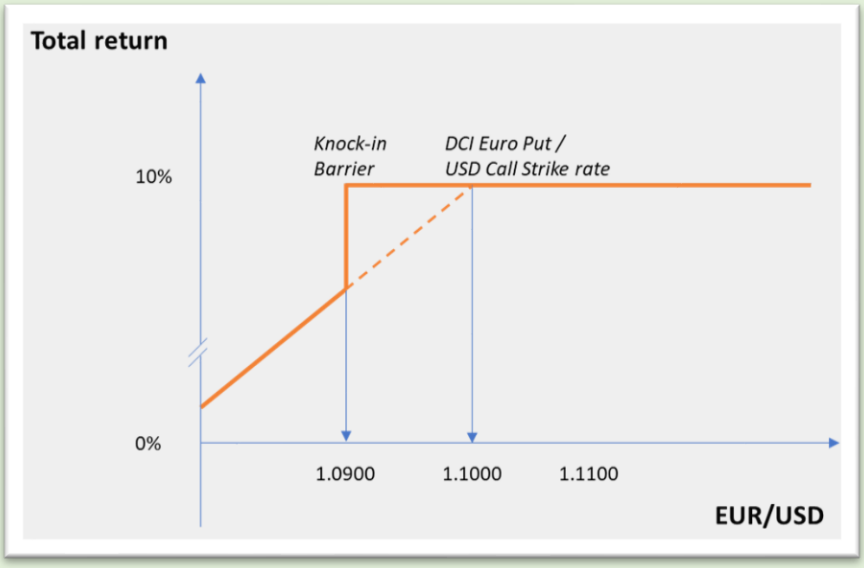
b. If the knock-in barrier is breached, the structure converts into a standard DCI using the existing call strike price.

A Knock-In DCI (Appearing DCI)

Client's view: EUR/USD downside limited.

EUR is expected to strengthen slightly.

Initial investment : USD Deposit
Current spot price : 1.1150
EUR put/USD call strike : 1.1000
Knock-in barrier level : 1.0900
Maturity / Expiry : 3 months
Maximum yield offered : 9.00%
Money market rate : 1.50%



- If the EUR/USD spot rate stays above the knock-in barrier of 1.0900, the knock-in conversion is not activated. The client will receive the principal and interest of 9.00% in USD at expiry.
- If at any time during the 3 months EUR/USD < 1.0900, the knock-in feature will be activated i.e., the features and parameters governing a standard DCI.

To do this trade, the client must have the view that the spot would stay relatively unchanged within a range, or the investing currency USD would have weakened slightly. The client would have an opinion that some volatility may occur causing the EUR/USD to weaken, but not to the knock-in barrier level of 1.09.

Assuming the DCI has been knocked-in, at maturity, if the spot rate is below 1.1000, the structure will pay out principal and interest in EUR. So effectively upon knocked-in, the investor will receive the principal and interest in EUR unless the EUR/USD recovers and strengthens above the 1.1000 strike level. If the spot recovers above 1.1000 at expiry, the principal and interest will be paid in USD.

7.4 Structured Products

7.4.1 Accumulators

An Accumulator is an investment option for investors who hold a positive view of the selected underlying. The accumulator allows the investor to accumulate the selected underlying at a discounted forward price relative to the spot price.

The investor is long a series of call options on the underlying share that allows the investor to buy the underlying share if the price of the underlying shares is at or higher than the forward price.

The purchase of the call options, if financed by the simultaneous shorting of a series of put options on the underlying shares, obliges the investor to buy the underlying share if the price of the underlying share is lower than the forward price.

The investor has the potential to profit from the underlying provided the accumulated shares trades above the forward rate.

The accumulated underlying share during the accumulation period (Number of Shares) is delivered to the investor's account on a regular basis on the scheduled settlement dates, usually once a month. Investor will need to arrange funding to pay the required settlement amount, computed as (Number of Shares x Forward Price). Investor can only sell the accumulated underlying share after the shares are delivered to him/her.

Accumulators may be structured with a knock-out feature. Upon the occurrence of the knock-out event, the structure will terminate immediately.

There are 4 key elements in a basic accumulator:

a. The barrier, below which, triggers accumulation of the underlying asset. This is also the knock-out barrier, wherein, once breached, the contract is early terminated;

b. The set discounted currency exchange rate for the accumulation (this is the strike price);

c. The schedule days when the price or currency rate will be observed; and

d. The size of each accumulation per observation. In some contracts, there will be step-up features increasing accumulation (i.e., leverage effect) when the spot rate crosses a particular level (strike price).

Accumulators are high-risk derivative structures that have acquired a reputation for mis-selling and client suits. While some banks have labelled this product as suitable for clients seeking yield optimisation, the risks associated with this product go well beyond most products offering just optimisation.

In accumulators, the underlying assets are purchased at a discount to the initial price until the price reaches a barrier level (which is higher than the initial price):

i. Above the barrier level, the investor will not accumulate any of the underlying asset (e.g., currency).

ii. At or below the barrier level, the investor will accumulate the fixed amount of the currency at the set discount rate.

Figure 7.4.1 is an illustration on how accumulators are used for currency investments.

Figure 7.4.1 – Use of Currency Accumulator

Currency Accumulator

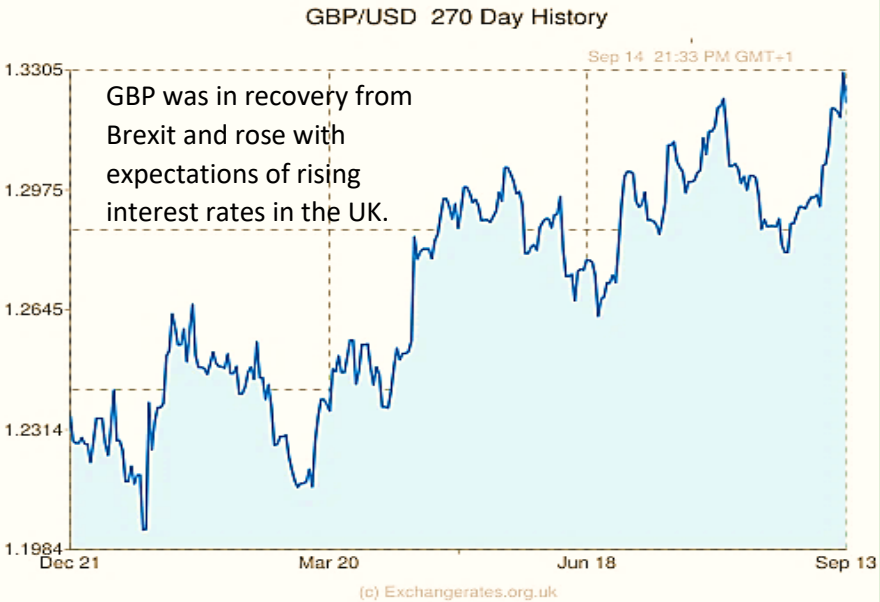
Your client expects the GBP/USD exchange rate to stay in a range and move sideways for the next 6 months. The client does not mind holding GBP because he flies regularly to London, he has property investments in London and his daughter is completing her university degree in the UK. He has decided to use an accumulator for his GBP/USD investment.

Product Features and Assumptions

Initial currency spot price for GBP/USD	:	1.2275 (2nd January 2017) (in 4 decimal place)
Fixed price set for conversion is 95% of initial price	:	USD 1.1661
Barrier level is set at 105% of the initial price	:	USD 1.2889 (below which triggers conversion)

If the spot GBP/USD on a particular fixing day is at or above barrier level	:	GBP is not accumulated
If the spot GBP/USD on a particular fixing day is below the barrier level	:	GBP 5,000 is accumulated
Tenor for settlement	:	After 6 months
Number of relevant fixing days per month	:	20 days per month (January - June)
Upfront investment required for 120 (i.e.; 6 x 20) fixing days	:	5,000 x 120 x USD1.1661 = USD 699,660

The accumulator will examine the closing at the exchange rate for 20 predetermined fixing dates.



Scenario 1: Suppose GBP/USD trades below 1.2889 for all fixing dates over 6 months (i.e., 120 days).

Accumulated GBP 600,000 (i.e., GBP 5,000 x 120 fixing days) will be delivered to you after 6 months at an exchange rate of GBP/USD 1.1661 (i.e., 95% of the initial spot-on 2 January). GBP has strengthened against the USD over the 6 months. The exchange rate on 30 June is GBP/USD 1.3027. So, your client would have accumulated gain of (1.3027 - 1.1661) x GBP 600,000 = **GBP 81,960**.

Scenario 2 - GBP/USD trades above the trigger price for half the number of days

Accumulated GBP 300,000 (i.e., GBP 5,000 x 60 fixing days) will be delivered to you after 6 months. The client would have secured a gain of (1.3027 - 1.1661) x GBP 300,000 = GBP 40,980. USD 699,660 x (60/120) = USD 349,830 after 6 months.

Scenario 3 – Real Market Conditions

In this scenario for the period 2 January – 30 June 2017, the client benefited from the rise in GBP/USD as he accumulated GBP at the discounted rate. There were only 24 days where GBP/USD was above the barrier level at 105% of the initial price of USD1.2889 (where no conversion takes place) and 96 days of accumulating GBP.

Scenario 4 – Risk of Losses (e.g., 1.1650)

Now suppose GBP/USD is falling while the client is accumulating GBP. When GBP/USD goes below the 95% discounted exchange rate of GBP/USD 1.1661, the client will be accumulating losses.

While seemingly simple in its trading approach and attractive because of the discounted price level, the risks associated with accumulators are high because the investor is obligated to accumulate the underlying asset at prices higher than the prevailing market price as it falls.

Leveraged accumulators have step-up features and knock-out barriers where the investor has to buy the underlying asset multiple times when the market price falls below the discounted price. There is no capital guarantee. In addition, investors cannot continue accumulating the underlying asset if the market price of the underlying asset rises above a designated barrier level (knock-out barriers) which is typically higher than the initial price. These features are explained below:

a. **Step-up features** – With this feature, the amount accumulated will rise multiple times (e.g., 2 times to 5 times) if the market price falls below a certain level. Hence, the client will be obliged to accumulate more of the falling currency (or in the case of equity, more of a falling stock). Some banks offer margin financing for accumulators, which can magnify the size of gains or losses, making the product even riskier for both the client and the financing bank. Once a client has bought an accumulator, it may be costly for them to unwind the position or to create an offsetting position. Furthermore, when credit concerns arise, banks may decide to lower their allowable lending value for certain asset classes or for certain investment activities which may force clients to cover their positions or to sell-off their investment and incur substantial losses.

b. **Knock-out barriers** – An accumulator is comprised of long in-the-money (ITM) calls plus short out-of-the-money (OTM) puts. Since ITM calls cost more than OTM puts, up and out Knock-out (KO) barriers can be placed to lower the cost of the ITM calls. Hence, this caps the gains from the calls. For Leveraged Accumulator, more than 1 x OTM put is often sold, the KO barriers can be shifted further from the strike and/or strike can be reduced further. These barriers represent a lower cost alternative but may work against clients under certain circumstances. Advisors should approach this type of products with caution, especially in the context of suitability.

7.4.2 Decumulators

A decumulator is the reverse of an accumulator. For decumulators, the investor is a writer of call options to the counterparty (the bank). The investor is taking on the obligation to sell a certain amount of the underlying asset on a regular basis at the contracted price (for stocks) or exchange rate (for currency).

Investors buy decumulators in expectation of a slightly bearish market. Like accumulators, there will usually be a knock-out level which will terminate the obligation to sell once the underlying asset price reaches that level. For currencies, a decumulator has almost a mirror image to an accumulator - every transaction involves a currency pair where the sale of one side of the currency pair involves the purchase of the alternate currency. Therefore, decumulators are less popular than accumulators.

7.4.3 Currency Notes and Baskets

We will be covering currency notes and baskets together with the other types of structured notes in Chapter 8.

7.5 Summary

1. A structured product is a financial product designed to provide investors with more tailored investments alternatives linked to a specific risk /reward profile or a hedging strategy options that addresses a view about the direction or volatility of an underlying asset or index. It is usually made up of a combination of two or more financial instruments, with one or more being a derivative structure created by an issuer (usually a financial institution).

2. The bulk of structured products are traded over-the counter, with banks being counterparties to various types of individual and institutional investors. Over the years, some tailored structured products have become so popular and standardized that they have been featured on some exchanges.

3. When helping clients invest in structured products, investment representatives should consider the client's risk appetite, investment experience and knowledge, liquidity needs, investment horizon, product risks, product restrictions and suitability and the need to make proper disclosures for transparency.

4. Structured products often face market risk, liquidity risk, reinvestment risk, counterparty risk, currency risk and leverage risk.

5. The features within a structured product can be negotiated and tailored by banks for their clients but there is no free lunch. There are trade-offs. When an investor sells a call or put to the issuer, the higher yield offered by the issuer is obtained because of the closer proximity of the strike price of the call and put price to the current market price provided a greater likelihood for the issuer to exercise to their benefit. Likewise, increase or decreasing the maturity will also provide greater length of time and opportunity for the price of the underlying to reach the strike price. Finally, the yield will also increase with greater leverage and volatility.

6. In structured products, the **participation rate** refers to the level of exposure to the movement of the underlying asset or index. This participation or exposure to the underlying can be 100% or less than 100%.

7. Structured products provide the following benefits:
   i. Customization of features and payoff.
   ii. Potential optimizing of yield and returns.
   iii. A means of creating partial or full capital preservation.
   iv. Opportunity to participate fully or partially or in an augmented manner to price movement of the underlying.
   v. Opportunity to enjoy the benefits of leveraging.
   vi. Hedging against certain events.
   vii. Provide an alternative means of rebalancing a portfolio without having to actually buy or sell securities in the portfolio.

8. Common structured products include **dual currency investments, triple currency investments, equity linked notes, accumulators, decumulators, callable step-up notes, callable range accrual notes, variable maturity range accrual notes** and **constant proportion portfolio insurance instruments.**

9. Structured products are available to investors in a variety of **wrappers**. Common wrappers include deposits, notes and bonds, funds, warrants, insurance-linked policies (ILPs), exchange traded funds (ETFs) and exchange-traded notes (ETNs). Variations of wrappers also enable financial intermediaries to focus on their clients' needs in various market segments.

10. **Structured deposits** are a debt obligation of the bank. It is essentially a combination of a deposit and an underlying financial instrument, where the return is linked to the performance of the underlying financial instrument. In Singapore, the distinguishing feature of structured deposits is that the principal sum, with or without interest, has to be repaid in full at maturity.

11. Structured deposits can only be issued by banks and must meet the definition of "deposit" as defined under the Banking Act 1970. However, they are not covered by the Deposit Insurance Scheme under the Deposit Insurance Act 2005 (now under the Deposit Insurance and Policy Owners' Protection Schemes Act 2011). If an investor makes an early withdrawal of the deposit, part of the return and/or principal may be lost.

12. If an investor holding currency deposits (e.g., USD) earning money-market rates believes that the USD will appreciate against another currency (e.g., JPY), he can work with a financial institution, sacrificing some or all of the money market returns for a period to set up a structure with exposure to USD/JPY appreciation for the period through a call option. If there are enough money-market returns sacrificed, he can also buy a put at a USD/JPY level that protects the value of the initial deposit in case he is wrong about the direction of USD. This would then represent a structured deposit with a call and a put option sold by a bank to a client.

13. A variation called **"knock-outs,"** or **"kick-outs"** can be introduced to enhance yields of a structure beyond what is available in the market as long as the underlying stays within a price range or above or below a set price level. However, if the underlying hits a price level, a reduced payout or even zero payout will be the result. This can be done using a knock-out feature. Effectively, the client receives a higher yield by selling a call or put option to the financial institution, but if a certain price level is crossed on the upside (using a call) or on the downside (using a put), options are exercised by the bank and the interest earned are automatically reduced.

14. In the double-barrier binary options (range currency deposit), the investor earns a significantly higher yield as long as the underlying stays within the range between the two barriers, but if either of the barriers are breached, the structure pays out a significantly lower yield than what the market would offer. Effectively, an investor in a double-barrier binary option, there are only two outcomes – one with a higher return within the range and one with a lower return outside the range. Hence the outcome is binary.

15. In the range accrual currency deposit, the investor earns the higher return for the number of days the underlying stays within the range and receives the lower return (or zero) for the days the underlying is outside the range. The range accrual currency deposit structure makes the risks for the investor lower, and he faces fewer extreme outcomes compared to the double-barrier binary option (range currency deposit).

16. "A one-touch lock-in" structured deposit requires the sacrifice of money market returns earned on a deposit to purchase a structure offering a much higher return only if the level is reached at any time during the life of the structure. This involves a call option or put option bought to allow the investor the higher rate of return once the level is touched. If the level is never touched, the investor earns the lower rate paid which will usually be lower than the money market rate for the deposit.

17. In a **dual currency investment (DCI),** the investor buys a structured product which combines a long position from a money market deposit with a short position on a call on the original reference currency (or effectively a put on the opposite currency used in the exchange. Remember that selling call on USD/JPY is similar to selling a put on JPY/USD). The investor earns a higher yield than what is available in-the-money market as long as the reference exchange rate does not weaken below a certain set level at

maturity. If it does weaken and stays below the predetermined strike level at maturity, the capital invested is paid back in the opposite currency. In a USD/JPY DCI, the higher yield earned may be totally offset by the fall of JPY against USD.

18. A **knock-out DCI (disappearing DCI)** is simply a standard DCI with a knock-out barrier level set out-of-the money. Once the spot price touches the knock-out price level, the embedded option in the DCI will deactivate or disappear, hence the name "disappearing DCI". The investor will receive the principal and interest in the original invested reference currency regardless of where the spot level goes to eventually at expiry. Since this knock-out feature is beneficial to the client, all things being equal except the knock-out feature, the yield obtained is typically lower than the standard DCI.

19. A **knock-in DCI or appearing DCI** (also sometimes known as "kick-in DCI") combines a money market deposit with a knock-in barrier level set in-the-money. If the underlying spot rate does not breach this knock-in barrier, then the DCI is not activated, and the investor gets back his principal and interest in the original reference currency invested at maturity. If the knock-in barrier is breached, the structure becomes a standard DCI with the existing call strike price and there is a risk of currency conversion.

20. **Accumulators** provide investors with exposure to capital appreciation of the price of the underlying asset price, which is accumulated, compared to the initial discounted price or rate. Hence, investors are effectively seeking capital gains or participation in the performance of the underlying asset and consequently also takes on the risk of losses. Accumulators allow investors to accumulate fixed amounts of underlying assets (usually equity shares or foreign exchange) daily or as specified. The risk to clients is the accumulation of a rapidly falling security and incurring more and more losses especially in the case of **Leveraged Accumulators** which have step-up features and knock-out barriers where the investor has to buy the underlying asset multiple times when the market price falls below the discounted price.

21. **Decumulators** are the opposite of accumulators where the investor is a writer of call options to the counterparty (the bank). The investor is taking on the obligation to sell a certain amount of the underlying asset on a regular basis at the contracted price (for stocks) or exchange rate (for currency).

Chapter 7: Appendix (Non-Examinable Content)

A7.1 Deposits with Standard Option and Step-Down Option

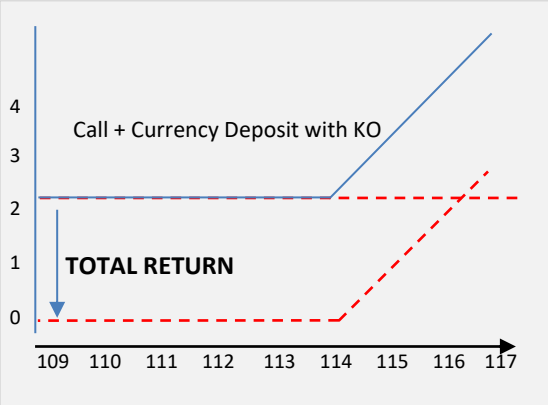
A variation to a structured deposit with a standard option, is not to have an absolute knock-out feature which reduces returns once touched, regardless of where the spot price lies at the end at maturity. Instead, we create a condition at the same knock-out level which when touched, would cause the entire pay-off profile to step down. Consequently, it does not limit returns to the minimum yield.

Pease refer to example – Deposits with Standard Option and Step-Down Option.

Deposits with Standard Option and Step-Down Option

For simplicity, suppose we set the strike at 114.50 with everything else remaining almost similar. There are two possible outcomes:

- If 114.5 > USD/JPD < 109.00, the yield is fixed at 2.00%.
- If USD/JPY > 114.50 and never touches 109.00 before expiry, then investors will get a higher yield than 2.00%. The higher the spot USD/JPY is at close, the higher the yield, then investors will get a yield higher than 2.00%.



If spot USD/JPY ≤ 109.00 any time before expiry, the whole payoff will shift down by 2.00%. All closing rates at or below USD/JPY 114.50 will get 0% (no capital loss) but if the spot rate rises past USD/JPY 114.50 at close, positive returns can be earned (e.g., about 116.50, the payoff will be about the same as the money market rate).

The underlying embedded option will include a standard call option, a lock-out (no-touch) option with the price of the lock out payout positively related to the distance between the initial spot and step-down strike and negatively related to the volatility of the exchange rate.

A7.2 Step Up One-Touch Digital Option

This structure is used when the upside potential is somewhat limited or less likely, with a higher probably of achieving something in the middle range. An investor can create a series of one-touch options with different strike prices in the same direction (e.g., rising or falling) using the same capital over time. As each strike level is breached, the return rises. The total interest return is the sum of the cumulative return from the strike prices touched. This widens the range of participation in a trend.

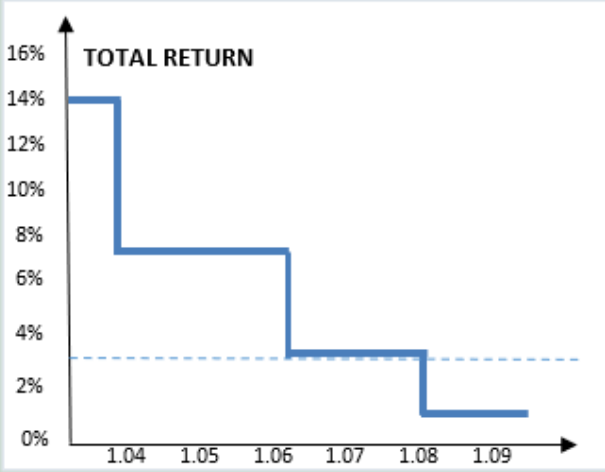
The spot range is divided into two or more sub-ranges and all the trading ranges correspond to different interest rate returns. If the spot closes beyond the furthest range, the investor receives the maximum yield (this yield will be the sum total of the yield from all the ranges).

Step Up One-Touch Digital Option

Suppose an investor or trader is bearish EUR/USD (i.e., positive about USD but negative on EUR).

Current spot price	: 1.08
One touch strike price	: 1.04
Maturity / expiry	: 3 months
If Spot ≤ 1.04	: 14.00%
If 1.04 < Spot < 1.06	: 7.50%
If 1.06 < Spot < 1.08	: 3.50%
If 1.08 < Spot	: 1.00%
Money market rate	: 1.75

The components are tied to spot performance:



- If spot < EUR/USD 1.04, the client gets 14%.
- If spot EUR/USD ranges from 1.04 to 1.06, the return is 7.5%.
- If spot is between EUR/USD 1.06-1.08, the return is 3.5%.
- If spot > EUR/USD 1.08, then the return will be the minimum which is 1%. The principal is untouched.

Traders can actively work to set up multiple payout outcomes based on their expected trend and find the cheapest or most cost-effective means to do this.

Chapter 8: Structured Notes

Learning Objectives
The candidate should be able to: ✓ Explain the basic characteristics and features of a structured note. ✓ Describe range accrual notes and other types of yield enhancement note structures. ✓ Describe equity linked notes, credit linked notes and other option embedded note structures. ✓ Describe exchange-traded notes, index-linked notes and other market access note structures. ✓ Explain the different type of risk exposures for investors in equity linked notes, credit linked notes, bond linked notes and structured deposits. ✓ Explain what type of investors would buy structured notes. ✓ Understand the contents of the required documents when issuing and investing in structured notes.

8.1 What is a Structured Note?

A structured note is a debt instrument / debenture, whose return characteristics, such as coupon amount or market value, are linked to the performance of other underlying instruments. The underlying instruments may include equities, indices, interest rates, credit default swaps, collateral debt obligations, or a basket of instruments.

While the underlying instruments are generally used as a reference for the note's performance, typically the note holder does not have claim over the underlying instruments. These structures usually have one or more embedded options or derivatives or may employ other strategies using other derivatives.

As a debenture (i.e., senior unsecured debt), a structured note is governed under the Securities and Futures Act 2001 (SFA) which requires the issuance of a prospectus unless the note is offered only to accredited Investors, institutional clients or for a minimum consideration of S$200,000 (or its equivalent in a foreign currency).

Hence, they are mostly sold over-the-counter by banks to other financial institutions, investment fund managers, corporations, and to accredited investors[36].

A common structure adopted by note issuers is to combine a straight debt instrument with one or more derivatives as follows:

a. **Principal Component** – For structured notes without collateral, investors depend solely on the note issuer for repayment of principal. For structured notes with collateral, the funded component can be in the form of zero-coupon or corporate bonds, or other types of securities. Principal repayment is often not guaranteed.

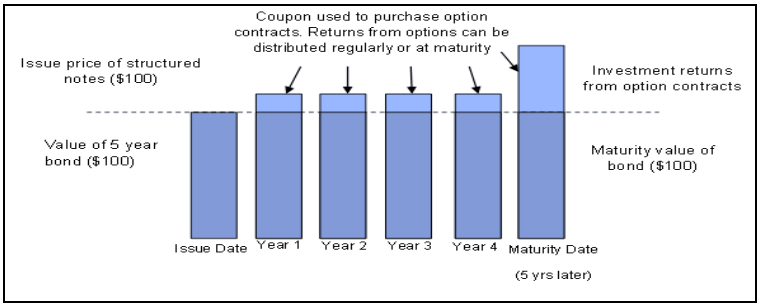
b. **Return Component** – The derivative component provides exposure to the chosen asset class. It may be in the form of selling an option in exchange for a fixed premium or taking a long position in the option.

There are also structures that link both the coupon payments and/or principal repayment to the performance of the underlying instrument. The structured note issuer could also enter into a derivatives contract or swap transaction with another party. The performance of the derivative instrument or swap transaction will have a direct impact on the overall return on the structured note.

The conditions affecting the payout of the coupons and principal under the structured note will be listed in the product offering documents. Investors should review the product offering documents to ensure that they fully understand the information in the documents before making any investment decisions.

Figure 8.1 shows a 5-year bond tied together with an option contract. Structured notes, like bonds, may give periodic coupons and lump-sum redemption at maturity. However, their returns will fluctuate depending on the performance of the underlying financial instruments.

Figure 8.1 - Example of a Structured Note



[36] Under Section 4A of the SFA, an Accredited Investor is defined as:
(i) An **individual**: (A) Whose net personal assets exceed in value $2 million (or its equivalent in a foreign currency) or such other amount as the Authority may prescribe in place of the first amount; or (B) Whose financial assets (net of any related liabilities) exceed in value $1 million (or its equivalent in a foreign currency) or such other amount as the Authority ay prescribe in place of the first amount, where "financial asset" means (BA) a deposit as defined in section 4B of the Banking Act 1970; (BB) an investment product as defined in section 2(1) of the Financial Advisers Act 2001; or (BC) any other assets as ay be prescribed by regulations made under section 341; or (c) whose income in the preceding 12 months is not less than $300,000 (or its equivalent in a foreign currency) or such other amount as the Authority may prescribe in place of the first amount;
(ii) A **corporation** with net assets exceeding $10 million in value (or its equivalent in a foreign currency) or such other amount as the Authority may prescribe, in place of the first amount, as determined by: (A) the most recent audited balance-sheet of the corporation; or (B) where the corporation is not required to prepare audited accounts regularly, a balance-sheet of the corporation certified by the corporation as giving a true and fair view of the state of affairs of the corporation as of the date of the balance-sheet, which date shall be within the preceding 12 months.
(iii) the trustee of such trust as the Authority may prescribe, when acting in the capacity; or
(iv) such other person as the Authority may prescribe.

8.2 Issuing Structured Notes

Structured notes are usually issued by banks, or a Special Purpose Vehicle (SPV) set up by banks.

Under direct issuance, the bank issues the notes to the note holders (investors). As the funds are raised directly by the bank, the debt will be reflected on its balance sheet as a liability. This debt is a direct obligation of the bank to its creditors and the investor bears the credit risk of the bank.

The issuer of a structured note may also enter into a swap transaction with another institution to derive the cash flows in order to pay coupons to investors (if any) and make redemptions on maturity or an early redemption date. In the case where the issuer passes on the credit risk of the swap counterparty to the investors, the investor bears the credit risk of both the note issuer and the swap counterparty.

A SPV is a separate legal construct set up by the bank for the purpose of the transaction. When a SPV is used to issue the note, they will use the proceeds to buy fixed income securities such as zero-coupon bonds, debt securities, collaterised debt obligations (CDOs) or other asset-backed securities, either to preserve the principal capital (capital preservation product) or to use them as collaterals against the options or credit insurance which they sell to the market (yield enhancement product).

The assets and liabilities of an SPV are usually not reflected on the bank's balance sheet – they are referred to as **off-balance sheet items** from the bank's perspective. In the event of default, the noteholders will first make a claim on the SPV's assets and may not have recourse to the bank which set up the SPV.

8.3 Types of Structured Notes

There are three main types of Structured Notes:

a. Those offering some capital preservation focus on yield enhancement/optimisation;

b. Those focused on yield enhancement/optimisation with no capital preservation; and

c. Those focused on capital gains from the price movement of the underlying with no capital preservation.

Asset Class / Purpose	Optimizing Yield/Returns (Partial or Full Capital Preservation)	Optimizing Yield/Returns (No Capital Preservation)	Capital Gain Participation/Performance (No Capital Preservation)
Equity		• Equity Linked Notes (ELNs) • Bull ELNs Callable ELNs • Worst of ELNs	• Performance Linked Basket / Notes ○ Without loss limits ○ With loss limits • Accumulators
Bonds (Credit)		• First to default credit notes • Credit Linked Notes	

Interest Rate	• Range Accrual Notes (RAN) • Callable Range Accrual Notes (CDRAN) or Variable Maturity Range Accrual Notes (VMRAN).	• Inverse Floater Notes • Callable Step-Up Notes	• Exchange-Traded Notes (ETNs) • Index-Linked Notes Participatory Notes (PNs)
Foreign Exchange	• Callable FX Linked Notes • FX Range Accrual Notes	• Dual Currency Deposits • FX Linked Barrier Notes	• FX Accumulators • FX knock-in / knock-out Notes • Tracker Certificates • Bonus Certificates

8.4 Structures with Yield Enhancement and Some Capital Preservation

Banks have designed and crafted yield enhancement structures that offer clients the opportunity to generate additional income and enhanced yield on the initial investment, when compared to money market yields. However, the potential extra yield offered comes with bearing additional risks. Most, if not all, of these 'yield enhancing' strategies have one thing in common - the potential extra yield comes from the income generated by writing or selling options. In these cases, clients of banks bear the additional risks associated with being an option writer rather than an option buyer.

Capital guaranteed structures generally provide relatively low yield and offer protection of the notional capital only if the product is held till maturity. There are still other types of risk associated with capital guaranteed products. The features of such structured products are listed in Table 8.4.

Table 8.4 - Features of Structured Notes that Provide Yield Enhancement with Some / Nominal Capital Preservation (or Partial Capital Preservation Depending in Yield)

Structured Notes with Yield Enhancement and Nominal Capital Preservation	Features
Types of clients	Risk-averse, low risk tolerance
Purpose	Seeking higher yield than the money market with some capital preservation
Common underlying asset	Money-market rates, LIBOR, SIBOR, Swap Rates (SOR)
Benefits and gains	• Slightly higher yield secured • Partial or full notional capital preservation when held till maturity

Risks and losses	• Loss of interest (zero or lower interest rate on deposit may occur) • Potential loss in the value of the principal investment if redeemed before maturity. Value of the note can fall due to interest rate and currency volatility during the life of the note. No nominal capital loss if held till maturity • Coupon reinvestment risks • Liquidity risk since the client is reliant on the OTC counterparty
Obligations	To provide specified amount of deposit for the structure
Types of structures	Various types of Range Accrual Notes

8.4.1 Range Accrual Notes (RANs)

A RAN is a structured note where the investor receives a target level of return if a reference index moves within an agreed range, failing which the investor will receive lesser to no interest. However, the principal will not be affected.

The reference index can be a stock index (e.g., Dow Jones Industrial Average (DJIA)) or an interest rate benchmark (e.g., LIBOR, SIBOR or Singapore SOR (Swap Offer Rate)). For example, if the reference index was the 3-month LIBOR, the interest would be accrued at the higher rates offered only on days when the 3-month LIBOR trades are within a given range. On days when LIBOR trades outside that range, no interest would be accrued.

Interest is usually accrued and calculated on a daily basis for the days when the reference index falls within the range stipulated in the RAN's terms and conditions. In essence, the RAN's final interest payout depends on observed movements of the reference index. As long as the reference index falls within the range, the investor will get a fixed or floating coupon. If the reference index closes outside of the predefined range, the investor will receive lesser to no interest.

A general expression for the payout of a range accrual note is:

Reference index inside the range: **Payout = P1 x (d/N)**

Reference index outside the range: **Payout = P2 x (N-d)/N (or zero)**

where:
N = Total number of observations or days within a period
d = Total number of observations or days when the index is inside range
P1 = Payout when the index is inside the range
P2 = Payout when the index is outside the range

The observation period can be daily, weekly or monthly (usually daily). It is also more common for the payout to be zero if the index is outside the agreed range.

The key terms of a range accrual note are:

a. **Reference Index** – The index observed for determining any payout;

b. **Range** – The range of values of the reference index within which coupon will be accrued;

c. **Inside Range Payout** – Coupon earned by investor if the reference index falls inside the range;

d. **Outside Range Payout** – Coupon earned by investor if the reference index falls outside the range;

e. **Observation Period** – Time period over which returns are observed; and

f. **Settlement Period** – The coupon payment interval.

Figure 8.4.1 - Trading Range of a RAN

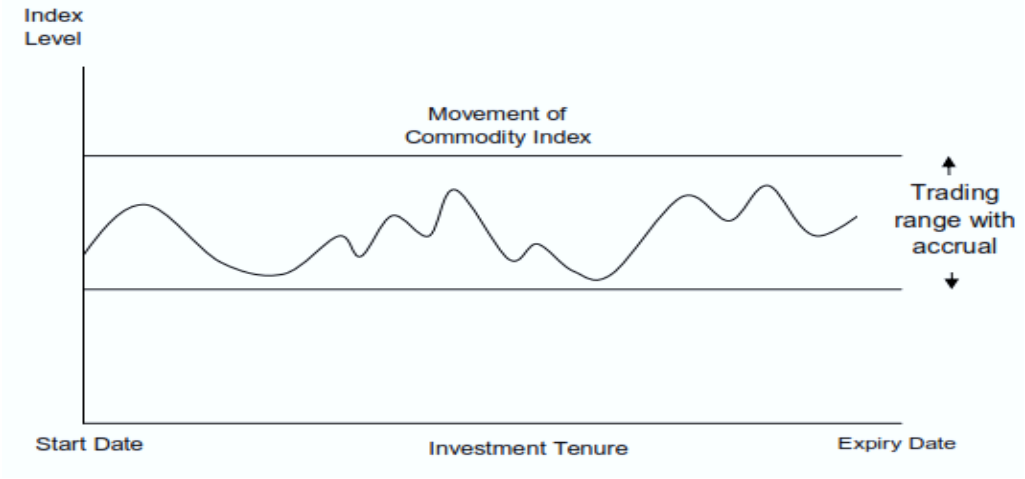


Figure 8.4.1 shows a hypothetical example of the market movement in the underlying asset's reference index over the investment period. The two horizontal lines indicate the lower and upper levels that the index must fall in to earn a return for the investor.

RAN Linked to 3-Month SGD Swap Offer Rate (SOR)

Description	Notes in SGD with yield linked to 3-month SGD SOR
Nominal investment	SGD 100,000
Term	1 year
Coupon	• 5% p.a. daily for each day the 3-month SGD SOR closes within 1% to 1.5% range • 0% if 3-month SGD SOR closes outside the range
Maturity	100% principal sum subject to the credit of issuer, ABC bank
Coupon payment	Semi-annually
Basis of coupon rate calculation	ACT/360 (where ACT = Actual number of days)

Coupon fixing dates	22 Jun 2013 and 22 Dec 2013

Scenario:

- From 23 Dec 2012 to 22 Jun 2013, the number of days 3-month SOR closes within the range is 110 days out of 182 calendar days.
- From 23 Jun 2013 to 22 Dec 2013, the number of days the 3-month SOR closes within the range is 160 days out of 183 calendar days.

Compute the coupon payment fixed on 22 Jun and 22 Dec 2013.

Hence:

- 1st Coupon paid on 22 Jun 2013 = 110/182 x 5%/2 = 1.51%
- 2nd Coupon paid on 22 Dec 2013 = 160/183 x 5%/2 = 2.19%

Accrual coupons will be payable to the note investors for the number of days the SOR falls within the index reference range of 1.0 to 1.5%.

8.4.2 Callable Daily Range Accrual Notes (Variable Maturity Range Accrual Notes)

A **callable daily range accrual note (CDRAN)** involves the addition of the call feature on specific future days where the issuer has the right to redeem the note (typically at par). This call option is valuable to the issuer and exposes the investor to re-investment risk. By giving the issuer this option to call the note at different points in the future, the investor should be compensated with a higher coupon rate or yield on the CDRAN. Like other range accrual notes it allows us to take a view on how an index price level will behave.

For each day that an investor is correct in anticipating the movement of an index (e.g., LIBOR, EURIBOR, SOR), interest will accrue. CDRANs typically become callable only after an initial "lockout" period. The note then become callable on designated coupon payment dates. When the note is called, the investor receives the principal back and stops receiving coupons. The callable nature changes the effective maturity or tenor of the instrument, hence the CDRAN is also called a **variable maturity range accrual note (VMRAN).**

8.4.3 Inverse Floating Rate Notes (or Inverse Floaters)

This is a structured note that pays coupons which are inversely linked to a floating interest rate index. The initial coupon is much higher than a bank deposit rate. An inverse floating rate note, or simply an inverse floater, is a type of bond or other type of debt instrument whose coupon rate has an inverse relationship to its reference rate (usually the near-term interest rate (e.g., 3-month LIBOR). With an inverse floater, as interest rates rise the coupon rate falls.

If interest rates go up, the coupon will then be reduced by the leveraged increase of the interest rate, which is determined by a leverage factor. The coupon is expressed as:

Coupon = [X% — Leverage x Floating Rate Index]

The note can also be structured so that the principal is not affected, and the coupon is subject to a floor:

Coupon = Max [X% - Leverage X Floating Rate Index, Min Coupon]

8.4.4 Callable Step-Up Notes

A **callable step-up note (CSUN)** is a capital guaranteed instrument that offers fixed coupons that can increase (step-up) over the life of the instrument. Given the risk to the issuer for such step-up of coupons, the instrument is callable by the issuer on specific dates during the life of the note. The risk for investors is the issuer risks and an investment grade rating (BBB+ or higher) associated with bonds is desired.

CSUNs became more popular for investors in a rising interest rate environment. For issuers, if the likelihood of further interest rate hike begins to wane (e.g., due to lower inflationary concerns), there is good reason for calling the note to avoid paying the higher step-up coupon rates, and the likelihood of early redemption rises. Table 8.4.4 below shows an example of a callable step-up fixed rate note.

Table 8.4.4 – Example of a Callable Step-up Fixed Rate Note

Issuer	**An American Bank**
Credit Rating	BBB+
Size of Issue	USD 20m (estimated pre-launch issue size)
Trade Date	30 May 2018
Minimum Investment (Denomination)	USD 25,000 (USD 1,000)
Maturity Date	23May 2024
Tenor	6 years
Semi-annual Coupon May, Nov 2018 -– 2019	3.00% annualized
Semi-annual Coupon May, Nov 2020 -– 2021	4.00% annualized
Semi-annual Coupon May, Nov 2022	5.00% annualized
Semi-annual Coupon May, Nov 2023	6.00% annualized

8.5 Structured Notes with Yield Enhancement (No Capital Preservation)

A short option strategy combines either the selling a put or call options with a note or deposit offering investors enhanced yield than what is available in the market. While such products provide a fixed return that is higher than what the market would allow, they do often come with downside risks and do not carry capital preservation. The risk-return distributions in some cases can skew against the investor since positive returns are capped on one side and the full capital loss in the worst-case scenario (though unlikely) is possible on the other side.

8.5.1 Equity Linked Notes

One of the most popular yield enhancing structured products amongst investors is the **equity linked note (ELN)**. An ELN is a structured product that seeks to provide investors with better yields than what is normally available in the market, as long as the underlying equity or index does not breach the strike price at maturity.

While marketing brochures often state that an ELN's performance is driven by performance of the underlying equities, an ELN has a totally different payoff profile from the underlying equity. Unlike investing in equities where the objective is focused on capital appreciation and dividends, ELNs are yield enhancement instruments with higher-than-market but capped returns on the upside while bearing a potential for downside capital loss if the underlying stock prices fall sharply. In Singapore and Hong Kong, ELNs are typically offered with no principal guarantee (no capital preservation).

a. For a "Bull" ELN", investors sell a put option which can enhance the yield of the note or deposit. If the price of underlying asset closes at or above the strike price on the valuation date, the investor receives the promised yield. However, if the underlying asset closes below the strike price at maturity, the put sold is exercised and the investor has to buy the underlying stock at the strike price (or the cash equivalent value is calculated). The overall returns will depend on how much of the return promised from the note is offset by the decline of the underlying asset below the issue price.

b. In a "Bear" ELN, investors sell a call option which can enhance the yield of the deposit if underlying equity price closes at or below the strike price on the valuation date. If it closes above the strike, the ELN requires that the investor sells the underlying at the agreed strike price to the financial institution realizing the equivalent net gain or loss.

c. For a "Strangle" ELN: investors sell both a call and put option at different strike prices, which can enhance the yield of the deposit if underlying equity price closes at or above the lower strike but at or below the upper strike prices. Bear ELNs and Strangle ELNs are less frequently offered so we will focus on the more popular Bull ELN. For the purposes of the SIP Module, you will only need to focus on Bull ELNs.

d. There is also a variation to the Bull ELN involving a few stocks (usually 2-4 stocks) called the "worst of" ELN. In this structure, the investor would be writing puts on 3 or 4 stocks for the higher yield on the ELN. Please refer to Section 8.5.3 for details on this product.

An ELN can be structured to allow for physical or cash settlement at maturity. The key terms of an ELN are:

i. **Issuer** – The note issuer can be a financial institution or a SPV;

ii. **Reference Financial Asset** – The underlying asset linked to the put option;

iii. **Strike Price** – The underlying asset's price level, below which the option will be exercised;

iv. **Price of Note** – A discount to the face value of the note. The discount calculates the yield of the note if the put option is not exercised;

v. **Mode of Settlement** – Cash or physical settlement; and

vi. **Fixing Date** – The day when the closing price level of the underlying asset / financial instrument is used to determine the payout of the ELN at maturity. It is usually 2 business days from the maturity date.

8.5.2 Bull Equity Linked Notes

The Bull ELNs is the most popular type of ELNs sold. Bull ELNs offer a higher fixed rate of return than what is available for typical notes and deposits in the period. In exchange, the investor sells an embedded put option in the structure to the financial institution, which allows the institution to sell underlying asset back to the client at a particular put strike price (or to receive its equivalent value) if it has been breached at maturity. If the strike is not breached, the client will enjoy the higher rate of return from the note without any need to receive equities.

Investors who buy Bull ELNs with equity share settlement are typically comfortable about accumulating the shares at the put strike level as a longer-term investment even if it falls below the put strike level. The choice of stocks used in the ELN is important and investors have to be positive about the long-term prospects of the underlying company. Some ELNs are structured to realize cash proceeds at maturity with no opportunity to convert to equities.

Though ELNs provide higher returns than returns available in the market, as shown in the payoff profile (Figure 8.5.2(b)), an ELN is structured with a fixed maximum return. If the underlying share price rises considerably, the investor will still receive the fixed return through the ELN and does not participate in the full upside potential of the underlying asset. Therefore, if an investor is very bullish about the potential of a stock, it may be better for him to invest directly in the underlying shares. If he is mildly bullish in the near term and sees modest to flat performance on the upside, the Bull ELN would be a potential yield enhancer.

Can we change the structure to secure some protection from downside risk? Yes, the investor can buy a lower strike put option to cap the downside risk, but this will come with additional cost which must be paid. This is usually achieved by sacrificing some upside returns or buying a put separately from the ELN.

The investor also bears the credit risk of the issuer or the debt securities backing the SPV. A credit event may trigger an early termination of the ELN, and the investor will be paid the liquidation value of the various components of the ELN, which could be substantially below the principal investment.

An ELN may have little or no liquidity. The investor should be prepared to hold the ELN for the entire tenor. Furthermore, if a secondary market exists, the mark-to-market value of the ELN will fluctuate depending on various factors, such as the movement in the underlying share price or index, its volatility, the performance of the fixed income investment, and the creditworthiness of the issuer.

Table 8.5.2 – Features of a Structured Note with Yield Enhancement / Optimisation (With No Capital Preservation)

Structured Notes with Yield Enhancement / Optimisation (With No Capital Preservation)	Features
Types of clients	Medium to high risk.
Purpose	• Seeking higher yield and to optimize return. • Willingness to acquire the particular stock at a specific discounted price level from the initial market price.

Structured Notes with Yield Enhancement / Optimisation (With No Capital Preservation)	Features
Common underlying assets	Individual stocks that are very liquid, indices, or a basket of a few stocks.
Benefits and gains	• Potentially significantly higher yield than otherwise available from the market. • An opportunity to set a price level for the underlying stock to fall beyond a level at which the investor is comfortable about owning the stock. • Flexibility and choice of underlying stocks to use for the underlying to structure the ELN. In general, the ELN should be created for the stocks that client is comfortable owning for the long term anyway (e.g., blue- chip stocks).
Risks and losses	• Potential loss of invested capital. • Taking delivery of equity that has fallen in value and reflecting a book loss or having to realise the loss at maturity if cash settlement is specified in a particular contract (e.g., when underlying is an equity index). • The risks stem from: – Market risk associated with the equities. – Foreign Currency risk if investing in a foreign stock, basket of stocks forming the underlying (USD, HKD, etc). – Issuer Credit Risk since the structure is issued by the particular financial institution. – The structure being usually sold without capital preservation in Asia. – Liquidity Risk. Since the structure is sold in the OTC market by financial institutions, there is little or no liquidity in the secondary market. – Early redemption will usually result in losses.
Obligations	To take delivery of the equity or to pay the equivalent of the loss (from the fall in equity price or fall in equity index price level).

Figures 8.5.2(a) and 8.5.2(b) show the structure and payoff profile of an ELN.

Figure 8.5.2(a) – Structure of an ELN

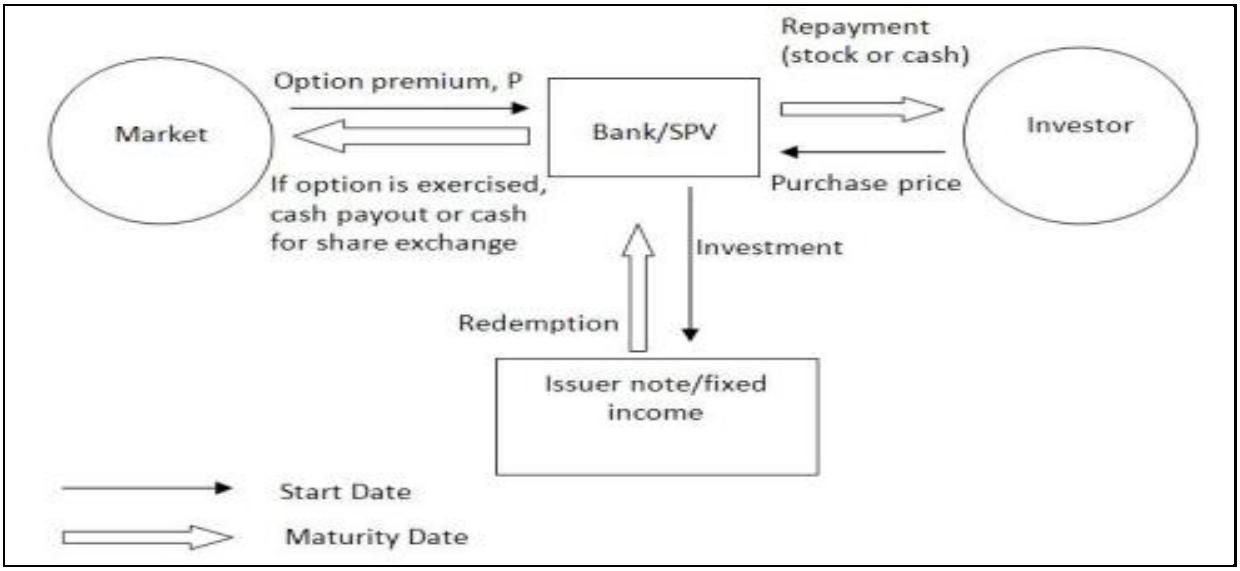
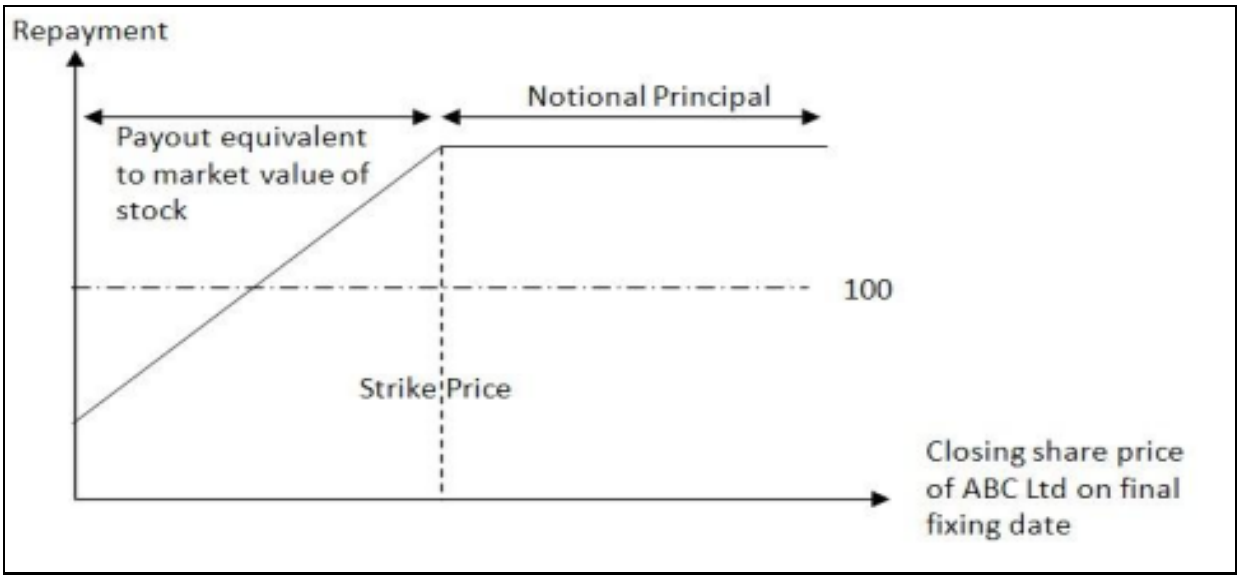


Figure 8.5.2(b) - Payoff of an ELN



Standard Equity Linked Notes (Bull ELNs)

Suppose it is Friday, 11 May, and an investor is interested in investing in Apple shares (APL USD 188.59) for the longer term. However, the share price has run up recently, and he is waiting for a pullback to accumulate this blue-chip stock. He is willing to accumulate it when it weakens by 6-10%. He approaches a financial institution for an ELN on Apple shares.

The 6-month (16 Nov 2018) out-of-the money put option on NASDAQ for APL with a strike at $174.52 was priced at $6 on 11 May 2018[37]. The current spot price of Apple shares is $188.59.

The minimum investment for this 6-month ELN is set at USD 1 million and an equivalent money market yield is 2.51688% per annum (p.a.) Note that an ELN is typically priced on a discounted basis.

Based on the strike price ($174.52), a USD 1 million investment can buy about 5,730 Apple shares. If not exercised, discounted ELN pays USD 1 million at maturity, else ELN pays 5,730 Apple Shares.

The ELN can be created by writing 5,730 put options on Apple shares which generates $6 x 5,730 = $34,380 in premium. The premium can be placed in a money market deposit for 6 months, earning interest of USD 1,034,380 x $(1.0251688)^{½}$ = USD 1,034,380 X 1.0125 = USD 1,047,309.75 (rounded to the nearest dollar).

This provides a yield of (USD 1,047,309.75/ USD 1 million) - 1 = 4.730% for 6 months, or an annualized return of $(1.0473^2 – 1) \times 100\% = (1.0968 – 1) \times 100\% = 9.68\%$. So, from the bank's perspective, they can offer the client a return of 9.68% or lower given the interest earned on the amount invested.

The price issued for the 6-month Bull ELN would be quoted at a discount to the face value of USD 1 million i.e.; $1/(1.0968)^{1/2} = 95.49\%$, assuming the bank does not take a spread on the note.

Product description	Structured note linked to Apple share price
Reference underlying asset	Apple shares
Current spot price	$188.59
Strike price (92.54% of Spot)	$174.52
Principal amount	$1 million
Issue price (95.49%)	$954,900
Yield offered	9.68%
Maturity value	$1 million or 5,730 Apple shares
Maturity date	16 November 2018
Valuation date	14 November 2018
Participation rate	100%
Tenor	6 months

Scenario Analysis

At maturity, there could be 3 scenarios:

1. **Scenario 1**: Apple share price rises by 12% to $211.22 at maturity. The put will not be exercised, and the investor will receive USD 1 million, i.e.; the nominal face value at maturity, earning 9.68% (p.a.) yield on the USD 954,900 initial investment. If he had bought Apple shares directly, he would have a return of 12% (after 6 months) but now he only received 9.68%, annualized.

2. **Scenario 2:** Apple shares fall by 5% to $179.16 at maturity but this is still higher than the strike price of $174.52 so put is not exercised and the investor received 9.68% annualized yield.

[37] For Singapore and Hong Kong stocks, instead of put options we can also use put options of local stocks matched by the closing date, maturity and put warrant strike price.

3. **Scenario 3**: Apple shares fall by 10% to $169.73 at maturity which is lower than the strike price of $174.52. The put is exercised, and the investor buys 5,730 shares at $174.52. The investor has an unrealised book loss of -2.74% i.e.; (169.73/174.52 - 1) x 100%.

   If this was a cash equivalent settlement, then he would have purchased the investment at $174.52 x 5,730 shares = $999,999.60, and he would be receiving USD 169.73 x 5,730 = $972,552.90. The realized loss would be -2.74% or $27,446.70.

Theoretically the downside for the Bull ELN is the entire notional amount of the note (there is no capital preservation), any stock, even Apple shares can go to zero, but if the right stocks are selected, the likelihood of this worst-case scenario is very low.

8.5.3 "Worst of" Equity Linked Notes

A "worst of" ELN is linked to more than one underlying share or indexes. Investors sell a put option on the basket of shares and are obliged to buy the worst performing stock in the basket at its strike price if the closing price of the worst performing stock is below it strike price. The worst performing stock or index is determined by comparing the closing price on the fixing date against its initial price in percentage terms.

The option premium is determined by the underlying shares. Typically, the higher the correlation between the underlying shares, the lower the put option premium as the chances of option exercise will be lower. If correlation is 1, there is no difference between a worst-of put option and a put option of a single underlying. Similarly, if the underlying shares are negatively correlated, it increases the probability of option exercise resulting in higher put option premium.

The option premium is also determined by the volatility of the underlying assets. The higher the volatility, the higher the option premium as the probability of option exercise is higher.

The "worst of" ELNs could incorporate options such as knock-in barrier and/or knock-out barrier.

The following are two examples of "worst of" ELNs – one with fixed coupon yield and one will variable coupon yield.

"Worst of" Equity Linked Notes or Laggard ELNs

Your client has some deposits and would like to enhance returns. She likes exposure to equities and does not mind owning A, B or C shares for the long term. She expects the market to remain relatively stable with a modest upside.

A "worst of" ELN can be set up with a knock-out on the upside and knock-in on the downside. The client is long a strip of digital calls options for the coupon and sells a knock-in worst-of put option to the bank.

On valuation dates, if all the shares A, B and C are at or above their set knock-out price level, the client will receive 100% of the principal in cash, together with the latest coupon payment and the note is redeemed. The note is knocked out. If stocks A, B, and C stay below their knock-out price level, the note continues towards maturity.

If the underlying shares A, B and C never trade below the knock-in levels, the client will never receive the shares. If any of the shares A, B and C perform below the knock-in levels and strike levels, the client will receive the worst performing share at the valuation date.

The details are given below:

Details of Underlying Shares in a Laggard ELN	Stock A	Stock B	Stock C
Initial Reference Spot and Knock-out (SGD)	13.34	12.43	3.12
Current Price level at Issuance	13.26	12.21	3.09
Strike Price (95% of Initial Reference Spot SGD)	14.00	13.05	3.28
Knock-in (85% of Initial Reference Spot SGD)	11.87	10.92	2.76

Laggard ELN Features	
Issue Date	15 May 2018
Valuation Dates	• 1st Val (15 June 2018) • 2nd Val (16 July 2018) • Last (14 August 2018) (3-month tenor)
Principal Amount:	SGD 200,000.00
Annualized Rate of Return Offered	• 11.35% (based on interest earned on proceeds from puts, calls, notes)

Scenarios:

1. On either valuation dates 15 June or 16 July, if all 3 share prices close at or above their knock-out price level, the client gets back the principal + coupon payment and the note early terminates.

2. If during the 3 months' period till maturity of note, all the 3 shares stay above the knock-in price level and below the knock-out level, the note will be redeemed at principal amount + coupon payment.

3. If during the 3 months' period, all 3 shares did not get knocked out and any of the 3 shares fall below the knock-in price level during the tenor, but all 3 shares close at or above the **strike price level,** the client gets back principal amount + coupon payment.

   In both scenarios 2 and 3, the client received the principal amount plus 3 coupon payments over the 3 months. The difference is that the share price that touched the knock-in level is required to end above the **strike price level**. If share C dips to SGD2.76 briefly in July, it must recover above SGD3.09 for non-conversion by the final valuation date on 14 August to ensure that no conversion to stocks takes place.

4. If any of the 3 shares fall below the knock-in level during the tenor and close below the strike price level, on the final valuation date, the client will receive the shares of the worst performing stock + coupon payment. (Note that the worst performing stock at the final valuation need not necessarily be the one that touched the knock-in level).

   In scenario 4, assume that:

- Share C dips briefly below the knock-in level of SGD2.76 and recovers to SGD3.00 below the strike level (down 3.8% from the initial reference spot level).
- Share B stays above both the strike and knock-in level at SGD12.50 on the final valuation date.
- Share A does not touch the knock-in level but at valuation it is priced at SGD12.00 (down 10.0% from the initial reference spot level).

Therefore, Share A would be the worst performing stock that would be converted. The worst stock is the one that has fallen the most at the final valuation date even if it did not touch its knock-in level. The conversion price used is the strike price level SGD13.26. The client would pay SGD13.26 per share for Share A, even if it is worth less at the valuation (down 5% from the initial reference spot level). The calculation would be SGD200,000/SGD13.26 = 15,082 shares with the remainder paid back in cash. The client would still receive the total of 3 coupons at valuation date.

Worst of Equity Linked Notes or Laggard ELNs (with Variable Cumulative Coupon)

Your client has some deposits and would like to enhance returns. She likes exposure to equities and does not mind owning Stock A and Stock B and is prepared to hold either stock in her portfolio for the longer term. You can structure the following "Worst of Equity" ELN structure for her.

Features of Laggard ELN	
Nominal price	USD 200,000.00
Callable price	100% of the closing price of the respective stocks on the trade date of the note
Conversion price	90% of the closing price of the respective stocks on the trade date of the note
Initial Price of Stock A	USD 30.00, Callable Price USD 30.00, Conversion Price = USD 27.00
Initial Price of Stock B	USD 55.00, Callable Price USD 55.00, Conversion Price = USD 49.50
Tenor	2 years
Call dates	Semi-annual (4 dates)
Coupon if called	1st call date: 8% if worst performer ≥ callable price; otherwise 0%
	2nd call date: 16% if worst performer ≥ callable price, otherwise 0%
	3rd call date: 24% if worst performer ≥ call price, price, otherwise 0%
Maturity date	4th call date: 32% if worst performer ≥ call price, price, otherwise 0%
Variable coupon	• At maturity: Worst performer's closing price divided by conversion price – 100%

	• If not called: if the callable price > worst performer's closing price at maturity ≥ conversion price

Over the 2-year period, before maturity at the call dates, the following scenarios are possible:

1. **Scenario 1 –** If the laggard stock (worst performing stock) is below its call price, the note pays zero. Each time the note is not called, it is paying zero and continues to the next coupon period.

	Price at 1st Call Date (USD)	Initial Price or Call Price (USD)	+/-%
Stock A	29.50	30.00	-1.67%
Stock B	56.00	55.00	+1.82%

Stock A is the laggard, and it is below its callable price, so the note pays zero coupon.

2. **Scenario 2 -** If both Stock A and Stock B are above the call prices, the note is called, and we have early redemption at 100%. The final coupon payment for the period is paid out.

	Price at 2nd Call Date (USD)	Initial Price or Call Price (USD)	+/-%
Stock A	32.00	30.00	+6.67%
Stock B	56.50	55.00	+2.73%

Stock B is the laggard, but it is above its call price, so the note of USD 200,000.00 is called and your client will be paid USD 232,000.00 representing the 16% return.

At maturity, the following scenarios are possible:

3. **Scenario 3 -** If ELN is not called during first 3 call dates, it would not have received any coupon payments at those call dates. At maturity, and if the laggard stock is above the call price, the cumulative coupon will be paid with the principal at maturity.

	Price at Final Call Date (USD)	Initial Price or Call Price (USD)	+/-%
Stock A	32.00	30.00	+6.67%
Stock B	65.00	55.00	+18.18%

Stock A is the laggard. It is above its call price, so the repayment of the note USD 200,000.00 will be accompanied by a cumulative coupon of 32% = USD 264,000.00

4. **Scenario 4:** The ELN is not called during the first 3 call dates, and the laggard stock closes below the call price but above the conversion price at maturity. A variable coupon rate will be calculated based on the spot price over the conversion price of the laggard stock. Conversion price of stock A is USD 27.00 and conversion price of stock B is USD 49.50.

	Price at Final Call Date (USD)	Initial Price or Call Price (USD)	+/-%
Stock A	29.16	30.00	-2.80%
Stock B	60.00	55.00	+9.09%

Stock A is the laggard. Stock A's price at the final valuation date is below the initial price or call price. The variable coupon rate = (29.16/27.00) – 1) x 100 = 8.00%. The client will get back USD 216,000.00 (i.e., USD 200,000.00 x 1.08), representing principal and coupon earned.

5. **Scenario 5:** The worst-case scenario is if the note is not called over the 4 periods over the 2 years, so no coupons are paid, and the client has to buy the worst performing stock. If the laggard closes below the call price on all call dates and below call and price and conversion price at maturity, the client will take delivery of the laggard stock. Conversion price of Stock A is $27.00 and conversion price of Stock B is $49.50.

	Price at Final Call Date (USD)	Initial Price or Call Price of A	+/-%
Stock A	26.00	30.00	-13.33%
Stock B	60.00	55.00	+9.09%

Stock A is the laggard stock. The USD 200,000.00 will be converted to USD 200,000.00/27.00 = 7,407.41 shares or 7,407 shares with the remainder paid in cash. There are no coupon interest payments, hence no interest is earned for this investment. Instead, the client incurred a paper loss = (USD26 - USD27) x 7,407 = - $7,407, or -3.70% loss over the initial investment.

8.5.4 Risks of Equity Linked Notes

Buying an ELN usually involves selling an out-of-the-money put on the underlying shares, but aggressive investors can choose to sell at-the-money puts. If the underlying stock trades below the put strike price, the investor is obliged to buy the underlying shares above the market price of the share on maturity. If the closing share price remains below the put strike price at the final fixing date, the investor is exposed to the downside movement in the underlying share price.

The investor also bears the credit risks of the issuer or the debt securities backing the SPV. A credit event can trigger an early termination of the ELN, and the investor will then be paid the liquidation value of the various components of the ELN, which could be substantially below the principal investment.

An ELN may have little, or no liquidity and investors should be prepared to hold it for the entire tenor. Furthermore, if a secondary market exists, the mark-to-market value of the ELN will fluctuate depending on a number of factors such as the movements in the underlying share price or index, its volatility, dividends payments, the performance of the fixed income investment, and the issuer's credit worthiness.

A "worst of" ELN is like a standard ELN, except that it is linked to more than one underlying share or index and its return depends on the performance of the worst performing underlying share in the basket. For a "worst of" ELN, the investor is exposed to the downside risk of the worst performing underlying security, even if the closing prices for all other underlying securities are above their strike prices on the fixing date. This ELN pays a higher yield (i.e., deeper discount) compared to a standard ELN because it is a much riskier investment. The risk to the downside is not just based on a single stock or single stock index, but multiple stocks or indices.

8.6 Credit Linked Notes (CLNs)

A CLN is another type of yield enhancement structured product. As the name suggests, it has exposure to the credit markets. In a CLN, the issuer offers credit insurance known as **credit default swaps (CDS)** linked to a particular company (referred to as "reference entity" in the term sheet). It has a fixed income investment, which is used as collateral against the credit insurance sold in the market.

A CDS is akin to providing credit insurance on a reference entity. It is a contract between two counterparties, whereby the "buyer" pays periodic payments (insurance premium) to the seller in exchange for the right to a payout if there is a credit default of the “reference entity”. CDS are OTC-traded products. Payment is made upon a credit event.

A credit event would typically be (amongst others):

a. Bankruptcy;

b. Failure to make payment;

c. Debt Restructure (adverse change);

d. Repudiation;

e. Moratorium;

f. Obligation Acceleration; or

g. Obligation Default.

The bankruptcy of the underlying company issuing the bond and its failure to make payments are quite clear as credit events.

Debt Restructuring refers to a change in the terms of the debt, which causes the debt to be less favourable to debt holders. Common examples of debt restructuring include a decrease in the principal amount to be paid, a decline in the coupon rate, a postponement of payment obligations, a longer maturity time, or a change in the priority ranking of payment.

When a credit event occurs, a notice will be issued by the buyer or seller to specify the protection amount that will be paid by seller either through physical settlement or cash settlement. For physical settlement, the issuer of the CDS (the protection seller) will buy the distressed loans or bond from the protection buyer. This is the delivery obligation of the CDS issuer (protection seller).

In a CLN, the investor sells the credit insurance (via the CDS embedded in the CLN) to the CLN issuer (typically a bank) in return for a higher yield. Upon the occurrence of a credit event, the CLN investor will receive the asset of the defaulted reference entity, instead of the original invested cash principal. If there was no credit event during the life of the CLN, the investor will receive the cash principal (plus spread from credit insurance) at maturity.

In the event of a credit default, there are 2 ways to settle the CDS/CLN contract:

i. Physical settlement - The bank, being both the seller of the CDS and issuer of the CLN, pays the CDS buyer the principal amount of the CDS in cash and receives a debt obligation (i.e., a bond) of the reference entity that is now in default. Note that the bank had originally bought the CDS (i.e., bought credit insurance) from the CLN investors. In this scenario, the bank acts as an intermediary for the CDS, where the CDS bought from the original CLN investors is sold to a third party as described above. The bank then transfers the defaulted bond (received from the third party CDS buyer) to the CLN investors. The CLN investors end up owning the bond of the defaulted reference entity, which would have a value determined by the market, typically, this would be at a deep discount.

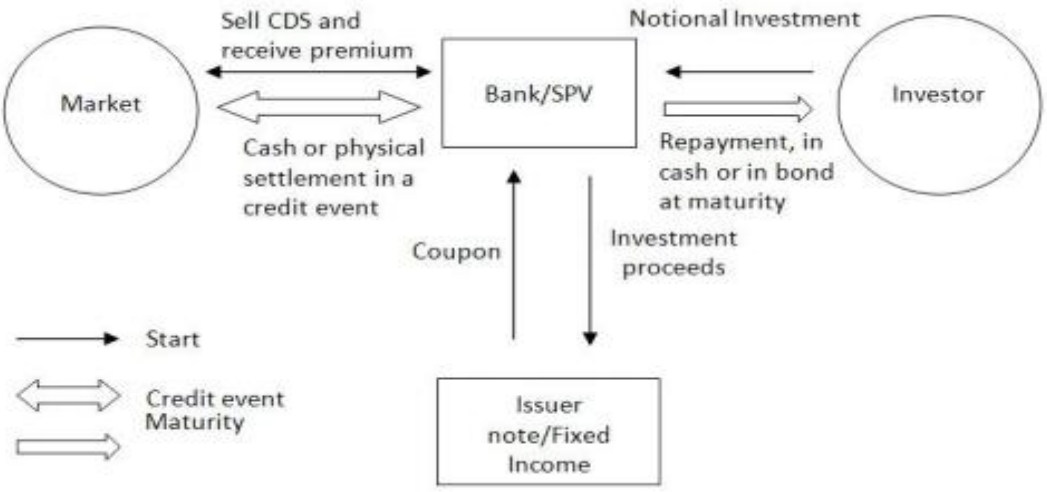
ii. Cash settlement - The bank pays the CDS buyer the difference between the par value and market price of a specified debt obligation of the defaulted reference entity, typically determined in an auction. In return, the bank will pass on this loss to the CLN investors. In essence, the CLN investors will bear a loss equivalent to that difference.

The key terms of a CLN are:

- **Issuer** – The entity issuing the note, which may be a financial institution or SPV;
- **Reference Entity / Entities** – Company / companies which are linked to the CDS;
- **Mode of Settlement** – Cash or physical settlement of reference entity in a credit event;
- **Type of CDS** – Single reference entity default or "first to default" for multiple reference entities;
- **Early Termination** – Event that may trigger an early redemption of the note;
- **Coupon** – Periodic interest rate payable to note holder; and
- **Coupon Payment Period** – Time interval between coupon payments.

Figure 8.6 depicts the structure of a CLN.

Figure 8.6– Structure of a Credit-Linked Note



Credit Linked Note (CLN)

Description	Note linked to the credit of ABC Ltd
Reference entity	ABC Ltd
Type of CDS	Single credit of ABC Ltd
Mode of settlement	Physical settlement with ABC Ltd bond
Issuer	XYZ Bank, rated single-A

Early termination trigger	Credit default of "reference entity"
Term	5 years
Coupon	4% p.a. payable annually, unless early termination of the note
Maturity	100% of principal if there is no credit event on the reference entity

Scenario Analysis

1. **Positive scenario: No credit default of the reference entity**

   In this scenario, the CLN holder will receive the 4% coupon annually and 100% of the principal investment at maturity.

2. **Negative scenario: Credit default of the reference entity**

   In this scenario, the issuing bank will use the cash to exchange for the reference bond of ABC Ltd at face value, as agreed in the CDS. The note will be redeemed and the CLN holder will receive the reference bond. The CLN holder can choose to hold the bond or sell in the market immediately upon receiving the bond. However, he is then likely to suffer a substantial loss, as it is unlikely for a defaulted bond to be trading close to its par value.

8.6.1 "First to Default" Credit Linked Note

In some CLN structures, the issuer offers credit insurance on multiple credits but on a first to default basis. This structure is much riskier as compared to a standard CLN. If any of the credits default (first one to default), cash held with the issuer is used to compensate the CDS buyer by cash or physical settlement as discussed earlier. In such an event, the investor is likely to lose a substantial part or all of the principal investment.

8.6.2 Risks of Credit Linked Notes

As the name suggests, this product has exposure to the credit markets. A CLN is another type of 'yield enhancement' structured product. It has a fixed income investment, which is used as collateral against the credit insurance sold in the market.

In a CLN, the investor provides credit insurance known as CDS linked to a particular company (referred to as "reference entity"). A CDS is akin to providing credit insurance on a reference entity. It is a contract between two counterparties, whereby the "buyer" pays periodic payments (insurance premium) to the seller, in exchange for the right to a payout if there is a credit default of the "reference entity". A CDS is an OTC derivative.

Like an ELN, a CLN investor receives higher-than-market rates reflecting the risk associated with the sale of the CDS linked to the reference entity. In a CLN, investor is exposed to two different credit risks:

a. Credit of the note issuer or bond(s) held by the SPV as collateral for the CDS; and

b. Credit of the "reference entity" linked to the credit default swap.

In "First to default" CLNs, the investor sells CDS insuring credits of multiple reference entities, on a first to default basis. This structure is a much riskier than a standard CLN. If any of the reference entities defaults (first one to default), cash held with the CLN issuer will be used to compensate the CDS buyer by cash or physical settlement. In such an event, the investor is likely to lose a substantial part of his principal investment.

8.6.3 Bond-Linked Notes (BLNs)

A BLN is also a yield enhancement structured product. It embeds a short put into a bond. The investor may end up owning a bond at maturity like a CLN described above. However, there are some differences between the two structured products:

a. A CLN sells a CDS on a reference entity, whereas a BLN sells a put option on a bond which has a strike price;

b. The CLN's payout depends on whether there is a credit event on the reference entity, whereas the payout of a BLN is dependent on the price of a bond;

c. Besides a credit event, the price of a bond can be affected by other factors such as a credit downgrade, widening spreads and volatile interest rates; and

d. In a BLN, the investor may end up owning a bond even if there is no credit event in the reference credit / bond.

8.6.4 Risk of Bond Linked Notes

A BLN embeds a short put on a bond. Potentially, investor may end up owning a bond at maturity, similar to a CLN. However, the difference between the two structured products is that for a CLN, a CDS is sold on a reference entity, while a BLN involves selling a put option on a bond which has a strike price.

Like a CLN, a BLN is also a 'yield enhancement' structured product. The payout of a CLN depends on whether there is a credit event on the reference entity while the payout of a BLN depends on the price of a bond. Besides a default, the bond price can be affected by other credit events such as credit downgrades, widening spreads and volatile interest rates. A BLN investor may end up owning a bond even if no credit default has occurred in the reference credit / bond.

8.7 Market Access

Some structured notes which give investors access to various markets and investment opportunities are:

a. Exchange-Traded Notes (ETNs);

b. Index-Linked Notes;

c. Participatory Notes; or

d. Access Notes.

8.7.1 Exchange-Traded Notes

ETNs have no fixed periodic coupon payments, and the principal is at risk. The price of the ETN is a function of both the credit quality of the issuer, and the performance of the underlying market benchmark or strategy.

ETNs are structured notes that are listed and traded on an exchange. This is the main difference between ETNs and unlisted structured notes, so ETN investors have greater liquidity, even though both share the same risk related to the issuer's credit quality.

ETNs are similar to the exchange-traded funds (ETFs). Both products are traded on exchanges and track an underlying market, and both products give retail investors access to hard-to-reach investments and markets, such as currencies and frontier emerging markets. However, there are differences between ETFs and ETNs:

a. ETFs hold a proportional stake in the financial products that the ETF tracks, offering instant diversification. When investors buy ETFs, they are investing in a fund which holds multiple assets. ETNs are debt instruments, backed by the credit worthiness of the issuer;

b. ETFs do not have fixed maturity and investors can buy or sell ETFs just like stocks on an exchange. ETNs have maturity dates. When an investor holds an ETN until the maturity date, he/she receives a one-time payment based on the performance of the underlying asset, index or strategy. If the investor wishes to liquidate the position sooner, he can sell the ETN on the exchange;

c. Tax treatment of ETFs and ETNs can be different in the different tax jurisdictions; and

d. As ETFs are investment funds, regulations require that they must have a trustee, who is independent from the fund manager. ETNs are issued by banks and there is no regulatory requirement for a trustee, although some ETNs do have them.

8.7.2 Index-Linked Notes

Index-linked notes are debt securities for which the coupon payments and/or the principal are linked to the movements of a market index or an asset price. In exchange for slightly less upside potential, index-linked notes may incorporate a minimum return at maturity in addition to the original principal.

Index-linked notes can be structured with either:

a. A periodic return and a specified participation rate (for example, an annual average performance with 100% participation); or

b. A periodic cap (for example, a 10% annual cap on the participation returns).

Below is an example of the term sheet of an index-linked note with principal preservation:

Index-Linked Note with Principal Preservation

The index-linked note aims to provide 100% principal preservation at maturity plus a return on principal equal to the greater of:

- 4% minimum total return at maturity; or
- 85% participation in the STI Index Annual Average Performance.

Issuer	**ABC Structured Products Ltd**
Rating	A
Market measure	Straits Times Index 3278

STI Index	3,278 (as of 18 May 2014)
Denomination	SGD 1,000.00
Settlement date	18 May 2014
Investment Amount	SGD 1 million
Maturity date	15May 2022 (8-year term to maturity)
Redemption amount	At maturity, each note will pay the greater of: a. Principal x (100% + 4%); b. Principal x (85% + Average Performance).
Average performance	The average performance is computed as the Average index value divided by the initial index value, minus one.
Initial index value	The closing value of the index on the settlement date.
Average index value	The arithmetic mean of the closing values of the index on the annual anniversary dates of the notes.

Scenario Analysis
At maturity:

Scenario 1:
If the average performance of the STI index rises by 5%, the investor would receive a return of $42,500 (i.e., 4.25% return over 8 years):

a) Return : $1,000,000.00 X 4% = $40,000.00
b) Participation : $1,000,000 X 85% participation rate X 5% rise in index = $42,500
(final redemption is the greater of (a) and (b))

Scenario 2:
If the average performance of the STI index falls by 5%, the investor would receive a return of $40,000 (i.e., 4% return over 8 years).

8.7.3 Participatory Notes

Participatory Notes (PNs) are offshore derivative instruments which are not registered with the securities regulator of its underlying assets' jurisdiction and are not listed. In the context of the Indian Equity market, for example, PNs are issued by India-based securities firms to foreign investors. The underlying assets are stocks, and any dividends or capital gains would flow through to the holders of the PNs.

8.7.4 Access Notes

Access notes are financial instruments that are sold directly to the public by the issuing corporation. It could be in the form of equities or bonds used to raise funds by corporations. Investors who buy into access notes would usually hold the instrument until maturity and collect coupon payments in the interim.

8.8 Investors of Structured Notes

Investors buy structured notes mainly because they want yield enhancement and market access. They must have a greater risk appetite for the higher risk that comes with higher yield. More importantly, investors must understand the products' strategies and how the products help to achieve their investment objectives.

8.8.1 Assessing Product Suitability for Retail Investors

When assessing the suitability of structured notes for investors, the selling financial institution must be clear about the investors' objectives and make full disclosure of the potential risks that may prevent investors from achieving their objectives. For example, retail investors generally require some degree of principal guarantee. If the structured note does not achieve the required level of principal guarantee, the financial institution must explain the risk of principal loss or unlimited downside and ensure that the retail investors fully understand the risks.

If the product does not suit the investors' profiles in terms of their wealth levels and risk tolerance, the financial institution must advise investors of this. However, if the investors insist on trading the product, the financial institution must have the necessary documentation and have the investors' acknowledgement on it.

8.8.2 Factors for Investors to Consider in the Sales Process

Investors must be aware of the implications of higher risk-return structures and agree to take on additional risks to meet their investment objectives. In addition, they should at least consider the following before choosing to invest in a structured note:

a. **Liquidity –** The amount invested usually will not be fully redeemable during the tenure of the structured note. Early withdrawal will result in the loss of part or the entire principal invested;

b. **Risk** – Structured notes have higher risks due to the different layers involved. Investors should understand the higher risks involved and have the risk appetite for it;

c. **Returns –** Returns are tied to various underlying instruments in the structured note. Investors should be aware that their returns will vary according to the performance of the underlying instruments and there are possibilities that the instruments' principal values will be affected;

d. **Terms and conditions –** Investors should at least review the terms and conditions, which will explain their obligations and risks that they are subjected to; and

e. **Fees and charges –** Investors should be informed of fees and charges before investing in the structured notes. The management fee is likely to be higher than standard financial instruments due to the note's complexity. For any underlying instruments used, it is likely that a transaction cost will be incurred for every transaction. Investors need to understand how these costs are passed on to them.

8.9 Summary

1. A structured note is a debt instrument / debenture, whose return characteristics, such as coupon amount or market value, are linked to the performance of other underlying instruments. The underlying instruments may include equities, indices, interest rates, credit default swaps, collateral debt obligations, or a basket of instruments.
2. Structured notes are usually issued by banks, or a Special Purpose Vehicle (SPV) set up by banks. Under direct issuance, the bank issues the notes to the note holders (investors). As the funds are raised directly by the bank, the debt will be reflected on its balance sheet as a liability. This debt is a direct obligation of the bank to its creditors and the investor bears the credit risk of the bank.
3. An SPV is a separate legal construct set up by the bank for the purpose of the transaction. When a SPV is used to issue the note, they will use the proceeds to buy fixed income securities to preserve the principal capital or to use them as collaterals against the options or credit insurance which they sell to the market. The assets and liabilities of a SPV are usually not reflected on the bank's balance sheet – they are off balance sheet items for a bank. In the event of default, the note holders will first make a claim on the SPV's assets and may not have recourse to the bank which set up the SPV.
4. There are three main types of Structured Notes:
    - Those offering some capital preservation focus on yield enhancement/optimisation;
    - Those focused on yield enhancement/optimisation with no capital preservation; and
    - Those focused on capital gains from the price movement of the underlying with no capital preservation.
5. Structured deposits are similar to structured notes as they are a debt obligation of the bank. However, structured deposits are not debentures but a type of deposit which can only be issued by banks and must meet the definition of "deposit" as defined under the Banking Act 1970.
6. Most structured products with 'yield enhancing' strategies have one thing in common - the potential extra yield comes from the income generated by writing or selling options. Clients of banks bear the additional risks associated with being an option writer rather than an option buyer.
7. Capital guaranteed structures generally provide relatively low yield and preservation of the notional capital only if the product is held till maturity. There are still other types of risk associated with capital guaranteed products.
8. A range accrual note **(RAN)** is a yield enhancement product in which the investor receives a target level of return if a reference index moves within an agreed range. If the index moves beyond the stipulated range, the investor will receive lesser or no interest, although the principal will not be affected.
9. A callable range accrual note **(CDRAN)** involves the addition of the call feature on specific future days where the issuer has the right to redeem the note. By giving the issuer this option to call the note at different points in the future, the investor should be compensated with a higher coupon rate or yield on the CDRAN.
10. An inverse floating rate note, or simply an inverse floater, is a type of bond or other type of debt instrument whose coupon rate has an inverse relationship to its reference rate (usually the near-term interest rate e.g., 3-month LIBOR).
11. A callable step-up note **(CSUN)** is a capital guaranteed instrument that offers fixed coupons that can increase (step-up) over the life of the instrument. Given the risk to the issuer for such step-up of coupons, the instrument is callable on specific dates during the life of the note. The risk for investors is the issuer

risk and call risk (or reinvestment risk) as interest rates decrease. CSUNs become more popular for investors in a rising interest rate environment.

12. An equity linked note **(ELN)** is a structured product that seeks to provide investors with better yields than what is normally available in the market as long as the underlying equity or index does not breach the strike price at maturity.

13. Unlike investing in equities where the objective is focused on capital appreciation and dividends, ELNs are yield enhancement instruments with high-than-market but capped returns on the upside while bearing potential for downside capital loss if the underlying stock price falls sharply.

14. A **Bull ELN** employs an issuer's note and a short put, linked to a stock index or a particular stock. It can be structured to allow for physical or cash settlement at maturity. There is also the Bear ELN which involves the sale of call options and Strangle ELN where the investor sells both a put and a call to enhance the yield of the ELN. The Bull ELN is the most popular type of ELN sold and are synonymous with the general reference to ELNs.

15. Bull ELNs offers a higher fixed rate of return than what is available in the period. In exchange, the investor sells an embedded put option in the structure to the financial institution which allows the institution to sell the underlying back to the client at a particular put strike level (or to receive its equivalent cash value) breached at maturity. If the strike is not breached, the client will enjoy the higher rate of return from the note without any need to receive equities. Investors who buy Bull ELNs with equity share settlement are typically comfortable about accumulating the shares at the put strike level as a longer-term investment.

16. For a "**worst of**" ELN, investors are exposed to the downside risk of the worst performing underlying security, even if all the other underlying securities' closing prices are above their strike prices on the fixing date. In this structure, the client has essentially sold call options (knock-out options) and put options (kick-in options) to the bank and the premium earned comes in the form of higher yield than what's available in the market. This ELN pays a higher yield (i.e., deeper discount) and/or has a lower strike level as compared to a standard ELN because it is much riskier, with exposure to a decline of not just a single stock or index, but multiple stocks or indices.

17. A credit linked note (CLN) has exposure to the credit markets in which the investor sells credit insurance via credit default swaps (CDS) on a particular company. It has a fixed income investment which is used as collateral against the credit insurance sold in the market. Some CLNs have CDS insuring multiple credits on a first-to-default basis. This note is riskier because if any of the credits default (first one to default), cash with the issuer will be used to compensate CDS buyers by cash or physical settlement. CLN investors receive a higher yield than an equivalent money market rate, which reflects the risks associated with the sale of the CDS. Investors is exposed to two different credit risks – the credit of the note issuer or held by the SPV as collateral for the CDS, and the credit of the "reference entity" linked to the CDS.

18. A bond linked note **(BLN)** embeds a short put on a bond. The BLN sells a put option on a bond and the payout is dependent on the price of the bond and its price may be affected by a credit event.

19. Market access notes give investors access to various markets and investment opportunities. They include exchange-traded notes (ETNs), index-linked notes participatory notes (PNs) and access notes.

20. Investors buy structured notes for yield enhancement and market access. Investors should have a greater risk appetite for the higher risk that comes with higher yield. Banks selling structured notes must be clear about investors' objectives and make full disclosure of the potential risks that may prevent investors from achieving their objectives.

21. A structured note must be mark-to-market periodically during the life of the note, so that the note holder is aware of the market value. The market pricing must be obtained from an independent source. Issuers should also ensure that timely and meaningful ongoing disclosures are provided to investors.

22. The bank selling the structured notes should provide a Prospectus and Product Highlights Sheet for every issue of the structured note. The Product Highlights Sheet highlights the key terms and risks of the note using simple and concise language and is a complement to the prospectus.

Chapter 8: Appendix (Non-Examinable Content)

A8.1 Spread CDRANs

A **spread CDRAN** is an instrument that pays out a return based on the spread between the two interest rates of different maturities which is used as the index (i.e., the index is a differential between the two rates). Long term rates are typically higher than short term rates. Spread CDRAN pays a higher-than market coupon for each day when the differential between the two rates (say 30-year rate vs 10-year rate) is positive or zero. Therefore, it may not pay when there is an inverted yield curve. The benefit of this instrument is that the instrument can generate returns as long as the longer rates stay above the near-term rates.

A8.2 Variations of RANs

With product innovation, some range accrual structures are now combined with an equity linked structure or DCI structure. Due to the low interest rate environment, we are also seeing new products combined with credit to increase the yield for investors.

In a **daily range accrual equity linked structure**, the coupon during the life of the note is subjected to the daily range accrual conditions, while the payment on maturity depends on an equity linked note structure. The principal paid on maturity for these RANs are not guaranteed.

There are also notes which link the range accrual condition to multiple underlying assets, for example, a basket of 3 underlying shares. In such cases, the structure will only accrue coupons if all 3 underlying shares fall within the agreed range, thereby greatly increasing the risk of not accruing coupons on certain days.

A8.3 Documentation Required for Structured Notes

The bank selling the structured notes to retail investors should provide a Prospectus and Product Highlights Sheet for every structured note issue. The Product Highlights Sheet highlights the key terms and risks using simple and concise language and is a complement to the Prospectus.

If the notes are offered to institutional or accredited investors, the note issuer is exempted from providing the Prospectus or Product Highlights Sheet.

A8.4 Contents of the Product Highlights Sheet[38]

The Product Highlights Sheet highlights the product's key features and risks. It should clearly disclose required information in the format as prescribed in the Guidelines on the Product Highlights Sheet for Offers of Debt

[38] Practice Note on the Product Highlights Sheet (SFA PN-02) (https://www.mas.gov.sg/regulation/practice-notes/practice-note-on-the-product-highlights-sheet) and Guidelines on the Product Highlights Sheet for Offers of Debt Securities, Hybrid Instruments and Equity Securities (SFA13-G13) (https://www.mas.gov.sg/regulation/guidelines/guidelines-on-the-product-highlights-sheet-for-offers-of-debt-securities-hybrid-instruments-and-equity-securities-sfa-13g13.

Securities, Hybrid Instruments and Equity Securities under the SFA. It should not contain any information that is not in the Prospectus and not contain any false or misleading information.

The Product Highlights Sheet is applicable for offers of:

a. Plain vanilla debentures (Debt Securities);

b. Preference shares, perpetual securities or convertible bonds (Hybrid Instruments); and

c. Ordinary shares, units in a real estate investment trust or units in a business trust (Equity Securities).

The Product Highlights Sheet should comply with the following:

i. Depending on the investment products offered (Debt Securities, Hybrid Instruments, Equity Securities), the length should not exceed the maximum page limits set out in the Guidelines. Diagrams, illustrations and definitions, count towards the page limit;

ii. Text should be in a font size of at least 10-points Times New Roman; and

iii. Information should be set out in a clear and simple language that investors can easily understand. Avoid using technical terms and disclaimers. If technical terms are unavoidable, issuers should attach a glossary to explain these technical terms.

The Guidelines provide templates of Product Highlights Sheet for investment products offered (Debt Securities, Hybrid Instruments, Equity Securities).

The Product Highlights Sheet should contain the following information (as per the relevant template):

1. Product name & summary information, including:
    - Product type / classification;
    - Issue price;
    - Name of issuer;
    - Name of trustee (if applicable);
    - Description of the investment product (Debt Securities, Hybrid Instruments, Equity Securities);
    - Issue & maturity date; and
    - Credit rating, capital guarantees feature and name of guarantor (if applicable).
2. Investment suitability:
    - Return objectives;
    - Risk of loss of principal;
    - Recommended investment tenure; and
    - Other key characteristics which will help the investor make an informed decision.
3. Key features:
    - Background information on the issuer;
    - Key financial information;
    - Underlying securities' key features;
    - Events likely to have a material effect;
    - Use of proceeds.
4. Key risks, including but not limited to:
    - Business related risks;

- Legal, regulatory and enforcement risks;
- Market risks;
- Credit risks;
- Liquidity risks; and
- Other pertinent risks.

5. If there is a risk that an investor may lose all of his principal investment, this must be emphasised with bold or italicised formatting.
6. Definitions.
7. Contact information.

Practice Note on the Products Highlight Sheet and weblink to the Guidelines for Product Highlights Sheet, issued by MAS, are provided in Appendix B.

Chapter 9: Structured Funds

Learning Objectives
The candidate should be able to: ✓ Explain what structured funds are and its key components. ✓ Understand common terms that are used for structured funds. ✓ Describe how structured funds differ from other funds and financial products. ✓ Describe fund structures, investment strategies and common types of funds available to investors. ✓ Explain the role of fund managers and fund trustees. ✓ Describe key documents used in the management and sale of structured funds. ✓ Explain the benefits of structured funds to investors. ✓ Explain what exchange traded funds (ETFs) are, the range of ETFs available and how they differ from traditional funds. ✓ Describe features of the global ETF market and various governance frameworks for funds and ETFs. ✓ Explain the differences between the direct and synthetic replication methods for creating structured ETFs. ✓ Describe structured ETFs investors and the investment strategies they use. ✓ Explain the different types of risks for investors in structured funds and structured ETFs.

9.1 Structured Funds[39]

Structured funds are investment portfolios that combine financial instruments, such as equities and fixed income securities and/or derivatives, to achieve specific risk-return profiles or cost savings. While it typically

[39] Note that the term "structured fund" only applies for the purposes of the CMFAS CM-SIP Examination.

comprises a combination of securities and derivatives components, it is also possible for a structured fund to invest only in derivatives.

There are a wide range of structured funds available to match different risk profiles and investment objectives, such as the desire for downside protection, provision of regular interest payouts, and so on. Fund managers involved purely in traditional asset classes still exist. However, over the last 10 years, an increasing number of new funds launched carry strategies that require the use of derivatives and alternative investments for diversification as a means of hedging, creating downside protection, and to take advantage of short-term investment opportunities. Some portfolio managers also prefer to use funds and derivatives to lower the cost of management and diversification.

Today, it is common practice to include the use of derivatives as a means of hedging and this is specified in the portfolio mandates. As investors look for better performing funds, new strategies, the use of algorithms and computer-program driven trading has also raised the popularity of exchange traded funds (ETFs) and structured funds as a means of executing strategies. Hence, what was considered unique in the past is now more and more commonly practised nowadays.

Structured funds can range from simple to highly complex funds and cover a wide range of risk exposures. The funds are created through financial engineering, by combining underlying securities (e.g., shares, bonds, notes, warrants or indices) with derivatives (e.g., options, forwards, swaps or collateralised debt obligations (CDOs)/credit default swaps (CDS)[40]).

Structured funds that offer some level of capital preservation are designed to earn a market-linked return, while having some level of protection during a financial market decline. However, capital preservation will not apply if the issuer becomes insolvent and cannot fulfil its obligations. Therefore, there is no full capital guarantee since the fund will still face counterparty risks.

Most investments that offer some level of capital preservation have fixed maturity schemes and are designed to be held to full maturity. For structured funds with capital preservation, the maturity period is usually more than 3 years. The capital is preserved only with respect to 100% of the capital invested less upfront sales charge and applies only if the investor holds it till maturity. Disclosure of limitations can be found in the structured fund's prospectus.

Structured funds are governed by the Code on Collective Investment Schemes (Code on CIS) under the Securities & Futures Act 2001 (SFA). Structured funds are packaged and can only be managed by an asset management company that has a capital markets services (CMS) licence. Fund managers have a fiduciary duty to look after investors' best interests. They therefore have a responsibility to ensure that they get the best prices when executing trades for the funds. In addition, structured funds have to provide regular net asset values (NAVs) as required under the Code on CIS, while structured notes and structured deposits are not required to do so. Many new funds launched over the last 10 years are actually open-end structured funds but are not marketed or classified as structured funds.

There are two major types of structured funds:

a. Based on derivatives (refer to Section 9.1.5 Structured Funds Using Derivatives); and

b. Based on techniques from portfolio insurance (refer to Section 9.1.6 Structured Funds using the Constant Proportion Portfolio Insurance Technique).

[40] Collateralised Debt Obligations are securities that are backed by a pool of assets such as non-mortgage loans, mortgage loans or bonds. Credit Default Swaps are swaps that are designed to shift credit risk between counterparties.

9.1.1 Common Terminology

This section explains common terminology related to structured funds.

Asset Allocation. This is the process of dividing investments into different kinds of assets such as equities, fixed income, money market, commodities, real estate, etc. Asset allocation aims to optimize the risk/reward trade-off based on an individual or institution's investment objectives, risk tolerance and investment horizon.

Assets under Management (AUM). This is the market value of assets that an asset management company manages on behalf of its investors. AUM is widely regarded as a measure of success in the asset management industry. AUM can grow or decline due to both capital appreciation/losses and money inflow to/outflow from the fund.

Closed-End Fund. A closed-end fund is a publicly traded investment company that raises a fixed amount of capital through an initial public offering (IPO). The fund is then structured, listed and traded like a stock on a stock exchange. A closed-end fund stock represents an interest in a specialized portfolio of securities that is actively managed by an investment advisor, and which typically concentrates on a specific industry, geographic market, or sector. The prices of a closed-end fund fluctuate according to market forces (supply and demand) as well as the changing values of the securities in the fund's holdings. Also known as a "closed-end investment".

Collective Investment Scheme (CIS). A CIS is an arrangement that enables a number of investors to pool their assets and have these professionally managed by an independent manager. Section 2(1) of the SFA has a detailed definition of CIS.

Constant Proportion Portfolio Insurance (CPPI). CPPI is a trading strategy designed to ensure that a fixed minimum return is achieved either at all times or more typically, at a set date in the future. The strategy involves continuously re-balancing the investment portfolio between performance assets and safe assets using a set formula or mathematical algorithm. CPPI is totally rule-based and non-discretionary.

The principal is preserved by adjusting the exposure to the performance assets so that the underlying portfolio (i.e., the mix of safe assets and performance assets) can absorb a defined decrease in value before its value falls below the level required to achieve principal preservation. This asset allocation process allows investors to participate in rising performance assets while preserving capital in downward trends. On a daily basis, the CPPI provider determines the allocation between cash and performance assets to ensure that at least 100% of capital is returned at maturity.

Cushion. This is the portion of the fund's assets that can be exposed to risk without jeopardizing the preservation feature. For example, if a fund is worth SGD 100 and needs SGD 80 to honour the preservation, it can expose SGD 20 (the cushion) to risk, in order to boost returns.

Custodian. A separate legal entity (from the asset management company) which manages and safekeeps investors' securities. In some countries, the custodian is appointed by the trustee. The custodian is paid a fee for keeping "custody" of securities and their role may differ from a trustee, who has additional duties of safeguarding the investors' interests and ensuring compliance with investment guidelines. The custodian and trustee should be independent from the fund manager.

Drawdown. This is the peak-to-trough reduction in a fund's assets during a specific period of an investment. It is usually quoted as the percentage between the highest and lowest NAV. This figure helps to measure the financial risk of investing in a fund.

Excess Returns. This is the portfolio's return in excess of the benchmark, e.g., an index.

Index Fund. This is a type of fund with a portfolio constructed to match or track a market index. Investing in an index fund is a form of passive investing which entails lower management expenses.

ISIN. This stands for International Securities Identification Number. This is a unique international code which identifies the issue of any security.

Management Fee. This is the compensation for the fund managers in lieu of their time and expertise. Management fee structures vary from fund to fund but are typically based on a percentage of AUM.

Minimum Subscription Amount. During an offering period, a fund would not accept orders for less than a minimum subscription amount. This minimum amount/number of shares can be different for distinct share classes of the fund.

Multiplier. The multiplier is the coefficient used to multiply the cushion to calculate the portfolio's risky asset exposure (with reference to CPPI).

Net Asset Value (NAV). This is the fund's price per share/unit. The per share dollar amount of the fund is obtained by dividing the value of total assets in the portfolio, less any liabilities (called net assets), by the number of fund shares outstanding.

Open-End Fund. A type of mutual fund that does not have restrictions on the amount of shares the fund will issue. If demand is high enough, the fund will continue to issue shares no matter how many investors there are. Open-ended funds also buy back shares when investors wish to sell. The majority of mutual funds or unit trusts are open-ended. By continuously selling and buying back fund shares, these funds provide investors with a very useful and convenient investing vehicle.

Option. An option is a financial contract that offers the buyer the right, but not the obligation, to buy or sell a security or other financial asset at an agreed price. The option can be exercised either on a specific date or any date during a certain period of time, depending on the terms of the option.

Share Class. This refers to the classification applied to shares and fund units. Each share class has different characteristics, e.g., fees charged, class of investors, dividend treatment, etc. For example, based on the dividend policy, share classes could be either capitalization (i.e., the shares do not distribute dividends but re-invest them in the fund) or distribution (i.e.; the shares distribute dividends).

Sharpe Ratio. This is the measure used to judge risk-adjusted performance. The Sharpe Ratio is calculated by subtracting the risk-free rate from the portfolio return and dividing the result by the standard deviation of portfolio returns.

SICAV. This stands for Société D'investissement à Capital Variable. The term can be translated as 'investment company with variable capital'. SICAV is an open-ended collective investment scheme and common in western European countries especially Luxembourg, Switzerland, Italy, Spain, etc.

Total Expense Ratio (TER). This is a measure of the total costs associated with managing and operating an investment fund such as a mutual fund, e.g., management fees and other expenses such as legal fees, auditor fees and other operational expenses. Trading fees, interest expense and brokerage fees may not be included in the TER. The TER is calculated by dividing the total cost of the fund by the fund's total assets. It is generally denoted as a percentage and can vary from year to year.

Total Return. The total return of an asset is defined as the sum of income generated by the asset and any capital gains. In other words, total return includes interest, capital gains, dividends and distributions realized over a given period of time.

Total Return Swap (TRS). A swap agreement in which one party makes payments based on a set rate, either fixed or variable, while the other party makes payments based on the total return of an underlying asset.

Trackers. They are also known as "index funds". Trackers are passively managed investment funds that track the market performance of chosen benchmark indices. Investing in an index fund is a form of passive investing.

Trailer Fee. If an asset manager uses distributors / salespersons / representatives providing financial advisory services (collectively referred to as “distributors”) to sell his funds to end investors, the fee payable to such distributor(s) by the asset manager is known as the trailer fee. This is paid in lieu of the distributor’s ongoing investment advice and services to the investors.

Upfront Sales Subscription Charge. The fund might levy this charge on the subscription during and after the offering period, which will be reverted to the distributor.

UCITS. This stands for **Undertakings for the Collective Investment of Transferable Securities**. It was first adopted in 1986 (UCITS I) and the UCITS rules apply to investment funds marketed to retail investors. UCITS has brought costs down for fund providers because it means they no longer had to create a new investment vehicle for each country in which they intended to market the product.

A UCITS-compliant (and registered in its home state) fund can be freely marketed (passported) to the public in all 30 countries of the European Economic Area (EEA). UCITS has become extremely valued for many reasons - compliance, risk controls, distribution and state-of-the-art administration. UCITS funds can be marketed globally and have also become successful in Asia and Latin America. In July 2012, the European Commission published a proposal for a draft UCITS Directive, known as ‘UCITS VI’.

Umbrella Fund. This refers to a single legal entity which has several distinct sub-funds. The sub-funds of an umbrella fund are traded as individual investment funds. SICAV is one such arrangement.

Underlying Asset. This refers to the asset that a fund with a direct investment policy seeks to track. It is normally one or more indices or basket of securities. For a fund with an indirect investment policy, it is the asset(s) to which the fund return is linked.

Underlying Securities. This refers to the transferable securities which constitute the underlying asset.

Value-at-risk (VaR). VaR is a statistical technique used to measure and quantify the level of financial risk within the investment portfolio over a specific time. VaR is measured in three variables: the amount of potential loss, the probability of that amount of loss, and the time frame. VaR is used by investment and risk managers to measure and control the level of risk which the fund undertakes. The manager's job is to ensure that risks are not taken beyond the level at which the fund can absorb the losses of a probable worst outcome.

Volatility. Volatility is a statistical measure of the dispersion of returns for a given asset or fund. It refers to the amount or risk or uncertainty about the changes in the size of its financial value. Volatility can either be measured by using the standard deviation or variance between returns of the asset or fund. A higher volatility means that a security's value can potentially be spread out over a larger range of values. (Refer to Chapters 2-4 for further details on volatility.)

Zero-Coupon Bond. This is a fixed income instrument that does not pay any interest during its tenor. This instrument is sold at a discount from its face value and renders profit at maturity when the bond is redeemed for its full-face value.

9.1.2 How Are Structured Funds Different from Traditional Funds?

With many of the new funds and asset management portfolios using derivatives in their strategy, the lines of distinction are getting blurred, but we can still make some general comparisons with traditional asset allocation funds:

a. While there are structured funds that can operate indefinitely over many years, most closed-end structured funds carry fixed maturity dates where the eventual accumulated asset value at maturity (e.g., 2-5 years) will be distributed to unit holders at the end of the period. Most traditional funds tend to be open-end and are expected to continue operating for as long as they can, and redemptions even in periods of gains in the short term are often discouraged by fund managers who want clients to stay invested for the long term;

b. Structured funds aim to replicate the underlying asset or to provide a synthetic return linked to the underlying asset of the fund by incorporating derivatives. Allocation is typically static or rule-based and the investment view can be long, short or market neutral; and

c. Structured funds involve exposure to a wider variety of risks including credit or counterparty risk, equity market risk, foreign exchange risk, interest rate risk, market volatility risk, political risk or any combination of these risks. There may also be further risks due to additional counterparties such as correlation risk. Counterparty risk is more present in structured funds as compared to traditional mutual funds.

9.1.3 How Are Structured Funds Different from Trackers?

Trackers are used to simply replicate the performance of their benchmark. On the other hand, structured funds can choose to deviate in their asset allocation given the outlook and model recommendations and they:

a. Can be used to realize various anticipated market views;

b. Have variable levels of exposure that adjusts systematically as markets change and are optimized based on expectations for market direction and volatility; and

c. Carry out adjustments and optimizations to deliver the promised level of capital preservation and participation (if any) in the underlying asset.

9.1.4 Components of a Structured Fund

Table 9.1.4 spells out the four key considerations for the setting up of a structured fund.

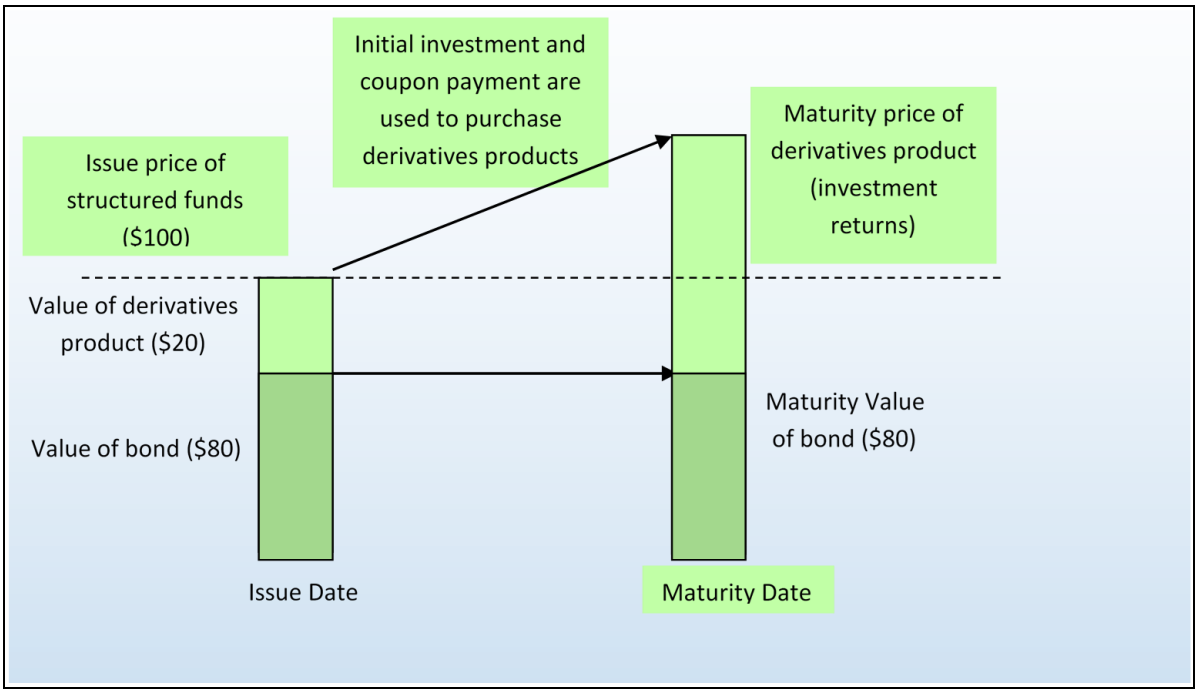
Table 9.1.4 – Key Considerations for Structured Fund Set-Up

Underlying Assets and Derivatives Used	Maturity Period	Payout Structure	Market Outlook
Foreign exchange. **Bonds / Fixed income / Credit.** **Equities.** **Commodities.**	1 year. 2-5 years. More than 5 years. No maturity (e.g., for some open-ended structured funds).	Regular distribution of fixed or variable coupon payment. Participative returns based on the outcome of the underlying asset(s).	Investors will have different views about the outlook for the market – bullish, bearish or neutral view which will determine the

Interest rates. **Proprietary indices.** **Real estate.** **(And any combination with derivatives instruments).**		(Or a combination of both of the above).	strategy for the structured fund.

Once all the input parameters have been defined, a structured fund can be established accordingly. An illustration of a structured fund is shown in Figure 9.1.4.

Figure 9.1.4 - Example of a Structured Fund



For example, an S&P 500 index structured fund preserves 80% of its principal. It will invest 80% of its funds in fixed income products with a low probability of falling below the principal amount. Instead of equities, the rest of the fund is invested in derivatives that are exposed to the S&P 500 index. Investors will gain as the S&P 500 index advances and the fund will be unlikely to lose more than 20% of its principal value if the index falls.

Structured exchange-traded funds (ETFs) are funds that are traded in real-time like shares and managed according to techniques aimed at pursuing returns that are related to the performance of an index or reference asset. Other innovative ETFs are available to investors, and offer returns through strategies which:

- Participate in a more than proportional manner in the performance of an index (e.g., through leveraged ETFs).
- Participate inversely in the movements of an index (e.g., short ETFs).
- Preserve the portfolio value, while participating in the increase of an index (e.g., protective put ETFs); or
- Involve a combination with derivatives.

Unlike structured deposits and notes, where the fees are usually embedded into the pricing of the products (e.g., through a theoretical lower yield or lower participation rate in the performance of the underlying asset), a structured fund usually has separate fees such as fund management and administration fees.

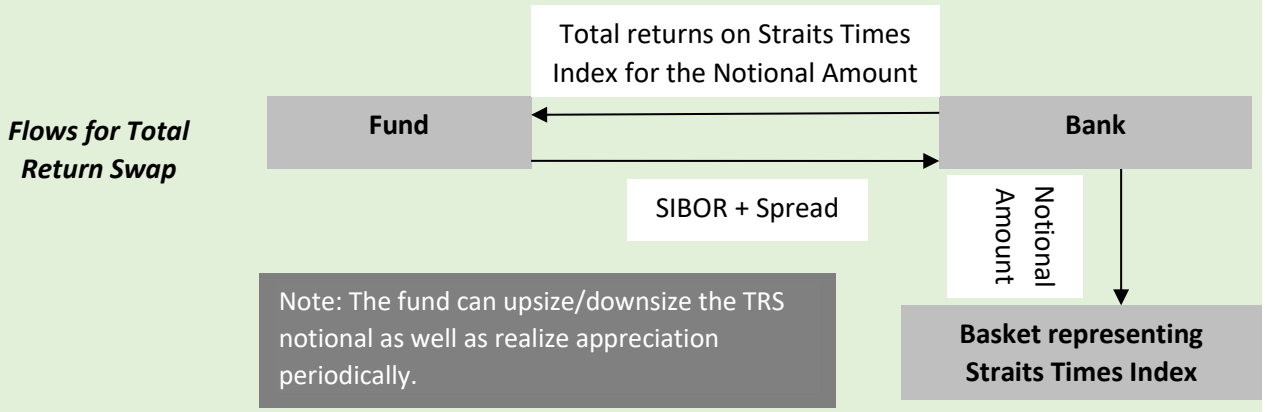
9.1.5 Structured Funds Using Derivatives

Derivatives can be used to fully replicate the price movement of an underlying security or index. Funds with indirect exposure to the underlying asset can track or replicate the underlying asset's performance synthetically by using derivatives rather than holding the underlying assets directly.

In some structured funds, derivatives take up 100% of the composition (e.g., Total Return Swaps) or it can be used in combination with existing assets to enhance returns while limiting the downside risk (e.g., Zero Plus Option Strategy which combines fixed income with options).

Total Return Swap (TRS)

A common derivative transaction, the TRS, uses an indirect approach. To illustrate, assume that a fund wants to match the performance of the Straits Times Index (STI). The fund has many ways of tracking the index - simply buy the underlying 30 company shares which make up the STI with the corresponding weightings, or more realistically trade the STI futures. This strategy requires some ongoing active management, either by frequently rebalancing the stock portfolio and reinvesting the dividends, or by rolling over the futures close to maturity.



Alternatively, the investor may decide to enter into a TRS providing the STI total return every 6 months for the next 5 years based on the notional amount. In exchange, the investor would agree to pay a rate of SIBOR plus (for example, 30 basis points = SIBOR + 30bps) per annum. In this way, the fund would have achieved its objective of matching the return of the index at a spread cost of 30bps.

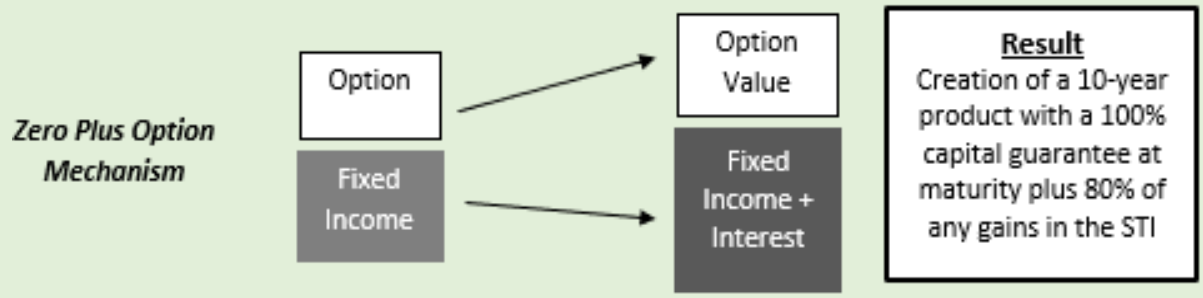
Zero Plus Option Strategy

Assume that a fund's investment objective is to create a 10-year investment product that aims to maintain 100% of the capital invested, plus a share in any gains in the STI. This is achieved by:

- Investing a portion of the initial investment funds in fixed income assets, e.g., 80% of the capital amount is invested in a zero-coupon bond. This zero-coupon bond, worth 80% of face value today, will

have an expected value of 100% in 10 years' time. (Note that the growth in the zero-coupon bond value is a projection and is not guaranteed); and

- Investing the remaining 20% in a 10-year call option on the STI. The participation share in the rise/fall of the index depends on the call option price and the 20% capital available for investment. If the call option price is 25% when the fund has 20% to invest, it means that the share of participation in the STI is 20/25 = 80%.



9.1.6 Structured Funds using the Constant Proportion Portfolio Insurance Technique

The CPPI Technique is a strategy which aims to maximize exposure to the performance of the underlying securities whilst ensuring that a percentage (e.g., 80-100%) of the invested capital is guaranteed as long as it is held till maturity.

It follows a trading strategy which is designed to allow investors to participate in promising but risky investments while limiting the risk of loss during falling markets. The strategy involves continuously re-balancing the investment portfolio between performance assets and safe assets using a set formula or mathematical algorithm. CPPI is totally rule-based and non-discretionary. The weighting of these two components is changed dynamically depending on market conditions and expectations. When the values of the risky assets are in a downtrend, the exposure to risky assets is reduced automatically, according to the rules, in favour of fixed income assets and vice versa.

If the portfolio insurance is just a disciplined process of allocating assets between fixed income and riskier assets where no derivatives are involved, the fund is not a structured fund. However, with the addition of derivatives like put options, downside risk can be limited in the CPPI approach of investing.

A structured product (fund or note) using a CPPI strategy dynamically allocates funds between a "risk free" reserve asset (i.e.; a zero-coupon bond) and a risky underlying active asset (i.e.; an equity index, equity fund or hedge fund).

Under CPPI, a floor value of the total portfolio is calculated at regular intervals. For a CPPI product, principal preservation is a key consideration and setting the appropriate floor value allows this objective to be achieved. When the portfolio value drops to the floor, funds are allocated to the risk-free asset, so that the investor can still receive the principal investment when the structured product matures. To better secure the principal, the manager may buy insurance from another financial institution (usually a bank) to guarantee the principal sum at maturity, typically for an annual fee of about 1%.

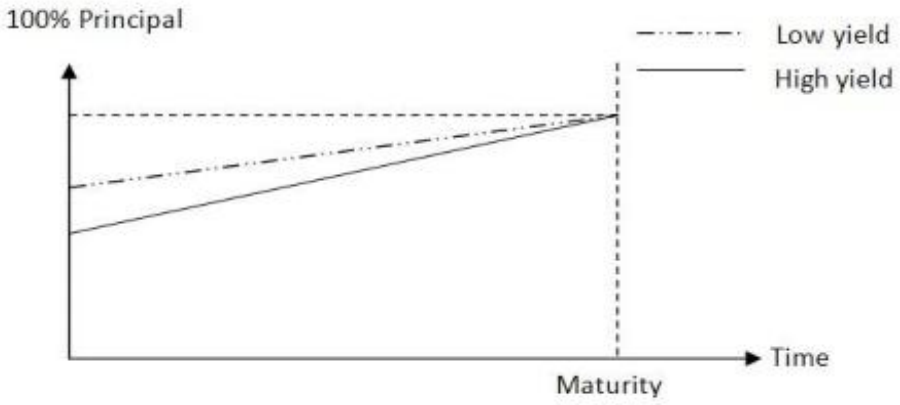
Please refer to **Appendix A** for a detailed illustration of the workings of the CPPI strategy under different scenarios (intervals).

9.1.6.1 How CPPI Works

a. **Asset Allocation.** Adjustment is done periodically, usually once a month. The difference between the total portfolio value and the floor value is known as the cushion. If the cushion is large, the manager can afford to take more risk, i.e.; higher allocation to risky assets, and vice versa.

b. **Floor Value.** The floor value changes during the life of the structured product, depending on the remaining term to maturity and the yield curve at each point in time. As the investment moves towards maturity, the floor value approaches 100%[41] of the principal sum (see Figure 9.1.6.1(a).

Figure 9.1.6.1(a) - Floor Values of a CPPI



c. **Crash Size**. This is the percentage drop in underlying asset value in any one period in a crash scenario.

d. **Multiplier.** This is used to determine the amount to be allocated to the risky asset component of the overall portfolio:

Multiplier = 1 / Crash Size

For example, if the manager is of the view that the largest percentage crash for the chosen risky asset is 20%, then the multiplier is equal to 1/20%, or 5.

e. **Risky Asset.** This is the component of the total portfolio which generates the upside returns. The amount allocated is driven by the multiplier and floor value:

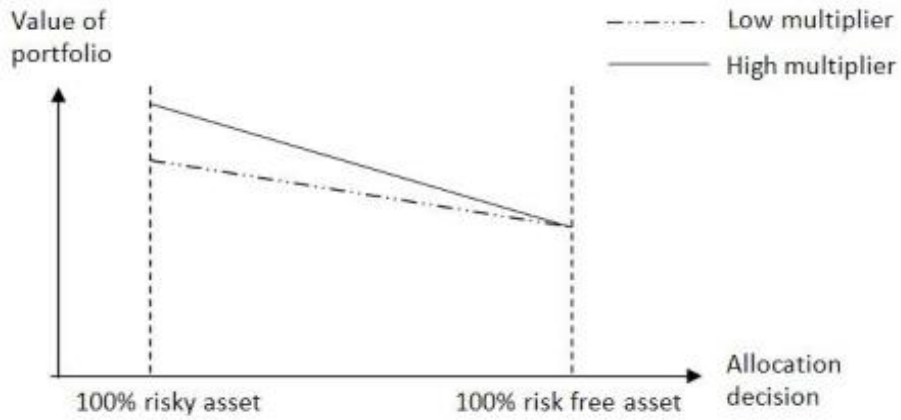
Allocation to Risky Asset = Multiplier x Cushion Value

where Cushion Value = Total portfolio – Floor Value

For example, if the cushion value is 15%, and the multiplier is 5, then the allocation to the risky asset is equal to 75%.

[41] If the fund aims to protect 100%. Some start below 100% e.g. 80% and lock in the highest NAV in the first N years. Eg, Final redemption = Max (80%, highest NAV in first N years, final NAV). This might not always be 100%.

Figure 9.1.6.1 (b) - Asset Allocation



Example – Structured Fund

Issuer	The Structuring Bank Ltd
Base Currency	USD
Term	5 years
Return	Variable, depending on volatility & performance of underlying index
Maturity	100% principal guaranteed at maturity
Guarantee Fee	1% p.a.
Management Fee	1.5% p.a.
Value of note at issue	USD100,000
Minimum Index allocation	0%
Maximum index allocation	120%
Strategy	CPPI

Time Period	Good Performance	Weak Performance	Capital Guarantee
At Issue	100	100	-
End of year 1	110	95	-
End of year 2	105	90	-
End of year 3	120	98	-
End of year 4	125	100	-
End of year 5	135	95	100

When the underlying index performs well, the allocation to the active underlying assets in the portfolio will be high. At maturity the returns generated will be the maturity value of the underlying which is USD 135/- in the case of the good performance scenario. In any given year, despite how well the asset classes are doing, only a maximum of 120% can be invested in the underlying. In the case of weak performance scenario, the portfolio is capital guaranteed at maturity. There is still an opportunity cost of potential income that was not realised in this CPPI.

9.1.6.2 Risks of Products that Use CPPI Strategy

It is important to highlight to the investor that such a CPPI strategy assumes the following characteristics of the underlying asset:

a. Asset that appreciates in value over time;
b. Low volatility; and
c. Limited drawdown as measured by percentage decline from peak to trough.

According to the CPPI asset allocation process discussed above, there is a higher allocation to the risky asset as its value appreciates, and vice-versa. Hence, a CPPI strategy may result in a situation where the investor has to buy high and sell low in a range bound market. If there is a prolonged period of range bound trading, there is also a higher chance that the portfolio may drop to the floor value, forcing the manager to allocate the entire fund into the risk-free asset, forgoing any subsequent appreciation in the underlying asset.

A CPPI product's overall fees are much higher than a traditional investment product, such as structuring, sales and guarantee fees, as well as management fees. The investment returns of the product are likely to be affected significantly by these fees.

This product also does not have much liquidity in the secondary market, so investors must be prepared to hold the product till maturity.

9.1.7 Dynamic Proportion Portfolio Insurance (DPPI)

DPPI is similar to CPPI. Unlike CPPI, however, the multiplier is a variable instead of a constant number. DPPI is also known as Dynamic Portfolio Insurance (DPI).

For example, in a DPPI strategy, the amount invested in the risky asset is allowed to fluctuate within a band of 3-5 times of the cushion value. The fund manager starts by allocating an amount that is 4 times the cushion value to the risky asset. When this amount falls outside the band, the portfolio is rebalanced by adjusting the risky asset multiple back to the starting point of 4 times.

9.1.8 Overview of Common Fund Structures

We highlight some examples of common investment themes for structured funds.

9.1.8.1. Funds with Features that Aim to Preserve Capital Invested

Depending upon the proportion of capital that the funds aim to conserve, the remaining amount can be allocated into risky assets or derivatives to synthetically access risky assets.

9.1.8.2 Formula Funds

Formula funds are funds where the final payout depends on a pre-defined, rule-based formula. It starts off with an objective based on an anticipated view and the final payout depends upon the realized market outcome. These products have the advantage of knowing from the start how investments would be allocated which depends on the pre-determined formula. Typically, such funds can be set up to track indices.

9.1.8.3 Capitalized / Distribution Funds

Capitalized funds (also known as accumulating or reinvesting funds) are funds where dividends from investments are automatically re-invested back into the fund. Distributing funds (also known as income funds) on the other hand pays out any dividends received to investors on a periodic basis.

9.1.8.4 Indirect Investment Policy Funds (Swap-based Funds)

These funds provide investors with a return (either on such payout date(s) and/or at the maturity date) linked to the performance of the underlying asset. To gain exposure to the underlying asset, the fund will not invest directly (and/or fully) in the underlying asset. Instead, the fund may invest part or all of the net proceeds of any issue of shares in one or more derivative transaction(s) in accordance with the "Investment Restrictions" set out in the prospectus. The investor's return depends on the underlying asset's performance and the performance of the derivative used to link the net proceeds from the issue of shares to the underlying asset.

The fund may also invest in a hedging asset in accordance with the guidelines set out as investment restrictions as stated in the prospectus and exchange all or part of the performance and/or income of this hedging asset for a performance linked to the underlying asset. This exchange of performances and/or income will be obtained with derivative instruments, which will be used in accordance with the limits set out under the Investment Restrictions in the prospectus. The underlying assets will be based on a passive strategy (typically a financial index or a rules-based strategy) or an active strategy according to which the real or notional basket comprising the underlying asset is actively managed in accordance with the Investment Restrictions.

9.1.9 Examples of Structured Funds

Most popular structured funds have equities as their underlying asset. Other offerings include commodities or hybrids (e.g., a combination of equities and commodities). Typically, investors choose to receive fixed coupons for certain period. After that, there could be a period of variable coupons or payouts linked to the performance of the underlying asset(s). A high fixed coupon would typically affect the variable coupon or the final payout as the participation ratio has to be kept low to accommodate payment of high fixed coupons.

Formula Fund with a Commodity Index as the Underlying Asset

ABC Fund is an agriculture-related and open-end fund (in USD) which tracks an agricultural index that has a diversified exposure to the agricultural sector. The index provides access to 7 liquid commodities in the agriculture sector – corn, wheat, soybean, sugar, cotton, coffee and cocoa. The weight of each agricultural subcomponent is fixed and reset annually. This fund requires low initial investment and offers daily liquidity at NAV. The agriculture index employs the ABC Optimum Yield rolling mechanism to generate "alpha".

The periodical rolling of expiring futures contract with a fixed term, which is often used in index construction, can be detrimental to returns potentials and even trigger losses. So instead of defining fixed roll periods, the "optimal yield" approach chooses to maximize the roll returns in backwardation markets and minimize the rolling losses in contango markets:

- In backwardation markets, forward prices are lower than spot prices. The downward slope of the price curve creates profits when contracts are rolled over. The "optimal yield" methodology aims to maximize the rolling profits.
- In contango markets, forward prices are higher than spot prices. The upward slope of the price curve causes losses when contracts are rolled over. The "optimal yield" methodology aims to minimize the rolling losses.

Hence, the formula-based asset allocation of this fund provides alpha or enhanced return by offering a more efficient roll methodology instead of a fixed roll period. As a result, in the ABC Fund factsheet, the fund manager reports that the agricultural index has outperformed the S&P GSCI[42] Agriculture Index Total Returns Index by an average of 8.38% per annum (excluding fees and commissions).

A Fund with Option-based and Structured Payoffs

DEF Fund Manager is the investment manager for DEF Fund (in SGD). The fund's objective is to provide a return linked to a basket of 25 stocks from the infrastructure, utilities and real estate sectors, and a quarterly potential dividend payment. The fund's investment strategy involves:

- Notionally holding stocks that may distribute dividends in the next quarter and also benefit from potential capital appreciation.
- Selling call options on each stock to receive premium income; and
- Buying put options on each stock as a stop loss mechanism.

Hence DEF Fund uses derivatives to its advantage for hedging and income generation, besides directly buying the underlying assets for the fund.

9.1.10 Risks of Structured Funds

Structure funds cover a broad variety of investments. The characteristics and risk-return profile of a specific fund primarily depends on the underlying assets in the fund, such as equities, fixed income, commodities, real estate and other asset classes.

Risk in a fund is also influenced by the investment style and strategies employed, such as long only, short, long-short, and other techniques that are used by various traditional, hedge fund and alternative asset managers. Funds that use leverage and derivatives as part of their investment strategy will have a higher risk profile. Currency risk would also be relevant for funds with foreign exchange positions and that hold assets which are outside the home market in their portfolio. Finally, a fund's operation and administration can contribute to its overall risk profile. This includes the fund's structure, legal domicile, governance, fee structure, and risk management policy.

[42] S&P GSCI refers to the Standard & Poor's Goldman Sachs Commodity Index. The S&P GSCI is a composite index of commodity sector returns representing an unleveraged, long-only investment in commodities futures that is broadly diversified across the spectrum of commodities.

9.1.11 Roles & Responsibilities of Fund Managers and Trustees

9.1.11.1 Fund Manager

The roles of the fund manager include:

a. Managing the assets of the CIS in line with the defined investment objective and within the investment restrictions as stated in the offering document.

b. Preparing the semi-annual accounts, annual accounts, semi-annual reports and annual reports. These reports are to be furnished to the trustee so that the trustee can have them audited before dissemination to the unit holders; and

c. Taking responsibility for the creation and the redemption of units.

The fund manager manages and invests the fund's assets and is responsible for the fund's performance while the trustee holds the property on behalf of the unit holders and ensures the fund manager carries out his duties in accordance with the trust deed.

9.1.11.2 Fund Trustee

The trustee must be independent of the fund manager and its main role is to look after the interests of the unit holders. The trustee's main responsibilities include:

a. Working in a fiduciary capacity and being accountable to the investors;

b. Ensuring that the fund manager manages the CIS in accordance with the investment objective and restrictions as laid out in the trust deed and prospectus;

c. Ensuring that the accounting records are kept properly, and the CIS is audited after which the unit holders receive the semi (within 2 months from the end of the period covered by the accounts or reports) and annual reports (within 3 months from the end of the period covered by the accounts or reports) in time;

d. Taking legal ownership of all assets in the CIS and ensuring that these assets are held independently from the fund management company;

e. If there are any breaches, informing MAS within 3 business days after becoming aware of the breach; and

f. Minimizing the risk of mismanagement by the fund manager and ensuring that the fund is managed in the interest of the unit holders.

9.1.11.3 Potential Conflicts of Interest

Sometimes the different entities involved in various functions of the fund may be part of the same group of companies and may have a conflict of interest with the fund investors. For example, the fund management company's affiliates may act as swap counterparties or valuation agents, so they do not have a fiduciary role to act in the best interests of unit holders.

The fund manager could address conflicts of interests in various ways. For instance, some funds are structured such that the fund manager's Board of Directors would need to ensure that each of the affiliated entities undertakes to resolve any conflicts of interest fairly. Possible measures to ensure that the interests of investors or unit holders are protected include using Chinese walls or different reporting lines.

9.1.12 How to Manage Underlying Risk Exposure in a Structured Fund

A structured fund's risk can be financial, operational or legal. The fund manager uses internal controls and risk procedures at the fund level to control the various risks as follows:

a. Operational processes like the assignment of orders, control of execution prices, transaction settlement, valuation control, reconciliation of cash and securities are checked frequently. Typically, administrative agents and the fund's investment managers (if any) perform these tasks internally. Apart from mandatory frequent reporting to the manager, the administrative agent also deploys independent risk controls, e.g., various automated IT software. IT systems also help to automate investment decisions for rule-based structured funds.

b. A close eye is kept on financial risks such as market risk, using financial techniques such as stress-testing, VaR and duration. If required, exposure ranges by asset type, volatility, credit-risk and diversification limits are also calculated to monitor financial risk. In case a derivative transaction involves any counterparty such as swap counterparty, then the counterparty can be asked to carry out financial risk controls at the level of the derivative transaction.

c. Managers often have dedicated legal teams to avoid legal risk(s). The manager also ensures that the structured fund follows the set-out investment restrictions.

d. The trustee must be independent of the fund manager; and

e. The funds' auditor audits the processes annually. Regular audits are also done for each entity involved (e.g., the management company, investment manager, administrative agent and custodian).

9.1.13 Typical Fund Documentation and Terms

9.1.13.1 Fund Prospectus

A fund's prospectus is a booklet that will identify and discuss everything from the fund's objectives and its past performance to a description of the fund manager and the fees associated with the fund. This booklet also contains contact information so that investors can get in touch with the managers and representatives of the fund to get their questions answered. The prospectus is updated from time to time and investors are always advised to go through the latest version.

A prospectus would consist of the following information:

a. Basic information which includes introductory information such as:
   i. Name of the fund,
   ii. Date of registration and expiry date of prospectus,
   iii. Place of constitution,
   iv. Trust deed and supplemental deeds,
   v. Semi-annual reports, semi-annual performance statements or audited financial statements,
   vi. Disclaimers, and
   vii. Whether the scheme is constituted outside Singapore.

b. Information on the manager:
   i. Name and address of the manager,

ii. Track record of manager.

c. Name and address of the representative.

d. Name of trustee.

e. Information on other parties:

i. Name of investment advisor,

ii. Name of registrar and where the register can be inspected,

iii. Name of auditor,

f. Structure of the scheme

i. Standalone fund or umbrella fund,

ii. Details required for a fund of funds, a feeder fund or a sub-managed fund,

g. Investment objectives focus and approach.

h. Facts of inclusion:

i. Fees and charges,

ii. Fee table,

i. Risks:

i. General risks,

ii. Specific risks.

j. Subscription of units:

i. Subscription procedure,

ii. Minimum subscription amount,

iii. Initial purchase price and initial offer period,

iv. Dealing deadline and pricing basis,

v. Numerical example under pricing system adopted,

vi. Confirmation of purchase,

vii. Minimum fund size,

viii. Return of contributions,

ix. Classes of units.

k. Regular Savings Plan (RSP):

i. Details of RSP,

ii. Statement that investors may cease participation in the RSP.

l. Realisation of units:

i. Realisation procedure,

ii. Minimum holding amounts and minimum realisation amount,

iii. Dealing deadline and pricing basis,

iv. Numerical example under pricing system adopted,

v. Payment of realisation proceeds.

m. Switching of units:

i. Switching procedure.

n. Obtaining prices of units:

i. How to obtain price information.

o. Suspension of dealings:

i. Circumstances.

p. Performance of the scheme:

i. Past performance of the scheme,

ii. Comparison between the scheme and another CIS or an index,

iii. Comparison between the scheme and other forms of investment,

iv. Performance of the manager or sub-manager,

v. Future performance of the scheme,

vi. Performance of benchmark,

vii. Expense ratio,

viii. Turnover ratio.

q. Details of Soft Dollar Commissions/ Arrangements.

r. Description of conflicts of interest and how they will be resolved or mitigated.

s. Reports:

i. Financial year-end and distribution of reports.

t. Specialised CIS:

i. Warning statements and/or additional information for specialised CIS.

u. Queries and Complaints:

i. Telephone contact.
ii. Other Material Information.

9.1.13.2 Product Highlights Sheet

This is a summary of the key features and risks of a structured fund. Appendix B shows extracts of MAS' Practice Note for the Product Highlights Sheet.

9.1.13.3 Trust Deed

A trust deed is a legal document that sets out the terms and conditions governing the relationship between investors, the fund manager and the trustee. It describes the investment objectives of the fund, and the obligations and responsibilities of the fund manager and trustee.

The trustee is independent of the fund manager and acts as the custodian of the fund's assets. The trustee ensures that the fund is managed according to the trust deed to minimise the risk of mismanagement by the fund manager.

9.1.14 What Types of Investors Invest in Structured Funds

Both retail and institutional investors can subscribe to structured funds, and not all schemes are offered to both types of investors. A fund scheme could float different classes of shares, depending upon the type of investors subscribing to the scheme. These differences are mainly due to the specific features with respect to sales, conversion, redemption charge, minimum subscription amount and dividend policy are different for different classes of investors.

Schemes offered only to institutional investors are not restricted schemes; restricted schemes are those offered only to accredited investors, other relevant persons, and those who subscribe with SGD 200,000 or more per transaction[43]. In case of a retail offer of scheme, a prospectus in compliance with the SFA must be lodged and registered with the MAS. In the case of an offer to accredited investors and other relevant persons, a prospectus is not required. In the case of an offer to institutional investors, an information memorandum is not required.

MAS requires financial institutions and fund distributors to put in place formal policies and procedures to assess the nature of a new investment product and assess its suitability for targeted customer segments before distributing the product. The investment representative providing financial advice also has a responsibility in advising on suitability of product to the investor.

Investors are advised to keep relevant considerations (risk/return structure, legal and tax) in mind before investing in any structured fund. The fund's prospectus will contain relevant appendices that will describe the details of the structured fund. Investors should look at the relevant sections of the prospectus for information on the fund. Investors are advised to seek independent advice to assess the risk level and suitability of the fund before making their investment decisions.

9.1.15 After Sales

9.1.15.1 Where to Get NAV

The NAV per share of each class within each structured fund is generally made available by the Administrative Agent. The fund issuer might arrange for the publication of NAV in one or more leading financial newspapers where the funds are distributed to public and may notify the relevant stock exchanges in cases where the shares of the fund are listed. NAV can also be accessed at sources like Bloomberg, Reuters and the website of the fund issuer using the fund ISIN or name.

9.1.15.2 How to Get Information on Performance

a. **Semi-annual Accounts and Reports to Unit Holders**

The fund issuer sends semi-annual and annual reports to their unit holders. The annual financial statements of the funds are to be audited by Independent Auditors. These reports inform the unit holders about:

i. **Statement of Net Assets** – Account of assets and liabilities of the company and NAV per share for existing share classes of the fund.

[43] Please refer to the Securities and Futures Act (SFA) for the specific definition of "accredited investor" and "institutional investor".

ii. **Statement of Changes in Net Assets** - A financial statement that shows how the net assets of fund have changed over past two reporting periods. The fund's income adds to its net assets while dividend payments and shares redemption subtract from net assets.

iii. **Statement of Investments** - This statement lists out the details of the investment portfolio of the company. Similar information is provided for each of the funds issued by the company.

iv. **Changes in the Number of Shares for Each Share Class of the Fund** - The statement which that shows the number of shares issued, number of shares redeemed, and thereby reconciles the opening and closing number of shares for each class of the fund.

b. **Investment Manager Report**

The fund issuer will usually publish a report by the fund manager which details the performance of underlying assets and the performance of each share class of the fund. This report also presents the fund's AUM figures, annualized volatility for different share classes and a performance outlook.

c. **Factsheet**

This is a concise document that highlights key information related to the fund - launch date, investment manager information, key features of the product, asset allocation, performance figures and the various applicable fees.

d. **Monthly Performance Report**

This report highlights:

i. **Principal Terms of the Fund** - issuer's name, launch date, name of management company, initial NAV, management fee per annum;

ii. **Overview of the Fund** – fund's investment policy and brief description of fund's portfolio methodology; and

iii. **Performance** - returns of the fund; typically, Year-to-Date (YTD) returns, 1-month returns, 6-month returns, 1-year returns, 3-year returns and returns since launch of the fund. Risk analysis such as volatility calculations, Sharpe Ratio and VaR calculations are also highlighted.

9.2 Structured Exchange Traded Funds

9.2.1 What are Exchange Traded Funds?

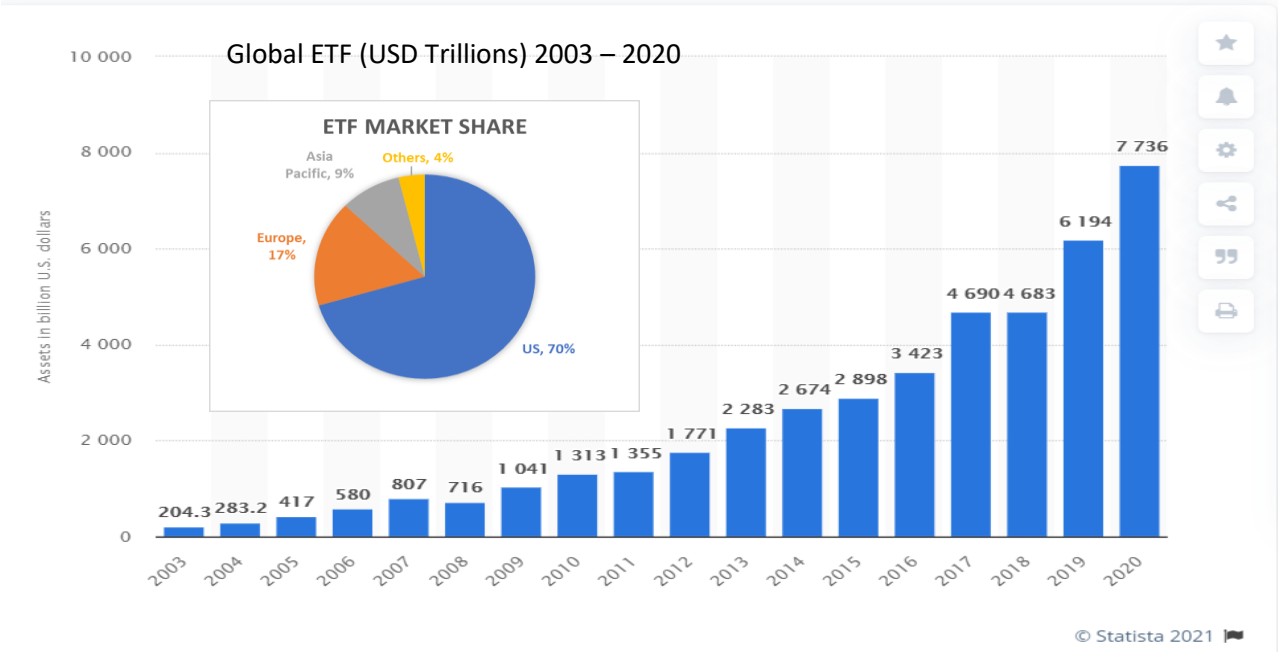
Exchange Traded Funds (ETFs) are extensively covered in the CMFAS CM-EIP Examination. An ETF is an open-end investment fund that aims to replicate the performance of a published market index. It is listed and traded on a stock exchange, and investors may buy or sell ETF units to participate in the performance of the underlying index which tracks equities and various other asset classes. The equity indices can be by country, region, or sector-specific and based on emerging markets, developed markets. Investors typically use ETFs to gain exposure to all the constituents of the index, thereby diversifying their portfolio without having to purchase individual assets underlying the index. Today, there are a wide range of ETFs that do more than track stock price trends. Some track volatility, some track only the daily price movement, and some are leveraged to produce 2-3x the performance of a particular index. For some ETFs, performance is more dependent on the movement of the options and futures market. No longer are ETFs solely focused on providing investors with a

more diversified low risk exposure. Some ETFs are designed specifically for amplifying performance and providing inverse performance to underlying equities, commodities and their indices. ETFs can be used for portfolio hedging as well for taking directional speculative exposure.

9.2.2 The Global ETF Market

The global ETF industry assets surpassed USD7.7 trillion in 2020 and is expected to continue growing aggressively for the next 5 years. The US is expected to be the dominant market for ETFs and accounts for roughly 70% of the assets under management (see Figure 9.2.2).

Figure 9.2.2 – The Global ETF Market



9.2.3 Types of ETFs

ETFs can be broadly classified into equity, fixed income, cash, alternatives, currency and commodities ETFs.

a. **Equity** – Categories of equity ETFs include:

   i. Global Markets,
   ii. Capitalisation Size (large, mid or small),
   iii. Industry Sectors,
   iv. Country-based,
   v. Emerging markets, and
   vi. Investment Styles/Themes (such as active investing, dividend-paying and Shariah-compliant).

b. **Fixed Income** – There are also ETFs which solely focus on fixed income securities and some categories include:

   i. Government Securities,
   ii. Corporate Securities,
   iii. Credits,
   iv. High yield Securities,

v. Mortgaged- back Securities, and
vi. Emerging markets Bonds.

c. **Cash** – Cash ETFs have their holdings in short term money market instruments. These ETFs hold money market securities, such as Treasury Bills, Municipal and Corporate Bonds with short maturities that are less than a year.

d. **Currency** – Currency ETFs track the performance of a basket of currencies from developed or emerging markets.

e. **Commodities** – Commodity ETFs track broad commodity indices, e.g., S&P GSCI (Goldman Sachs Commodity Index) and RICI (Rogers International Commodity Index). They also track other commodity sub-indices, such as energy, livestock, precious metals, industrial metals and agriculture. Such ETFs can also be based on the commodities futures or forwards. A commonly known ETF is the SPDR Gold.

f. **Indices** – Index ETFs are usually passively managed funds where the manager will replicate most of the securities within a particular index.

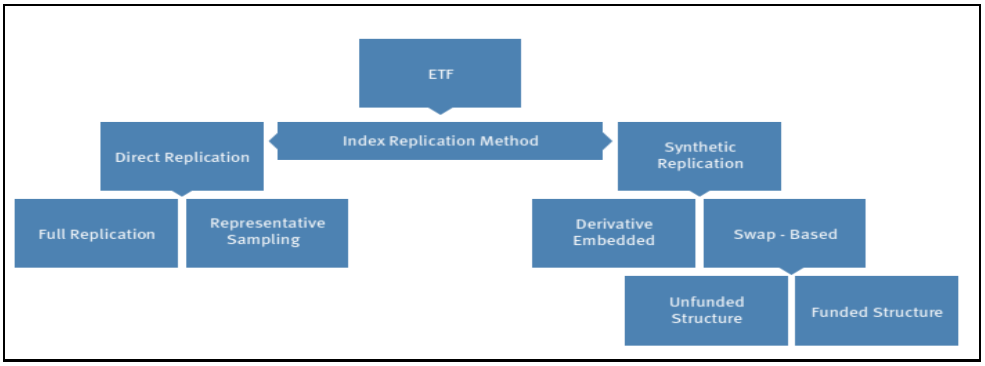
g. **Alternative Assets** – ETFs that focus on alternative asset mainly invest in listed hedge funds. These ETFs provide a secondary market for less liquid hedge funds.

h. **Exotic ETFs** – Examples include leveraged ETFs and inverse ETFs, which operate differently from regular ETFs. ETFs which adopt leveraged strategies aim to hold a constant amount of leverage during the investment time frame, such as a 2:1 or 3:1 ratio. Inverse ETFs seek to replicate the performance of a specified index but in the opposite direction. They are also known as short or bear ETFs. Investors in inverse ETFs seek to profit from a decline in the value of the underlying benchmark, which is similar to holding short positions or using a combination of advanced investment strategies to profit from falling prices.

9.2.4 Replication Methodologies

Different ETF providers can adopt different replication methods to track the performance of an underlying index. The two common types of index replication methods are direct replication and synthetic replication (see Figure 9.2.4). Under each method, there are several variants based on the specific techniques and instruments employed in the ETF.

Figure 9.2.4 – ETF Replication Methods



Source: SGX

The more commonly known replication method, the **direct replication method**, replicates the performance of an index by investing in the index's constituents. It is also known as a "cash based" ETF or "physically replicated" ETF. Direct replication comes in two different forms:

a. Full replication; and
b. Representative sampling method.

Synthetic replication ETFs use derivatives and/or OTC transactions to replicate the index's performance without directly holding the underlying assets. They further branch out into:

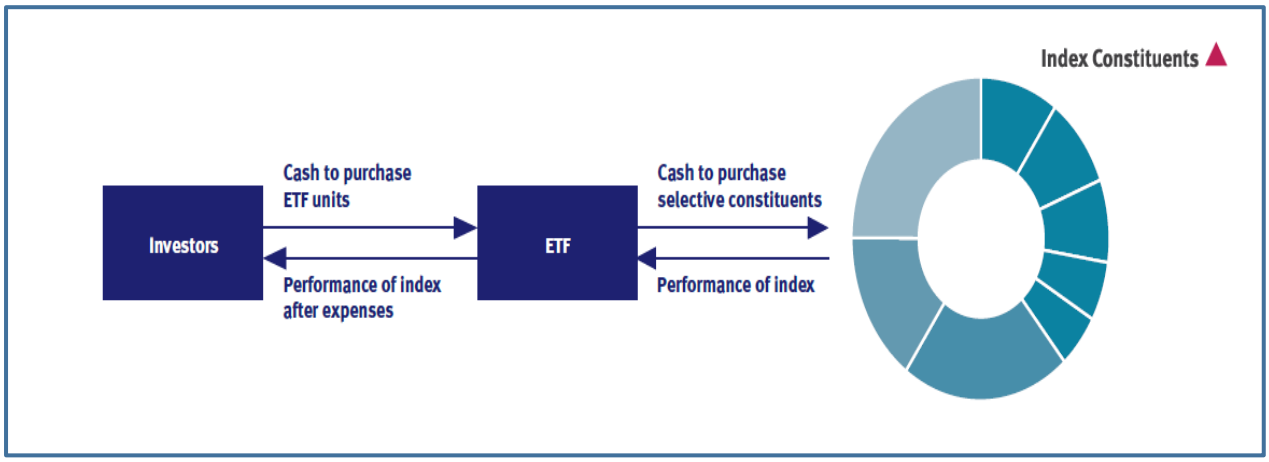
a. Derivative embedded; and
b. Swap-based.

Each of these replication methods is explained as follows:

9.2.4.1 Direct Replication (Full Replication)

Full replication is where an ETF would hold all assets underlying an index in the exact proportion with the index. This ETF will directly hold all constituents within the index to replicate the performance of an index. The returns of the ETF's portfolio, after expenses, will be passed on to investors according to the number of ETF units held (see Figure 9.2.4.1).

Figure 9.2.4.1 – Direct Replication (Full Replication)

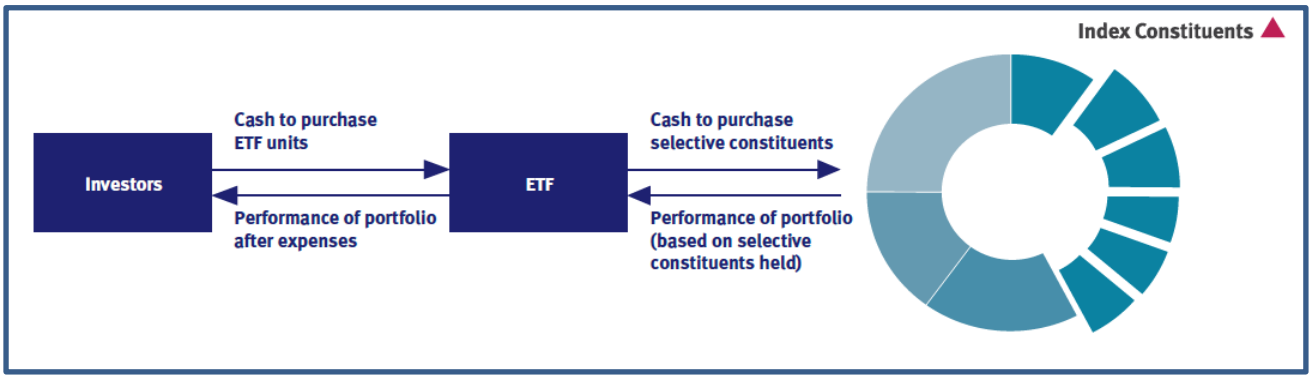


Source: SGX

9.2.4.2 Direct Replication (Representative Sampling)

In this method, an ETF will invest in a subset of index constituents, usually the more dominant underlying, to reduce number of securities held while tracking the index as closely as possible. The representative sampling method is generally used when some of the components of the index tracked are not liquid or easily available (see Figure 9.2.4.2).

Figure 9.2.4.2 - Direct Replication (Representative Sampling)



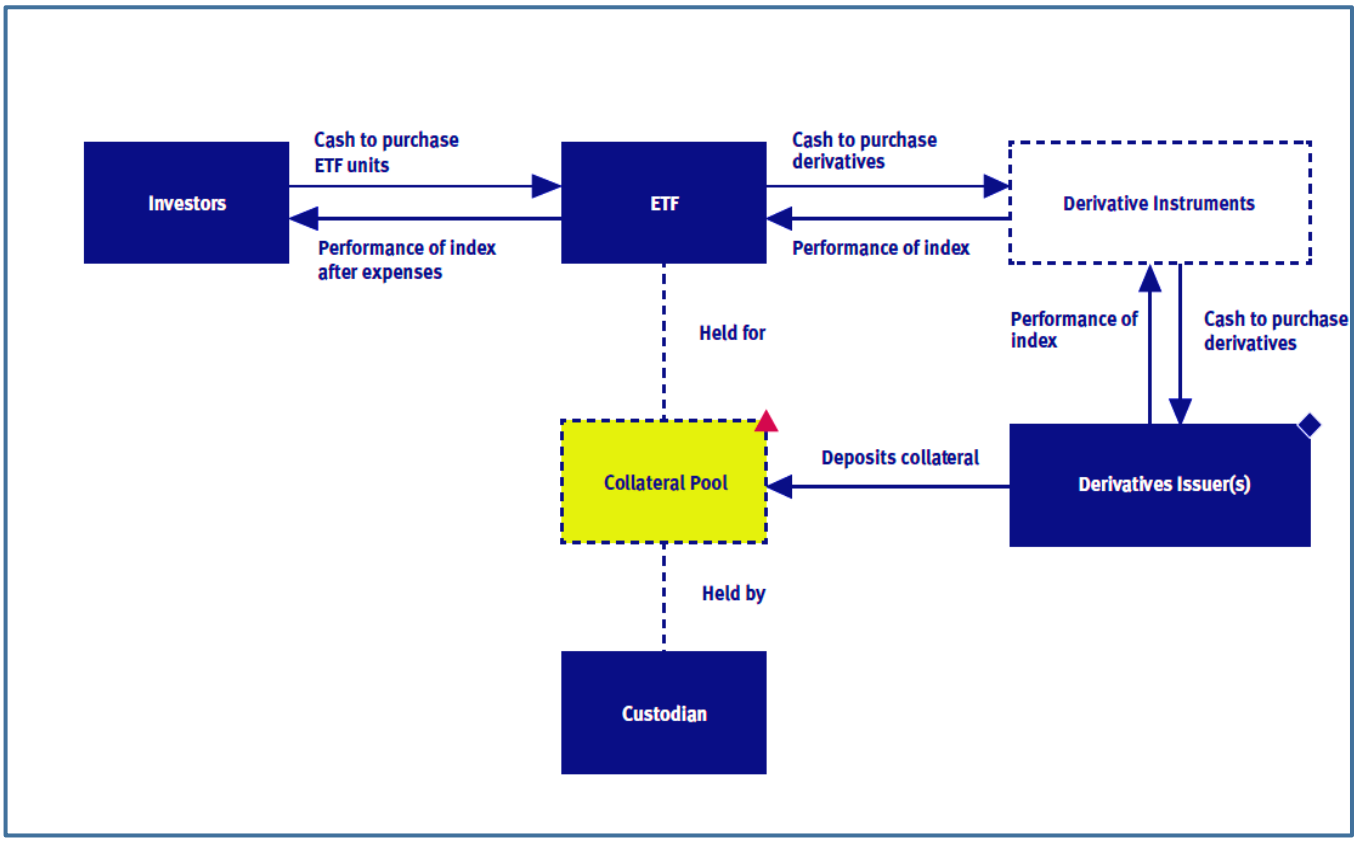
Source: SGX

This ETF will hold a subset of the constituents of the index to replicate the performance of an index as closely as possible. The returns of the ETF's portfolio, after expenses, will be passed on to investors according to the number of ETF units held.

9.2.4.3 Synthetic Replication (Derivative Embedded)

In this method, the ETF purchases futures, options and other derivative instruments from other counterparty(ies) in exchange for the index performance (see Figure 9.2.4.3).

Figure 9.2.4.3 - Synthetic Replication (Derivative Embedded)



Source: SGX

A derivative embedded ETF purchases derivative instruments from a third party and these instruments are contractual obligations that bind the issuer(s) to deliver the index performance to the fund. Investors therefore rely on the credit worthiness of the derivative issuer(s) to deliver the performance of the index to the fund. Derivatives can be used to gain exposure to markets that cannot be accessed by directly investing in the underlying asset. For example, some ETFs track restricted markets such as China and India which have foreign investment or tax limitations, so the ETFs may invest in market access products such as participatory notes (PNs / P-notes) to gain access to the underlying securities. P-notes are derivatives by third party providers which provide a return linked to an underlying security in the index, as well as carrying the credit risk of the issuer.

A derivative embedded ETF must comply with the net counterparty exposure requirements under the Code on CIS or UCITS, which allows a maximum of 10% net counterparty exposure. With this maximum net counterparty exposure, investors risk losing only up to 10% of the fund's value if there is a counterparty default. Typically, the derivative issuer will deposit collateral for the 90% balance with a third counterparty custodian, and the collateral will be owned by the ETF's trustee.

9.2.4.4 Synthetic Replication (Swap-Based)

Swap-based ETFs pay a fee or the collateral performance to a third counterparty in exchange for the index's performance. Swap-based ETFs listed on SGX must also comply with the 10% net counterparty exposure requirement under either the Code of CIS or UCITS. In order to comply with the requirement, the swap counterparty must collateralise the ETF, which could be done through two methods - fully funded structure or unfunded structure. The funded and unfunded swap structure is a combination of both funded and unfunded swap arrangements. The ETF enters into a swap agreement with a third party to obtain the performance of the index. The swap will then be collateralised through the adoption of both kinds of swap arrangements:

a. **Unfunded swap** - where the manager uses proceeds from the sale of ETF units to purchase collateral which will be pledged to the ETF (see Figure 9.2.4.4(a)); and

b. **Funded swap** - where the swap counterparty deposits collateral with a third-party custodian and pledges the collateral in favour of the ETF (see Figure 9.2.4.4(b)).

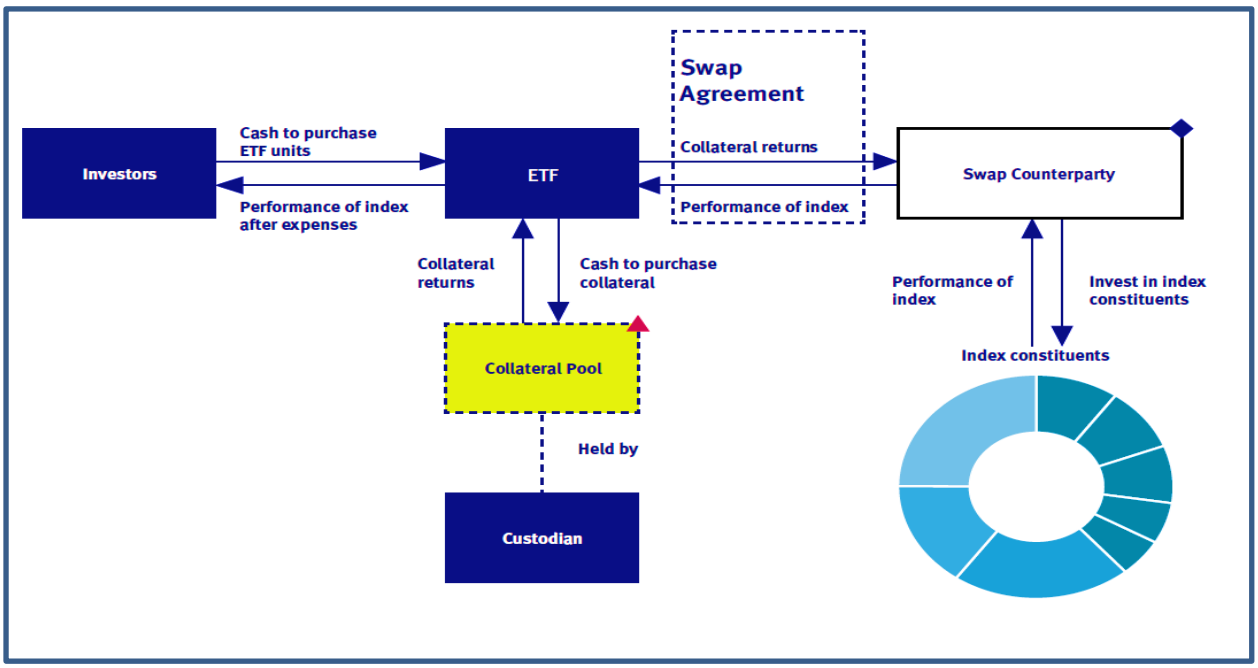
Both methods aim to limit the ETF's overall exposure to the swap counterparty to 10% of the ETF's NAV.

(a) **Unfunded Swap-Based ETF**

The unfunded swap-based ETF enters into a swap agreement with a swap counterparty to obtain the performance of the index. Under the swap agreement, the ETF will use the proceeds from the sale of units, to purchase and hold a pool of collateral placed with a third-party custodian. The returns generated by the collateral held by the ETF will then be exchanged with the swap counterparty in return for the performance of the index.

Due to the limit imposed on net counterparty exposure, the daily value of the collateral pool must be rebalanced on a daily basis to ensure that its value is at least 90% of the NAV of the ETF. In the event of a counterparty default, the ETF may choose to liquidate this pool of collateral to repay its investors.

Figure 9.2.4.4(a) - Unfunded Swap-Based ETF

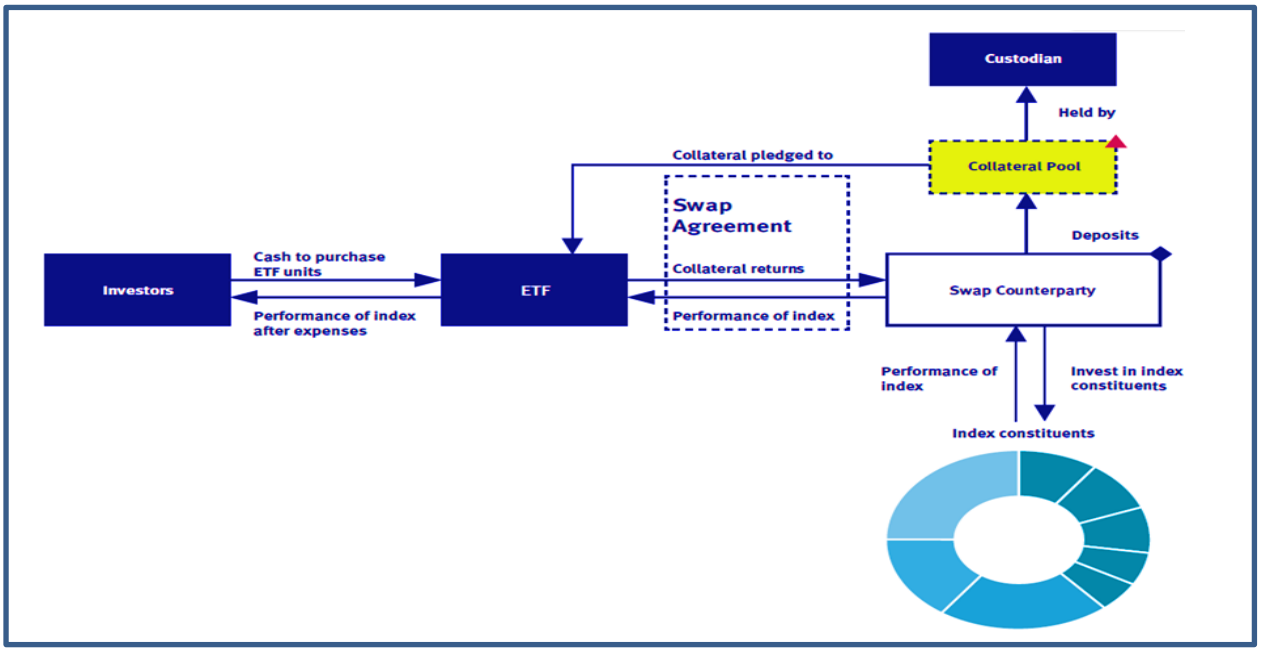


Source: SGX

(b) **Funded Swap-Based ETF**

Similar to the unfunded structure, the fully funded swap ETF will be required to deliver performance of the collateral it holds to the swap counterparty in exchange for the performance of the index. However, in this structure, the ETF transfers its sale proceeds to the swap counterparty that will purchase a pool of collateral to be placed with a third-party custodian. This pool of collateral will also be pledged in favour of the ETF, to be liquidated in the event of a default.

Figure 9.2.4.4(b) – Fully Funded Swap-Based ETF

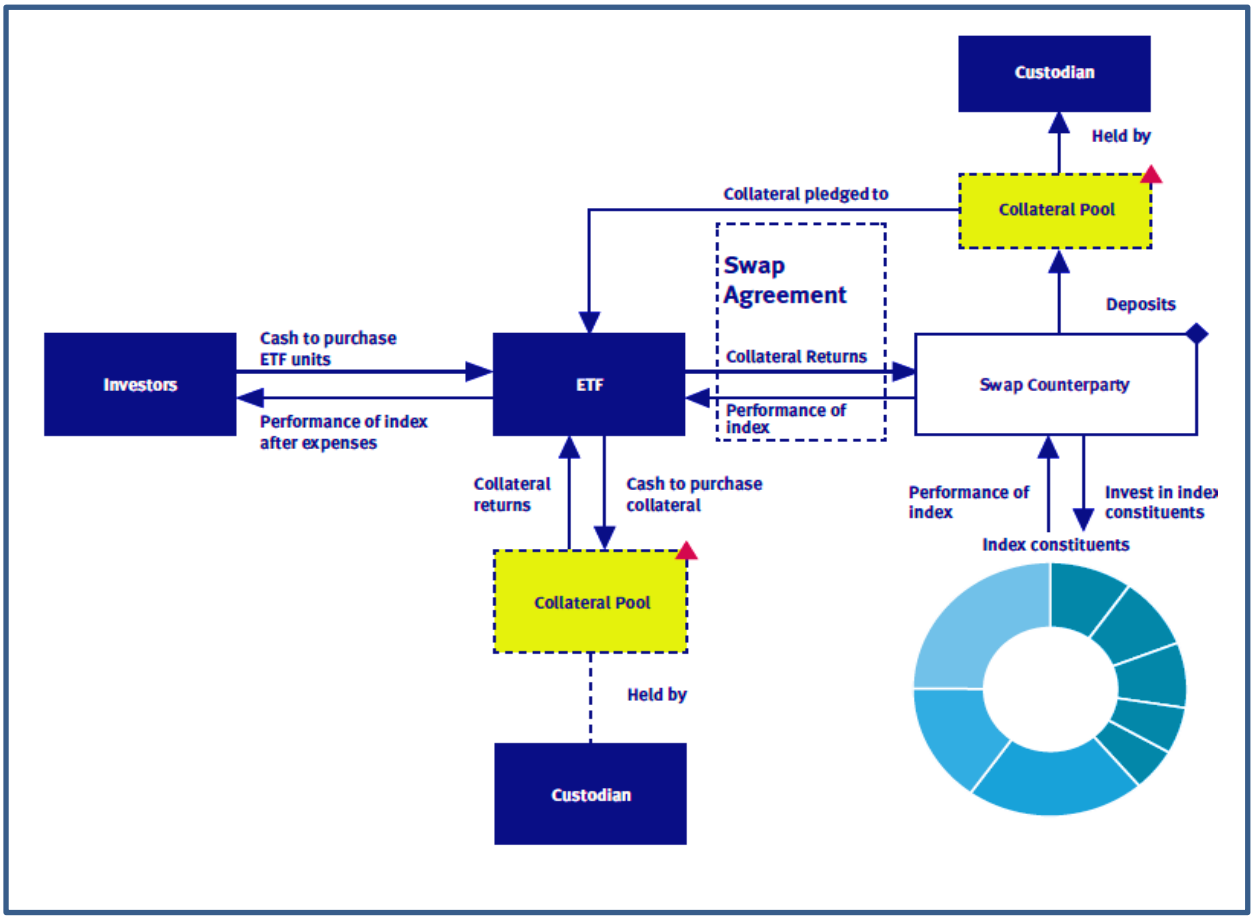


Source: SGX

(c) **Funded and Unfunded Swap-Based ETF**

The structure is a combination of both funded and unfunded swap arrangements. The ETF enters into a swap agreement with a third party to obtain the performance of the index. The swap will then be collateralised through the adoption of both kinds of swap arrangements, (a) the unfunded swap, where the manager would use proceeds from the sale of ETF units to purchase collateral which will be pledged to the ETF; and (b) the funded swap, where the swap counterparty would deposit collateral with a third party custodian and pledge the collateral in favour of the ETF. Both methodologies aim to limit the ETF's overall exposure to the swap counterparty to 10% of the ETF's NAV (see Figure 9.2.4.4(c)).

Figure 9.2.4.4(c) – Funded and Unfunded Swap-Based ETF



Source: SGX

9.2.5 ETF Pricing & Valuation Issues

9.2.5.1 Bid-Ask Spreads[44]

In the creation and redemption process, an ETF will be required to acquire or dispose of underlying index securities in order to mimic the performance of the index. If the underlying index securities are illiquid and thinly traded, the availability of underlying will decrease, causing the bid-ask spread of the underlying to

[44] Bid-ask spread is determined by the difference of the price that a buyer is willing to pay for an asset and the price that a seller is willing to sell.

widen. The failure of the creation and redemption will cause ETF units to be traded at a premium or discount, and consequently, the tracking error will increase.

9.2.5.2 Net Asset Value

Investors generally rely on the NAV of an ETF to estimate its fair value, which is calculated once at the end of each trading day. The NAV is published on the issuer's website and the exchange on which it is traded daily. It is important to note that the NAV of an ETF differs from the price at which it is traded. As an ETF trades like a stock, it is quoted at bid and ask prices on the stock exchange. It could be traded at a price higher than its NAV (at a premium) or below the NAV (at a discount).

9.2.5.3 Total Expense Ratio (TER) / Annual Fee

The TER is an annual fee for ETFs and is expressed as a percentage or in basis points (bps). The TER includes the management fee and fees paid to cover administrative, licence and other operational costs of the fund. It is calculated and accrued daily in the NAV calculations and directly deducted from the ETF on a running basis. As ETFs are passive instruments, they typically charge lower annual management fees compared to traditional actively managed fund products.

9.2.5.4 Tracking Error

Tracking error is a measure of how the value of the ETF may deviate from the value of the underlying which it tracks. Tracking error arises from:

a. **Transaction costs** – the cost of buying and selling of securities, or fees of swap transactions within the portfolio will be charged to the ETF, thereby causing a reduction in performance from its underlying;

b. **Portfolio holdings** – if the assets held in the portfolio are different from the underlying assets tracked, the replicated performance may not be the same as the underlying;

c. **Replication methodology** – synthetic replication will improve an ETF's ability to track, as it is the swap counterparty's obligation to deliver the performance of the underlying, subject to a small fee for the swap arrangement. However, this will expose the ETF to the swap counterparty's default risk;

d. **Types of investments –** an ETF that buys into futures contracts will be required to roll them repeatedly (where the ETF must sell an expiring contract to buy the next contract at a different price). The ETF is exposed to rollover risk. If price of the next contract is higher, also known as a contango effect, a loss is recorded. If the price of the next contract is lower, a backwardation effect again is recorded. The gains and losses will affect the performance of the ETF, causing tracking error;

e. **Cash holdings** – such as dividends from the securities held by an ETF, may be held as cash in the ETF until the payout date, and investors will lose out on the possible gain from reinvesting or the interest payment of the holdings / cash drag; and

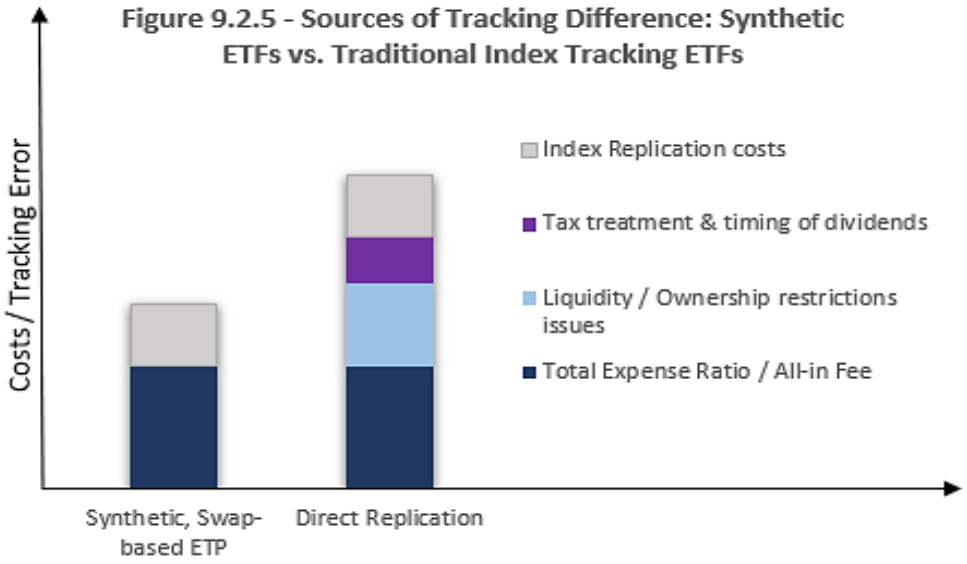
f. **Additional features** – an ETF that conducts currency hedging may incur hedging costs in a volatile market, leading to a reduction in the ETF's performance.

Tracking errors refer to the disparity in performance between an ETF and its underlying index. Refer to Figure 9.2.5. In general, tracking error is:

a. Lower for synthetic replication ETFs than physical replication ETFs.

b. Higher for benchmarks with higher volatility such as emerging markets.

Figure 9.2.5 - Sources of Tracking Difference: Synthetic ETFs vs. Traditional Index Tracking ETFs



Source: Financial Times

9.2.6 ETFs vs Exchange-Traded Notes

Although Exchange-Traded Notes (ETNs) share several of the characteristics of ETFs in that both structures are traded on an exchange and provide a return linked to the performance of an underlying benchmark, ETNs are debt securities issued by banks which are not subject to the requirements of funds such as diversification rules. The debt is typically senior and unsecured. Investors are exposed to the counterparty risk of the issuer.

Exchange-traded commodities (ETCs) are debt securities issued by banks to track the performance of single or physical commodities. ETCs are a special type of ETN which are backed by securities or physical assets such as gold. Under European fund regulations, funds cannot be exposed to underlying assets in single or physical commodities, thus such products are a popular way to gain exposure to commodities.

The VIX ETN is based on the Chicago Board of Exchange (CBOE) volatility index, a widely referenced indicator of U.S. equity market volatility. These ETNs hold long positions in the VIX futures contracts that roll daily. There are several variations of the ETNs, based on the different tenors of the underlying futures contracts. The performance of VIX ETNs can be very volatile, with a high risk of divergence in the ETN's performance from actual movements in the VIX.

9.2.7 ETFs vs Mutual Funds / Unit Trusts

The key differentiating factor between ETFs and the other popular fund formats, such as mutual funds and closed-end funds, is that ETFs offer investors two sources of liquidity:

a. Primary liquidity via the creation and redemption process; and
b. Secondary liquidity through the trading of shares on exchange.

ETF investors can buy or sell existing shares of an ETF in the secondary market throughout the trading day. In addition, investors can buy newly created or redeem existing ETF shares in the primary market with an Authorized Participant (AP) or market-maker who sits between the ETF issuer and investors. The creation / redemption process can be done in cash or in-kind with the underlying securities.

The ability to buy and sell shares of an ETF intraday is one of the biggest differences between mutual funds and ETFs. Investors can only purchase shares of a mutual fund directly from the fund company or via a distributor at the end of the day NAV. However, investors can buy and sell an ETF at any time during market hours at the bid/ask price quoted on the exchange. There is no difference in the secondary trading of ETFs on the exchange, regardless of whether they are physical or swap-based.

9.2.8 ETF & Fund Governance in US and Europe

The first US-listed ETFs, launched in 1993, were benchmarked to the S&P 500 index and adopted a physical replication structure. Since then, most US ETFs have adopted a physical replication structure using full, representative, or optimised sampling.

ETFs in the US were launched under the provisions of the Investment Company Act of 1940 (1940 Act). The 1940 Act established the terms, conditions, and procedures applicable to collective or pool securities investments products offered to US investors. US ETFs are organized as a management investment company (a mutual fund-type structure) or as a unit investment trust and fall under the jurisdiction of the US Securities and Exchange Commission (SEC).

The emergence of synthetic ETFs in Europe reflects differences in the regulatory environments of the U.S. and Europe. Implementation of the UCITS III fund regulations from 2001 onwards resulted in the development of synthetic replication ETFs in Europe. Most ETFs in Europe adopt a swap-based replication structure.

Under the UCITS regulations, it became possible for ETFs to invest in instruments such as swaps, subject to certain counterparty risk exposure limits. Instead of holding all or a representative sample of the index constituents, swap-based ETFs use index swaps or equity-linked swaps to replicate the index performance. Among other investment restrictions, the UCITS regulations stipulate that an ETF is not allowed to invest more than 10 % of its prevailing NAV in derivative instruments including swaps issued by a single counterparty. In other words, the amount that the swap counterparty owes to the fund is limited to a maximum of 10 % of the fund's prevailing NAV.

Under the UCITS III guidelines, the swaps mark-to-market value cannot exceed 10% of the fund's NAV on a daily basis. Depending on the swap counterparty involved, this UCITS limit on counterparty exposure can be decreased to less than 10% voluntarily.

9.2.9 ETF Investors and Investment Strategies

9.2.9.1 Type of Investors

ETFs are embraced by self-directed retail investors who recognise the benefits of a liquid, convenient investment tool which can be used to diversify their investment portfolio in a cost-efficient manner. ETFs are also used by institutional investors for asset allocation, cash flow management, hedging as well as other trading strategies.

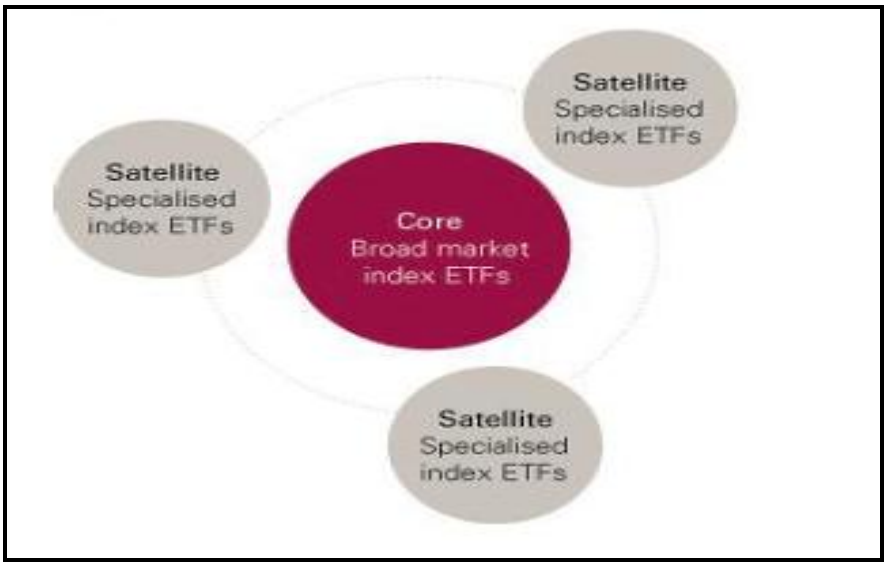
a. **Investment Strategies**

i. **Core-Satellite Approach**

Individual investors and portfolio managers using the core-satellite approach often use ETFs as the core portfolio. ETFs can also be used for tactical reasons, such as gaining exposure to a specific market

rapidly while searching for specific securities to invest in. They can be used to provide for thematic exposures (for example, selected dividend-paying stocks). Refer to Figure 9.2.9.1(a).

Figure 9.2.9.1(a) - Core-Satellite Investment Approach



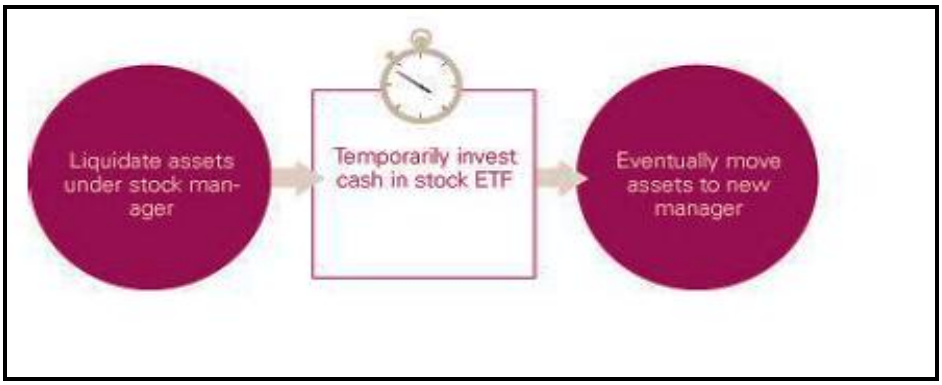
Source: Vanguard

b. **Managing Portfolio Imbalances and Cash Positions**

As asset values change over time, an investor's carefully considered asset allocation may shift out of balance, which could jeopardise the investor's chances of meeting a goal. ETFs can quickly and easily correct those imbalances when buying and selling from existing holdings is not practical.

For investors who have a large, temporary cash position such as a bonus or other windfall, or who are transitioning between asset managers, ETFs can be a good option. Such a cash position may tilt an investor's portfolio away from its targeted asset allocation. Over extended periods, that position can lead to performance shortfalls relative to benchmarks or financial goals. Rather than leaving cash uninvested even for a short time, investing a temporary cash position in ETFs can counter the potential for performance shortfalls. Refer to Figure 9.2.9.1(b).

Figure 9.2.9.1(b) – Managing Portfolio Imbalances and Cash Positions



Source: Vanguard

9.2.10 Benefits of ETFs

There are several reasons why investors trade and invest in ETFs:

a. Broad range of asset classes available – e.g., equities, fixed income, cash, alternatives, currency and commodities. ETFs can be a good diversification vehicle as they create exposure to a basket or a large number of securities;

b. ETFs offer liquidity and flexibility as they are listed on exchanges and can be traded any time the market is open. In addition, they are priced continuously throughout the day;

c. ETFs generally incur relatively lower transaction costs and are also a cost-effective way for investors to have a diversified market exposure. As they are passive instruments, ETFs typically charge lower annual management fees compared to traditional actively managed fund products;

d. Some exchanges allow ETFs to be shorted, just like equities. As there are risks involved in shorting ETFs, typically only sophisticated investors or traders and hedge fund managers engage in shorting ETFs; and

e. As ETFs are passive instruments, they aim to replicate the performance of the underlying index one to one. As such, the ETF's performance should be similar to the benchmark index. One of the main advantages of replicating an index through swaps is that the tracking error risk of the ETF, before fees, is passed to the swap counterparty. As such, swap-based ETFs have been embraced by investors who want to minimize the tracking error of the ETF versus the benchmark.

9.2.11 Risks of ETFs

Several key risks are associated with ETFs.

9.2.11.1 Market Risk

ETFs including leveraged and inverse ETFs are designed to track the performance of certain indices, market sectors, or assets classes such as equities, bonds and commodities. Investors must be prepared to bear the risk and volatility associated with the underlying index or asset class. A major adverse market move can result in the ETF being shut down. A few inverse VIX ETNs/ETFs were shut down after VIX futures spiked more than 100% in 2018. For those remaining, even when VIX went back to the original levels, investors would have still lost at least 70% of their investments[45].

9.2.11.2 Tracking Error

Tracking errors refer to the disparity in performance between an ETF and its underlying index. Tracking errors can arise due to factors such as the impact of transaction fees and expenses incurred to the ETF, changes in composition of the underlying index, index replication costs resulting from liquidity and ownership restrictions on the underlying, cash drag, and the structured ETF manager's replication strategy. In general, tracking error is lower for synthetic replication than physical replication ETFs. It could also be greater for ETFs which have benchmarks that have inherently higher volatility, such as emerging markets.

9.2.11.3 NAV Trading at Discount or Premium

ETFs may be traded at a discount or premium to NAV. This price discrepancy may be caused by supply and demand factors or imbalances during periods of high market volatility and uncertainty. Discrepancies also arise

[45] Refer to CNBC news article published on 6 Feb 2018: https://www.cnbc.com/2018/02/06/the-obscure-volatility-security-thats-become-the-focus-of-this-sell-off-is-halted-after-an-80-percent-plunge.html

for ETFs tracking specific markets or sectors that are subject to direct investment restrictions (e.g., China A-shares) or for ETFs trading in a different time zones from where the underlying assets it tracks are domiciled.

9.2.11.4 Foreign Exchange Risk

Investors of ETFs which have underlying assets or exposures not denominated in their home currency are also exposed to exchange rate risk. Currency rate fluctuations can adversely affect the underlying asset value, also affecting the ETF price.

9.2.11.5 Liquidity Risk

Designated market-makers and liquidity providers facilitate trading in ETFs. Although most ETFs are supported by one or more market-maker, there is no assurance that active trading will be maintained. If the designated market-maker defaults or ceases to fulfil their role, investors may not be able to buy or sell the product.

9.2.11.6 Counterparty Risk Involved in ETFs with Different Replication Strategies

a. Physical replication ETFs that invest directly in the underlying assets may be exposed to counterparty risk if the ETF engages in securities lending of the funds' assets. The ETF may suffer losses if the borrowers default or fail to honour their contractual commitments.

b. Synthetic replication ETFs are exposed to counterparty risk of the swap dealers and derivative issuers. The ETFs may suffer losses if any of the dealers/issuers default or fail to honour their contractual commitments. ETFs which make use futures are also exposed to basis risk and roll risk.

9.2.12 ETFs After Sales

Major banks and dealers are involved in providing liquidity and in secondary trading of ETFs on exchanges. Many exchanges in Asia and Europe have designated ETF liquidity providers who are obligated to quote bid and offer prices continuously during market hours subject to maximum spread and minimum quantity requirements. The bid/offer spread of an ETF is driven by several factors including hedging cost. The hedging cost reflects how efficiently an Authorised Participant (AP) or market-maker can access the underlying market. Under physical replication, the ETF accesses the underlying market directly by buying the physical index constituents. Under synthetic replication, the swap counterparty accesses the underlying market to hedge its index exposure. Thus, irrespective of the ETF's replication method, the ETF's liquidity is dictated by access to and liquidity of the underlying market.

The open-ended nature of ETFs means that on any given trading day, APs or market-makers can create sufficient shares to meet the settlement requirements resulting from their trading activity on the stock exchange. Subscriptions/redemptions by market-makers and APs with the management company are typically done in large sizes. ETF trades done at NAV are not reflected in the exchange-traded volumes of an ETF. Thus, the liquidity of an ETF is measured by the liquidity of the constituent of its underlying index rather than the trading volume of the ETF on the stock exchange.

ETFs share many similarities with closed-end funds in that both are listed and traded on an exchange and can be bought and sold at any time during market hours. However, there are also some significant differences between closed-end funds and ETFs, which are open-ended. Closed-ended funds are launched via an IPO process and trade with a fixed number of shares outstanding. The fixed number of shares means that the relationship between the price of the fund and its NAV, which equals the sum of the fund assets minus its liabilities, is determined by market demand. As such, closed-end funds can and generally do trade at a premium or discount to NAV.

The trading price of an ETF refers to the prices that an ETF can be bought and sold while trading on an exchange. Under normal circumstances, the trading price of an ETF should be close to its NAV, which is the estimated NAV of the ETF intraday. This reflects the fact that arbitrageurs can take advantages of discount/premiums of the ETF relative to the index by buying or selling the ETF versus the underlying securities. For example, if the price of an ETF is trading at a premium relative to the securities in the index, an arbitrageur can purchase the underlying basket of shares, deliver the basket to create new shares of the ETF and sell the ETF in the market to realize a profit. The open-ended feature of ETFs ensures that premium/discounts in the secondary market can be corrected through the activities of market participants who are able to subscribe for or redeem ETF shares in the primary market.

There are some situations where the trading price may differ from the NAV, in particular for ETFs that provide access to restricted markets such as China A-shares. Persistent premiums or discounts in ETF prices reflect inefficiencies in the underlying markets. Furthermore, prices are likely to be farther away from NAVs when an ETF is not trading in the same time zone as its underlying assets. For example, Asian ETFs will typically trade closer to their NAVs during Asian trading hours than when they are trading in the U.S. where market-makers will trade at a risk discount or premium.

9.3 Summary

1. A **structured fund** is a fund that combines various financial instruments to achieve specific risk/return profiles or cost/savings objectives that may not be achievable in the marketplace. While most structured funds typically have combinations of equity, fixed income securities and derivatives, some funds invest only in derivatives.

2. Structured funds are created through the process of financial engineering and can range from simple to highly complex funds, covering a wide range of risk exposures. Funds offering a degree of capital preservation, can potentially give investors the opportunity to earn a market-linked investment return, while getting back their initial investment at the maturity if financial markets decline.

3. Unlike traditional mutual funds which rely on the fund manager's expertise and active decisions on making asset allocations, structured funds aim to replicate the underlying asset or to provide a synthetic return linked to the underlying asset by incorporating derivatives, using static or rule-based allocation decisions.

4. While **trackers** or **index funds** are used to simply replicate the performance of their benchmarks, structured funds can be much more versatile. A structured fund can be used to act on various projected market scenarios, to systematically adjust investment exposures to market developments, and to optimize on the desired level of capital preservation and participation in the underlying.

5. The 4 main components that go into the building of a structured fund are the **underlying assets, tenor / maturity, payout structures** and **market views**. Once all the input parameters have been defined, a structured fund can be established.

6. Structured funds can invest directly into the underlying asset to gain direct exposure. Funds can also get indirect exposure by tracking or replicating the underlying asset's performance synthetically by using derivatives. A **total return swap** is an example of a fund made up entirely of derivatives instruments.

7. The two types of structured funds, based on the financial techniques used, are funds based on derivatives and funds based on techniques derived from portfolio insurance (such as CPPI).

8. The fund manager's primary role is to manage the fund assets in line with defined investment objectives and investment restrictions. The fund trustee is independent of the fund manager and has a fiduciary responsibility to the investors. The trustee holds the assets on behalf of the fund unit holders and ensures the fund manager carries out its duties according to the trust deed.

9. The risks in a structured fund are financial, operational or legal in nature. The fund manager uses internal controls and risk procedures at the fund level to monitor and manage the various risks. Financial techniques such as stress-testing, **value-at-risk (VaR)** and **duration** can be applied to manage financial risks such as market risk. The auditor conducts annual audits for each of one of the fund entities - the management company, the investment manager, the administrative agent and the custodian.

10. A **fund prospectus** is a document that will identify and discuss everything from the fund's objectives and past performance, to a description of the fund manager and the fees associated with the fund. The product highlights sheet is a product-specific summary of the key features and risks of the structured fund. The trust deed is a document that sets out the terms and conditions governing the relationship between investors, the fund manager and the trustee.

11. Retail and institutional investors can subscribe to structured funds. Not all schemes are offered to both classes of investors and a fund scheme could float different classes of shares, depending upon the type of investors subscribed to the scheme. For a retail offer of a scheme, a prospectus in compliance with the SFA must be lodged and registered with the MAS. For institutional investors, accredited investors and other relevant persons, a prospectus is not required.

12. Investors can get information on the **Net Asset Value (NAV)** from leading financial newspapers, financial newswires like Bloomberg and Reuters, and the website of the fund issuer. Investment performance information is available from the fund factsheets, monthly performance reports, investment manager report, semi-annual and annual accounts to investors.

Chapter 10:
Contracts for Differences (CFDs)

Learning Objectives
The candidate should be able to: ✓ State what are the features and characteristics of CFDs. ✓ Explain the advantages of CFDs over other investment products. ✓ Describe the trading models and mechanisms for CFDs. ✓ Discuss the global trading opportunities and strategies for CFDs.

10.1 History and Origin of Contracts for Differences

Contracts for Differences (CFDs) originated in London in the early 1990s where they were first traded over-the-counter (OTC) in the equity swap market. Institutional investors used them as a cost-effective way of hedging their equity exposures. Other players included arbitrageurs and hedge fund managers using market neutral trading strategies, where they found CFDs to be an effective way to take positions if they had a negative outlook on the market, or on a particular share, without the need to borrow stock to short sell.

CFDs enabled investors to take long and short equity positions as well as to benefit from leverage on their investments. Even though there was no actual change of equity ownership with CFDs, investors got exposure to the real-time performance of the underlying share price, were entitled to share dividends, and faced the impact of other corporate actions. However, CFDs investors were not entitled to voting rights.

In the late 1990s, CFDs were made available to private clients and retail investors. The rapid growth and popularity with individual investors were greatly aided by the introduction of online trading. Individual investors could trade using the internet via brokerage platforms into the London Stock Exchange.

From its origins in the UK, CFDs caught on with other European countries, particularly with investors in Belgium, Germany, and Ireland. Today, CFDs are also available in the Australian and Asian markets, where they are popular with investors in Hong Kong, Singapore, and South Korea. Due to regulatory restrictions, CFDs are not available in the US.

10.2 What Are CFDs?

CFDs are derivative products, and the contract involves two parties - the buyer and the seller. The buyer or seller of a CFD contract, who has a positive or negative view of the market respectively, takes a position on asset price movements without direct ownership of the underlying asset. Upon maturity, there is a cash settlement whereby the seller pays the buyer the difference between the opening price and closing price of the contract (or vice versa if the market moves against the buyer).

The terms of a CFD contract include the following:

a. **Underlying asset** – it can be a stock, index, commodity, currency, etc;

b. **Price of the underlying asset** – this is established when the contract commences;

c. **Number of units** of the underlying asset specified in the contract;

d. **Expiry period**, if any; and

e. **Settlement currency**.

10.3 Characteristics and Features of CFDs

The main characteristics and features of CFDs are:

a. **Leverage** - The margin requirement is a fraction of the underlying asset value. For a small outlay, the investor can get a larger market exposure. For example, if the margin requirement is 10%, the potential gains can be magnified by a factor of 10. However, on the downside, any loss will also be magnified;

b. **Flexibility** - It is easy to take long or short positions to exploit opportunities in both rising and falling markets;

c. **Low Cost** - Brokerage commissions and finance charges tend to be low and highly competitive. In Singapore, there is no stamp duty to be paid;

d. **Transparency** - The underlying asset or index is known at the outset and can be easily tracked as the CFDs follows the underlying price very closely;

e. **Ease and Simplicity** - Investors can trade via a personal broker or online, with online trading being widely popular with investors. CFDs providers are brokerage firms who are members of exchanges, with robust CFDs online trading platforms that provide real-time information. Once the investor opens an account and has capital available, all he needs is internet access to trade in CFDs;

f. **No Expiry Date** - An open position can be rolled over and held as long as the investor wishes. The investor can close out his position when he so desires;

g. **Access to Asset Classes** - Investors can trade in CFDs on many individual stocks, indices, government treasuries, foreign exchange, commodities, real estate and ETFs. New asset classes and market indicators are always emerging, and these products are offered to CFDs investors, such as market volatility indices and weather futures**;**

h. **Access to Global Investments** - CFDs are not limited to domestic Singapore securities. The investor can access many equities, indices, investment classes and sectors that are traded worldwide on various exchanges.

i. **Cash Settlement -** CFDs are cash settled products as there is no delivery of any underlying instrument, such as a share certificate. The settlement involves the cash difference between the buy price and the sell price; and

j. **Corporate Actions -** For equity CFDs, the investor holding a long position will get cash dividends on the underlying stock, and participate in share splits and other corporate actions, just as if the investor owns the physical stock (the opposite applies to an investor with a short position). The only major difference is that the CFD investor is not entitled to any voting rights.

10.4 CFDs Business Models

There are three main business models used by CFDs providers in global markets:

a. **Market-Maker** - The CFDs provider makes a market in CFDs and quotes its own bid-ask prices based on the underlying asset.

b. **Direct Market Access (DMA)** - The buy/sell order goes through the CFDs provider's platform. The investor gets direct access to the market in which the underlying asset is traded, and its market price determines the price paid by the investor for the CFD contract. This is the predominant approach of brokerage firms who are CFDs providers in Singapore.

C. **Exchange-Traded CFDs** - These CFDs are traded on an exchange just like listed equities. Orders are placed through brokers who have access to the exchange.

10.5 Advantages of CFDs over Other Instruments

	Shares	CFDs	Options	Warrants
Ease of Trading	Easy – most investors start with shares	Easy – just like actual shares	Different from share trading – can be complex for beginners	Different from share trading – can be complex for beginners
Price Movement	Actual price movement	Similar to shares	Affected by many factors, not just share price	Affected by many factors, not just share price
Trade on Margin	No – unless margin facility is obtained	Yes	Yes	No
Trade Long or Short	Process is complex and costly	Yes	Yes	No
Dividend Payment	Yes	Yes	No	No

	Shares	CFDs	Options	Warrants
Expiry Date	No	No	Yes – can expire worthless	Yes
Stop-Loss Orders	Yes	Yes	No	No
Spread	Actual price	Similar to shares	Wider than underlying instrument	Wider than underlying instrument
No. of Instruments Traded	Usually limited to domestic equities	Accessible to other exchanges and global CFDs	Limited - to few shares	Limited - to few shares

10.6 Trading Mechanisms of CFDs

10.6.1 Margin

To initiate a CFD trade, the investor does not have to pay the full underlying value. He pays a fractional sum in cash, called the "initial margin". The initial margin rate is computed as a percentage of the overall value of the underlying asset and varies depending on the type of security, asset class, country regulations and the specific requirements of the CFDs provider.

Example 1

A CFD on Company A's shares requires a 10% initial margin. The share price is $2 and the total value for 10,000 shares is $2 x 10,000 = $20,000. As the initial margin required is 10%, the investor puts up a cash deposit of only $2,000 when he opens his position. This effectively gives the investor a leverage of 10x the invested sum.

Example 2

An investor wants to take a long position in US Crude Oil, which is trading at $105. A commodity CFD requires 20% initial margin. The value for 100 barrels of US Crude Oil will be 100 x $105 = $10,500. The initial margin requirement is 20% x $10,500 = $2,100. This effectively gives the investor a leverage of 5x the invested sum.

10.6.2 Mark-to-Market

The CFD's value in the investor's account varies with the price of the underlying asset in real-time. At the end of each trading day, any gain (or loss) is added to (or subtracted from) the investor's account. This practice is called "mark-to-market".

10.6.3 Margin Call

If the balance in the investor's account falls below a stipulated level called the "maintenance margin", the investor will receive a "margin call" to top up his account in order to restore the account balance to an acceptable level (back to the level of the "initial margin"). If the investor is unable to meet the margin call, the CFDs provider will proceed with "liquidation".

The CFDs provider must inform the investor when the CFD account is opened about the margin requirements, margin limits and liquidation procedures. The investor must be familiar with these procedures which are stated in the Risk Disclosure Statement (RDS). The RDS must be presented to and confirmed with a signed acknowledgment by the investor at the time of establishing the CFD trading account.

10.6.4 Liquidation

The CFDs provider may liquidate the investor's position in the event that his account balance is below the level of the maintenance margin, and if funds to restore the margin position have not been received within the stipulated time frame. The CFDs provider will proceed with forced selling of the investor's CFDs holdings. The liquidation may be for the investor's entire CFDs holdings or partial positions, to the extent that the number of contracts that are force sold is sufficient to cover the shortfall and restore the margin level. The investor is liable for any shortfall, loss and expense as a result of the liquidation.

10.6.5 Orders and Stops

An order can be placed before, during and after trading hours. Orders placed after trading hours will be processed for execution when the market opens for trading on the following market day. All orders are considered to be day orders unless stated otherwise. All CFDs orders are done based on the bid-ask price of the underlying asset.

10.6.5.1 Orders

In Singapore, there are three types of orders for equity CFDs:

a. **Limit Order**

This is an order to transact at a limit price or better. The investor is in a queue to transact at that price or better.

i. For a buy order, the investor will pay the price (ask) that is the limit price or lower.

ii. For a sell order, the investor will get the price (bid) that is the limit price or higher.

Example – Limit Order

Company B's shares are currently trading at a bid of $10.00 and an offer of $10.02. If a limit order is placed to buy at $10.00 or lower, the order will be placed in a queue and remain there at the specified price level until it is filled, amended or cancelled. If a limit order is placed to sell at $10.00 or more, the order will be placed in a queue and remain there at the specified price level until it is filled, amended or cancelled.

b. **Market-to-Limit Order**

This is an order to buy or sell equity CFDs at the market price when the order is placed and executed. There is no guarantee that the quoted price at the time the order is placed will be achieved as the price may be different when the order is executed. When the order is executed, it may be filled completely or partially. If partially filled, any unfilled part will remain open in the market at the price where the previous part of the order was filled.

Example - Market-to-Limit Order

Company C's shares are currently trading at a bid of $10.00 and an offer of $10.02. There is a market-to-limit buy order to purchase Company C's shares at $10.02. If the order quantity is not completely filled, the remaining part of the order will stay in the market at $10.02 and can be kept or cancelled. Market-to-limit orders can only be placed during market hours.

c. **Market Order**

This is an order to buy or sell a share at the current market price. The order will be done at the best price that the market can offer at the particular point in time when the order is keyed in.

i. For a buy order, the order will be matched against the seller who has the lowest ask price.

ii. For a sell order, the order will be matched against the buyer who has the highest bid price.

10.6.5.2 Contingent Orders (Stop-Loss/Stop Entry)

Contingent orders are orders that will automatically be generated when specific market conditions are met, such as Stop-Loss or Stop Entry order types.

a. **Stop-Loss Order**

This is an order that is contingent upon the stop price being reached. This order is used by traders to cut a losing position at a price worse than the current prevailing price or to protect an existing profit (Trailing Stop) once a pre-determined price is hit.

Example – Stop-Loss Order

A trader buys 1,000 CFDs of Company D at $10.00 and places a contingent order to sell at $9.50 (stop limit $9.45). If the market falls to a last traded price of $9.50, a sell stop-limit order becomes a market order until all the CFDs are sold or until it reaches a traded price of $9.45.

A stop-loss trading strategy that can mitigate some of the risks involved in trading CFDs. Most CFDs trading platforms usually allow investors to set a stop-loss price at which the open position will be automatically closed out.

The use of a stop-loss strategy can be risky. Even if a stop-loss price has been set, the CFDs provider may not always execute the stop-loss order at the price. This can happen in volatile market conditions where there may be price jumps and there is no transaction taking place at the set order price. Some brokerage firms offer an additional premium service called "guaranteed stop-loss" whereby the stop-loss will always be executed when the underlying asset reaches the investor's specified set price.

b. **Stop Entry**

An investor can also use contingent orders to place Stop Entry orders. When exiting a short CFD position, a stop order (Buy-Stop) can be used to set a stop-loss target on the position at a price higher than the current prevailing price. A long CFD investor can place a Buy-Stop order to enter the market at a higher price than the current prevailing price. Using a Stop in this instance allows entry into the market when the investor believes that the share price has moved through a breakout point in its trading range and will then continue to rise. The reverse applies for a Sell-Stop.

Example – Stop Entry

Company E is currently trading at $20.80, with the trader holding an open position of 1,000 Company E's CFDs at $21.00. Using a Stop-Loss order, the trader places a stop sell at $20.50 (stop limit $20.46). A limit must be placed and can be no more than 10 ticks away (i.e., $20.40) from the stop price of $20.50, based on a tick size of $0.01.

10.6.6 Financing

Apart from the margin and mark-to-market requirements, CFDs are subject to other transaction costs, such as daily financing charges, commissions and Goods and Services Tax (GST). There is also a charge whenever there is rollover of a contract.

10.6.6.1 Financing Charge

Investors incur a financing cost (or derive a benefit) for their long (or short) CFDs positions. As CFDs require only a margin payment, investors are essentially borrowing to enjoy the exposure of going long in the underlying shares.

The interest charges that investors pay reflect the cost of borrowing and are usually applied at an agreed rate based on some prevailing interest rate benchmark. Interest is computed on a daily basis and charged for the number of days the position is open.

For short positions, the investors may receive interest in lieu of the deferring sale proceeds. In addition, CFDs providers may charge interest for the borrowing costs incurred in allowing the 'short' transactions.

10.6.6.2 Commission

Dealer firms charge a commission for the CFDs transactions, which is usually a percentage of the total value of the underlying asset, subject to a minimum sum. Investors should find out the commission rates from their CFDs providers, before trading in CFDs.

10.6.6.3 GST

The commission charged by the dealer firm is subject to GST.

Example – CFD Costs

An investor buys a CFD contract on Company F. The price of the share on opening day is $2.00. After 10 days, the share price rises to $2.20. What is the net profit and total expenses incurred?

Company:	Company F
Quantity:	10,000
Commission:	0.4%
GST (Commission):	8%
Financing:	5% per annum

Total Value of Purchase	
= 10,000 x $2.00	$20,000.00
Commission	
= 10,000 x $2.00 x 0.4%	$80.00
GST on Commission (8%)	$6.40
Total Transaction Cost (Buy)	$86.40

Total Value of Sale	
= 10,000 x $2.20	$22,000.00
Commission	
= 10,000 x $2.20 x 0.4%	$88.00
GST on Commission (8%)	$7.04
Total Transaction Cost (Sell)	$95.04
Financing Interest	
= $20,000 x 5.0% ÷ 360	
= $2.778 per day	
For 10 days = $2.778 x 10	$27.78
All Expenses	
Transaction Costs (Buy)	$86.40
Transaction Costs (Sell)	$95.04
Financing Interest	$27.78
= Total Expenses Incurred	$209.22
Profit & Loss	
Total Sales Value	$22,000.00
- Purchase Cost	$20,000.00
- Total Expenses Incurred	$209.22
= Net Profit	$1,790.78

10.6.6.4 Contract Rollover Charges

CFDs have an expiry date which is determined by the CFDs provider. Upon expiry, the investor can continue to maintain his position by rolling over the contract. As the exact procedure may vary according to the

underlying asset class of the CFDs and the market in which it is transacted, investors should find out from their CFDs providers the specific details for extending their positions and if any rollover fees are charged.

10.6.7 Going Short

The following examples illustrate how investors can use CFDs contracts to "go short" on the underlying stock.

Example 1 - Going Short

An investor shorts a CFD contract on Company G. The price of the share on opening day is $3.30. After 5 days, the share price falls to $3.10. The margin requirement for the share is 20%. What is the net profit and investment returns (ROI)?

Company:	Company G
Quantity:	10,000
Commission:	0.40%
Margin requirement:	20%
GST (Commission):	8%
Financing:	6% per annum

Total Value of Sale	
= 10,000 x $3.30	$33,000.00
Commission	
= 10,000 x $3.30 x 0.4%	$132.00
GST on Commission (8%)	$10.56
Total Transaction Cost (Sell)	$142.56

Profitable (Positive) Outcome

Total Value of Purchase	
= 10,000 x $3.10	$31,000.00
Commission	
= 10,000 x $3.10 x 0.4%	$124.00
GST on Commission (8%)	$9.92
Total Transaction Cost (Buy)	$133.92
Financing Interest	
= $33,000 x 6.0% ÷ 360	
= $5.50 per day	
For 5 days = $5.50 x 5	$27.50
All Expenses	
Transaction Costs (Buy)	$133.92
Transaction Costs (Sell)	$142.56
Financing Interest	$27.50
= Total Expenses Incurred	**$303.98**

Profit & Loss	
Total Sales Value	$33,000.00
- Purchase Cost	$31,000.00
- Total Expenses Incurred	$303.98
= Net Profit	**$1,696.02**
Capital Investment*	$6,600.00
Return on Capital Invested	**25.70%**

*20% margin requirement x Total Sales Value of 10,000 shares at $3.30

Example 2 – Going Short

An investor shorts a CFD contract on Company H. The price of the share on opening day is $3.30. After 5 days, the share price rises to $3.45. The margin requirement for the share is 20%. What is the net profit and investment returns (ROI)?

Company:	Company H
Quantity:	10,000
Commission:	0.40%
Margin requirement:	20%
GST (Commission):	8%
Financing:	6% per annum

Total Value of Sale	
= 10,000 x $3.30	$33,000.00
Commission	
= 10,000 x $3.30 x 0.4%	$132.00
GST on Commission (8%)	$10.56
Total Transaction Cost (Sell)	$142.56

Loss (Negative) Outcome	
Total Value of Purchase	
= 10,000 x $3.45	$34,500.00
Commission	
= 10,000 x $3.45 x 0.4%	$138.00
GST on Commission (8%)	$11.04
Total Transaction Cost (Buy)	$149.04
Financing Interest	
= $33,000 x 6.0% ÷ 360	
= $5.50 per day	
For 5 days = $5.50 x 5	$27.50

All Expenses	
Transaction Costs (Buy)	$149.04
Transaction Costs (Sell)	$142.56
Financing Interest	$27.50
= **Total Expenses Incurred**	$319.10
Profit & Loss	
Total Sales Value	$33,000.00
- Purchase Cost	$34,500.00
- Total Expenses Incurred	$319.10
= **Net Profit**	($1,819.10)
Capital Investment*	$6,600.00
Return on Capital Invested	-27.56%

*20% margin requirement x Total Sales Value of 10,000 shares at $3.30

10.6.8 Corporate Actions

10.6.8.1 Cash Dividends

Dividends will be credited or debited based on the investor's CFDs position. Investors with long positions in equity CFDs will receive a dividend credit, while those with short positions will have the dividend amount debited from their accounts. The exact timing of this entry/adjustment (ex-date, payment date or other) can vary, depending on the country of the underlying asset. The investor should check with the CFDs provider on the exact terms for handling cash dividends.

Dividend entries and adjustments to an investor's CFDs account are usually made in the declared currency of the underlying shares. If the investor is trading outside his domestic market, he must check with the CFDs provider on handling the multi-currency transactions.

10.6.8.2 Share Splits and Reverse Splits

The quantity and price adjustment will be made in the investor's account to reflect the market equivalent transaction.

10.6.8.3 Scrip Dividends, Bonus Issues and Rights Issues

CFDs investors may or may not be entitled to such corporate actions. For non-cash dividends, bonus issues and rights issues, investors may not receive the entitlements and the CFDs provider may require them to close all open positions before the ex-date.

10.6.9 Comparing CFDs with Futures

Table 10.6.9 lists the differences between CFDs and futures contracts:

Table 10.6.9 – Differences between CFDs and Futures Contracts

	CFDs	Equity Futures
Trading Platform	Mostly OTC (except exchanged traded CFDs) so it leads to counterparty risk	On exchanges so no counterparty risk
Maturity & Expiry Date	Extend/Rollover for as long as investor wishes (subject to the CFDs provider's policy on corporate actions)	Fixed maturity dates when contracts expire
Financing Cost	Yes - explicitly computed & added for duration of holding period	Yes - implicit and embedded in quoted price
Dividends	Entitled	Not entitled
Margin & Leverage	Yes	Yes

10.7 Global Trading Opportunities

10.7.1 Equity CFDs

An investor can trade CFDs on thousands of individual shares listed on the major global exchanges and take long or short positions in a cost-effective manner and with absolute price transparency, as the underlying asset is known and observable.

10.7.2 Index and Sector CFDs

Besides the main domestic Singapore indices (Straits Times Index and SiMSCI), an investor can trade CFDs on numerous global and sector indices. Examples include:

a. Equity indices: USA (S&P 500, DJIA 30), UK (FTSE 100), Europe (CAC 40, DAX 30), Japan (Nikkei 225), Australia (ASX 200), Hong Kong (Hang Seng 40), China (A 50).

b. Industry sectors: Technology (Nasdaq 100), Energy, Financials, Healthcare, Utilities.

10.7.3 Currencies

CFDs trade on a wide range of currencies, including the Australian dollar, Canadian dollar, European euro, Chinese renminbi, Japanese yen, Swiss franc, UK pound and US dollar.

10.7.4 Commodities

Choices of commodities CFDs are extensive and include all the major commodity categories such as energy (with positions on US-NYMEX and UK-ICE crude oil-based contracts being highly popular), industrial metals, precious metals, and agricultural soft commodities.

10.7.5 Government Treasuries

These include government bonds including US, UK, Japan and Euro sovereign securities.

10.7.6 CFDs on Exchange-traded Funds (ETFs)

With the rapid growth of ETFs, there are CFDs available on ETFs such as SPDR Gold ETF, and many others.

10.7.7 Others

New categories include CFDs on indices which capture market volatility (VIX) and real estate. CFDs investors can possibly trade in all these markets, with one account on a single platform, depending on the services offered by the various CFDs providers.

10.8 CFDs Trading Strategies

10.8.1 Trading for Dividends (Dividend Stripping / Capture)

A long position in an equity CFD gives the investor the benefit of receiving dividends declared by the company. Conversely, the investor who has a short position, has to pay back the amount of dividends declared. An investor can profit from positions in good yielding stocks by timing the trade of the equity CFDs to his advantage.

When the share goes ex-dividend, its price is theoretically supposed to fall by the dividend amount and there is no net difference in the total value to the shareholder. However, share prices do not always behave this way. If sentiment is positive or the stock is rising due to positive momentum, the share price may not fall by the full amount of the dividend and may even get back to its pre-dividend level quickly.

An astute CFDs trader can exploit such market anomalies and make a tidy profit by capturing the share dividend. He buys the CFD before the ex-dividend date and closes out his position shortly thereafter before the share price has fully adjusted to the theoretical ex-dividend price. Such an investor will look out for companies that have good dividend payouts, monitor them, and select the right share offering opportunities for a dividend stripping trade.

10.8.2 Trading in Pairs

CFD pairs trading involves taking positions in two CFDs - one long and one short. The objective of this strategy is to exploit a perceived deviation in the underlying share prices from historical norms and the likelihood of price reversion to the normal trend in the near future.

This strategy is "market-neutral" as the direction of the overall market should not affect the investment position's overall outcome. The intention is to remove the market risk, as the long and short positions will neutralize each other. It is not necessary for each position to yield a profit. The net result will be positive even if only one leg is profitable and exceeds a potential loss on the other leg. The investor's gain is based primarily on stock selection as overall outperformance depends on the returns on the long position (undervalued) over the short position (overvalued).

A pairs trading strategy is not without risk or costs. Sometimes, anomalies between the overvalued and undervalued underlying shares may persist for long periods without converging and the arbitrage profits may not materialize quickly. The markets could also move against the investor, where one or both sides of the pairs trade may shift in the wrong direction. As a result, the loss may overwhelm the profitable leg of the trade and leave the investor with a negative return overall.

Since the investor is taking two positions – one long and the other short – the trade will involve double commissions and finance charges.

There are numerous strategies and variations possible for trading equity pairs:

10.8.2.1 Industry Sector Plays

This is a play on two companies in the same sector, one that is undervalued or has a positive outlook (long position), and the other overvalued or having a negative outlook (short position). For example, a pairs trade can be taken in Capitaland and City Developments in the property sector.

10.8.2.2 Statistical Arbitrage

This is based on quantitative analysis, where the investor identifies two companies that have a strong historical price correlation, but recent short-term market movements indicate a deviation from the trend. If the investor believes there will be "mean reversion" and the price patterns will return to the historical norm, he can execute a pairs trade with the two equity CFDs.

10.8.2.3 Merger Arbitrage

After a merger announcement, there is usually much volatility in the share prices of the target company and acquiring company. A trader can take a long position in the target company, if he expects further upside price movement, and short the acquirer's stock on the other leg of the trade.

The main risk in merger arbitrage strategies is "deal risk". This is the risk that the announced deal may fail and not be completed. This leaves the investor facing potential loss on both the long and short legs of the position, as the share price of the target company may decline to the pre-bid traded price.

10.8.2.4 Equity Indices

Selecting individual stocks for pairs trading allows an investor to seek profit while limiting overall risk but finding good candidates for this strategy may not be easy and can be time consuming. Furthermore, the selected stocks may not be very liquid. Another way to play the equity market is through equity indices. The indices can be easier to monitor for trends and price movements while offering deeper market liquidity to the trader.

For example, an investor may be looking at the Standard & Poor's 500 (S&P 500) and Nikkei 225 equity indices, which capture the performance of the US and Japanese equity markets respectively. Data analysis over a 5-year period indicates a high degree of correlation in the daily and weekly changes between the two, which

have been relatively consistent. Recent trading patterns have diverged from historical trends due to technical factors, but the two markets are expected to converge again. Hence, index CFDs can be utilised to capitalise on this expected convergence.

10.9 Summary

1. CFDs are derivative products in which the buyer or seller of a CFD contract takes positions on asset price movements without direct ownership of the underlying asset. Upon maturity, there is a cash settlement and the seller pay the buyer the difference between the opening price and closing price of the contract (or vice versa if the market moves against the buyer).

2. The main features of CFDs are leverage, flexibility, low cost, transparency, simplicity, no expiry date, access to various asset classes, access to global markets, and cash settlement. For equity CFDs, the investor holding a long position will get cash dividends and is subjected to corporate actions but is not entitled to any voting rights.

3. There are three main business models used by CFDs providers in global markets: market-maker, direct market access (DMA) and exchange-traded. With DMA, the buy/sell order goes through the CFDs provider's platform, and the investor gets direct access to the market in which the underlying asset is traded. It is the predominant method used by CFDs providers in Singapore.

4. The CFDs providers must inform the investor at the time of establishing the CFDs account about margin requirements, margin limits and liquidation procedures. The investor must be familiar with the procedures stated in the Risk Disclosure Statement (RDS) and confirmed with a signed acknowledgment.

5. All orders are considered to be day orders unless otherwise stated. All CFDs orders are done based on the bid-ask price of the underlying asset. In Singapore, there are three types of orders for equity CFDs: Limit Order, Market-to-Limit Order and Market Order.

6. CFDs are subject to other transaction costs, such as daily financing charges, commissions and Goods and Services Tax (GST). There is also a charge imposed whenever there is a contract rollover.

7. CFDs give investors access to global trading opportunities, and to take long or short investment positions in a cost-effective manner. Not limited to domestic Singapore securities, the investor can access many equities, indices, investment classes and sectors traded worldwide on various exchanges. There are also CFDs on government securities, currencies, commodities, ETFs and alternative asset classes such as market volatility and weather futures.

8. One equity CFD trading strategy is to take positions in good yielding shares to capture dividends by timing the trades. Another is a market-neutral strategy called pairs trading. This involves taking a long and a short position to exploit market deviation in the underlying share prices, with expectations of price reversion to normal historical trends.

Chapter 11:
Key Products and Investment Risks for Derivatives and Structured Products

Learning Objectives
The candidate should be able to: ✓ Understand the risks inherent in basic derivatives products: ✓ Explain the specific types of risks inherent in futures. ✓ Explain the specific types of risks inherent in options. ✓ Explain the specific types of risks inherent in warrants. ✓ Explain the risks of investing in Contracts for Differences (CFDs). ✓ Discuss why structured products have more complex types of risks. ✓ Explain what type of trading controls and risk management processes can be implemented by financial organisations.

11.1 Investment Risks for Derivatives and Structured Products

The basic generic risks considerations for all investment products were covered in the CMFAS CM-EIP Examination. This chapter focuses on the risks related to derivatives and structured products.

11.1.1 Risks in Trading Futures

11.1.1.1 Basis Risk

The basis of a futures contract is the difference between the futures price and the cash / spot price:

Basis = Cash or Spot Price - Futures Price

Before maturity, the futures price for later delivery may differ substantially from the current spot price. If the asset to be hedged and the underlying security in the futures contract are the same, the basis will be zero on the maturity date of the contract.

If the futures contract and asset are liquidated before maturity, the hedger is confronted with basis risk as the futures price and cash-spot price do not move in perfect lockstep at all times before delivery date. In this case, gains and losses on the contract and the asset do not need to exactly offset each other.

Basis risks arise because of several factors:

a. The asset, whose price is to be hedged, may not be exactly the same as the asset underlying the futures contract;

b. The hedger may be uncertain about the exact date when the asset will be bought or sold;

c. The hedge may require the futures contract to be closed out well before its expiration date; and

d. The size and unit of measurement of futures contracts may not correspond with the size of the underlying hedged position. As a result, a position may be under-hedged or over-hedged.

11.1.1.2 Liquidity Risk

Although futures contracts are traded on exchanges, there is no assurance that liquidity will always be available. Unless an investor is willing to take delivery of the underlying asset (if the contract cannot be cash settled), it is important to bear in mind as the only way for an investor to close his futures position is to offset it with another futures contract.

Liquidity could also dry up near the maturity of the futures contract. Liquidity can also decline in a rapidly rising or falling market. For example, if the futures price has risen or fallen by the maximum allowable daily limit and there are no counterparties willing to transact, there will be no trades. In such a “lock limit” market, an investor will not be able to offset his existing position.

Some useful indicators to gauge the level of liquidity for a particular futures contract and for particular delivery months are:

a. Trading volume; and

b. Open interest (the number of open futures contracts not yet closed off by an offsetting trade or not yet delivered).

11.1.1.3 Leverage Risk

Futures are traded using margin financing and due to the high degree of leverage in futures contracts, a relatively small price movement may generate significant profits or losses through the leverage effect. The loss in percentage terms for the futures contracts will be greater than the percentage price movement in the underlying asset.

When entering into futures transactions, the investor has to put up a cash amount which is the “initial margin”. If the market moves against the investor or the margin levels are increased by the broker, the investor will have to deposit additional funds at short notice to maintain his position. If the investor fails to comply with

the request for additional funds within the specified time, the broker may liquidate the position and the investor will be liable for any resulting shortfall in the account.

In adverse market situations, the investor may suffer a total loss of the initial margin as well as face the possibility of further losses. The financial risk is not limited to the initial margin amount but there is exposure to the full downside of a fall in the value of the underlying asset. Therefore, investors may find themselves very quickly out-of-the-money if they make wrong predictions on the direction of the market.

11.1.2 Risks in Trading Options

Option holders (buyers) and option writers (sellers) assume many different types of risks.

11.1.2.1 General Risks Assumed by Option Holders

a. Option holders run the risk of losing the entire premium paid for the option in a relatively short period of time. This is because options constantly lose time value and become completely worthless at expiration if the option expires out-of-the-money. If an option holder does not sell or does not exercise his option prior to expiration, he will lose the option premium paid. This risk increases if the option is out-of-the money as expiration draws near.

b. In theory, European options are more restrictive as compared to American style options. In the absence of a secondary market, European options can only realise their value upon expiration. This lack of flexibility means it is more difficult to make profits - not only does the option holder need to correctly predict the price movement of the underlying asset, but he also needs to correctly predict this at the exact time of expiration. In practice, American options may not have an active secondary market, making their behaviour similar to European options. With a declining time value, American options are usually sold before maturity. When there is no active secondary market, option prices are determined by market players from the option intrinsic value.

c. Specific exercise provisions of a specific option contract may create additional risks. This needs to be carefully monitored especially in the case of over-the-counter (OTC) options.

11.1.2.2 General Risks Assumed by Option Writers

a. Writers of naked call or put options are exposed to unlimited losses if the underlying asset rises or drops respectively.

b. As American options may be exercised by the option holder any time before expiry, options writers have no control over the timing and magnitude of their liabilities. Even if a market is unavailable or they cannot execute a closing transaction, options writers still have obligations to fulfil under the options sold.

11.1.2.3 General Risks Assumed by Both Option Holders and Option Writers

a. Option trading can be volatile and relatively small price movements on the underlying asset could generate significant profits or losses. Therefore, investors may find themselves very quickly out of the money if they make incorrect predictions on the direction of the market.

b. As options are very versatile and can be combined to create different strategies, the complexity of some option strategies may be a source of significant risk. Similar to a stock exchange, an options

exchange has the right to halt trading of any option. When this happens, investors are prevented from realising value.

c. Investors who trade internationally traded options assume additional risks due to the differences in timing and foreign currency exposure.

11.1.3 Risks of Trading Equity Index Options

Well-known equity indices like the S&P 500, STI Index and the Nikkei 225 are widely followed by large investors. They represent some of the key global and regional equity markets, and their constituent stocks are actively traded. Other equity indices may track a small or niche group of company shares or may cover illiquid markets. Lack of liquidity in the underlying markets will expose the investor to greater risks. If the investor trades in OTC options on equity indices, he or she will also face counterparty risk.

11.1.4 Risks of Trading Bond Options

The risk exposure in trading bond options is invariably linked to the risks in bonds. Volatility in bond markets is usually quite moderate, especially when compared to commodities and foreign exchange. However, the risk for specific bonds can be significant, especially for bonds or other fixed income securities with low credit ratings. Downgrades by credit rating agencies can very quickly increase the yields required by investors and hence affecting prices.

In financial terminology, credit spread refers to the difference in yield between a corporate bond and a government bond (such as Treasuries in the U.S and Singapore Government Securities in Singapore). It is assumed that government bonds will not default and so, have no default risk. However, corporate bonds carry a risk of default, and the yield difference between a corporate bond and government bond of the same maturity is the risk premium investors require for taking on the extra credit risk inherent in the corporate bond. In reality, there have been sovereign defaults in several countries, illustrating that government bonds of specific countries may not be risk free.

Investors can monitor how credit spreads for a category of bonds change over time. By comparing historical credit spreads to current levels, investors can determine if bonds are “cheap” (wide credit spread) or “expensive” (tight credit spread) and look at the spreads to figure out where bond markets may be headed.

Interest rate changes for different types of bonds do not always move in sync. When the markets are driven by economic and political uncertainty, global investors tend to park their money in what is still perceived to be safe US Treasuries. In such a “flight to safety”, the yield of US Treasuries will drop while the credit spreads of the riskiest bonds will widen the most. Hence, in such times of uncertainty, the absolute level of the Treasury yields may fall but the other bond credit spreads can widen. Macroeconomic factors also play a role in determining sovereign bond yields and prices. A country with poor economic fundamentals may come under the scrutiny of “bond vigilantes’” as demonstrated by how global investors pushed down prices of bonds from Greece, Spain and Portugal during the Eurozone debt crisis in 2010.

Besides credit risk, other factors which contribute to the credit spread premium include market liquidity, embedded options in bonds (i.e., embedded call or put options), and terms and conditions of individual bonds, such as change of control and event risk.

11.1.5 Risks of Trading Currency Options

Trading in currency / foreign exchange option is risky as the notional value involved is extremely large. Furthermore, as the foreign exchange market is open round the clock, every day except weekends, traders need to be able to monitor their positions or risk being caught out by fast moving events. Managing counterparty risk is especially important if one has a position in an OTC currency option, as options contracts are fulfilled by actual delivery of the currencies.

11.1.6 Risks of Trading Warrants

There are two types of warrants, company issued warrants and structured warrants issued by financial institutions, typically banks.

A company warrant is essentially a long-term option issued by a company. If the company share price does not reach the strike level before the warrant expires, the investor will not exercise the warrant and its value would drop to zero. A warrant holder does not have any voting or dividend rights. If the warrant is in-the-money and is exercised, investors will have to accept that their shareholdings may be diluted.

Structured warrants are issued based on various underlying instruments such as single stocks, equity indices, investment funds and commodities. As such, the investor is exposed to the specific market and investment risks of the underlying instruments. These warrants are leveraged products which offer benefits of gearing. In adverse market conditions, a fall in the underlying share price may lead to a larger percentage loss in the value of the warrants. As structured warrants are issued by third party banks, the investor has exposure to issuer risk and the credit standing of the issuing bank.

11.1.7 Risks of Investing in Contracts for Differences (CFDs)

A CFD is a leveraged product as it is traded on margin. The initial margin is normally a percentage of the contract value and is used as security for a much larger exposure to the underlying assets (for example, 10% on an underlying investment of $10,000 means that the initial margin is $1,000).

11.1.7.1 Leverage Risk

With the margin set up in CFDs, a small change in the market price for the underlying asset can have a significant impact on the investor's trading returns, as the gains or losses on a leveraged product are magnified.

Based on daily mark-to-market and market conditions, the CFDs provider may have to make a "margin call" and request that the investor put up more money at short notice. If the investor is unable to meet the margin call, the position may be forced sold.

Excessive price movements may result in the loss of the initial margin and also expose the investor to further losses if the value of the underlying asset falls significantly (conversely, if the price rises significantly and the investor has a short CFD position).

11.1.7.2 Market Volatility (Underlying Asset)

An investor takes a long CFD position on an underlying asset with the intention of making a profit when the price of the asset rises. An investor with a negative view of the market may go short and sell a CFD on an

underlying asset, expecting the price to fall. As markets are uncertain, it can move against the investor's expectations and lead to losses.

As a CFD is linked to an underlying asset, such as a company share, equity index or foreign exchange rate, the value of the CFD investment is affected by market conditions affecting the underlying asset. If the underlying asset is illiquid or suspended from trading, it will be more difficult to trade the CFD, or to close out the CFD if the investor wishes to liquidate.

Closing a CFD position while the underlying asset is subject to a corporate action may result in delays to the investor receiving any proceeds, making a credit entry to his account or even the possibility of not being able to get any price quotes to close out his position.

11.1.7.3 Counterparty Risk

Counterparty risk is the exposure one faces when dealing with another entity (counterparty) in a transaction. When buying or selling a CFD, the only asset the investor trades is a contract issued by the CFDs provider. Hence, the CFDs provider becomes the counterparty. In the event the CFDs provider is unable or unwilling to meet its contractual obligation due to financial or other difficulties, this will have an impact on the investor. This risk is existent in almost all CFDs with the exception of exchange traded CFDs.

11.1.7.4 Liquidity Risk

An investor must deal with liquidity risk when trading in any market. If there are not enough trades being made in the market for an underlying asset due to a lack of liquidity, the investor may be unable to trade CFDs on that asset.

The CFDs provider may either decline to fill the investor's trade, or only agree to process the trade at an inferior price even if he already has an open CFD position on that underlying asset. This may leave the investor with an open CFD position that he is unable to close.

For international assets, changes in the economic, political and regulatory environments in a country or region can affect markets. Demand and supply for equities and commodities may rise or fall suddenly which makes the investor's position highly volatile.

11.1.7.5 Currency Risk

Transactions in a foreign market or in a foreign currency-denominated instrument that is different from the investor's base currency may give rise to foreign exchange risk. The exchange rates of foreign currencies fluctuate and will impact the investment performance by either reducing actual profits or worsening losses.

The underlying asset of a CFD (e.g., shares) may be traded in markets outside Singapore and this may expose the investor to additional risks. These markets may not have the same regulatory framework as Singapore and may not protect investors' interests to the same degree. Foreign markets also carry the risk of foreign exchange controls.

11.1.7.6 Financing Costs

When evaluating CFD investments and computing final returns, the investor must take into account all related costs involved: brokerage commissions, account maintenance fees and financing costs. As CFDs are leveraged investments, the CFDs provider will charge the investor interest for providing him with financing. The industry practice is for interest to be charged on the full amount of the value of the CFD position, computed on a daily

basis and based on a spread over LIBOR or another market benchmark, as disclosed and agreed to between the CFDs provider and investor.

The reference market rate is a floating rate, and the client is exposed to interest rate movements. If interest rates rise, the financing cost for the CFD investor goes up. This raises the total holding cost of his investment position and erodes the eventual investment returns. The CFD investor must take into account the impact of interest rate volatility on his CFD investments.

11.1.7.7 Risks Associated with Different CFDs Models (Direct Market Access vs Market-Maker)

In a Direct Market Access (DMA) model, the pricing is transparent as the CFDs trade is based on the underlying assets' market price, which is directly observable. In a market-maker model, liquidity and efficient pricing will depend on the CFDs provider's ability to make a market by ensuring adequate supply to meet investor demand, and to keep a tight bid-ask spread so that CFDs investors can trade in a cost-efficient manner. However, discretion on the bid-ask spread lies with the CFDs provider and is not transparent to investors.

11.2 Key Risks of Structured Products

As a structured product involves the combination of two or more underlying financial instruments, the risk profile is usually more complicated than a standard financial product. It is critical to find out what risks the structured product is exposed to, and how these risks could affect the investor's overall risk-return profile.

11.2.1 Market Risk

Market risk is essentially systemic risk which affects the entire market. The main sources of market risk are changes in macroeconomic factors such as interest rate, inflation rate, employment rate and recession, and microeconomic drivers such as corporate earnings, which can affect company share prices. Market risk is also affected by political factors, such as government instability and policy uncertainty.

In "Zero Plus" option based structured products, and other variations which have embedded options, the products are exposed to the market risk of the underlying derivatives which drive the returns component. Suppose an investor seeking to enhance yield using an aggressive structure buys a zero-coupon bond linked to the performance of a securities index. The investor buys a normal zero-coupon bond and shorts a securities index option. The return component from the investors' perspective is the fixed option premium received for selling an index option, which is essentially selling the protection against the market risk of the securities index.

If the protection the investor sold is against the increase of securities index price, then the investor is selling a call option on the index. If the protection is against the decrease of index price, then the investor is selling a put option on the index. If the option the investor sold becomes "in-the-money", the investor will be required to pay out the "in-the-money" amount to the option buyer. So the investor, being the option seller, is taking a view that the option will stay out-of-the-money so that it will not be exercised against them.

The following examples illustrate the market risk impact of various products.

11.2.1.1 Structured Product with Shorting of Call Option on Securities Index (selling protection against the increase of a securities index)

Figure 11.2.1.1 - Payoff of a Call Option on Expiration

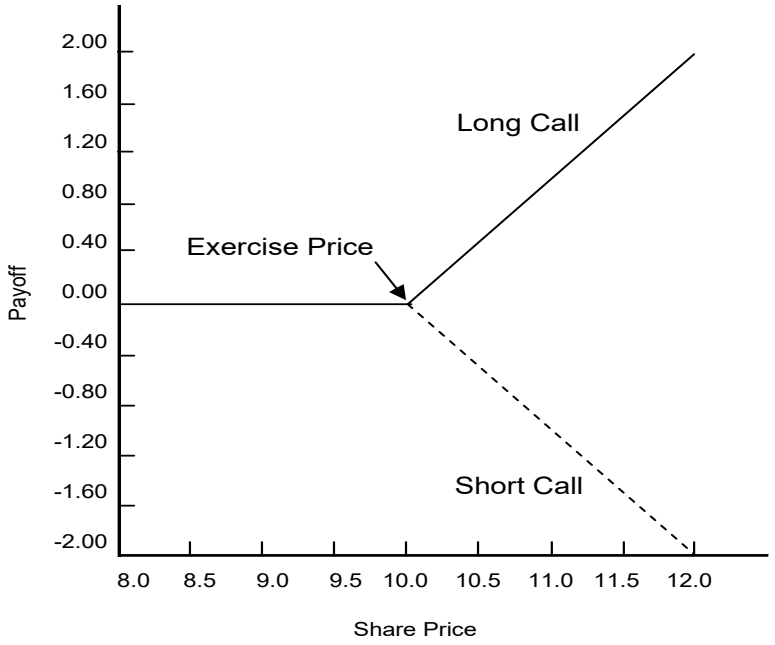


Figure 11.2.1.1 illustrates the payoff for a call option.

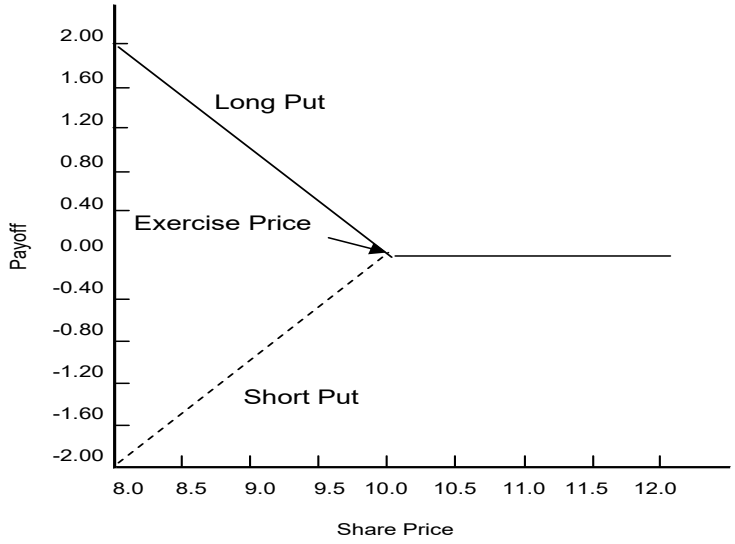
For the investors of the structured products who are effectively selling a call option (bottom half of diagram) on an underlying asset (i.e., selling a protection against the increase in price of the underlying asset), they are required to pay the option buyer as the price of the underlying increases beyond the strike or exercise price. The loss amount is in proportion to the difference between the underlying price share price and the exercise price. As the upside value of the underlying asset is theoretically unlimited, the call writer could potentially be faced with unlimited losses.

For short calls, the loss is mitigated when the investor has a covered call position, or a call spread strategy. For covered calls, the investor holds the requisite quantity of the underlying asset which he can deliver when the option is exercised. For call spreads, the maximum loss will be the difference in the strike prices of the long and short calls in the spread, multiplied by the number of the option contracts in the spread strategy.

11.2.1.2 Structured Product with Shorting of Put Option on Securities Index (selling protection against the decline of a securities index)

For a structured product shorting a put option on a particular securities index, the option will be in-the-money when value of the index decreases. When a market event occurs, such as a recession or a major financial scandal, it is likely that the value of the index will fall. The put option will be in-the-money and the investor of the structured product, being the put option seller, will have to pay out to the option buyer, resulting in a reduction in his investment return (see Figure 11.2.1.2)

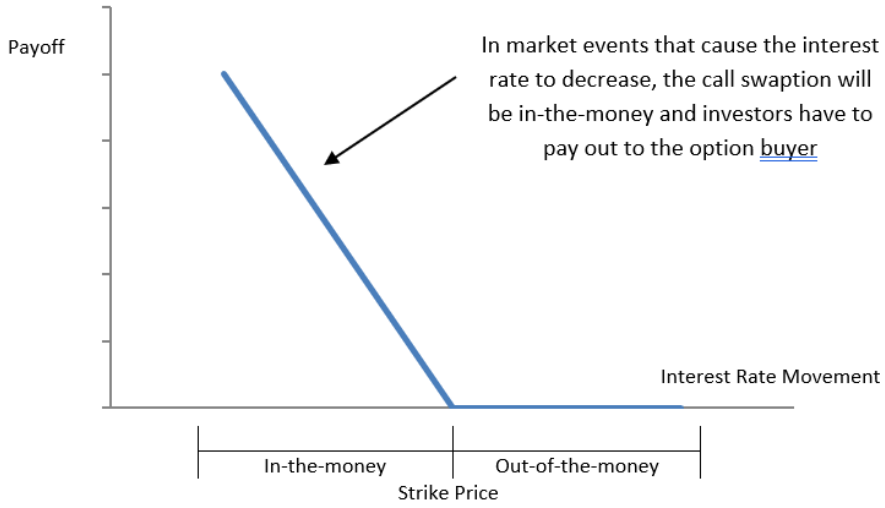
Figure 11.2.1.2 - Payoff of a Put Option on Expiration



11.2.1.3 Structured Product with Shorting of Interest Rate Call Swaption

Figure 11.2.1.3 involves a receive-fixed swap option (protection against interest rate decrease). A receive-fixed swap option is also known as a call swap option. The seller of an interest rate call swap option (commonly known as a swaption) will have to pay out a fixed rate and receive a floating rate when the option is exercised by the buyer of the swap.

Figure 11.2.1.3 - Interest Rate Call Swaption Payoff Diagram for Option Buyer

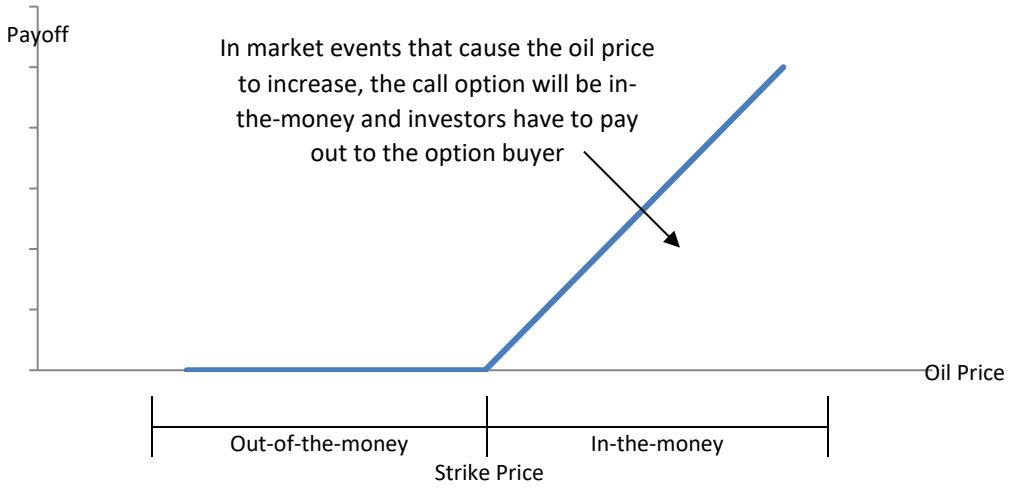


If market events result in a decrease in interest rates below the strike, the swaption payoff diagram will be in-the-money for the option buyer as he will receive a pre-determined fixed rate and pay a declining floating rate. The investors of the structured product, being the option seller, will have to pay out to the option buyer. As such, they are effectively selling a protection against a decrease in interest rates.

11.2.1.4 Structured Product with Shorting of Call Option on Commodities Index (selling protection against increase in prices of the commodities)

In a structured product with shorting of a call option on a commodity index (such as the S&P/GS Commodity Index), the value of the option will be greatly affected by the movement in the prices of the underlying commodities (see Figure 11.2.1.4).

Figure 11.2.1.4 - Call Option Payoff Diagrams for Option Buyer



In the event of political instability or military actions (such as conflict in the Middle East), oil supplies may be disrupted, their prices will increase, and the call option will be in-the-money. Investors of the structured product, being the call option seller, will have to pay out to the option buyers.

As the above examples have illustrated, the market risk profiles of various structured products are similar regardless of the underlying instruments. The specific market risk depends on the respective underlying instruments, particularly options.

11.2.2 Risks of Constant Proportion Portfolio Insurance (CPPI) Strategy Products

In the CPPI strategy asset allocation process, there is greater allocation made to the risky asset class as its value appreciates, and vice-versa. CPPI may result in a situation where the investor ends up buying high and low in a range bound market.

In market situations such as a prolonged period of a range bound prices, after the occurrence of a major deleveraging event, and when there has been a sharp drop in asset prices, there is a higher chance that the portfolio value may drop to the floor value. When this occurs, it would force the manager to allocate the entire fund into the risk-free asset. Once that happens, there shall no longer be any participation for any subsequent appreciation in the value of the underlying asset.

It is important to highlight to the investor that with a CPPI such a strategy assumes the following characteristics of the underlying asset:

a. Asset that appreciates in value over time;
b. Low volatility; and
c. Limited drawdown as measured by percentage decline from peak to trough.

11.2.3 Credit and Counterparty Risks

A structured product is exposed to the credit risk of the issuer. This would include special purpose vehicles (SPVs), which are separate legal entities usually created by banks to manage the issuance of the securities. If these separate entities are not guaranteed by the parent, investors will have no recourse if the issuer defaults. Investors should check and know exactly the credit rating and standing (e.g., credit outlook, credit watch and guarantee status) of the issuer before investing in the product.

Where there is a swap agreement between the issuer and another institution, the investor should understand who bears the risks. In some swap structures, the investor may bear the credit risk of both the issuer and the swap counterparty. For notes issued by SPVs and involving credit default swaps (CDS), it is important to be sure about the credit quality of the pool of underlying debt securities and reference entities linked to the SPVs.

11.2.4 Types of Credit Default

An investor of a structured product is exposed to the credit risk of the issuer and the risk involved when the underlying financial instruments involve a credit product. In the event of a credit default with the underlying, the borrower is unlikely to repay the loan or fulfil the contractual obligations on the credit product. This would in turn impact the value of the structured product adversely.

The International Swaps and Derivatives Association (ISDA)[46] classifies credit default as follows:

a. Bankruptcy;
b. Obligation Acceleration ;
c. Obligation Default ;
d. Failure to Pay;
e. Repudiation/Moratorium;
f. Restructuring.

As markets evolve, investors, banks and regulators have to adapt to the new realities. In 2014, the ISDA implemented new Credit Derivatives Definitions which constituted a major reform of the terms governing the treatment of credit derivatives to address the issues these products have been facing for the past decades.

Example of a Credit Event

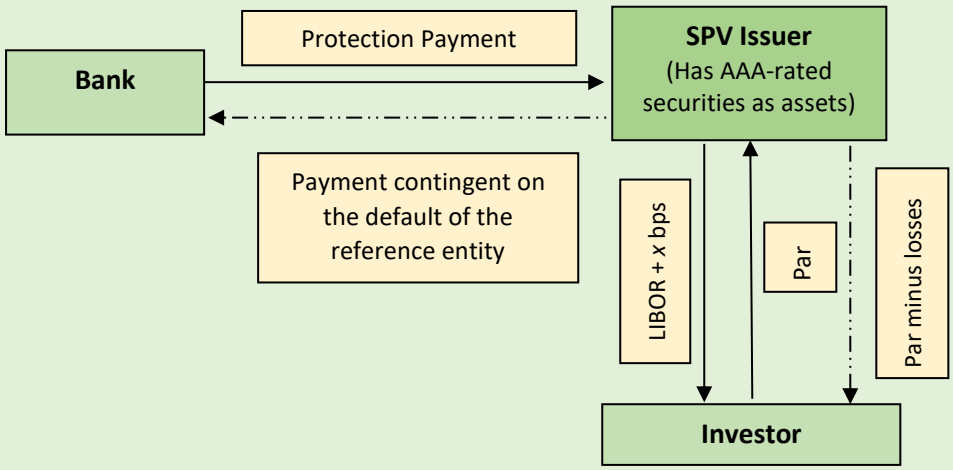
An investor with a relatively conservative investment approach buys a structured product comprising a zero–coupon bond with a long position in an option. This approach has very low market risk compared to structures which has an embedded short position in options. The underlying fixed income instrument (bond) of a structured product is subject to credit risk. There is a possibility that the bond may default, resulting in the structured product being unable to redeem the underlying bond or fulfil the coupon payments to the investor.

[46] International Swaps and Derivatives Association (ISDA); www.isda.org

Example – Credit Derivatives

An investor of a structured product enters into a CDS that guarantees the credit worthiness of a reference entity. In the event of a default situation of the reference entity, the investor is required to pay out to the issuer since the investor had sold credit protection on the reference entity. Figure 11.2.4 shows the cash flows involved in the CDS transaction.

Figure 11.2.4 - Cash Flows Involved in a CDS Transaction



The investor buys notes issued by a Special Purpose Vehicle (SPV). The SPV uses the proceeds to buy AAA-rated securities. The investor receives enhanced interest (LIBOR + *x* bps). He is selling credit protection to the SPV issuer. The SPV in turn sells credit protection to the bank (by entering into a CDS) and receives protection payment. This payment received by the issuer will be used to subsidise the enhanced interest paid to the investor. If the reference entity defaults, the SPV issuer, being the protection seller to the bank, has to liquidate the AAA-rated securities and pay the bank. The investor, being the protection seller to the issuer, pays the contingent payment to the SPV and recovers the remaining sum (par minus losses).

11.2.5 Liquidity Risks and Secondary Market Trading

Liquidity risk is a result of the lack of marketability. Structured products are usually designed for investors who are willing to hold the investment to maturity. As they are customised products tied to a specific risk / reward profile, there is typically a limited secondary market making it hard for investors to sell it before maturity date.

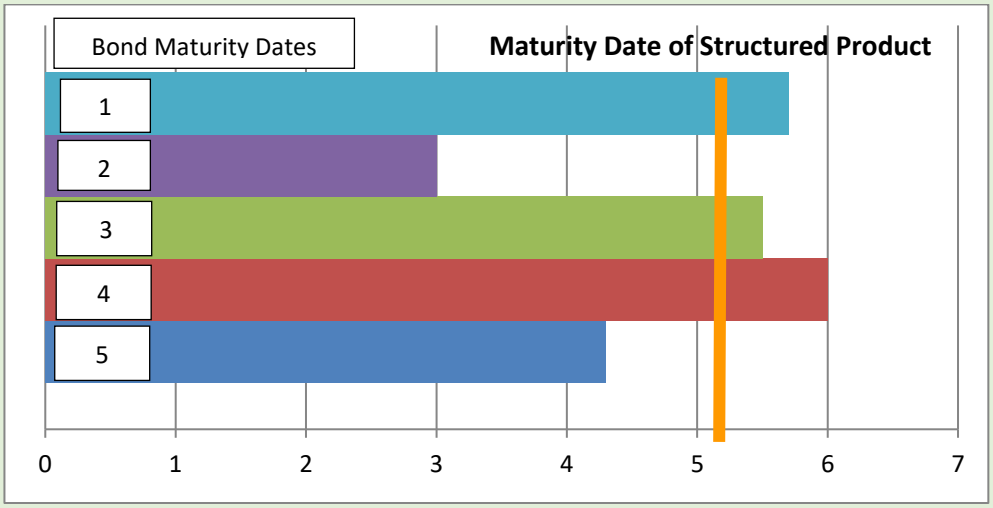
Investors should be aware if a lock-up period exists in the structured product. A lock-up requires that the investor's funds cannot be pre-maturely withdrawn until the expiry of the lock-up date, which could range from a few months to a few years. Another reason why structured products is illiquid is that the underlying derivatives components such as options (especially exotic options) and CDS may not be actively traded. Investors may also lose some part or even a substantial part of the principal sum if they were to liquidate the investment before maturity.

The issuer or its affiliates usually provide a secondary trading market for the structured product after issuance, even though they may not be obliged to do so. This is important and clients should be advised to assess the issuing bank for its quality, commitment, as well as its track record of providing liquidity for secondary market trading. In spite of the bank's commitment to provide secondary trading, investors should be aware that the structured product's price ultimately depends on prevailing market conditions and levels/prices of the

underlying instruments, such as yield, terms of maturity and the mark-to-market values of any underlying derivatives.

Another example is when there is a mismatch of maturity date between the structured product and the underlying assets, for example, a basket of high-yield bonds. In this case, the high-yield bonds with maturity longer than the structured product have to be sold when the structured product matures. When that happens, the structured product is subject to liquidity risk as a buyer for the bond may not be readily available. In adverse conditions, the bond could be sold at a huge discount, resulting in a reduction to the investment returns.

Figure 11.2.5 - Comparing Bond Maturity Dates with the Maturity Date of a Structured Product



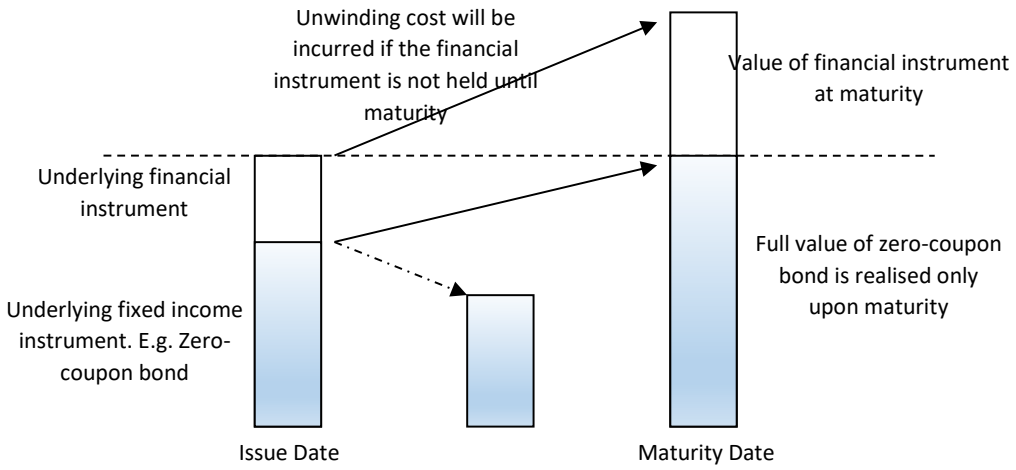
In Figure 11.2.5, Bonds 1, 3 and 4 will have to be sold in the market when the structured product matures. As such, these bonds will be exposed to liquidity risk.

Liquidity risk may however be mitigated if the products used are widely traded or are exchange-traded instruments. For example, if the S&P 500 index futures contract is used in the return component of a structured product, buying or selling this contract on maturity date would not be a problem given its high liquidity.

11.2.6 Early Termination Risk

Most structured products are designed such that the full extent of the investment returns will only be realised upon maturity. Using a zero-coupon bond for some principal preservation as an example, the principal preservation feature will only be realised upon maturity. If there is early withdrawal, the bond might be sold at a discount and hence the value of the structured product will be adversely affected. Similarly, with the underlying derivatives, in the event of an early withdrawal, there is a risk of incurring unwinding costs of the embedded derivatives contract. Figure 11.2.6 illustrates the early termination risk if a bond is sold before maturity.

Figure 11.2.6 - Early Termination Risk



If the bond is sold before maturity date, it is likely to be sold at a discount

11.2.7 Structure Risk

With numerous possible combinations of underlying financial instruments, a structured product has no fixed or standardized form. As such, the benefits and liabilities of the structured product are highly dependent on how each product is structured. For example, in structured products that use financial instruments that do not limit losses, the entire amount invested could be lost.

Figure 11.2.7(a) below illustrates interest rate swap options (commonly referred to as **swaptions**). If the structured product involves shorting an interest rate call swaption, investors of the structured product are liable to pay out a fixed rate when the option is exercised (i.e., selling protection against decrease in interest rates) while receiving the floating rate in the swap. In normal cases, the loss to the swaption seller is limited to the fixed rate when the floating rate declines to zero when the option is exercised by the swaption buyer. In rare market conditions where negative interest rates occur, the investor may have paid the floating interest leg of the swaption as well.

Figure 11.2.7(a) - Receive-Fixed Interest Rate Call Swaption

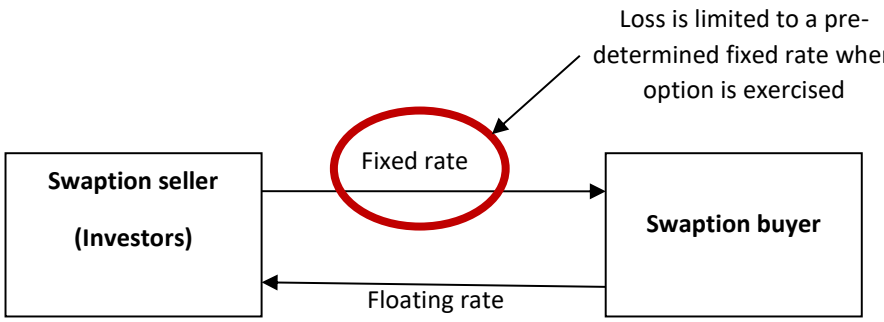
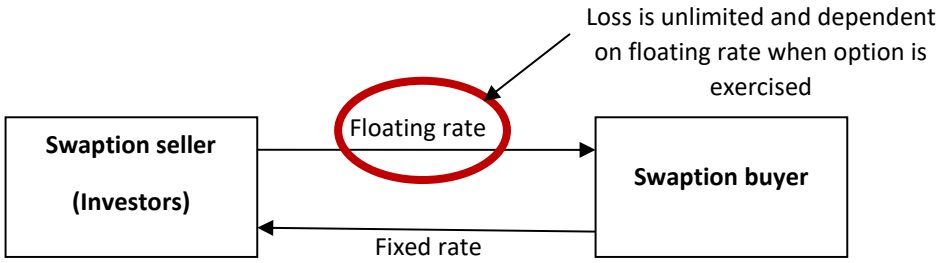


Figure 11.2.7(b) shows a pay-fixed swap option (protection against interest rate increase). If the structured product involves shorting an interest rate put swaption, investors are liable to pay out a floating rate when the option is exercised, as shown below. The pay-fixed swaption buyer will exercise the swaption when the market rate is higher that the strike rate. The losses to the swaption seller are unlimited in this case and dependent on how high the floating rate is when the option is exercised.

Figure 11.2.7(b) - Pay-Fixed Interest Rate Swaption

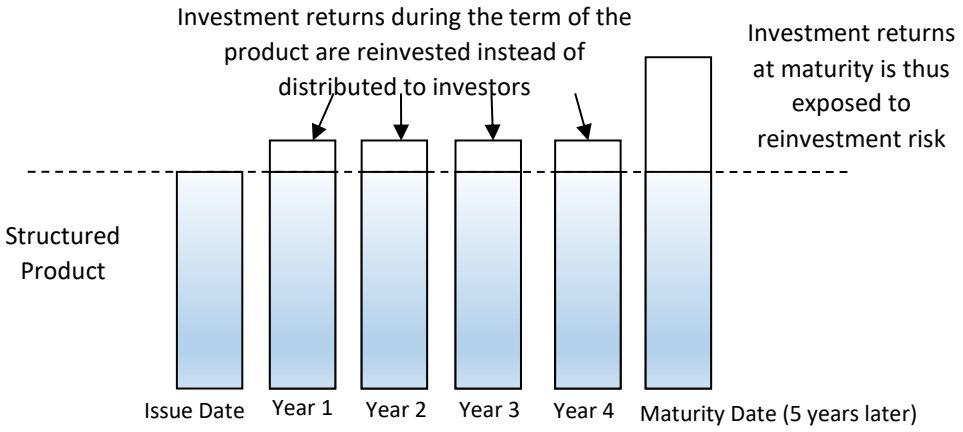


11.2.8 Reinvestment Risk

Reinvestment risk refers to the risk that investors have to reinvest their bond proceeds – coupons and/or principal – at an interest rate below their yield-to-maturity (YTM). If interest rates fall, bond proceeds can only be reinvested at the prevailing *lower* interest rates. Hence, the bond return will be lower than the YTM.

Reinvestment risk is greatest for bonds with large coupons, higher coupon payment frequency, long maturities, selling at a premium. Zero-coupon bonds have no reinvestment risk, while callable bonds have high reinvestment risk (see Figure 11.2.8).

Figure 11.2.8 - Reinvestment Risk



11.2.9 Volatility Risk

This applies to bonds and instruments with embedded options, such as callable bonds, putable bonds and convertible bonds. Their prices are affected not just by interest rates but also by changes in volatility. For example, the value of a call option increases when interest rate volatility increases. This will reduce the value of the callable bond because the bond holders will have sold a call option component to the issuer.

11.2.10 Correlation Risk

Correlation risk refers to the likelihood of an event of any kind having a direct impact on another. A structured product is exposed to correlation risk when it involves underlying instruments that are correlated with each other.

For example, a structured product has a CDS for a basket of companies that are highly correlated to one another. Some scenarios that can cause high correlation include companies in the same industry (such as banking and financial services) because they have the same customers, or the companies which have credit relationships with one another. Such structured products have high correlation risk because when one of the companies defaults, it is likely that the other companies are also affected.

11.2.11 Foreign Exchange Risk

In many cases, some of the underlying financial instruments purchased are not denominated in the same currency as the structured product. The investment return of the structured product will depend on changes in the exchange rates for the various transactions that were carried out.

11.2.12 Legal Risk

The way which legal contracts relating to the financial instruments are drafted also has great impact on the investment. For example, if the issuer defaults, the investor does not automatically have ownership or legal rights to the underlying assets. In most structured products, investors only own units of the structured products and not the underlying assets. However, depending on how the product is structured and the agreement is drafted, investors may have the first right to the instruments before other creditors of the issuer. Investors would suffer losses in situations where other creditors of the issuer have priority over the underlying instruments.

11.2.13 Transactional Risk

This refers to the risk resulting from the time difference between the commencement of a financial instrument contract and settlement of the contract. This is especially so when there is a difference in currencies between structured products and the financial instruments that it will enter into. After the commencement of a financial instrument contract, the pricing, terms and conditions are usually fixed. Investors of the structured product may have to bear any price fluctuations that occur between the contract commencement date and the settlement date.

11.2.14 Risk of Mis-selling (Incongruence to Investment Strategy)

Most structured products are created based on specific investment objectives. However, when the structured products are formed with underlying financial instruments, the purpose of the structured product may not be obvious especially when the product is sold to investors after the issue date.

The product's risks may not be adequately disclosed by the issuer. There is a potential for investors to misunderstand the investment strategy and hence, not achieve their investment objectives. Therefore, the product highlights sheet, prospectus and private placement memorandum are important documents to ensure proper risk disclosure. Distributors of structured products and financial advisers play an important role in guiding and educating investors on the risks involved in the product.

For example, investors who want a fixed income product with "100% principal preservation" may misconstrue a structured product with underlying bonds used as collaterals in a CDS as a suitable investment vehicle. However, they may not be aware that the risk is vastly different, and their investment objectives may not be met in a default situation when the collaterals are used for compensation against the CDS.

11.3 Trading Risk Control and Management

There are three main types of risks involved in trading:

a. Market risks,
b. Credit risks, and
c. Settlement risks.

The personnel in the front, middle and back office must be diligent to proactively monitor and manage these risks. Depending on whether futures or options are involved, the risk metrics used are different.

11.3.1 Managing Market Risk for Futures

Market risk is the probability of loss faced by any investment which could result from changes in market variables such as interest rates, foreign exchange rates, asset prices or commodity prices.

11.3.1.1 Setting Limits

For futures, several types of limits can be set to measure and control the level of market risk exposure.

a. **Open Contracts Limit**

This is determined after studying the contract's volatility and estimating the expected maximum loss. More volatile contracts are subject to a smaller open contract limit. Different open contracts limit can be set for outright and spread trades. The total overall open contracts that can be held by one firm or entity should be limited to a certain percentage of the Gross Open Interest[47] of each contract in an exchange to avoid liquidity problems. Some exchanges may also impose limits on speculative positions by one entity.

b. **Maximum Loss Limit**

Maximum loss limit is set for each trader/trading group in terms of weekly and monthly maximum losses. The trading position will have to be squared off if the loss limit is hit. The trader who has triggered the maximum loss limit will be suspended from trading until further approval is given.

c. **Maturity Limit**

Maturity limit is set for trading purposes to limit losses from poor liquidity. The reason for doing so is because liquidity is normally better for the near-month contracts and thins as we move out to the further months. As further contracts normally have lower liquidity, they are more volatile and in times of adverse conditions, these contracts may be difficult to unwind. Putting a limit on maturity can reduce the risks of such an eventuality.

d. **Stress Test Limit**

A limit is set based on the risk tolerance level for the investment portfolio or trading activity. The test should incorporate all the relevant inputs which impact the various underlying asset classes of the futures that are traded. The decision on the level of acceptable risk will be based on the statistical results generated for various market scenarios and conditions of duress.

[47] Gross Open Interest is defined as the total number of contracts that have not been offset by opposite futures transactions.

11.3.1.2 Duration and Modified Duration for fixed income instruments

Fixed income instruments and related derivatives are exposed to interest rate risk. When dealing with fixed income investments, investors must understand some key concepts:

a. **Basis Point (bps)** - A basis point is 1/100th of a percentage point. Yield differences between bonds are quoted in bps. For example: Bond A's yield is 3.21% while the Bond B's yield is 3.50%. The yield difference between the two bonds is 29 bps.

b. **DV01 (dollar value per basis point)** - This is the dollar change in price with respect to the yield of a fixed-income security or a swap. It is also known as dollar duration, PV01 (present value of a 01), or PVBP (price value of a basis point).

c. **Duration** – This is the weighted average time to maturity of a bond in present value terms. It considers the size and timing of cash flows over the bond's life. The earlier the investor recoups his investment in the bond, the shorter the bond's duration.

d. **Modified duration** – This is a related to (c) and measures the bond price's sensitivity to changes in interest rates. A higher modified duration implies that a security is more interest rate sensitive. It reflects the greater risks of longer-dated exposures compared to shorter-dated ones. Modified duration increases with:

   i. Increase in time to maturity;
   ii. Fall in coupon rates;
   iii. Fall in bond yields.

For fixed income securities, DV01 (dollar value per basis point) is a better measure for interest rate risk as it reflects the dollar, rather than the percentage change, in price with respect to the yield of a fixed-income security or a swap. The objective of hedging a fixed income position with futures contracts is to insure that as the underlying security loses value, the futures hedge compensates for this loss by gaining a comparable amount.

Changes in Bond Price

$\Delta P/P = - D \times \Delta YTM$
Where - D = Modified Duration

A bond has a modified duration of 2. The bond yield changes by 100 basis points. How much will the price of the bond change?

$\Delta P/P = - D \times \Delta YTM$
$\Delta P/P = - 2 \times 100$
$= - 200$ bps (or - 2%)

The price of the bond will fall by 2%.

Dollar Value Per Basis Point (DV01) or Price Value of a Basis Point (PVBP)

DV01 or PVBP = Price change that occurs if a bond's yield changes by 1 bps.

Suppose Bond X is currently priced at $980 and its PVBP = $1.
If the yield of the bond increases by 5bps, what is the price now?

Since PVBP = $1, the bond price will fall by 5bps x $1 = $5
The new price is: $980 - $5 = $975 (price & yield move inversely).

11.3.1.3 Value at Risk (VaR)

VaR is widely used for risk measurement on a specific portfolio of financial assets. The measurement incorporates volatility and correlation between market factors into the risk assessment to arrive at a potential loss amount or value at risk. As mentioned previously, volatility is the most common risk measure and volatility increases in falling markets and declines in rising markets. An asset can also become volatile because its price suddenly jumps higher. Investors are not distressed by gains but are very concerned with the risk of losing money.

By assuming investors care about the odds of a really big loss, VaR answers the question, "What is my worst-case scenario?" or "How much could I lose in a really bad month?" It measures the threshold value such that the probability that the mark-to-market loss on the portfolio over the given time horizon exceeds this value.

A VaR statistic has three components:

a. A relatively high level of confidence (typically either 95% or 99%);

b. A time period (a day, a month or a year); and

c. An estimate of investment loss (expressed either in dollar or percentage terms).

For example, if a portfolio has a 1-day 95% VaR of SGD 1 million, it means that there is a 5% probability that the portfolio will fall in value by more than SGD 1 million over a 1-day period. In other words, a loss of SGD 1 million or more on this portfolio is expected on 1 day out of 20 days.

Hence, market participants can specify an acceptable level of VaR by calculating the maximum loss expected on an investment, over a given time period and given a specified degree of confidence. By setting a lower VaR, the portfolio size has to be lowered and/or highly correlated risks have to be reduced.

11.3.2 Managing Market Risk for Options

Options carry multiple risk parameters, so precise risk control requires specifying and setting limits for the various dimensions measured by different option Greeks - Delta, Gamma, Vega, Theta and Rho.

11.3.2.1 Delta and Gamma

Delta (price reaction of the option against a change in the underlying asset price) limits are normally set in terms of USD or the local currency amount. Gamma (change in delta or the second order price derivative) is usually measured against a specified movement in the spot price (such as one standard deviation). Gamma can be very volatile when the option is trading near the strike price and when it is close to maturity. A portfolio has negative Gamma when it is net short options. Because of its sensitivity to the option strike price and maturity, Gamma limits are typically set for different tenors.

11.3.2.2 Vega

Vega is a major trading risk in options. It is measured by the change in value of an option, i.e., the premium over a 1% change in volatility of the underlying. It is represented by the standard deviation number in the

options pricing formula. Limits are placed in terms of the maximum loss that would be tolerated given certain volatility movements in either direction.

11.3.2.3 Theta

Theta measures the potential loss through time decay. This is limited by potential loss over a specified period, such as a 1-day period.

Gamma, Vega and Theta are sometimes controlled together by setting the maximum loss for all three combined. For short option positions, the positive effects of Theta are automatically offset against the negative effects of Gamma.

11.3.2.4 Rho

Rho measures the impact of a change in interest rates on the option price. Limits for Rho can be set in terms of delta amounts in different periods through the portfolio or by reference to expected losses on movement of a currency interest rate. Executing a swap transaction (such as floating for fixed rate) can reduce rho risks.

There are other market risk limits that can be implemented to control risk, such as:

a. Maturity concentration control;
b. Total book size;
c. Premium payment control;
d. Intra-day control.

11.3.3 Managing Credit Risk for Options

11.3.3.1 Counterparty Risk

For OTC options, the counterparties' credit risks are greater and are predefined for individual counterparties. The limits are defined according to the credit rating of the institution in terms of settlement amount and contract tenor.

The parties normally sign a legal document, called the Credit Support Annex (CSA), which defines the terms under which collateral is posted or transferred between swap counterparties, to mitigate the credit risk arising from the derivative positions.

There can also be premium payment limits imposed to avoid financial institutions from constructing a synthetic loan through options.

11.3.3.2 Clearing Broker Risk

As margins must be maintained with the clearing broker, the credit risk arises from the placement of such funds with the broker.

11.3.3.3 Exchange Risk

An exchange and its associated clearing house are the counterparty to any exchange-traded futures contract. Although the clearing house has committed lines of credit from commercial banks, its own capitalization, and mutual liability arrangements among the clearing members, it nevertheless has to make good any contractual demands even if its counterparties fail.

11.3.4 Managing Settlement Risk for Options

Control of settlement risk is primarily the responsibility of the financial institution's back office. Timely and accurate confirmations of trades are essential to help ensure there is prompt settlement of transactions. For OTC options, confirmation should include details like the style of option, expiry date and time, premium payment date, settlement date and strike price. For currency options, details such as the call currency and amount, and the put currency and amount, should be included.

11.4 Summary

1. The risks in trading futures are **basis risk, liquidity risk** and **leverage risk**. Basis risks can arise because the hedge instrument is not exactly the same as the asset underlying the futures contract, the exact date when the asset will be bought/sold is uncertain, the hedge contract is closed before its expiration date, or the size of the futures contract does not correspond with the underlying hedged position, resulting in a position that is under-hedged or over-hedged.

2. The risks in trading CFDs include leverage, market volatility of the underlying asset, counterparty risk, liquidity risk, deal risk, currency risk and financing cost.

3. Liquidity risk arises when it is difficult to trade futures contracts on the exchange. Liquidity could dry up in a rapidly rising or falling market when a particular futures contract has risen/fallen by the maximum allowable daily limit and there are no counterparties willing to transact. Useful indicators to gauge the level of liquidity are **trading volume** and **open interest**.

4. Futures contracts are traded using margins and this gives rise to **leverage risk.** A relatively small price movement may generate significant profits or losses and investors may find themselves very quickly out of the money if they make wrong bets on the direction of the market.

5. A major risk faced by both option holders (buyers) and option writers (sellers) is the option price **volatility** i.e., relatively small price movements on the underlying asset could generate significant losses very quickly. The complexity of some option strategies may also be a source of risk. If an exchange halt trading of any option, investors cannot liquidate their positions. International investors assume additional risks due to the differences in timing and foreign currency exposure.

6. Option holders run the risk of losing the entire premium paid for the option in a relatively short period of time. European option holders are restricted as they can only realise their value upon expiration. For OTC options, specific exercise provisions of an option contract may create additional risks.

7. Option writers of naked calls or puts options are exposed to unlimited losses if the underlying asset rises or drops respectively. For American options, option writers effectively lose control over the timing and magnitude of their liabilities.

8. Some equity indices may track only a very small or niche group of company stocks or may cover illiquid markets. Trading options in such equity indices may be very volatile and expose the investor to **counterparty default risk if they are traded in the OTC market.**

9. The risk exposure in trading bond options is linked to the risks inherent in bonds. Volatility in bond markets is usually quite moderate in comparison to asset classes but the risk related to specific bonds can

be significant, especially to lower rated corporate bonds. Macroeconomic factors also affect risk and impact sovereign bond yields and prices.

10. Trading in currency options is risky as the notional value involved is extremely large. As foreign exchange (FX) markets are open round the clock, traders need to be able to monitor their positions or risk being caught out by fast moving events. If an investor has an OTC currency option, managing counterparty risk is important.

11. Structured products involve the combination of two or more underlying financial instruments, so their risk profiles are more complicated than a standard financial product. Understanding the individual components' features affects the structured product investor's overall risk-return profile.

12. CPPI strategy products may result in a situation where the investor ends up buying high and selling low, particularly in a prolonged period of a range bound market trading.

13. **Market risk** is essentially systemic risk which affects the entire market. The main sources of market risk include changes in interest rate, inflation rate, recession and political instability. The general market risk profile of many structured products is similar, with specific aspects of market risk affecting underlying instruments differently, particularly options.

14. A structured product is exposed to counterparty risk. In the event of a default by the counterparty, it would impact the value of the structured product adversely. Where there is a swap agreement between the issuer and another institution, the investor bears the credit risk of both the issuer and the swap counterparty. For structured notes issued by a special purpose vehicle (SPV) and involving credit default swaps (CDS), it is important to be sure about the credit quality of the underlying debts securities and entities linked to the SPV.

15. Liquidity risk is a result of the lack of marketability. Structured products are designed for investors who are willing to hold the investment to maturity and there is a limited secondary market. Investors should be aware of any lock-up period in the structured product. Structured products can also be as illiquid as the underlying derivatives components such as options and credit default swaps may not be actively traded.

16. **Early termination risk** is the risk that in the event of an early withdrawal or liquidation of a structured product, the underlying asset is likely to be sold at a discount and hence the value of the structured product will be affected adversely.

17. **Structure risk** arises as structured products have no fixed or standardized form due to the numerous possible combinations of underlying financial instruments. The benefits and liabilities of the structured product are highly dependent on how each product is structured and it is important to understand the structure to appreciate the potential downside loss.

18. **Reinvestment risk** is the possibility of having to reinvest your bond proceeds – coupons and/or principal – at an interest rate below your yield-to-maturity (YTM). Reinvestment risk is greatest for bonds with large coupons, higher coupon payment frequency, long maturities, and selling at a premium.

19. **Volatility risk** applies to bonds and instruments with embedded options, such as callable bonds, putable bonds and convertible bonds, as their prices are affected by changes in interest rates and in volatility. A rise in volatility affects the value of a call option, which will reduce the value of the callable bond because the bond holders have sold the call option component to the issuer.

20. **Correlation risk** refers to the likelihood of an event of any kind having a direct impact on another. A structured product is exposed to correlation risk when it involves underlying instruments that are correlated with each other.

21. Financial instruments that are not denominated in the same currency as the structured product are exposed to **foreign exchange risk**. The investment return of the structured product can be affected, depending on changes in the exchange rates for the various transactions that were carried out.

22. **Legal risk** is the exposure to legal contracts relating to the structured product. Ownership or legal rights to the underlying financial instruments in the event of default by the issuer do not automatically go to the investors as investors only own units of the structured products, not the underlying financial instruments.

23. **Transactional risk** refers to the risk resulting from the time difference between the commencement of a financial instrument contract and settlement of the contract. This is prevalent when there is a difference in currencies between structured products and the financial instruments that it will enter into.

24. The **risk of mis-selling** is related to the possibility for investors to misunderstand the investment strategy and hence, not achieve their investment objectives. The prospectus and product highlights sheet are important documents to ensure there is proper risk disclosure.

25. There are three main types of trading risks - **market risks, credit risks** and **settlement risks**. The personnel in the front, middle and back office must be diligent to proactively monitor and manage these risks. Depending on whether futures or options are involved, the risk metrics used are different.

26. One approach to managing market risk for futures is setting limits. This could include open contracts limit, maximum loss limit, maturity limit and stress test limit.

27. For fixed income instruments and related derivatives, **interest rate risk** can be managed by hedging the fixed income position with futures contracts. To establish the correct hedge, various methods can be used to measure the underlying asset exposure - **the DV01 (dollar change in price)** or **PVBP (price value of a basis point), duration** and **modified duration**.

28. **Value at risk (VaR)** is used for risk measurement on a specific portfolio of financial assets. VaR incorporates volatility and correlation between market factors into the risk assessment to arrive at a potential loss amount or value at risk. A VaR statistic has three components: a relatively high level of confidence, a time period and an estimate of investment loss.

29. Options carry multiple risk parameters, and the correct form of risk management control requires specifying and setting limits for the option Greeks, i.e., Delta, Gamma, Vega, Theta and Rho.

30. **Delta** (price reaction of the option against a change in the asset price) limits are normally set in terms of USD or the local currency amount. **Gamma** (change in delta or the second order price derivative) is usually measured against a specified movement in the spot price (such as one standard deviation).

31. **Vega** is measured by the change in value of an option, i.e., the premium over a 1% change in volatility in the underlying. Limits are then placed in terms of the maximum loss that would be tolerated given certain movements in volatility, in either direction.

32. **Theta** measures the potential loss through time decay. Gamma, Vega and Theta are sometimes controlled together by setting the maximum loss for all three combined because this way, the positive effects of theta are automatically offset against the negative effects of Gamma.

33. **Rho** measures the impact of a change in interest rates on the option price. Limits for Rho can be set in terms of delta amounts in different periods or by reference to expected losses on movement of a currency interest rate. Executing a swap transaction (such as floating for fixed rate) can reduce Rho risks.

34. Other market risk limits that can be implemented to control risk are **maturity concentration control**, **total book size**, **premium payment control** and **intra-day control**.

35. Counterparties' credit risks for OTC options are greater and limits are predefined for individual counterparties, defined according to the credit rating of the institution in terms of settlement amount and contract tenor. There can also be premium payment limits imposed for clearing broker risk and exchange risk.

Chapter 12: Case Studies

Learning Objectives
The candidate should be able to apply the learning from Chapters 1-11 to: ✓ Provide possible solutions to meet different clients' needs and objectives, profile, and requirements. ✓ Deal with different market scenarios affecting clients' investments. ✓ Help clients make a decision on actions that need to be taken and possible solutions.

Whether you are a representative with a financial advisory firm, a relationship manager at a private bank, a derivatives trader or a trading/investment representative at a securities brokerage, your job would involve dealing with a wide range of clients who are looking to you to provide various financial services. Your range of clients could include individuals, or local or foreign institutions. As a finance professional, you must have good knowledge of the products you are dealing in so that you can provide the necessary information to your clients. As an investment representative, it is important to understand the clients' financial needs, investment objectives, risk tolerance and any personal investment preferences.

The following sections examine several case studies of different structured products. In each case, the product features, underlying components, payoff scenarios, advantages and disadvantages are discussed. The intention of the case studies is to gain better understanding and insight as to how these structured products work and appreciate different possible performance outcomes under various market scenarios. In doing so, you will have the confidence in your ability to engage your clients when you discuss structured products with them.

Case Study 12.1 Enhanced Yield with Currency Investments

Mrs Goh is a high net worth individual who has most of her investments in three currencies (USD, GBP and SGD).

Assume that it is 26 September 2017. Mrs Goh has observed that the USD has weakened against most currencies this year, and that the USD has fallen by more than 6% against the SGD. She would like to get a higher yield on her USD deposits to offset some of the exchange rate loss. She is hoping for a quick rebound in USD in the short term but believes that the USD will weaken against SGD and GBP over the longer term. She would have no problem if some of her USD deposits are converted to SGD or GBP.

Mrs Goh would like to keep her currency deposits as investments in currency. She is tired of the low deposit rates and is looking to generate a higher yield on her currency investment. She is looking at a dual currency investment (DCI) using either the USD/SGD and GBP/USD currency pair. Details shown below:

In this case, although Mrs Goh is a Singaporean, she will be using USD as the base currency for her investments for both currency pairs. The bank will usually structure conversion to the weakening currency. So, if the USD weakens against both, there is no conversion. Hence you will note that the strike prices are set in such a way that they will take effect on weakening from the current spot rates to the strike price level.

Calculating Expected Returns for the USD/SGD Dual Currency Investment (DCI)

Assume that Mrs Goh invests in the 11% USD/SGD 3-month DCI with an initial investment of USD 1 million from her USD account. Suppose the details of the DCI are as shown below:

	USD/SGD
Current spot rate	1.3569
Tenor	3 months
Annualized return offered by the bank	11%
Strike price (for conversion from USD)	1.3800
Equivalent minimum amount	USD 1 million

Question 1

How much SGD will Mrs Goh receive if the spot rate at expiry is 1.3300?

a. SGD 990,271.74.
b. SGD 1,000,000.00.
c. SGD 1,027,500.00.
d. SGD 1,366,575.00.

Rationale / Workings:

Since the spot rate at expiry is 1.33, there is no conversion. Mrs Goh will receive her initial investment of 1 million plus interest of 11% in USD. Interest = USD 1 million X 11% X 3/12 = USD 27,500. Total Principal + Interest = USD 1,027,500.00

Mrs Goh does not mind receiving SGD, to convert to SGD we have USD 1,027,500 X 1.33 = SGD 1,366,575.00.

Question 2

Assume the spot rate at expiry is 1.38. How much USD will be in Mrs Goh's account?

a. USD 1,380,000.00.
b. USD 1,120,000.00.
c. USD 1,300,000.00.
d. USD 1,027,500.00.

Rationale / Workings:

If USD/SGD is 1.38 at expiry, the bank will not exercise the put option. Mrs Goh will receive principal USD 1 million plus interest is 11% per annum. This is a 3-month DCI, so she will receive (USD 1 million) + (USD 1 million x 3/12 months X 11%) = USD 1,027,500.00.

Question 3

Since Mrs Goh does not mind receiving SGD, how much SGD will be in her account if the spot rate at expiry is 1.41?

a. SGD 1,438,500.00.
b. SGD 1,027,500.00.
c. SGD 1,417,950.00.
d. SGD 1,448,775.00.

Rationale / Workings:

If USD/SGD @ expiry is 1.41, it means that USD has strengthened, and SGD has weakened. There is a conversion at 1.38. Mrs Goh does not mind receiving her principal and returns in SGD. From Question 2, we know that Total Principal plus Interest = USD 1,027,500.00. Converting USD 1,027,500 to SGD at strike price of 1.38, Mrs Goh will receive SGD 1,417,950.00.

Learning Outcomes (Chapter 7 – Structured Deposits and Other Structured Products)

- Explain the key components and features of structured deposits.
- Understand the main strategies used in structured deposits.
- Discuss the risk-return trade-off for specific structured deposits.

Understanding the USD/SGD DCI from the Bank's Perspective

Let's review the same situation from the side of the bank, which is often not visible to the client.

From various exchanges we can get a 3-month quote on the USD put price for 3 months at 1.3800. Suppose we find out the 3-month USD call/SGD put struck at 1.3800 can be sold at the price of 0.0250.

USD/SGD spot price	: 1.3569
Strike price	: 1.3800
USD 3-mth money market rate	: 1.3919 %
SGD 3-mth money market rate	: 1.1258 %

Question 4

Should the bank offer the USD/SGD DCI to Mrs Goh?

a. Yes, because the contract delivers a 2.86% return over the 3-month period to the client.

b. Yes, because the contract yields an annualized return that is higher than the annual interest of 11% offered by the bank to the client.

c. No, because the contract delivers a yield that is higher than the 1.1258% SGD 3-month interest that the bank can obtain from the market.

d. No, because the contract delivers a yield that is higher than the 1.3919% USD 3-month interest that the bank can obtain from the market.

Rationale / Workings:

To create the structure, we place USD 1 million (the equivalent of SGD 1,356,900) in a money market deposit for 3 months at 1.3919%. Mrs Goh should also receive a premium on the option (0.0250 x USD 1 million = USD 25,000) for selling the put option.

Total proceeds = (USD 1,025,000) x (1+ (0.013919/4) =USD 1,028,566.74 after 3 months.

So, the total interest accrued + put proceeds should generate a return of USD 1,028,566.74 / USD 1 million = 1.0286%, or 2.86% for 3 months. Using the money market convention, this is equal to 2.86% / 3 x 12 = 11.44% per annum.

Therefore, the annualized rate of return for the bank is at 11.43%, which is higher than 11% offered to Mrs Goh. This will be a profit to the bank and the bank should offer the USD/SGD DCI to Mrs Goh.

Learning Outcomes (Chapter 7 – Structured Deposits and Other Structured Products)

- Understand the main strategies used in structured deposits.
- Discuss the risk-return trade-off for specific structured deposits.

Case Study 12.2 Taking Advantage of Short-Term Downside Volatility

It is the 3rd quarter of 20X1. The markets are uncertain if the Federal Reserve will start rapidly raising interest rates to fight inflation. Your client, Brian, believes that global equities will be sold off ahead of the upcoming December FOMC meeting. The market seems poised for a correction, but longer-term fundamentals point towards more room for the equities market to rise in 20X2. Brian is convinced that the markets will adopt a risk-off approach for the next 2 months and safe haven asset classes will dominate. He wants to take advantage of the increased volatility in the markets.

Brian is an aggressive investor and risk-taker, who would like to "bet" on a major correction of the U.S. market and possibly global markets within the next 2 months. Brian is asking for ideas to take advantage of the likely developments.

12.2.1 What Should Professional Investment Representatives or Advisors Do with Clients Before Making Any Recommendations?

a. **Confirm risk willingness**. You will need to clearly brief and document Brian's suitability and understanding of the types of investments that he is about to invest in and that the risks of potential

losses that can be significant. There should be some acknowledgement and sign-off on product suitability and client's willingness and understanding for taking on such risks.

b. **Confirm financial obligations, capacity and overall exposure.** Check on the limits and buffers, discuss the financing position (margins) and extent to which Brian should take on risk. Ensure that the credit limits and funds are available so that he does not over commit. Clients may be trading with several brokers and banks and may have concentration risks on some securities. As the investment specialist who is advising Brian, you should seek to get a broad picture of his overall exposure.

c. **Confirm client's level of investment knowledge and experience.** Some clients may be overconfident to the point of arrogance to their own detriment. While the best form of training for some clients may be to experience the pain of losses, it is your role and responsibility as an investment advisor or specialist to provide good advice, words of caution and guidance to Brian. It is important that you educate and help Brian understand the various products and investment strategies along with the risks involved, and where possible you should send him educational materials to help him better understand the products, strategies and risks.

d. **Confirm internal credit limits and policies.** During times of high volatility, the credit limits and policies will often tighten. Lending values on portfolios can be suddenly reduced to the disadvantage of the investor. Hence, you must be aware of the internal policies, processes and limits.

e. **Set buffers and do not maximize leverage.** When clients are speculative in nature and are risk-takers (such as Brian), it is important that you advise him to take precautionary measures such as setting buffers and not trade at his limits before the investment positions are taken, to help him manage his exposures.

f. **Educate the client on trading limits, price limits, offsetting positions.** When Brian begins to trade, his trading should be accompanied with a discipline of setting buying and selling limits and stop loss positions. Cutting losses might be encouraged from time to time but this is ultimately Brian's decision. Alternatively, you could advise Brian on available offsetting trades that can help him close out or offset existing positions. This is important for long-short trades for futures, options, structured warrants, ETFs, DLC and leveraged ETFs.

g. **Encourage profit taking, stay alert and keep abreast of losses.** It is important that you help Brian to learn how to set targets to take profits and to keep abreast of accumulated losses. While his initial and original positions may start out positive, the potential risks and disasters will emerge in the course of trading and investing. Therefore, staying alert and keeping abreast of his positions, especially losses, will help you better assist and advise Brian when managing negative positions.

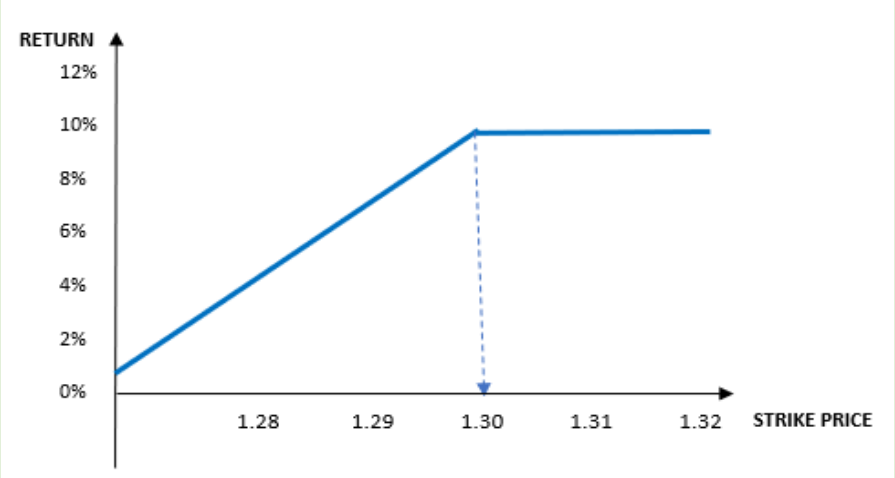
h. **Keep the client updated on new market and research information.** Having a habit of keeping alert on market developments overnight, good research analysis, market outlook and investment strategy presentations will be helpful in keeping yourself and your clients informed of critical factors driving market movement at different times.

Calculating Expected Returns for the GBP/USD 3-month Dual Currency Investment (DCI)

Assume that Mrs Goh believes that the GBP/USD exchange rate will weaken i.e., the USD will strengthen against the GBP in the near term (quick rebound expected).

She has decided to invest in a standard GBP/USD DCI with the following details:

	GBP/USD
Current spot rate	1.3414
Tenor	3 months
Annualized return offered by the bank	10%
Strike price (for conversion from USD)	1.3000
Equivalent minimum amount	USD 1 million



Question 5

How much USD will be in Mrs Goh's account if the spot rate at expiry is 1.2800?

a. USD 1,000,000.00.
b. USD 1,009,230.76.
c. USD 1,025,000.00.
d. USD 1,032,230.77.

Rationale/ Workings:
If the GBP/USD is < 1.3000, payment will be converted at strike level and total amount payable will be USD Principal + USD Interest = [(USD 1 million/1.3) X 1.28] + [(USD 25,000/1.3) X 1.28] = USD 984,615.38 + USD 24,615.38 = USD 1,009,230.76.

* Note that USD Interest earned for 3 months using the money market convention = 10% X USD 1 million X 3/12 months = USD 25,000.00

Question 6

What will be the annualized return for Mrs Goh if the spot rate at expiry is 1.26?

a. 1.62%.
b. 1.39%.
c. 2.62%.
d. -2.62%.

Rationale/ Workings:

If change spot rate at expiry to 1.26. [(USD 1 million/1.3) X 1.26] + [(USD 25,000.00/1.30) X 1.26] = USD 969,230.77 + USD 24,230.77 = USD 993,461.54. Rate of Return using money market convention= [(USD 993,461.54/ USD 1million) – 1] X 12 / 3 = -0.02615 = - 2.62%.

Learning Outcomes (Chapter 7 – Structured Deposits and Other Structured Products)

- Explain the key components and features of structured deposits.
- Understand the main strategies used in structured deposits.
- Discuss the risk-return trade-off for specific structured deposits.

Understanding the Roles and Responsibilities of Professional Investment Advisors

Question 1

As Brian's advisor, which of the following actions should you check and confirm before you execute any trades on his behalf? (Select all options that apply)

a. Brian's understanding of the risk of each of the investment instruments.
b. Potential changes in internal limits as market conditions change.
c. Brian's positions with other financial services institutions.
d. Brian's willingness to trade at his maximum leverage levels.

Rationale / Workings:

Option A is correct. Advisors must check & confirm the investment knowledge and experience of their clients before making any recommendations. See Section 12.2.1, point (c).

Option B is correct. Advisors must be aware of internal limits and policies, as such limits will often tighten when the market is more volatile. See Section 12.2.1, point (d).

Option C is correct. Advisors should try to get a broad picture of their client's overall exposure, as a precautionary measure and to ensure that clients do not trade beyond their levels of available funds. See Section 12.2.1, point (b).

Option D is incorrect. Aggressive or more risk-loving clients (such as Brian) should be encouraged to take precautionary measures rather than maximise their leverage. See Section 12.2.1, point (e).

Learning Outcomes (Chapter 11 – Key Products & Investment Risks for Derivatives & Structured Products)

- Discuss why structured products have more complex types of risks.
- Explain what type of trading controls and risk management processes can be implemented by financial organisations.

Controls and Risk Management Processes

Question 2

Which of the following can help to mitigate or manage Brian's risk? (Select all options that apply)

a. Limits and buffers.
b. Margins.
c. Shorting the VIX futures exchange-traded fund (ETF).
d. Educating the client.

Rationale / Workings:

Options A and B are correct. Limits, buffers and margins can help clients to manage their risk exposures. See Section 12.2.1, point (iv) and (e).

Option C is incorrect. Brian is expecting volatility to increase, so he should be buying the VIX futures ETF. See Section 12.2.2 (Table).

Option D is correct. Advisors should try to educate their clients on trading limits, price limits, offsetting positions and to practise discipline during trading. See Section 12.2.1, point (f).

Learning Outcomes (Chapter 9 – Structured Funds)

- Understanding structured ETFs and how it can be used as an investment strategy.
- Understanding the different types of trading controls and risk management process implemented by financial organisations.
- Describe structured ETFs investors and the investment strategies they use.

12.2.2 What Products and Investments Could You Recommend to Brian?

There are several possible product solutions that can be used but there are different costs, obligations and return outcomes that will determine their suitability for Brian.

Table 12.2.2 provides various product solutions and ideas to capitalize on potential correction in the markets. Each market event is different, and the actual outcome and timing of price movements of different asset classes will be different, depending on the market and investor responses.

Table 12.2.2 – Possible Instruments to Be Used When Anticipating a Major Correction

Product and Strategy	Positives/Benefits	Negatives/Shortcomings & Risks
i. Buy Put Options on futures indices - MSCI, DJIA, Nasdaq,	• Put options can be a low-cost approach to take a short position against futures indices (or against key high beta stocks that will react to	• Put options will decline in value over time even if market is flat or if the decline of the underlying is modest and stays above the strike price.

Product and Strategy	Positives/Benefits	Negatives/Shortcomings & Risks
and some other global indices.	most of the market downturn). • Put options provide the right but not the obligation to execute. Hence, the choice and timing are determined by the put option holder.	• Put options will become worthless at expiry if the underlying remains above strike price.
ii. Buy Put Warrants on the market indices.	• Put warrants can be a low-cost approach to take a short position against the market. Put warrants are more popular in Singapore and Hong Kong but less popular in the US. • Warrants, like options provide the right to sell and need to be exercised by the warrant holder.	• Put warrants will have an expiry date like options and will also decline in value over time as it approaches expiry unless the price of the underlying falls below the strike price. • Warrants will also become worthless on expiry.
iii. Short Futures Contracts on Indices.	• More direct exposure to any correction of the indices. Immediate daily settlement, hence, there is a disciplined squaring of positions on the exchange every day. This means you can realize gains (or losses) daily. Exposure can be offset easily on the exchange.	• Higher cost and cash outlay. • Unlike a long position in options, futures are outstanding contracts which have obligations of delivery. • When the market rises sharply, the short futures contracts will fall in value and the losses need to be settled daily.
iv. Buy Daily Leveraged Certificates with Short Position (3x – 5x) on the Hang Seng Index or MSCI Singapore.	• Directly inverse 3-5x augmented gains from the movement of the Hang Seng Index and MSCI Singapore. • Maximize potential gains and exposure to downward price correction.	• Highly volatile and meant mainly for traders and high-risk investors. Losses are magnified when investor is position on wrong side of market movement. Investors can lose almost all their capital if they are on the wrong side of the market movement.
v. Short ETN and ETFs linked to market indices (or for more aggressive play buy some inverse.	• For inverse ETFs/ETNs and inverse leveraged ETFs /ETNs[48], there are augmented gains from the inverse	• Cost and cash outlay are higher than options. Investors can lose almost all of their capital if they are on the wrong side of the market movement.

[48] Inverse ETFs (short funds) seek to deliver the opposite of the performance of the index or benchmark they track. Inverse leveraged ETFs (also known as "ultra short" funds) seek to achieve a return that is a multiple of the inverse performance of the underlying index.

Product and Strategy	Positives/Benefits	Negatives/Shortcomings & Risks
leveraged ETN and ETFs) ETF remain listed on the exchange until it retired by the exchange or issuer compared to ETN that has a set maturity date (e.g., 5 years).	relationship with selected equity market indices. • Maximize potential gains and exposure to downward price correction.	• Sensitivity to market indices may vary because there are usually a few ETFs and ETNs tracking the same index offered by different providers. • The derivatives products (e.g., futures, options) used by the different ETFs will have different maturities resulting in different price actions. • Large hedge funds can actively try to influence the price movements of ETFs and ETNs because of their size. Hence the price movements of ETNs and ETFs can vary from expectations.
vi. Long Volatility Futures Indices ETF (e.g., UVXY).	• When markets correct sharply, the volatility index usually rises. ETFs that track the volatility indices will also rise. • Since such ETFs can be augmented by leveraging (e.g., 3x). The gains from volatile price movement can be quite significant.	• Such ETFs that long volatility is speculative. They perform well due mainly to the spike up in volatility when markets are uncertain. • The leverage impact can lead to significant losses. • Since the VIX futures market is in Contango 70% of the time, there is a natural tendency for long volatility products to trend downward, even in flat markets, due to the roll of contracts from month to month. • Long term holdings in VXX, UVXY and TVIX are not viable and anyone trading volatility needs to be willing to "cut and run" to preserve capital in a losing trade. • A prospectus for a volatility product may include a statement "this product will head to zero".

It should be noted that strategies that seek to capitalize on a significant market correction will always be more difficult to execute because the timing, extent and duration of corrections are extremely difficult to predict. Managers, employees, shareholders, investors, governments, and other stakeholders all want to see growth in the value of companies reflected in the equity market – which is why we have witnessed the ongoing rise in the equity market over time with only the occasional period of correction.

Such short positions against the market cannot be maintained over the long term. Hence instruments that seek to short the market tend to be short-term oriented and need to be rolled over to sustain the short position. Traders and speculators tend to execute such strategies when they see a potential sustainable

period of volatility, caused by a market downturn, over a few weeks. Demand for such instruments will build once the market believes that bad news, volatility and negative sentiment are building, accompanied by the initial market price declines and downward momentum. Unfortunately, by this time, puts, put warrants and short futures positions would also become costlier and would only make sense if the downtrend is sustainable for a period.

Disciplined traders who execute such strategies will seek to close out their position once their target return levels have been reached and roll over selected positions, until there is a clear reversal of the trend. Less disciplined traders who accumulate larger positions for too long or at the wrong time and risk being caught in reversals when markets recover. Another group of investors who will actively take advantage of such instruments, are those holding long positions on the market and want to hedge against short term downward price movements. The short positions assumed would then provide some offsetting gains during market downturns.

Possible Products or Strategies for Anticipating a Downward Market Correction and Increased Volatility

Question 3

Which of the following possible strategies would be appropriate for Brian? (Select all options that apply)

a. Long a put option.
b. Short a futures contract.
c. Short a leveraged inverse Daily Leveraged Certificate (DLC) contract.
d. Short a volatility futures ETF.

Rationale / Workings:
Option A is correct. See Section 12.2.2 (Table), point (i).

Option B is correct. Shorting a futures contract is a higher cost strategy than buying put options, but it has a more direct exposure to any correction of the indices. See Section 12.2.2 (Table), point (iii).

Option C is incorrect. Since Brian is expecting the market to move downwards, so he should either short a DLC or long an inverse DLC. See Section 12.2.2 (Table), point (iv).

Option D is incorrect. Brian is expecting the market to move downwards and volatility to rise, so he should long / buy a volatility futures ETF. See Section 12.2.2 (Table), point (vi).

Learning Outcomes (Chapter 3- Futures Strategies)

Explain how to identify and manage financial risks using futures.

Learning Outcomes (Chapter 11 – Key Products & Investment Risks for Derivatives and Structured Products

- Understand the risks inherent in basic derivative products.

Case Study 12.3 Portfolio Protection

Assume it is now 22 September 2017. A portfolio manager, William, has benefited from investing in a few blue-chip shares in Singapore. He owns Capital Land, DBS and OCBC shares.

William does not foresee a major correction but would like to protect the remainder of his portfolio against anything more than a 7% decline. He thinks that current markets valuations are slightly overstretched in the near term and will be susceptible to corrections. He would like to keep the rest of these stocks for the longer term since they are blue chips but at the same time, he would like to reduce downside risk for the portfolio and hopefully get some additional yield.

The year-to-date performance of the portfolio of stocks is shown in Table 12.3.

Table 12.3.1 – Year-to-Date Performance of the Stock Portfolio

Stocks	2 Jan (SGD)	22 Sep (SGD)	YTD Change %	No. of Shares	Total Value (SGD)
OCBC	9.18	11.15	+21.5%	3 million	33,450,000
DBS	17.67	20.61	+16.6%	2 million	41,220,000
Capital Land	3.09	3.61	+16.8%	5 million	18,050,000
Total Portfolio					**92,720,000**

The details of related structured put warrants for DBS, OCBC and Capital Land are shown in Table 12.3.2.

Table 12.3.2 – Details of DBS, OCBC and Capital Land Structured Put Warrants as at 22 Sept 2017

	Structured Warrant Name	Option Price (SGD)	Conversion Ratio	Exercise Price (SGD)	Stock Price (SGD)	Breakeven Price Level (SGD)
A	DBS MB EPW180115	0.085	9 : 1	19.55	20.61	18.79 = 19.55 - (0.085 x 9)
B	DBS MB EPW171204	0.143	9 : 1	21.00	20.61	19.71 = 21.00 - (0.143 x 9)
C	OCBC BK MB EPW171201	0.025	9 : 1	10.50	11.15	10.27 = 10.50 - (0.025 x 9)
D	OCBC BK MB EPW180301	0.101	9 : 1	11.40	11.15	10.49 = 11.40 - (0.101 x 9)
E	CAPITALAND MB EPW171002	0.013	5.5 : 1	3.05	3.61	2.98 = 3.05 - (0.013 x 5.5)

Assuming a 7% decrease in stock prices, the stock prices would be as follows: DBS $19.17, OCBC $10.37, and Capital Land $3.36. The two structured warrants that would have the possibility of being in-the-money are (B) DBS MB EPW171204 and (D) OCBC BK MB EPW180301. We want to pick the warrants with put exercise prices that are closest to current prices. Furthermore, both structured warrants selected will give some protection for about 2 months until early December.

William could also wait until another favourable set of structured warrants is issued in the next few months to establish a proper hedge to cover the rest of the year until January (i.e.; assuming issuers / market makers of structured warranted offer attractive enough pricing and strike price levels for the warrants).

Price and Time Protection from Structured Put Warrants

Question 1

Which combination provides more price protection?

a. A, B and C.
b. A, C and E.
c. B, D and E.
d. C, D and E.

Rationale / Workings:

Structured put warrants B and D (from DBS and OCBC respectively) are both in-the-money (i.e.; the strike price is higher than the current spot price). Structured warrant E is included because it is the only structured put warrant for Capitaland and we should have some protection for the Capitaland stocks in William's portfolio. Structured put warrant E is also less out-of-the-money than the other out-of-the-money put warrants (A and C).

Question 2

Which combination provides more time protection?

a. A and D only.
b. A, D and E.
c. B, C and E.
d. A, B, C and D.

Rationale / Workings:

Structured put warrants A and D have the longest time to expiry, so they provide the most time protection. Structured warrant E is included because it is the only structured put warrant for Capitaland and we should have some protection for the Capitaland stocks in William's portfolio.

Learning Outcomes (Chapter 5 – Structured Warrants and Daily Leveraged Certificates)

- Explain the features, benefits and risks associated with structured warrants.
- Identify and compute the various elements in the valuation of structured warrants.

12.3.1 How Would Volatility of the Underlying Stocks Influence William's Decisions?

Another factor that we have not considered is the volatility of the underlying stocks. The higher volatility of the underlying stock, the higher the likelihood of the strike price being reached and hence the higher the warrant price. The volatility could also influence William's willingness to select the warrants.

12.3.2 What Can William Do on the OTC Market to Optimize His Returns and Still keep His Stock Portfolio?

a. William can lend the stocks to an investment bank for a fee of say 4-5% on the value of the portfolio. This generates a return to offset any downside for the next 12 months. Since William intends to hold on to these stocks for the long term anyway, he can lend his portfolio for the additional 4-5%. This can be carried out through brokerage and bank operations.

b. William has enjoyed a double-digit gain for some of his blue chips and is expecting a modest market correction. He can work with an investment bank to buy a put option with strike price set at 3-4% lower than the current price for his stocks. For example, if OCBC is at $11.15, the put strike price can be set at $10.80. If the price moves up or stays above $10.80, William can choose not to exercise the put option. If the price falls below the put strike price level of 10.80, he can decide to exercise the put option and sell at the strike levels and rebuild his portfolio with the same stocks at lower market prices. This is known as a protective put or portfolio insurance. The same type of protective put can be constructed for the two other shares DBS and Capital Land. However, William has to evaluate the cost of buying enough put options to cover his portfolio.

12.3.3 Can William Protect His Portfolio with Futures Contracts?

Suppose the market starts falling and William wants to respond quickly to provide some hedging for his specific portfolio. Would shorting the MSCI Singapore Futures Index help?

The 3 stocks owned account for only 31% weight of the MSCI Singapore Index, so the portfolio's correlation with the index will only be partial.

Portfolio Protection with Futures Contracts

Question 3

Which of the following offers protection for William's portfolio? (Select all options that apply)

a. Short MSCI SG Futures.
b. Buy an equity linked note (ELN).
c. Buy a tailored put option on the OTC market.
d. Lend the shares and get interest.

Rationale / Workings:

Option A is correct. Since William owns DBS, OCBC and Capitaland shares which account for about 31% weight of the MSCI Singapore Index, so there is some correlation with the MSCI SG Futures. See Section 12.3.3 for more information.

Option B is not correct. William should sell / short ELN in a downward market.

Option C is correct. William can work with the investment bank to buy a put option with strike price set at 3% - 4% lower than the current price for his stock.

Option D is correct. William can lend the stock to the investment bank for a fee of 4% - 5% on the value of the portfolio. This generates a return to offset any downside for the next 12 months.

Learning Outcomes (Chapter 3 – Futures Strategies)

- Describe and demonstrate how to develop and manage hedging programmes for various cash positions.

Learning Outcomes (Chapter–7 - Structured Deposits and Other Structured Products)

- Understand the main strategies used in structured deposits.
- Discuss the risk-return trade-off for specific structured deposits.

Understand structured ETFs and how it can be used as an investment strategy

Question 4

Which of the following portfolio protection strategy is accompanied by immediate settlement obligations?

a. **Short MSCI SG Futures.**
b. Tailored long put, OTC.
c. Lend the shares and get interest.
d. Tailored short call, OTC.

Rationale / Workings:

Option A is affected by immediate obligation because it is a daily settlement.

Option B is incorrect, the client is long the put option and once the option is paid, the holder has the right but not the obligation to exercise.

Option C is incorrect because the client is the lender so there is no short-term margin, immediate obligation, like the investment bank that requires a margin.

Option D is incorrect, the client is short the call option and has the obligation to deliver the underlying shares only when the option buyer exercises.

Learning Outcomes (Chapter 3 – Futures Strategies)

- Understand the terms used in futures contract specifications and futures has mark-to-market obligations.

Case Study 12.4 ELNs Offering Yield and Buying Blue Chips at a Discount

Your client, Mr Sunjoyo, is interested in accumulating more OCBC shares. However, the market has already risen significantly and many blue-chip stocks in Singapore enjoyed double-digit growth over the last 12

months. Mr Sunjoyo wants to acquire more OCBC shares around 12.50 but is waiting for a major market pull-back to accumulate the shares.

The current price of OCBC shares is SGD 13.30 and Mr Sunjoyo is interested in accumulating OCBC shares. Table 12.4.1 sets out the feature of the Equity-Linked Note structure for Mr Sunjoyo.

Table 12.4.1 – Features of Equity-Linked Structured Note Linked to OCBC Shares

Product Description	Product Features
Reference underlying asset	OCBC shares
Nominal investment	SGD 1 million
Purchase price of ELN	975,000
Maturity	100% notional sum or SGD 1 million
Annualized yield	10.66%
Tenor	3 months
Current OCBC share price	SGD 13.30
Strike price	SGD 12.50 (80,000 shares)

Calculating Annualised Returns and Break-Even Levels of the Equity Linked Note (ELN)

Question 1

If OCBC's share price at expiry of the ELN is SGD 12.40, what would be Mr Sunjoyo's annualized return/yield?

a. 0.00%.
b. 10.66%.
c. 7.16%.
d. - 6.01%.

Rationale / Workings:

If OCBC's share price is SGD 12.40, which is below the strike price of SGD 12.50 (with overall share value > purchase price), Mr Sunjoyo will get the equivalent of 80,000 OCBC shares. This is equal to (80,000 x SGD12.40) = SGD 992,000 which is still higher than the purchase price of SGD 975,000 which gives Mr Sunjoyo a return of (SGD 992,000 – SGD 975,000)/ SGD 975,000 = 1.7435% over 3 months.
The annualised return will be $(1.017435)^4 - 1 = 7.16\%$.

Note that although the share price is below the strike price, a positive return can still be generated as long as the overall equivalent conversion value is greater than initial investment in the ELN.

Question 2

What is Mr Sunjoyo's annualised return / yield if the share price is SGD 12 at expiry?

a. 12.18%.
b. 7.16%.
c. - 6.29%.
d. 1.54%.

Rationale / Workings:

If OCBC's share price falls to SGD 12 which is the same as the strike price SGD12.00 (with overall share value < purchase price), Mr Sunjoyo will receive an equivalent value of 80,000 OCBC shares, which is worth SGD 960,000 (80,000 X $12). Taking the equivalent cash means realizing the losses. Return = (SGD 960,000 – SGD 975,000)/ SGD 975,000 = - 1.53846% Annualized return = $-(1.0153846)^4$ = - 6.29%.

The worst-case scenario is if the value of the OCBC shares falls to zero and the loss will be 100% of the invested capital but this is highly unlikely in this case.

Question 3

What is the break-even price level for Mr Sunjoyo for the ELN?
(Answer is to be rounded to the nearest 2 decimal places)

a. SGD 12.50.
b. SGD 13.30.
c. SGD 13.48.
d. SGD 12.19.

Rationale / Workings:

The breakeven price level for OCBC is SGD 975,000/ 80,000 = SGD 12.1875 or SGD 12.19. At this price level there are no losses except for the transaction costs.

Learning Outcomes (Chapter 8 – Structured Notes)

- Explain what the basic characteristics and features of a structured note are.
- Describe the basic characteristics and features of an equity linked note.
- Explain the different types of risk exposures for investors in equity linked notes.

Case Study 12.5 More Yield in Quiet Currency Markets with Some Degree of Capital Preservation

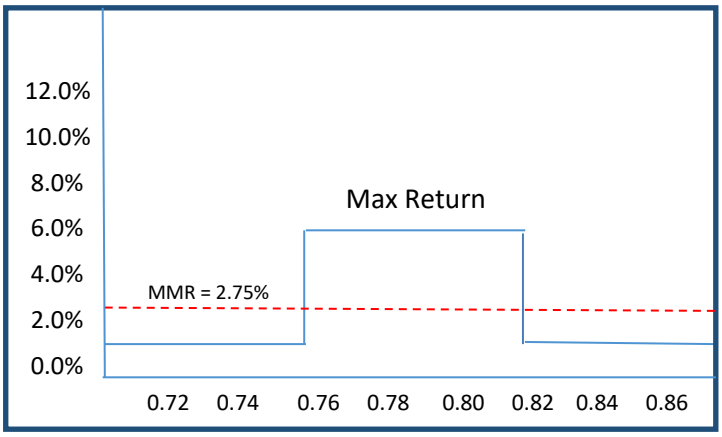
Your client, Robert, believes that the AUD/USD will be range bound for the next 3 months between AUD/USD 0.7600-0.8200. He has 2 million to invest and was offered the range accrual note to secure a higher yield.

Table 12.5.1 – Features of AUD/USD Range Accrual Note

Maturity	3 months
Underlying Spot	AUD/USD
Range	0.7600 - 0.8200
Current Spot Reference	0.7830

Maximum Yield offered	6.00%
Minimum Yield	1.00%
Money Market Rate (MMR)	2.75%

Figure 12.5.2 – Payoff of AUD/USD Range Accrual Note



In this range accrual note, as long as the currency AUD/USD stays within the range of 0.7600-0.8200, the bank will pay 6% for the days the currency is within range. If the exchange rate falls outside the range i.e., below 0.7600 or above 0.8200, Robert will only receive 1% p.a. for this period.

Calculating the Yield of the Range Accrual Note

Question 1

What will be the annualized yield if the exchange rate stays within the range for 2 months?

a. 2.67%.
b. 4.32%.
c. 6.00%.
d. 1.00%.

Rationale/ Workings:

If the exchange rate stays within range for 2 months, Robert will get 6% x 2 months / 3 months = 3.99%.
If the exchange rate is outside the range for 1 month, he will get 1% x 1 month / 3 months = 0.33%.
Therefore, Robert will receive 3.99% + 0.33% = 4.32% per annum for 3 months.

Learning Outcomes (Chapter 8 – Structured Notes)

- Explain what the basic characteristics and features of a structured note are.
- Describe range accrual notes and other types of yield enhancement note structures.

Suppose Robert is unhappy and wants a higher maximum yield of 10% or to lower the cost of the digital option spread. What can you do to offer this higher return, or to lower the price of the digital call spread option?

To offer a cheaper price for the structure or to offer a higher maximum yield, one or more of the following can be carried out:

- Narrow the range (e.g., AUD/USD 0.7650-0.8150);
- Shift and skew the range to more to one side (e.g., AUD/USD 0.7400 - 0.8000);
- Stretch the maturity period (e.g., extend the maturity to 6 months); or
- Sacrifice more minimum yield (e.g., offer 0.5% if spot rate rests outside the range at maturity).

Increasing the Yield or Lowering the Cost of the Range Accrual Note / Digital Option Spread

Question 2

What strategies can help you offer Robert a cheaper structure or a higher maximum yield? (Select all options that apply)

a. Widen the range.
b. Skew the range to either side.
c. Reduce the minimum yield requirement.
d. Reduce the maturity period.

Rationale / Workings:

Option A is incorrect. The wider the range, the more likely the exchange rate will fall within the range and the option will be more expensive.

Option B is correct. Skewing the range to one side (with the spot nearer to the barrier) will increase the likelihood of the exchange rate falling outside the range, Hence the range accrual deposit structure will be cheaper, and the client's potential yield will be higher.

Option C is correct. Reducing the minimum yield requirement will make the range accrual deposit structure cheaper.

Option D is incorrect. Reducing the maturity period will make the range accrual deposit structure more expensive, not cheaper.

Learning Outcomes (Chapter 7 – Structured Deposits and Other Structured Products)

- Understand the main strategies used in structured deposits.
- Discuss the risk-return trade-off for specific structured deposits.
- Explain the different client needs in using structured products.

Suppose Robert is very confident that the AUD/USD will stay within the range. He doesn't want you to change the range in any way and he wants to keep the maturity range of 3 months. He also wants to have

a minimum of 1.00% return. So effectively everything above in (1) stays the same. How else can you change the structure so as to offer a higher rate?

To offer a higher maximum yield without the earlier changes offered, we can offer the double no-touch version of this range accrual note. When we offer the double no-touch version, we are not looking only at the exchange rate at expiry, we are looking at the performance of the spot rate throughout its life. If the spot rate touches any of the two barriers during the life of the double-no touch range accrual note, Robert will be knocked out and only payment of 1% will be made.

Double No-Touch Range Accrual Note

Question 3

Assume that Robert was wrong in his prediction and the currency AUD/USD stays outside the range at expiry. How much would he receive from the bank?

a. 2.67%.
b. 4.33%.
c. 6.00%.
d. 1.00%.

Rationale / Workings:

In a range accrual note, as long as the currency AUD/USD stays within the range of 0.7600-0.8200 at expiry, the bank will pay 6%. If the exchange rate falls outside the range i.e., below 0.7600 or above 0.8200, Robert will only receive 1%.

Learning Outcomes (Chapter 8 – Structured Notes)

- Explain what the basic characteristics and features of a structured note are.
- Describe range accrual notes and other types of yield enhancement note structures.

Case Study 12.6 Equity Accumulator

Your client, Harry, received a proposal to buy an accumulator on ABC Ltd (ABC), a blue-chip stock which he is comfortable owning. There are mixed equities research views about ABC, and you do not think that this is a good stock because you see a structural change in the industry that might cause ABC's stock price to fall. Harry points out that the offer looks interesting because the accumulation of the strike price over time is at a 10% discount from its current price.

Discuss the risks and features of the accumulator with Harry.

12.6.1 Product Description

An accumulator is a structured note that allows Harry to purchase a pre-determined quantity of a reference stock (i.e., ABC stock) at regular intervals. The product involves a long-call and a short-put on ABC stock with the same strike price.

The accumulator can be structured as an OTC option where Harry is only required to put up a margin (i.e., unfunded basis) or wrapped as a structured note (fully funded). Unfunded accumulators are more commonly purchased by investors.

For accumulators, the key terms that Harry should know are:

i. **Reference Stock** – The share that the investor is 'accumulating';

ii. **Strike Price** – The price at which the investor buys the reference stock;

iii. **Knock-out Price** – The share price which triggers termination of the accumulator;

iv. **Full Observation Period** – The tenor of the note;

v. **Settlement Period** – Time interval for settlement, e.g., daily or month-end; and

vi. **Quantity of Shares** – The quantity of reference stocks to be accumulated per observation.

Table 12.6.1 – Features of Equity Accumulator for ABC Ltd

Product Variables	Details
Reference share	ABC Ltd
Tenor / observation period	1 year (250 business days)
Spot reference price	SGD 1.10
Strike price	SGD 1.00
Knock-out barrier	SGD 1.30
Number of shares per daily observation	10,000
Settlement	Daily observation, monthly settlement

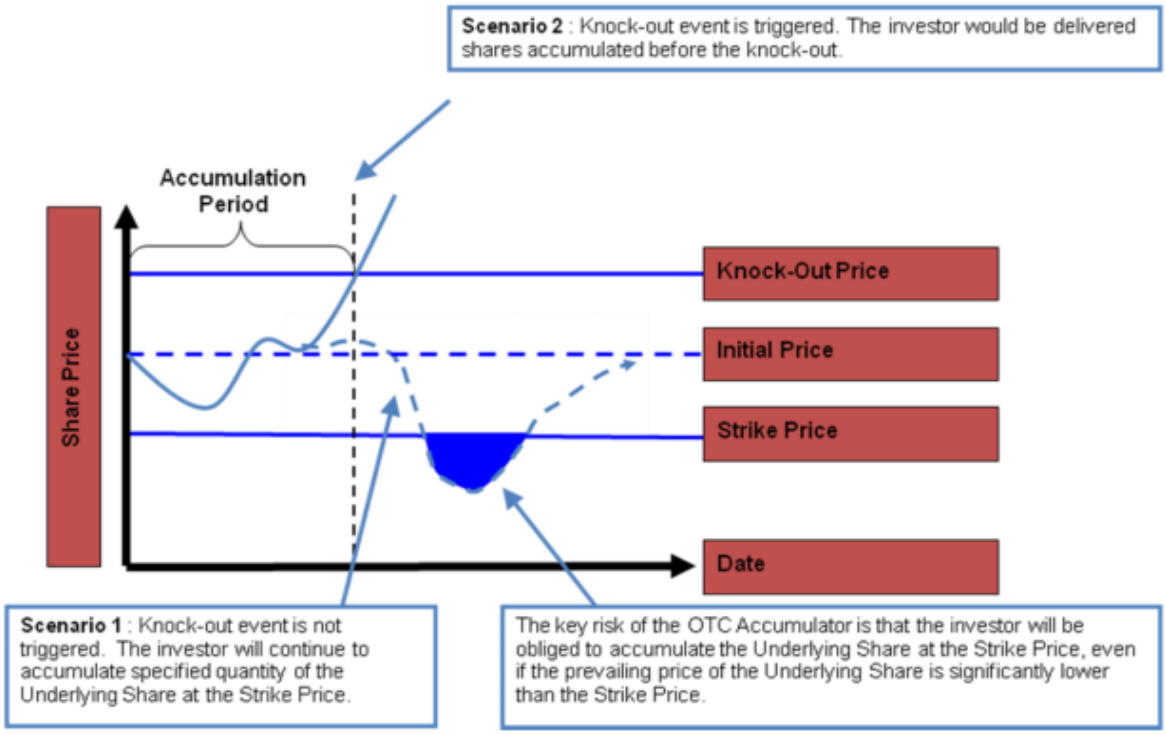
12.6.2 Mechanism of the Note

Most accumulators have knock-out barrier. If the daily closing price of ABC shares is at or above the knock-out barrier, the derivative agreement will be terminated. With such a feature, the upside (quantities purchased at discounted price) for Harry is limited by the knock-out barrier. As long as the daily closing price of ABC shares is below the knock-out barrier, he will accumulate a predetermined quantity of the shares daily and take delivery periodically, e.g., monthly.

The purchase price (strike price) is fixed at the beginning. It usually represents a discount to the market price of ABC shares when Harry enters into the OTC accumulator agreement with the bank. The discounted strike price reflects the risk that Harry will bear as there is a requirement to purchase ABC stock at the strike price throughout the tenor of the agreement. This condition applies even if, on any purchase date, the market price of the shares has dropped below the strike.

In a "1X2 gear" scheme with a knock-out barrier accumulator, Harry must buy 2X the predefined quantities of ABC stock when the price is below the strike price. However, if the share price is above the strike and below the knock-out barrier, Harry can buy only 1X of the predefined quantities of ABC shares. If the closing price of ABC shares is at or above the knock-out barrier, the agreement will be terminated immediately. Harry has to pay for and take delivery of any shares accumulated at settlement date, which is usually 2 or 3 business days after termination of agreement.

Figure 12.6.2 – Illustration of an Accumulator



12.6.3 Scenario Analysis

i. ***Best case scenario - The closing price of the ABC Ltd.'s shares stays above the strike price and is below the knock-out barrier on each business day during the tenor of the agreement.*** In this scenario, Harry will accumulate 10,000 shares at a price of SGD 1 per share on each business day and will pay for and receive the accumulated shares on a monthly basis.

If the prevailing market price of the shares is above SGD 1 when Harry receives the shares, he can sell the accumulated shares immediately in the market and realise a gain. Harry may also choose to hold the shares if he thinks that ABC's share price will head higher.

ii. ***Moderate case scenario - The closing price of ABC Ltd.'s shares stays above the strike price but on one trading day during the tenor, the closing price is at or above the knock-out barrier.*** In this case, Harry will accumulate 10,000 shares at a price SGD 1 per trading day up to but excluding the day on which the closing price is at or above the knock-out barrier. When that happens, the agreement will be terminated.

If the prevailing share price is above SGD 1 when Harry receives the shares, he can sell the accumulated shares in the market and realise a gain. However, the gain is limited as he will not be able to purchase the maximum number of underlying shares at the strike price, which is probably at a discount to the prevailing share price.

iii. ***Worst case scenario - The closing price of ABC's shares is zero for each business day during the tenor.*** In this case, Harry will have to purchase 10,000 shares per day for 250 business days, or 2.5 million ABC Ltd shares at S$1.00 per share. He will not be able to sell the shares as they are valueless and incur a marked-to-market loss of SGD 2.5 million, the maximum loss under this agreement.

If the accumulator operates on a non-funded basis (i.e.; margin facilities), the bank will probably ask Harry to top up margin to the full extent, i.e.; SGD 2.5 million, much earlier before maturity of the agreement.

Computing the Number of Shares Accumulated and Cost for an Equity Accumulator

Question 1

Assume that ABC share price stays above the strike price for 12 days but on trading day 13, the closing price reaches the knock-out barrier. How many shares will be delivered to Harry on the settlement month?

a. 230,000 shares.
b. 130,000 shares.
c. 120,000 shares.
d. 220,000 shares.

Rationale / Workings:

If the closing price of ABC shares is above the strike price and below the knock-out barrier throughout the tenor of the agreement, Harry will accumulate 10,000 shares daily at the price of SGD 1.

In this scenario, we see that ABC's share price stays above the strike price but touches the knock-out barrier on day 13. Harry will thus accumulate 10,000 shares at a price SGD 1 per trading day up to but excluding day 13, when the knock-out barrier is touched, and the agreement is terminated.

Number of shares that Harry will accumulate = (13 – 1) x 10,000 shares = 120,000 shares.

Question 2

Assume that ABC share price stays above the strike price for 20 days in the 1st month. In the 2nd month, ABC share price stays above the strike price but closes at the knock-out barrier on trading day 13.

What is the total cost of the accumulated shares?

a. SGD 320,000.
b. SGD 330,000.
c. SGD 299,000.
d. SGD 130,000.

Rationale / Workings:

Number of shares accumulated = (20 + 13 - 1) X 10,000 shares = 320,000 shares.

Cost of accumulated shares = 320,000 X SGD 1 = SGD 320,000.

Question 3

What is Harry's gain at the settlement day?

a. SGD 96,000.
b. SGD 156,000.

c. SGD 46,000.
d. SGD 196,000.

Rationale / Workings:

From Table 12.6.1, the 'mark-to-market' knock-out barrier price is at SGD 1.30. The value of the accumulated shares is SGD 416,000 (i.e., no. of accumulated shares x knock-out price = 320,000 X SGD 1.30). The total cost of the shares = SGD 320,000.

Harry's gain is the difference between the mark-to-market value of the shares @ SGD1.30 and the cost of the shares @ SGD1. Calculation = SGD 416,000 – SGD 320,000 = SGD 96,000.

Learning Outcomes (Chapter 7 – Structured Deposits and Other Structured Products)

- Discuss the risk-return trade-off for specific structured deposits.
- Explain the different client needs in using structured products.

12.6.4 Investment Risks of Accumulators (with a knock-out barrier option)

The investment risks of accumulators are discussed below:

a. There is no capital preservation feature. If the closing price of ABC shares is below the knock-out barrier at any time during the tenor of the accumulator, Harry has to buy the underlying shares at the strike price for the tenor even if the market price of the underlying shares remains substantially below the strike price for most or all of the tenor, which could result in a substantial loss.

b. If the closing price is at or above the knock-out barrier on any day during the tenor of the accumulator, the accumulator will terminate immediately, and Harry will not be able to purchase ABC shares at the strike price. Hence, Harry's potential gain is limited by the knock-out barrier.

c. Before investing in accumulators, Harry should consider whether he would like to own ABC stock at the strike price, which may be higher than the prevailing market price when the accumulation takes place. Harry must also be aware of the maximum quantities of shares which he is required to acquire throughout the tenor of the accumulator, and in the worst-case scenario, where the share price dropped to zero value, the maximum loss he will bear.

d. Harry is not allowed to terminate the accumulator before maturity without the bank's consent. If the bank agrees to terminate at the Harry's request, he will be required to pay the "break" costs which can be substantial. Therefore, Harry should consider his level of certainty that ABC shares will trade within range between the strike and the knock-out price till its maturity.

e. Harry is exposed to larger potential losses if it is a 1X2 gear accumulator where he is required to buy 2X the predefined quantities of reference share when the share price trades below the strike price.

12.6.5 Other Advantages and Disadvantages of Accumulators

12.6.5.1 Advantages

a. If, during the tenor of an accumulator, the prevailing underlying share price remains above the strike price but below the knock-out barrier, Harry can realize a gain by acquiring the pre-defined number of underlying shares at the strike price and selling these shares at the market price upon receipt.

b. For unfunded accumulators that provide leveraging / margin facilities, Harry may not be required to make an initial payment.

12.6.5.2 Disadvantages

a. Leverage works for and against investor. For a 1X2 geared accumulator with a knock-out barrier, the risk-reward is skewed as the gain is limited by the knock-out barrier while the loss will be further magnified by the 2X predefined quantity that Harry is required to buy when the share price is below the strike price.

b. The bank which offers an unfunded accumulator will have a margin monitoring system to track the volatility of the ABC share price and the mark-to-market value of the accumulator. If there is a significant fall in ABC's share price, margin calls or liquidation of the underlying shares will be enforced if Harry is not able to top up the margin within a stipulated time. In this situation, the bank may also, at its discretion, decide to close out the accumulator and/or initiate any other steps as defined in the credit facility agreement. Harry will bear all the costs and expenses incurred from closing out of the accumulator.

Case Study 12.7 Taking Advantage of Volatility in the Commodities Market

You have an accredited investor, Brandon, who is interested in trading gold, silver and the USD because of the upcoming volatility expected over the next few weeks.

Gold prices corrected when the U.S. Federal Reserve (Feds) first indicated that it would start reducing its balance sheet by reducing new U.S. Treasury issues as they mature, potentially causing a decrease in money supply and effectively moving interest rates up. Rising interest rates usually result in a stronger USD and weaker gold and silver prices. The Feds suggested that further rate hikes are on the cards next year.

On the flipside, physical gold demand has trended upwards despite the last two US interest rate hikes and given the uncertainties about the current U.S. government and an equities market which is perceived to be overdue for correction, gold as a safe-haven currency may benefit from the flight to safety in the coming months. At the same time, Brandon noticed that gold was reacting more to USD than silver with gold trading at a wider premium over silver.

Figures 12.7(a) - Chart for EUR/USD

From Mar - Sep 201X, EUR/USD has risen 14% 1.04946 to 1.20046



Figures 12.7(b) - Chart for Gold/USD

Gold has appreciated against the USD despite the last two interest rate hikes, reaching USD 1,358/Troy oz before correcting to USD 1,296.78 on Sept 21 – 22nd.



SD

Figures 12.7(c) - Chart for Silver

Silver has been mostly between USD16-18 for most of the year



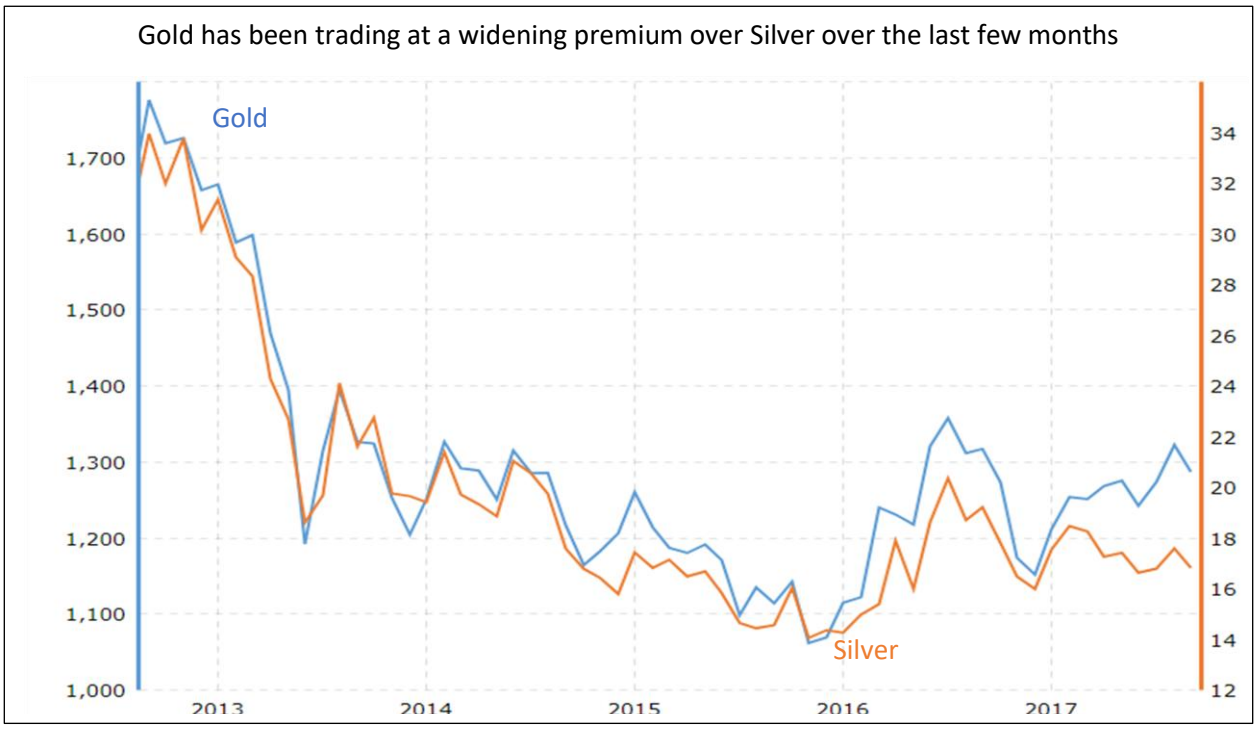
Figures 12.7(d) - Chart for Gold

Gold has appreciated against the USD despite the last two interest rate hikes, reaching USD 1,358/Troy oz before correcting to USD 1,296.78 on Sept 21 – 22nd.



SD

Figures 12.7(e) - Chart on Gold Trading at a Widening Premium over Silver



Suppose a major set of economic data is expected to be released in two days which can drive prices either way. The Federal Open Market Committee (FOMC) is also meeting in 3 days and their comments will either raise or lower the probability of a further rate hike at the end of the year. Hence, Brandon is expecting a major rise or fall of the USD and a similarly significant fall or rise in the prices of precious metals.

Options Spreads Strategies for Higher Expected Volatility

Question 1

What strategy could Brandon adopt, based on his views of the currency market?

a. Long put gold option.
b. Long gold straddle.
c. Short call gold option.
d. Short gold straddle.

Rationale / Workings:

Brandon is interested in trading gold.

Option A may work if the Feds raises the probability of a further rate hike at the end of the year.

Option B is correct, given the uncertainty of a major rise or fall of the USD.

Option C works only if the Feds raises the probability of a further rate hike in at the end of the year.

Option D is wrong as a short straddle position benefits only in a stable market.

Question 2

The straddle could be structured as:

a. Long call and Long put with strike price @ 1.2005.
b. Long call @ 1.2005 strike price and Long put @ 1.0495 strike price.
c. Long call @ 1.0495 strike price and short put @ 1.2005 strike price.
d. Short call @ 1.2005 strike price and long put @ 1.0495 strike price.

Rationale/Workings:

Option A is correct. A straddle is long a call and long a put at the same strike price. It always has a positive payoff.

Option B is incorrect because this is a strangle not a straddle.

Option C is incorrect because the strategy has no payoff if spot is in between 1.0495 and 1.2005 and will result in a negative payoff if spot is below 1.0495 at option expiry, unlike a straddle.

Option D is incorrect because the strategy will result in a negative payoff if spot is above 1.2005 at option expiration, unlike a straddle.

Question 3

What combination of options spreads strategies can Brandon undertake based on his investment outlook?
(Select all options that apply)

a. Combine a bull call gold spread and a bear put gold spread.
b. Combine a bear call gold spread and a bear put gold spread.
c. Combine a bull call gold spread and a bull put gold spread.
d. Combine a bull call gold spread and a bear call gold spread.

Rationale / Workings:

Option A is correct. A bull call spread will profit when the gold price goes up, and a bear put spread will profit when the gold price falls. Hence Brandon will be able to take advantage of the higher volatility and uncertainty on both sides.

Option B is incorrect. A bear call spread, and bear put spread will both profit when the gold price goes down.

Option C is incorrect. A bull call spread, and bull put spread will both profit when the gold price goes up.

Option D is correct. A bull call spread will profit when the gold price goes up, and a bear call spread will profit when the gold price falls.

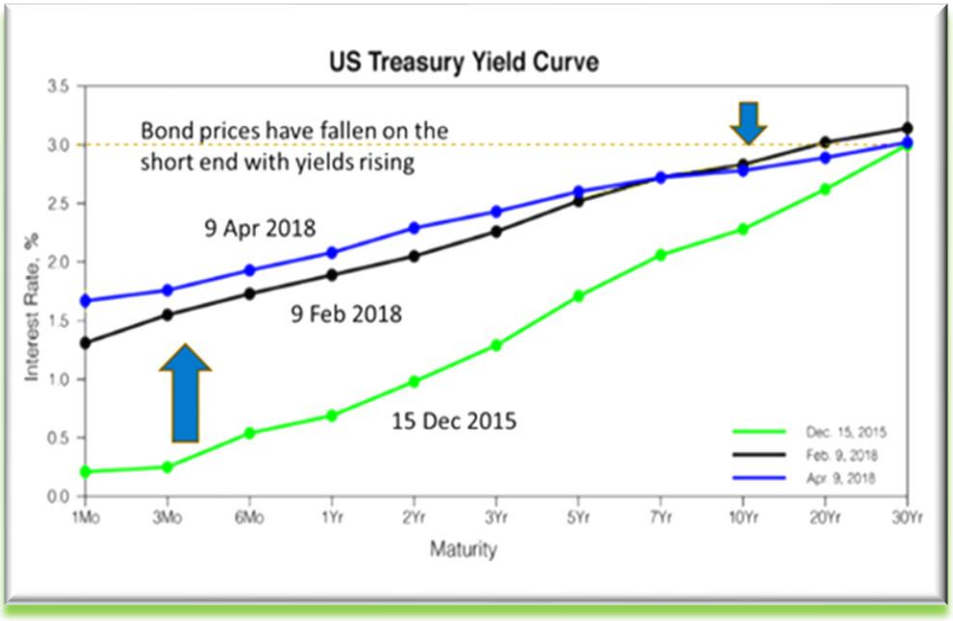
Learning Outcomes (Chapter 4 – Options)

- Explain how to use neutral, bullish and bearish trading strategies based on the investor's outlook.
- Explain how to use options to hedge an investor's underlying position.

Case Study 12.8 Taking Advantage of Varying Interest Rate Movements

Tracy is a bond portfolio manager who notices that the US Treasury Yield Curve is flattening out (i.e., the spread between long-term and short-term interest rates is decreasing). She expects the curve to invert slightly over the next few months into the next year with short rates rising faster than long rates. This example is shown in Figure 12.8.

Figure 12.8 – Flattening of the US Treasury Yield Curve



Tracy would like take advantage of the difference in the pace of change along the yield curve using a constant maturity swap (CMS), also known as a yield curve swap.

A CMS is a variation of the regular interest rate swap in which the floating portion of the swap is reset periodically against the rate of a fixed maturity instrument. Unlike a vanilla interest rate swap (IRS), where the floating rate (usually LIBOR) is reset against the fixed rate of the IRS, in a CMS, the floating rate (usually LIBOR) is reset against the rate of a long-term instrument with a fixed and constant maturity. The duration of a CMS is unchanged throughout its life, unlike a vanilla IRS. Hence, CMS can be used by investors who have a view on the direction of the shape of the yield curve; steepening or flattening.

Understanding How To Read the Yield Curve

Question 1

What trend is exhibited by Figure 12.8?

a. Interest rates are not expected to move much.
b. A falling interest rate environment.
c. A rising interest rate environment.
d. Insufficient information to answer the question.

Rationale / Workings:

Option C is correct. Figure 12.8 shows that interest rates are rising over time (2015, 2018, 2019). The yield curves also flatten over time.

Using Interest Rate Swaps to Take Advantage of Varying Interest Rate Movements

Question 2

What can Tracy do to take advantage of how yields are changing? (With the client expecting a flattening out of the yield curve, what can the client do to take advantage of how yields are changing?)

a. Swap the long end for the short end (Pay the long end, receive the short end).
b. Swap the short end for the long end (Pay the short end, receive the long end).
c. No action is needed given the current environment.
d. Swap out of the interest rate market.

Rationale / Workings:

Based on Tracy view, she expects the curve to flatten, and the curve to invert slightly over the next few months to the next year with shorts rates rising faster than long rates.

Option A is correct, as short-term rates are expected to rise faster than long term rates.

Option B is incorrect, as short-term rates are expected to rise and Tracy should not be selling short-term and buying (receiving) the long-term interest rates.

Option C is incorrect, as Tracy wants to take advantage of the yields changing and earns a profit.

Option D is incorrect, as she has a view on the yield curve and should act accordingly.

Learning Outcomes (Chapter 2 – Futures, Forwards & Swaps)

- Explain how to use interest rate swaps, foreign exchange swaps, currency swaps, equity swaps and credit default swaps.
- Describe and demonstrate how to develop and manage hedging for short term and long-term interest rate risks.

Appendix A: Constant Proportion Portfolio Insurance (CPPI) Strategy

CPPI Parameters

Investment amount	: $100
Crash Size	: 20%
Multiplier	: 1/20% = 5
Risky Asset	: S&P Index Fund
Tenor	: 5-years

Asset allocation on Day 1

Floor value	: 85% (or cushion 15%)
Multiplier	: 5
Investment in risky asset	: 5 X 15% = $75
Investment in risk free asset	: $100 - $75 = $25

Asset allocation at the end of Period 1

Assume that the new floor value is 86% of initial principal sum. At this point, assuming the price of the risky asset is unchanged, the manager will have to adjust the allocation to the risky asset as per the new floor value below:

Floor value	: 86% (or cushion of 14%)
Multiplier	: 5
Investment in risky asset	: 5 X 14% = 70
Investment in risk free asset	: 100 - 70 = 30
Total value of CPPI structure	: $100

Scenario 1: Value of portfolio falls to 86%

Assume the price of the risky asset falls by the full crash size of 20%. The value of the risky asset would have fallen from $70 to $56. The total portfolio value at this point would have been $86 computed as follows:

Investment in risky asset = $56
Investment in risk-free asset = $30
Total value of CPPI structure = $86

Because the structure's total value is now at $86 which equals the floor value, the entire allocation to the risky asset will be liquidated and re- allocated to the risk-free asset. When this happens, investors will no longer enjoy further price appreciation in the S&P index fund and will only receive their principal sum at maturity.

Scenario 2: Value of the portfolio falls to 96%

The risk asset's price falls by 5.71%, such that the risky asset value declines from $70 to $66. The allocation to the risky asset is 5 x (96% - 86%), or 50%. The manager will have to sell part of the $66 investment in the risky asset to adjust the asset allocation to $50 in the risky asset and $46 in the risk-free asset.

	Before Re-allocation	After Re-allocation	Change
Risky Asset	$66	$50	-$16
Risk Free Asset	$30	$46	+$16
Total Asset	$96	$96	

Scenario 3: Value of portfolio rises to $110

The risk asset's price rises by 14.29% such that its value increases from $70 to $80. Allocation of the risky asset equals 5*(110% - 86%), or 120%, and all the funds will be allocated to the risky asset as follows:

	Before Re-allocation	After Re-allocation	Change
Risky Asset	$80	$110	+$30
Risk Free Asset	$30	$0	-$30
Total Asset	$110	$110	

In some CPPI structures, the manager may be permitted to take on leverage, and further increase the allocation to the risky asset to meet the allocation as follows:

	Before Re-allocation	After Re-allocation	Change
Risky Asset	$110	$120	+$10
Risk Free Asset	$0	-$10	-$10
Total Asset	$110	$110	

If the risky asset suffers a price fall equalling the 20% crash size after taking on leverage, such that the value falls from $120 to $96, the reallocation will be:

	Before Re-allocation	After Re-allocation	Change
Risky Asset	$96	$0	-$96
Risk Free Asset	-$10	$86	+$96
Total Asset	$86	$86	

Notice that the 20% fall in price of the risky asset has taken the CPPI structure back to the floor of 86%. The manager will have to liquidate all $96 of the risky asset, pay off the leverage of $10, and allocate the remaining $86 into the risk-free asset.

These scenario analyses and asset reallocation will be carried out periodically. For example, in Scenario 2 – if the portfolio value at the end of period 2 (risk asset value + risk-free asset value) rises to 105%, the manager will then sell the risk-free asset and reallocate the sales proceeds to the risky asset, according to the above formula.

Appendix B:
MAS Practice Note on the Product Highlights Sheet[49]

1 Purpose of this Practice Note

1.1 This Practice Note is issued by the Monetary Authority of Singapore (the "Authority") pursuant to section 321 of the Securities and Futures Act (Cap. 289) ("SFA") to provide guidance to issuers[1] and their professional advisers in preparing a Product Highlights Sheet.

1.2 This Practice Note is to be read in conjunction with the provisions relating to the preparation and publication of a Product Highlights Sheet in the Securities and Futures (Offers of Investments) (Securities and Securities-based Derivatives Contracts) Regulations 2018 and Securities and Futures (Offers of Investments) (Collective Investment Schemes) Regulations 2005 (collectively, the "Regulations"), and is not intended to override or replace any provisions in the Regulations. Issuers and their professional advisers shall ensure that all requirements in the Regulations are complied with.

2 Submission and Dissemination of Product Highlights Sheet

2.1 The Product Highlights Sheet should be submitted to the Authority at the time of the lodgment of the prospectus with the Authority.

2.2 The Product Highlights Sheet should be made available together with the prospectus. If the issuer makes the prospectus available online, the Product Highlights Sheet should also be made available online together with the prospectus.

3 Guidance to Issuers on Preparing the Product Highlights Sheet

3.1 The format of the Product Highlights Sheet set out in the Regulations serves as a minimum standard. Issuers must adhere to the format illustrated, including the tabular structure, the yellow Pantone 109C on the right edge of the document and the prescribed headings and sub-headings. Other colours cited in the illustration are optional, although encouraged, in order to aid readability. Issuers may add additional sub-headings to enumerate points in a long section. Issuers should include any additional key information that is important for investors to understand the product.

[1] In the case of CIS, "issuer" refers to the "responsible person" as defined under the SFA to be –
(a) in the case of a scheme which is constituted as a corporation, the corporation; or
(b) in the case of a scheme which is not constituted as a corporation, the manager for the scheme.

[49] Extracts taken from **MAS Practice Note on the Product Highlights Sheet (SFA PN-02)**; Issue Date: 8 June 2018. (practice-note-on-the-product-highlights-sheet-sfa-pn-02.pdf (mas.gov.sg) Please also refer to **MAS Guidelines on the Product Highlights Sheet for Offers of Debt Securities, Hybrid Instruments and Equity Securities (SFA 13-G13)** (Guidelines on the Product Highlights Sheet for Offers of Debt Securities, Hybrid Instruments and Equity Securities [SFA 13-G13] (mas.gov.sg)).

3.2 Notes to guide issuers in preparing their Product Highlights Sheets are presented as italicised statements in the square brackets in the templates. Examples are also presented. These notes and examples are neither prescriptive nor exhaustive. Issuers should decide on the information to be disclosed in the Product Highlights Sheet so as to highlight key features and risks of their respective investment product to investors.

3.3 Issuers should answer the questions prescribed in the format of the Product Highlights Sheet in clear and simple language that investors can easily understand. Issuers should avoid using legal, financial or technical business jargon (collectively, "technical jargons") that may not be easily understood by lay persons. Where such terms are unavoidable, issuers should attach a glossary to the Product Highlights Sheet to explain these technical jargons.

3.4 Issuers may include references to websites or to corresponding sections of the prospectus which set out additional information for investors in the right-hand column of the Product Highlights Sheet. Issuers are encouraged to include links to online copies of disclosure documents, educational resources or explanatory material. However, key information should be clearly disclosed in the Product Highlights Sheet and issuers should not merely make reference to information in other sources, such as the prospectus.

3.5 The use of diagrams such as infographics, graphs, charts, flowcharts, tables or numerical explanations to explain structures or payoffs of the investment products to investors is encouraged.

3.6 Diagrams may be inserted within the main body of the Product Highlights Sheet if appropriate. The issuer's corporate logo or trade mark may also be inserted.

3.7 Issuers should ensure:

a) a margin of at least 2.54cm on all four sides of the page;
b) at least single line spacing before and after each line of text for the content; and
c) left and right cell margins of at least 0.19cm within the tabular structure in the Product Highlights Sheet.

The presentation of the information should not hinder readability.

3.8 Issuers should refrain from including disclaimers in the Product Highlights Sheet.

4 Product Naming and Other Matters

4.1 Issuers should ensure there is clear product naming. The name of the product should be descriptive of the nature of the product. It should not give investors a misleading view of the true nature and risks of the product.

4.2 Issuers should not produce a publication which resembles or may otherwise be confused with a Product Highlights Sheet, if the publication is not a Product Highlights Sheet required under the SFA.

Appendix C: Futures Contracts Specifications

SGX EUR/ CNH FX Futures

Contract Size	EUR 100,000
Ticker Symbol	EC
Contract Months	13 monthly followed by the next 8 quarterly month
Tick Value	CNH 10
Trading Hours (Singapore Time)	T Session: Pre - Opening: 7.10 am - 7.23 am Non - Cancel: 7.23 am - 7.25 am Opening: 7.25 am - 5.55 pm Pre - Closing: 5.55 pm - 5.59 pm Non - Cancel: 5.59 pm - 6.00 pm T + 1 Session: Pre - Opening: 6.05 pm - 6.13 pm Non - Cancel: 6.13 pm - 6.15 pm Opening: 6.15 pm - 5.15 am Pre - Closing: NA Non - Cancel: NA
Trading Hours on Last Day	7.25 am - 4.00 pm (or 3.00 pm during EU daylight saving time)
Minimum Price Fluctuation	CNH 0.0001
Daily Price Limit	Nil
Last Trading Day	2 business days prior to the 3rd Wednesday of the contract expiry month.
Settlement Basis	Cash Settlement (CNH)
Final Settlement Price	Cross calculation of: (i) midpoint between the bid and ask prices of the WM/Reuters Intraday Spot Rates for EUR/USD published at 4.00 Singapore time (or 3.00 pm Singapore time during EU daylight saving time), and

	(ii) the midpoint between the bid and ask prices of the WM/Reuters Intraday Spot Rates for USD/CNH published at 4.00pm Singapore time (or 3.00pm Singapore time during EU daylight saving time); and the result rounded to 4 decimal places
Position Accountability / Position Limit	Position limit is not applicable to this contract. However, a person owning or controlling more than 5,000 contracts net long or net short in all contract months combined, or such position as the Exchange may prescribe from time to time with prior notification, shall provide, in a timely fashion, upon request by the Exchange, information regarding the nature of the position, trading strategy, and hedging information if applicable.
Negotiated Large Trad	Minimum 20 lots
Bloomberg Ticker Code	XEDA<CURNCY>CT
Refinitiv Ticker Code	0#SER:

SGX 10-year Full-sized Japanese Government Bond Futures

Contract Size	JPY 100 million Notional 10-year JGB with 6% coupon
Ticker Symbol	JG
Contract Months	5 quarterly months in the March, June, September, December cycle
Trading Hours (Tokyo time)	T Session: Pre-Opening: 7.10 am -7.23 am Non-Cancel: 7.23 am -7.25 am Opening: 7.25 am - 5.10 pm Pre-Closing: 5.10 pm - 5.14 pm Non-Cancel: 5.14 pm - 5.15 pm T+1 Session: Pre-Opening: 5.20 pm - 5.28 pm Non-Cancel: 5.28 pm - 5.30 pm Opening: 5.30 pm - 5.15 am
Trading Hours on Last Day	7.25am – 2.15pm
Quality Specification	JPY 0.01 per JPY 100 notional value (JPY 10,000)
Minimum Price Fluctuation	JPY 0.01 per JPY 100 notional value (JPY 10,000)
Daily Price Limit	Nil
Last Trading Day	One business day preceding the Osaka Exchange's (OSE) 10-year JGB futures contract determined from the beginning of their evening session on SGX's Last Trading Day.

Settlement Basis	Cash Settlement
Final Settlement Price	Based on the Official Opening Price of OSE's 10-year JGB futures contract on TSE's last trading day
Position Accountability / Position Limit	Position limit is not applicable to this contract. However, a person owning or controlling more than 2,500 * contracts net long or net short in all contract months combined, or such position as the Exchange may prescribe from time to time with prior notification, shall provide, in a timely fashion, upon request by the Exchange, information regarding the nature of the position, trading strategy, and hedging information if applicable. * The limit is the overall position limit for both mini-JGB and JGB contracts.
Bloomberg Ticker Code	BIA <COMDTY>CT
Refinitiv Ticker Code	1SJG:<F3>
Price Information (Vendor: Ticker)	SIX Financial: JGyym, 344 Quick: 801.n/SGXE Moneyline Telerate: SG@JF/av Future Source: SJG
Fungibility Mechanism	Position and margin offset in the ratio of 1 SGX JGB contract to 10 Mini JGB contracts

SGX Nikkei 225 Index Futures

Contract Size	JPY 500 x Nikkei 225 Index Futures Price	
Ticker Symbol	NK (Outright, Strategy Trades), NKTI (Trade-At-Index-Close)	
Contract Months	6 nearest serial months & 32 nearest quarterly months	
Trading Hours (Singapore time)	<u>T Session:</u>	
	Pre -Opening	7.15 am -7.28 am
	Non -Cancel Period	7.28 am -7.30 am
	Opening	7.30 am - 2.25 pm
	Pre-Closing	2.25 pm- 2.29 pm
	Non-Cancel Period	2. 29 pm - 2.30 pm
	<u>T+1 Session:</u>	
	Pre -Opening	2.45 pm – 2.53 pm
	Non -Cancel Period	2.53 pm – 2.55 pm
	Opening	2.55 pm – 5.15 am
	Pre-Closing	NA
	Non-Cancel	NA

Trading Hours on Last Day	Same as T Session trading hours
Minimum Price Fluctuation	Outright: 5 index points (JPY 2500) Strategy Trades: 1 index point (JPY 500), Trade-At-Index-Close: 0.25 index point (JPY 125)
Daily Price Limit	Whenever the price moves by 10% in either direction, from the previous day's daily settlement price, trading at or within the price limit of +/- 10% is allowed for the next 5 minutes. Thereafter when the price moves by 15% in either direction, from the previous day's daily settlement price, trading at or within the price limit of +/-15% is allowed for the next 5 minutes. After this cooling-off period has elapsed, there will be no price limits for the remainder of the trading day. There shall be no price limits for the expiring contract month on the Last Trading Day. *The Exchange will signal the commencement of the Cooling Off Period when there is an unsatisfied bid or offer at the prevailing Upper Price Limit or Lower Price Limit respectively for a continuous period of more than ten seconds. The Exchange retains the discretion to take into account any other factors in the market in determining the commencement of the Cooling Off Period
Last Trading Day	The day before the second Friday of the contract month
Settlement Basis	Cash Settlement
Final Settlement Price	The Special Nikkei 225 Index Quotation based on the opening prices of each component issue in the Nikkei 225 Index on the business day following the last trading day.
Position Accountability / Position Limit	Position limit is not applicable to this contract. However, a person owning or controlling more than 10,000 contracts net long or net short in all contract months of SGX Nikkei Stock Average Futures and Options, SGX Mini Nikkei Stock Average Futures and SGX USD Nikkei Stock Average Futures combined in position delta, or such position as the Exchange may prescribe from time to time with prior notification, shall provide, in a timely fashion, upon request by the Exchange, information regarding the nature of the position, trading strategy, and hedging information if applicable.
Negotiated Large Trade	Minimum 5 lots
Underlying Index Bloomberg Ticker	NKY<Index>
Bloomberg Ticker Code	Outright, Strategy Trades: NIA<INDEX>CT Trade-At-Index-Close: AKTA<INDEX>CT
Refinitiv Ticker Code	Outright, Strategy Trades: 1SSI:<F3> Trade-At-Index-Close: 0#SSITI:
Price Information (Vendor: Ticker)	Outright, Strategy Trades CQG : ZNA ComStock: 2sNKNmy Future Source: SNIA MoneylineTelerate: SG@NAyym or Pg 7257 Quick: 101.n/SGXE SIX Financial: NKyym,344

	Trade-At-Index-Close SIX Financial: NKmyy_TAIC,344 (T Session), NKmyy_TAIC,812 (T+1 Session) Colt Technology: NKTImyy
Other Contract Specification Details	Mutual Offset Trading with CME: Eligible for Outright and Strategy Trades executed on screen. Not eligible for Negotiated Large Trades (NLT) and Trade-At-Index-Close (TAIC) Trades.

SGX FTSE Taiwan Index Future

Contract Size	USD 40 x SGX FTSE Taiwan Index Futures Price
Ticker Symbol	TWN (Outright, Strategy Trades) TWNTI (Trade-At-Index-Close)
Contract Months	2 nearest serial months and 12 quarterly months on a March, June, September and December cycle
Trading Hours (Singapore time)	<u>T Session:</u> Pre -Opening 8.30 am - 8.43 am Non -Cancel Period 8.43 am - 8.45 am Opening 8.45 am - 1.45 pm Pre-Closing 1.45 pm - 1.49 pm Non -Cancel Period 1.49 pm - 1.50 pm <u>T+1 Session:</u> Pre -Opening 2.05 pm - 2.13 pm Non -Cancel Period 2.13 pm - 2.15 pm Opening 2.15 pm – 5.15 am Trading Hours on Last Day is the same as T Session trading hours
Trading Hours on Last Day	Same as T Session trading hours
Minimum Price Fluctuation	Outright: 0.25 index points (US$10), Strategy: 0.25 index point (US$10), NLT: 0.01 index point (US$0.4), Trade-At-Index-Close: 0.05 index point (US$2)
Daily Price Limit	Whenever the Initial Price Limits are reached, i.e.; the price moves by 10% in either direction from the previous day's Daily Settlement Price (DSP), a Cooling Off Period is triggered where trading within the Initial Price Limits shall continue for a period of ten minutes After the Cooling Off Period is over, the Final Price Limits, i.e., the price moves by 15% in either direction from the previous day's DSP, come into effect and shall apply for the rest of the trading day. No trades are allowed to take place at prices beyond the Final Price Limits.

	There shall be no price limits on the Last Trading Day for the expiring contract.
Last Trading Day	The Last Trading Day shall be the second last Taiwan Business Day of the Contract Month.
Settlement Basis	Cash Settlement
Final Settlement Price	The Final Settlement Price shall be the average of the FTSE Taiwan RIC Capped Index values on the Last Trading Day taken at 1-minute intervals during the last 25 minutes of trading on the FTSE Taiwan RIC Capped Index preceding the commencement of the closing auction session, and the closing index value. The Final Settlement Price shall be rounded to two (2) decimal places.
Position Accountability / Position Limit	Position limit is not applicable to this contract. However, a person owning or controlling more than 10,000 contracts net long or net short in all contract months combined, or such position as the Exchange may prescribe from time to time with prior notification, shall provide, in a timely fashion, upon request by the Exchange, information regarding the nature of the position, trading strategy, and hedging information if applicable.
Negotiated Large Trade	Minimum 50 lots
Underlying Index Bloomberg Ticker	FTCRTWRP Index
Underlying Index Refinitiv Ticker	FTFTCRTWNT
Bloomberg Ticker Code	Outright, Strategy Trades:: TWTA <Index> Trade-At-Index-Close : TATA<INDEX>
Refinitiv Ticker Code	Outright, Strategy Trades: 0#STWN: (Combined session) 0#STWNam: (T Session) 0#STWNpm: (T+1 Session) Trade-At-Index-Close : 0#STWNTI: (Combined session) 0#STWNTIam: (T Session) 0#STWNTIpm: (T+1 Session)
Price Information (Vendor: Ticker)	Outright, Strategy Trades: Trading Technologies Ticker Symbol : TWN Market Prizm Ticker Symbol : TWN Combined Session : TWNmyy Activ Financial Ticker Symbol : TWN Six Financial

	Ticker Symbol : TWNmyy T Session : TWNmyy,344 T+1 Session : TWNmyy,812 CQG Ticker Symbol : TWN Trade-At-Index-Close: Activ Financial Ticker Symbol : TWNTI Combined Session: TWNTImyy Six Financial Ticker Symbol : TWNTI T Session : TWNmyy_TAIC, 344 T+1 Session : TWNmyy_TAIC, 812 CQG Ticker Symbol : TWNTI Colt Technology Services Ticker Symbol : TWNTI Combined Session: TWNTImyy

SGX MSCI Singapore Index Futures

Contract Size	SGD 100 x SGX MSCI Singapore Index Futures Price	
Ticker Symbol	SGP	
Contract Months	2 nearest serial months & 4 quarterly months on March, June, September, December cycle	
Trading Hours (Singapore time)	<u>T Session:</u> Pre -Opening Non -Cancel Period Opening Pre-Closing Non-Cancel Period <u>T+1 Session:</u> Pre -Opening Non -Cancel Period Opening	 8.15 am -8.28 am 8.28 am -8.30 am 8.30 am -5.20 pm 5.20 pm -5.24 pm 5.24 pm -5.25 pm 5.40 pm – 5.48 pm 5.48 pm – 5.50 pm 5.50 pm – 5.15 am
Minimum Price Fluctuation	0.05 index point (SGD 5)	

Daily Price Limit	±10% and ±15% from previous day's settlement price (5 minutes cooling off period when limit is reached). Thereafter no price limits for rest of the day. No price limits for expiring contract on its Last Trading Day.
Last Trading Day	The second last business day of the contract month (A business day is defined as a day on which the Singapore stock market is open for trading).
Settlement Basis	Cash Settlement
Final Settlement Price	the Final Settlement Price shall be the official closing value of the MSCI Singapore Free IndexSM for the Last Trading Day, rounded to two decimal places.
Trading Hours on Last Day	T Session: Pre - Opening: 8:15 am - 8:28 am Non - Cancel: 8:28 am - 8:30 am Opening: 8:30 am - 5:20 pm Pre - Closing: 5:20 pm - 5:24 pm Non - Cancel: 5:24 pm - 5:25 pm
Position Accountability / Position Limit	Position limit is not applicable to this contract. However, a person owning or controlling more than 5,000 contracts net long or net short in all contract months combined, or such position as the Exchange may prescribe from time to time with prior notification, shall provide, in a timely fashion, upon request by the Exchange, information regarding the nature of the position, trading strategy, and hedging information if applicable.
Negotiated Large Trade	Minimum 50 lots
Underlying Index Bloomberg Ticker	SIMSCI <Index>
Underlying Index Refinitiv Ticker	SSG: <F3>
Bloomberg Ticker Code	QZA <INDEX> CT
Refinitiv Ticker Code	SSG:<F3> or 0#SSG:
Price Information (Vendor: Ticker)	CQG: ZG SIX Financial: T-session: SGPmyy,344 T+1 session: SGPmyy,812

SGX Straits Times Index Futures

Contract Size	SGD 10 x Straits Times Index Futures Price

Ticker Symbol	ST
Contract Months	2 nearest serial months and 4 quarterly contract months on March, June, September, December cycle
Trading Hours (Singapore time)	<u>T Session:</u> Pre-Opening 8.15 am - 8.28 am Non-Cancel Period 8.28 am - 8.30 am Opening 8.30 am - 5.20 pm Pre-Closing 5.20 pm - 5.24 pm Non-Cancel Period 5.24 pm - 5.25 pm <u>T+1 Session:</u> Pre -Opening 5.40 pm – 5.48 pm Non -Cancel Period 5.48 pm – 5.50 pm Opening 5.50 pm – 5.15 am Pre-Closing NA Non-Cancel Period NA
Trading Hours on Last Day	Same as normal trading hours (as above)
Last Trading Day	The second last business day of the expiring contract month (A business day is defined as a day on which the Singapore stock market is open for trading).
Daily Price Limits	+10% and +15% from previous day's settlement price (5 minutes cooling off period when limit is reached). Thereafter no price limits for rest of the day. No price limits for expiring contracts on its Last Trading Day.
Settlement Basis	Cash Settlement
Final Settlement Price	Final Settlement Price shall be the official closing value of the Straits Times Index for the Last Trading Day, rounded to two decimal places.
Position Accountability / Position Limit	Position limit is not applicable to this contract. However, a person owning or controlling more than 5,000 contracts net long or net short in all contract months combined, or such position as the Exchange may prescribe from time to time with prior notification, shall provide, in a timely fashion, upon request by the Exchange, information regarding the nature of the position, trading strategy, and hedging information if applicable.
Negotiated Large Trade	Minimum 50 lots
Underlying Index Bloomberg Ticker	FSSTI<index>
Bloomberg Ticker Code	SDA <INDEX>CT Sdmy <INDEX>CT
Refinitiv Ticker Code	SST:<F3> (combined)

Minimum Price Fluctuation	1 index point (SGD 10)
Price Information (Vendor: Ticker)	CQG: STI SIX Financial: 3, STyym, 344 Quick: FST.n/SGX InTV: Page 404 Moneyline Telerate: Pg 22536 / 22537 or SG@Styymm Standard & Poor's Comstock: 2sSTmy Spot @STY

Note: The specifications of futures contracts are updated as at Apr 2022. Candidates may wish to refer to the latest contract specifications provided by SGX at https://www.sgx.com/derivatives/products?code=all.

Appendix D: Formulae Sheet

Please note that this Formulae Sheet will be provided during the CMFAS SIP Exam. The examination might not cover all formulas provided.

Futures

Cost of Carry Model

$F_{0,t} = S_o (1 + C_{0,t})$

Price of Short-Term Interest Rate Futures

$P = 100 \times (1 - R)$

Price of Long-Term Interest Rates Futures

$$\left\{1 + \frac{d_1 r_1}{360}\right\} x \left\{1 + \frac{d_2 r_2}{360}\right\} = \left\{1 + \frac{DR}{360}\right\}$$

$$r_2 = \frac{360}{d_2} x \left\{\frac{(360 + DR)}{(360 + d_1 r_1} - 1\right\}$$

where:
d_1 = number of days in the 1st period
d_2 = number of days in the 2nd period
D = number of days for the entire period
r_1 = interest rate for the 1st period
r_2 = interest rate for the 2nd period
R = interest rate for the entire period

Interest Rate Parity Theory

$$F = S \times \frac{1 + R_c(\frac{n}{360})}{1 + R_b(\frac{n}{360})}$$

where:
F = forward (futures) rate
S = spot rate
R_b = annualised interest rate of base currency
R_c = annualised interest rate of counter-currency
N = number of days

Pricing Equity Index Futures

Futures price = Spot (cash) price + Financing cost - Income from stock

Or

Futures price = Spot price + Interest - Dividend

$$F = S \times \left[1 + r\frac{t}{360}\right] - D$$

where:
F = fair value of futures
S = equity index or spot price
r = annualised financing rate (money market yield)
t = time to expiry
D = value of dividends paid before expiry

Futures Strategies

Target Rate for Hedge

Target Rate for Hedge = Futures Rate + Target Rate Basis

Or V_t = Ending security price + Futures gain = $S_t + (F - F_t)$

$Vt = F + (S_t - F_t)$ = Initial futures price + Ending basis

Where:
V_t = Value of the hedged position at time t
F = Initial futures price
S_t = Security or spot price at time t
F_t = Futures price at time t

Hedging Long-Term Interest Rate Risk

$$\text{Hedge ratio} = \frac{\text{PVBP of hedge security}}{\text{PVBP of most deliverable bond}} \times \text{Conversion factor for most deliverable bond}$$

where PVBP = Change in price for a single point change in yield

Hedge Ratio

$$\textbf{Hedge Ratio (h)} = \frac{\textbf{Change in Security Price}}{\textbf{Change in Futures Price}} \textbf{ or } h = \frac{\Delta V - \Delta S}{\Delta F}$$

Where:
V = S + h x F
V = Value of hedged position
S = Security price
F = Futures price

For a complete or delta-neutral hedge (ΔV = 0), the hedge ratio will be:

$$h = -\frac{\Delta S}{\Delta F} \text{ or Number of contracts} = -\text{Hedge ratio} \times \frac{\text{Loan Value}}{\text{Contract Size}}$$

Hedging Equity Risks

$$N = \frac{V_P}{F \times T} \times \beta$$

where:
N = Number of contracts
V_P = Current value of portfolio

F = Current futures quote
T = Value per tick
β = Beta of portfolio

Options

Option Price = Intrinsic Value + Time Value

- For a call: Intrinsic Value = Current Market Price – Option Strike Price
- For a put: Intrinsic Value = Option Strike Price – Current Market Price

Time Value of Option = Option Price – Intrinsic Value

Put-call parity

C + PV (X) = p + S

where:

c = current price or market value of the European call
X = option strike price
PV (X) = present value of the strike price of European call discounted from the expiry date at the risk-free rate
p = current price or market value of the European put
S = the current market value of the underlying share

Delta = Change in option price/Change in underlying price

$\Delta = \partial V/\partial S$

where V is the option price and S is the underlying price.

Gamma = $\partial^2 V/\partial S$

Warrants and Other Investment Products

Gearing ratio = Share price / (Warrant price x n)

Effective gearing = Delta x Gearing

Delta = n x ∂WP /∂S

where:

Delta = Rate at which warrant price changes with change in share price
∂WP = Change in warrant price
∂S = Change in share price
n = Conversion ratio

Intrinsic Value

Call Warrant: Intrinsic Value = MAX {0, (S - X)/n}

Put Warrant: Intrinsic Value = MAX {0, (X - S)/n}

Where n is the conversion ratio

Conversion Price

Call Warrant: Conversion Price = X + nWP

Put Warrant: Conversion Price = X – nWP

Where n is the conversion ratio

Premium

Call Warrant Premium ($) = nWP + X – S

Put Warrant Premium ($) = nWP - X + S

Cash Settlement

Call Warrant: Cash settlement per warrant = (S – X)/n

Put Warrant: Cash settlement per warrant = (X – S)/n

Adjustment for Dividends

New Exercise Price = Old Exercise Price X Adjustment Factor

Adjustment Factor = (P – SD – ND) / (P – ND)

New Conversion Ratio = Old Conversion Ratio X Adjustment Factor

where P is the last cum-date closing price of the underlying
SD is the Special Dividend per Share
ND is the Normal Dividend per Share

Adjustment for Share Cancellation/Buy Back

New Exercise Price = Old Exercise Price x Adjustment Factor 1 x Adjustment Factor 2

Adjustment Factor 1 = A / B

Adjustment Factor 2 = (P – CD) / P

New Conversion Ratio = Old Conversion Ratio x Adjustment Factor 1 x Adjustment Factor 2

where A = Existing shares
B = Number of shares on an ex-basis
P = the last cum-date closing price of the underlying
CD = Cash distribution per share held immediately prior to the capital reduction/buyback

Adjustment for Share Split

New Exercise Price = Old Exercise Price X Adjustment Factor

New Conversion Ratio = Old Conversion Ratio X Adjustment Factor

where:

Adjustment Factor = A / B

A = Existing shares

B = Number of shares on an ex-basis

Convertible Bonds

Conversion Value = Market Price of Share x Conversion Ratio

Minimum Value of Convertible Bond = MAX { Conversion Value, Straight Value}

$$\text{Market conversion premium ratio (\%)} = \frac{\text{Market Conversion premium per share}}{\text{Share Price}}$$

$$\text{Premium over straight value} = \frac{\text{Market price of convertible bond}}{\text{Straight value}} - 1$$

$$\text{Premium payback period} = \frac{\text{Market conversion premium}}{\text{Income differential per share}}$$

where:

$$\text{Income differential} = \frac{\text{Coupon} - (\text{Conversion ratio} \times \text{Dividend per share})}{\text{Conversion ratio}}$$

Convertible bond value = Straight bond value + Value of call option on stock

Structured Products

Constant Proportion Portfolio Insurance (CPPI)

Cushion Value = Total Portfolio - Floor Value

Multiplier (M) = 1 / Crash Size

Amount Invested in Risky Assets = M x Cushion Value

Amount Invested in Risk-Free Assets = Total Portfolio – Amount Invested in Risky Assets

Structured Notes

Range Accrual Notes (RANs)

Reference index inside range: Payout = P1 x (n/N)

Reference index outside range: Payout = P2 x (N - n)/N

Where:

N = Total number of observations within a period
n = Total number of observations when the index is inside range
P1 = Payout when the index is inside the range
P2 = Payout when the index is outside the range

Inverse Floater Note

Coupon = [X% — Leverage x Floating Rate Index]

Coupon if subject to a floor: Coupon = Max [X% — Leverage x Floating Rate Index, Min Coupon]

Comparison of Different Types of Structured Products

Participation Rate = Discount Sum / Call Premium

Knock-Out Products

Gearing Ratio = Underlying Asset Price / (Price of CBBC x Conversion Ratio)

Theoretical Price of CBBC

$$\text{Bull Contract} = \frac{(\text{Underlying Asset Price} - \text{Strike Price}) + \text{Financial Cost}}{\text{Conversion Ratio}}$$

$$\text{Bear Contract} = \frac{(\text{Strike Price} - \text{Underlying Asset Price}) + \text{Financial Cost}}{\text{Conversion Ratio}}$$

Residual Value

For Category N-CBBCs, there is no residual value if a call event occurs.

R-Category CBBC Residual Values:

$$\text{Bull Contract} = \frac{\text{Settlement Price} - \text{Strike Price}}{\text{Conversion Ratio}}$$

$$\text{Bear Contract} = \frac{\text{Strike Price} - \text{Settlement Price}}{\text{Conversion Ratio}}$$

Where Mandatory Call Event Settlement Price =

- Bull contract: Not lower than the minimum trading price of underlying asset between the period of MCE up to the next trading session.
- Bear contract: Not lower than the maximum trading price of underlying asset between the period from the MCE up to the next trading session.

Maturity Value

$$\text{Bull Contract} = \frac{\text{Settlement Price} - \text{Strike Price}}{\text{Conversion Ratio}}$$

$$\text{Bear Contract} = \frac{\text{Strike Price} - \text{Settlement Price}}{\text{Conversion Ratio}}$$

where MCE Settlement Price = Closing price of the underlying asset on settlement day

Appendix E: Review Questions

Candidates should note that the sole purpose of the Review Questions is to familiarize candidates with the scope and general nature of the examinations, and the format of the examination questions.

The Review Questions are not intended to be used as preparatory study material for the examinations, nor do the questions cover all the material tested in the examination.

The correct answer for each question is in **bold**.

Chapter 2 – Futures

1. There are certain risks involved in hedging in the forward or futures market. Which of the following is **TRUE?**

    a. **Futures contracts have lower counterparty risk, while forward contracts carry the credit risk of the counterparty**.
    b. Futures contracts have basis risk, whereas forward contracts do not.
    c. Forward contracts have the risk of the exchange.
    d. Forward contracts have interim partial settlements.

2. Advantages of futures contract over OTC instruments include:

    a. More transparency in pricing.
    b. Market liquidity.
    c. Reduced credit risk.
    d. Increased credit risk.

    - (a) and (c).
    - (b) and (c).
    - **(a), (b) and (c).**
    - (a), (b) and (d).

3. What is the economic importance and impact of the Futures Market?

    a. Futures are not standardized contracts therefore not readily available.
    b. Futures market is both a less active and important source of vital market information.
    c. **Participants can use contracts to leverage on long or short positions.**
    d. Participants cannot use contracts to leverage on long positions.

4. Spot GBP/USD is trading at 1.2422. Given the annualised US interest rates at 1.799 and annualised British interest rates at 0.70444. What should be the theoretical price for a 1-year forward contract?

   a. **1.2557.**
   b. 1.2487.
   c. 3.1743.
   d. 0.9989.

5. What are the factors affecting basis in the market of a futures contract?

   a. Cost / return of Carry.
   b. Non-difference in coupons between fixed income instruments.
   c. Market rates versus administrated rates.
   d. Yield curve changes.

   - (a) and (d).
   - **(a), (c) and (d).**
   - (a), (b) and (c).
   - (b) and (c).

Chapter 3 – Futures Strategies

6. Using bond futures to hedge a bond portfolio is more complicated because in the case of US T-bonds futures, there are certain options embedded for the sellers and these include delivery timing and choice of deliverable bonds. As a result, we would expect that the futures pricing is _____ than the implied by the cash and carry arbitrage.

   a. **Lower.**
   b. Higher.
   c. Same.
   d. Not enough information provided.

7. Which are the main investment and trading strategies used in the futures market?

   a. Outright trades.
   b. Hedging.
   c. Spread trades.
   d. Bid trades.

   - **(a), (b) and (c).**
   - (b), (c) and (d).
   - (a), (c) and (d).
   - (a), (b), (c) and (d).

8. A local company wishes to take a 6-month USD 50 million loan and the prevailing the 6-month USD lending rate is 2.15%. The September Eurodollar futures contract is trading at 98.15. At the time of entering the hedge, the price of September Eurodollar futures is 98.65. The Eurodollar futures contract size: $1,000,000 with each Eurodollar futures contract represents a 3-month LIBOR period (0.25 years).

   Assuming that the interest rate on the loan is correlated one to one with the Eurodollar, and the company intends to fully hedge its interest rate exposure using September futures. Calculate the number of contracts that will be needed to execute the hedge.

   a. 25
   b. 50
   c. **100**
   d. 150

9. Calculate the hedge ratio required to hedge the interest rate on a 6-month loan for USD 100 million. Assume that the interest rate on the loan is correlated one-to-one with the Eurodollar rate of the nearest 90-day Eurodollar futures contract. Assume a stack hedge is employed. The Eurodollar futures contract size: $1,000,000 with each Eurodollar futures contract represents a 3-month LIBOR period (0.25 years).

   a. 200
   b. 20
   c. **2**
   d. 0.2

10. A player in the financial markets who sells contracts to manage his cash market exposure while holding the underlying stock is a/an:

    a. Arbitrageur.
    b. **Hedger.**
    c. Market maker.
    d. Speculator.

Chapter 4 – Options

11. There are positive relationships between the value of a call option and _____.

    a. Interest rates.
    b. Time to expiration of the call option.
    c. Dividend rate of the underlying stock.
    d. Volatility of the underlying stock.

    - (a) and (c).
    - (a) and (b).
    - (a), (c) and (d).
    - **(a), (b) and (d).**

12. The exercise price of a call option is $50. The price of the call is $6, and it expires in 3 months.

    The current share price is $54. What is the intrinsic value?

    a. $50.
    b. **$4.**
    c. $54.
    d. $6.

13. Which of the following statement is TRUE about Theta and Vega?

    a. Theta measures the change in asset price with respect to time while Vega measures the volatility of the option's portfolio.
    b. Vega measures the change in asset price with respect to time while Theta measures the volatility of the option's portfolio.
    c. Vega is positive for long calls and short puts.
    d. **Vega is positive for a long volatility investment position with call or put options.**

14. Which of the following statements about zero cost options are TRUE?

    a. Since the option is zero cost, it is possible to buy as much as possible to take advantage of its zero-cost structure without having to worry about the risk associated with it.
    b. Zero cost structures give unlimited profits and limited losses.
    c. A zero-cost collar is defined as buying a protective put and selling an out-of-the-money covered call.
    d. A zero cost (costless collar) option is sold with a strike price where the premium received is equal to the premium paid for the put purchased.

    - (b) and (c).
    - (a) and (d).
    - (a) and (c).
    - **(c) and (d).**

15. A trader buys a Nikkei 225 futures contract at 10,000, a put struck at 9500 (premium is 200) and sells a call struck at 10500 (premium is 190). What is the synthetic position created by the trader?

    a. **Bull spread.**
    b. Bear spread.
    c. Ratio put spread.
    d. Short call.

Chapter 5 – Structured Warrants and Daily Leveraged Certificates

16. Which of the following will increase the value of both a call option and a put option?

    a. Increase in the exercise price.

b. **Increase in volatility.**
c. Increase in interest rates.
d. Decrease in time to maturity.

17. The warrant premium is affected by:

a. The length of time before expiration date.
b. The size of the cash dividends paid on the underlying share.
c. The intrinsic value of the warrant.
d. The credibility of the individual buyer.

- (a), (b) and (d).
- **(a), (b) and (c).**
- (b), (c) and (d).
- (a), (b), (c) and (d).

18. A call warrant with an exercise price of $1.00 is trading at $1.20. One warrant converts to one share. What is its conversion price?

a. $1.00.
b. $1.20.
c. **$2.20.**
d. $0.20.

19. The warrant is trading at $0.40. The exercise price is $0.80. The share price is $2.40. What is its gearing? (Assume that the warrant is company-issued; in practice they all have a conversion ratio of 1, unless otherwise stated.)

a. 2X.
b. **6X.**
c. 3X.
d. 5X.

20. Company A's warrant is trading at $0.35; its exercise price is $0.50 while the share price is $0.65. What is its intrinsic value?

a. $0.80.
b. **$0.15.**
c. $0.30.
d. $1.00.

21. The market price of Company B's common share is $5 and the exercise price of the warrant is $3.50. Two warrants can be converted into one common share. The warrant price is currently trading at $2. What is the warrant premium?

a. 10%.
b. **50%.**
c. 57%.
d. 100%.

Chapter 6 – Barrier Options, Binary Options and Callable Contracts

22. For a single "knock out", up-and-out call barrier option, the barrier level is:

a. Below the current spot price of the asset.
b. Equal to the current spot price of the asset.
c. Equal to the strike price.
d. Above the current spot price of the asset.

23. Which of the following statements are CORRECT?

a. For a Category-N callable bull/bear contracts (CBBC), there is no residual value when a mandatory call event occurs
b. For a Category-R CBBCs, there is always a positive residual value when a mandatory call event occurs
c. The Category-N CBBC call price is always equal to the strike price
d. The Category-R CBBC call price is always different from the strike price

- (a), (b) and (c).
- (b), (c) and (d).
- **(a), (c) and (d).**
- (a), (b) and (d).

24. In a double "knock-out" barrier option, the barrier levels _____.

a. For both barriers are higher than the strike price for a call option.
b. For both barriers are lower than the strike price for a call option.
c. Will vary as there will be two strike prices with one barrier level for each strike price.
d. Will vary as there will be one barrier level above the strike price and another barrier level below the strike price.

25. All the following are common features of callable bull/bear contracts (CBBC) and structured warrants except one. Which characteristic is TRUE for CBBC products but NOT for structured warrants?

a. Call price specified.
b. Fixed maturity period.
c. No margin requirement.
d. Traded on an exchange.

26. What are the DISADVANTAGES of barrier options?
a. Low risk of loss of premium if barrier conditions are breached.
b. High risk of loss of premium if barrier conditions are breached.

c. Investor is exposed to counterparty risk on the issuer which arises when dealing in OTC products.
d. Barrier options are not over-the-counter (OTC) products which investor do not pay option premium.

- (a) and (c).
- **(b) and (c).**
- (c) and (d).
- (b) and (d).

Chapter 7 – Structured Deposits and Other Structured Products

27. What should investment professionals consider when helping clients invest in structured products?

a. Risk appetite.
b. Investment experience and knowledge.
c. Investment horizon.
d. Liquidity needs.

- (a), (b) and (c).
- (a), (c) and (d).
- (b), (c) and (d).
- **(a), (b), (c) and (d).**

28. What would be the purposes and uses of OTC Structured Products?

a. Structured products can be customized to meet needs that cannot be fulfilled by standard traditional investments.
b. Structured products provide optimization of yield and returns.
c. Structured products are an alternative rebalancing investment to the direct purchase of the underlying asset.
d. Structured products allow clients to gain leveraged exposure that can potentially augment gains using a smaller investment outlay.

- (a), (b) and (c).
- **(a), (b), (c) and (d).**
- (b), (c) and (d).
- (a), (c) and (d).

29. Which of the following are structured products?

a. Accumulators.
b. Callable range accrual notes.
c. Step-up digital options.
d. Equity linked notes.

- (a), (c) and (d).
- (b), (c) and (d).

- **(a), (b), (c) and (d).**
- (a), (b) and (d).

30. Which of the following instruments can be used as the principal component of a structured note?

a. AAA rated bond.
b. Common shares.
c. Range accrual deposit.
d. Zero-coupon or corporate bonds.

- **(a), (c) and (d).**
- (b), (c) and (d).
- (a) and (d).
- (b) and (c).

31. A structured product offered a minimum 100% return of capital at the end of the investment term, plus 80% of any rise in the FTSE100 index. If the index rose by 40% over the period than the investor would receive back his initial capital in full plus an additional return of 32%. What is the participation rate?

a. 32%.
b. 104%.
c. 80%.
d. 100%.

Chapter 8 – Structured Notes

32. The underlying instruments of a structured note could include products that are related to movements of _____.

a. Interest rates.
b. Equities.
c. Indices.
d. Maturity dates.

- **(a), (b) and (c).**
- (b), (c) and (d).
- (a), (c) and (d).
- (a), (b), (c) and (d).

33. A range accrual note is a structured note where a client receives a higher return if:

a. The reference index is within a certain range.
b. The reference index is beyond a certain range.
c. The coupon rate is within certain range.
d. The coupon rate is beyond certain range.

34. Which structured note would typically be structured with a principal preservation feature?

    a. Range accrual note.
    b. Dual currency investment.
    c. Inverse floater note.
    d. Exchange-traded note.

35. Which of the following statements describe a structured note?

    a. A structured note is a debt instrument, whose return characteristics are linked to the performance of other underlying instruments.
    b. A structured note is governed under the Securities and Futures Act.
    c. A structured note has return characteristics that are not linked to the performance of other underlying instruments.
    d. A structured note is governed by the Guidelines of Outsourcing.

    - (a) and (d).
    - (b) and (d)
    - (a), (b) and (c).
    - **(a) and (b).**

36. Investors buy structured notes mainly because they want yield enhancement and market access. Which of the following factors should an investor consider before investing?

    a. Own liquidity needs.
    b. Terms and condition of the structured note.
    c. Fees and charges of the structured note.
    d. Risk and return.

    - (a), (c) and (d).
    - (a), (b) and (c).
    - (b), (c) and (d).
    - **(a), (b), (c) and (d).**

Chapter 9 – Structured Funds

37. Which of the following statements are TRUE about structured funds?

    a. A combination of equity and fixed income securities and/or derivatives, which achieve specific risk-return profiles or cost savings.
    b. Structured funds range from simple to highly complex funds and cover a wide range of risk exposures.
    c. Structured funds offer some capital preservation and are not designed to earn a market linked return.
    d. Structured funds are not governed by the Code on Collective Investment Schemes under the Securities and Futures Act.

    - **(a) and (b).**

- (b), (c) and (d).
- (a), (b) and (d).
- (a), (b), (c) and (d).

38. A Structured Fund combines both _____ and _____ to provide investors with a degree of both capital preservation and capital appreciation.

    a. Fixed income products; derivatives.
    b. Fixed income products, equity products.
    c. Alternative assets; derivatives.
    d. Commodities; equity products.

39. Which of the following parameters can be used to judge the performance of any fund?

    a. Sharpe ratio.
    b. Total expense ratio (TER).
    c. Underlying asset.
    d. Share class.

    - (a) and (b).
    - (a), (c) and (d).
    - **(a), (b) and (c).**
    - (a), (b), (c) and (d).

40. What is the ADVANTAGE of investing in a structured fund using the constant proportion portfolio insurance (CPPI) strategy?

    a. It has a capital preservation feature.
    b. In a range bond market, the manager would be able to buy low and sell high.
    c. It has lower fees as compared to a direct investment in the underlying asset.
    d. It may not have a 100% exposure to the underlying asset in day one.

Chapter 10 - Contracts for Differences (CFDs)

41. If an investor has a negative view on a stock and believes the stock price will fall. What products will allow him to profit from this view?

    a. Callable bull/bear contracts or CFDs only.
    b. CFDs or options only.
    c. Options or callable bull/bear contracts only.
    d. **CFDs, options or callable bull/bear contracts.**

42. Which statement regarding CFDs and equity futures is INCORRECT?

    a. CFDs have counterparty risk but equity futures do not.
    b. CFDs can be extended by the investor for as long as the investor wishes but equity futures have a fixed maturity date.
    c. CFDs have implicit financing cost as part of its price whereas the financing cost for equity futures is explicitly computed.
    d. CFDs are entitled to dividends and equity futures have no dividend entitlements.

43. What are terms of a CFD?

    a. Underlying asset.
    b. Price of the underlying asset.
    c. Number of units of the underlying asset specified in the contract.
    d. Volatility of the currency.

    - (a), (b) and (d).
    - (a) and (d).
    - (b), (c) and (d).
    - **(a), (b) and (c).**

44. An investor using a CFD trading strategy involving dividend capture should look for companies _____.

    a. That are start-ups and early growth stage.
    b. That do stock splits and issue scrip dividends regularly.
    c. **That have strong earnings and a high payout ratio.**
    d. Whose price performance is strongly driven by macroeconomic factors.

45. Which of the following statements about CFDs and futures contract is TRUE?

    a. **CFDs are mostly traded OTC whereas equity futures are traded on exchanges.**
    b. Both CFDs and equity futures have counterparty risk.
    c. Both CFDs and equity futures are entitled to dividends.
    d. CFDs have margin and leverage whereas equity futures have no margin and leverage.

Chapter 11 – Key Products and Investment Risks for Derivatives and Structured Products

46. Trading in a CFD in an international stock outside the investor's home country has:

    a. No financing cost.
    b. No leverage risk.
    c. Market risk.
    d. Credit risk.

    - (a) and (c)

- (b) and (d).
- **(c) and (d).**
- (a) and (b).

47. Which of the following types of credit default are classified under the International Swaps and Derivatives Association?

    a. Restructuring.
    b. Failure to pay.
    c. Obligation acceleration.
    d. Obligation deceleration.

- **(a), (b) and (c).**
- (b) and (c).
- (a) and (d).
- (a), (b) and (d).

48. Which of the following factors contribute to Basis Risk?

    a. The asset, whose price is to be hedged, may not be exactly the same as the asset underlying the futures contract.
    b. The hedger may be uncertain about the exact date when the asset will be bought.
    c. The hedge may require the futures contract to be closed out well before its expiration date.
    d. The size and unit of measurement of futures contracts may not correspond with the size of the underlying hedged position.

- (a), (b) and (d).
- (a), (c) and (d).
- (b), (c) and (d).
- **(a), (b), (c) and (d).**

49. What are the key risks involved in investing in structured products?

    a. Market risk.
    b. Liquidity risk.
    c. Reinvestment risk.
    d. Currency risk.

- (a) and (b).
- **(a), (b), (c) and (d).**
- (b) and (c).
- (a) and (c).

50. Market risk will be affected by:

    a. Macroeconomic factors such as interest rates or inflation rates.
    b. Volatility in currency rates.
    c. Political factors such as government instability.
    d. The sudden departure of the CEO of a large corporation.

    - **(a), (b) and (c).**
    - (b), (c) and (d).
    - (a), (c) and (d).
    - (a), (b), (c) and (d).

Appendix F: Essential Readings

1. **General – Investments, Options**

   - Bodie, Zvi, Kane, Alex and Marcus, Alan*; Investments;* McGraw-Hill (9th Edition, 2010)

2. **Foreign Exchange**

   - Hicks, Alan; Managing Currency Risk Using Foreign Exchange Options (International Treasury Management Series); CRC Press (2000)
   - Walmsley, Julian; Foreign Exchange and Money Markets Guide (Frontiers in Finance Series); John Wiley and Sons (2000)

3. **Fixed Income**

   - Fabozzi, Frank J. and Leibowitz, Martin L; Fixed Income Analysis (CFA Institute Investment Series); Wiley (2007)
   - CME Group, Labuszewski, John W., Nyhoff, John E., Peterson, Paul E. and Co, Richard; *The CME Group Risk Management Handbook: Products and Applications;* Wiley Finance (2010)

4. **Futures, Options and Derivatives**

   - Chance, Don; *Analysis of Derivatives;* Association for Investment Management and Research (2002)
   - Hull, John C.; *Options, Futures & Other Derivatives;* Prentice Hall (9th Edition, 2014)
   - Kolb, Robert and Overdahl, James A.; *Financial Derivatives: Pricing And Risk Management, The Robert W. Kolb Series in Finance;* John Wiley & Sons (2010)
   - Kolb, Robert and Overdahl, James A.; *Futures, Options, and Swaps;* Blackwell Publishing (2007)
   - Wilmont, Paul; *Derivatives - The Theory and Practice of Financial Engineering*; John Wiley & Sons (1998)

5. **Structured Products**

   - Blumke, Andreas; *How To Invest In Structured Products – A Guide For Investors & Asset Managers;* Wiley Finance, John Wiley & Sons (2009)

- Lancaster, Brian, Schultz, Glenn and Fabozzi, Frank J.; *Structured Products and Related Credit Derivatives: A Comprehensive Guide for Investors (Frank J. Fabozzi Series);* John Wiley & Sons (2008)
- Swiss Structured Products Association (https://sspa.ch/en)

6. Structured Funds & Structured ETFs

- Fabozzi, Frank J., Davis, Henry A., and Choudhry, Moorad; *Introduction to Structured Finance (Frank J. Fabozzi Series);* Wiley Finance, 2006

7. Other Sources

- Monetary Authority of Singapore (https://www.mas.gov.sg)
- Moneysense (www.moneysense.gov.sg)
- Singapore Exchange (www.sgx.com)

Appendix G: Glossary[50]

A

Absolute returns The returns achieved over a certain time period. Absolute returns do not consider the risk of the investment or the returns achieved by similar investments.

Accounts payable Money owed by a company to suppliers that have extended the company credit.

Accounts receivable Money owed to a company by customers who purchase on credit.

Accredited Investor Is a person or a business entity considered to be financially savvy and has higher risk tolerance with privileged access to complex and risky products.

Additional margin The margin or amount that will be required to top up a securities or derivatives trading account, if the balance falls below the maintenance margin level.

Alpha Outperformance relative to a relevant market benchmark.

Alternative investments A diverse asset class that typically includes private equity, real estate, and commodities. It provides an alternative to traditional investments, such as debt and equity securities.

American option An option contract that can be exercised at any time before expiration.

Amortisation The process of expensing the costs of intangible assets over their useful lives.

Annualizing returns A geometric average to show what an investor would earn over a period of time if the annual return was compounded. An annualized total return provides only a snapshot of an investment's performance.

Annuity The exchange of an initial amount for a fixed number of future payments of a certain amount.

Appraisal Assessment or estimation of the value of an asset, such as real estate, that is subject to certain assumptions, which may not always be realistic.

Appreciation A situation in which a currency is getting stronger relative to other currencies.

Arbitrage An opportunity to make money by taking advantage of a price difference between two markets.

Arbitrage pricing theory (APT) A multi-factor asset pricing model based on the idea that an asset's returns can be predicted using the linear relationship between the asset's expected return and a number of macroeconomic variables that capture systematic risk. It is a useful tool for analyzing portfolios from a value investing perspective, in order to identify securities that may be temporarily mispriced.

Ask price The price at which a dealer is willing to sell an asset or a security, typically qualified by a maximum quantity (ask size). Also called *offer price*.

Assets Items that have value and include real assets and financial assets.

Asset allocation The process to determine the proportion of a portfolio to hold in various asset classes or the proportion of a portfolio held in various asset classes.

Asset-backed securities (ABS) Financial securities created by securitisation whose associated payments and value are backed by a pool of underlying assets,

[50] The definitions in this Glossary are sourced from Investopedia, The International Swaps & Derivatives Association (ISDA).

such as car loans, credit card receivables, bank loans, or airplane leases.

Asset class A broad grouping of similar types of investments, such as shares, bonds, real estate, and commodities.

Asset managers See *investment managers*.

Asset turnover A measure that indicates the volume of revenues being generated by the assets used in the business, or how effectively the company uses its assets to generate revenue.

At-the-money option An option is at-the-money when there is no profit from exercising the option. Also see - *in-the-money option* and out-*of-the-money option.*

B

Balance of payments Record that tracks transactions between residents of one country and residents of the rest of the world over a period of time, usually a year.

Balance sheet A statement of the company's financial position at a specified point in time; essentially, it shows the company's assets, liabilities or debt, and owner-supplied capital or equity.

Banks Financial institutions that collect deposits from savers and transform them into loans to borrowers.

Banker's acceptance An instrument representing a promised future payment by a bank. The payment is accepted and guaranteed by the bank as a time draft to be drawn on a deposit.

Barrier option A type of option or derivative where the payoff depends on whether or not the underlying asset has reached or exceeded a predetermined price (or barrier level).

Barriers to entry Obstacles, such as licences, brand loyalty, or control of natural resources, that prevent competitors from entering the market.

Basket warrant A warrant with that has a basket of underlying shares as the underlying asset.

Basis point A measure equal to 0.01% or 0.0001.

Basis swap A transaction in which one party pays periodic amounts of a given currency based on a floating rate and the other party pays periodic amounts of the same currency based on another floating rate, with both rates reset periodically.

Basis trading An arbitrage trading strategy that aims to profit from small basis point changes in value between two securities or derivatives instruments.

Bear spread An options strategy that is designed to profit from fall in the price of the underlying security. A bear spread can be created by either call or put options. See *Bear call spread, or Bear put spread.*

Bear call spread A bearish options strategy that is created with call options.

Bear put spread A bearish options strategy that is created with put options.

Bearish strategy This is used when the investor expects the underlying asset price to move downwards.

Benchmark A comparison portfolio (e.g., the S&P 500 Index).

Beta A generic term for market risk, systematic risk, or non-diversifiable risk.

Bid price The price at which a dealer is willing to buy an asset or a security, typically qualified by a maximum quantity (bid size).

*Blue chip C*ompanies known for quality, reliability and the ability to weather downturns and operate profitably in the face of adverse economic conditions, which helps to contribute to their long record of stable and reliable growth

Board of directors A group of people whose job is to monitor the company's business activities on behalf of its shareholders.

Bond A formal contract that represents a loan from an investor (bondholder) to an issuer. The contract describes the key terms of the debt obligation, such as the interest rate and the maturity.

Bond forward A transaction in which one party agrees to pay an agreed price for a specific amount of a bond an issuer or a basket of bonds of several issuers at a future date, and the other party agrees to pay a price for the same amount of the same bond to be set on a specific date in the future.

Bond futures A derivative contract which gives the futures contract holder the obligation to purchase or sell a bond on a specified date at a predetermined price.

Bond option A transaction in which one party grants to the other party (in consideration for a premium payment) the right, but not the obligation, the purchase a specific amount of a bond of an issuer, at a specified strike price.

Bond swaps Bond swaps consist of selling one debt instrument and using the proceeds to purchase another debt instrument.

Bonus Issue An issue of additional shares to shareholders instead of a dividend, in proportion to the shares already held.

Book values Balance sheet values of a company's assets, liabilities, and equity.

Breakeven point The number of units produced and sold at which the company's profit is zero—that is, revenues exactly cover total costs. If the company sells less units than the breakeven point, it will suffer a loss. In contrast, if it sells more units than the breakeven point, it will make a profit.

Brokerage services Trading services provided to clients who want to buy and sell securities; they include not only execution services (that is, processing orders on behalf of clients) but also investment advice and research.

Brokers Trading services providers who act as agents and, in exchange for a commission, arrange trades by finding sellers for their clients who want to buy and buyers for their clients who want to sell.

Bull spread An options strategy that is designed to profit from a rise in the price of the underlying security. A bull spread can be created with either call options or put options. See *Bull call spread, or Bull put spread.*

Bull call spread A bullish options strategy that is created with call options.

Bull put spread A bullish options strategy that is created with put options.

Bullish strategy This is used when the investor expects the underlying asset price to move upwards.

Business cycles Economy-wide fluctuations in economic activity.

Business trust An unincorporated commercial organization in which assets are held in trust by trustees, who have been appointed to manage the assets of one or more beneficiaries.

Butterfly spread A market neutral options strategy that consists of a combination of a bull spread and a bear spread.

Buyouts funds Private equity investment strategy that consists of financing established companies that require capital to restructure and facilitating a change of ownership.

C

Call market Market in which trades can be arranged only when the market is called, which is usually once a day.

Call option The right (but not the obligation) to buy an underlying at the exercise price until the option expires.

Call spread An options spread strategy that is created with call options.

Call warrant See *Structured warrant.*

Calendar spread See *Horizontal spread.*

Callable bond A bond that provides the issuer with the right to buy back (call) the bond from bondholders prior to the maturity date at a pre-specified price.

Cap A transaction where one part pays a single or periodic fixed amount, and the other party pays periodic amounts of the same currency based on the excess, if any, of a specific floating rate (in the case of an interest rate cap), rate or index. The floating rate is reset periodically.

Capital account A component of the balance of payments that reports capital transfers between domestic entities and foreign entities, such as debt forgiveness or the transfer of assets by migrants entering or leaving the country.

Capital Asset Pricing Model (CAPM) It describes the relationship between systematic risk and expected return for assets, particularly stocks. CAPM is widely used throughout finance for pricing risky securities and generating expected returns for assets given the risk of those assets and cost of capital.

Capital markets See *financial markets*.

Capitalisation-weighted indices Indices for which the weight assigned to each security depends on the security's market capitalisation—that is, the market price of the security multiplied by the number of units

outstanding of the security. Also called *cap-weighted*, *market-weighted*, or *value-weighted indices*.

Capitalised Classifying a cost as generating long-term economic benefits and reporting it as an asset rather than charging it as an expense to current operations.

Capped warrant See *Discount certificates.*

Cash Flow Statement A financial statement that summarizes the amount of cash and cash equivalents the go in and out of a company.

Caveat emptor The onus is on the buyer to do the due diligence check in a buy and sell transaction.

Central counterparty clearing (CCP) A financial institution (e.g., Singapore Exchange Derivatives Clearing Pte Ltd) that takes on counterparty credit risk between parties to a transaction and provides clearing and settlement services for trades in foreign exchange, securities, options, and derivative contracts.

Central Provident Fund (CPF) A compulsory comprehensive savings plan for working Singaporeans and permanent residents primarily to fund their retirement, healthcare, and housing needs.

Certificate of Deposit (CD) A bank-issued investment instrument where a bank typically agrees to repay the principal plus interest on a fixed maturity date. CDs are typically negotiable, i.e.; they can be sold on the secondary market, allowing the principal and accrued interest to be redeemed before maturity.

Clearing All activities that occur from the arrangement of the trade to its settlement.

Clearing houses Trading services providers that arrange for final settlement of trades.

Closed-end funds Pooled investment vehicles that have a fixed number of shares and thus do not issue or redeem shares on demand. Investors who want to buy or sell closed-end funds must trade with investors willing to sell or buy these funds.

Coincident indicators Measures of economic activity that are intended to measure the current state of the economy rather than the past or to predict the future. Coincident indicators have a tendency to change at the same time as the economy measured as a whole.

Collar A collar transaction is a combination of a cap and a floor where one party pays the floating rate on the cap, and the other party pays the floating rate on the floor. Also see *caps* and *floors.*

Costless collar See *Zero cost collar.*

Collateral Specific assets (generally a tangible asset) that a borrower pledges to a lender to secure a loan.

Collateralised debt obligation (CDO) A structured financial product that pools together cash flow-generating assets and repackages the asset pool into discrete tranches that can be sold to investors.

Collective Investment Scheme Is an investment scheme wherein several individuals come together to pool their money for investing in a particular asset(s) and for sharing the returns arising from that investment.

Commercial paper (CP) A short-term, unsecured promissory note usually issued in bearer form, ie. it is a negotiable instrument. By issuing the paper, the issuer promises to pay the bearer the face value of the paper on a fixed maturity date. CP is usually issued at a discount.

Commodity forward A transaction in which one party agrees to buy a specified quantity of a commodity at a future date at an agreed price, and the other party agrees to pay a price for the same quantity to be set on a specified date in the future.

Commodity funds Mutual funds that seek to track the underlying price of various commodities and natural resources. This can include assets such as corn, copper, oil, wheat and gold.

Commodity option A transaction in which one party grants to the other party (in consideration for a premium payment) the right, but not the obligation, to buy (in the case of a call) or sell (in the case of a put) a specified quantity of a commodity at a specified strike price.

Commodity swap A type of swap agreement where the floating price based on an underlying commodity is swapped for a fixed price of the same underlying commodity over a specified period.

Commodity warrant A warrant with that has a commodity as the underlying asset.

Commodities Physical products, such as precious and base metals (e.g., gold, copper), energy products (e.g., oil), and agricultural products that are typically consumed (e.g., corn, cattle, wheat) or used in the manufacture of goods (e.g., lumber, cotton, sugar).

Common shares Also known as common stocks, ordinary shares, or voting shares, it is the main type of equity security issued by a company. It represents an ownership stake in the company.

Company warrant A warrant that is issued by listed companies to existing shareholders, usually as a "sweetener" attached to an offering of bonds or rights issues for shares. Company warrants are essentially long-dated "American-style" call options, be exercisable at any time during the life of the warrants.

Compound interest Interest that is calculated on principal and interest; it assumes reinvestment of interest received. Compound interest is often referred to as interest on interest.

Compound option An option on an option.

Condor spread A market neutral options strategy that limits both gains and losses while seeking to profit from either low or high volatility. A condor spread is similar to a butterfly spread in that it also consists of 4 options, but the maximum profit zone for a condor is much wider than that for a butterfly, although the trade-off is a lower profit potential.

Conflict of interest When either the employee's personal interests or the employer's interests conflict with the interests of the client. Conflicts of interest can also arise when employee's and employer's interests conflict.

Consumer price index Constructed by determining the weight (or relative importance) of each product and service in a typical household's spending in a particular base year and then measuring the overall price of that basket of goods from year to year.

Contracts for differences (CFDs) A contract between two parties where differences in the settlement between the open and closing trade prices are settled in cash. There is no delivery of physical goods or securities with CFDs.

Contract specifications An agreement that sets out the terms & conditions under which an investment will be made. For futures contracts, the contract specifications list out the key information such as contract size and value, tick size, trading hours, denomination of underlying asset, etc.

Conversion ratio The number of warrants needed to be exercised to buy or sell one unit of the underlying security.

Convertible bond A bond that offers the bondholder the right to convert the bond into a pre-specified number of shares of common stock of the issuing company.

Correlation A measure of the strength of a relationship between two variables; essentially, two variables are correlated when a change in one variable is always accompanied by a change in another variable. Variables can be positively or negatively correlated.

Correlation coefficient A number between –1 and +1 that measures the consistency or tendency for two variables to move in tandem with each other.

Counterparty risk Risk that one of the parties to a contract will fail to honour the terms of the contract.

Coupon rate The interest rate for a bond. The bond's coupon rate multiplied by its par value equals the annual interest owed to the bondholders.

Covered call In a covered call options strategy, an investor holding a long position in an underlying asset then writes (sells) call options on that same asset.

Credit default swap (CDS) A contract that protects the buyer against a loss of value in a debt security or index of debt securities. The contract will specify under what conditions the other party has to make payment to the buyer of the CDS.

Credit rating Assessment of the credit quality of a bond based on the creditworthiness of the issuer.

Credit rating agencies Investment research service providers that specialise in providing opinions about the credit quality of bonds and of their issuers.

Credit risk For a lender, the risk of loss caused by a borrower's failure to honour the contract and make a promised payment in a timely manner.

Credit spread The difference between a risky bond's yield and the yield on a government bond with the same maturity.

Currency forward An agreement that locks in the exchange rate for the purchase or sale of a currency on a future date. Currency forwards are traded in the over-the-counter markets.

Currency forward An agreement that locks in the exchange rate for the purchase or sale of a currency on a future date. Currency forwards are traded in the over-the-counter markets.

Currency future An exchange-traded futures contract that specifies the price in one currency at which another currency can be bought or sold at a future date.

Currency risk The risk associated with the fluctuation of exchange rates; also called *foreign exchange risk*.

Currency option An option contract that gives the holder the right, but not the obligation, to buy or sell currency at a specified exchange rate. Also known as a *foreign exchange option.*

Currency swap The exchange of debt service obligations denominated in different currencies.

Currency translated warrant A warrant that is based on an underlying index, where the settlement and trading currency is different from the currency of the underlying index.

Current account A component of the balance of payments that indicates how much a country consumes and invests (outflows) with how much it receives (inflows). It includes three components, the goods and services account, the income account, and the current transfers account.

Current account balance The sum of the goods and services, income, and current transfers accounts.

Current account deficit A negative current account balance.

Current account surplus A positive current account balance.

Current assets Short-term assets that are expected to be converted into cash, used up, or sold within the current operating period (usually one year), such as cash, inventories, and accounts receivable.

Current liabilities Short-term liabilities that must be repaid within the next year.

Current ratio A liquidity ratio calculated as current assets divided by current liabilities.

Current yield The annual coupon payment divided by the current market price.

Custodians Typically, banks and brokerage firms that hold money and securities for safekeeping on behalf of their clients.

D

Daily leveraged certificate (DLC) A short-term and high-risk exchange-traded derivative that allows investors to take amplified long or short positions on an underlying asset (typically an index).

Dark pools Type of alternative trading venue that does not display orders from clients to other market participants.

Dealers Trading services providers who participate in their clients' trades and stand ready to buy or sell when their clients want to sell or buy, providing liquidity and profiting when they can buy securities for less than they sell them. Also called *market makers*.

Debenture *A* type of debt instrument that is not secured by collateral and usually has a term greater than 10 years.

Debt-to-equity ratio A measure of financial leverage that indicates the extent to which debt is used in the financing of the company.

Debt-to-total assets ratio An indicator of a company's financial leverage where the company's total assets are financed by creditors.

Debt security A debt instrument, such as a government bond, corporate bond, certificate of deposit, municipal bond or preferred stock, that can be bought or sold between two parties and has basic terms defined.

Default A situation in which the bond issuer fails to make the promised payments.

Deflation A persistent and pronounced decrease in prices across most products and services in an economy.

Delta Measures the change in option price vs a change in the price of the underlying asset.

Delta In the context of warrants, delta measures the change in the warrant price vs a change in the price of the underlying asset.

Demand The quantity of a product or service buyers are willing and able to buy at a given price.

Delivery date The final date by which the underlying commodity for a futures or forward contract must be delivered for the terms of the contract to be fulfilled.

Deposit Insurance A scheme that protects depositors' money with a member bank for up to a certain threshold per depositor per bank. All full banks and finance companies in Singapore are members of the scheme.

Depository Receipt A security issued by a financial institution that represents an economic interest in a foreign company. The financial institution holds the foreign company's shares in custody and issues depositary receipts against the shares held. These depositary receipts trade like common stock on the local stock exchange.

Depreciation The process of allocating the cost of an asset over the asset's estimated useful life; a situation in which a currency is getting weaker relative to other currencies.

Depreciation expense The amount of depreciation allocated each year and reported on the income statement as an expense.

Derivatives Contracts (agreements to do something in the future) that derive their value from the performance of an underlying asset, event, or outcome.

Designated market maker (DMM) A dealer or party who is responsible for providing quotes and liquidity for the products that been issued.

Diagonal spread An options trading strategy that involves buying and selling multiple options of the same underlying security but difference strike prices and expiration dates.

Direct investments Purchase of securities issued by companies, governments, and individuals and of real assets, such as real estate, art, or timber.

Discount certificates Structured product that give the investor an opportunity to buy an underlying asset at a discount to the market price. Investors buy the discount certificate in the expectation that the price of the underlying asset at expiry will be above a specific cap level. Otherwise, the holder will receive the market value of the underlying asset. And if the price exceeds the cap level, no more income will be earned, as the cap level prevents any further gains.

Discount rate The rate used to calculate the present value of some future amount.

Discounted Cash Flow (DCF) Analysis A valuation method used to estimate the value of an investment based on it's future cash flows. In a DCF, future cash flows are estimated and discounted to their respective present values. They are then added to the discounted terminal value to reach an estimated company value.

Discrete data Data that show observations only as distinct values.

Discretionary relationships Relationships that permit the service provider to act on behalf of the client.

Distressed Private equity investment strategy that focuses on purchasing the debt of troubled companies that may have defaulted or are on the brink of defaulting.

Distribution The set of values that a variable can take, showing their observed or theoretical frequency of occurrence.

Diversification The practice of combining assets and types of assets with different characteristics in a portfolio for the purpose of reducing risk.

Dividend per share The amount of cash dividends the company pays for each share outstanding.

Dividend Discount Model (DDM) A mode of determining whether a stock is over- or undervalued. According to the DDM, when dividend is assumed to grow at a constant rate, the price of a stock can be estimated by dividing the annual dividend by the value of the required rate of return less the growth rate.

Dividend Payout Ratio A measure of the portion of the earnings that are paid out in dividends

Dividend yield The ratio of a company's annual dividend compared to its share price.

Document A piece of written, printed, or electronic matter that provides information or evidence or that serves as an official record.

Dollar-weighted Rate of Return A measure of the performance of an investment. The dollar-weighted rate of return is calculated by finding the rate of return that will set the present values of all cash flows equal to the value of the initial investment.

E

Earnings per share The amount of income earned during a period per share of common stock; net income divided by the number of shares outstanding.

Earnings yield (E/P) It is the percentage of how much a

company earned per share. This is the inverse of P/E ratio.

EBITDA Earnings before interest, taxes, depreciation, and amortization, is a measure of a company's overall financial performance and is used as an alternative to simple earnings or net income in some circumstances.

Economic growth The percentage change in real output (real GDP) for an economy.

Economic indicator A measure that offers insight regarding economic activity.

Economics The study of production, distribution, and consumption or the study of choices in the presence of scarce resources.

Economies of scale Cost savings arising from a significant increase in output without a simultaneous increase in fixed costs.

Effective gearing In the context of warrants, effective gearing is calculated by multiplying the delta of a warrant by its gearing.

Efficient Frontier A set of optimal portfolios that offer the highest expected return for a defined level of risk or the lowest risk for a given level of expected return.

Efficient Market Hypothesis (EMH) An investment theory that that hypothesise that share prices reflect all information at any given time and in a liquid market.

Electronic Communication Network Is a computerized system that automatically matches buy and sell orders for securities in the market and connects major brokerages and individual traders for direct trades without a middleman.

Elliott Wave Theory A form of technical analysis that finance traders use to analyse financial market cycles and forecast market trends by identifying extremes in investor psychology, highs and lows in prices, and other collective factors.

Enterprise value **(EV)** is a measure of a company's total *value*, often used as a more comprehensive alternative to equity market capitalization. EV includes in its calculation the market capitalization of a company but also short-term and long-term debt as well as any cash on the company's balance sheet.

Enterprise Value to Sales (EV/S) ratio A financial ratio that compares the total value of the company to its sales. Some analysts regard EV/S as a superior metric to Price-to-sales.

Equity Assets minus liabilities; the shareholders' (owners') investment in the company.

Equity futures contracts A futures contract that allows investors to buy (long) or sell (short) equities or shares for future delivery at an agreed price which can be higher or lower than the current price.

Equity index option An option contract that gives the option holder the right to buy / sell a specific stock index at a specified price by a specified expiration date.

Equity linked note A type of structured product that consists of a debt instrument, usually a bond, and selling a put option on an underlying equity security (e.g., a single stock, basket of stocks or an equity index). The final payout is based on the return of the underlying equity security.

Equity security Represents ownership interest held by shareholders in an entity (a company, partnership or trust), realized in the form of shares of capital stock.

Eurobond International long-term debt securities with maturities over one year denominated in any Eurocurrency. International distribution is a key feature, and they are usually in bearer form, but the bonds can be issued in any currency or any interest basis.

European Market Infrastructure Regulation The body of European legislation for the regulation of over-the-counter derivatives.

European option An option contract that can only be exercised on the expiration date.

Exchange rate The rate at which one currency can be exchanged for another.

Exchange-traded funds (ETFs) Pooled investment vehicles that are typically passively managed to track a particular index or sector and that trade continuously as common stocks on exchanges or through dealers.

Exchange-traded notes (ETNs) Unsecured debt securities, similar to bonds, that track an underlying index of securities and trade on a major exchange like a stock. Unlike bonds, ETNs do not pay interest payments.

Exercise price Specified in an options contract, the price to trade the underlying in the future. Also called the *strike price*.

Exotic options Refer to non-standard types of options contracts. The opposite of Plain vanilla options.

***Expense ratio* (ER),** also sometimes known as the management *expense ratio* (MER), measures how much of a fund's assets are used for administrative and other operating *expenses.*

Expenses The cost of using up company resources to earn revenues. Typical expenses include operating expenses (such as cost of sales; selling, general, and administrative expenses; and depreciation expenses); interest expenses; and income taxes.

Exports Products and services that are produced within a country's borders and then transported to another country.

F

Fair value Value that reflects the amount for which an asset could be sold in an arm's length transaction between willing and unrelated parties.

Fed Fund Rate The interest rate at which banks and other depository institutions lend money to each other, usually on an overnight basis.

Federal Reserve aka The Feds, is the central banking system of the United States of America

Financial Advisers Act (FAA) Provides guidance to financial advisers about FAA regime and covers topics including licensing requirements, exemptions, representative notification framework, business conduct and reporting requirements. The Monetary Authority of Singapore is the regulatory body responsible for the administration of the FAA.

Financial assets Claims on other assets; for example, a share of stock represents ownership in a company or a claim to some of the company's assets and earnings.

Financial capital Money provided to individuals, companies, and governments to finance their needs.

Financial institutions Financial intermediaries, such as banks and insurance companies, whose role is to collect money from savers and to invest it in financial assets.

Financial markets Places where buyers and sellers can trade securities; also called equity markets.

Fiscal policy The use of taxes and government spending to affect the level of aggregate expenditures.

Fixed costs Costs that do not fluctuate with the level of output of the company.

Fixed rate bonds A bond with a finite life that offers a coupon rate that does not change over the life of the bond. Also known as *straight bonds*.

Floating rate bonds A bond with a finite life that offers a coupon rate that changes over time. Also known as *variable-rate bonds*.

Floor A transaction in which one party pays a single or periodic amount, and the other party pays periodic amounts of the same currency based on the excess of a specified per annum rate (in the case of an interest rate floor), rate or index.

Foreign exchange market A highly integrated decentralised network in which currencies are traded.

Foreign exchange risk See *Currency risk*.

Foreign exchange (FX) warrant A warrant that has an exchange rate between two currencies as the underlying asset.

Forward contract An agreement between two parties in which one party agrees to buy from the seller an underlying at a later date for a price established at the start of the contract.

Forward market Foreign exchange market in which currencies are traded now but delivered at some future date.

Forward rate The exchange rate for forward market transactions.

Foundations Grant-making institutions funded by gifts and by the investment income that they produce.

Fraud Intentional deception, such as deliberately causing or falsely reporting losses to collect insurance settlements.

Free Cash Flow to the Firm (FCFF) The amount of cash flow from operations available for distribution after depreciation expenses, taxes, working capital, and investments are accounted for and paid. FCFF is essentially a measurement of a company's profitability after all expenses and reinvestments.

Front office Client-facing activities that provide direct revenue generation, such as sales, marketing, and customer service activities.

Funds of funds Investment vehicles that invest in other funds.

Future value The amount to which a payment or series of payments will grow by a stated future date.

Futures contract An agreement that obligates the seller, at a specified future date, to deliver to the buyer a specified underlying in exchange for the specified futures price.

G

Gamma Measures the change in delta vs change in the price of the underlying asset. See the definition of *Delta.*

GDP per capita Gross domestic product divided by the population; a measure of average output or income per person.

Gearing In the context of warrants, gearing indicates how many more warrants can be bought with a certain amount of capital compared to buying the underlying share.

Generally Accepted Accounting Principles aka **GAAP** A common set of accepted accounting principles, standards, and procedures that companies and their accountants must follow when they compile their financial statements.

Geometric average The average compounded return for each period; the average return for each period assuming that returns are compounding.

Global Investment Performance Standards (GIPS) A set of voluntary standards used by investment managers to ensure the full disclosure and fair representation of their investment performance.

Goodwill An intangible asset that arises when a company purchases another company and pays more than the fair value of the net assets (assets minus liabilities) of the purchased company.

Gross domestic product The total value of all final products and services produced within an economy in a given period of time (output definition), or equivalently, the aggregate income earned by all households, all companies, and the government within an economy in a given period of time (income definition). Nominal GDP uses current market values. Real GDP adjusts for changes in price levels.

Gross profit Sales minus the cost of sales.

Growth capital Private equity investment strategy that usually focuses on financing companies with proven business models, good customer bases, and positive cash flows and profits but that need additional capital to support their growth plans.

H

Hedge The usage of financial instruments or derivatives to reduce or eliminate risk caused by fluctuations in the prices of commodities, securities, or currencies.

Hedge ratio A ratio that compares the value of a position protected through the use of a hedge with the size of the entire position itself.

Hedge funds Private investment pools that investment managers organise and manage. They are characterised by their availability to only a limited number of investors, agreements that lock up the investors' capital for fixed periods, and performance-based managerial compensation.

High-yield bonds See *non-investment-grade bonds*.

Horizontal spread An options trading strategy that involves options of the same underlying stock, same strike prices, but different expiration month. It can be created with either all calls or all puts. Also known as *calendar spread* or *time spread*.

Hyperinflation Price increases so large and rapid that people find it difficult to purchase products and services.

I

In-the-money option -An option is in-the-money when there is a positive profit from exercising the option. Also see *at-the-money option* and *out-of-the-money option.*

Immunization A hedging strategy that matches the duration of assets and liabilities.

Implied volatility The market's estimate regarding the warrant's underlying price volatility for the duration of the life of the warrant.

Imports Products and services that are produced outside a country's borders and then brought into the country.

Income statement A financial statement that identifies the profit or loss of a company during a given time period, such as one year.

Index fund A portfolio of securities structured to track the returns of a specific index called the benchmark index.

Index warrant A warrant that has an index as the underlying asset.

Industrial production A measure of economic output by the following three segments of an economy: manufacturing, mining, and utilities.

Inflation The percentage increase in the general price level from one period to the next; a sustained rise in the overall level of prices for products and services.

Inflation risk The risk associated with inflation.

Information ratio A reward-to-risk ratio defined as the portfolio's mean active return (the difference in average return between the portfolio and its benchmark) over its active risk (tracking error).

Initial margin The percentage of the purchase price of a security or derivative that must be covered by cash or collateral when setting up the margin account.

Initial public offering (IPO) The first issuance of common shares to the public by a formerly private corporation.

Insider trading Trading while in possession of material non-public information.

Institutional investor A nonbank person or organization that trades securities in large quantities or dollar amounts that it qualifies for preferential treatment and lower commissions

Insurance companies Financial institutions that help individuals and companies offset the risks they face; also among the largest investors.

Insurance company A company that sells insurance contracts (policies) that provide payments in the event that losses occur.

Intangible assets Assets lacking physical substance, such as patents and trademarks.

Inter-commodity spread A futures or options trade that aims to take advantage of the value differential between two or more related commodities.

Inter-delivery spread A futures or options trade that aims to take advantage of the value differential between two contracts in the same commodity but with different delivery months.

Inter-market spread A futures or options trade that aims to take advantage of the value differential between two contracts that trade on different exchanges.

Interest Payment for the use of borrowed money.

Interest rate cap See *Caps.*

Interest rate collar An options strategy that uses interest rate options to hedge an investor's exposure to interest rate fluctuations. The investor sells an interest rate floor / cap to collect the premium, to fund the purchase of an interest rate cap / floor.

Interest rate floor See *Floors.*

Interest rate option -An option that allows the holder to benefit from changes in interest rates. Interest rate options are option contracts on the rate of bonds like U.S. Treasury securities or Eurodollar futures.

Interest rate risk -The risk associated with decreases in bond prices resulting from increases in interest rates.

Interest rate swap A forward contract that usually involves the exchange of a fixed interest rate for a floating rate, or vice versa, to manage interest rate risk or exposure.

International Financial Reporting Standards (**IFRS)** A set of common rules so that financial statements can be consistent, transparent and comparable around the world. IFRS are issued by the International Accounting Standards Board (IASB).

Interest coverage ratio A debt ratio and profitability ratio used to determine how easily a company can pay interest on its outstanding debt.

Inventory turnover A ratio showing how many times a company has sold and replaced inventory in a given period.

Intra-commodity spread A futures or options trade that aims to take advantage of the value differential between two contracts in the same commodity.

Intra-delivery spread A futures or options trade that aims to take advantage of the value differential

between two contracts in the same commodity and the same delivery month.

Intra-market spread A futures or options trade that aims to take advantage of the value differential between two contracts that trade on the same exchange.

Intrinsic value The perceived or calculated value of an asset, an investment, or a company. For options, intrinsic value refers to the option's "moneyness", i.e.; the potential profit or loss from the exercise of an option at a point in time.

Inventories The unsold units of production on hand.

Investment banks Financial intermediaries that assist companies and governments raise capital and can provide other services, such as strategic advisory services, brokerage and dealing services, and research services; also known as merchant banks.

Investment companies that exist solely to hold investments on behalf of their shareholders, partners, or unitholders, including mutual funds, hedge funds, venture capital funds, and investment trusts.

Investment-grade bonds Bonds rated BBB– or higher by Standard & Poor's and Fitch or Baa3 or higher by Moody's.

Investment managers Investment professionals who receive authority from their clients to trade securities and assets on their behalf. Also called *asset managers*.

Investment policy statement (IPS) A written planning document describing a client's investment objectives—return requirements and risk tolerance—over a relevant time horizon, along with constraints that apply to the client's portfolio. The IPS serves as a guide to what is required and acceptable in the investment portfolio.

Investment risk The risk associated with investing that arises from the fluctuation in the value of investments.

Investment vehicles Assets, such as securities and real assets, created by the investment industry to help investors move money from the present to the future, with the hope of increasing the value of their money.

Issuers Typically companies and governments that sell securities to investors in exchange for cash to raise money.

J

J curve The graphical representation of net cash flow (that is, the cash distributions net of carried interest minus the sum of the capital calls and management fees) for limited partners. It shows that net cash flows are negative in the early years of a fund but turn positive toward the end of a fund's life.

Jensen's Alpha A risk-adjusted performance measure that represents the average return on a portfolio or investment, above or below that predicted by the capital asset pricing model (CAPM), given the portfolio's or investment's beta and the average market return.

Junk bonds See *non-investment-grade bonds*.

K

Knock-in / knock-out products A type of barrier option that begins to function like a normal option contract, after certain a price level/trigger has been reached.

L

Lagging indicators Turning points that signal a change in economic activity after output has already changed.

Leading indicators Turning points that signal changes in the economy in the future, and thus are considered useful for economic prediction and policy formulation.

Legal risk The risk that an external party could sue an organisation for breach of contract or other violations.

Leverage ratio Measurements of a company's abilities to service its debts.

Liability A monetary obligation of a company as a result of previous events.

Limit price Ceiling price for a buy order and floor price for a sell order. A trade cannot be arranged at a price higher than the specified limit price when buying or a price lower than the specified limit price when selling.

Limited liability Liability that does not exceed an investor's initial contribution of capital. For example, shareholders are protected by limited liability, which means that higher claimants—particularly debt investors—cannot recover money from the personal assets of the shareholders if the company's assets are insufficient to fully cover their claims.

Liquidity risk The risk that an asset or security cannot be bought or sold quickly without a significant concession in price.

London Interbank Offered Rate (LIBOR) The most widely used reference rate, defined as the average interest rate that banks charge each other in the London interbank market. Also called Libor.

Long positions Positions for which investors own assets or securities. These positions increase in value when prices rise.

Long-term debt Money borrowed from banks or other lenders that is to be repaid during periods of greater than one year.

M

Macroeconomics The branch of economics that studies the economy as a whole.

Maintenance margin . The amount of funds or equity that must be maintained in the margin account after the securities or derivatives were purchased and must be maintained with the broker at all times.

Management fees Fees that limited partners must pay general partners to compensate them for managing investments. Management fees are typically set as a percentage of the amount the limited partners have committed rather than the amount that has been invested.

Mark-to-market The process of the daily settling of gains and losses due to changes in the market value of a security or derivative contract.

Market equilibrium The price where quantity demanded equals quantity supplied. At the equilibrium price, demand and supply in the market are balanced, and neither buyers nor sellers have an incentive to try to change the price, all other factors remaining unchanged.

Market makers See *dealers*.

Market risk The risk caused by changes in market conditions affecting prices.

Maturity date Date when the borrower must repay the amount borrowed.

Microeconomics The branch of economics that studies how individuals and companies make decisions to allocate scarce resources.

Model risk The risk arising from the use of models and associated with inappropriate underlying assumptions, the unavailability or inaccuracy of historical data, data errors, and misapplication of models.

Modified Dietz Method A measure of the ex post performance of an investment portfolio in the presence of external flows.

Monetary Authority of Singapore (MAS) Acts as Singapore's central bank and is responsible for the conduct of monetary policy, the issuance of currency, the oversight of payment systems and serving as banker to and financial agent of the Government of Singapore.

Monetary policy Actions taken by a nation's central bank to affect aggregate output and prices through the money supply or credit.

Money laundering A process in which criminals use financial services to transfer money from illegal operations to other legal activities; the money becomes "clean" in the process.

Money market The part of financial market that deals in the borrowing and lending of short-term loans for less than a year.

Money market funds Special class of open-ended mutual funds that investors view as uninsured interest-paying bank accounts. Unlike other open-ended mutual funds, regulators permit money market funds to accept deposits and satisfy redemptions at a constant price per share if they meet certain conditions.

Moneyness Refers to the potential profit or loss from the immediate exercise of an option. Terms such as *in-the-money (ITM), out-of-the-money (OTM)* and *at-the-money (ATM)*, describe the moneyness of options.

Mortgage backed securities (MBS) Similar to asset backed securities, MBS are created by securitisation except that MBS are backed by a pool of underlying mortgages.

Multiplier effect An initial increase (decrease) in spending produces an increase (decrease) in GDP and consumption greater than the initial change in spending.

N

Net asset value (NAV) Total net value of a fund (the value of all assets minus the value of all liabilities) divided by the fund's current total number of shares outstanding.

Net book value Calculated as the gross value of an asset minus accumulated depreciation, where accumulated depreciation is the sum of all reported depreciation expenses for the particular asset.

Net exports The difference between exports and imports of goods and services; also called balance of trade or trade balance.

Net income The difference between revenue and expenses; income available to distribute to shareholders.

Net present value The present value of future cash flows minus the cost of the investment, or the present value of cash inflows minus the present value of cash outflows.

Net profit margin A profitability ratio that indicates how much (percentage) of each monetary unit of revenue is left after all costs and expenses are covered.

Non-deliverable forwards (NDFs) Forward contracts where two parties agree to take on opposite sides of a transaction for a set amount of money. NDFs are usually short-tern, cash-settled and the notional amount is never exchanged, hence the name "non-deliverable." The profit or loss is calculated on the notional amount of the agreement by taking the difference between the agreed-upon rate and the spot rate at the time of settlement.

Non-investment-grade bonds Bonds rated BB+ or lower by Standard & Poor's and Fitch and Ba1 or lower by Moody's. Also called high-yield bonds or junk bonds.

Non-tariff barriers A range of measures, such as certification, licensing, sanctions, or embargoes, that make it more difficult and expensive for foreign producers to compete with domestic producers.

Normal distribution A symmetrical distribution in which the mean, median, and mode are the same value. The distribution is completely described by its mean and variance (or standard deviation).

Notional value Refers to the value of the underlying asset in a derivatives trade, or monetary value upon conversion. It can be the total value of a position, how much value a position controls, or an agreed-upon amount in a contract.

O

Offer prices See *ask prices*.

Open-ended mutual funds Pooled investment vehicles that have the ability to issue or redeem (repurchase) shares on demand. When investors want to invest in a mutual fund, the fund issues new shares in exchange for cash that the investors deposit. When existing investors want to withdraw money, the fund redeems the investors' shares and pays them cash.

Open market operations Activities that involve the purchase and sale of government bonds by a central bank.

Operating income Income generated by the company from its usual business activities before taking into account financing costs and taxes. It is often referred to as earnings before interest and taxes (**EBIT**).

Operating profit margin A profitability ratio calculated as operating income divided by revenue.

Operational risk The risk of losses from inadequate or failed people, systems, and internal policies and procedures, as well as from external events that are beyond the control of the organisation but that affect its operations.

Opportunity cost The cost of any activity measured in terms of the value of the best alternative that is not chosen; the value that investors forgo by choosing a particular course of action. For example, the cost of not having money to invest, spend, or hold; the cost of giving up opportunities to use money.

Option contract An agreement in which the buyer of the option has the right, but not the obligation, to trade the underlying asset.

Options on futures An option contract that gives the holder the right to buy / sell a futures contract on an underlying asset, at a specified price.

Option premium The amount paid by the option buyer to the option seller, at the initiation of the option contract, to compensate option sellers for their risk.

Orders Instructions that people who want to trade give brokers or trading venues and that specify what security to trade, whether to buy or sell, and how much should be bought or sold. They also have other instructions attached to them, such as order execution, exposure, and time-in-force instructions.

Out-of-the-money option An option is out-of-the-money when there is a negative profit from exercising the option. Also see *in-the-money option* or *at-the-money option.*

Over-the-counter (OTC) markets Another name for quote-driven markets dating from when securities were literally traded over a counter in the dealer's office.

P

Par value The stated value or face value of a security; the amount the investor would be entitled to receive in a liquidation scenario, which also serves as the principal value on which coupon payments are calculated.

Passive investment managers Managers who follow a buy-and-hold approach and seek to match the return and risk of a benchmark.

Penny Stock A common stock valued at less than one dollar, and therefore highly speculative.

Performance attribution The process of identifying the source of a fund manager's return. Potential sources include asset allocation, sector selection, stock selection, and currency exposure.

Performance measurement The process of measuring the performance of investments, including the calculation of reward-to-risk ratios.

Perpetual bond A bond with no maturity date

Physical capital The means of production; tangible goods such as equipment, tools, and buildings.

Plain vanilla option Also known as a standard option. See *Option contract*.

Policies Principles of action adopted by a company.

Political risk The risk of a change in the ruling political party of a country leads to changes in policies that can affect everything from monetary and fiscal policies to investment incentives, public investments, and procurement.

Pooled investments Indirect investment vehicles in which investors pool their money together to gain the advantages of being part of a large group. The resulting economies of scale can significantly improve investment returns.

Portfolio The group of assets in which savings are invested.

Portfolio management The act and process of management investment portfolios, which involves making decisions about investment mix and policy, matching investments to objectives, asset allocation and balancing risk against performance.

Portfolio Theory A mathematical framework for assembling a portfolio of assets such that the expected return is maximized for a given level of risk.

Position Quantity of an asset or security that a person or institution owns or owes.

Preferred stock aka **preference shares**; a type of equity security that ranks between debt securities and common stock. It typically does not carry voting rights but has priority over common stock in the receipt of dividends.

Present value The present discounted value of future cash flows.

Price-earnings Growth ratio A valuation metric that factors in growth expectations and helps determine whether a stock is under- or overvalued given its expected growth rate.

Price-to-book value ratio The ratio of a company's share price to its book value per share to determine whether a stock is appropriately priced relative to the value of its assets.

Price-to-earnings ratio The ratio of a company's share (market) price to its earnings per share.

Price-to-cash flow ratio A metric to determine whether a stock is appropriately priced using the market value per share divided by the operating cash flow per share.

Price-to-sales ratio A metric to determine whether a stock is appropriately priced. It is calculated by dividing market value per share by revenue per share.

Price-weighted index Indices for which the weight assigned to each security is determined by dividing the price of the security by the sum of all the prices of the securities.

Price discovery The process or act of price setting as market participants absorb and react to the flow of market information.

Primary market Markets in which issuers, such as companies and governments, sell their securities to investors.

Principal The money initially lent on which interest is paid.

Private equity A type of alternative investment that consists of investing in private companies—that is, companies that are not listed on a stock exchange.

Private placement Type of primary market transaction in which companies sell securities directly to a small group of qualified investors, usually with the assistance of an investment bank.

Procedures What the company must do to achieve a desired outcome.

Processes Individual steps that a company must take, from start to finish, to achieve a desired outcome. They divide procedures into manageable actions.

Proprietary trading Direct trading by traders with their clients rather than by arranging trades with others on behalf of their clients.

Prospectus Document that discloses the investment policies, deposit and redemption procedures, fees and expenses, and past performance statistics associated with an investment vehicle.

Protective put A risk management or hedging options strategy that involves buying the underlying asset (e.g., a stock or share) and purchasing a put option with a strike price.

Put option The right (but not the obligation) to sell the underlying at the exercise price until expiration.

Put-call parity The relationship that exists between European call and put prices for the same underlying stock, strike price and expiration month. The put-call parity formula is: C + PV (X) = p + S

Put spread An options strategy that is created with put options.

Put warrant See *Structured warrant.*

Q

Quick ratio A liquidity ratio that indicates a company's ability to satisfy current liabilities with its most liquid assets.

R

Ratio spread An options trading strategy that buys/sells option contracts in specified ratios.

Real assets Physical assets, such as land, buildings, machinery, cattle, and gold.

Real estate Land and buildings. Also a type of alternative investment that consists of direct and indirect investments in land and buildings.

Real estate equity funds Often open-ended funds that hold investments in hundreds of commercial properties.

Real estate investment trusts (REITs) Public companies that mainly own, and in most cases operate, income-producing real estate.

Rebalancing The process of adjusting the weights of the constituent securities in an index or the weights of assets in a portfolio.

Recession A period during which real GDP decreases (i.e.;, negative growth) for at least two successive quarters, or a period of significant decline in total output, income, employment, and sales lasting more than a few months.

Receivables turnover ratio An accounting measure used to quantify a company's effectiveness in collecting its receivables or money owed by clients.

Regulations Rules that set standards for conduct and carry the force of law.

Reinvestment risk Risk that in a period of falling interest rates, the periodic coupon payments received during the life of a bond and/or the principal payment received from a bond that is called early must be reinvested at a lower interest rate than the bond's original coupon rate.

Relative returns The difference between holding-period returns (absolute returns) and returns on a benchmark over the same holding period.

Repackaging In the context of derivatives, repackaging refers to combining derivatives and other financial instructions (e.g., debt instruments) to provide customised structured products.

Repurchase Agreements Also known as a repo, or a sale and repurchase agreement, is the sale of securities together with an agreement to buy them back at a future date, in return for a short-term loan. The repo seller, who is receiving the loan, is offering securities as collateral against the loan.

Reserve currency A currency held in significant quantities by many governments and institutions as part of their foreign exchange reserves.

Reserve requirement The requirement for banks to hold reserves in proportion to the size of deposits.

Residual claimants Investors whose claims rank last. Common shareholders are residual claimants. In the event of a company's liquidation, common shareholders share proportionately in the remaining company assets after all other claimants have been satisfied.

Retail Investor An individual investor who purchases securities for their own personal account rather than on behalf of organization. They typically trade in much smaller amounts than accredited or institutional investors.

Retained earnings The accumulated net income that is retained by the company rather than distributed to its owners (shareholders) as dividends.

Return on assets (ROA) A profitability ratio that indicates a company's net income generated per monetary unit invested in total assets.

Return on equity (ROE) A profitability ratio calculated as net income divided by average shareholders' equity.

Revenues The amount charged (and expected to be received) for the delivery of products or services in the ordinary activities of a business.

Reverse Stock Split A process by which shares of corporate stock are effectively merged to form a smaller number of proportionally more valuable shares.

Rho Measures the change in option price vs change in interest rates.

Rights Issue An issue of shares offered at a special price by a company to its existing shareholders in proportion to their holding of old shares

Risk The effect of uncertain future events on an organisation or on the outcomes the organisation achieves.

Risk appetite An organisation's willingness to take on risk, which depends on its attitude toward risk and on its risk culture.

Risk management An iterative process used by organisations to support the identification and management of risk (or uncertainty) and reduce the chances and/or effects of adverse events while enhancing the realisation of opportunities and the ability to achieve company objectives.

Risk tolerance The level of risk an organisation or person is able and willing to take on.

S

Seagull strategy A moderately bearish options strategy that involves 3 legs, either two call options and a put option or two puts and a call.

Secondary market Market in which investors trade securities and contracts with each other but not with the original security issuer.

Securities Financial assets that can be traded, such as shares and bonds.

Securities and Futures Act 2001 (SFA) The primary source of securities law that regulates activities and institutions in the securities, futures and derivatives industry in Singapore, including dealing in capital markets products, clearing facilities, and related matters. The Monetary Authority of Singapore is the regulatory body responsible for the administration of the SFA.

Securitisation Creation of new financial products by buying and repackaging securities or other assets; the creation and issuance of new debt securities that are backed by a pool of other debt securities.

Settlement Activity that consists of the final exchange of cash for securities following a trade.

Settlement risk The risk that when settling a transaction, a firm performs one side of the deal but the counterparty does not complete its side of the deal as agreed, often because it has declared bankruptcy.

SGX Catalist The first sponsor-supervised listing platform in Asia and caters to the needs of smaller or fast-growing companies.

SGX Mainboard A listing platform that caters to the needs of more established companies with higher entry and listing requirements in the Singapore Exchange.

Share buyback See *share repurchase*.

Share repurchase A transaction in which a company uses its cash to buy back its own shares from existing

shareholders. This transaction reduces the number of shares outstanding.

Shareholders The owners of shares (stock) of a corporation.

Sharpe ratio A reward-to-risk ratio defined as the excess portfolio return (portfolio return minus risk-free return) over the standard deviation of portfolio returns.

Short positions Positions for which investors sell assets or securities that they do not own, a process that involves borrowing the assets or securities, selling them, and repurchasing them later to return them to their owner. These positions increase in value when prices fall.

Sinking Fund Provision A provision in some bond indentures requiring the issuer to put money aside to repay bondholders at maturity.

Skewed A distribution is skewed when the bulk (majority) of the values (possibly including the median) lie either to the right or to the left of the mean; the distribution is not symmetrical.

Sovereign risk The risk that a foreign government will not repay its debt because it does not have either the ability or the willingness to do so.

Sovereign wealth funds Funds created by governments to invest surpluses for the benefit of current and future generations of their citizens.

Specific (non-market) risk Refers to the risk that is specific to a particular company or industry. Also known as *non-systematic risk* or *residual risk.*

Speculate In finance, the act of speculating, or speculative trading, involves conducting a financial transaction with a view on the future direction of the market. The transaction has the risk of losing value but also holds the expectation of a significant gains.

Spot market Foreign exchange market in which currencies are traded now and delivered immediately.

Spot rate The exchange rate for spot market transactions.

Stack A futures stack uses the deferred contract months to match the rollover dates or tenor on a cash loan. Also see *Strip*.

Standard deviation A measure, in the same units as the original data, of the variability, volatility, or dispersion of a data set around the average value of that data set (i.e.; the arithmetic mean). It is the positive square root of the variance.

Statement of cash flows A financial statement that identifies the sources and uses of cash over a time period and explains the change in the company's reported cash balance during the period.

Stock dividend A transaction in which a company distributes additional shares of its common stock to shareholders instead of cash. This transaction increases the number of shares outstanding but does not affect the company's value because the stock price decreases accordingly.

Stock split A transaction in which a company increases the number of shares outstanding. For example, in a two-for-one stock split, the company doubles the number of shares outstanding, and the stock price is halved, but the company's value is unaffected.

Straddle A (long) straddle is a market-neutral options strategy that involves buying a call option that is at-the-money and buying a put option that is at-the-money. The straddle investor expects the market to experience significant volatility in the near term.

Strangle (long) A (long) strangle is a market-neutral options strategy that involves buying a call option that is out-of-the-money and buying a put option that is out-of-the-money. The strangle investor expects the market to experience significant volatility in the near term.

Strangle (short) A (short) strangle is a market-neutral options strategy that involves selling a call option that is out-of-the-money and selling a put option that is out-of-the-money. The strangle investor expects the market to experience lower significant volatility in the near term, and the underlying asset price to stay range bound.

Strategic asset allocation The long-term mix of assets that is expected to achieve the client's long-term objectives, given the client's investment constraints.

Strike price See *exercise price*.

Strip A futures strip is the buying or selling of futures contracts in sequential delivery months traded as a single transaction.

Structured deposit Unlike structured notes, structured deposits are a type of deposit and are not debentures (i.e.; unsecured debt). A structured deposit is essentially a combination of a deposit and an underlying financial instrument, where the return is linked to the

performance of the underlying financial instrument. In Singapore, the principal sum of structured deposits has to be repaid in full at maturity, with or without interest.

Structured fund A type of fund that is designed to provide investors with some degree of capital preservation or capital appreciation. Structured funds typically combine both equity, fixed income and derivative components.

Structured note A debt obligation that contains an embedded derivative component with characteristics that adjust the security's risk/return profile.

Structured product, also known as a market-linked investment, is a pre-packaged structured finance investment strategy based on a single security, a basket of securities, options, indices, commodities, debt issuance or foreign currencies, and to a lesser extent, derivatives. Structured products are designed to provide investors with more tailored investment alternatives linked to a specific risk /reward profile or a hedging strategy options that addresses a view about the direction or volatility of an underlying asset or index.

Structured warrant A form of structured investment products issued by a third-party financial institution over a wide range of assets, including the shares of an un-related listed company, a basket of companies' shares or an index. Structured warrants can be either *call warrants* or *put warrants.*

Swaps Derivatives in which two parties swap cash flows or other financial instruments over multiple periods (months or years) for mutual benefit, usually to manage risk.

Swaptions (also known as *swap options*) An option contract to enter into a swap agreement with another party.

Synthetic position A synthetic position refers to a financial product or structured product that is engineered to simulate other instruments while altering key characteristics.

Systematic risk Also known as market risk, it is the risk created by general economic conditions that affect all risky investments. Systematic risk factors include changes in macroeconomic conditions, interest rate risk, and political risk, among others.

Systemic failure Failure of the financial system as a whole, including loss of access to credit and collapse of capital markets.

T

Tactical asset allocation The decision to deliberately deviate from the strategic asset allocation in an attempt to add value based on forecasts of the short-term relative performance of asset classes.

Tariffs Taxes (duties) levied on imported goods and services, which allow governments not only to establish trade barriers but also to raise revenue.

TED spread The difference between the price of the 3-month futures contract for US Treasuries and 3-month contracts for Eurodollars, with the same expiration month.

Technical analysis A trading discipline applied to evaluate investments and identify trading opportunities in price trends and patterns on charts.

Term sheet See *Contract specifications.*

Term structure of interest rates The relationship between the yields to maturity offered by government bonds and the maturities of these government bonds.

Terminal value, or horizon value, determines the value of a business or project beyond the forecast period when future cash flows can be estimated**.**

Theta Measures the change in option price vs change in time to maturity.

Time value (of an option) The additional amount an investor is willing to pay over the current intrinsic value. Investors are willing to pay this because an option could increase in value before its expiration date.

Time-weighted rate of return A measure of investment performance in which the overall measurement period is divided into sub-periods. The timing of each individual cash flow is identified and then defines the beginning of the sub-period in which it occurs.

Time spread See *Horizontal spread.*

Total asset turnover ratio A measure on the value of a company's sales or revenues relative to the value of its assets. It can be used as an indicator of the efficiency with which a company is using its assets to generate revenue.

Tracking error The standard deviation of the differences between the deviation over time of the returns on a

portfolio and the returns on its benchmark; a synonym of active risk.

Trade balance See *net exports*.

Trade bills A bill of exchange that is used to pay for goods.

Trade deficits When the value of exports is lower than the value of imports.

Trade surplus When the value of exports is higher than the value of imports.

Transaction costs Costs associated with trading, which include explicit costs (mainly brokerage commissions) and implicit costs (bid–ask spreads, price impact, and opportunity costs).

Treasury Bill (**T-Bill**) A short-term debt obligation backed by the U.S. Treasury Department with a maturity of one year or less. T-bills are usually sold in denominations of $1,000.

Treasury bond (T-bond) A marketable, fixed-interest US government debt security with a maturity of more than 10 years. T-bonds make interest payments semi-annually, and the income received is only taxed at the federal level.

Treasury futures Futures contracts that have Treasury bills or Treasury bonds as the underlying asset.

Treasury options Options contracts that have Treasury bills or Treasury bonds as the underlying asset.

Treynor ratio A reward-to-risk ratio defined as the excess portfolio return (portfolio return minus risk-free return) over the beta of portfolio returns.

Trust Deed A legal document that defines the *trust* such as the trustee, beneficiaries, settlor and appointer, and the objectives and terms and conditions of the trust.

U

Underwriters The (lead) investment bank or brokerages guarantee the sale of the securities at an offering price that is negotiated with the issuer of a public offering.

Unit trust An unincorporated fund comprising investors' money, which is invested in a variety of financial assets to meet the unit trust's objectives set by a fund manager.

V

Value at risk An estimate of the minimum loss of value that can be expected for a given period within a given level of probability.

Variable costs Costs that fluctuate with the level of output of the company.

Variable-rate bonds See *floating-rate bonds*.

Variance A measure of dispersion that is equal to the standard deviation squared (i.e.; the standard deviation multiplied by itself).

Vega Measures the change in option price vs change in volatility.

Venture capital Private equity investment strategy that consists of financing "start-up" companies that exist merely as an idea or a business plan.

Vertical spread An options trading strategy that involves buying and selling multiple options of the same underlying security and same expiration date, but at different strike prices.

Voting rights The rights of shareholders to vote—for example, to elect the members of the board of directors.

W

Warrant An equity-like security that entitles the holder to buy a pre-specified amount of common stock of the issuing company at a pre-specified stock price prior to a pre-specified expiration date.

Warrant premium The difference between the warrant price and intrinsic value.

Weighted Average Cost of Capital (WACC) The rate that a company is expected to pay on average to all its security holders to finance its assets.

Y

Yield curve Term structure of interest rates presented in graphical form.

Yield to maturity The discount rate that equates the present value of a bond's promised cash flows to its market price.

Z

Zero-coupon bonds Bonds that do not offer periodic interest payments during the life of the bond. The only cash flow offered by a zero-coupon bond is a single payment equal to the bond's par value to be paid on the bond's maturity date.

Zero Cost Collar A protective options strategy that involves buying a put option and selling an out-of-the-money call. The call option is sold with a strike price where the premium received is equal to the premium paid for the put purchased. Also see *Costless collar*.

Zero strike warrant A type of warrant that is essentially a synthetic stock, and has an exercise price of zero, which means that it is always in-the-money. The cash settlement and price of this warrant is equal to the closing price of the underlying stock.



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